

# Geometric $q$ -Fabius–Rvachev Frontiers

Exponent sequences, sinc products, atomic splines,  
and parameter deformations of the Fabius–Rvachev system

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## Provenance

This volume is the single consolidated frontier document for the geometric  $q$ -deformation of the Fabius–Rvachev system. Parts I–VII continue the seven-part synthesis assembled on 28 August 2026; Parts VIII–XII absorb the five further reports listed below, which previously stood as separate packages under `q-fabius-parameter-deformations/` and `geometric-sinc-and-exponent-families/`. Labels, citation keys, and asset paths are mechanically prefixed per part, `p1:–p12:`, with `p0:` reserved for the common framework of Part 0. Shared definitions, normalizations, and background results are stated once in Part 0; each absorbed part’s own preliminaries fix only its local letters and record the exact dictionary to the canonical normalization. The absorbed sources are deleted from the working tree once merged: git history is the archive, and the SHA-256 digests below pin the absorbed snapshots.

## Sources absorbed into Parts I–VII

- `Fabius_Newton_Rvachev_Frontier_Report/Exponent_Sequence_Newton_Rvachev_Frontiers.tex`  
SHA-256 `2c068211b3e5f85f410976f89eda3616b3d48c1e1817dabdded0db5e41429a22`
- `fabius_frontier_results/q_richardson_sinc_frontier.tex`  
SHA-256 `3b612cf8bbc34c4bfa9b3fbefbee14c618ff2edbbe84bc38d7cbe6baece8fcf4`
- `finite_sinc_products_report/finite_sinc_products_report.tex`  
SHA-256 `e01c3a2716d1c28ba6982510870775490981056b3f97b2a2b3f79002b2ef11b1` (eighth wave, folded in 2026-08-28)
- `Rvachev_Piecewise_Approximation_Fourier_Images/Rvachev_Piecewise_Approximation_Fourier_Images.tex`  
SHA-256 `e70c2dd20b1ad71d44096d9c5c462596e36be202833bdbb5235efd4fba8355fd` (ninth wave; merged editorially into Part IV)
- `rvachev_fourier_frontier_report/rvachev_piecewise_fourier_frontier.tex`  
SHA-256 `23acde817ac19802ffed9cc19e80a80368d70315f87edf41e9fdc3f54c2a970b` (ninth wave; merged editorially into Part IV)
- `fabius_finite_products_frontier/finite_sinc_frontier.tex`  
SHA-256 `e7d436a8a236e6c04f5b7a56dd846ea42aada67c12b0e2ff4dc03f45d3b31e48` (tenth wave; Part V)
- `atomic_sinc_splines_report_package/atomic_sinc_splines_report.tex`  
SHA-256 `1c65e30feb6589de33512ce659b87261f240c538813c454efc7d5eef4be54f6f` (thirteenth wave; Part VI — translation and frontier expansion of Rvachev’s Chapter 3)
- `Atomic_Functions_Beyond_Dyadic_Report/Atomic_Functions_Beyond_Dyadic.tex`  
SHA-256 `021d14ffcac62fa18a9a11ba48dce4bb1c58f8da9ffa6b7c1f47f659f112c8b6` (fourteenth wave; merged editorially into Part VI — second independent reconstruction of the same chapter, contributing the fractal-string/tube-formula, degree-law, parameter-limit, general-base Gamma-zeta Laplace, and Fup-ladder layers)
- `Rvachev_Atomic_Functions_Report/Atomic_Functions_Rvachev_Report.tex`  
SHA-256 `6a7f769197b695ce6f73510ecbead96997e75ae3856e5372e93950169b96a662` (fifteenth wave; merged editorially into Part VI — third independent reconstruction of the same chapter, contributing the signed gap leading coefficients, derivative equimeasurability with the full  $L^p$  ladder, the classical  $\text{Fup}_n$  hierarchy with its triangular reconstruction mask and Gaussian regime, and the edge pantograph equations)

- `Fabius_Q_Connections_Report/fabius_q_connections_report.tex`  
SHA-256 5e45c732e77a72616c37d5d4d0617b065bb43fe2b525631a6bb2e589cd869f29 (sixteenth wave, “Beyond the Dyadic Fabius Web”; merged editorially into Part VII with its seventeenth-wave twin)
- `Signed_Reciprocal_q_Fabius_Frontiers/Signed_Reciprocal_q_Fabius_Frontiers.tex`  
SHA-256 ba843c7a4e0cc250e889c398ab5b72f9ade276f6ad9587b56cf22a4a5e7e2db8 (seventeenth wave; merged editorially into Part VII with its sixteenth-wave twin — signed and reciprocal parameter orbits of the geometric-uniform family)
- `Atomic_Functions_Beyond_Dyadic_Report-2/Atomic_Functions_Beyond_Dyadic.tex`  
SHA-256 c560d350d80655759a463c7930b8d1131137d214c40d834f3b7e95d2ca253b11 (revised fourteenth-wave edition; merged into Part VI — Orlicz/rearrangement equimeasurability and the spectral Stieltjes–Wigert bridge; shipped the Russian source scan `source_rvachev_scan.pdf`, SHA-256 ad4b0078dc1d5de133e5f2bf8f673ea080197c16a624d4fec7cb5126c59c6320, and its raw OCR transcription `source_rvachev_ocr.tex`, SHA-256 c1a5074f2b3099f5d48b166ab43bc2726139282fe174537c60bc1ae4010be46b; both were deleted after their recoverable content was merged and checked — the repair ledger is in Part VI’s concordance appendix, and git history archives the files)
- `Atomic_Functions_Beyond_Dyadic_Report-3/Atomic_Functions_Beyond_Dyadic.tex`  
SHA-256 8b846f7e6e32efd926bcd770ae1aa195ba051d1db81b7ee45857e8eb4d07ba51 (second revised fourteenth-wave edition; merged into Part VI — the Mellin law of the distance to  $K_a$  with the shifted complex-dimension pole lattice and the critical logarithmic gap scale)
- `Atomic_Functions_Rvachev_Report_Package/Atomic_Functions_Rvachev_Report.tex`  
SHA-256 4db6ab0d19ecab7ef9291910548f5a26aaf9d5380b9e7be094586f351eeb01f1 (revised fifteenth-wave edition; merged into Part VI — the  $q$ -Gaussian derivative Gram kernel with Pochhammer determinants and sharp theta Riesz bounds, the log-Weibull jet-intermittency law, and the proof of the uniform all-orders  $\text{Fup}_n$  Edgeworth expansion)
- `Atomic_Functions_Rvachev_Expanded_Report/Atomic_Functions_Rvachev_Expanded_Report.tex`  
SHA-256 690c7991e45e29dfe4e3123837a4e1955aaa690967f47d6fdcfce6b5e0c61236 (expanded fifteenth-wave edition, audit-aware — it explicitly marks previously merged layers as inherited baseline; its new layer is merged into Part VI: the closed Gaussian-binomial Gram–Schmidt orthogonalization with Rogers–Szegő identification, explicit Cholesky and inverse Gram, the wrapped-heat-kernel circle model, the MacMahon determinant constant with triple-product Riesz bounds, and the overlap-regime theta conjecture; its figures and data are kept under `assets/Atomic_Functions_Rvachev_Expanded_Report/`)
- `Atomic_Functions_Beyond_Dyadic_Expanded/Atomic_Functions_Beyond_Dyadic_Expanded.tex`  
SHA-256 6bfb9891baf7acb68e22e30797b7c7c0ba91d84dfaf77b8ce695d16838ea7f3f6 (expanded fourteenth-wave edition, audit-aware; its new layer is merged into Part VI: the physical-space Stieltjes–Wigert differential ladder  $\Upsilon_{a,n} = P_{a,n}(\mathcal{L})h_a$  with closed norms,  $q$ -binomial derivative expansion, and three-term operator recurrence, the derivative-jet Gram determinants of both parities, the autocorrelation germ with zero Taylor radius, and the explicit-spectral-null-modes conjecture; it re-shipped byte-identical copies of the recorded source scan and OCR, again not retained; figures and data under `assets/Atomic_Functions_Beyond_Dyadic_Expanded/`)
- `Atomic_Functions_Beyond_Dyadic_Frontiers/Atomic_Functions_Beyond_Dyadic`

c\_Frontiers.tex

SHA-256 972e76445ebbae00cf90b9396b8413a65565292d088437a185727c80d63c2ed1 (second expanded fourteenth-wave edition, audit-aware; its new layer is merged into Part VI: the exact derivative-energy factorization with Bernoulli-convolution limit,  $W_\infty$  rate, exact Hausdorff support rate, and Rényi/Shannon entropy laws, plus the overlap-regime derivative-energy conjecture; it likewise re-shipped byte-identical copies of the source scan and OCR, again not retained; figures and data under assets/Atomic\_Functions\_Beyond\_Dyadic\_Frontiers/)

- Atomic\_Functions\_Rvachev\_Report\_Expanded/Atomic\_Functions\_Rvachev\_Report\_Expanded.tex

SHA-256 d6426d4e7b609d9b97be8dc14c87ea8c56178dcda15d0c1ee5dfb158cb79c2cc (expanded fifteenth-wave edition, audit-aware; its new layer is merged into Part VI: the two-term jet tail with the sharp exponential-Orlicz threshold, the Schur-minor strict total positivity of the  $q$ -Gaussian kernel, the exact inverse Gram–Schmidt transform, and the dual-tower and staircase-limit-set conjectures; it re-shipped byte-identical copies of the source scan, the OCR, and the two previous editions of its lineage (SHA-256 all previously recorded), none retained; figures and data under assets/Atomic\_Functions\_Rvachev\_Report\_Expanded/)

- Atomic\_Functions\_Rvachev\_qBinomial\_Frontiers/Atomic\_Functions\_Rvachev\_qBinomial\_Frontiers.tex

SHA-256 671b10aa47754317dfa0cab5bc24b7217cb7aeff5a34b224ffcbba00b08b7732 (second expanded fifteenth-wave edition, audit-aware; its new layer is merged into Part VI: the nodal-polynomial reading of the Gram–Schmidt residuals with the explicit two-sided inverse and positive Cholesky, the minimum-phase theta whitening filter with the exact identity  $|A_q|^2 \vartheta_3 = (q^2; q^2)_\infty$ , the highest-jet partial-theta law with the joint jet–distance transform, and the finite-section, partial-theta-pole, and graph-directed-prediction conjectures; it likewise re-shipped byte-identical scan/OCR copies, not retained; figures and data under assets/Atomic\_Functions\_Rvachev\_qBinomial\_Frontiers/)

## Sources absorbed in the present consolidation

- geometric-sinc-and-exponent-families/Fabius\_Rvachev\_Frontier\_Report/fabius\_frontier\_report.tex  
SHA-256 beb6ac1072d0c59df3ca5ab0d9950cb7d31c218ca6c66108a095d072193b3e31 (negative-parameter affine duality, the exact negative dyadic law with Thue–Morse derivative peaks, log-concavity and dominant-digit plateaux, residue-class multiplication formulas, reciprocal-base normalization, and the quantitative central limit theorem at  $|q| \uparrow 1$ ; the twin of Part VII)
- geometric-sinc-and-exponent-families/Cyclotomic\_q\_Fabius\_Rvachev\_Frontier/cyclotomic\_q\_fabius\_frontier.tex  
SHA-256 5074d72a4be4bbdba0e4ccd79ed409ca79b8a43d9744b67be82d028155b27065 (root-of-unity radial expansions, the sign of the cyclotomic action and the natural-boundary theorem, cyclotomic blow-ups to all orders, spectral zero condensation, Bell/moment Appell collapse, polyharmonic heat symbols, and the two inverse branches)
- q-fabius-parameter-deformations/fabius\_q\_frontiers\_report/q\_fabius\_parameter\_frontiers.tex  
SHA-256 f875dfca180556d923b6e38ef4151160f4e8d6a689ac9c684c521e62a8bc813d (the exact parameter-flow kernel and its martingale coupling, the midpoint  $q$ -integral form of the scaled cumulant transform, an all-orders midpoint Euler–Maclaurin expansion, the large-deviation principle as  $q \uparrow 1$ , the Lambert- $W_{-1}$  rate boundary, and a uniform all-orders Edgeworth expansion)
- q-fabius-parameter-deformations/Continuous\_Parameter\_Edgeworth\_and\_q\_Gevrey\_Frontier/fabius\_frontier\_report.tex

SHA-256 e9d99619992f78050326249272b18f5941f659dea0f022522b23ec218953d5bf (all-orders Edgeworth expansion in Schwartz seminorms, CDF and Cornish–Fisher inversion, moderate deviations and the compact-scale large-deviation principle, Lambert- $W$  inversion at the limiting support edge, and quadratic  $q$ -Gevrey Denjoy–Carleman regularity for the atomic family)

- `q-fabius-parameter-deformations/Fabius_Flat_Parameter_Response_Dynamics/fabius_frontier_report.tex`

SHA-256 41fc1d2406689c89ea65452f6e4917eff9ccb2c8c75c1c0db421b763a4e5f010 (joint smoothness with flat nonanalytic parameter fronts, the first  $q$ -response and its tangent measure, a contractive susceptibility resolvent, exact refinement equations for the tangent CDF, the rational shifted-Legendre spectrum, and a conditional global smooth Schröder coordinate with a logarithmic Böttcher height)

# Part 0. Common Framework, Normalizations, and Shared Background

## 0.1 Purpose of this part

Each of the twelve numbered parts of this volume was written as an independent report, and each therefore opened by fixing the same objects for itself: the geometric-uniform family, its centering and scaling, its moment generating and characteristic functions, the Bernoulli convention behind its cumulants, and the sinc and Fourier normalizations. Those openings did not agree. Three different affine normalizations of one random variable appear across the volume, two different sinc arguments, and the same letters  $X_q$  and  $Y_q$  denote different — though affinely related — objects in different parts.

This part removes that redundancy. It states the shared material once, with complete proofs, in one fixed normalization, and then gives the exact dictionary from each numbered part’s local convention to the canonical one. Every later part now opens with a short conventions paragraph pointing here rather than with a re-derivation. Nothing in the mathematics of the absorbed reports is changed by this; what changes is that a formula carried from one part into another can be transported along a dictionary entry proved in [section 0.5](#) instead of by inspection.

Three points of usage. First, results are numbered  $0.n.m$  inside this part and  $N.n.m$  inside numbered Part  $N$ ; a reference of the form “[theorem 0.4.2](#)” therefore always resolves here. Second, all labels in this part carry the prefix  $p0$  :, continuing the per-part prefixing scheme described in the provenance section. Third, the notation is that of the shared corpus contract `docs/fabius-notation.tex` and of the notation catalogue; symbols introduced only for this volume are declared where they first appear and are collected in [section 0.9](#).

## 0.2 The canonical geometric-uniform family

**Definition 0.2.1** (Canonical geometric-uniform family). Let  $(U_j)_{j \geq 0}$  be independent random variables, each uniform on  $[0, 1]$ , and put  $V_j := U_j - \frac{1}{2}$ , so that each  $V_j$  is uniform on  $[-\frac{1}{2}, \frac{1}{2}]$  and symmetric. For real  $q$  with  $|q| < 1$  set

$$X_q := (1 - q) \sum_{j=0}^{\infty} q^j U_j, \quad X_q^\circ := X_q - \frac{1}{2} = (1 - q) \sum_{j=0}^{\infty} q^j V_j. \quad (0.2.1)$$

Write  $G_q(x) = \mathbb{P}(X_q \leq x)$  for the distribution function and  $g_q = G'_q$  for the density where it exists, and let

$$\sigma_q := \sqrt{\text{Var}(X_q)}, \quad X_q^* := \frac{X_q^\circ}{\sigma_q} \quad (0.2.2)$$

for the standardized variable. We call [\(0.2.1\)](#) the *canonical normalization*;  $X_q$  is the *uncentered*,  $X_q^\circ$  the *centered*, and  $X_q^*$  the *standardized* member of the family at parameter  $q$ .

The choice of  $[0, 1]$  for the seed variables is what makes  $q = 1/2$  reproduce the bounded Fabius function on the nose ([proposition 0.2.5](#)); the two competing normalizations that use seeds uniform on  $[-1, 1]$  differ from  $X_q^\circ$  by the single factor 2, recorded in [table 1](#).

**Proposition 0.2.2** (Convergence, mean, and support). *For every real  $q$  with  $|q| < 1$  the series in [\(0.2.1\)](#) converges absolutely almost surely and in every  $L^p$ ,  $1 \leq p < \infty$ . Moreover  $\mathbb{E}X_q = \frac{1}{2}$ , the law of  $X_q^\circ$  is symmetric about 0, and*

$$\text{supp}(X_q) = \begin{cases} [0, 1], & 0 \leq q < 1, \\ \left[-\frac{a}{1-a}, \frac{1}{1-a}\right], & q = -a, 0 < a < 1. \end{cases} \quad (0.2.3)$$

In both cases the support is a compact interval with midpoint  $\frac{1}{2}$ .

*Proof.* Since  $0 \leq U_j \leq 1$ , the partial sums are dominated termwise by  $|1 - q| \sum_j |q|^j = |1 - q| / (1 - |q|) < \infty$ , so the series converges absolutely for every realization;  $L^p$  convergence follows from the uniform bound  $|X_q| \leq |1 - q| / (1 - |q|)$  and dominated convergence. Because the weights  $(1 - q)q^j$  sum to 1 and  $\mathbb{E}U_j = \frac{1}{2}$ , monotone convergence gives  $\mathbb{E}X_q = \frac{1}{2}$ . Each  $V_j$  is symmetric and the  $V_j$  are independent, so  $-X_q^\circ$  has the same law as  $X_q^\circ$ .

For the support, write  $c_j(q) = (1 - q)q^j$ . The map  $(u_j) \mapsto \sum_j c_j(q)u_j$  is continuous from  $[0, 1]^{\mathbb{N}_0}$  with the product topology (compact) onto its image, and the image is the set of all such sums, hence the interval  $[\sum_j c_j(q)^-, \sum_j c_j(q)^+]$  with  $c^\pm = \max(\pm c, 0)$ ; every intermediate value is attained because the weights may be varied continuously. The law charges every nonempty relatively open subset of that interval, since the  $U_j$  have full support in  $[0, 1]$ . For  $0 \leq q < 1$  all  $c_j(q) \geq 0$  and  $\sum_j c_j(q) = 1$ , giving  $[0, 1]$ . For  $q = -a$  with  $0 < a < 1$ ,

$$\sum_{j \text{ even}} c_j(-a) = (1 + a) \sum_{i \geq 0} a^{2i} = \frac{1 + a}{1 - a^2} = \frac{1}{1 - a}, \quad \sum_{j \text{ odd}} c_j(-a) = -\frac{a(1 + a)}{1 - a^2} = -\frac{a}{1 - a},$$

which is (0.2.3). The two endpoints average to  $\frac{1}{2}(\frac{1}{1-a} - \frac{a}{1-a}) = \frac{1}{2}$ , and for  $0 \leq q < 1$  the midpoint of  $[0, 1]$  is again  $\frac{1}{2}$ , consistently with the symmetry of  $X_q^\circ$ .  $\square$

*Lean status.* Partial. Absolute convergence of the defining series is `Fabius.summable_norm_geometricUniformWeight` and `Fabius.hasSum_geometricUniformWeight`, with the sum continuous and measurable in the sample point (`Fabius.continuous_geometricUniformSeries`, `Fabius.measurable_geometricUniformSeries`); the support statement for  $0 \leq q < 1$  is `Fabius.geometricUniformSeries_mem_Icc` together with `Fabius.geometricUniformDistribution_support_eq_range`, and the symmetry of the centered law is `Fabius.geometricUniformSeries_reflect`, in distribution-function form `Fabius.geometricUniformCDF_reflection` (all in `GeometricUniformLaw.lean` and `GeometricUniformCDF.lean`). Not formalized as such: the  $L^p$  convergence clause, the mean  $\mathbb{E}X_q = \frac{1}{2}$  — the corpus states the median form `Fabius.geometricUniformCDF_one_half`, which is what the reflection gives — and the support for negative  $q$ .

**Proposition 0.2.3** (Affine fixed point and its uniqueness). *Let  $U$  be uniform on  $[0, 1]$  and independent of  $X'_q$ , an independent copy of  $X_q$ . Then*

$$X_q \stackrel{\text{law}}{=} (1 - q)U + qX'_q. \quad (0.2.4)$$

*Conversely, for every real  $q$  with  $|q| < 1$ , the law of  $X_q$  is the unique probability measure with finite first moment satisfying (0.2.4).*

*Proof.* Splitting off the term  $j = 0$  in (0.2.1) gives  $X_q = (1 - q)U_0 + q((1 - q) \sum_{j \geq 0} q^j U_{j+1})$ , and the bracket is an independent copy of  $X_q$ ; this is (0.2.4).

For uniqueness, let  $\mathcal{P}_1(\mathbb{R})$  carry the Wasserstein-1 metric  $W_1$ , under which it is complete. Define  $T(\nu) = \text{Law}((1 - q)U + qZ)$  with  $Z \sim \nu$  independent of  $U$ . If  $Z \sim \nu$  and  $Z' \sim \nu'$  are coupled optimally, then  $((1 - q)U + qZ, (1 - q)U + qZ')$  — with the same  $U$  — is a coupling of  $T(\nu)$  and  $T(\nu')$ , whence

$$W_1(T(\nu), T(\nu')) \leq \mathbb{E} |q(Z - Z')| = |q| W_1(\nu, \nu').$$

Thus  $T$  is a contraction with ratio  $|q| < 1$  and has exactly one fixed point in  $\mathcal{P}_1(\mathbb{R})$ . By the first paragraph  $\text{Law}(X_q)$  is a fixed point, and it has finite first moment by [proposition 0.2.2](#).  $\square$

*Lean status.* Both clauses are formal. The fixed-point identity is `Fabius.geometricUniformDistribution_selfSimilar`, with the convolution form `Fabius.geometricUniformDistribut`



ion\_conv in GeometricUniformDictionary.lean; the converse — that it characterizes the law — is Fabius.eq\_geometricUniformDistribution\_of\_selfSimilar in GeometricUniformUniqueness.lean, for every probability measure on  $\mathbb{R}$  satisfying the same push-forward equation, at every ratio with  $|q| < 1$ .

*Remark 0.2.4.* [proposition 0.2.3](#) needs no support or density hypothesis, only a finite first moment; this is the form used in [appendix 2](#) and in Part VIII. The finite-moment restriction cannot simply be dropped: without it the contraction argument has no complete metric to run in.

**Proposition 0.2.5** (The Fabius–Rvachev point). *At  $q = \frac{1}{2}$  the distribution function  $G_{1/2}$  is the bounded Fabius function  $F_{\text{bd}}$ , and the density is*

$$g_{1/2}(x) = 2 \operatorname{up}(2x - 1), \quad x \in [0, 1], \quad (0.2.5)$$

where  $\operatorname{up}$  is the Rvachev up-function, normalized to be supported on  $[-1, 1]$  with  $\int \operatorname{up} = 1$ .

*Proof.* At  $q = \frac{1}{2}$ , (0.2.4) reads  $X_{1/2} \stackrel{\text{law}}{=} \frac{1}{2}U + \frac{1}{2}X'_{1/2}$ , which is exactly the defining self-similarity of the Fabius distribution; by [proposition 0.2.3](#) the solution with finite mean is unique, and  $F_{\text{bd}}$  is by definition the distribution function of that solution. For the density,  $X_{1/2}^\circ = X_{1/2} - \frac{1}{2}$  satisfies  $X_{1/2}^\circ = \sum_{j \geq 0} 2^{-(j+1)} V_j$  with  $V_j$  uniform on  $[-\frac{1}{2}, \frac{1}{2}]$ ; scaling by 2 turns this into  $2X_{1/2}^\circ = \sum_{j \geq 0} 2^{-j} V'_j$  with  $V'_j$  uniform on  $[-1, 1]$ , i.e. into the standard Rvachev series, whose density is  $\operatorname{up}$ . Undoing the two affine steps gives (0.2.5).  $\square$

*Lean status.* The identification of the distribution function is formal: at  $q = \frac{1}{2}$  the law is the classical dyadic weighted-sum law (Fabius.weightedSumDistribution\_eq\_geometricUniformDistribution\_one\_half, ProbabilityRepresentation.lean), whose distribution function is the bounded Fabius function (Fabius.weightedSumDistribution\_real\_Iic\_eq\_fabiusReal, IncompleteBetaMaster.lean). The density display (0.2.5) is not stated in that form; what the corpus carries about the density is the general Fabius.geometricUniformDensity with its continuity, nonnegativity and compact support in  $[0, 1]$  (GeometricUniformCDF.lean).

*Remark 0.2.6* (Degenerate and boundary parameters). At  $q = 0$  the family degenerates to  $X_0 = U_0$ , uniform on  $[0, 1]$ , and every product formula below reduces to its single  $j = 0$  factor. As  $q \uparrow 1$  the variance (0.4.5) tends to 0 and  $X_q \rightarrow \frac{1}{2}$  in probability, while the standardized variable  $X_q^*$  converges in law to the standard Gaussian; the sharp form of that statement, with all correction orders, is the subject of Parts X and XI. As  $q \downarrow -1$  the support (0.2.3) escapes to  $\mathbb{R}$ ; Parts VII and VIII treat that regime through the reciprocal germ rather than by a limit.

### 0.3 Transforms, the two sinc conventions, and the Mahler equation

Throughout the volume  $\operatorname{sinc}_{\text{rad}}$  and  $\operatorname{sinc}_\pi$  are the two normalized cardinal sines of the corpus contract,

$$\operatorname{sinc}_{\text{rad}} z = \frac{\sin z}{z}, \quad \operatorname{sinc}_\pi z = \frac{\sin(\pi z)}{\pi z}, \quad \operatorname{sinc}_{\text{rad}} 0 = \operatorname{sinc}_\pi 0 = 1, \quad (0.3.1)$$

both entire and even. We write  $E$  for the normalized exponential difference quotient and  $\operatorname{sinhc}$  for the hyperbolic cardinal sine,

$$E(z) = \frac{e^z - 1}{z}, \quad E(0) = 1, \quad \operatorname{sinhc} z = \frac{\sinh z}{z}, \quad \operatorname{sinhc} 0 = 1, \quad (0.3.2)$$

so that  $E(z) = e^{z/2} \operatorname{sinhc}(z/2)$  and  $\operatorname{sinhc} z = \operatorname{sinc}_{\text{rad}}(iz)$ . The Fourier convention is the angular one of the contract,

$$\widehat{f}_{\text{ang}}(\omega) = \int_{\mathbb{R}} e^{-i\omega x} f(x) \, dx, \quad f(x) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{i\omega x} \widehat{f}_{\text{ang}}(\omega) \, d\omega, \quad (0.3.3)$$



so that the characteristic function of a variable  $Z$  with density  $f$  is  $\varphi_Z(t) = \widehat{f}_{\text{ang}}(-t)$ . Individual parts that state their inversion formula in the opposite sign convention are listed in [table 1](#); since every density occurring in this volume is real and symmetric about its own mean, the two conventions differ only by that recorded sign and never by a factor.

**Proposition 0.3.1** (Product formulas for the transforms). *For real  $|q| < 1$  the moment generating function of  $X_q$  is entire and*

$$M_q(t) := \mathbb{E}e^{tX_q} = \prod_{j=0}^{\infty} E((1-q)q^j t), \quad t \in \mathbb{C}, \quad (0.3.4)$$

while the characteristic function of the centered variable is

$$\varphi_q(t) := \mathbb{E}e^{itX_q^\circ} = \prod_{j=0}^{\infty} \text{sinc}_{\text{rad}}\left(\frac{(1-q)q^j t}{2}\right), \quad t \in \mathbb{C}. \quad (0.3.5)$$

Both products converge locally uniformly on  $\mathbb{C}$ , and (0.3.5) converges locally uniformly in  $(q, t)$  on  $\{|q| < 1\} \times \mathbb{C}$ .

*Proof.* For  $U$  uniform on  $[0, 1]$  and  $c \in \mathbb{C}$ ,  $\mathbb{E}e^{tcU} = \int_0^1 e^{tcu} du = E(ct)$ , and for  $V$  uniform on  $[-\frac{1}{2}, \frac{1}{2}]$ ,  $\mathbb{E}e^{itcV} = \text{sinc}_{\text{rad}}(ct/2)$ . Independence gives the corresponding finite products for the partial sums  $S_N = (1-q) \sum_{j \leq N} q^j U_j$ .

For the passage to the limit fix  $R > 0$  and  $|t| \leq R$ . Both  $E(w) - 1$  and  $\text{sinc}_{\text{rad}}(w) - 1$  are  $O(|w|)$  and  $O(|w|^2)$  respectively as  $w \rightarrow 0$ , uniformly on compact sets, so with  $w_j = (1-q)q^j t$  we have  $\sum_j |E(w_j) - 1| < \infty$  and  $\sum_j |\text{sinc}_{\text{rad}}(w_j/2) - 1| < \infty$ , uniformly for  $|t| \leq R$ ; for the second sum the bound is  $O(R^2 |1-q|^2 \sum_j |q|^{2j}) = O(R^2 |1-q|^2 / (1-|q|^2))$ , which is bounded on  $|q| \leq \rho < 1$ . The Weierstrass criterion for infinite products then gives locally uniform convergence, with the joint statement in  $(q, t)$  for (0.3.5). Since  $S_N \rightarrow X_q$  almost surely and  $|S_N|$  is bounded by the deterministic constant of [proposition 0.2.2](#), dominated convergence identifies the limits with  $M_q$  and  $\varphi_q$ . Entirety of  $M_q$  follows from the local uniform convergence of the product of entire factors.  $\square$

**Proposition 0.3.2** (Mahler equation and the centered germ). *With  $M_q$  as in (0.3.4),*

$$M_q(t) = E((1-q)t) M_q(qt), \quad M_q(0) = 1. \quad (0.3.6)$$

The centered germ  $C_q(t) := e^{-t/2} M_q(t)$  satisfies  $C_q(-t) = C_q(t)$ ; consequently all odd cumulants of  $X_q^\circ$  vanish. Finally, if  $q^n \neq 1$  for every  $n \geq 1$ , then (0.3.6) together with the normalization  $M_q(0) = 1$  determines the formal power series  $M_q$  uniquely.

*Proof.* Splitting the factor  $j = 0$  out of (0.3.4) and reindexing gives (0.3.6). Since the weights sum to 1 and  $E(-z) = e^{-z} E(z)$ ,

$$M_q(-t) = \prod_{j \geq 0} E(-(1-q)q^j t) = e^{-t \sum_j (1-q)q^j} \prod_{j \geq 0} E((1-q)q^j t) = e^{-t} M_q(t),$$

which is exactly  $C_q(-t) = C_q(t)$ . An even function has vanishing odd Taylor coefficients, and the cumulant generating function  $\log C_q$  is then even as well, so the odd cumulants of  $X_q^\circ$  vanish.

For the last claim write  $M_q(t) = \sum_{n \geq 0} a_n(q) t^n$  with  $a_0 = 1$ . Comparing coefficients of  $t^n$  in (0.3.6), and using  $E(z) = \sum_{r \geq 0} z^r / (r+1)!$ , gives

$$(1-q^n) a_n(q) = \sum_{r=1}^n \frac{(1-q)^r q^{n-r}}{(r+1)!} a_{n-r}(q). \quad (0.3.7)$$

When  $q^n \neq 1$  this determines  $a_n$  from  $a_0, \dots, a_{n-1}$ .  $\square$

*Remark 0.3.3* (Coordinates used by the Lean development). The formalization names the geometric sinc product in the frequency coordinate  $\text{geometricSincProduct}(q, u) = \prod_{j \geq 0} \text{sinc}_{\text{rad}}(\pi q^j u)$ . Comparing with (0.3.5),

$$\varphi_q(t) = \text{geometricSincProduct}\left(q, \frac{(1-q)t}{2\pi}\right), \quad (0.3.8)$$

so a manuscript frequency  $t$  and the Lean product variable differ by the factor  $2\pi/(1-q)$ . At  $q = \frac{1}{2}$  the further identity  $\text{centeredSincProduct}(z) = \text{rvachevFourierProduct}(z/2\pi)$  records the corresponding conversion for the Rvachev product; the relevant declarations are `Fabiús.geometricSincProduct` and `Fabiús.centeredSincProduct_eq_rvachevFourierProduct`. Part IX states the same dictionary in its own variable, and Part II uses the  $\text{sinc}_\pi$  form; [table 1](#) lists both.

## 0.4 Bernoulli numbers, exact cumulants, and the Bell bridge

Throughout, the Bernoulli numbers are those of the convention

$$\frac{z}{e^z - 1} = \sum_{n \geq 0} B_n \frac{z^n}{n!}, \quad B_0 = 1, B_1 = -\frac{1}{2}, B_2 = \frac{1}{6}, B_4 = -\frac{1}{30}, B_6 = \frac{1}{42}, \quad (0.4.1)$$

so that  $B_{2m+1} = 0$  for  $m \geq 1$ . Every part of this volume uses this convention; none uses the “even-index-only” variant.

**Lemma 0.4.1** (Logarithmic cardinal-sine series). *For  $|z| < 2\pi$ ,*

$$\log \frac{\sinh(z/2)}{z/2} = \sum_{m \geq 1} \frac{B_{2m}}{2m} \frac{z^{2m}}{(2m)!}, \quad (0.4.2)$$

and equivalently, for  $|w| < \pi$ ,

$$\log \sinh w = \sum_{m \geq 1} \frac{2^{2m-1} B_{2m}}{m (2m)!} w^{2m}. \quad (0.4.3)$$

Both radii are exact.

*Proof.* From (0.4.1),  $\frac{z}{e^z - 1} + \frac{z}{2} = \frac{z}{2} \coth \frac{z}{2} = \sum_{m \geq 0} B_{2m} \frac{z^{2m}}{(2m)!}$ , the odd terms cancelling because  $B_1 = -\frac{1}{2}$  and  $B_{2m+1} = 0$ . Hence

$$\frac{d}{dz} \log \frac{\sinh(z/2)}{z/2} = \frac{1}{2} \coth \frac{z}{2} - \frac{1}{z} = \frac{1}{z} \left( \frac{z}{2} \coth \frac{z}{2} - 1 \right) = \sum_{m \geq 1} B_{2m} \frac{z^{2m-1}}{(2m)!}.$$

Integrating from 0, where the left side vanishes, gives (0.4.2). Substituting  $z = 2w$  turns  $\frac{B_{2m}}{2m} \frac{(2w)^{2m}}{(2m)!}$  into  $\frac{2^{2m-1} B_{2m}}{m (2m)!} w^{2m}$ , which is (0.4.3). The function  $\sinh(z/2)/(z/2)$  is entire, equals 1 at 0, and has its zeros exactly at  $z = 2\pi i k$ ,  $k \in \mathbb{Z} \setminus \{0\}$ ; the principal logarithm is therefore holomorphic on the disc of radius  $2\pi$  and no larger, and the same computation in the variable  $w$  gives radius  $\pi$ .  $\square$

*Lean status.* Formal coefficientwise rather than analytically: `Fabiús.sinhDivLogCoefficient_eq_bernoulli` in `SinhDivBernoulliLog.lean` identifies every coefficient of the formal logarithm of  $\sinh(z/2)/(z/2)$  with  $B_{2m}/(m (2m)!)$ , which is (0.4.2) as an identity of formal power series. The proof is deliberately free of any analytic logarithm, branch choice, or convergence statement, so the disc of validity  $|z| < 2\pi$  is not part of it.

**Theorem 0.4.2** (Exact cumulants of the family). *Let  $|q| < 1$  be real. Then all odd cumulants of  $X_q^\circ$  vanish and, for every  $m \geq 1$ ,*

$$\kappa_{2m}(X_q^\circ) = \frac{B_{2m}}{2m} \frac{(1-q)^{2m}}{1-q^{2m}}. \quad (0.4.4)$$

*In particular*

$$\sigma_q^2 = \text{Var}(X_q) = \frac{B_2}{2} \frac{(1-q)^2}{1-q^2} = \frac{1-q}{12(1+q)}, \quad (0.4.5)$$

*and the standardized cumulants are*

$$\lambda_{2m}(q) := \kappa_{2m}(X_q^*) = \frac{B_{2m}}{2m} 12^m \frac{(1-q^2)^m}{1-q^{2m}}, \quad m \geq 2, \quad \text{for real } |q| < 1. \quad (0.4.6)$$

*Proof.* The cumulant generating function of  $V_j$  is  $\log \mathbb{E} e^{zV_j} = \log(\sinh(z/2)/(z/2))$ , so by [lemma 0.4.1](#) the cumulants of a centered uniform are  $\kappa_{2m}(V) = (2m)! \cdot \frac{B_{2m}}{2m(2m)!} = \frac{B_{2m}}{2m}$  and  $\kappa_{2m+1}(V) = 0$ . Cumulants are additive over independent summands and homogeneous of degree  $n$  in order  $n$ , so for  $|z|$  small

$$\log \mathbb{E} e^{zX_q^\circ} = \sum_{j \geq 0} \log \mathbb{E} e^{z(1-q)q^j V_j} = \sum_{m \geq 1} \frac{\kappa_{2m}(V)}{(2m)!} z^{2m} (1-q)^{2m} \sum_{j \geq 0} q^{2mj},$$

the interchange being legitimate because the double series converges absolutely for  $|z(1-q)| < 2\pi$ . Summing the geometric series gives (0.4.4), and  $m = 1$  with  $B_2 = \frac{1}{6}$  gives (0.4.5) after cancelling  $1-q$ . Finally, since  $\kappa_{2m}$  is homogeneous of degree  $2m$ ,

$$\lambda_{2m}(q) = \frac{\kappa_{2m}(X_q^\circ)}{\sigma_q^{2m}} = \frac{B_{2m}}{2m} \frac{(1-q)^{2m}}{1-q^{2m}} \left( \frac{12(1-q^2)}{(1-q)^2} \right)^m = \frac{B_{2m}}{2m} 12^m \frac{(1-q^2)^m}{1-q^{2m}},$$

which is (0.4.6). For  $m = 1$  the same formula returns  $\lambda_2 = 1$ , as it must.  $\square$

**Proposition 0.4.3** (Bell bridge and rationality). *Let  $\mathcal{B}_n^{\text{exp}}$  denote the complete exponential Bell polynomial. Then for every  $n \geq 0$*

$$\mathbb{E}(X_q^\circ)^n = \mathcal{B}_n^{\text{exp}}(\kappa_1, \dots, \kappa_n), \quad \kappa_i = \kappa_i(X_q^\circ), \quad (0.4.7)$$

*and consequently every centered moment of  $X_q$  is a rational function of  $q$  with rational coefficients, of the form  $P_n(q)/\prod_{m=1}^{\lfloor n/2 \rfloor} (1-q^{2m})$  for a polynomial  $P_n \in \mathbb{Q}[q]$ ; the odd centered moments vanish.*

*Proof.* Identity (0.4.7) is the standard moment–cumulant relation, valid whenever the moment generating function is analytic near 0, which holds by [proposition 0.3.1](#). Each  $\kappa_{2m}$  in (0.4.4) is a rational function of  $q$  with denominator  $1-q^{2m}$  and rational numerator, and  $\mathcal{B}_n^{\text{exp}}$  is a polynomial with rational coefficients in  $\kappa_1, \dots, \kappa_n$  involving only  $\kappa_i$  with  $i \leq n$ ; clearing denominators gives the stated form. Odd cumulants vanish by [proposition 0.3.2](#), and  $\mathcal{B}_{2k+1}^{\text{exp}}$  evaluated on a cumulant vector with vanishing odd entries is 0.  $\square$

**Remark 0.4.4** (Where the volume uses this). (0.4.4) is the single source of the Bernoulli–Bell cumulant layer that Parts VII–XII each set up separately: Part VIII in the  $[0, 1]$  scale, Parts IX and X in the  $[-1, 1]$  scale (with the resulting factor  $2^{2m}$ ), and Part XI in the standardized scale (0.4.6). The scale factors are exactly the entries of [table 1](#), and [lemma 0.5.1](#) makes the transport explicit.

## 0.5 The normalization dictionary

**Lemma 0.5.1** (Affine transport). *Let  $Z$  be a random variable with finite exponential moments near 0 and let  $W = \alpha Z + \beta$  with  $\alpha \neq 0$ . Then*

$$\varphi_W(t) = e^{i\beta t} \varphi_Z(\alpha t), \quad M_W(t) = e^{\beta t} M_Z(\alpha t), \quad (0.5.1)$$

*and the cumulants satisfy  $\kappa_1(W) = \alpha \kappa_1(Z) + \beta$  and  $\kappa_n(W) = \alpha^n \kappa_n(Z)$  for  $n \geq 2$ . If  $Z$  has density  $f_Z$ , then  $W$  has density  $f_W(x) = |\alpha|^{-1} f_Z((x - \beta)/\alpha)$ .*

*Proof.* The transform identities are immediate from the definitions, and the cumulant statement follows by taking logarithms:  $\log M_W(t) = \beta t + \log M_Z(\alpha t)$ , whose  $t^n/n!$  coefficient is  $\alpha^n \kappa_n(Z)$  for  $n \geq 2$ . The density formula is the change of variables.  $\square$

table 1 records, for each numbered part, the letters that part uses for the geometric-uniform variable and its transform, and the exact relation to the canonical objects of definition 0.2.1 and proposition 0.3.1. Reading a formula out of one part and into another means composing the two corresponding rows through lemma 0.5.1.

Table 1: Normalization dictionary. “Seed” is the law of the independent variables summed;  $X_q, X_q^\circ$ , and  $X_q^*$  are as in definition 0.2.1.

| Part               | Local symbol        | Seed        | Relation to the canonical family                 |
|--------------------|---------------------|-------------|--------------------------------------------------|
| 0                  | $X_q, X_q^\circ$    | Unif[0, 1]  | canonical, definition 0.2.1                      |
| I–VI               | (finite products)   | —           | prefix products; see remark 0.5.3                |
| VII                | $Y_q$               | Unif[0, 1]  | $Y_q = X_q$                                      |
| VIII               | $X_q, Y_q$          | Unif(0, 1)  | $X_q = X_q, Y_q = X_q^\circ$                     |
| IX                 | $Y_q$               | Unif[−1, 1] | $Y_q = 2X_q^\circ$                               |
| X                  | $X_q$               | Unif[−1, 1] | $X_q = 2X_q^\circ$                               |
| XI                 | $X_r, Z_h$          | Unif[0, 1]  | $X_r = X_r, Z_h = X_r^*, r = e^{-h}$             |
| XII                | $X_q, Z_q$          | Unif[0, 1]  | $X_q = X_q, Z_q = X_q^\circ$                     |
| <i>Transforms.</i> |                     |             |                                                  |
| VIII               | $\psi_q(t)$         | —           | $= \varphi_q(t)$                                 |
| IX                 | $\Phi(q, z)$        | —           | $= \varphi_q(2z)$                                |
| X                  | $\phi_q(t), M_q(t)$ | —           | $\phi_q(t) = \varphi_q(2t), M_q(t) = C_q(2t)$    |
| XI                 | $\Psi_h(t)$         | —           | $= \varphi_r(t/\sigma_r), a_h = \sqrt{3(1-r^2)}$ |
| XII                | $\Phi_q(t)$         | —           | $= \varphi_q(t)$                                 |

**Proposition 0.5.2** (Correctness of the dictionary). *Every entry of table 1 is exact. In particular:*

- (i) *Parts IX and X use seeds uniform on  $[-1, 1]$ , and their variable is  $2X_q^\circ$ ; consequently their cumulants are  $2^{2m}$  times (0.4.4), namely  $\frac{2^{2m-1} B_{2m}}{m} \frac{(1-q)^{2m}}{1-q^{2m}}$ , and their variance is  $\frac{1-q}{3(1+q)}$ , four times (0.4.5).*
- (ii) *Part XI’s normalizing constant satisfies  $a_h = (1-r)/(2\sigma_r)$ , so  $\Psi_h(t) = \varphi_r(t/\sigma_r)$  and Part XI’s  $\lambda_{2m}(r)$  agrees with (0.4.6).*
- (iii) *Part IX’s product  $\Phi(q, z) = \prod_{j \geq 0} \text{sinc}_{\text{rad}}((1-q)q^j z)$  equals  $\varphi_q(2z)$ , and Part X’s  $\phi_q$  is the same function of the same argument.*

*Proof.* (i) If  $U^\pm$  is uniform on  $[-1, 1]$  then  $U^\pm = 2V$  with  $V$  uniform on  $[-\frac{1}{2}, \frac{1}{2}]$ , so  $(1-q) \sum_j q^j U_j^\pm = 2X_q^\circ$ . lemma 0.5.1 with  $\alpha = 2, \beta = 0$  multiplies  $\kappa_{2m}$  by  $2^{2m}$ ; applying this to (0.4.4) gives  $2^{2m} \frac{B_{2m}}{2m} \frac{(1-q)^{2m}}{1-q^{2m}} = \frac{2^{2m-1} B_{2m}}{m} \frac{(1-q)^{2m}}{1-q^{2m}}$  and, at  $m = 1$ ,  $4\sigma_q^2 = \frac{1-q}{3(1+q)}$ .

(ii) By (0.4.5),  $\sigma_r^2 = \frac{(1-r)^2}{12(1-r^2)}$ , so  $\sigma_r = \frac{1-r}{2\sqrt{3(1-r^2)}}$  and

$$\frac{1-r}{2\sigma_r} = \frac{1-r}{2} \cdot \frac{2\sqrt{3(1-r^2)}}{1-r} = \sqrt{3(1-r^2)} = a_h.$$

Hence  $\varphi_r(t/\sigma_r) = \prod_j \text{sinc}_{\text{rad}}(\frac{(1-r)r^j}{2\sigma_r} t) = \prod_j \text{sinc}_{\text{rad}}(a_h r^j t) = \Psi_h(t)$ , and the standardized cumulants agree because both sides are the cumulants of the same variable. As a check,  $\sum_{j \geq 0} (a_h r^j)^2 = \frac{3(1-r^2)}{1-r^2} = 3$ , the normalization identity used in Part XI.

(iii) By (0.3.5),  $\varphi_q(2z) = \prod_j \text{sinc}_{\text{rad}}((1-q)q^j z)$ , which is the displayed product. The same computation applies verbatim to Part X’s  $\phi_q$ ; and  $M_q^X(t) = \mathbb{E} e^{2tX_q^\circ} = e^{-t} M_q(2t) = C_q(2t)$ .  $\square$

*Remark 0.5.3* (Finite products and the prefix parts). Parts I–VI work mainly with *finite* products  $\prod_{j=0}^{N-1} \text{sinc}_{\text{rad}}(\cdot)$  and with the exponent sequences attached to them, rather than with the law of  $X_q$  itself. The relation is that the  $N$ -fold product is the characteristic function of the truncation  $(1-q) \sum_{j < N} q^j V_j$ , rescaled by whichever normalization that part fixes; Part V makes the convex-order comparison between truncations and the limit precise. Those parts also use the  $\text{sinc}_\pi$  convention of (0.3.1) in places where a unit-period Fourier variable is natural; since  $\text{sinc}_\pi z = \text{sinc}_{\text{rad}}(\pi z)$ , the two differ only by the argument scaling recorded here.

## 0.6 $q$ -series conventions

The  $q$ -series notation is that of the corpus contract, with every base an explicit argument. For  $n \in \mathbb{N}_0$ ,

$$(a; q)_n = \prod_{k=0}^{n-1} (1 - aq^k), \quad (a; q)_\infty = \prod_{k \geq 0} (1 - aq^k) \quad (|q| < 1), \quad (0.6.1)$$

$$[m]_q = 1 + q + \cdots + q^{m-1} = \frac{1 - q^m}{1 - q} \quad (q \neq 1), \quad [m]_1 = m, \quad (0.6.2)$$

and the Gaussian binomial coefficient is

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{(q; q)_n}{(q; q)_k (q; q)_{n-k}}, \quad 0 \leq k \leq n. \quad (0.6.3)$$

Two facts are used repeatedly and are recorded here rather than in each part. First,  $[2m]_q$ -type denominators are exactly the ones appearing in (0.4.4): since  $1 - q^{2m} = (1 - q) [2m]_q$ , that identity may be written

$$\kappa_{2m}(X_q^\circ) = \frac{B_{2m}}{2m} \cdot \frac{(1 - q)^{2m-1}}{[2m]_q}, \quad (0.6.4)$$

which is the form in which Parts VII, VIII, and X state it. Second, the outer product  $(a; q^2)_\infty$  with nome  $q^2$  is the one that appears in the spectral factorizations of Parts VII and IX; its locally uniform convergence for every complex  $|q| < 1$ , including  $q = 0$ , is formalized as `Fabiús.hasProdLocallyUniformly_geometricSincProduct_complexQPochhammerInf`.

## 0.7 Status vocabulary

The volume marks the epistemic status of every displayed assertion, using one vocabulary throughout:

- **proved here** — a complete proof appears in this volume.
- **derived from audited corpus theorems** — deduced here from results proved elsewhere in the audited corpus, with the citation given.
- **existing repository input** — quoted from the audited corpus as an input, not reproved.
- **numerical check** — supported only by the deterministic numerical experiments of the reproducibility packages under `assets/`.
- **conditional** — proved subject to an explicitly stated hypothesis that is not itself established here.
- **conjectural/open** or **conjectural** — conjectural.

Environments named *Conjecture* and *Open problem* are never used for assertions that are proved somewhere in the volume; where an absorbed report conjectured something that another absorbed report proves, the conjecture has been demoted to a remark carrying a forward reference, and those demotions are listed in [section 0.8](#).

## 0.8 What each part contributes

### Parts I–VI

The finite-product side: exponent sequences and the Newton basis (I),  $q$ -binomial Richardson acceleration (II), the dyadic box-spline inverse transforms (III), Fourier images of the repeated-integration approximants (IV), exact transport geometry and convex order (V), and atomic splines beyond the binary point (VI).

### Part VII

Signed and reciprocal parameter orbits: the Klein-four sign calculus, the reciprocal germ outside the unit circle, and the spectral  $q^2$ -Pochhammer factorization.

### Part VIII

The negative-parameter affine duality in full, the exact negative dyadic law, log-concavity and dominant-digit plateaux, residue-class multiplication formulas, and the quantitative central limit theorem at  $|q| \uparrow 1$ .

### Part IX

Complex parameters at roots of unity: radial expansions, the natural-boundary theorem, cyclo-tomic blow-ups, and inverse branches.

### Part X

Parameter flow as a martingale transport, convex order along the flow, the midpoint  $q$ -integral form of the scaled cumulant transform, and large deviations with the Lambert- $W_{-1}$  boundary.

### Part XI

The continuous-parameter Edgeworth and Cornish–Fisher theory, moderate deviations, and quadratic  $q$ -Gevrey regularity.

### Part XII

The response of the family to the parameter at  $q = \frac{1}{2}$ : flat nonanalytic fronts, the tangent measure, the susceptibility resolvent, and the smooth Schröder–Koenigs dynamics, whose linearization hypothesis is now proved.

### 0.8.1 Conjectures settled inside the volume

The status rule of [section 0.7](#) forbids a *Conjecture* or *Open problem* environment for an assertion proved somewhere in the volume. Merging the six reports settled four such items, each by a result that another report had proved independently; they are recorded here so that a reader comparing the parts with their standalone sources can see what moved and why.

- Part VIII conjectured the uniform first Edgeworth expansion of the standardized density. Part XI proves the all-orders, polynomially weighted, termwise-differentiable expansion for the same standardized variable ([theorem 5.1](#), with the identification of the two variables in [proposition 0.5.2](#)); the conjecture is now [proposition 8.5](#), proved by specialization.
- Part VIII’s all-orders Bernoulli–Bell–Hermite conjecture is the statement of [theorem 5.1](#) together with [proposition 5.2](#); it is now the remark [remark 10.1](#), pointing there.
- Part IX’s appendix left the all-orders fixed-frequency radial expansion as a conjecture; it is now [theorem 6.4](#), proved once the multiplier estimates of that part were made uniform in the summation index.
- Part III posed as an open problem the sharp leading constants of the prefix-product error for general base; Part VI proves them for every base  $a \geq 2$ , and Part III now points there instead.

- Part XII’s standalone source made its smooth Schröder–Koenigs coordinate conditional on a linearization hypothesis it called (K). That hypothesis is now proved in Part XII itself ([theorem 6.4](#)): the flat contraction is linearized by a  $C^\infty$  conjugacy with unit derivative at the fixed point, so every statement that was conditional on (K) is unconditional.

One conjecture was settled the other way. Part X conjectured that all coefficients of its central rate series are positive; exact rational reversion through order 28 shows the coefficient of  $x^{24}$  is negative,  $c_{12} = -3032147123502547968/3562005918173828125$ , confirmed independently to sixty digits from the exact saddle. The conjecture is refuted and replaced by the remark [remark 5.2](#), which records the sign pattern and the radius of convergence  $0 < R_I \leq 1$ . The remaining entropy-deficit conjecture of Part VIII stays conjectural and is cross-referenced to the matching conjecture of Part XI, whose second-order constant agrees with it in exact rationals.

## 0.8.2 Result census

[table 2](#) counts the numbered environments of each part. It is mechanically derived from the source by `drafts/scripts/part_result_census.py`, so it can be regenerated rather than maintained by hand, and it exists to make one thing visible at a glance in a volume of this length: which parts are carrying open problems and which are carrying proofs. The last column counts proof environments, not proved statements: a corollary proved in one line inside its statement, or two results discharged by a single argument, will not match one for one.

Table 2: Numbered environments per part.

| Part  | Thm | Prop | Lem | Cor | Def | Conj | Prob | Proofs |
|-------|-----|------|-----|-----|-----|------|------|--------|
| 0     | 1   | 7    | 2   | 0   | 1   | 0    | 0    | 10     |
| I     | 20  | 0    | 1   | 5   | 1   | 4    | 3    | 21     |
| II    | 12  | 2    | 1   | 2   | 2   | 0    | 0    | 16     |
| III   | 6   | 2    | 0   | 2   | 0   | 4    | 3    | 8      |
| IV    | 13  | 1    | 0   | 3   | 1   | 1    | 0    | 13     |
| V     | 13  | 1    | 2   | 3   | 1   | 4    | 0    | 17     |
| VI    | 40  | 6    | 1   | 13  | 1   | 19   | 0    | 52     |
| VII   | 15  | 1    | 0   | 3   | 1   | 3    | 4    | 16     |
| VIII  | 6   | 4    | 2   | 5   | 1   | 2    | 1    | 17     |
| IX    | 15  | 6    | 6   | 4   | 3   | 2    | 7    | 30     |
| X     | 6   | 3    | 0   | 1   | 0   | 4    | 4    | 9      |
| XI    | 9   | 2    | 3   | 1   | 1   | 8    | 3    | 15     |
| XII   | 12  | 4    | 2   | 5   | 1   | 4    | 3    | 23     |
| Total | 168 | 39   | 20  | 47  | 14  | 55   | 28   | 247    |

Two readings are worth stating. First, the volume asserts 274 results of proved kind against 55 conjectures and 28 open problems: it is a proof document with a frontier register attached, not a problem list. Second, the distribution is uneven by design — Part VI carries a third of everything because it is itself a threefold editorial merge of independent reconstructions of one source chapter, while Parts X and XI are deliberately lean, having had their duplicated layers merged into one another and their shared setup moved here.

## 0.9 Register of volume-local symbols

The following symbols are introduced by this part and are used with the stated meaning throughout the volume; every other symbol is either a shared corpus command or is declared locally in the part that uses it, under the declaration schema of the notation catalogue.



| Symbol            | Meaning                                                                |
|-------------------|------------------------------------------------------------------------|
| $X_q$             | canonical geometric-uniform variable, <a href="#">definition 0.2.1</a> |
| $X_q^\circ$       | its centering $X_q - \frac{1}{2}$                                      |
| $X_q^*$           | its standardization $X_q^\circ/\sigma_q$                               |
| $G_q, g_q$        | distribution function and density of $X_q$                             |
| $\sigma_q$        | standard deviation, <a href="#">(0.4.5)</a>                            |
| $M_q$             | moment generating function of $X_q$                                    |
| $C_q$             | centered germ $e^{-t/2}M_q(t)$                                         |
| $\varphi_q$       | characteristic function of $X_q^\circ$                                 |
| $\lambda_{2m}(q)$ | standardized cumulants, <a href="#">(0.4.6)</a>                        |
| $E(z), \sinh z$   | $(e^z - 1)/z$ and $\sinh(z)/z$                                         |

*Warning 0.9.1* (Three near-identical phi's). The letter  $\varphi$  carries three distinct meanings in this volume, and they are distinguished only by their decoration.

- (a)  $\varphi_q$ , with a parameter subscript, is the canonical characteristic function [\(0.3.5\)](#) of  $X_q^\circ$ .
- (b)  $\phi_q$  in Part X is the characteristic function of that part's variable  $2X_q^\circ$ , hence  $\phi_q(t) = \varphi_q(2t)$  and *not*  $\varphi_q(t)$ . The two are typeset with different glyphs,  $\phi$  against  $\varphi$ , but the safe reading is the dictionary row, not the glyph.
- (c)  $\varphi$  with no subscript, in Part XI, is the standard normal density  $e^{-x^2/2}/\sqrt{2\pi}$ .

This is a document-local overload of the kind governed by the collision register of the notation catalogue; it is recorded there under this volume's namespace.



## Part I

# Exponent-Sequence and Newton-Basis Frontiers for the Fabius–Rvachev System

## Abstract

The audited PROVEIt corpus already contains the classical Fabius and Rvachev functions, exact dyadic arithmetic, the full Thue–Morse formula atlas, Gaussian  $q$ -binomial interpolation, all-orders endpoint expansions, lower-branch Lambert- $W$  coordinates, Bell- and Bernoulli-polynomial coefficient calculus, sharp Fourier decay, the integer-zero spectrum of the Rvachev transform, a spectral determinant, rational arithmetic rays, half-integer quadrature identities, logarithmic  $q$ -Richardson acceleration, and a Pascal-weighted hierarchy of compactly supported densities. This report avoids repackaging those results. Instead it identifies a single master object that contains them as special coordinates and exposes several new deductions.

For a nonnegative integer exponent sequence  $a = (a_h)_{h \geq 0}$  with  $A(1/2) < \infty$ , where  $A(q) = \sum_{h \geq 0} a_h q^h$ , define

$$\Phi_a(z) = \prod_{h \geq 0} \operatorname{sinc}_\pi \left( \frac{z}{2^h} \right)^{a_h}, \quad \operatorname{sinc}_\pi z = \frac{\sin \pi z}{\pi z}.$$

It is the characteristic function of an explicit compactly supported convolution of uniform variables. We prove a universal dictionary in which the same generating function  $A$  controls: the integer-zero multiplicity  $m_a(n) = \sum_{h \leq \nu_2(n)} a_h$ ; the spectral zeta  $\mathcal{Z}_a(s) = \zeta(s)A(2^{-s})$ ; every even cumulant  $\kappa_{2j} = B_{2j}A(4^{-j})/(2j)$ ; the Fourier-lobe sign; exact digital zero counts; and the order of Prouhet cancellation of each finite sign block. The zero divisor, the spectral zeta quotient, and the complete even-cumulant sequence each determine  $a$  uniquely.

The cumulant samples  $A(4^{-m})$  admit a new exact reconstruction mechanism. Lagrange extrapolation at geometric nodes has Gaussian- $q$ -binomial weights, an exact transformed series, fixed alternating sign, bounded  $\ell^1$  condition number, and certified integer brackets. Thus Richardson extrapolation becomes a stable tomography algorithm for the hidden dyadic exponent sequence.

For polynomial exponent sequences, the Newton expansion  $P(h) = \sum c_r \binom{h}{r}$  gives the finite factorization  $\Phi_P = \prod_r \Phi_{r+1}^{c_r}$  through the audited Pascal–Rvachev hierarchy. Negative Newton coefficients are allowed: whenever  $P(h) \geq 0$  on the nonnegative integers, the resulting alternating quotient is nevertheless entire and positive definite. The even family

$$P_d(h) = \binom{h-1}{d}, \quad d = 0, 2, 4, \dots,$$

is especially striking. It satisfies

$$\Psi_d(z) = \operatorname{sinc}_\pi(z) \Phi_{d+1}(z/2) = \prod_{r=0}^d \Phi_{r+1}(z)^{(-1)^{d-r}},$$

has support  $[-1, 1]$ , zero multiplicity  $1 + \binom{\nu_2(n)}{d+1}$ , and converges in total variation to the uniform law. Its transition across the two discontinuities of the limiting box density has a Gaussian boundary layer with an explicit Berry–Esseen rate. For  $d = 2$  this supplies the concrete positive-definite quotient  $\Phi_1 \Phi_3 / \Phi_2$ .

Finally, a Mellin-reflection identity relates the endpoint Laplace transform directly to the reflected spectral zeta. For polynomial weights it explains the complete hierarchy of logarithmic powers and periodic coefficients, gives a generalized lower-Lambert saddle, and yields the leading inverse-endpoint law. Reasonable conjectures and a detailed research and Lean-formalization program conclude the report. Numerical experiments were independently implemented at high precision; the commented Python source and generated data accompany the paper.

# 1 Audit boundary, source stratification, and novelty discipline

## 1.1 Pinned corpus

The repository changed several times on the audit date, so the source boundary is the exact commit printed on the title page. Its recursive documentation tree contains 67 distinct `.tex` paths: five living synthesis/reference documents, thirty-two frozen historical sources, twenty-three active frontier drafts (including twelve Fourier-decay dossiers), and seven transcribed or corrected source papers. Every path is listed in [appendix A](#). Historical revisions were read as mathematical lineages rather than as independent evidence; the archive’s provenance ledger identifies what was superseded, merged, or deliberately frozen.

The living primary exposition separates the bounded distribution function, the compactly supported Rvachev bump, and the signed global extension, and confines itself to claims with matching Lean declarations. The consolidated frontier volume and its current descendants contain the non-formalized but mathematically developed material. In particular, the audit treated the following as prior art rather than as discoveries of this report:

- the integer zero set and multiplicity  $1 + \nu_2(n)$  of the classical Rvachev transform;
- its spectral determinant, spectral zeta, heat trace, local zero jets, and symmetric  $q$ -integer deformation;
- the full signed Thue–Morse formula atlas, Prouhet cancellation, Mahler products, rarefied sums, Mellin–Dirichlet theory, and automatic-sequence machinery;
- exact dyadic values of  $F$ , half-base  $q$ -Pochhammer and Gaussian-binomial formulas, Lagrange interpolation, and several existing Richardson transforms;
- the corrected lower-Lambert endpoint expansion, its nonconstant periodic factor, all-order Bell-polynomial coefficient construction, and the inverse Fabius expansion;
- sharp geometric, arithmetic, metric, and envelope results for the infinite sinc product;
- cyclotomic rational rays, half-integer spectral identities, exact zero jets, and the Pascal hierarchy  $\Phi_r(z) = \prod_h \text{sinc}_\pi(z/2^h)^{\binom{h}{r-1}}$ ;
- logarithmic  $q$ -Richardson acceleration of truncated sinc products and phase-locked Richardson extraction of the endpoint periodic term;
- positive Gaussian and Lobatto tail quadrature, exact ordinary-prefix error laws, generalized gamma-convolution duality, and the existing Gaussian rank limit for the Pascal hierarchy.

## 1.2 What is new relative to the pinned tree

Targeted path, phrase, and formula searches found no counterpart of the following theorem package in the audited snapshot:

1. the arbitrary nonnegative exponent-sequence monoid and its single generating-function dictionary for zero spectra, cumulants, digital signs, and endpoint Mellin transforms;
2. exact digital zero-count defects and generalized Prouhet order for an arbitrary exponent parity word, including the first surviving power moment;
3. equivalence and constructive recovery of the exponent sequence from either the zero divisor or all even cumulants;
4. certified Gaussian- $q$ -binomial Richardson tomography of the exponent coefficients;
5. the finite-refinement-rank classification by rational ordinary generating functions;

6. the Newton-basis factorization of every nonnegative integer-valued polynomial weight, including positive-definite alternating quotients of Pascal transforms;
7. the normalized family  $P_d(h) = \binom{h-1}{d}$  for even  $d$ , its sparse zero spectrum, automatic sign selector, exact total-variation law, and Gaussian boundary layer;
8. the general spectral-endpoint Mellin reflection formula  $\mathcal{M}[\log L_a](w) = -\Gamma(w)\zeta(w+1)Z_a(-w)/\zeta(-w)$ ;
9. the polynomial-weight universality theorem and its generalized Lambert- $W_{-1}$  endpoint coordinate.

A targeted external search located the classical ingredients separately—the original Fabius–Rvachev analysis and arithmetic, lacunary Thue–Morse products, automatic sequences, Berry–Esseen bounds, rational and integer-valued generating functions, Mellin harmonic sums, exponential Tauberian transfer, and Lambert inversion [33, 31, 9, 42, 40, 93, 55, 49, 48, 52]—but not this combined classification. Accordingly, “new” below always means new to the pinned corpus and apparently nonstandard; it is not an assertion of exhaustive worldwide priority.

---

**Proof-status convention.** Results explicitly labeled **proved here** are proved in this manuscript. Results labeled **derived from audited corpus theorems** use a theorem already developed in the audited frontier corpus (most often the sharp Pascal-hierarchy Fourier or endpoint asymptotic) and then add a new deduction. Items labeled **conjectural/open** are conjectures or proposed programs.

A fourth, orthogonal label is now carried alongside these three. Wherever a statement of this report—or an exactly reusable core of one—already has a machine-checked counterpart in the repository’s Lean library, that declaration is cited inline in typewriter type—for instance `Fabius.weightedScaleMultiplicity`—together with its module under `Analysis/FabiusFunction/Lean/FabiusFunction/`. Roughly a dozen such citations appear below, alongside the *Post-snapshot Lean crosswalk* paragraphs. In most of them the formal statement is *more general* than the one printed here (arbitrary base instead of 2, arbitrary coefficient ring or module instead of  $\mathbb{R}$ , arbitrary geometric ratio instead of  $q \in (0, 1)$ ). Each citation says which. In particular the finite rational geometric-Lagrange kernel used in the Gaussian- $q$  rows—closed weights, all residual moments and their sign, and the finite exact  $\ell^1$  norm—is formalized at arbitrary ratio  $0 < q < 1$ , so the  $q = 1/4$  instance is covered, and the denominator-free principal specialization with the complete positive-degree geometric-moment identities behind [lemma 5.1](#) is formal as well (see the crosswalk paragraphs there). The post-snapshot module `AnalyticSeriesFilter.lean` now also proves the convergent-series analogue. Its finite filter acts diagonally under Mathlib’s *unconditional* `Summable` hypothesis on each weighted sampled series, splits into an exact finite head and infinite tail, and specializes geometric and geometric-Lagrange rows to the complete Gaussian residual tail, including the shifted-scale form used in (5.6). The hypothesis is sharp at a zero-weight node: the weighted series is then identically zero, and no convergence of the corresponding unweighted sample is required. It does not cover a series convergent only in its natural order at a boundary point, and by itself proves neither positivity, signs, bounds, nor uniform convergence.

---

**Post-snapshot analytic extensions.** The current tree now supplies the previously missing *infinite-product* instantiation. `AnalyticMoments.lean` names the even mode `Fabius.rvachevFourierMomentTerm`, proves its constant and scaling laws `Fabius.rvachevFourierMomentTerm_zero` and `Fabius.rvachevFourierMomentTerm_scale`, and proves genuine unconditional summation by `Fabius.hasSum_rvachevFourierMomentTerm`. `RvachevQBinomialFilter.lean` then proves the corresponding series for the standalone product, its power-scaling law, and the exact Gaussian residual identity. Generically  $c, z \in \mathbb{C}$ ,  $p \in \mathbb{N}_0$ ,  $q = c^2$ , and the sole hypothesis is injectivity of  $j \mapsto q^j$  on  $\{0, \dots, p\}$ ; no contraction, realness, positivity, or global nonvanishing is assumed. At  $c = 1/2$ ,  $q = 1/4$ , the theorem is assumption-free. This does not yet identify the finite prefixes  $P_{b,n}$ , their quotient or Bell coefficients, and it supplies none of the report’s analytic error signs, bracketing or quantitative bounds, uniform/derivative convergence, or asymptotics.

An independent reusable layer is now present in `FractionalVolterraCalculus.lean` and `FabiusFractionalVolterra.lean`. Positive affine scale changes on an ordered real interval are formalized. For  $\alpha \leq 0$ , affine covariance is an identity for the totalized Lean interval-integral definition; a classical Riemann–Liouville or integrability interpretation is asserted only for positive order. Gamma-normalized order raising assumes real  $\alpha > 0$ , a continuous Banach-valued primitive, an interval-integrable derivative, and the right derivative `HasDerivWithinAt` on the open interval. Its boundary-free form yields the exact dyadic shift for the signed Fabius extension when  $x \geq 0$ . The total causal Rvachev fractional primitive has its support-truncated integral, positive-natural bridge, and positive-order semigroup formalized on  $x \geq -1$ ; the same module gives the bounded Fabius shift when  $0 \leq x \leq 1$  and the up–Fabius bridge from base  $-1$  when  $x \geq -1$ . Complex orders and powers, Caputo or Riemann–Liouville derivative operators, weighted-monomial or iterated identities, negative- $x$  signed branches, shifted endpoint lattices, endpoint moments, transform/tail and piecewise/refinement formulas, and inverse/quantile formulas remain frontier. These post-snapshot series and fractional APIs were checked at source checkpoint 149332f9d.

These labels concern representation only, not correctness: report-specific probability and finite-prefix/Bell instantiations still require their own bridges. In particular the sign and bound in (5.7)–(5.8), Fourier envelopes for the Pascal hierarchy of rank at least two, Mellin contour displacement, exponential Tauberian transfer, Berry–Esseen for non-identically-distributed summands, total-variation and Wasserstein distances, and the automatic-sequence machinery have no Lean counterpart here. The infinite condition-number limit is formalized separately in `LimitConditionNumber.lean` by `Fabius.tendsto_sum_abs_qToeplitzWeight`; its divergence as  $q \rightarrow 1^-$  is the literal filter theorem `Fabius.tendsto_qConditionNumberLimit_atTop_at_one_left`.

## 2 The exponent-sequence convolution monoid

### 2.1 Admissible sequences

Throughout, the normalized entire sinc is

$$\operatorname{sinc}_\pi z = \frac{\sin \pi z}{\pi z}, \quad \operatorname{sinc}_\pi 0 = 1. \quad (2.1)$$

It is the Fourier transform, under the kernel  $e^{-2\pi i z x}$ , of the uniform probability density on  $[-1/2, 1/2]$ .

**Definition 2.1** (Admissible exponent sequence). A sequence  $a = (a_h)_{h \geq 0}$  is *admissible* if

$$a_h \in \mathbb{N}_0, \quad A(q) := \sum_{h \geq 0} a_h q^h, \quad A(1/2) < \infty.$$

Its generalized Rvachev transform is

$$\Phi_a(z) = \prod_{h=0}^{\infty} \operatorname{sinc}_{\pi}\left(\frac{z}{2^h}\right)^{a_h}. \quad (2.2)$$

For each  $h$  and  $1 \leq j \leq a_h$ , let  $U_{h,j}$  be independent and uniform on  $[-1/2, 1/2]$ , and put

$$X_a = \sum_{h \geq 0} \sum_{j=1}^{a_h} 2^{-h} U_{h,j}, \quad R_a = \frac{1}{2} A(1/2). \quad (2.3)$$

Absolute convergence is deterministic because the sum of the radii is  $R_a$ .

**Theorem 2.2** (Probability, analyticity, support, and convolution). *For every admissible  $a$ :*

1.  $X_a$  is supported on  $[-R_a, R_a]$  and has characteristic function  $\Phi_a$ ;
2. the product in (2.2) converges locally uniformly on  $\mathbb{C}$ , hence is entire;
3.  $\Phi_{a+b} = \Phi_a \Phi_b$ , and the corresponding probability law is the convolution of the laws for  $a$  and  $b$ ;
4. if  $\sum_h a_h = \infty$ , then  $X_a$  has a compactly supported  $C^\infty$  density; if only finitely many  $a_h$  are nonzero, its law is a finite uniform box spline.

All four assertions are **proved here**. At the probability-law level, only the classical case  $a_h \equiv 1$  is represented: `Fabius.rvachevFourier_eq_product` (`FourierProduct.lean`) identifies the Fourier transform of the Rvachev up-function with the sinc product (2.2) at  $a_h \equiv 1$ , in the same normalization (2.1), and `Fabius.ProbabilityRepresentation.rvachevUp_eq_weightedSum_probability_global` (`ProbabilityRepresentation.lean`) is the matching dyadic-uniform-series representation, stated in the Fabius normalization  $\sum_{h \geq 0} 2^{-h-1} V_h$  on  $[0, 1]$ , which is  $(X_a + R_a)/2$  in the notation used here.

The transform itself is now formal at a general admissible  $a$ , in `GeneralizedRvachevProduct.lean`: `Fabius.generalizedRvachevProduct` is  $\Phi_a$  in the normalization (2.1), and `Fabius.generalizedSincFactors_multipliable` gives convergence of the product at every  $z \in \mathbb{C}$  — including the points where factors vanish, since multipliability in Mathlib’s sense is unconditional convergence of the net of finite subproducts and admits the limit 0. Assertion (iii),  $\Phi_{a+b} = \Phi_a \Phi_b$ , is `Fabius.generalizedRvachevProduct_add`; the finite-support case of (iv) is `Fabius.generalizedRvachevProduct_eq_prod`; the classical case is recovered as an identity, not merely a specialization (`Fabius.generalizedRvachevProduct_one`). The zero divisor is formal in both directions —  $\Phi_a(z) = 0$  exactly when  $z = k2^h$  for a nonzero integer  $k$  and some  $h$  with  $a_h \neq 0$  (`Fabius.generalizedRvachevProduct_eq_zero_iff`) — as are the one-step refinement law (`Fabius.generalizedRvachevProduct_two_mul`) and its all-orders iterate (`Fabius.generalizedRvachevProduct_two_pow_mul`).

Assertion (ii) is formalized as well, in `GeneralizedRvachevEntire.lean`: the product converges locally uniformly on every ball (`Fabius.hasProdLocallyUniformlyOn_generalizedRvachevProduct`) and  $\Phi_a$  is therefore entire (`Fabius.generalizedRvachevProduct_differentiable`). With that in hand the zero divisor is complete —  $\Phi_a$  vanishes exactly at the nonzero integers  $n$  with  $m_a(|n|) > 0$ , to order exactly  $m_a(|n|)$ , and nowhere else (`GeneralizedZeroDivisor.lean`).

The characteristic-function reading of (i) now has three elementary necessary checks, though not positive-definiteness or the statement itself:  $\Phi_a(0) = 1$ , continuity from entirety, and

$$\|\Phi_a(x)\| \leq 1 \quad \text{for every real } x$$

(`Fabius.norm_generalizedRvachevProduct_ofReal_le_one`, `GeneralizedRealBound.lean`), which had been missing for arbitrary admissible weights. The classical constant-weight case

already appears as `Fabius.norm_rvachevFourierProduct_1e_one` in `BaselineDecay.lean`. The generalized bound is attained at  $x = 0$ , so it is sharp.

What remains not formalized is the probabilistic content proper: the support statement (i) — that  $R_a = A_a(1/2)/2$  is the actual support radius rather than only the deterministic radius bound — the random variable  $X_a$  and the convolution reading of (iii), and the smoothness dichotomy (iv). Assertion (iii) is formal as an identity of products but not as a statement about laws. The modules prove the algebraic, convergence and analytic skeleton on which the probabilistic layer would rest, and their docstrings say so explicitly.

*Proof.* Independence gives the finite-prefix characteristic products. The variables are bounded and  $\sum_{h,j} |2^{-h} U_{h,j}| \leq R_a$ , so the finite sums converge almost surely and in every  $L^p$ , and their characteristic functions converge to that of  $X_a$ .

Locally,  $\log \operatorname{sinc}_\pi w = \mathcal{O}(w^2)$ . Since  $\sum_h a_h 4^{-h} = A(1/4) < \infty$ , the logarithms of the factors converge normally on compact sets avoiding a harmless choice of local branch; standard infinite-product theory then gives an entire limit. Addition of exponent sequences is multiplication of products, which is exactly convolution of independent sums.

If infinitely many individual factors occur, fix  $M$  of them. On the real axis,  $|\operatorname{sinc}_\pi(\xi/2^h)| \leq \min(1, 2^h/(\pi|\xi|))$ . The chosen  $M$  factors give  $|\Phi_a(\xi)| \leq C_M(1 + |\xi|)^{-M}$ , while all remaining factors have modulus at most one. Thus  $\Phi_a$  is rapidly decreasing; Fourier inversion gives a  $C^\infty$  density. With finitely many factors one obtains a finite convolution of interval densities, hence a box spline.  $\square$

*Remark 2.3.* The classical Rvachev transform corresponds to  $a_h \equiv 1$ . The audited Pascal hierarchy corresponds to  $a_h = \binom{h}{r-1}$ . The point of [theorem 2.2](#) is not merely that both products exist, but that the collection of all such laws is a commutative convolution monoid whose algebra is the coefficientwise additive monoid of exponent sequences.

### 3 Integer zeros, spectral zeta, digital defects, and lobe signs

#### 3.1 The universal zero-divisor formula

Euler’s product for the normalized sinc gives

$$\operatorname{sinc}_\pi z = \prod_{k \geq 1} \left(1 - \frac{z^2}{k^2}\right).$$

An integer  $n \neq 0$  is a zero of the scale- $h$  factor precisely when  $2^h \mid n$ . This immediately yields the following master theorem.

**Theorem 3.1** (Zero divisor and spectral zeta). *Let  $a$  be admissible and define*

$$m_a(n) = \sum_{h=0}^{\nu_2(n)} a_h, \quad n \geq 1. \quad (3.1)$$

Then

$$\Phi_a(z) = \prod_{n \geq 1} \left(1 - \frac{z^2}{n^2}\right)^{m_a(n)} \quad (3.2)$$

with locally uniform convergence. In particular,  $\operatorname{ord}_{z=\pm n} \Phi_a = m_a(n)$ . For  $\Re s > 1$ ,

$$\mathcal{Z}_a(s) := \sum_{n \geq 1} \frac{m_a(n)}{n^s} = \zeta(s) A(2^{-s}). \quad (3.3)$$



Both statements are [proved here](#). The multiplicity function (3.1) is formalized as `Fabius.weightedScaleMultiplicity` (`WeightedScaleMultiplicity.lean`), and in two directions more generally than it is used here: the base 2 is replaced by an arbitrary  $b$ , and the weights  $a_h$  may lie in an arbitrary additive commutative monoid rather than in  $\mathbb{N}_0$ . The canonical form (3.2) is formalized at a general admissible  $a$ , in `GeneralizedCanonicalForm.lean`:

$$\Phi_a(z) = \prod_{n \geq 1} \left(1 - \frac{z^2}{n^2}\right)^{m_a(n)}$$

(`Fabius.generalizedRvachevProduct_eq_canonical`). The regrouping reuses, verbatim, the fibre equivalence already built for the classical case (`Fabius.dyadicFactorEquiv`, `SincCanonicalProduct.lean`): the map  $(h, r) \mapsto (r+1)2^h$  has fibre  $\{h : h \leq \nu_2(n)\}$  over  $n$ . The only change is that each factor now carries the exponent  $a_h$ , so the fibre product is a product of  $\nu_2(n) + 1$  powers of the same base and collapses to a single power whose exponent is the prefix sum — which is exactly  $m_a(n)$ . That is why the multiplicity function was the right object to formalize first. The classical statement is recovered as a corollary (`Fabius.generalizedRvachevProduct_one_of_canonical`).

The locally uniform convergence the volume states alongside this display, and the consequence  $\text{ord}_{z=\pm n} \Phi_a = m_a(n)$  that the volume draws from it, are both formalized at a general admissible  $a$ , in `GeneralizedRvachevEntire.lean`. The convergence is `Fabius.hasProdLocallyUniformlyOn_generalizedRvachevProduct`, on every ball  $\|z\| < R$ ; `HasProdLocallyUniformlyOn` is by definition the locally uniform convergence of the finite subproducts, so this is the volume’s clause verbatim rather than a surrogate for it. Holomorphy on each ball (`Fabius.differentiableOn_generalizedRvachevProduct_ball`) and hence entirety (`Fabius.generalizedRvachevProduct_differentiable`) follow, and with them the order statement in both halves:

$$\text{ord}_{z=n} \Phi_a = m_a(n) \quad \text{and} \quad \text{ord}_{z=-n} \Phi_a = m_a(n), \quad n \geq 1,$$

as `Fabius.analyticOrderAt_generalizedRvachevProduct_pos` and `Fabius.analyticOrderAt_generalizedRvachevProduct_neg_pos`. The hypothesis  $n \geq 1$  is not a limitation of the proof but of the statement:  $\Phi_a(0) = 1$ , so the order at the origin is 0, whereas  $m_a(0) = a_0$  may be anything.

Two remarks on how this was reached, both of which sharpen the volume. First, the algebraic input was already in place before this module — the factorization  $\Phi_a(z) = (1 - z^2/n^2)^{m_a(n)} R_n(z)$  with  $R_n(\pm n) \neq 0$  (`Fabius.generalizedRvachevProduct_eq_canonicalComplexFactor_mul`, `Fabius.canonicalCofactor_natCast_ne_zero`) — so what was genuinely missing was analyticity of the cofactor  $R_n$ , supplied here as `Fabius.canonicalCofactor_differentiable`. Second, the reflected half needs no evenness of  $\Phi_a$ : the mirror factorization  $1 - z^2/n^2 = (z+n) \cdot (-(z-n)/n^2)$  (`Fabius.canonicalComplexFactor_eq_add_mul`) is the same one line of algebra as its sibling, so both halves of  $\text{ord}_{z=\pm n}$  are proved directly and the appeal to  $\Phi_a(-z) = \Phi_a(z)$  (`Fabius.generalizedRvachevProduct_neg`, `LobeSignNegative.lean`) is not needed.

Third, the order statement now has a companion that completes it into a divisor. `GeneralizedRvachevProduct.lean` pins the zero set in scale coordinates,  $\exists h, a_h \neq 0$  and  $\exists k \neq 0, z = k2^h$ , and its header recorded that the translation into arithmetic coordinates was not carried out and the multiplicities not computed. `GeneralizedZeroDivisor.lean` does both:

$$\Phi_a(z) = 0 \iff \exists n \in \mathbb{Z}, n \neq 0, z = n, m_a(|n|) \neq 0$$

(`Fabius.generalizedRvachevProduct_eq_zero_iff_int`), and  $\text{ord}_{z=n} \Phi_a = m_a(|n|)$  at every nonzero integer, both signs in one statement (`Fabius.analyticOrderAt_generalizedRvachevProduct_int`). Since  $\Phi_a$  is entire, the two together are its complete divisor: it vanishes precisely on  $\{n \in \mathbb{Z} : n \neq 0, m_a(|n|) > 0\}$ , to exactly that order, and nowhere else in  $\mathbb{C}$  — in particular nowhere off the real axis (`Fabius.generalizedRvachevProduct_ne_zero_of_not_int`). The bridge

between the two coordinate systems is arithmetic rather than analytic:  $m_a(m)$  is a sum of naturals over  $h \leq \nu_2(m)$ , so it is nonzero exactly when some  $a_h$  with  $2^h \mid m$  is (`Fabius.weightedScaleMultiplicity_two_ne_zero_iff`).

The divisor now also has an exact inverse. The zero-definition, six-theorem module `GeneralizedRvachevIdentifiability.lean` proves, for every admissible exponent sequence,

$$\text{ord}_{z=2^n} \Phi_a = \sum_{h=0}^n a_h$$

(`Fabius.analyticOrderAt_generalizedRvachevProduct_two_pow`), decodes  $a_0$  from the order at  $z = 1$  and every  $a_{n+1}$  by the consecutive first difference (`Fabius.exponent_zero_eq_toNat_analyticOrderAt_generalizedRvachevProduct`, `Fabius.exponent_succ_eq_toNat_analyticOrderAt_generalizedRvachevProduct`), and therefore proves both dyadic-order injectivity (`Fabius.exponentSequence_eq_of_analyticOrderAt_two_pow_eq`) and full-product rigidity (`Fabius.generalizedRvachevProduct_eq_iff`). Its arithmetic core `Fabius.weightSequence_eq_of_weightedScaleMultiplicity_base_pow_eq` works at every base  $b > 1$  and for weights in any additive cancellative commutative monoid. The sample indexed by  $n = 0$  is  $z = 1$ , not the origin: every generalized product equals 1 at  $z = 0$ .

At the constant weight the sharper integer-indexed form remains available: `Fabius.analyticOrderAt_generalizedRvachevProduct_one` reads  $\text{ord}_{z=m} \Phi_1 = m_1(|m|)$  for every nonzero integer  $m$ , chaining `Fabius.analyticOrderAt_rvachevFourierProduct_int` (`IntegerZeroAnalyticOrder.lean`) with the exponent identity. The spectral zeta (3.3) is now formalized as well, in `SpectralZetaWeighted.lean`, for real  $s > 1$  and at an arbitrary base  $b > 1$ , with the  $\zeta$  factor left as its own  $p$ -series rather than identified with `riemannZeta` (`Fabius.tsum_weightedScaleMultiplicity_succ_rpow`). Three hypotheses are carried and none is hidden:  $1 < s$ , nonnegativity of the weights, and summability of  $A(b^{-s})$  — the last genuinely not automatic, since  $a$  may grow arbitrarily fast. The rearrangement needs nothing about the canonical product: the fibre of  $(h, k) \mapsto b^h k$  over  $n$  is  $\{h : b^h \mid n\}$ , whose weight sum is exactly  $m_a(n)$ , and it is the formalization of  $m_a$  at arbitrary base — noted above — that makes this reachable. The unit-weight case is recorded as a guard (`Fabius.tsum_padicValNat_succ_rpow`). What (3.3) still lacks is narrower than the product: the corpus proves it for real  $s > 1$  with the first factor as its own  $p$ -series, so the extension to complex  $s$  and the identification of that factor with `riemannZeta` are the two remaining steps.

*Proof.* The scale- $h$  sinc contributes one zero at  $n$  for every representation  $n = 2^h k$  with  $k \geq 1$ , and it occurs with multiplicity  $a_h$ . Summing over  $h \leq \nu_2(n)$  gives (3.1). Moreover

$$\sum_{n \geq 1} \frac{m_a(n)}{n^2} = \sum_{h \geq 0} a_h \sum_{k \geq 1} \frac{1}{(2^h k)^2} = \zeta(2) A(1/4) < \infty,$$

so the genus-zero canonical product converges normally. The same rearrangement for  $\Re s > 1$  gives

$$\sum_{n \geq 1} \frac{1}{n^s} \sum_{2^h \mid n} a_h = \sum_{h \geq 0} a_h 2^{-hs} \zeta(s).$$

□

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**Spectral–probabilistic dual use of  $A$ .** The spectral side samples  $A$  at  $2^{-s}$ , while the probability side will sample it at  $4^{-j}$ . The entire zero divisor and every even cumulant are therefore two transforms of one coefficient sequence, not merely analogous formulas.

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### 3.2 Exact counting and a generalized digital defect

Let

$$M_a(N) = \sum_{n=1}^N m_a(n). \quad (3.4)$$

For  $N = \sum_{j \geq 0} b_j(N)2^j$ , define the tail weights

$$\tau_j(a) = \sum_{k \geq 1} a_{j+k} 2^{-k}. \quad (3.5)$$

**Theorem 3.2** (Exact zero count and digit defect). *For every positive integer  $N$ ,*

$$M_a(N) = \sum_{h \geq 0} a_h \left\lfloor \frac{N}{2^h} \right\rfloor, \quad (3.6)$$

$$= NA(1/2) - \rho_a(N), \quad \rho_a(N) = \sum_{h \geq 0} a_h \left\{ \frac{N}{2^h} \right\}, \quad (3.7)$$

$$= NA(1/2) - \sum_{j \geq 0} b_j(N) \tau_j(a). \quad (3.8)$$

This is **proved here**. The floor form (3.6) is formalized as `Fabius.sum_range_weightedScaleMultiplicity` (`WeightedScaleMultiplicity.lean`), over the finite range  $h < N$  that already contains every nonzero term, and again for an arbitrary base  $b \neq 1$  and weights in an arbitrary additive commutative monoid, so the Lean statement is strictly more general than (3.6). The defect form (3.7) is now formalized as well, in `DigitDefectCounting.lean`: with  $A(1/2) = \sum_h a_h 2^{-h}$  (`Fabius.dyadicWeightValue`) and  $\rho_a(N) = \sum_h a_h \{N/2^h\}$  (`Fabius.dyadicDefect`), the identity  $\sum_h a_h \lfloor N/2^h \rfloor = NA(1/2) - \rho_a(N)$  holds under the single hypothesis that  $\sum_h |a_h| 2^{-h}$  converges (`Fabius.tsum_mul_natCast_div_two_pow_defect`), and the cumulative reading is `Fabius.sum_range_weightedScaleMultiplicity_real`. Boundedness of  $a$  is not needed: since  $\{N/2^h\} \leq N/2^h$ , the same summability dominates both series. At  $a_h \equiv 1$  the identity reduces to Legendre's  $\sum_h \lfloor N/2^h \rfloor = 2N - \text{wt}_2(N)$ , carried as a regression test (`Fabius.sum_range_div_two_pow_add_digitSum`, `Fabius.dyadicDefect_one`). The per-scale digit expansion  $\{N/2^h\} = \sum_{j < h} b_j(N) 2^{j-h}$  that the bit form (3.8) rests on is also formal (`Fabius.fract_natCast_div_two_pow_eq_sum_range`), and so is the exchange of summation that converts it into (3.8), in `DigitDefectBitForm.lean`. The tail weight  $\tau_j(a) = \sum_{k \geq 1} a_{j+k} 2^{-k}$  is `Fabius.dyadicTailWeight`, well defined by `Fabius.summable_dyadicTailWeight_terms` and pinned against an off-by-one in the index convention by the halving recursion  $\tau_j = (a_{j+1} + \tau_{j+1})/2$  (`Fabius.dyadicTailWeight_succ`) together with  $\tau_j(1) = 1$  (`Fabius.dyadicTailWeight_one`); the resolved defect  $\rho_a(N) = \sum_j b_j(N) \tau_j(a)$  is `Fabius.dyadicDefect_eq_sum_range_testBit` in windowed form and `Fabius.dyadicDefect_eq_tsum_testBit` unconditionally, and (3.8) itself is `Fabius.sum_range_weightedScaleMultiplicity_bit`. The exchange needs no Fubini on a triangular index set: only finitely many digits of  $N$  are nonzero, so a fixed window reduces it to a finite sum of convergent series. As a regression test the module re-derives  $\sum_j b_j(N) = \text{wt}_2(N)$  from the new machinery and lands it on the corpus's audited Legendre count (`Fabius.tsum_testBit_eq_digitSum`).

*Proof.* Count the multiples of  $2^h$  among  $1, \dots, N$  and sum their contributions  $a_h$ . Subtracting the fractional parts from  $N \sum_h a_h 2^{-h}$  gives (3.7). Finally,

$$\left\{ \frac{N}{2^h} \right\} = \sum_{j=0}^{h-1} b_j(N) 2^{j-h};$$

absolute convergence permits exchanging the sums and yields (3.8). □

For  $a_h \equiv 1$ , one has  $A(1/2) = 2$  and  $\tau_j = 1$ , recovering the audited identity  $M_a(N) = 2N - \text{wt}_2(N)$ . For general  $a$ , the ordinary binary weight is replaced by a weighted suffix transform of the exponent sequence.

### 3.3 Lobe signs and generalized Thue–Morse characters

For  $x \in (N, N+1)$ , every canonical-product factor with index  $n \leq N$  is negative and all remaining factors are positive. Hence

$$\text{sgn } \Phi_a(x) = (-1)^{M_a(N)}. \quad (3.9)$$

Modulo two,  $\lfloor N/2^h \rfloor \equiv b_h(N)$ , so the sign is a digit character.

**Theorem 3.3** (Generalized lobe sign). *Put  $\bar{a}_h = a_h \bmod 2$  and*

$$\varepsilon_a(N) = (-1)^{\sum_{h \geq 0} \bar{a}_h b_h(N)}. \quad (3.10)$$

*Then, away from the integer zeros,*

$$\boxed{\text{sgn } \Phi_a(x) = \varepsilon_a(\lfloor x \rfloor)}. \quad (3.11)$$

Moreover  $\varepsilon_a$  is a 2-automatic sequence if and only if the parity word  $(\bar{a}_h)_{h \geq 0}$  is eventually periodic. Both statements are **proved here**. At  $a_h \equiv 1$  the sign law (3.11) is formalized as `Fabijs.rvachevFourierProduct_eq_thueMorse_sign_mul_norm (ThueMorseLobeSign.lean)`, in the equivalent form  $\Phi(x) = \varepsilon_N \|\Phi(x)\|$  on the lobe  $(N, N+1)$ . The general parity character (3.10) is now formalized as `Fabijs.parityCharacter (ParityCharacter.lean)`. Because  $b_h(N) = 1$  exactly when  $h$  lies in the bit support of  $N$ , the exponent is a plain sum of weights over that support, and the corpus’s Boolean-cube machinery applies verbatim: `Fabijs.parityCharacter_const_one` recovers the Thue–Morse sign at  $a_h \equiv 1$ , `Fabijs.parityCharacter_mod_two` confirms that only the parity word  $(\bar{a}_h)$  enters, and `Fabijs.prod_one_add_neg_one_pow_eq_sum_parityCharacter` gives the weighted product identity  $\prod_{j < m} (1 + (-1)^{a_j} z^{2^j}) = \sum_{n < 2^m} \varepsilon_a(n) z^n$  over an arbitrary commutative ring — which at  $a_h \equiv 1$  is the finite Thue–Morse product identity. The automaticity criterion is formal too, in `ParityCharacterKernel.lean`, and the assessment that it needed automatic-sequence theory was too pessimistic: taking the standard finite-2-kernel characterization as the definition, the criterion is purely combinatorial. Writing  $\mathcal{K}(f) = \{n \mapsto f(2^k n + r) : r < 2^k\}$ , one has

$$\mathcal{K}(\varepsilon_a) \text{ finite} \iff (\bar{a}_h)_{h \geq 0} \text{ eventually periodic}$$

(`Fabijs.finite_twoKernel_parityCharacter_iff`). The engine is the volume’s own kernel-shift identity (3.12) (`Fabijs.parityCharacter_two_pow_mul_add`), which rests on the bit-support split of  $2^k n + r$ : read at  $r = 0$  it places every shifted character in the kernel, and read in general it makes every kernel element  $\pm$  a shifted character, so the kernel is covered by one image together with its negation. The converse direction uses the probe  $\varepsilon_{S^k a}(2^j) = (-1)^{\bar{a}_{k+j}}$  (`Fabijs.parityCharacter_shiftSeq_two_pow`) to turn equal characters into equal shifted parity words. What the library still lacks is the equivalence between a finite 2-kernel and an automaton definition; that is imported from the literature as the meaning of the word, not proved. The constant weight is recorded as a guard (`Fabijs.finite_twoKernel_thueMorseSign`): the weight word is periodic and the kernel finite, while the corpus’s `Fabijs.thueMorseSign_not_eventually_periodic` says the sign sequence is not eventually periodic — the criterion constrains the weight word, not the character. The sign law (3.11) splits in two, and its arithmetic half — the observation, made here in one line, that  $\lfloor N/2^h \rfloor \equiv b_h(N)$  modulo two, so that the exponent (3.9) is a digit character — is formal in `DigitCharacterCongruence.lean`:

$$M_a(N) = \sum_h a_h \left\lfloor \frac{N}{2^h} \right\rfloor \equiv \sum_{h \in \text{supp}_2(N)} a_h \pmod{2}, \quad \text{hence } (-1)^{M_a(N)} = \varepsilon_a(N)$$

(*Fabius.sum\_div\_two\_pow\_mul\_mod\_two*, *Fabius.neg\_one\_pow\_sum\_div\_two\_pow*); the bridge is simply that the  $h$ -th bit of  $N$  is  $\lfloor N/2^h \rfloor \bmod 2$ , and at the constant weight the statement becomes Legendre’s count read modulo two (*Fabius.neg\_one\_pow\_sum\_div\_two\_pow\_const\_one*). The analytic half is formal as well, in *LobeSignLaw.lean*, so (3.9) and (3.11) are theorems — on the open lobes. On  $N < x < N + 1$  the canonical form splits into a finite prefix of factors of index  $n \leq N$ , each with a negative base, and a tail of positive ones, so

$$\operatorname{sgn} \Phi_a(x) = (-1)^{M_a(N)} = \varepsilon_a(N)$$

(*Fabius.generalizedRvachevProduct\_ofReal\_eq\_parityCharacter\_mul\_norm*). The statement is carried across to  $\Phi_a$  itself, not merely to a real surrogate, and the constant-weight case reproduces the corpus’s existing Thue–Morse lobe sign.

The integer points are covered too, in *CanonicalIntegerPoint.lean*. At  $x = n \geq 1$  the canonical factor of index  $n$  is  $0^{m_a(n)}$ , so when  $m_a(n) = 0$  it is 1 — neither negative nor vanishing — and the same prefix/tail argument gives the same formula (*Fabius.generalizedRvachevProduct\_n\_atCast\_eq\_parityCharacter\_mul\_norm*); when  $m_a(n) \neq 0$  the product vanishes, which is the volume’s exclusion (*Fabius.generalizedRvachevProduct\_natCast\_eq\_zero\_of\_ne\_zero*). So (3.11) is formalized at every real  $x > 0$  at which no factor of  $\Phi_a$  vanishes, which is the volume’s own scope on that range.

Two residues are stated rather than glossed. At  $x = 0$  no factor vanishes and  $\Phi_a(0) = 1$ , but the hypothesis  $m_a(0) = 0$  need not hold (indeed  $m_a(0) = a_0$ ), so that point is closed by a separate lemma (*Fabius.canonicalRealProduct\_zero\_eq\_one*) rather than by the theorem above. Negative arguments are carried by evenness of  $\Phi_a$  (*Fabius.generalizedRvachevProduct\_neg*, *LobeSignNegative.lean*), and there the law reads  $\varepsilon_a(\lfloor |x| \rfloor)$  rather than  $\varepsilon_a(\lfloor x \rfloor)$ : on  $-(N + 1) < x < -N$  the reflected point lies in the lobe  $(N, N + 1)$ , so the sign is  $\varepsilon_a(N)$ , while  $\lfloor x \rfloor = -(N + 1)$  is not even a natural number. We write  $\lfloor x \rfloor$  because we have  $x \geq 0$  in mind.

Assembled, these pieces give the law on the whole axis with no hypothesis at all, in *LobeSignComplete.lean*, in which  $\Phi_a$  is read on the real axis through its canonical product (3.2):

$$\Phi_a(x) = 0 \quad \text{or} \quad 0 < \varepsilon_a(\lfloor |x| \rfloor) \Phi_a(x), \quad x \in \mathbb{R}$$

(*Fabius.parityCharacter\_mul\_canonicalRealProduct\_dichotomy*), the first alternative occurring exactly when  $|x|$  is a positive integer of nonzero multiplicity (*Fabius.canonicalRealProduct\_eq\_zero\_iff\_of\_nonneg*). There is no third possibility and no remaining gap; what made the assembly possible is the complete zero set of *GeneralizedZeroDivisor.lean*, without which the exceptional set could be excluded but not identified.

*Proof.* The sign formula follows from (3.9), (3.6), and reduction modulo two. For the automaticity criterion, let  $S$  denote the left shift of the parity word. If  $0 \leq r < 2^k$ , binary concatenation gives

$$\varepsilon_a(2^k n + r) = \varepsilon_a(r) \varepsilon_{S^k a}(n). \quad (3.12)$$

Thus an eventually periodic parity word has only finitely many shifts and therefore gives a finite 2-kernel. Conversely, equality of two shifted sign sequences forces equality of the corresponding shifted parity words, because  $\varepsilon_{S^k a}(2^j) = (-1)^{\bar{a}_{k+j}}$ . A finite 2-kernel therefore implies a finite shift orbit of the parity word, which is equivalent to eventual periodicity. This is the present Walsh-character specialization of the standard finite-kernel characterization of automatic sequences [40].  $\square$

### 3.4 Finite sign products and exact Prouhet order

For  $L \geq 0$ , let

$$O_L(a) = \{0 \leq h < L : a_h \text{ is odd}\}, \quad \omega_L(a) = |O_L(a)|. \quad (3.13)$$

**Theorem 3.4** (Generalized Prouhet block). *The length- $2^L$  sign polynomial factors as*

$$T_{a,L}(z) := \sum_{n=0}^{2^L-1} \varepsilon_a(n) z^n = \prod_{h=0}^{L-1} \left(1 + (-1)^{a_h} z^{2^h}\right). \quad (3.14)$$

*Its zero at  $z = 1$  has exact order  $\omega_L(a)$ . Consequently*

$$\sum_{n < 2^L} \varepsilon_a(n) n^k = 0, \quad 0 \leq k < \omega_L(a), \quad (3.15)$$

*and the first surviving power moment is*

$$\sum_{n < 2^L} \varepsilon_a(n) n^{\omega_L(a)} = (-1)^{\omega_L(a)} \omega_L(a)! 2^{L-\omega_L(a) + \sum_{h \in O_L(a)} h}. \quad (3.16)$$

This is [proved here](#). Its sharp-moment core is formalized, and in greater generality: `Fabius.sum_powerset_neg_one_pow_pow_card` (`ThueMorseSparseProuhet.lean`) evaluates

$$\sum_{T \subseteq S} (-1)^{|T|} \left(x + \sum_{j \in T} w_j\right)^{|S|} = (-1)^{|S|} |S|! \prod_{j \in S} w_j$$

for an arbitrary finite index set  $S$ , arbitrary step weights  $w_j$  in an arbitrary commutative ring, and every offset  $x$ . Taking  $S = O_L(a)$  and  $w_h = 2^h$  produces exactly the constant  $(-1)^{\omega_L(a)} \omega_L(a)! 2^{\sum_{h \in O_L(a)} h}$  of (3.16), independently of  $x$ , and the companion `Fabius.sum_powerset_neg_one_pow_eval_of_degree_lt` is the vanishing (3.15) for every polynomial of degree below  $|S|$ ; `Fabius.sum_submask_neg_one_pow_pow` is the binary specialization in which  $S$  is the bit support of an integer. The factorization (3.14) is formal as well, and over an arbitrary commutative ring rather than over  $\mathbb{Z}[z]$ : it is `Fabius.prod_one_add_neg_one_pow_eq_sum_parityCharacter` (`ParityCharacter.lean`), the same identity already cited in [theorem 3.3](#), read from right to left. The free-even-layer step is formal as well, in `GeneralizedProuhetBlock.lean`, so both boxed displays (3.15) and (3.16) are now theorems (`Fabius.sum_range_two_pow_parityCharacter_mul_pow_eq_zero` and `Fabius.sum_range_two_pow_parityCharacter_mul_pow`). The mechanism is that  $\varepsilon_a$  cannot see the even layers: splitting  $T \subseteq \{0, \dots, L-1\}$  into  $T \cap O_L(a)$  and its complement, the sign depends only on  $|T \cap O_L(a)|$  while the value  $\sum_{j \in T} 2^j$  splits into an odd-layer part and an even-layer offset, so the block sum is a sum over even-layer offsets of a core that does not depend on the offset — which is precisely why the factor  $2^{L-\omega_L(a)}$  appears and nothing more. The decomposition itself (`Fabius.sum_powerset_range_split`) is stated over an arbitrary additive commutative monoid for an arbitrary summand depending on the two parts. Both boxed displays are in fact extremal cases of a single statement, formalized here in that stronger form: for every integer polynomial  $p$  of degree at most  $\omega_L(a)$ ,

$$\sum_{n < 2^L} \varepsilon_a(n) p(n) = p_{\omega_L(a)} \cdot (-1)^{\omega_L(a)} \omega_L(a)! 2^{L-\omega_L(a)} 2^{\sum_{h \in O_L(a)} h},$$

with  $p_\omega$  the coefficient of  $z^\omega$  and no nonzeroness hypothesis anywhere — the zero polynomial and the case  $\omega_L(a) = 0$  are included (`Fabius.sum_range_two_pow_parityCharacter_mul_eval`). The vanishing (3.15) is that coefficient being zero below the threshold, and (3.16) is  $p = z^{\omega_L(a)}$ , where it is one. A constant-weight corollary recovers the classical Prouhet–Thue–Morse evaluation (`Fabius.sum_range_two_pow_thueMorseSign_mul_pow`). The exact order of the zero of  $T_{a,L}$  at  $z = 1$  is formal as well, in `ProuhetBlockZeroOrder.lean`, so this theorem is now formalized in all three of its displays:

$$\text{ord}_{z=1} T_{a,L} = |O_L(a)|$$

(`Fabius.rootMultiplicity_prouhetBlockPoly`) — and so no zero at all when every layer below  $L$  is even, a case the statement covers. The mechanism is the factorization read in  $\mathbb{Z}[z]$ : an odd layer

contributes  $1 - z^{2^h}$ , a simple zero, while an even layer contributes  $1 + z^{2^h}$ , which is 2 at  $z = 1$ ; root multiplicity is additive over the product, so the order counts the odd layers. The classical case is recorded separately (`Fabius.rootMultiplicity_prouhetBlockPoly_const_one`).

*Proof.* Expand the product by choosing the binary digit  $b_h \in \{0, 1\}$  independently. At  $z = 1$ , an odd exponent contributes a factor  $1 - z^{2^h}$  with a simple zero, while an even exponent contributes  $1 + z^{2^h} = 2$ . Therefore the zero order is exactly  $\omega_L$  and its leading coefficient in powers of  $z - 1$  is

$$(-1)^{\omega_L} 2^{L - \omega_L + \sum_{h \in O_L} h}.$$

The derivatives  $T_{a,L}^{(k)}(1)$  are the signed falling-factorial moments. Vanishing for  $k < \omega_L$  implies vanishing of ordinary moments of those degrees. At the first surviving degree, lower falling-factorial terms have already vanished, so multiplying the leading coefficient by  $\omega_L!$  gives (3.16).  $\square$

*Remark 3.5.* The usual Thue–Morse Prouhet theorem is the case  $a_h \equiv 1$ , for which  $\omega_L = L$ . Here the cancellation order is not tied to the block length alone: it is exactly the number of odd spectral layers present before scale  $L$ .

## 4 Bernoulli cumulants, Bell moments, and identifiability

### 4.1 The same generating function at $q = 4^{-j}$

For  $U \sim \text{Unif}[-1/2, 1/2]$ ,

$$\log \mathbb{E} e^{tU} = \log \frac{\sinh(t/2)}{t/2} = \sum_{j \geq 1} \frac{B_{2j}}{2j} \frac{t^{2j}}{(2j)!}. \quad (4.1)$$

**Theorem 4.1** (Cumulant transform). *For an admissible exponent sequence,*

$$\kappa_{2j}(X_a) = \frac{B_{2j}}{2j} A(4^{-j}), \quad \kappa_{2j+1}(X_a) = 0. \quad (4.2)$$

Equivalently,

$$\log \mathbb{E} e^{tX_a} = \sum_{j \geq 1} \frac{B_{2j} A(4^{-j})}{2j} \frac{t^{2j}}{(2j)!}. \quad (4.3)$$

This is [proved here](#).

*Proof.* Cumulants add under independent sums and scale homogeneously. The level- $h$  contribution to the  $2j$ th cumulant is  $a_h 2^{-2jh} B_{2j}/(2j)$ . Summing over  $h$  gives  $A(4^{-j})$ .  $\square$

**Corollary 4.2** (Bell-polynomial moments). *The even moments are*

$$\mathbb{E} X_a^{2n} = \mathcal{B}_{2n}^{\text{exp}}(0, \kappa_2, 0, \kappa_4, \dots, 0, \kappa_{2n}), \quad (4.4)$$

or explicitly

$$\mathbb{E} X_a^{2n} = (2n)! \sum_{\substack{j_1, \dots, j_n \geq 0 \\ \sum_{r=1}^n r j_r = n}} \prod_{r=1}^n \frac{1}{j_r!} \left( \frac{B_{2r} A(4^{-r})}{2r(2r)!} \right)^{j_r}. \quad (4.5)$$

This is a direct, established Bell-polynomial conversion applied to the new master cumulant formula. The conversion itself is formalized: (4.4) is `Fabius.completeBellPolynomial (MomentCumulantAlgebra.lean)`, the explicit partition sum (4.5) is `Fabius.partitionExpSum (ExponentialPartition.lean)`, and the two are proved to agree by `Fabius.partitionExpSum_eq_expCoeff (ExponentialBell.lean)`; all three hold in an arbitrary commutative  $\mathbb{Q}$ -algebra, so the formal statement is considerably more general than this application. Its cumulant input [theorem 4.1](#) is not formalized, so the corollary is not yet formally available as stated.



## 4.2 Zero spectrum and cumulants determine one another

Normalize the even cumulants by

$$c_j(a) = \frac{2j}{B_{2j}} \kappa_{2j}(X_a) = A(4^{-j}). \quad (4.6)$$

**Theorem 4.3** (Four equivalent encodings). *The following data determine one another uniquely:*

1. the exponent sequence  $(a_h)_{h \geq 0}$ ;
2. the integer-zero multiplicity function  $m_a(n)$ ;
3. the quotient  $\mathcal{Z}_a(s)/\zeta(s) = A(2^{-s})$ ;
4. the normalized even-cumulant samples  $(c_j(a))_{j \geq 1}$ .

Constructively,

$$a_v = m_a(2^v) - m_a(2^{v-1}), \quad m_a(2^{-1}) := 0, \quad (4.7)$$

$$a_0 = \lim_{j \rightarrow \infty} c_j, \quad (4.8)$$

$$a_v = \lim_{j \rightarrow \infty} 4^{vj} \left( c_j - \sum_{h=0}^{v-1} a_h 4^{-hj} \right), \quad v \geq 1. \quad (4.9)$$

This is **proved here**.

*Proof.* The first formula is the first difference of the prefix sum  $m_a(2^v) = \sum_{h \leq v} a_h$ . The spectral-zeta equivalence follows from (3.3). For the cumulants,  $c_j = A(4^{-j})$ , and  $4^{-j} \rightarrow 0$  inside the disk of analyticity of  $A$ . Equality of all samples for two sequences gives an analytic difference with infinitely many zeros accumulating at the interior point 0, hence the generating functions coincide. The explicit limits follow by peeling off one Taylor coefficient at a time; the remaining tail converges absolutely because  $A(1/4) < \infty$ .  $\square$

**Lean crosswalk.** Items 1 and 2, including the constructive recovery formula (4.7), are machine-checked exactly. Together, `GeneralizedZeroDivisor.lean` and the zero-definition, six-theorem `GeneralizedRvachevIdentifiability.lean` identify  $m_a(2^n)$  with the analytic order of  $\Phi_a$  at  $z = 2^n$ , recover the head at  $z = 1$ , recover each successor exponent by an exact first difference, prove that equality of all dyadic order samples forces equality of the exponent sequences, and prove  $\Phi_a = \Phi_c \iff a = c$  for two admissible sequences. Thus the zero-order/exponent half of this theorem, and the stronger full-function rigidity statement, are exact Lean results. The spectral-zeta quotient in item 3 has formalized component infrastructure, but its identifiability equivalence as packaged here is still Partial. The cumulant identity and coefficient-peeling/analytic-sample uniqueness from  $A(4^{-j})$  in item 4, as well as construction and identifiability of the associated probability law, also remain Partial in Lean. The manuscript proof status **proved here** is unchanged; this paragraph records only the machine-checked boundary.

*Remark 4.4.* The limits in (4.8)–(4.9) are mathematically exact but numerically wasteful. The next section replaces them by finite, sign-certified,  $q$ -binomial extrapolants.

## 5 Certified Gaussian- $q$ -binomial Richardson tomography

The coefficient limits in theorem 4.3 are exact, but they use one cumulant at a time and amplify roundoff by powers of  $4^j$ . The geometric nature of the samples admits a much better reconstruction. The result is not merely an acceleration formula: positivity of the hidden coefficients supplies an alternating sign theorem and hence certified integer brackets.

### 5.1 Lagrange weights on a geometric grid

Fix  $q \in (0, 1)$ . We use

$$(q; q)_r = \prod_{k=1}^r (1 - q^k), \quad \begin{bmatrix} r \\ j \end{bmatrix}_q = \frac{(q; q)_r}{(q; q)_j (q; q)_{r-j}}.$$

For  $r \geq 0$  and  $0 \leq j \leq r$ , let

$$w_{r,j}(q) = \prod_{\substack{0 \leq k \leq r \\ k \neq j}} \frac{-q^k}{q^j - q^k}. \quad (5.1)$$

These are the Lagrange interpolation weights which evaluate, at the origin, a polynomial known at  $1, q, \dots, q^r$ .

**Lemma 5.1** (Closed Gaussian-binomial weights). *One has*

$$w_{r,j}(q) = (-1)^{r-j} \frac{q^{\binom{r-j+1}{2}}}{(q; q)_j (q; q)_{r-j}} = \frac{(-1)^{r-j} q^{\binom{r-j+1}{2}}}{(q; q)_r} \begin{bmatrix} r \\ j \end{bmatrix}_q. \quad (5.2)$$

Consequently,

$$\sum_{j=0}^r w_{r,j}(q) q^{jh} = \delta_{h0} \quad (0 \leq h \leq r). \quad (5.3)$$

For every integer  $h \geq r + 1$  the first nonvanishing value is

$$\sum_{j=0}^r w_{r,j}(q) q^{jh} = (-1)^r q^{\binom{r+1}{2}} \begin{bmatrix} h-1 \\ r \end{bmatrix}_q. \quad (5.4)$$

This is **proved here**. The weights (5.1) are formalized as `Fabiús.geometricLagrangeWeight` (`GeometricLagrange.lean`); the annihilation (5.3) is `Fabiús.sum_geometricLagrangeWeight_mul_pow`, with the normalization at  $h = 0$  recorded as the genuine boundary value  $0^0 = 1$ . The division-free forward and reversed Gaussian numerators are supplied by `GeometricQBinomialLagrange.lean`, while `Fabiús.sum_geometricLagrangeWeight_mul_pow_of_pos` (`GeometricResidualMoments.lean`) gives (5.4) for every  $h > 0$ ; above-diagonal zero extension contains the range  $1 \leq h \leq r$  as well. These results hold over an arbitrary field whenever  $q^0, \dots, q^r$  are distinct, so positivity and the hypothesis  $q \in (0, 1)$  enter only in later sign arguments. The denominator-free principal specialization  $H_n(1, q, \dots, q^r) = \begin{bmatrix} n+r \\ r \end{bmatrix}_q$  is proved over commutative semirings in `GeometricCompleteHomogeneous.lean`.

*Proof.* Separating the factors with  $k < j$  and  $k > j$  in (5.1) gives (5.2). The annihilation relations are the exactness of Lagrange interpolation for the monomials  $1, x, \dots, x^r$ .

For completeness, let  $H_h(x_0, \dots, x_r)$  denote the complete homogeneous symmetric polynomial of degree  $h$ . The divided difference of  $x^h$  at  $x_0, \dots, x_r$  is  $H_{h-r}(x_0, \dots, x_r)$ . The interpolation remainder at  $x = 0$ , with  $x_j = q^j$ , therefore gives

$$\sum_{j=0}^r w_{r,j} q^{jh} = (-1)^r q^{0+1+\dots+r} H_{h-r-1}(1, q, \dots, q^r).$$

The principal specialization identity  $H_n(1, q, \dots, q^r) = \begin{bmatrix} n+r \\ r \end{bmatrix}_q$  yields (5.4); see, for example, the standard finite  $q$ -binomial calculus in [21].  $\square$



**Post-snapshot Lean crosswalk.** The source-level individual quotient is the already compiled theorem `geometricLagrangeWeight_eq_gaussianBinomial_div` in `GeometricQBinomialLagrange.lean`; it assumes a field and injectivity of  $j \mapsto q^j$  on  $\{0, \dots, r\}$ , and deliberately uses the numerator `gaussianBinomialqr(r-j)`. The symmetric-index notation in (5.2) is now justified directly by the denominator-free generic symmetry theorems `gaussianBinomial_add_symm` and `gaussianBinomial_symm_via_completeHomogeneous` in `GeometricCompleteHomogeneous.lean`. The same current six-theorem module contains `completeHomogeneousEval_geometric`, which proves

$$H_n(1, q, \dots, q^r) = \begin{bmatrix} n+r \\ r \end{bmatrix}_q$$

over every commutative semiring, without division, distinctness, nonzeroness, or an exclusion of roots of unity. More literally, its right-hand side is the recursive term `gaussianBinomialq(n+r)r`, which agrees with the displayed quotient in the present field setting. Over a commutative ring, `sum_weight_mul_geometric_pow_of_pos` proves more generally that any row  $a_0, \dots, a_r$  with

$$\sum_{j=0}^r a_j (q^j)^d = 0^d \quad (0 \leq d \leq r)$$

satisfies

$$\sum_{j=0}^r a_j (q^j)^h = (-1)^r q^{\binom{r+1}{2}} \begin{bmatrix} h-1 \\ r \end{bmatrix}_q \quad (h > 0).$$

The Lagrange specialization is `sum_geometricLagrangeWeight_mul_pow_of_pos`; it gives both (5.3) and (5.4), with mass one at degree zero supplied separately.

The scaled exact-row theorem is genuinely stronger than merely factoring a unit from the nodes. If exactness is assumed directly on  $c, cq, \dots, cq^r$ , then for  $s \geq 0$

$$\sum_{j=0}^r a_j (cq^j)^{r+1+s} = c^{r+1+s} (-1)^r q^{\binom{r+1}{2}} \begin{bmatrix} r+s \\ r \end{bmatrix}_q;$$

this is `sum_weight_mul_scaled_geometric_pow_succ_add`, valid even when  $c$  is zero or a zero divisor. For the field-valued Lagrange row and the shifted nodes  $q^{m+j}$ , `sum_geometricLagrangeWeight_mul_shifted_pow_of_pos` gives, for every  $h > 0$ ,

$$\sum_{j=0}^r w_{r,j}(q) (q^{m+j})^h = (-1)^r q^{mh + \binom{r+1}{2}} \begin{bmatrix} h-1 \\ r \end{bmatrix}_q.$$

The finite theorems alone make no convergence claim. Their analytic companion `AnalyticSeriesFilter.lean` does supply the exact unconditionally summable Gaussian tail, including the shifted Lagrange row, under summability of each nonzero-weight sample. It supplies neither the one-sided sign nor the numerical remainder bound used below.

## 5.2 Exact transformed series and sign certification

Let

$$\mathcal{R}_{r,m}[A] := \sum_{j=0}^r w_{r,j}(q) A(q^{m+j}), \quad m \geq 1. \quad (5.5)$$

Scaling all interpolation nodes by  $q^m$  leaves the weights unchanged.

**Theorem 5.2** (Exact  $q$ -Richardson tomography). *Suppose  $A(x) = \sum_{h \geq 0} a_h x^h$  converges at  $x = q^m$ . Then*

$$\mathcal{R}_{r,m}[A] = a_0 + (-1)^r q^{\binom{r+1}{2}} \sum_{h \geq r+1} a_h \begin{bmatrix} h-1 \\ r \end{bmatrix}_q q^{mh}. \quad (5.6)$$

If  $a_h \geq 0$ , then

$$(-1)^r (\mathcal{R}_{r,m}[A] - a_0) \geq 0. \quad (5.7)$$

Moreover,

$$0 \leq (-1)^r (\mathcal{R}_{r,m}[A] - a_0) \leq \frac{q^{\binom{r+1}{2}}}{(q; q)_r} \left( A(q^m) - \sum_{h=0}^r a_h q^{mh} \right). \quad (5.8)$$

All three statements are **proved here**. The finite exactness half is formalized: `Fabiús.geometricLagrange_richardson_exact` (`GeometricLagrange.lean`) proves that the weights (5.1) annihilate the first  $r$  geometric error modes of a sample block beginning at an arbitrary index and return the constant term exactly. It is stated for samples in an arbitrary module over an arbitrary field, with no positivity hypothesis, no separate  $q \neq 0$  condition beyond finite-node injectivity, and no lower bound on the starting index, so it covers more than the real nonnegative-coefficient setting used here. The full transformed tail (5.6) is now the shifted specialization `Fabiús.geometricLagrangeAnalyticSeriesFilter_shifted` (`AnalyticSeriesFilter.lean`) under unconditional summability of the sampled series. Only nonzero Lagrange weights create an unweighted convergence obligation; a zero-weight node is discharged identically. This theorem excludes a conditionally-only boundary series and does not formalize the positivity argument, the sign (5.7), or the bound (5.8).

*Proof.* Insert the Taylor series of  $A$  into (5.5); absolute convergence permits interchanging the two sums. The terms  $1 \leq h \leq r$  vanish by (5.3), while (5.4) gives the remaining terms. The sign follows because Gaussian binomial coefficients are positive for  $0 < q < 1$ . Finally,

$$\begin{bmatrix} h-1 \\ r \end{bmatrix}_q = \frac{\prod_{k=0}^{r-1} (1 - q^{h-1-k})}{(q; q)_r} \leq \frac{1}{(q; q)_r},$$

which proves (5.8). □

**Corollary 5.3** (Certified integer brackets). *If the coefficients are nonnegative integers, then for any odd  $r_-$  and even  $r_+$ ,*

$$\lceil \mathcal{R}_{r_-,m}[A] \rceil \leq a_0 \leq \lfloor \mathcal{R}_{r_+,m}[A] \rfloor. \quad (5.9)$$

Once the two rounded endpoints coincide,  $a_0$  has been recovered with a proof certificate. For the Fabiús–Rvachev cumulants, the grid parameter is  $q = 1/4$  and the data are  $A(q^m) = 2m\kappa_{2m}/B_{2m}$ .

### 5.3 Conditioning does not deteriorate with order

High-order extrapolation is often unstable because its weights become enormous. The geometric grid behaves unusually well.

**Theorem 5.4** (Exact  $\ell^1$  condition number). *The weight norm is*

$$\sum_{j=0}^r |w_{r,j}(q)| = \frac{(-q; q)_r}{(q; q)_r} = \prod_{k=1}^r \frac{1 + q^k}{1 - q^k}. \quad (5.10)$$

It increases to the finite limit  $(-q; q)_\infty / (q; q)_\infty$ . At the cumulant grid  $q = 1/4$ ,

$$\frac{(-1/4; 1/4)_\infty}{(1/4; 1/4)_\infty} = 1.96926035366826943194719369697 \dots \quad (5.11)$$

This is **proved here**. Identity (5.10) was initially formalized only at the half base  $q = 1/2$ , as a strict specialization of what is stated here: `Fabius.sum_abs_discreteLimitWeight (FabiusDiscreteLimitToeplitz.lean)` evaluates the  $\ell^1$  norm of the  $r$ th Toeplitz row as  $(-1/2; 1/2)_r / (1/2; 1/2)_r$ , and `Fabius.discreteLimitWeight_eq_geometricLagrangeWeight` identifies that row, after the reversal  $j \mapsto r - j$ , with the weights (5.1). General  $q$  is now formalized too, in `GeneralQConditionNumber.lean`, and the hypotheses turn out to split three ways. The signed generating identity needs no hypothesis at all — it is a direct reading of `Fabius.finite_qBinomial_theorem` at the argument  $qz$  (`Fabius.sum_qToeplitzNumerator_mul_pow`, `Fabius.sum_qToeplitzWeight_mul_pow`), so the half base is entirely incidental there. The strict alternation  $(-1)^j w_q(n, j) > 0$  needs  $0 < q < 1$  (`Fabius.neg_one_pow_mul_qToeplitzWeight_pos`), and the absolute-value identity  $\sum_j |w_q(n, j)| = (-q; q)_n / (q; q)_n$  needs only  $0 \leq q < 1 - q = 0$  is admissible — (`Fabius.sum_abs_qToeplitzWeight`). Both order hypotheses are shown necessary by counterexample rather than asserted: at  $q = -1/2$  the alternation fails and the two sides are 1 and  $1/3$ , and at  $q = 2$  they are 3 and  $-3$  (`Fabius.sum_abs_qToeplitzWeight_ne_of_neg`, `Fabius.sum_abs_qToeplitzWeight_ne_of_one_lt`). As a machine-checked consistency check the two established half-base theorems are re-derived from the general ones (`Fabius.sum_abs_discreteLimitWeight_of_qToeplitz`, `Fabius.sum_range_discreteLimitWeight_mul_pow_of_qToeplitz`). The passage to the limit in  $r$  is formal too, in `LimitConditionNumber.lean`: for  $0 \leq q < 1$  both families  $j \mapsto 1 - aq^j$  are multipliable (`Fabius.multipliable_one_sub_mul_pow`), the finite symbols converge to the infinite ones (`Fabius.qPochhammerInf`, `Fabius.tendsto_finiteQPochhammerIn`), and  $(q; q)_\infty > 0$  (`Fabius.qPochhammerInf_self_pos`), so the condition numbers converge to  $(-q; q)_\infty / (q; q)_\infty$  (`Fabius.tendsto_sum_abs_qToeplitzWeight`). Positivity of the denominator is the crux, and it is obtained from the summability of the logarithms rather than from a Weierstrass product bound, which unpeeled is vacuous past  $q = 1/2$  — itself recorded as a theorem (`Fabius.weierstrass_bound_vacuous_at_three_quarters`) rather than as a remark. The failure of a uniform-in- $q$  bound is now effective as well: the limit is at least  $1/(1 - q)$  (`Fabius.one_div_one_sub_le_qConditionNumberLimit`), and in fact tends to  $+\infty$  as  $q \rightarrow 1^-$  (`Fabius.tendsto_qConditionNumberLimit_atTop_at_one_left`), with  $q = 999/1000$  exhibited (`Fabius.thousand_le_qConditionNumberLimit`), and it equals 1 exactly at  $q = 0$  (`Fabius.one_lt_qConditionNumberLimit`, `Fabius.qConditionNumberLimit_zero`). So the half-base constant 16 has no general- $q$  analogue, and that is a theorem. The numerical value (5.11) at the cumulant grid is now certified as well, though only to finitely many places: `CumulantGridConstant.lean` proves the kernel-checked rational enclosure

$$1.969260353 < \frac{(-1/4; 1/4)_\infty}{(1/4; 1/4)_\infty} < 1.969260354$$

(`Fabius.qConditionNumberLimit_quarter_enclosure`), pinning the first nine decimals, with a coarser cut-14 companion at seven and a guard enclosing the denominator symbol alone to nine (`Fabius.qPochhammerInf_quarter_enclosure`). The reusable content is the machinery rather than the number: the prefix/tail split (`Fabius.qPochhammerInf_eq_prefix_mul_tail`) exhibits the tail as again a  $q$ -Pochhammer symbol at the same base with coefficient scaled by  $q^N$ , so each tail estimate is proved once rather than once per cut, and the estimates carry an abstract deficit budget rather than a threshold on  $q$  — there is no  $q < 1/2$  anywhere — so they serve any base below 1 once the cut is taken large enough. The remaining twenty quoted digits are not proved.

*Proof.* Use (5.2), replace  $j$  by  $r - j$ , and apply the finite  $q$ -binomial theorem:

$$\sum_{k=0}^r q^{\binom{k+1}{2}} \begin{bmatrix} r \\ k \end{bmatrix}_q = \prod_{k=1}^r (1 + q^k) = (-q; q)_r.$$

Division by  $(q; q)_r$  gives (5.10). Both infinite products converge for  $|q| < 1$ . □

*Remark 5.5.* The bounded condition number is a structural advantage over extrapolation on an arithmetic grid. It does not remove the need for accurate high cumulants, but it prevents the interpolation layer itself from introducing an exponentially growing amplification factor.

## 5.4 Recovering the entire exponent sequence

After  $a_0, \dots, a_{v-1}$  have been certified, define the deflated generating function

$$A^{(v)}(x) = x^{-v} \left( A(x) - \sum_{h=0}^{v-1} a_h x^h \right) = \sum_{k \geq 0} a_{v+k} x^k. \quad (5.12)$$

Applying [theorem 5.2](#) to  $A^{(v)}$  recovers  $a_v$  with the same alternating certificate.

**Algorithm 5.6** (Cumulant-to-spectrum tomography). Given certified intervals for the normalized cumulants  $c_m = A(4^{-m})$ :

1. set  $q = 1/4$  and  $v = 0$ ;
2. subtract the already recovered polynomial  $\sum_{h < v} a_h q^{mh}$  from each  $c_m$  and multiply by  $q^{-mv}$ ;
3. evaluate one odd- and one even-order transform ([5.5](#));
4. round outward as in ([5.9](#));
5. increase  $m$  or the orders until the integer interval collapses;
6. increment  $v$  and repeat.

The truncation estimate ([5.8](#)), combined with interval arithmetic, turns the procedure into a fully rigorous recovery algorithm.

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**Relation to prior Richardson work.** The corpus already contains Gaussian- $q$ -binomial acceleration of finite sinc products, dyadic value sequences, spline approximants, and phase-locked endpoint expansions. The new point here is different: the input is the sequence of *even cumulants*, the unknowns are the integer multiplicities of dyadic uniform layers, and nonnegativity turns the Lagrange remainder into alternating spectral certificates.

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## 6 Shift algebra, refinement systems, and rational classification

### 6.1 The shift is the dilation operator in disguise

Let  $(Sa)_h = a_{h+1}$  and  $\Delta = S - I$ . For signed integer sequences we use the notation  $\Phi_b$  for the analytic germ at the origin obtained by exponentiating  $\sum_h b_h \log \operatorname{sinc}_\pi(z/2^h)$ ; negative entries may make its global continuation meromorphic, but every identity below is unambiguous near zero.

**Theorem 6.1** (Master shift–refinement identity). *For every admissible nonnegative sequence,*

$$\boxed{\Phi_a(2z) = \operatorname{sinc}_\pi(2z)^{a_0} \Phi_{Sa}(z).} \quad (6.1)$$

More generally, for every  $m \in \mathbb{N}_0$ ,

$$\Phi_a(2^m z) = \left( \prod_{h=0}^{m-1} \operatorname{sinc}_\pi(2^{m-h} z)^{a_h} \right) \Phi_{S^m a}(z). \quad (6.2)$$

Since  $S^m = (I + \Delta)^m$ , one also has the finite-difference factorization

$$\Phi_{S^m a}(z) = \prod_{r=0}^m \Phi_{\Delta^r a}(z)^{\binom{m}{r}}. \quad (6.3)$$

The final identity is initially an identity of analytic germs; all apparent poles cancel whenever the left side is entire. These statements are **proved here**. The exponent-level content of (6.1)–(6.2)—that passing to  $S^m a$  exposes the first  $m$  layers and shifts the rest—is formalized as `Fabijs.weightedScaleMultiplicity_base_pow_mul` (`WeightedScaleMultiplicity.lean`), for an arbitrary base  $b > 1$  and weights in an arbitrary additive commutative monoid. The one-step identity is `Fabijs.generalizedRvachevProduct_two_mul`, and the complete variable-exponent iteration is `Fabijs.generalizedRvachevProduct_two_pow_mul` (`WeightLinearityProducts.lean`). Under exactly the admissibility hypothesis  $\sum_h a_h/2^h < \infty$ , the latter theorem holds for every  $m \in \mathbb{N}_0$  and every complex  $z$ : its finite product over  $0 \leq h < m$  is precisely the exposed prefix  $\text{sinc}_\pi(2^{m-h}z)^{a_h}$ , while the infinite tail is  $\Phi_{S^m a}(z)$ . At  $m = 0$  the prefix is empty,  $S^0 a = a$ , and the statement reduces to  $\Phi_a(z) = \Phi_a(z)$ . Admissibility of the tail at every depth is recorded separately as `Fabijs.summableShiftExponent_iterate`. The classical identities `Fabijs.rvachevFourierProduct_two_mul` and `Fabijs.rvachevFourierProduct_two_pow_mul` (`SincProductShells.lean`) are thus the  $a_h \equiv 1$  instances; the generic fixed-factor lemmas `Fabijs.tprod_geom_scale` and `Fabijs.tprod_geom_scale_pow` (`GeometricScaleProducts.lean`) remain the underlying scale-product pattern rather than the formalization boundary. The finite-difference factorization (6.3) is now formalized as an identity of products, in `WeightLinearityProducts.lean`, and by supplying the passage the last sentence of this paragraph calls for — once, and in the generality that sentence describes rather than for this display alone. If a weight is any  $\mathbb{N}_0$ -linear combination of admissible weights,  $a_h = \sum_{i \in s} c_i w_{i,h}$  over an arbitrary finite index set, then

$$\Phi_a(z) = \prod_{i \in s} \Phi_{w_i}(z)^{c_i}$$

at every  $z$  (`Fabijs.generalizedRvachevProduct_linearCombination`), the proof being induction over  $s$  on `Fabijs.generalizedRvachevProduct_add` and the scaling law  $\Phi_{k \cdot a} = \Phi_a^k$  (`Fabijs.generalizedRvachevProduct_natMul`), with admissibility closed under the same combination. Our display is the instance  $s = \{0, \dots, m\}$ ,  $c_r = \binom{m}{r}$ , recorded as `Fabijs.generalizedRvachevProduct_shift_factorization` together with the identification  $S^m a = (h \mapsto a_{h+m})$  (`Fabijs.shiftExponent_iterate`).

One hypothesis is genuinely needed and is exactly the germ caveat above. The coefficients and the weights  $w_i$  are natural numbers, because  $\Phi_a$  is defined only for  $a$  valued in  $\mathbb{N}_0$ ; a combination with a negative coefficient names no transform at all. So the Lean statement is the factorization wherever the differences  $\Delta^r a$  are nonnegative, which is precisely the case in which our own display needs no passage through germs. The signed finite-difference case — where the apparent poles cancel because the left side is entire — is not formalized. This germ issue does not affect the nonnegative admissible shift-refinement identity above, whose all- $m$  analytic product statement is formalized.

The exponent-level content is formal as well, in `ShiftDifferenceWeights.lean`: the identity  $S^m = (I + \Delta)^m$  that drives it is the Gregory–Newton formula, and for a weight sequence valued in any additive commutative group

$$a_{h+m} = \sum_{r \leq m} \binom{m}{r} (\Delta^r a)_h$$

(`Fabijs.weight_shift_eq_sum_fwdDiff`), together with the inverse alternating form (`Fabijs.weight_fwdDiff_eq_sum_shift`). These coefficients are moreover the only ones: applying  $\Delta^s$  at 0 to a Newton combination returns  $c_s$  and annihilates every other basis vector, so the Newton

basis is free and not merely spanning (*Fabius.fwdDiff\_iter\_newtonCombination\_zero*, *Fabius.newtonCombination\_injOn*, *NewtonCoefficientUniqueness.lean*).

So the split that this paragraph once described as a gap — combinatorics formal, passage to products missing — is closed on both sides, and the passage turned out to be cheaper than the combinatorics: it is one induction on *Fabius.generalizedRvachevProduct\_add*, needing nothing about the particular identity being transported.

*Proof.* The first formula follows by separating the  $h = 0$  factor and reindexing the rest:

$$\prod_{h \geq 0} \operatorname{sinc}_{\pi}(2z/2^h)^{a_h} = \operatorname{sinc}_{\pi}(2z)^{a_0} \prod_{k \geq 0} \operatorname{sinc}_{\pi}(z/2^k)^{a_{k+1}}.$$

Iteration gives (6.2). Finally, exponent sequences add when their transforms multiply, and the binomial identity for the shift gives  $S^m a = \sum_r \binom{m}{r} \Delta^r a$ .  $\square$

*Remark 6.2.* This formula explains why finite differences, Newton interpolation, and refinement equations continually appear together in the Fabius corpus. The coincidence is not merely an analogy:  $S$  acts on the exponent sequence exactly as dilation acts on the sinc product.

## 6.2 When the refinement tower closes

Write

$$\mathcal{V}(a) = \operatorname{span}_{\mathbb{C}}\{S^j a : j \geq 0\}.$$

The following classical sequence-theoretic equivalence acquires a direct analytic meaning in this setting.

**Theorem 6.3** (Rational generating functions are precisely finite refinement systems). *For a complex sequence  $a$  of at most exponential growth, the following are equivalent:*

1.  $\dim \mathcal{V}(a) < \infty$ ;
2.  $a$  satisfies a constant-coefficient linear recurrence from some point onward;
3. its ordinary generating function  $A(q) = \sum_{h \geq 0} a_h q^h$  is rational.

When these conditions hold, the logarithmic transforms  $L_j(z) = \log \Phi_{S^j a}(z)$  span a finite-dimensional space  $\mathcal{L}$ , and the dilation  $z \mapsto 2z$  maps  $\mathcal{L}$  into  $\mathcal{L} + \mathbb{C} \log \operatorname{sinc}_{\pi}(2z)$ : the  $L_j$  obey a finite inhomogeneous refinement system with forcing term  $\log \operatorname{sinc}_{\pi}(2z)$ . Conversely, any finite-dimensional shift-closed exponent model has rational  $A$ . This is **proved here** as a synthesis of the standard rational-series theorem with [theorem 6.1](#).

*Proof.* The equivalence of the first three statements is the usual companion-matrix characterization of rational ordinary generating functions [55]. If  $S^d a = \sum_{j < d} \gamma_j S^j a$ , then  $L_d = \sum_{j < d} \gamma_j L_j$  as analytic germs, while

$$L_j(2z) = a_j \log \operatorname{sinc}_{\pi}(2z) + L_{j+1}(z).$$

Thus every dilated  $L_j$  lies in  $\mathcal{L} + \mathbb{C} \log \operatorname{sinc}_{\pi}(2z)$ , which is the asserted inhomogeneous refinement system. The forcing term is genuinely outside  $\mathcal{L}$ : for  $a \equiv 1$  one has  $d = 1$  and  $\mathcal{L} = \mathbb{C} \log \Phi_a$ , which is singular exactly at the nonzero integers, whereas  $\log \operatorname{sinc}_{\pi}(2z)$  is singular at  $z = \pm \frac{1}{2}$  as well. The converse is the same statement read on the exponent vectors.  $\square$

**Corollary 6.4** (Pole dictionary for rational weights). *If  $A$  is rational, then*

$$\frac{\mathcal{Z}_a(s)}{\zeta(s)} = A(2^{-s}) \tag{6.4}$$

is a finite combination of terms whose poles lie on vertical arithmetic progressions obtained by taking logarithms of the poles of  $A$ . Poles of  $A$  at roots of unity produce log-periodic terms in Mellin asymptotics; poles away from the unit circle produce additional power-law or exponentially modulated scales. The exact transseries classification suggested by this observation is formulated in [conjecture 12.3](#).



## 7 Polynomial weights and Newton factorization through the Pascal hierarchy

### 7.1 The audited Pascal–Rvachev transforms

For  $r \geq 1$ , put

$$\Phi_r(z) := \prod_{h \geq 0} \operatorname{sinc}_\pi \left( \frac{z}{2^h} \right)^{\binom{h}{r-1}}. \quad (7.1)$$

The case  $r = 1$  is the classical Rvachev transform. The pinned corpus already develops this Pascal hierarchy: compact support and smoothness, refinement ladders, zero divisors, spectral zeta functions, cumulants, Gaussian rank limits, Fourier envelopes, and Mellin endpoint structure. We use those theorems as a basis and ask which other exponent products can be factored through finitely many  $\Phi_r$ .

### 7.2 Integer-valued polynomials are exactly the finite Newton sector

Let  $P \in \mathbb{Q}[h]$  be integer-valued on  $\mathbb{N}_0$  and nonnegative there. Its Newton expansion is

$$P(h) = \sum_{r=0}^d c_r \binom{h}{r}, \quad c_r = \Delta^r P(0) \in \mathbb{Z}. \quad (7.2)$$

The coefficients  $c_r$  need not be nonnegative even though every  $P(h)$  is.

**Theorem 7.1** (Newton–Rvachev factorization). *Let  $d = \deg P$  and define  $\Phi_P = \Phi_{(P(h))}$ . Then*

$$A_P(q) = \sum_{h \geq 0} P(h) q^h = \sum_{r=0}^d c_r \frac{q^r}{(1-q)^{r+1}}, \quad (7.3)$$

$$\boxed{\Phi_P(z) = \prod_{r=0}^d \Phi_{r+1}(z)^{c_r}}. \quad (7.4)$$

Although the right side may be written as a quotient, it is an entire positive-definite function because the unreduced exponent sequence  $P(h)$  is nonnegative. Its support radius, integer-zero multiplicity, spectral zeta, and even cumulants are

$$R_P = \frac{1}{2} A_P(1/2) = \sum_{r=0}^d c_r, \quad (7.5)$$

$$m_P(n) = \sum_{r=0}^d c_r \binom{\nu_2(n) + 1}{r+1}, \quad (7.6)$$

$$\mathcal{Z}_P(s) = \zeta(s) \sum_{r=0}^d c_r \frac{2^s}{(2^s - 1)^{r+1}}, \quad (7.7)$$

$$\kappa_{2j}(X_P) = \frac{B_{2j}}{2j} \sum_{r=0}^d c_r \frac{4^j}{(4^j - 1)^{r+1}}. \quad (7.8)$$

All formulas are **proved here**. Of these, (7.6) is formalized one Newton component at a time: `Fabiús.weightedScaleMultiplicity_choose` (`weightedScaleMultiplicity.lean`) proves  $m(n) = \binom{\nu_2(n)+1}{r+1}$  for the single Pascal weight  $h \mapsto \binom{h}{r}$ , for an arbitrary base, its hockey-stick input being `Fabiús.inclusivePrefixSum_choose`. The assembly of those components into the signed sum over  $r$  is formal too, in `NewtonMultiplicityAssembly.lean`: it is nothing but linearity of



the prefix sum in its weight sequence, so (7.6) holds over an arbitrary Finset of Newton indices, at an arbitrary base, and over any semiring rather than only over  $\mathbb{Z}$  (`Fabiús.weightedScaleMultiplicity_newton_finset`, `Fabiús.weightedScaleMultiplicity_newton`, and the literal display `Fabiús.weightedScaleMultiplicity_newton_two`). The Gregory–Newton reading of the coefficients,  $a_m = \sum_{r \leq m} \binom{m}{r} (\Delta^r a)_0$ , is `Fabiús.weight_eq_sum_choose_fwdDiff`, and that it forces the coefficients is now formal too, in `NewtonCoefficientUniqueness.lean`. The Newton basis is triangular under  $\Delta$ , so applying  $\Delta^k$  at 0 to a Newton combination reads off the  $k$ -th coefficient and kills every other basis vector:

$$\Delta^k \left( \sum_{r \in s} \binom{\cdot}{r} c_r \right) (0) = \begin{cases} c_k, & k \in s, \\ 0, & k \notin s, \end{cases}$$

over an arbitrary finite index set and an arbitrary additive commutative group of coefficients (`Fabiús.fwdDiff_iter_newtonCombination_zero`). The  $k \notin s$  branch is what makes this an inversion rather than an evaluation — it says the basis is free, not merely spanning — and it is the half uniqueness needs (`Fabiús.newtonCombination_injOn`, and over  $\mathbb{R}$  for our own  $\Phi_P$  `Fabiús.fwdDiff_iter_newtonPoly_zero`, `Fabiús.newtonPoly_coeff_unique`). Running the Gregory–Newton expansion back through the inversion returns the iterated forward differences (`Fabiús.fwdDiff_iter_weight_zero`), so existence and uniqueness meet and  $\Delta^r P(0)$  is the unique choice, not merely a valid one.

Equations (7.3) and the arithmetic half of (7.5) are formal in `NewtonBasisGeneratingFunction.lean`, resting on `Mathlib`’s `choose-geometric` family. The crux is a single identity, and it is what makes the Newton basis the one adapted to the dyadic problem: at  $q = 1/2$  every Newton-basis generating function  $q^r / (1 - q)^{r+1}$  takes the value 2, independently of  $r$  (`Fabiús.pow_div_one_sub_pow_at_half`), so  $A_P(1/2) = 2 \sum_r c_r$  while the halved series  $A_P(1/2)/2 = R_P$  collapses to the plain coefficient sum (`Fabiús.tsum_newtonPoly`, `Fabiús.half_mul_tsum_newtonPoly_half`).

The arithmetic above computes  $A_P(1/2)$ ; the volume’s candidate support radius is  $R_P = A_P(1/2)/2$ . The matching analytic bound is formal and needs no probability: in the  $e^{-2\pi izx}$  convention, a law carried by  $[-R, R]$  has a characteristic function of exponential type at most  $2\pi R$ , and the product directly satisfies

$$\|\Phi_a(z)\| \leq \exp(\pi \|z\| A_a(1/2)) = \exp(2\pi R_a \|z\|), \quad A_a(1/2) = \sum_{h \geq 0} a_h 2^{-h} = 2R_a,$$

for every  $z \in \mathbb{C}$  (`Fabiús.norm_generalizedRvachevProduct_le_exp`, `GeneralizedExponentialType.lean`). So  $\Phi_a$  is entire of order at most one with type at most  $\pi A_a(1/2) = 2\pi R_a$ , exactly as the support normalization predicts. The converse half — that the type is no smaller, and that the law really is carried by  $[-R_a, R_a]$  — is the support statement of [theorem 2.2](#), a probabilistic fact about the associated random variable rather than a property of the product, and it remains unformalized.

At the constant weight this says something checkable. There  $A_1(1/2) = \sum_h 2^{-h} = 2$  and  $R_1 = 1$ , so  $\Phi$  has exponential type at most  $2\pi$  (`Fabiús.norm_rvachevFourierProduct_le_exp`, `ClassicalIGrowth.lean`), and Paley–Wiener reads type  $2\pi$  as support radius 1 — which is where the up-function actually lives, as `ProbabilityRepresentation.lean` establishes by a route that does not use any of this. The agreement is a check on the normalization rather than a new claim about it. The same module records  $\|\Phi(x)\| \leq 1$  on the real axis (`Fabiús.norm_rvachevFourierProduct_ofReal_le_one`), a compatibility corollary that recovers the existing `Fabiús.norm_rvachevFourierProduct_le_one` from `BaselineDecay.lean`.

The sharpness of the factor estimate is the whole difficulty and is worth recording, since the obvious routes all fail: any constant  $C > 1$  in  $\|\operatorname{sinc}_\pi w\| \leq C e^{\|w\|}$  compounds to  $C^{\sum_h a_h}$  across the factors and diverges whenever the weight has infinite total mass. Writing  $\sin$  as a difference of exponentials gives  $e^{3\|w\|}$ , which would inflate the type threefold. The constant 1 comes from comparing the sine series termwise against the hyperbolic sine,  $\|\sin w\| \leq \sinh\|w\|$ , followed by  $\sinh t \leq te^t$  — which is  $1 - 2t \leq e^{-2t}$  rearranged.

The spectral zeta (7.7) is formalized, in `NewtonSpectralZeta.lean`, and it is exactly the composite of two identities already in hand: the general-weight zeta (3.3) evaluated at the Newton weight, with  $A_P$  supplied by (7.3) at  $q = 2^{-s}$ . Clearing the negative powers turns each Newton term  $q^r/(1-q)^{r+1}$  into  $2^s/(2^s-1)^{r+1}$ , which is the display (`Fabiús.tsum_newtonPoly_spectral_zeta`). The composite is slightly stronger than the juxtaposition of its inputs: the summability hypothesis that the general zeta must carry — an arbitrary weight may grow too fast — is here discharged, since a polynomial weight is always admissible at a dyadic base (`Fabiús.summable_newtonPoly_rpow`).

The cumulants (7.8) are half formalized. Their analytic content needs no probabilistic model, and is proved in `GeneralizedSincZeta.lean`: on the central disk,

$$\Phi_a(z) = \exp\left(-\sum_{r \geq 1} \frac{\zeta(2r) A_a(4^{-r})}{r} z^{2r}\right), \quad \|z\| < 1$$

(`Fabiús.generalizedRvachevProduct_eq_cexp`), at every admissible  $a$ . The coefficient of  $z^{2r}$  in  $-\log \Phi_a$  is therefore  $\zeta(2r) A_a(4^{-r})/r$ , which is (7.8) with the random variable removed; what is still missing is the identification of  $\Phi_a$  as a characteristic function of  $X_a$ , not any further analysis. At the constant weight  $A_1(4^{-r}) = 4^r/(4^r-1)$  and the display is the corpus's existing `SincZetaDyadic.lean`.

Reaching the general weight needed one new tool, and it is reusable. The pair form of  $\Phi_a$  carries the exponents  $a_h$ , and the Euler log transform does not apply to a product of powers; adding the exponent to the transform would raise a branch question, since  $\log((1-x)^c) = c \log(1-x)$  can leave the principal branch. `WeightedEulerTransform.lean` avoids it by putting the multiplicity into the index rather than the exponent — over  $\sum_i \text{Fin}(c_i)$  the family is exponent-free, the existing transform applies verbatim, and both sides transfer back along the finite fibres (`Fabiús.tprod_one_sub_pow_eq_cexp_powerSum`). No logarithm of a power is taken anywhere. The scale direction then contributes  $\sum_h a_h(4^{-r})^h$  in place of the geometric series, which is exactly  $A_a(4^{-r})$ .

*Proof.* The generating function  $\sum_{h \geq 0} \binom{h}{r} q^h = q^r/(1-q)^{r+1}$  gives (7.3). Exponents add under multiplication, proving (7.4). At  $q = 1/2$ , every Newton-basis generating function has value 2, which gives (7.5). The hockey-stick identity  $\sum_{h=0}^v \binom{h}{r} = \binom{v+1}{r+1}$  proves (7.6); (7.7) and (7.8) follow by substituting  $q = 2^{-s}$  and  $q = 4^{-j}$  into (7.3).  $\square$

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**Positive-definite quotients.** Equation (7.4) gives a systematic source of nonobvious positive-definite quotients of compactly supported entire characteristic functions. Cancellation is certified at the level of dyadic sinc exponents, not guessed from zeros after the fact. This is substantially more flexible than taking ordinary convolutions, which only permits nonnegative exponents in the Pascal basis itself.

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### 7.3 Leading Fourier and endpoint orders

Let  $c_d = \Delta^d P(0) > 0$  and set

$$p = d + 2, \quad C_P = \frac{c_d}{(d+2)!(\log 2)^{d+1}}. \quad (7.9)$$

Lower Newton components have smaller logarithmic order, so the highest Pascal factor controls the leading scale.

**Theorem 7.2** (Leading Fourier and small-ball laws). *The sharp Pascal-hierarchy envelope theorem in the audited corpus implies*

$$\limsup_{|x| \rightarrow \infty} \frac{\log |\Phi_P(x)|}{(\log |x|)^{d+2}} = -C_P. \quad (7.10)$$

Let

$$Y_P = X_P + R_P \in [0, 2R_P], \quad G_P(x) = \mathbb{P}(Y_P \leq x).$$

The corresponding exponential Tauberian theorem of de Bruijn type [48, 38] gives

$$-\log G_P(x) \sim C_P (\log(1/x))^{d+2} \quad (x \downarrow 0). \quad (7.11)$$

Consequently the inverse endpoint satisfies

$$\log \frac{1}{G_P^{-1}(y)} \sim \left( \frac{\log(1/y)}{C_P} \right)^{1/(d+2)} \quad (y \downarrow 0). \quad (7.12)$$

These statements are **derived from audited corpus theorems** from the established Pascal hierarchy and the standard exponential Tauberian transfer.

**Remark 7.3.** For  $P(h) \equiv 1$ , one has  $d = 0$ ,  $c_0 = 1$ , and  $C_P = 1/(2 \log 2)$ , recovering the quadratic logarithmic scale of the classical Fabius endpoint. Increasing the polynomial degree increases the endpoint flatness from a quadratic to a higher logarithmic power.

#### 7.4 The generalized lower-Lambert coordinate

The leading saddle for inverse Laplace analysis solves

$$x = C_P p \lambda^{p-1} e^{-\lambda}. \quad (7.13)$$

It has the explicit lower-branch solution

$$\lambda_P(x) = -(p-1)W_{-1}\left(-\frac{1}{p-1}\left(\frac{x}{C_P p}\right)^{1/(p-1)}\right). \quad (7.14)$$

For  $p = 2$  this is the familiar lower-Lambert coordinate of the classical Fabius analysis; the branch notation and inversion conventions follow [52]. For  $p > 2$  it gives a single closed coordinate in which the polynomial-weight endpoint transseries should be organized; the Mellin calculation in [section 10](#) explains the periodic coefficients that decorate it.

Equation (7.14) is formalized, in `GeneralizedLambertCoordinate.lean`: writing  $q = p - 1$ , `Fabius.generalizedLambertCoordinate` is  $\lambda_P$  and `Fabius.generalizedLambertCoordinate_solves_saddle` is the assertion that it solves (7.13). The verification is pure algebra once  $We^W = z$  is available; nothing analytic beyond that is used. It is worth separating out the core, which carries the whole content and has no real powers in it: with  $w = W_{-1}(u)$  and  $\lambda = -qw$ ,

$$\lambda^q e^{-\lambda} = (-q)^q (we^w)^q = (-qu)^q \quad (\text{Fabius.pow\_mul\_exp\_neg\_lambert}).$$

Everything else is the choice of  $u$  making  $(-qu)^q$  equal to  $x/(C_P p)$ , and that is the only place the real power  $(x/(C_P p))^{1/q}$  enters, bringing its round-trip hypothesis  $x/(C_P p) \geq 0$  with it. The unit-rate bridge `Fabius.lowerPowerExponentialPhase_rate_one` in `PowerExponentialLambertFabius.lean` identifies this coordinate with the generic lower power-exponential phase. Write

$$P_q = (C_P p)(qe^{-1})^q.$$

For  $q \in \mathbb{N}_0 \setminus \{0\}$  and  $C_P p > 0$ , the intrinsic profile intervals now derive the branch premise:  $0 < x < P_q$  puts the Lambert argument in  $(-e^{-1}, 0)$ , while  $0 < x \leq P_q$  puts it in  $[-e^{-1}, 0]$ . These are respectively `Fabius.generalizedLambertCoordinate_argument_mem_Ioo` and `Fabius.generalizedLambertCoordinate_argument_mem_Ico`. On the endpoint-inclusive

interval  $(0, P_q]$ , `Fabijs.generalizedLambertCoordinate_solves_saddle_of_mem` proves (7.13), including the peak.

The generic two-branch package is stronger still. Let  $\lambda_{P,0}$  be the principal phase paired with the lower phase  $\lambda_P$ . The full-domain continuity declarations `Fabijs.principalPowerExponentialPhase_continuousOn_Icc` and `Fabijs.lowerPowerExponentialPhase_continuousOn_Ioc` in `PowerExponentialLambertCalculus.lean` give

$$\lambda_{P,0} \in C([0, P_q]), \quad \lambda_P \in C((0, P_q]).$$

They are obtained by composing the raw branch endpoint theorems with `Fabijs.powerExponentialLambertArgument_continuous`. Via the unit-rate bridge, `Fabijs.lowerPowerExponentialPhase_image_Ioc` gives the exact image  $\lambda_P((0, P_q]) = [q, \infty)$ , and `Fabijs.lowerPowerExponentialPhase_invOn` gives the two-sided `InvOn` law with  $\lambda \mapsto C_{PP}\lambda^q e^{-\lambda}$  on  $[q, \infty)$  and  $(0, P_q]$ . More completely, for  $x \in (0, P_q]$  and every  $\mu \geq 0$ ,

$$C_{PP}\mu^q e^{-\mu} = x \iff \mu = \lambda_{P,0}(x) \text{ or } \mu = \lambda_P(x).$$

This is the specialization of `Fabijs.powerExponentialSaddle_eq_iff_eq_principal_or_eq_q_lower`; on  $0 < x < P_q$  the two roots are distinct by `Fabijs.principalPowerExponentialPhase_ne_lowerPowerExponentialPhase`, both in `PowerExponentialLambertInverse.lean`. The restriction  $\mu \geq 0$  is part of the theorem and cannot be dropped for even  $q$ .

The inverse-profile calculus now extends one derivative further. For  $m \in \mathbb{N}_0 \setminus \{0\}$ ,  $A > 0$ , and  $\beta > 0$ , `PowerExponentialLambertCurvature.lean` proves that both smooth branches  $\lambda(x)$  of  $x = A\lambda^m e^{-\beta\lambda}$  satisfy, for  $0 < x < P_{m,A,\beta}$ ,

$$\lambda''(x) = \frac{\lambda(x)(m - (m - \beta\lambda(x))^2)}{x^2(m - \beta\lambda(x))^3}.$$

The exact `HasDerivAt` and derivative statements are `Fabijs.deriv_principalPowerExponentialPhase_hasDerivAt`, `Fabijs.deriv_deriv_principalPowerExponentialPhase`, `Fabijs.deriv_lowerPowerExponentialPhase_hasDerivAt`, and `Fabijs.deriv_deriv_lowerPowerExponentialPhase`. This generic module records the formula, not yet the square-root threshold or a branchwise strict-shape classification.

The compiled small-input theorem is expressed in the intrinsic variable

$$\epsilon_q(x) = \frac{1}{q} \left( \frac{x}{C_{PP}} \right)^{1/q}.$$

Specializing `PowerExponentialLambertAsymptotics.lean` to  $(m, A, \beta) = (q, C_{PP}, 1)$  gives

$$\lambda_{P,0}(x) \sim \left( \frac{x}{C_{PP}} \right)^{1/q}, \quad (7.15)$$

$$\lambda_P(x) \longrightarrow +\infty, \quad (7.16)$$

$$\lambda_P(x) + q(\log \epsilon_q(x) - \log |\log \epsilon_q(x)|) \longrightarrow 0 \quad (x \downarrow 0). \quad (7.17)$$

The exact generic declarations are `Fabijs.powerExponentialLambertEpsilon`, `Fabijs.principalPowerExponentialPhase_isEquivalent_rpow`, `Fabijs.tendsto_lowerPowerExponentialPhase_nhdsGT_zero_atTop`, `Fabijs.lowerPowerExponentialPhaseIntrinsicMain`, and `Fabijs.lowerPowerExponentialPhase_sub_intrinsicMain_tendsto_zero`.

At  $q = 1$  the lower coordinate collapses to  $-W_{-1}(-x/(C_{PP}))$  (`Fabijs.generalizedLambertCoordinate_one`), the shape used by the classical  $p = 2$  analysis. The totalized raw lower branch satisfies  $W_{-1}(z)e^{W_{-1}(z)} = z$  on  $[-e^{-1}, 0)$ , through `Fabijs.lowerLambertW_mul_exp_of_mem_Ico` and `Fabijs.lowerLambertW_unique_of_mem_Ico`. Its branch-point and

full-domain continuity are `Fabius.lowerLambertW_continuousWithinAt_branchPoint` and `Fabius.lowerLambertW_continuousOn_Ico`; the principal companion has `Fabius.principalLambertW_continuousWithinAt_branchPoint` and `Fabius.principalLambertW_continuousOn_Ici`. At zero, `Fabius.deriv_principalLambertW_zero` states  $W_0(0) = 1$ , while `Fabius.principalLambertW_isEquivalent_zero` states  $W_0(z) \sim z$ . The compiled `LambertWCurvature.lean` layer gives, for  $k = 0$  with  $z > -e^{-1}$ , and for  $k = -1$  with  $-e^{-1} < z < 0$ , the nonsingular form

$$W_k''(z) = -e^{-2W_k(z)} \frac{W_k(z) + 2}{(W_k(z) + 1)^3}.$$

The two derivative witnesses are `Fabius.deriv_principalLambertW_hasDerivAt` and `Fabius.deriv_lowerLambertW_hasDerivAt`; the displayed identities are `Fabius.deriv_deriv_principalLambertW` and `Fabius.deriv_deriv_lowerLambertW`. In particular `Fabius.deriv_deriv_principalLambertW_zero` gives  $W_0''(0) = -2$ , and `Fabius.strictConcaveOn_principalLambertW` proves strict concavity on the full closed domain  $[-e^{-1}, \infty)$ . On the lower branch, `Fabius.deriv_deriv_lowerLambertW_pos_iff`, `Fabius.deriv_deriv_lowerLambertW_eq_zero_iff`, and `Fabius.deriv_deriv_lowerLambertW_neg_iff` locate the unique inflection at  $-2e^{-2}$ ; the endpoint-inclusive strict convexity and right-hand strict concavity are `Fabius.strictConvexOn_lowerLambertW_left` and `Fabius.strictConcaveOn_lowerLambertW_right`.

**Lean crosswalk: raw Lambert branch-point geometry and leading scale.** The compiled raw endpoint layer is split between `LambertWBranchPointGeometry.lean` and `LambertWBranchPointAsymptotics.lean`. The former has exactly eight public declarations:

- the one-sided derivative divergences `Fabius.tendsto_deriv_principalLambertW_branchPoint_atTop` and `Fabius.tendsto_deriv_lowerLambertW_branchPoint_atBot`;
- the matching endpoint-secant divergences `Fabius.tendsto_principalLambertW_secantSlope_branchPoint_atTop` and `Fabius.tendsto_lowerLambertW_secantSlope_branchPoint_atBot`;
- failure of a finite right derivative, `Fabius.principalLambertW_not_differentiableWithinAt_branchPoint` and `Fabius.lowerLambertW_not_differentiableWithinAt_branchPoint`; and
- failure of ordinary differentiability of the totaled branches, `Fabius.principalLambertW_not_differentiableAt_branchPoint` and `Fabius.lowerLambertW_not_differentiableAt_branchPoint`.

Thus “vertical tangent” means the precise right-hand limits  $W'_0(z) \rightarrow +\infty$  and  $W'_{-1}(z) \rightarrow -\infty$ , together with the absence of a finite right derivative at the common endpoint.

The asymptotic module contributes one definition and eight theorems. Its intrinsic scale is `Fabius.lambertWBranchPointScale`, with positivity and exact square given by `Fabius.lambertWBranchPointScale_pos` and `Fabius.lambertWBranchPointScale_sq`. The exact squared-ratio limits are `Fabius.tendsto_principalLambertW_add_one_sq_div_branchPoint` and `Fabius.tendsto_lowerLambertW_add_one_sq_div_branchPoint`; their equivalence forms are `Fabius.principalLambertW_add_one_sq_isEquivalent_branchPoint` and `Fabius.lowerLambertW_add_one_sq_isEquivalent_branchPoint`. Finally, `Fabius.principalLambertW_add_one_isEquivalent_branchPoint` and `Fabius.lowerLambertW_add_one_isEquivalent_branchPoint` formalize only the signed leading laws

$$W_0(z) + 1 \sim q(z), \quad W_{-1}(z) + 1 \sim -q(z), \quad q(z) = \sqrt{2e(z + e^{-1})}.$$

These are first-order equivalents, not the stronger  $\mathcal{O}(z + e^{-1})$  remainder, the subsequent Puiseux coefficients, a coefficient recurrence, or convergence of the full Puiseux series. Nor are there yet generic power–exponential or classical Fabius-phase pullbacks of these endpoint results, including corresponding peak-threshold derivative or square-root theorems.

For the classical Fabius rate, the specialization  $(m, A, \beta) = (1, 1, \log 2)$  is named explicitly in `PowerExponentialLambertFabius.lean`: `Fabius.fabiusPrincipalLambertPhase` is the principal root and `Fabius.fabiusLambertPhase` the lower root. `Fabius.fabiusSaddle_eq_iff_eq_principal_or_eq_lower`, `Fabius.fabiusPrincipalLambertPhase_ne_fabiusLambertPhase`, `Fabius.fabiusPrincipalLambertPhase_continuousOn_Icc`, and `Fabius.fabiusLambertPhase_continuousOn_Ioc` specialize the dichotomy, strict-interior separation, and full-domain continuity. `Fabius.fabiusPrincipalLambertPhase_isEquivalent_id` gives the principal small- $x$  law; the generic lower divergence and intrinsic expansion specialize to the lower Fabius phase.

The specialization is now curvature-sharp as well. Put

$$x_* = \frac{2e^{-2}}{\log 2}, \quad P_* = \frac{e^{-1}}{\log 2}.$$

The definition `Fabius.fabiusLambertInflectionInput` is  $x_*$ , and `Fabius.fabiusLambe`



`rtPhase_inflectionInput` gives  $\lambda_-(x_*) = 2/\log 2$ . On  $0 < x < P_*$ , the exact formula is `Fabius.deriv_deriv_fabiusLambertPhase`. The three theorems `Fabius.deriv_deriv_fabiusLambertPhase_pos_iff`, `Fabius.deriv_deriv_fabiusLambertPhase_eq_zero_iff`, and `Fabius.deriv_deriv_fabiusLambertPhase_neg_iff` say that the lower phase curvature is respectively positive, zero, and negative before, at, and after  $x_*$  within that smooth interval. Its named exact zero is `Fabius.deriv_deriv_fabiusLambertPhase_inflectionInput`. Consequently `Fabius.strictConvexOn_fabiusLambertPhase_left` and `Fabius.strictConcaveOn_fabiusLambertPhase_right` give strict convexity on  $(0, x_*]$  and strict concavity on  $[x_*, P_*]$ . The principal companion has a stronger global statement: `Fabius.fabiusPrincipalLambertPhase_continuousOn_Iic` and `Fabius.strictConvexOn_fabiusPrincipalLambertPhase` cover the whole half-line  $(-\infty, P_*]$ , while `Fabius.deriv_deriv_fabiusPrincipalLambertPhase_pos` holds at every input below the peak and `Fabius.deriv_deriv_fabiusPrincipalLambertPhase_zero` evaluates the second derivative at zero as  $2\log 2$ .

The import order is `PowerExponentialLambert.lean`, `PowerExponentialLambertCalculus.lean`, `PowerExponentialLambertInverse.lean`, `PowerExponentialLambertAsymptotics.lean`, `LambertWCurvature.lean`, `LambertWBranchPointGeometry.lean`, `LambertWBranchPointAsymptotics.lean`, `PowerExponentialLambertCurvature.lean`, `PowerExponentialLambertFabius.lean`, and finally `PowerExponentialLambertFabiusCurvature.lean`. Thus the coordinate, continuity, exact nonnegative-root geometry, principal equivalence, lower divergence, intrinsic two-term lower expansion, interior second derivatives, complete classical Fabius curvature split, raw branch-point derivative geometry, and signed leading raw square-root laws have Lean counterparts. The generic curvature module does not yet export its square-root thresholds or strict-shape corollaries; the raw branch-point results have not yet been transported to the generic power-exponential phases or the classical Fabius phases and their peak thresholds. The generic small-input asymptotic result does not yet replace  $\epsilon_q$  by a cleaned  $L = \log(C_{pp}/x)$  expression and does not provide further terms or a full logarithmic series. Likewise the  $\mathcal{O}$ -refined and full convergent Puiseux packages remain outside this formal boundary, as does the endpoint transseries organized by the coordinate.

## 8 An alternating Newton family of positive-definite Pascal quotients

The abstract Newton factorization becomes more compelling when negative coefficients are unavoidable and the quotient simplifies to something not visibly positive definite.

### 8.1 Definition and three exact factorizations

For an even integer  $d \geq 0$ , define the generalized-binomial polynomial sequence

$$P_d(h) = \binom{h-1}{d} \quad (h \in \mathbb{N}_0). \quad (8.1)$$

Here  $\binom{-1}{d} = (-1)^d = 1$ . Thus

$$P_d(0) = 1, \quad P_d(1) = \cdots = P_d(d) = 0, \quad P_d(h) = \binom{h-1}{d} > 0 \quad (h \geq d+1). \quad (8.2)$$

The parity restriction on  $d$  is precisely what makes the  $h = 0$  exponent nonnegative.



**Theorem 8.1** (Alternating Newton family). *Let  $d \geq 0$  be even and put  $\Psi_d = \Phi_{P_d}$ . Then*

$$A_d(q) = 1 + \left( \frac{q}{1-q} \right)^{d+1}, \quad (8.3)$$

$$P_d(h) = \sum_{r=0}^d (-1)^{d-r} \binom{h}{r}, \quad (8.4)$$

$$\Psi_d(z) = \text{sinc}_\pi(z) \Phi_{d+1}(z/2) = \prod_{r=0}^d \Phi_{r+1}(z)^{(-1)^{d-r}}. \quad (8.5)$$

In particular,

$$\Psi_2(z) = \frac{\Phi_1(z) \Phi_3(z)}{\Phi_2(z)}. \quad (8.6)$$

Every expression in (8.5) is the characteristic function of a  $C_c^\infty$  probability density supported on  $[-1, 1]$ . These statements are **proved here**.

*Proof.* The positive-index part is

$$\sum_{h \geq 1} \binom{h-1}{d} q^h = q \sum_{k \geq 0} \binom{k}{d} q^k = \frac{q^{d+1}}{(1-q)^{d+1}},$$

and the exceptional  $h = 0$  term equals 1, proving (8.3). The finite identity (8.4) follows from the backward-shift formula for binomial polynomials, or by comparing coefficients in  $(1+t)^{h-1} = (1+t)^h / (1+t)$  through degree  $d$ . The first product formula separates  $h = 0$  and reindexes  $h = k + 1$ :

$$\Psi_d(z) = \text{sinc}_\pi(z) \prod_{k \geq 0} \text{sinc}_\pi\left(\frac{z/2}{2^k}\right)^{\binom{k}{d}} = \text{sinc}_\pi(z) \Phi_{d+1}(z/2).$$

The second is [theorem 7.1](#). Since the unreduced exponents (8.2) are nonnegative,  $\Psi_d$  is positive definite. Finally,  $A_d(1/2) = 2$ , so the support radius is one.  $\square$

**Corollary 8.2** (Concrete random-series model). *Let  $U \sim \text{Unif}[-1/2, 1/2]$  and let  $X_{d+1}$  have characteristic function  $\Phi_{d+1}$ , independently of  $U$ . Then*

$$Y_d := U + \frac{1}{2} X_{d+1} \quad (8.7)$$

has characteristic function  $\Psi_d$ . This exhibits the hidden positivity of the alternating Pascal quotient without any appeal to a zero-by-zero cancellation argument.

## 8.2 Sparse enhancement of the classical zero divisor

**Theorem 8.3** (Exact arithmetic spectrum of  $\Psi_d$ ). *For every positive integer  $n$ ,*

$$\text{ord}_{z=n} \Psi_d(z) = 1 + \binom{\nu_2(n)}{d+1}. \quad (8.8)$$

Its spectral zeta and even cumulants are

$$\mathcal{Z}_d(s) = \zeta(s) \left( 1 + \frac{1}{(2^s - 1)^{d+1}} \right), \quad (8.9)$$

$$\kappa_{2j}(Y_d) = \frac{B_{2j}}{2j} \left( 1 + \frac{1}{(4^j - 1)^{d+1}} \right), \quad \kappa_{2j+1}(Y_d) = 0. \quad (8.10)$$

In particular,

$$\text{Var}(Y_d) = \frac{1}{12} \left( 1 + 3^{-(d+1)} \right). \quad (8.11)$$

These are [proved here](#). The multiplicity (8.8) is now formalized in full, and as an order statement rather than only an exponent count, in `AlternatingNewtonFamily.lean`:

$$\text{ord}_{z=\pm n} \Psi_d(z) = 1 + \binom{\nu_2(n)}{d+1}, \quad n \geq 1$$

(`Fabius.analyticOrderAt_alternatingNewton`, `Fabius.analyticOrderAt_alternatingNewton_neg`). What made this reachable is the order of vanishing at a general admissible weight, `GeneralizedRvachevEntire.lean`, which did not exist when this paragraph was written; before it, only the arithmetic was available.

The arithmetic half is `Fabius.weightedScaleMultiplicity_alternatingNewton`, and it holds at every  $n$ , the convention  $\nu_2(0) = 0$  making both sides 1. The exceptional  $h = 0$  term, which the hockey stick does not cover, is handled by splitting it off the prefix sum; what remains is  $\sum_{k < \nu_2(n)} \binom{k}{d}$ , and that needs the hockey stick in exclusive form,  $\sum_{k < v} \binom{k}{d} = \binom{v}{d+1}$ , which the corpus already carries as `Fabius.sum_range_choose_eq_choose_succ` (`ThueMorseMoments.lean`), where it was proved for the binomial moments of the Thue–Morse signs. It is the form wanted here rather than the inclusive `Fabius.inclusivePrefixSum_choose`,  $\sum_{h \leq v} \binom{h}{d} = \binom{v+1}{d+1}$ , precisely because the  $h = 0$  term has already been removed: the inclusive form would need a  $v \geq 1$  side condition, while the exclusive one is correct at  $v = 0$  too, both sides vanishing. A consequence worth naming: at odd  $n$  the order is 1 for every  $d$  (`Fabius.analyticOrderAt_alternatingNewton_odd`).

One remark on the convention. Our  $\binom{-1}{d} = (-1)^d = 1$  is legitimate only because  $d$  is even, and that is the whole purpose of the parity restriction. In the Lean weight there is no convention to make: `Fabius.alternatingNewtonWeight` is defined by  $P_d(0) = 1$  and  $P_d(h+1) = \binom{h}{d}$ , so it is  $\mathbb{N}_0$ -valued for every  $d$ , and the parity enters only in the question of whether it agrees with our signed formula (8.4).

The spectral zeta (8.9) is formalized as well, in `AlternatingNewtonZeta.lean`, together with the generating function (8.3) it rests on. The latter,

$$A_d(q) = 1 + \frac{q^{d+1}}{(1-q)^{d+1}}, \quad |q| < 1,$$

is `Fabius.tsum_alternatingNewtonWeight_pow`: the  $h = 0$  term contributes the 1, and the rest is the corpus's Newton generating function `Fabius.tsum_choose_mul_pow` with the index shifted by one. Feeding it to the general weighted zeta at  $q = 2^{-s}$ , where  $q/(1-q)$  collapses to  $1/(2^s - 1)$ , gives

$$\sum_{n \geq 1} \frac{m_{P_d}(n)}{n^s} = \left( \sum_{n \geq 1} n^{-s} \right) \left( 1 + \frac{1}{(2^s - 1)^{d+1}} \right), \quad s > 1$$

(`Fabius.spectral_zeta_alternatingNewton`). As in `SpectralZetaWeighted.lean` the first factor is left as its own  $p$ -series rather than identified with `riemannZeta`, and  $s$  is real; both are limitations of the corpus's zeta, not of this weight. What this weight does contribute is that the summability hypothesis the general theorem must carry — an arbitrary weight may grow too fast — is here discharged, since  $P_d$  grows polynomially and is admissible at every  $s > 0$  (`Fabius.summable_alternatingNewtonWeight_rpow`).

The cumulants (8.10) are half formalized, in the same sense as (7.8) and for the same reason. Their analytic content is proved, in `AlternatingNewtonCumulantKernel.lean`: on the central disk,

$$\Psi_d(z) = \exp \left( - \sum_{r \geq 1} \frac{\zeta(2r) (1 + (4^r - 1)^{-(d+1)})}{r} z^{2r} \right), \quad \|z\| < 1$$

(*Fabiús.alternatingNewton\_eq\_cexp*), and the bracket is exactly the one in (8.10). So the coefficient of  $z^{2^r}$  in  $-\log \Psi_d$  already carries the cumulant bracket; what separates the two statements is the probabilistic reading of  $\Psi_d$  as the characteristic function of  $Y_d$ , not any further analysis. The identity is the composite of the general-weight expansion (*GeneralizedSincZeta.lean*) with the closed form of  $A_d$  (*AlternatingNewtonZeta.lean*), and the only work between them is a coercion, since both the weight and the ratio  $4^{-r}$  are real.

The variance (8.11) remains unformalized: it is the  $j = 1$  cumulant, so it needs exactly the probabilistic step that is missing.

*Proof.* Use [theorem 3.1](#) and the hockey-stick identity:

$$\sum_{h=0}^v P_d(h) = 1 + \sum_{h=1}^v \binom{h-1}{d} = 1 + \binom{v}{d+1}.$$

The other formulas are obtained by substituting  $q = 2^{-s}$  and  $q = 4^{-j}$  into (8.3); the variance is the  $j = 1$  case because  $B_2/2 = 1/12$ .  $\square$

**Example 8.4** (The first genuinely alternating quotient). For  $d = 2$ ,

$$P_2 = (1, 0, 0, 1, 3, 6, 10, 15, \dots), \quad A_2(q) = 1 + \frac{q^3}{(1-q)^3}.$$

Every odd integer and every integer with  $\nu_2(n) \leq 2$  remains a simple zero. The first extra multiplicity occurs at  $n = 8$ , where the order is 2; at  $n = 16$  it is 5; at  $n = 32$  it is 11. Yet the same entire function is the quotient  $\Phi_1 \Phi_3 / \Phi_2$ .

Every number in this example is checked against the general formula in *AlternatingNewtonFamily.lean*: the weight values (*Fabiús.alternatingNewtonWeight\_two\_values*), the simple-zero range (*Fabiús.weightedScaleMultiplicity\_alternatingNewton\_two\_of\_1e*), and the three multiplicities 2, 5, 11, which are  $1 + \binom{\nu_2(n)}{3}$  at  $\nu_2 = 3, 4, 5$ . Since the example was worked by hand and the formula proved independently, the agreement is evidence for both. The final sentence is the one part not formalized: reading  $\Psi_2$  as the quotient  $\Phi_1 \Phi_3 / \Phi_2$  needs a signed exponent, which  $\Phi_a$  does not accept.

### 8.3 Automatic lobe signs and sparse Prouhet order

For  $h \geq 1$ , Lucas' theorem gives

$$P_d(h) \equiv 1 \pmod{2} \iff d \text{ is a binary submask of } h-1. \quad (8.12)$$

The Lucas input to (8.12) is formal: *Fabiús.odd\_choose\_iff\_bitSupport\_subset* (*ThueMorseLucasSupport.lean*) states that  $\binom{n}{k}$  is odd exactly when the bit support of  $k$  is contained in that of  $n$ , which is the submask criterion used here; the identification of  $P_d(h)$  with the relevant binomial coefficient is not itself formalized. This parity sequence is periodic, hence the Fourier-lobe sign sequence of  $\Psi_d$  is 2-automatic by [theorem 3.3](#). More precisely, among a complete sufficiently long binary block of values of  $h-1$ , the forced one-bits of  $d$  reduce the density by a factor  $2^{-\text{wt}_2(d)}$ . That density count is now formal: for  $d < 2^m$ ,

$$\#\{n < 2^m : d \text{ a binary submask of } n\} = 2^{m-\text{wt}_2(d)}$$

(*Fabiús.card\_filter\_bitSupport\_subset*, *DigitalDensity.lean*), and combining it with the Lucas criterion gives the same count for the odd binomial coefficients across a row block (*Fabiús.card\_filter\_odd\_choose*). The subset bijection it rests on — that the subsets of  $t$  containing a fixed  $s$  are the subsets of  $t \setminus s$  with  $s$  adjoined — is proved there as well (*Fabiús.filter\_powerset\_subset*); Mathlib carries only the graded form of that bijection.

**Corollary 8.5** (Linear Prouhet order with a digital density). *Let*

$$\omega_d(L) = \#\{0 \leq h < L : P_d(h) \text{ is odd}\}.$$

*Then*

$$\omega_d(L) = 2^{-\text{wt}_2(d)} L + \mathcal{O}_d(1). \quad (8.13)$$

The length- $2^L$  lobe-sign block attached to  $P_d$  annihilates every power of degree  $< \omega_d(L)$ , and its first surviving power moment is given exactly by (3.16). Thus the Prouhet order remains linear in  $L$ , but its slope is the binary cylinder density selected by  $d$ . The Lucas input (8.12) is formalized as `Fabius.odd_choose_iff_bitSupport_subset` (`ThueMorseLucasSupport.lean`), in the submask form “ $\binom{n}{k}$  is odd exactly when the binary support of  $k$  is contained in that of  $n$ ”, with `Fabius.oddBinomialIndices_eq_image_powerset` parametrizing the odd positions of a Pascal row by the powerset of the bit support. The density count (8.13) is now formalized as well, in `SubmaskDensityAsymptotic.lean`, and with the implied constant exhibited rather than hidden inside the  $\mathcal{O}$ : writing  $\omega(d, L) = \#\{n < L : \binom{n}{d} \text{ odd}\}$  (`Fabius.submaskCount`, identified with that count by `Fabius.submaskCount_eq_card_filter_odd_choose`), one has the two-sided bound

$$\left| 2^{\text{wt}_2(d)} \omega(d, L) - L \right| \leq 2^m \quad \text{whenever } d < 2^m$$

(`Fabius.abs_two_pow_weight_mul_submaskCount_sub_le`), valid for every  $L$ , not merely along complete blocks; taking  $m = d$  makes the constant depend on  $d$  alone (`Fabius.abs_two_pow_weight_mul_submaskCount_sub_le_self`). Dividing by  $L$  gives the density itself as a limit,  $\omega(d, L)/L \rightarrow 2^{-\text{wt}_2(d)}$  (`Fabius.tendsto_card_filter_odd_choose_div_atTop`). The mechanism is periodicity: all bits of  $d$  lie below  $m$ , so adding a multiple of  $2^m$  leaves the submask condition unchanged (`Fabius.bitSupport_subset_two_pow_mul_add_iff`), and the count over  $[0, L)$  splits into  $\lfloor L/2^m \rfloor$  full blocks — each counted exactly by `Fabius.card_filter_bitSupport_subset` — plus a partial block dominated by a full one. What still stands between this and (8.13) as stated is only the identification of  $P_d(h)$  with the binomial coefficient, noted above as the remaining informal step.

*Proof.* In each block of  $2^k$  consecutive values of  $h - 1$ , where  $k$  exceeds the binary length of  $d$ , precisely  $2^{k-\text{wt}_2(d)}$  numbers contain all one-bits of  $d$ . Partial end blocks account for the bounded error. The Prouhet statement is [theorem 3.4](#).  $\square$

## 9 Gaussian boundary layers and convergence to a box density

The factorization  $Y_d = U + X_{d+1}/2$  makes the large- $d$  limit geometrically transparent. The Pascal variable collapses to zero after normalization, so the new smooth compactly supported density tends to the discontinuous uniform density. The collapse is Gaussian on its natural microscopic scale.

### 9.1 A quantitative central limit theorem for the Pascal component

Let  $X_r$  denote the centered random variable with characteristic function  $\Phi_r$ . It is the sum of  $\binom{h}{r-1}$  independent copies of  $2^{-h}U$ , with  $U \sim \text{Unif}[-1/2, 1/2]$ .

**Theorem 9.1** (Variance and Berry–Esseen rate). *For  $r \geq 1$ ,*

$$\sigma_r^2 := \text{Var}(X_r) = 3^{-(r+1)}, \quad (9.1)$$

$$\sum \mathbb{E}|\text{summand}|^3 = \frac{1}{4 \cdot 7^r}. \quad (9.2)$$

Hence any valid universal Berry–Esseen constant  $C_{\text{BE}}$  gives

$$\sup_{y \in \mathbb{R}} \left| \mathbb{P}\left(\frac{X_r}{\sigma_r} \leq y\right) - \Phi_G(y) \right| \leq C_{\text{BE}} \frac{3\sqrt{3}}{4} \left(\frac{3\sqrt{3}}{7}\right)^r. \quad (9.3)$$

This is [proved here](#).

*Proof.* Since  $\text{Var}(U) = 1/12$  and  $\mathbb{E}|U|^3 = 1/32$ ,

$$\begin{aligned}\text{Var}(X_r) &= \frac{1}{12} \sum_{h \geq 0} \binom{h}{r-1} 4^{-h} = \frac{1}{12} \frac{4^{-(r-1)}}{(1 - 1/4)^r} = 3^{-(r+1)}, \\ \sum \mathbb{E}|\text{summand}|^3 &= \frac{1}{32} \sum_{h \geq 0} \binom{h}{r-1} 8^{-h} = \frac{1}{32} \frac{8^{-(r-1)}}{(1 - 1/8)^r} = \frac{1}{4 \cdot 7^r}.\end{aligned}$$

Divide the latter by  $\sigma_r^3$  and apply the Berry–Esseen theorem for independent, not necessarily identically distributed summands [93].  $\square$

*Remark 9.2.* The standardized fourth cumulant is also exact:

$$\frac{\kappa_4(X_r)}{\sigma_r^4} = -2 \left( \frac{3}{5} \right)^{r+1}. \quad (9.4)$$

It decays faster than the Berry–Esseen upper bound of [theorem 9.1](#) – at the geometric rate  $3/5 = 0.6$  per unit of  $r$ , against  $3\sqrt{3}/7 = 0.7423 \dots$  – and predicts the first Edgeworth correction in [conjecture 12.1](#).

## 9.2 Boundary scaling of the smooth box approximants

Let  $f_d$  be the density of  $Y_d$ , and put

$$\tau_d = \frac{1}{2} 3^{-(d+2)/2}. \quad (9.5)$$

This is the standard deviation of  $X_{d+1}/2$ .

**Theorem 9.3** (Gaussian boundary layers). *For each fixed bounded interval of  $y$  values,*

$$f_d\left(-\frac{1}{2} + \tau_d y\right) = \Phi_G(y) + \mathcal{O}\left(\left(\frac{3\sqrt{3}}{7}\right)^{d+1}\right), \quad (9.6)$$

$$f_d\left(\frac{1}{2} + \tau_d y\right) = \bar{\Phi}_G(y) + \mathcal{O}\left(\left(\frac{3\sqrt{3}}{7}\right)^{d+1}\right), \quad (9.7)$$

where the implied constant is universal. Outside any neighborhood of the two endpoints of radius  $\rho_d$  with  $\rho_d/\tau_d \rightarrow \infty$ ,  $f_d$  converges uniformly to the box density  $\mathbf{1}_{[-1/2, 1/2]}$ . This is [proved here](#).

*Proof.* Let  $T_d = X_{d+1}/2$ . Convolution with the unit-height box gives the exact formula

$$f_d(x) = \mathbb{P}(x - 1/2 \leq T_d \leq x + 1/2). \quad (9.8)$$

At  $x = -1/2 + \tau_d y$ , the lower endpoint of the window lies below the support of  $T_d$  for all sufficiently large  $d$ , while the upper endpoint is  $\tau_d y$ . Thus the value is the CDF of  $T_d/\tau_d$  at  $y$ . The right endpoint is the complementary CDF. Apply (9.3). If  $x$  stays a distance much larger than  $\tau_d$  from  $\pm 1/2$ , the standardized threshold tends uniformly to  $\pm\infty$ , giving the final assertion.  $\square$

## 9.3 Exact global distances from the limiting box

**Theorem 9.4** (Exact  $L^1$  identity and transport bounds). *Let  $b(x) = \mathbf{1}_{[-1/2, 1/2]}(x)$ . Then*

$$\boxed{\|f_d - b\|_{L^1(\mathbb{R})} = \mathbb{E}|X_{d+1}|.} \quad (9.9)$$

Consequently,

$$\|f_d - b\|_1 \sim \sqrt{\frac{2}{\pi}} 3^{-(d+2)/2}, \quad (9.10)$$

$$d_{\text{TV}}(\text{Law}(Y_d), \text{Law}(U)) \sim \frac{1}{\sqrt{2\pi}} 3^{-(d+2)/2}, \quad (9.11)$$

$$W_2(\text{Law}(Y_d), \text{Law}(U)) \leq \frac{1}{2} 3^{-(d+2)/2}. \quad (9.12)$$

These statements are **proved here**.

*Proof.* For a fixed shift  $t \in [-1/2, 1/2]$ , the two unit intervals  $[-1/2, 1/2]$  and  $[t - 1/2, t + 1/2]$  overlap in length  $1 - |t|$ . Thus the shifted box puts mass  $|t|$  outside the unshifted box. Averaging over  $T_d = X_{d+1}/2$ , and using equality of the inside deficit and outside excess, gives

$$\|f_d - b\|_1 = 2\mathbb{E}|T_d| = \mathbb{E}|X_{d+1}|.$$

The normalized variables  $X_r/\sigma_r$  converge to a standard Gaussian by [theorem 9.1](#); their second moments are uniformly one, so the absolute first moments converge as well. This proves (9.10), and total variation is half the  $L^1$  distance. Finally couple  $Y_d = U + T_d$  with  $U$  itself and use  $W_2^2 \leq \mathbb{E}T_d^2 = \tau_d^2$ .  $\square$

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**A smooth-to-discontinuous limit with an exact microscopic profile.** The family  $f_d$  consists entirely of compactly supported  $C^\infty$  densities, yet it tends to a discontinuous box. The global error is exactly an absolute moment of a Pascal–Rvachev variable, while the local endpoint transition is a Gaussian survival function. The same Pascal hierarchy therefore controls both the positive-definite quotient and the singular limit it regularizes.

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## 10 Mellin reflection, periodic endpoint hierarchies, and Lambert inversion

### 10.1 The endpoint Laplace transform

Shift the random series to its left endpoint:

$$Z_a = X_a + R_a = \sum_{h \geq 0} \sum_{j=1}^{a_h} 2^{-h} V_{h,j}, \quad V_{h,j} \sim \text{Unif}[0, 1]. \quad (10.1)$$

Its Laplace transform is

$$L_a(t) = \mathbb{E}e^{-tZ_a} = \prod_{h \geq 0} \left( \frac{1 - e^{-t/2^h}}{t/2^h} \right)^{a_h}. \quad (10.2)$$

Set

$$\lambda(u) = \log \frac{1 - e^{-u}}{u}. \quad (10.3)$$

Then  $\log L_a(t) = \sum_h a_h \lambda(t/2^h)$ .

**Theorem 10.1** (Spectral–endpoint Mellin reflection). *Assume  $a_h$  has polynomial growth. On the strip  $-1 < \Re w < 0$ ,*

$$\int_0^\infty \lambda(u) u^{w-1} du = -\Gamma(w) \zeta(w+1), \quad (10.4)$$

$$\boxed{\int_0^\infty \log L_a(t) t^{w-1} dt = -\Gamma(w) \zeta(w+1) A(2^w)}. \quad (10.5)$$



Using the spectral zeta from (3.3), this becomes the reflection formula

$$\mathcal{M}[\log L_a](w) = -\Gamma(w)\zeta(w+1) \frac{\mathcal{Z}_a(-w)}{\zeta(-w)}. \quad (10.6)$$

Thus the endpoint asymptotic and the integer-zero divisor are two Mellin views of the same generating function. The contour-transfer interpretation follows the harmonic-sum Mellin framework of [49]. This is **proved here**. Neither (10.4) nor (10.5) is formalized. The library does prove the sibling real-variable integral `Fabius.bose_mellin_integral_zeta` (`MellinBose.lean`),  $\int_0^\infty x^{a-1} \log(1 - e^{-x}) dx = -\Gamma(a)\zeta(a+1)$  for real  $a > 0$ : the same  $\Gamma\zeta$  right-hand side, but for the bare kernel  $\log(1 - e^{-x})$  and on the half-line to the right of the strip in (10.4), so it is a companion identity and not (10.4) itself. The double pole of  $\Gamma(w)\zeta(w+1)$  at  $w = 0$  that governs the hierarchy below is regularized by `Fabius.tendsto_gammaZetaRegularProduct_finitePart` (`MellinFinitePart.lean`). The subtraction of  $\log u$ , the continuation to  $-1 < \Re w < 0$ , and the complex-variable form all remain to be supplied.

*Proof.* The kernel behaves as  $-u/2 + \mathcal{O}(u^2)$  at zero and as  $-\log u + \mathcal{O}(e^{-u})$  at infinity, so the stated strip is maximal for the elementary integral. Integration by parts followed by the classical Mellin integral for  $(e^u - 1)^{-1}$  gives (10.4). For (10.5), absolute convergence permits summation under the integral and the substitution  $u = t/2^h$  contributes the factor  $2^{hw}$ . Hence the layer sum is  $A(2^w)$ . The last formula follows from  $\mathcal{Z}_a(-w) = \zeta(-w)A(2^w)$ .  $\square$

## 10.2 Pole order is the endpoint logarithmic order

Suppose now that  $a_h = P(h)$  has degree  $d$  and Newton leading coefficient  $c_d > 0$ . Near  $w = 0$ ,  $A_P(2^w)$  has a pole of order  $d+1$ , while  $\Gamma(w)\zeta(w+1)$  has a double pole. At the nonzero dyadic lattice

$$\chi_k = \frac{2\pi i k}{\log 2}, \quad k \in \mathbb{Z} \setminus \{0\}, \quad (10.7)$$

$A_P(2^w)$  has poles of order  $d+1$  and the Gamma–zeta factor is regular.

**Theorem 10.2** (Polynomial and periodic hierarchy). *As  $t \rightarrow \infty$ , Mellin inversion of (10.5) gives an expansion of the form*

$$\log L_P(t) = -C_P(\log t)^{d+2} + \sum_{j=0}^{d+1} \alpha_j (\log t)^j + \sum_{k \neq 0} t^{-\chi_k} Q_k(\log t) + \mathcal{E}_P(t), \quad (10.8)$$

where  $C_P$  is (7.9), every  $Q_k$  is a polynomial of degree at most  $d$ , and the remainder is the exponentially small endpoint correction alone: for a polynomial weight the integrand  $-\Gamma(w)\zeta(w+1)A_P(2^w)$  has no pole with  $\Re w > 0$ , its poles lying at  $w \in \{0, -1, -2, \dots\}$  and on the lattice (10.7). Poles to the right of the imaginary axis do appear in the rational setting of [conjecture 12.3](#), where a denominator root  $\rho$  with  $|\rho| > 1$  sits at  $\Re w = \log |\rho| / \log 2 > 0$ . The Fourier coefficients of the periodic-polynomial part are explicit residues of  $-\Gamma(w)\zeta(w+1)A_P(2^w)$ .

In particular, degree  $d$  produces:

- a nonperiodic logarithmic polynomial of degree  $d+2$ ;
- log-periodic coefficient polynomials of degree  $d$ ;
- the generalized lower-Lambert saddle (7.14).

This is **proved hereat** the structural level; obtaining a chosen number of explicit coefficients is a finite residue calculation.



*Proof.* The pole at zero has order  $(d + 1) + 2 = d + 3$ , and a Mellin pole of order  $m$  produces a polynomial of degree  $m - 1$  in  $\log t$ . Each nonzero  $\chi_k$  has order  $d + 1$  and contributes  $t^{-\chi_k}$  times a polynomial of degree  $d$ . Since  $t^{-\chi_k} = \exp(-2\pi i k \log_2 t)$ , these terms are periodic in  $\log_2 t$  with polynomial amplitudes. The leading coefficient is inherited from the highest Newton component and agrees with the audited rank- $(d + 1)$  Pascal law, giving  $C_P$ .  $\square$

*Remark 10.3.* For the classical case  $d = 0$ , the periodic terms are scalar Fourier modes. For  $d = 1$  they carry affine functions of  $\log t$ ; in general they are degree- $d$  periodic polynomials. This is a genuine hierarchy rather than a cosmetic perturbation of the rank-one Fabius factor.

### 10.3 Generalized Bernoulli polynomials generate the residues

Define the Nörlund generalized Bernoulli polynomials [47] by

$$\left(\frac{u}{e^u - 1}\right)^\alpha e^{xu} = \sum_{n \geq 0} B_n^{(\alpha)}(x) \frac{u^n}{n!}. \quad (10.9)$$

Every Newton component of  $A_P(e^u)$  has the Laurent expansion

$$\frac{e^{ru}}{(1 - e^u)^{r+1}} = (-1)^{r+1} u^{-r-1} \sum_{n \geq 0} B_n^{(r+1)}(r) \frac{u^n}{n!}. \quad (10.10)$$

**Algorithm 10.4** (Bernoulli–Bell endpoint coefficient engine). For a polynomial weight  $P$  and a desired truncation order:

1. compute its integer Newton coefficients  $c_r$ ;
2. expand  $A_P(e^u)$  using (10.10);
3. multiply by the Laurent series of  $-\Gamma(w)\zeta(w + 1)$  with  $u = w \log 2$ ;
4. take the residue at  $w = 0$  for the nonperiodic logarithmic polynomial and at  $w = \chi_k$  for the periodic coefficient polynomials;
5. solve the saddle recursively in the coordinate (7.14);
6. exponentiate and compose the resulting formal series with complete Bell polynomials.

All ingredients are finite algebra once the desired order and Fourier modes are fixed.

---

**Bernoulli and Bell roles.** Ordinary Bernoulli numbers enter the cumulants through (4.2); generalized Bernoulli polynomials encode the higher-order poles of the rational Newton generating function; complete Bell polynomials convert logarithmic cumulants and saddle corrections into moments, amplitudes, and inverse-series coefficients. The three appearances are different faces of one coefficient calculus rather than unrelated special-function decorations.

---

### 10.4 Inverse endpoint consequence

Combining (10.8) with saddle inversion yields an all-orders formal expansion for  $G_P$  and its inverse in the generalized Lambert coordinate. The leading term was proved in theorem 7.2; the exact periodic-polynomial transseries requires the uniform contour estimates already carried out in the classical corpus to be lifted to higher-order poles. We isolate that analytic completion as conjecture 12.4 rather than silently promoting a formal residue calculation to a theorem.

## 11 Numerical experiments and reproducibility audit

The numerical work was designed to falsify algebraic mistakes, not to replace proofs. This part’s verification material is kept in the volume directory under `assets/Fabius_Newton_Rvachev_Frontier_Report/`, whose script `numerical_experiments.py` uses 100-digit arithmetic for the  $q$ -Richardson identities and deterministic FFT inversion for the boundary-layer curves. No Monte Carlo sampling is used. That directory also holds the generated CSV tables, verification log, and both figures.

### 11.1 Exact identity checks

The script independently verifies:

1. the closed Lagrange weights (5.2);
2. annihilation of  $1, q^h, \dots, q^{r^h}$  and the exact survivor (5.4);
3. the generating function (8.3) for  $d = 0, 2, 4, 6$  at several high-precision arguments;
4. the multiplicity formula (8.8);
5. the complete transformed series (5.6);
6. the condition-number product (5.10).

Every check passed at tolerances between  $10^{-65}$  and  $10^{-80}$ .

### 11.2 Tomography of the first hidden exponent

For the new  $d = 2$  family,

$$A_2(x) = 1 + \left( \frac{x}{1-x} \right)^3, \quad a_0 = 1, \quad q = 1/4.$$

Table 4 records  $\mathcal{R}_{r,1}[A_2] - a_0$ . The exact alternating sign is visible, and six interpolation steps reduce an initial  $3.7 \times 10^{-2}$  error to  $3.6 \times 10^{-16}$  without a large weight norm.

Table 4: Certified  $q$ -Richardson errors for  $A_2$  at  $m = 1, q = 1/4$ .

| Order $r$ | $\mathcal{R}_{r,1}[A_2] - 1$    | Predicted side |
|-----------|---------------------------------|----------------|
| 0         | $3.7037037037 \times 10^{-2}$   | upper          |
| 1         | $-1.1950617284 \times 10^{-2}$  | lower          |
| 2         | $6.9704584427 \times 10^{-4}$   | upper          |
| 3         | $-6.0938671972 \times 10^{-6}$  | lower          |
| 4         | $1.0586844464 \times 10^{-8}$   | upper          |
| 5         | $-4.0454865708 \times 10^{-12}$ | lower          |
| 6         | $3.5620070678 \times 10^{-16}$  | upper          |

The limiting interpolation condition number was recomputed as

$$1.96926035366826943194719369697 \dots,$$

in agreement with (5.11). This is small enough that the observed convergence reflects the exact tail annihilation rather than a cancellation between huge floating-point weights.

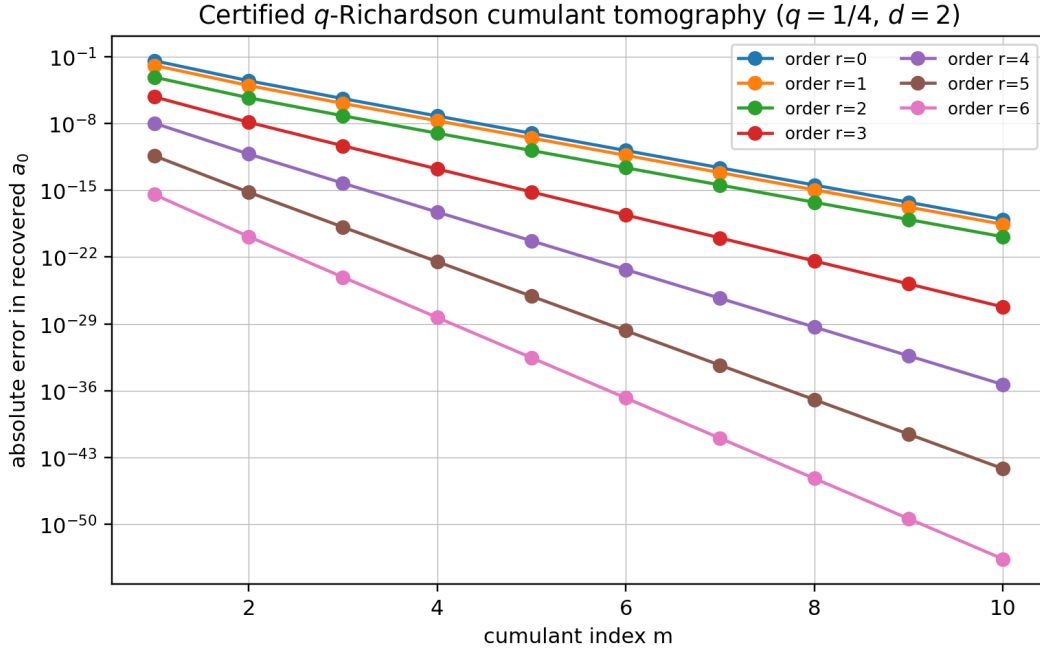


Figure 1: Absolute errors in the recovery of  $a_0$  from normalized cumulants  $A_2(4^{-m})$ . The CSV file retains the signs; the semilogarithmic plot shows the many orders of magnitude gained by increasing either the cumulant index  $m$  or the interpolation order  $r$ .

### 11.3 The sparse zero spectrum for $d = 2$

The generated file `assets/Fabiús-Newton-Rvachev-Frontier-Report/data/zero_multiplcities_d2.csv` compares the prefix sum of the original exponents with  $1 + \binom{\nu_2(n)}{3}$ . The first exceptional values are

| $n$ | $\nu_2(n)$ | zero order |
|-----|------------|------------|
| 8   | 3          | 2          |
| 16  | 4          | 5          |
| 24  | 3          | 2          |
| 32  | 5          | 11         |
| 40  | 3          | 2          |
| 48  | 4          | 5          |
| 64  | 6          | 21         |

The spectrum is therefore classical at almost every integer but develops rapidly increasing multiplicity on a sparse tower of highly divisible integers.

### 11.4 Deterministic boundary-layer inversion

For ranks  $r = 3, 5, 7, 9$  (equivalently  $d = 2, 4, 6, 8$ ), the script evaluates the exact truncated characteristic product of  $X_r$ , inverts it on a nonaliasing FFT interval, normalizes the resulting density, and compares its standardized CDF with the Gaussian. The largest observed CDF errors on  $[-4, 4]$  are shown in [table 5](#).

The errors decrease consistently with [theorem 9.1](#). The theorem’s universal Berry–Esseen bound is intentionally conservative; the exact standardized fourth cumulant (9.4) gives a better explanation of the observed first correction.

Table 5: Deterministic FFT check of the standardized Pascal CDF.

| Rank $r$ | family degree $d = r - 1$ | maximum CDF error on $[-4, 4]$ |
|----------|---------------------------|--------------------------------|
| 3        | 2                         | $7.6043 \times 10^{-3}$        |
| 5        | 4                         | $2.4624 \times 10^{-3}$        |
| 7        | 6                         | $9.3598 \times 10^{-4}$        |
| 9        | 8                         | $6.4752 \times 10^{-4}$        |

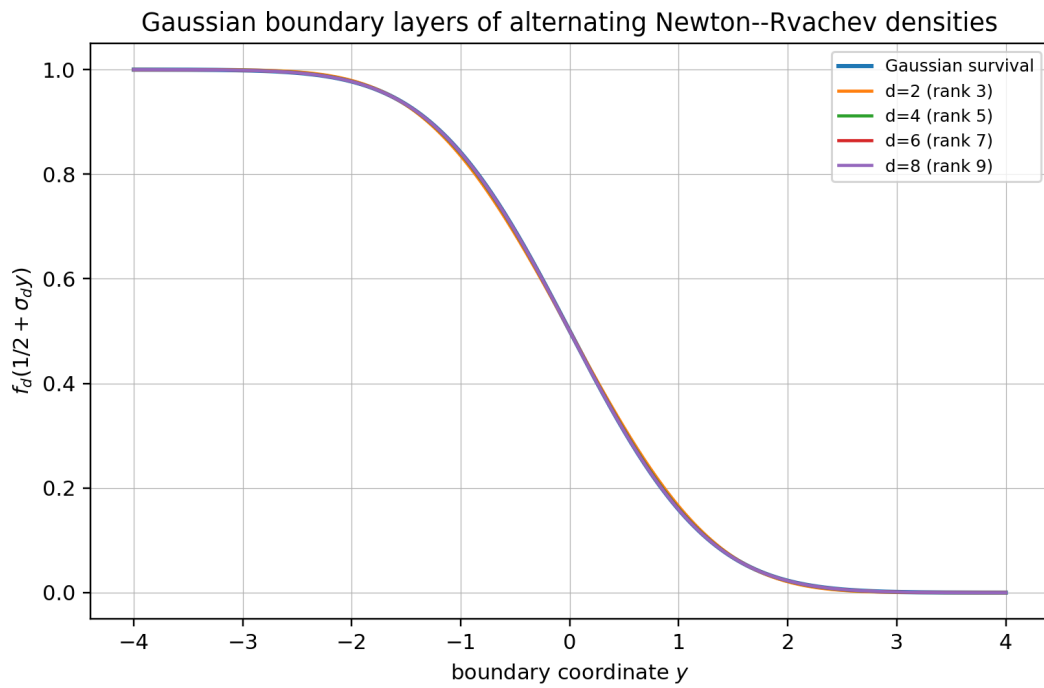


Figure 2: Right-edge boundary profiles  $f_d(1/2 + \tau_d y)$  compared with the Gaussian survival function  $1 - \Phi_G(y)$ . The curves are obtained by deterministic Fourier inversion of the Pascal–Rvachev characteristic products.

## 11.5 Reproduction command

From the extracted archive, the full experiment can be rerun with

```
python numerical_experiments.py --output-dir reproduced-data
```

The script checks its own exact identities with assertions before writing any figure. A failure therefore stops the run rather than silently producing visually plausible output.

## 12 Conjectures and frontier problems

The following statements are deliberately separated from the proved theorem package. Each is supported by the exact structure above, but each still requires a substantive analytic, optimization, or arithmetic argument.

**Conjecture 12.1** (Uniform Edgeworth boundary expansion). *Let  $r = d + 1$  and  $\tau_d$  be (9.5). For every fixed  $J$  and every fixed number of  $y$ -derivatives, the two boundary profiles admit a uniform expansion*

$$f_d\left(\frac{1}{2} + \tau_d y\right) = \bar{\Phi}_G(y) + \varphi(y) \sum_{j=1}^J E_j(y; r) + \mathcal{O}\left(\left(\frac{3^{J+2}}{4^{J+2} - 1}\right)^{r+1}\right), \quad (12.1)$$

where  $\varphi$  is the Gaussian density and each  $E_j$  is an explicit Hermite polynomial whose coefficients are Bell polynomials in the standardized cumulants. The first correction is governed by  $\kappa_4/\sigma^4 = -2(3/5)^{r+1}$ . The remainder gauge is the size of the first omitted group: the weight- $j$  group carries  $\kappa_{2(j+1)}/\sigma^{2(j+1)}$ , of exact size  $(3^{j+1}/(4^{j+1} - 1))^{r+1}$ , which for  $j \geq 2$  already exceeds  $(3/5)^{j(r+1)}$ .

**Why it is plausible.** The triangular array has bounded summands, an entire characteristic function, and standardized cumulants that decay exponentially with rank. A Fourier proof should be considerably cleaner than a generic Edgeworth theorem because the exact product provides uniform control in both the central and large-frequency regions.

**Conjecture 12.2** (Minimax property of alternating  $q$ -tomography). *Fix  $q, m, r$  and the cone*

$$\mathcal{C}_M = \left\{ A(x) = \sum_{h \geq 0} a_h x^h : a_h \geq 0, A(q^m) \leq M \right\}.$$

*Among all linear estimators using  $A(q^m), \dots, A(q^{m+r})$  and exact on polynomials of degree at most  $r$ , the Lagrange estimator (5.5) minimizes the worst one-sided error on  $\mathcal{C}_M$ . Moreover, consecutive odd/even transforms form the narrowest possible universal bracket under these constraints.*

**Why it is plausible.** The remainder kernel in (5.6) has one sign on the coefficient cone and is the unique kernel annihilating the first  $r$  monomials. That uniqueness is already formal: `Fabiús.eq_lagrangeEvalWeight_of_moments` (`GeometricLagrange.lean`) shows that on any finite family of distinct field-valued nodes the Lagrange row is the only row with the prescribed moments, with no nonzeroness or positivity assumption on the competing weights. A proof of the conjecture should be expressible as a finite-dimensional linear-programming duality or a Chebyshev-system extremal argument.

**Conjecture 12.3** (Rational refinement transseries). *Let  $A$  be rational, nonnegative coefficientwise, and analytic at the origin. Then the complete large- $t$  transseries of  $\log L_a(t)$  is determined functorially by the poles of  $A$ : each pole  $\rho$  of order  $m$  produces vertical Mellin lattices*

$$w = \frac{\log \rho + 2\pi i k}{\log 2}, \quad k \in \mathbb{Z},$$

*with degree- $(m - 1)$  logarithmic amplitudes. Poles on the unit circle yield bounded log-periodic phases; poles of other moduli yield additional power scales. No further frequencies occur.*

**Research route.** Start from (10.5); establish a common meromorphic continuation and vertical bounds for the rational factor, Gamma, and zeta; then shift the Mellin contour across a finite set of pole lattices. The main technical issue is uniform summability of the residue series when several pole lattices have the same real part.

**Conjecture 12.4** (All-orders generalized inverse-Fabius expansion). *For every nonnegative integer-valued polynomial  $P$  of degree  $d$ , the endpoint CDF  $G_P$  and its inverse admit all-orders expansions in the generalized lower-Lambert coordinate (7.14). Their coefficients are finite differential polynomials in the degree- $d$  periodic amplitudes of (10.8); equivalently, they are algorithmically generated from Nörlund Bernoulli polynomials, Gamma–zeta jets, and complete Bell polynomials. The expansion is uniform under bounded shifts of the Lambert phase.*

**Why it matters.** This would extend the corpus’s inverse Fabius theory from the rank-one quadratic logarithmic saddle to every polynomial Newton sector and would make the Lambert–Bernoulli–Bell connection exact rather than merely formal.

**Open problem 12.5** (The Newton positivity cone). *Classify the integer vectors  $(c_0, \dots, c_d)$  for which*

$$P(h) = \sum_{r=0}^d c_r \binom{h}{r} \geq 0 \quad (h \in \mathbb{N}_0).$$

*Describe the extreme rays, the indecomposable characteristic functions  $\prod_r \Phi_{r+1}^{c_r}$ , and the minimal Pascal numerator and denominator after all entire cancellations. Determine whether the alternating vectors  $((-1)^d, (-1)^{d-1}, \dots, 1)$  for even  $d$  generate distinguished extreme rays under a natural support normalization.*

**Open problem 12.6** (Arithmetic of generalized half-integer spectra). *For a rational generating function  $A$  with integer coefficients, classify the fields, traces, and norms of*

$$\Phi_a \left( k + \frac{1}{2} \right).$$

*The audited half-integer report handles the classical exponent sequence. The master zero/cumulant dictionary suggests that rational  $A$  should produce finite-dimensional cyclotomic trace recurrences, but the orders and positivity constraints will depend on the pole data of  $A$ .*

**Open problem 12.7** (Optimal smooth approximation of the box within the Newton family). *At fixed support and fixed highest Newton degree, choose a nonnegative integer-valued polynomial  $P$  to minimize one of*

$$\|f_P - b\|_1, \quad W_2(\text{Law}(X_P), \text{Law}(U)), \quad \sup_{|\xi| \leq T} |\Phi_P(\xi) - \text{sinc}_\pi(\xi)|.$$

*The family  $P_d(h) = \binom{h-1}{d}$  is natural because it appends one rescaled Pascal law to a box, but it is not yet known to be extremal under any of these criteria.*

## 13 Formalization and computational research program

The theorem package is unusually suitable for staged Lean formalization: its algebraic core is finite, and the analytic results can be added only after the exact identities are stable. A dependency-aware order is as follows.

### 13.1 Phase I: finite algebra and digital combinatorics

1. Define exponent prefixes and prove  $m_a(n) = \sum_{h \leq \nu_2(n)} a_h$  for finite products.
2. Prove the finite zero-count identity  $M_a(N) = \sum_h a_h \lfloor N/2^h \rfloor$  and the binary defect (3.8).
3. Define the generalized sign character and prove (3.12); formalize the automaticity equivalence for eventually periodic parity words.
4. Prove the finite sign-product factorization (3.14), its exact order at one, and the first surviving moment (3.16).
5. The geometric Lagrange weights, Gaussian-binomial closed form, exact principal specialization, survivor moments, finite residual sign, and exact finite norm are now formalized. The exact transformed series is also formalized under unconditional summability in `AnalyticSeriesFilter.lean`, and `RvachevQBinomialFilter.lean` applies it to the actual infinite Rvachev sinc product. The report’s finite-prefix quotient and Bell series, positivity, sign-certified bounds, and uniform or asymptotic estimates remain open.

These statements require only finite sums, polynomials, valuations, and existing  $q$ -binomial infrastructure. Several of them are in fact already discharged in the library, in greater generality than is needed here: step 1 and the floor identity of step 2 by `Fabius.weightedScaleMultiplicity` and `Fabius.sum_range_weightedScaleMultiplicity`; the sharp moment of step 4 by `Fabius.sum_powerset_neg_one_pow_pow_card`, together with its vanishing companion `Fabius.sum_powerset_neg_one_pow_eval_of_degree_lt`; and the annihilation and first surviving moment of step 5 by `Fabius.sum_geometricLagrangeWeight_mul_pow` and `Fabius.sum_geometricLagrangeWeight_mul_pow_succ`, with `Fabius.geometricLagrange_richardson_exact` already giving module-valued Richardson exactness. What genuinely remains in Phase I is the binary defect (3.8), the automaticity equivalence of step 3, the sign-product factorization (3.14) and its free-even-layer count, and the report-specific ordered sign and bound built on the now-formal general- $q$  survivor. The closed Gaussian-binomial form and every positive survivor moment are no longer Phase I obligations.

### 13.2 Phase II: probability and entire products

1. Construct  $X_a$  from finite uniform sums and pass to the bounded infinite series using the summability hypothesis  $A(1/2) < \infty$ .
2. Lift the formal product identity  $\Phi_{a+b} = \Phi_a \Phi_b$  to the convolution law for the probability measures. Local uniform convergence and the product identity themselves are already formalized.
3. Derive cumulants from the uniform Bernoulli expansion and moments from the existing Bell-polynomial library.
4. The zero-multiplicity stage of identifiability is now complete in `GeneralizedRvachevIdentifiability.lean`: dyadic analytic orders recover  $a_0$  and every successor by exact first differences, and equality of the full products is equivalent to equality of admissible exponent sequences. Formalizing uniqueness from the analytic samples  $A(4^{-j})$ , and connecting those samples to cumulants and a constructed probability law, remains open.
5. Build polynomial exponent sequences from Newton expansions and prove the entire cancellation behind (7.4).

### 13.3 Phase III: asymptotics and inverse functions

1. Reuse the corpus’s Mellin kernel theorem to prove (10.5) for polynomial weights.



2. Formalize pole orders and the leading coefficient  $C_P$  before attempting the full periodic residue expansion.
3. Transfer the existing Pascal-hierarchy Fourier envelope to a general polynomial by the Newton factorization and lower-order comparison.
4. Prove the leading small-ball and inverse laws. The library contains no de Bruijn–Voss exponential Tauberian transfer, so this step must begin by isolating that theorem as a reusable library component.
5. Only then lift the classical all-orders saddle and inverse-Lambert machinery to the higher-order pole hierarchy.

### 13.4 Certified computation

The tomography algorithm is a natural application for exact rationals and interval arithmetic. At  $q = 1/4$  all Lagrange weights are rational, as are the normalized cumulants of integer exponent sequences. A verified executable can therefore recover coefficients with no floating-point assumptions. For asymptotic experiments, Arb-style ball arithmetic would certify Mellin residues, Lambert branches, and boundary-layer errors. The supplied Python script is a reproducible exploratory implementation, not a substitute for that final certification layer.

## 14 Conclusion

The central result of this report is a change of viewpoint. Instead of treating the classical Rvachev transform, the Pascal hierarchy, its zero spectrum, its cumulants, and its endpoint asymptotics as separate constructions, one begins with the exponent generating function

$$A(q) = \sum_{h \geq 0} a_h q^h.$$

The same  $A$  then appears at four geometrically related arguments:

$$\begin{aligned} A(1/2) &: \text{twice the support radius,} \\ A(2^{-s}) &: \text{spectral zeta and zero divisor,} \\ A(4^{-j}) &: \text{even cumulants,} \\ A(2^w) &: \text{endpoint Mellin transform.} \end{aligned} \tag{14.1}$$

Its coefficient parity is the Fourier-lobe character, its prefix sums are zero multiplicities, its rationality is finite refinement rank, and its Newton coordinates are Pascal–Rvachev exponents.

Three consequences seem especially durable.

First, the spectral and probabilistic data are identifiable at the manuscript level: the zero divisor or the full even-cumulant sequence determines the hidden dyadic convolution law. Lean now verifies the zero-divisor side constructively and more sharply: the orders at  $1, 2, 4, \dots$  recover the exponent sequence by exact first differences, and the full generalized product determines and is determined by that sequence. The spectral-zeta quotient equivalence, cumulant and analytic-sample identifiability, and probability-law construction remain Partial in Lean. On the manuscript side, the Gaussian- $q$ -binomial Richardson transform makes cumulant recovery constructive and supplies certified alternating bounds.

Second, nonnegative integer-valued polynomials form a rich positive-definite quotient sector. Negative Pascal exponents are not pathologies when they recombine into a nonnegative scale sequence. The family  $P_d(h) = \binom{h-1}{d}$  gives the explicit quotient  $\Phi_1 \Phi_3 / \Phi_2$  at  $d = 2$ , sparse enhanced zero divisors, automatic lobe signs, and a smooth approximation to the box with an exact Gaussian edge profile.

Third, Mellin reflection makes the endpoint and spectrum literally the same meromorphic object viewed at  $s$  and  $-s$ . Polynomial degree becomes pole order; pole order becomes the power of the

endpoint logarithm; the dyadic pole lattice becomes the periodic coefficient hierarchy; and the leading saddle is solved by a generalized lower branch of Lambert  $W$ . Nörlund Bernoulli polynomials and complete Bell polynomials then supply the coefficient algebra required for an all-orders inverse theory.

These deductions are new relative to the pinned 67-source corpus, but they are not presented as the end of the subject. Their main value is that they turn a collection of striking special identities into a classification problem with exact algebraic coordinates, stable reconstruction algorithms, and a clear route to formal verification.

## A Complete recursive corpus inventory

The following path ledger is the authoritative source list used for the nonduplication audit. Every path is relative to `Analysis/FabijsFunction/docs/`. The pinned commit is `32d6d36c51d803289e6d6a0dc0c37753766eba47`. The count is 67. Frozen revisions were read for distinctive claims and provenance, but a repeated theorem in several revisions was counted once when determining novelty.

| No. | Relative path                                                                                                                      |
|-----|------------------------------------------------------------------------------------------------------------------------------------|
| 1   | <code>FabijsFunction_Mathematical_Glossary/FabijsFunction_Mathematical_Glossary.tex</code>                                         |
| 2   | <code>Fabijs_Function_and_Rvachev_Up/Fabijs_Function_and_Rvachev_Up.tex</code>                                                     |
| 3   | <code>Non_Elementarity_of_the_Fabijs_Function/Non_Elementarity_of_the_Fabijs_Function.tex</code>                                   |
| 4   | <code>fabius_lean_walkthrough/fabius_lean_walkthrough.tex</code>                                                                   |
| 5   | <code>semi-formalized-research-frontiers/semi-formalized-research-frontiers.tex</code>                                             |
| 6   | <code>archive/early-notes/FabijsAsymptotic/FabijsAsymptotic.tex</code>                                                             |
| 7   | <code>archive/early-notes/K-foldsummationoverthesignedThue-Morsesequence/K-foldsummationoverthesignedThue-Morsesequence.tex</code> |
| 8   | <code>archive/glossary/FabijsFunction_Mathematical_Glossary-2/FabijsFunction_Mathematical_Glossary.tex</code>                      |
| 9   | <code>archive/glossary/FabijsFunction_Mathematical_Glossary/FabijsFunction_Mathematical_Glossary.tex</code>                        |
| 10  | <code>archive/lean-walkthrough/Fabijs_Function_Lean_Walkthrough/Fabijs_Function_Lean_Walkthrough.tex</code>                        |
| 11  | <code>archive/lean-walkthrough/fabius_lean_walkthrough/fabius_lean_walkthrough.tex</code>                                          |
| 12  | <code>archive/primary-exposition/Fabijs_Function_and_Rvachev_Up-2/Fabijs_Function_and_Rvachev_Up.tex</code>                        |
| 13  | <code>archive/primary-exposition/Fabijs_Function_and_Rvachev_Up-3/Fabijs_Function_and_Rvachev_Up.tex</code>                        |
| 14  | <code>archive/primary-exposition/Fabijs_Function_and_Rvachev_Up/Fabijs_Function_and_Rvachev_Up.tex</code>                          |
| 15  | <code>archive/primary-exposition/Fabijs_and_Rvachev_Up-2/Fabijs_and_Rvachev_Up.tex</code>                                          |
| 16  | <code>archive/primary-exposition/Fabijs_and_Rvachev_up/Fabijs_and_Rvachev_up.tex</code>                                            |
| 17  | <code>archive/primary-exposition/Fabijs_function_and_up_function/Fabijs_function_and_up_function.tex</code>                        |
| 18  | <code>archive/primary-supplements/Fabijs_Function_Missing_Parts-2/Fabijs_Function_Missing_Parts.tex</code>                         |
| 19  | <code>archive/primary-supplements/Fabijs_Function_Missing_Parts/Fabijs_Function_Missing_Parts.tex</code>                           |
| 20  | <code>archive/q-calculus/Fabijs_q_Special_Functions/Fabijs_q_Special_Functions.tex</code>                                          |

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*Corpus inventory (continued)*

| No. | Relative path                                                                                                                                |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------|
| 21  | archive/q-calculus/fabius-q-special-functions/fabius-q-special-functions.tex                                                                 |
| 22  | archive/research-frontiers/Fabius_Dyadic_Asymptotic_Bridge/Fabius_Dyadic_Asymptotic_Bridge.tex                                               |
| 23  | archive/research-frontiers/Fabius_Dyadic_Formulae_and_Alternative_Representations/Fabius_Dyadic_Formulae_and_Alternative_Representations.tex |
| 24  | archive/research-frontiers/Fabius_Dyadic_Formulae_to_Asymptotics/Fabius_Dyadic_Formulae_to_Asymptotics.tex                                   |
| 25  | archive/research-frontiers/Fabius_Dyadic_q_Connections/Fabius_Dyadic_q_Connections.tex                                                       |
| 26  | archive/research-frontiers/Fabius_Thue_Morse_Convergence_Rate/Fabius_Thue_Morse_Convergence_Rate.tex                                         |
| 27  | archive/research-frontiers/Fabius_Thue_Morse_Convergence_Rates/Fabius_Thue_Morse_Convergence_Rates.tex                                       |
| 28  | archive/research-frontiers/Primary_Exposition_Gap_Register/Primary_Exposition_Gap_Register.tex                                               |
| 29  | archive/research-frontiers/Repeated_Integration_and_Rvachev_Up/Repeated_Integration_and_Rvachev_Up.tex                                       |
| 30  | archive/research-frontiers/Rvachev_Up_from_Repeated_Integration/Rvachev_Up_from_Repeated_Integration.tex                                     |
| 31  | archive/research-frontiers/Thue_Morse_Formula_Atlas-1/Thue_Morse_Formula_Atlas.tex                                                           |
| 32  | archive/research-frontiers/Thue_Morse_Formula_Atlas-2/Consolidated_Formula_Atlas_for_the_Thue_Morse_Sequence.tex                             |
| 33  | archive/research-frontiers/Thue_Morse_Formula_Atlas-3/Thue_Morse_Formula_Atlas.tex                                                           |
| 34  | archive/research-frontiers/non-formalized-research-frontiers/semi-formalized-research-frontiers.tex                                          |
| 35  | archive/standalone-studies/Non_Elementarity_of_the_Fabius_Function/Non_Elementarity_of_the_Fabius_Function.tex                               |
| 36  | archive/standalone-studies/Small_Argument_Asymptotics/Small_Argument_Asymptotics.tex                                                         |
| 37  | archive/standalone-studies/fabius-inverse-dyadic-closed-form/fabius-inverse-dyadic-closed-form.tex                                           |
| 38  | semi-formalized-research-frontiers/drafts/Fabius_Arithmetic_Rays_Frontier_Report/Fabius_Arithmetic_Rays_Frontier_Report.tex                  |
| 39  | semi-formalized-research-frontiers/drafts/Fabius_Half_Integer_Spectral_Frontier_Report/Fabius_Half_Integer_Spectral_Frontier_Report.tex      |
| 40  | semi-formalized-research-frontiers/drafts/Fabius_Rvachev_Frontier_Report-2/Fabius_Rvachev_Frontier_Report.tex                                |
| 41  | semi-formalized-research-frontiers/drafts/Fabius_Rvachev_Frontier_Report/Fabius_Rvachev_Frontier_Report.tex                                  |
| 42  | semi-formalized-research-frontiers/drafts/Fabius_Rvachev_New_Frontiers/Fabius_Rvachev_New_Frontiers.tex                                      |
| 43  | semi-formalized-research-frontiers/drafts/Fabius_Rvachev_Thue_Morse_Frontier_Results/Fabius_Rvachev_Thue_Morse_Frontier_Results.tex          |
| 44  | semi-formalized-research-frontiers/drafts/Spectral_Arithmetic_Pascal_Rvachev_Hierarchy/Spectral_Arithmetic_Pascal_Rvachev_Hierarchy.tex      |
| 45  | semi-formalized-research-frontiers/drafts/Thue_Morse_Formula_Atlas/Thue_Morse_Formula_Atlas.tex                                              |
| 46  | semi-formalized-research-frontiers/drafts/fabius_frontier_new_results/fabius_frontier_new_results.tex                                        |
| 47  | semi-formalized-research-frontiers/drafts/fabius_frontier_results_bundle/fabius_frontier_results.tex                                         |

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*Corpus inventory (continued)*

| No. | Relative path                                                                                                                                                                                        |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 48  | semi-formalized-research-frontiers/drafts/fabius_frontier_spectral_endpoint_report_bundle/fabius_frontier_spectral_endpoint_report.tex                                                               |
| 49  | semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay-2/Rvachev_Up_Fourier_Decay.tex                                                                           |
| 50  | semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay-3/Fourier_decay_of_Rvachev_up.tex                                                                        |
| 51  | semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay_Comparative_Audit/Rvachev_Up_Fourier_Decay_Comparative_Audit.tex                                         |
| 52  | semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay_Gentle_Guide/Rvachev_Up_Fourier_Decay_Gentle_Guide.tex                                                   |
| 53  | semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay_Second_Wave_Audit/Rvachev_Up_Fourier_Decay_Second_Wave_Audit.tex                                         |
| 54  | semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/asymptotic-decay-rate-of-an-infinite-product-of-sinc-functions/asymptotic-decay-rate-of-an-infinite-product-of-sinc-functions.tex |
| 55  | semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-1/rvachev_up_fourier_decay.tex                                                                           |
| 56  | semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-4/rvachev_up_fourier_decay.tex                                                                           |
| 57  | semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-5/Rvachev_Up_Fourier_Decay_Final.tex                                                                     |
| 58  | semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-6/Rvachev_Up_Fourier_Decay_Final_Synthesis.tex                                                           |
| 59  | semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-7/rvachev_up_fourier_decay_final.tex                                                                     |
| 60  | semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-8/rvachev_up_fourier_decay_final.tex                                                                     |
| 61  | papers/AIF_2017__67_2_637_0/AIF_2017__67_2_637_0-corrected.tex                                                                                                                                       |
| 62  | papers/AIF_2017__67_2_637_0/AIF_2017__67_2_637_0.tex                                                                                                                                                 |
| 63  | papers/arXiv-1502.06738v2/Lacunary_products.tex                                                                                                                                                      |
| 64  | papers/arXiv-1702.05442v1/09-Function.tex                                                                                                                                                            |
| 65  | papers/arXiv-1702.06487v3/157-Arithmetic-v3.tex                                                                                                                                                      |
| 66  | papers/arXiv-2211.13570v1/document.tex                                                                                                                                                               |
| 67  | papers/ubiq15/ubiq15-corrected.tex                                                                                                                                                                   |

## B Result ledger and dependency map

Table 7 records the logical status and principal inputs of the report’s main claims. It is intended to make later formalization and peer checking easier: a reader can separate finite algebra from analytic continuation, and new deductions from imported frontier theorems.

Table 7: Result and proof-dependency ledger.

| ID | Statement                                                                      | Status | Main inputs                                         |
|----|--------------------------------------------------------------------------------|--------|-----------------------------------------------------|
| A1 | Exponent-sequence probability law, support, entire product, convolution monoid | proved | uniform characteristic function; normal convergence |
| A2 | Zero multiplicity $m_a(n)$ and zeta $\zeta(s)A(2^{-s})$                        | proved | Euler product; Tonelli/rearrangement                |

| ID | Statement                                                          | Status              | Main inputs                                      |
|----|--------------------------------------------------------------------|---------------------|--------------------------------------------------|
| A3 | Digital zero defect and generalized lobe character                 | proved              | binary floors; parity                            |
| A4 | Automaticity iff the parity word is eventually periodic            | proved              | exact 2-kernel/shift identity                    |
| A5 | General Prouhet order and first surviving power moment             | proved              | finite product; falling factorials               |
| B1 | Cumulant transform $B_{2j}A(4^{-j})/(2j)$                          | proved              | uniform cumulants; additivity                    |
| B2 | Identifiability from zeros, zeta, or all even cumulants            | proved              | prefix differences; identity theorem             |
| B3 | Gaussian- $q$ -binomial tomography and one-sided remainder         | proved              | Lagrange interpolation; principal specialization |
| B4 | Bounded exact condition number $(-q; q)_r/(q; q)_r$                | proved              | finite $q$ -binomial theorem                     |
| C1 | Shift/refinement identity and rational finite-rank classification  | proved              | reindexing; rational-series theorem              |
| C2 | Newton factorization of nonnegative polynomial weights             | proved              | integer-valued polynomial calculus               |
| C3 | Leading Fourier, small-ball, and inverse laws                      | derived             | audited Pascal hierarchy; de Bruijn transfer     |
| D1 | Alternating family $P_d(h) = \binom{h-1}{d}$ and positive quotient | proved              | Newton identity; random-series factorization     |
| D2 | Sparse spectrum, exact cumulants, automatic Prouhet selector       | proved              | hockey-stick; Lucas theorem                      |
| D3 | Gaussian edge layer, exact $L^1$ law, TV and $W_2$ bounds          | proved              | Berry–Esseen; interval overlap                   |
| E1 | Mellin reflection between endpoint Laplace and spectral zeta       | proved              | Mellin kernel; geometric scaling                 |
| E2 | Polynomial/log-periodic pole hierarchy                             | proved structurally | Mellin residues; rational pole orders            |
| E3 | All-orders generalized inverse transseries                         | conjectural         | uniform higher-pole saddle analysis needed       |

**Lean crosswalk for B3–B4.** The finite weight, all-moment, residual-sign, and finite-norm identities in B3–B4 are formalized in `GeometricQBinomialLagrange.lean`, `GeometricLagrangeQBinomial.lean`, and `GeometricLagrangeQMoments.lean`. The exact unconditionally summable transformed series, including the shifted Gaussian tail, is formalized in `AnalyticSeriesFilter.lean`; zero-weight nodes require no convergence of their unweighted samples. The infinite condition-number limit on the whole range  $0 \leq q < 1$  is formalized in `LimitConditionNumber.lean` by `Fabijs.tendsto_sum_abs_qToeplitzWeight`, with left-limit divergence at one proved by `Fabijs.tendsto_qConditionNumberLimit_atTop_at_one_left`. The conditionally convergent boundary, the ordered one-sided sign and bound, the report’s finite-prefix exponent/Bell generating function, and the uniform estimates remain results of this report rather than claims about the current Lean corpus. The distinct actual infinite-product Rvachev instantiation is now formalized in `RvachevQBinomialFilter.lean`.

## C Reproducibility manifest

This part’s verification material is kept in the volume directory under `assets/Fabius_Newton_Rvachev_Frontier_Report/`. The paths below are relative to the volume root; the absorbed report’s own source and PDF were deleted when it was merged, and git history is their archive.

| Path                                                                                        | Purpose                                                     |
|---------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| <code>assets/Fabius_Newton_Rvachev_Frontier_Report/numerical_experiments.py</code>          | Commented high-precision and FFT verification program.      |
| <code>assets/Fabius_Newton_Rvachev_Frontier_Report/figures/richardson_tomography.png</code> | Figure used in <a href="#">figure 1</a> .                   |
| <code>assets/Fabius_Newton_Rvachev_Frontier_Report/figures/boundary_layers.png</code>       | Figure used in <a href="#">figure 2</a> .                   |
| <code>assets/Fabius_Newton_Rvachev_Frontier_Report/data/richardson_tomography.csv</code>    | Full signed extrapolation errors.                           |
| <code>assets/Fabius_Newton_Rvachev_Frontier_Report/data/boundary_layer_errors.csv</code>    | Deterministic standardized-CDF errors.                      |
| <code>assets/Fabius_Newton_Rvachev_Frontier_Report/data/zero_multiplicities_d2.csv</code>   | Direct and closed-form zero orders for the $d = 2$ family.  |
| <code>assets/Fabius_Newton_Rvachev_Frontier_Report/data/verification_log.txt</code>         | Precision settings, assertion status, and headline values.  |
| <code>assets/Fabius_Newton_Rvachev_Frontier_Report/README.md</code>                         | The absorbed package’s own run notes.                       |
| <code>assets/Fabius_Newton_Rvachev_Frontier_Report/requirements.txt</code>                  | Exact Python package versions used for the numerical audit. |

The volume itself is rebuilt from its own directory by running `pdflatex` on `geometric_q_fabius_frontiers.tex` three times, so that the table of contents and every cross-reference settle.

The numerical script fixes the precision and grid sizes explicitly. Its formulas use `numpy.sinc(x) = sin( $\pi x$ ) / ( $\pi x$ )`, matching the normalization in (2.1). The FFT interval is wider than the compact support, so periodic copies do not overlap. All exact checks run before figure generation.

## Part II

# q-Binomial Richardson Acceleration of Geometric Sinc Products

### Abstract

The Fourier transform of Rvachev’s compactly supported up-function is the geometric sinc product

$$P(\xi) = \prod_{\nu \geq 1} \text{sinc}_{\text{rad}}(\xi/2^\nu).$$

The repository corpus develops its large-frequency decay, its connection with the Fabius function and the signed Thue–Morse sequence, dyadic exact values, small-argument and inverse asymptotics, and several Richardson-type limits. This report studies a different asymptotic variable: the truncation level in the finite product  $P_n(\xi) = \prod_{\nu=1}^n \text{sinc}_{\text{rad}}(\xi/2^\nu)$ .

For a general geometric base  $b > 1$ , with  $q = b^{-2}$ , we prove, in the disc  $|q^n \xi^2| < \pi^2 b^2$ , the

total exact analytic factorization

$$P_{b,n}(\xi) = P_b(\xi) \exp \left( \sum_{k \geq 1} c_k(b) \xi^{2k} q^{kn} \right) = P_b(\xi) \sum_{r \geq 0} \alpha_r(b) \xi^{2r} q^{rn}.$$

Off the zero set of  $P_b$ , division gives the corresponding quotient form. The cumulants  $c_k(b)$  are explicit Bernoulli-number combinations and the coefficients  $\alpha_r(b)$  are complete Bell polynomials. Lagrange evaluation at the geometric nodes  $1, q, \dots, q^m$  then gives a universal extrapolant

$$\mathcal{R}_{m,n} P_b = \sum_{j=0}^m w_j^{(m)} P_{b,n+j}, \quad w_j^{(m)} = \frac{(-1)^{m-j} q^{(m-j)(m-j+1)/2}}{(q; q)_j (q; q)_{m-j}}.$$

On the same disc and off the zero set of  $P_b$ , its complete error series has the closed  $q$ -binomial form

$$\frac{\mathcal{R}_{m,n} P_b(\xi)}{P_b(\xi)} - 1 = (-1)^m q^{\binom{m+1}{2}} \sum_{r \geq m+1} \begin{bmatrix} r-1 \\ m \end{bmatrix}_q \alpha_r(b) \xi^{2r} q^{rn}.$$

Consequences include an exact sign law, non-asymptotic divided-difference bounds, a rigorous a posteriori error certificate, and a uniformly bounded stability constant  $\prod_{r=1}^m (1+q^r)/(1-q^r)$ . For  $q = 1/4$  this constant stays below 1.970, even as the order tends to infinity.

After inverse Fourier transformation, the same filter combines consecutive finite box splines. It matches the first  $2m$  moments of the limiting up-density *exactly at every truncation level*, and converges in every  $C^s$ -norm at rate  $q^{(m+1)n}$ . The first error profile is an explicit even derivative of the up-function. Translating to the Fabius function gives the same accelerated approximation, while inversion on compact interior intervals yields an explicit first correction for the inverse Fabius function. These statements are proved here and checked symbolically and numerically by the accompanying Python program. They are described as *repository-new*: the claim is that they do not occur in the audited `*.tex` corpus under the stated repository path, not that a priority search over the entire literature has been completed.

**Status of results.** Every result labelled theorem, proposition, lemma, or corollary below has a proof in this report. The numerical tables are reproducible from `assets/fabius_frontier_results/verify_frontier_results.py`. The denominator-free finite  $q$ -binomial theorem is formalized in `FiniteQBinomialCore.lean`. The exact elementary base-reversal, terminating-factor, and adjacent-ratio layer is in `QPochhammerElementaryIdentities.lean`, with root-safe cross-multiplied statements separated from field quotients carrying their explicit nonzero denominators. Meanwhile, `GeometricLagrange.lean` proves mass one, cancellation of the first geometric modes, the first surviving moment, and module-valued Richardson exactness. The already compiled `GeometricQBinomialLagrange.lean` identifies the individual weights with their Gaussian-binomial quotients under injectivity of the finite geometric node set. At post-snapshot source checkpoint `1e761ed77583e9870ceaeab2c74ac824094f0f3f`, `GeometricCompleteHomogeneous.lean` and `GeometricResidualMoments.lean` additionally formalize the denominator-free principal specialization and the complete positive-degree Gaussian  $q$ -moment formula, including common scales and shifted blocks. At the reconciled source checkpoint `f77ff7c296b99050107948f5aa2a9488aec01d05`, the rational layer is implemented in

```
GeometricQBinomialLagrange.lean
GeometricRichardson.lean
GeometricLagrangeWeights.lean
GeometricLagrangeQBinomial.lean
GeometricLagrangeQMoments.lean
```

These modules prove the finite rational  $0 < q < 1$  weight formulas, the  $W_m(q^r)$  and  $q$ -Pochhammer moment presentations, all residual moments and their sign, and the exact finite  $\ell^1$  norm, including dedicated  $q = 1/4$  corollaries. The generic transformed-error series is now formalized in `Analytic`



`SeriesFilter.lean` under unconditional summability, with nonzero-weight hypotheses only at samples that actually contribute.

The infinite condition-number limit is formalized too, in `LimitConditionNumber.lean`, which lets  $n$  grow in the closed form  $\sum_{j \leq n} |w_q(n, j)| = (-q; q)_n / (q; q)_n$  of `GeneralQConditionNumber.lean`: both families are multipliable, the finite symbols converge to  $(a; q)_\infty$ , and the condition numbers converge to  $(-q; q)_\infty / (q; q)_\infty$  (`Fabius.tendsto_sum_abs_qToeplitzWeight`) on the whole range  $0 \leq q < 1$ . The crux is positivity of the denominator, and it is worth recording how it is obtained, because the obvious route fails: the one-shot Weierstrass product bound has deficit budget  $q/(1 - q)$ , which already reaches 1 at  $q = \frac{1}{2}$ , and its vacuity at  $q = \frac{3}{4}$  is recorded as a theorem rather than as prose (`Fabius.weierstrass_bound_vacuous_at_three_quarters`). Positivity instead comes from summability of the logarithms, which needs no threshold at all (`Fabius.qPochhammerInf_pos`). The limit is bounded below by 1, attained exactly at  $q = 0$ , and by  $1/(1 - q)$  (`Fabius.one_div_one_sub_le_qConditionNumberLimit`); it tends to  $+\infty$  as  $q \uparrow 1$  (`Fabius.tendsto_qConditionNumberLimit_atTop_at_one_left`), and at  $q = 999/1000$  is already at least 1000.

The named even-moment coefficients and their Gaussian filter for the actual infinite Rvachev sinc product are now formalized. Identifying instead the report’s finite-prefix quotient  $P_{b,n}/P_b$  with its Bell series, and proving the resulting analytic signs, remainder and bracketing bounds, uniform spline limits, asymptotics, and inverse-function consequences, remain unformalized.

## 1 Audit scope and the non-duplication boundary

### 1.1 How the corpus was used

The source audit covered every LaTeX family in the repository documentation tree [80], including the current expository volumes, the formalization walkthrough, the non-elementarity article, the mathematical glossary, the consolidated frontier volume, the Thue–Morse formula atlas, all waves of the Fourier-decay discussion, the paper-source subdirectories, and the historical archive. The archive index was used to track renamed and byte-identical historical variants, so that an old statement was not mistaken for a new one merely because it appeared under a superseded filename. A source map is given in [appendix B](#).

The audit revealed four especially important boundaries.

- B1.** The main exposition and its supplements already contain the normalized infinite sinc product, the box-convolution construction of  $\text{up}$ , its relation to the Fabius function, dyadic identities,  $q$ -Pochhammer and  $q$ -binomial formulae, and the signed Thue–Morse derivative  $\text{comb}$ .
- B2.** The asymptotic documents already contain a corrected Lambert– $W$  endpoint scale, a periodic dyadic phase, and all-order small-argument expansions for the Fabius function and its inverse. Those results concern  $x \downarrow 0$ , not finite-product truncation  $n \rightarrow \infty$ .
- B3.** The Fourier-decay waves study  $|P(\xi)|$  as  $|\xi| \rightarrow \infty$ , including dyadic oscillation and exceptional frequencies. They do not develop the exact analytic expansion of  $P_{b,n}/P_b$  in the small geometric variable  $q^n$ , nor the extrapolation filter derived below.
- B4.** The frontier volume already has  $q$ -Richardson ideas for discrete Fabius spline rows and degree-dependent approximants. We therefore do not present a generic Richardson recipe as new. The new object here is the *Fourier-tail truncation sequence*  $P_{b,n}$ , for which the error coefficients, all  $q$ -moments, stability constant, moment matching, and inverse-function consequence can all be written exactly.

The external search performed for this report also found general literature on Richardson extrapolation and on powers of sinc expressed through Bell polynomials, but no source matching the combined construction in [appendices 3 to 5](#). This is supporting evidence only, not an exhaustive priority claim.

## 1.2 What is new in this report

The principal contributions are the following.

- N1.** A one-variable analytic tail function  $\Phi_b$  whose Taylor cumulants are rational Bernoulli combinations and whose Taylor coefficients are complete Bell polynomials.
- N2.** A q-Lagrange operator  $W_m(T_q) = \prod_{r=1}^m (T_q - q^r)/(1 - q^r)$  that annihilates the first  $m$  truncation modes exactly.
- N3.** An all-order q-binomial error identity, not merely a leading-order Richardson estimate.
- N4.** A closed stability constant with a finite infinite-order limit, and a successive-level error certificate requiring no knowledge of the infinite product.
- N5.** Exact matching of the first  $2m$  moments of the limiting Rvachev density by a signed combination of finite box splines, for every  $n$ .
- N6.** A  $C^\infty$  superconvergence theorem with an explicit first derivative profile, followed by an interior inverse-Fabius perturbation formula.
- N7.** A kernel-independent extension showing that the same q-filter accelerates any geometric product whose elementary factor has an even logarithmic Taylor series.

## 2 Geometric sinc products and finite Rvachev splines

Fix  $b > 1$  and set

$$q = b^{-2} \in (0, 1), \quad \text{sinc}_{\text{rad}} z = \begin{cases} \sin z/z, & z \neq 0, \\ 1, & z = 0. \end{cases}$$

Define

$$P_b(\xi) = \prod_{\nu=1}^{\infty} \text{sinc}_{\text{rad}}(\xi/b^\nu), \quad P_{b,n}(\xi) = \prod_{\nu=1}^n \text{sinc}_{\text{rad}}(\xi/b^\nu). \quad (2.1)$$

The convergence is locally uniform in  $\mathbb{C}$ , since  $1 - \text{sinc}_{\text{rad}}(\xi/b^\nu) = O(b^{-2\nu})$  on compact sets.

Let

$$h_{b,\nu}(x) = \frac{b^\nu}{2} \mathbf{1}_{[-b^{-\nu}, b^{-\nu}]}(x), \quad u_{b,n} = h_{b,1} * \cdots * h_{b,n}. \quad (2.2)$$

With the Fourier convention

$$\widehat{f}_{\text{ang}}(\xi) = \int_{\mathbb{R}} f(x) e^{-i\xi x} dx, \quad f(x) = \frac{1}{2\pi} \int_{\mathbb{R}} \widehat{f}_{\text{ang}}(\xi) e^{i\xi x} d\xi,$$

we have  $\widehat{h_{b,\nu}}_{\text{ang}}(\xi) = \text{sinc}_{\text{rad}}(\xi/b^\nu)$ , hence

$$\widehat{u_{b,n}}_{\text{ang}} = P_{b,n}. \quad (2.3)$$

The infinite convolution  $u_b = *_{\nu \geq 1} h_{b,\nu}$  is a probability density supported on  $[-1/(b-1), 1/(b-1)]$ , and  $\widehat{u_b}_{\text{ang}} = P_b$ . The product decays faster than every power, so  $u_b \in C_c^\infty(\mathbb{R})$ . At  $b = 2$ ,  $u_2$  is the normalized Rvachev up-function.

### 2.1 The finite Thue–Morse derivative comb

The bridge to the signed Thue–Morse sequence is already part of the repository background, but the exact normalization is useful for what follows. Write  $t_k = (-1)^{\text{wt}_2(k)}$ , where  $\text{wt}_2(k)$  is the binary digit sum.

**Proposition 2.1** (distributional Thue–Morse comb). *For  $n \geq 1$ ,*

$$D^n u_{2,n} = 2^{n(n-1)/2} \sum_{k=0}^{2^n-1} t_k \delta_{-1+(2k+1)2^{-n}}. \quad (2.4)$$

*Proof.* For  $a > 0$ ,

$$D \left( \frac{1}{2a} \mathbf{1}_{[-a,a]} \right) = \frac{1}{2a} (\delta_{-a} - \delta_a).$$

Differentiate every factor in (2.2). Choosing the right endpoint in the  $j$ -th factor contributes one minus sign. If the choices are encoded by the binary digits of  $k$ , the coefficient is  $(-1)^{\text{wt}_2(k)}$ , while the location is

$$-\sum_{j=1}^n 2^{-j} + 2 \sum_{j=1}^n \varepsilon_j 2^{-j} = -1 + 2^{-n} + (2^{1-n})k.$$

Finally  $\prod_{j=1}^n (2 \cdot 2^{-j})^{-1} = 2^{n(n-1)/2}$ . □

Thus every finite product in (2.1) is simultaneously a Fourier approximation and an integrated Thue–Morse object. The acceleration below acts on both descriptions at once.

### 3 The exact Bernoulli–Bell tail expansion

#### 3.1 A universal tail function

The key observation is that the two variables  $n$  and  $\xi$  enter the omitted tail only through  $q^n \xi^2$ .

**Definition 3.1.** For  $z$  near zero, define

$$\Phi_b(z) = \prod_{\ell=1}^{\infty} \frac{1}{\text{sinc}_{\text{rad}}(b^{-\ell} \sqrt{z})}. \quad (3.1)$$

The expression is independent of the sign of  $\sqrt{z}$ , and hence defines an ordinary analytic function of  $z$ .

**Theorem 3.2** (exact tail factorization). *The function  $\Phi_b$  is analytic in  $|z| < \pi^2 b^2$ , meromorphic in  $\mathbb{C}$ , and has its nearest singularity to the origin at  $z = \pi^2 b^2$ . For every  $n \geq 0$ ,*

$$P_{b,n}(\xi) = P_b(\xi) \Phi_b(q^n \xi^2). \quad (3.2)$$

*The identity is entire in  $\xi$ ; at common zeros it is interpreted by analytic continuation. Moreover,*

$$\Phi_b(z) = \frac{\Phi_b(qz)}{\text{sinc}_{\text{rad}}(b^{-1} \sqrt{z})}. \quad (3.3)$$

*Proof.* Cancel the first  $n$  factors in  $P_b/P_{b,n}$  and write  $\nu = n + \ell$ :

$$\frac{P_{b,n}(\xi)}{P_b(\xi)} = \prod_{\ell=1}^{\infty} \frac{1}{\text{sinc}_{\text{rad}}(\xi/b^{n+\ell})} = \Phi_b(q^n \xi^2).$$

The zeros of  $\text{sinc}_{\text{rad}}(b^{-\ell} \sqrt{z})$  occur at  $z = \pi^2 k^2 b^{2\ell}$ ,  $k \in \mathbb{Z} \setminus \{0\}$ . The nearest is  $\pi^2 b^2$ . Splitting off the  $\ell = 1$  factor and reindexing proves (3.3). □

### 3.2 Bernoulli cumulants

The classical expansion

$$\log \operatorname{sinc}_{\text{rad}} w = - \sum_{k=1}^{\infty} \frac{\zeta(2k)}{k\pi^{2k}} w^{2k}, \quad |w| < \pi, \quad (3.4)$$

turns the geometric tail into an explicitly summable cumulant series.

The analytic input in (3.4) is already machine checked. On the complex disk, `SincZetaSeries.lean` proves the branch-free exponential forms `Fabius.complexSinc_pi_mul_eq_cexp` and `Fabius.complexSinc_eq_cexp`; its real theorem `Fabius.log_sin_div_pi_mul` recovers the displayed logarithm literally. At the dyadic base, `SincZetaDyadic.lean` resums all geometric scales in `Fabius.rvachevFourierProduct_eq_cexp` and `Fabius.rvachevFourierProduct_eq_prefix_mul_cexp`, while `SincZetaRemainder.lean` supplies the certified remaining zeta tail. What is not yet packaged for this part is the general- $b$  reciprocal-tail coefficient sequence  $\alpha_r(b)$  and its concrete connection to the geometric analytic filter below.

**Theorem 3.3** (Bernoulli cumulants). *For  $|z| < \pi^2 b^2$ ,*

$$\Phi_b(z) = \exp \left( \sum_{k=1}^{\infty} c_k(b) z^k \right), \quad (3.5)$$

where

$$c_k(b) = \frac{\zeta(2k)}{k\pi^{2k}(b^{2k} - 1)} \quad (3.6)$$

$$= \frac{(-1)^{k+1} 2^{2k-1} B_{2k}}{k(2k)!(b^{2k} - 1)}. \quad (3.7)$$

In particular  $c_k(b) > 0$  for every  $k \geq 1$ .

*Proof.* Insert (3.4) into (3.1), interchange absolutely convergent sums, and use

$$\sum_{\ell=1}^{\infty} b^{-2k\ell} = \frac{1}{b^{2k} - 1}.$$

The Bernoulli form follows from Euler's formula for  $\zeta(2k)$ . Positivity is clear from (3.6).  $\square$

Combining theorems 3.2 and 3.3 gives the exact truncation expansion

$$\frac{P_{b,n}(\xi)}{P_b(\xi)} = \exp \left( \sum_{k \geq 1} c_k(b) \xi^{2k} q^{kn} \right). \quad (3.8)$$

This is an equality, not a merely formal asymptotic series, whenever  $|\xi| < \pi b^{n+1}$ .

### 3.3 Bell coefficients and their recurrence

Write

$$\Phi_b(z) = \sum_{r=0}^{\infty} \alpha_r(b) z^r, \quad \alpha_0(b) = 1. \quad (3.9)$$

**Proposition 3.4** (complete Bell-polynomial formula). *With  $\mathcal{B}_r^{\text{exp}}$  the complete exponential Bell polynomial in the notation of [proposition 0.4.3](#),*

$$\alpha_r(b) = \frac{1}{r!} \mathcal{B}_r^{\text{exp}}(1!c_1(b), 2!c_2(b), \dots, r!c_r(b)), \quad (3.10)$$

and equivalently

$$r\alpha_r(b) = \sum_{k=1}^r k c_k(b) \alpha_{r-k}(b). \quad (3.11)$$

Every  $\alpha_r(b)$  is strictly positive.

*Proof.* Equation (3.10) is the defining exponential generating identity for complete Bell polynomials. Differentiating (3.5) and comparing coefficients gives (3.11). Positivity follows inductively from  $c_k(b) > 0$ .  $\square$

For the Rvachev base  $b = 2$ , the first cumulants are

$$\begin{aligned} c_1 &= \frac{1}{18}, & c_2 &= \frac{1}{2700}, & c_3 &= \frac{1}{178605}, \\ c_4 &= \frac{1}{9639000}, & c_5 &= \frac{1}{478533825}, & c_6 &= \frac{691}{15688261338750}. \end{aligned}$$

The corresponding Bell coefficients are

$$\alpha_0 = 1, \quad \alpha_1 = \frac{1}{18}, \quad \alpha_2 = \frac{31}{16200}, \quad (3.12)$$

$$\alpha_3 = \frac{2347}{42865200}, \quad \alpha_4 = \frac{1904369}{1311675120000}, \quad \alpha_5 = \frac{3309760579}{88561680752160000}. \quad (3.13)$$

The exact rationals are useful because no numerical fitting is needed at any order.

## 4 q-Lagrange Richardson extrapolation

### 4.1 The filter polynomial

Let  $T_q$  be the dilation operator  $(T_q f)(z) = f(qz)$ . Define

$$W_m(t) = \prod_{r=1}^m \frac{t - q^r}{1 - q^r}, \quad W_0(t) = 1. \quad (4.1)$$

Thus  $W_m(1) = 1$  and  $W_m(q^r) = 0$  for  $1 \leq r \leq m$ . Expanding  $W_m(t) = \sum_{j=0}^m w_j^{(m)} t^j$  gives the following exact weights.

**Theorem 4.1** (geometric Lagrange weights). *For  $0 \leq j \leq m$ ,*

$$w_j^{(m)} = \prod_{\substack{0 \leq \ell \leq m \\ \ell \neq j}} \frac{-q^\ell}{q^j - q^\ell} = \frac{(-1)^{m-j} q^{(m-j)(m-j+1)/2}}{(q; q)_j (q; q)_{m-j}}, \quad (4.2)$$

where  $(q; q)_r = \prod_{s=1}^r (1 - q^s)$ . These are precisely the Lagrange weights that recover the value at zero from data at  $1, q, \dots, q^m$ .

*Proof.* The first expression is the Lagrange cardinal basis evaluated at zero. Separating factors with  $\ell < j$  and  $\ell > j$ , then collecting powers of  $q$ , gives the second expression. Alternatively, the q-binomial theorem expands (4.1) directly.  $\square$

**Lean boundary for the individual weights.** The source-level closed quotient predates the new all-residual tranche. Over a field, under  $\text{InjOn}(j \mapsto q^j, \{0, \dots, m\})$ , the declaration

`geometricLagrangeWeight_eq_gaussianBinomial_div`

in `GeometricQBinomialLagrange.lean` gives the numerator with the deliberate reverse index `gaussianBinomialqm(m-j)`. The displayed symmetric- index notation in (4.2) is now a theorem rather than a bare paper convention: `gaussianBinomial_add_symm` and `gaussianBinomial_symm_via_completeHomogeneous` in `GeometricCompleteHomogeneous.lean` prove it over every commutative semiring without division or cancellation. The same module also contains the following cleared-denominator identity:

`finiteQPochhammerIn_mul_geometricLagrangeWeight_eq.`

In the present real range  $0 < q < 1$ , the required finite-node injectivity is automatic. These pointwise weight results should not be attributed to the later principal-specialization module.

**Definition 4.2** (accelerated finite product). For  $m, n \geq 0$ , define

$$\mathcal{R}_{m,n}P_b(\xi) = \sum_{j=0}^m w_j^{(m)} P_{b,n+j}(\xi). \quad (4.3)$$

With  $z = q^n \xi^2$ , the tail factorization gives the compact operator identity

$$\frac{\mathcal{R}_{m,n}P_b(\xi)}{P_b(\xi)} = (W_m(T_q)\Phi_b)(z). \quad (4.4)$$

The right side is well defined as an analytic continuation even at common zeros.

## 4.2 A Romberg-style triangular algorithm

The factorization of  $W_m$  gives a recursion that avoids an explicit weighted sum.

**Algorithm 4.3** (q-Richardson table). Start with  $A_{0,n} = P_{b,n}(\xi)$ . For  $m \geq 1$ , form

$$A_{m,n} = \frac{A_{m-1,n+1} - q^m A_{m-1,n}}{1 - q^m}. \quad (4.5)$$

Then  $A_{m,n} = \mathcal{R}_{m,n}P_b(\xi)$ .

*Proof.* The polynomial recursion

$$W_m(t) = \frac{t - q^m}{1 - q^m} W_{m-1}(t)$$

becomes (4.5) after applying it to the shift  $n \mapsto n + 1$ . □

For  $b = 2, q = 1/4$ , the first filters are

$$\mathcal{R}_{1,n}P = \frac{4P_{n+1} - P_n}{3}, \quad (4.6)$$

$$\mathcal{R}_{2,n}P = \frac{P_n - 20P_{n+1} + 64P_{n+2}}{45}, \quad (4.7)$$

$$\mathcal{R}_{3,n}P = \frac{-P_n + 84P_{n+1} - 1344P_{n+2} + 4096P_{n+3}}{2835}. \quad (4.8)$$

The coefficients are exact rationals; the included program generates arbitrary orders.

### 4.3 The exact q-moment identity

**Lemma 4.4** (all geometric moments). *For every integer  $r \geq 0$ , let  $M_r^{(m)} = \sum_{j=0}^m w_j^{(m)} q^{jr}$ . Then*

$$M_r^{(m)} = W_m(q^r). \quad (4.9)$$

Consequently,

$$M_0^{(m)} = 1, \quad M_r^{(m)} = 0 \quad (1 \leq r \leq m), \quad (4.10)$$

and for  $r \geq m + 1$ ,

$$M_r^{(m)} = q^{mr} \frac{(q^{1-r}; q)_m}{(q; q)_m} \quad (4.11)$$

$$= (-1)^m q^{\binom{m+1}{2}} \begin{bmatrix} r-1 \\ m \end{bmatrix}_q. \quad (4.12)$$

*Proof.* Equation (4.9) follows from the coefficient definition of  $W_m$ . The roots give (4.10). For  $r > m$ , evaluate the product in (4.1):

$$W_m(q^r) = \prod_{\ell=1}^m \frac{q^r - q^\ell}{1 - q^\ell} = (-1)^m q^{\binom{m+1}{2}} \prod_{\ell=1}^m \frac{1 - q^{r-\ell}}{1 - q^\ell},$$

which is (4.12); (4.11) is the equivalent q-Pochhammer form.  $\square$

**Post-snapshot Lean status of the moment formula.** The principal specialization

$$h_s(1, q, \dots, q^m) = \begin{bmatrix} s+m \\ m \end{bmatrix}_q$$

is completeHomogeneousEval\_geometric; it holds over every commutative semiring, with no division or restriction on  $q$ . Its Lean right-hand side is the recursive term `gaussianBinomial(s+m, m)`; in the present real range this is the quotient denoted by  $\begin{bmatrix} s+m \\ m \end{bmatrix}_q$ . More generally, over a commutative ring, `sum_weight_mul_geometric_pow_of_pos` proves that any row  $a_0, \dots, a_m$  satisfying

$$\sum_{j=0}^m a_j (q^j)^d = 0^d \quad (0 \leq d \leq m)$$

has, for every  $r > 0$ ,

$$\sum_{j=0}^m a_j (q^j)^r = (-1)^m q^{\binom{m+1}{2}} \begin{bmatrix} r-1 \\ m \end{bmatrix}_q.$$

Here and in the two formulas below, the literal Lean coefficient is the recursive `gaussianBinomial`; the quotient notation is valid in this report's real range  $0 < q < 1$ . The above-diagonal vanishing of the recursive Gaussian coefficient includes all cancellations in (4.10); degree zero remains the separate mass-one theorem. For the Lagrange row, finite-node injectivity supplies the low moments and `sum_geometricLagrangeWeight_mul_pow_of_pos` is exactly the displayed formula.

The same checkpoint also records the scale rather than cancelling it. If a row is exact directly on  $c, cq, \dots, cq^m$ , meaning

$$\sum_{j=0}^m a_j (cq^j)^d = 0^d \quad (0 \leq d \leq m),$$

then `sum_weight_mul_scaled_geometric_pow_succ_add` gives, for  $s \geq 0$ ,

$$\sum_{j=0}^m a_j (cq^j)^{m+1+s} = c^{m+1+s} (-1)^m q^{\binom{m+1}{2}} \begin{bmatrix} m+s \\ m \end{bmatrix}_q.$$



This hypothesis is imposed after scaling and remains meaningful when  $c$  is zero or a zero divisor. In the field-valued shifted block used here, `sum_geometricLagrangeWeight_mul_shifted_pow_of_pos` specializes it to

$$\sum_{j=0}^m w_j^{(m)} (q^{n+j})^r = (-1)^m q^{nr + \binom{m+1}{2}} \begin{bmatrix} r-1 \\ m \end{bmatrix}_q \quad (r > 0).$$

These declarations are finite algebraic statements. The separate module `AnalyticSeriesFilter.lean` now establishes the corresponding finite-sum/ infinite-series interchange and exact Gaussian tail under unconditional summability of the contributing samples; a zero filter weight creates no unweighted summability obligation. This generic module does not cover conditionally-only boundary convergence or establish signs, bounds, asymptotics, or uniform convergence. The finite declarations in this paragraph were checked at source checkpoint `1e761ed77583e9870ceeaeb2c74ac824094f0f3f`.

**Current infinite-product bridge.** The current tree names

$$A_r(\xi) = \text{Fabiús.rvachevFourierMomentTerm } \xi^r$$

and proves that these modes unconditionally sum to the standalone Rvachev product. For  $c, \xi \in \mathbb{C}$ ,  $m \in \mathbb{N}_0$ ,  $q = c^2$ , and only the finite node hypothesis that  $j \mapsto q^j$  is injective on  $\{0, \dots, m\}$ , it proves

$$\sum_{j=0}^m w_j^{(m)} \text{Fabiús.rvachevFourierProduct}(c^j \xi) = 1 + \sum_{s=0}^{\infty} (-1)^m q^{\binom{m+1}{2}} \begin{bmatrix} m+s \\ m \end{bmatrix}_q A_{m+1+s}(\xi). \quad (4.13)$$

No contraction, realness, positivity, or global nonvanishing assumption on  $c$  is present. At  $c = 1/2$ , hence  $q = 1/4$ , the node hypothesis is discharged internally and the result is assumption-free.

The five public declarations of `RvachevQBinomialFilter.lean` are `Fabiús.hasSum_rvachevFourierMomentTerm_product`, `Fabiús.rvachevFourierMomentTerm_pow_scale`, `Fabiús.geometricLagrange_rvachevFourierProduct_eq_gaussian_tsum`, `Fabiús.quarterLagrange_rvachevFourierProduct_eq_gaussian_tsum`, and `Fabiús.geometricLagrange_rvachevFourier_eq_gaussian_tsum`; the last is the same identity for every bounded  $F$  satisfying `Fabiús.IsFabiús`. This is an exact filter of the *infinite* product. It is not a Lean identification of this report's finite  $P_{b,n}$ , the quotient  $P_{b,n}/P_b$ , or its coefficients  $\alpha_r(b)$ , and it proves none of the analytic sign, bracketing, error-bound, uniform/derivative-convergence, or asymptotic conclusions below.

#### 4.4 The complete error series

**Theorem 4.5** (exact q-binomial error). *Let  $z = q^n \xi^2$ , and suppose  $|z| < \pi^2 b^2$ . If  $P_b(\xi) \neq 0$ , then*

$$\frac{\mathcal{R}_{m,n} P_b(\xi)}{P_b(\xi)} - 1 = (-1)^m q^{\binom{m+1}{2}} \sum_{r=m+1}^{\infty} \begin{bmatrix} r-1 \\ m \end{bmatrix}_q \alpha_r(b) z^r. \quad (4.14)$$

*Without the nonvanishing hypothesis, the total statement is the multiplied identity*

$$\mathcal{R}_{m,n} P_b(\xi) - P_b(\xi) = P_b(\xi) (-1)^m q^{\binom{m+1}{2}} \sum_{r=m+1}^{\infty} \begin{bmatrix} r-1 \\ m \end{bmatrix}_q \alpha_r(b) z^r. \quad (4.15)$$

*At a common zero of  $P_b$  and all the finite products, both sides of (4.15) vanish. In particular, for fixed  $m$  and fixed  $\xi \neq 0$  with  $P_b(\xi) \neq 0$ , as  $n \rightarrow \infty$ ,*

$$\frac{\mathcal{R}_{m,n} P_b(\xi)}{P_b(\xi)} - 1 \sim (-1)^m q^{\binom{m+1}{2}} \alpha_{m+1}(b) \xi^{2m+2} q^{(m+1)n}. \quad (4.16)$$

*Proof.* Apply the total tail identity (3.2) separately to each  $P_{b,n+j}(\xi)$ , multiply by  $w_j^{(m)}$ , and sum. This gives

$$\mathcal{R}_{m,n}P_b(\xi) = P_b(\xi) \sum_{j=0}^m w_j^{(m)} \Phi_b(q^j z)$$

without division, including at zeros. Insert  $\Phi_b(z) = \sum_{r \geq 0} \alpha_r z^r$ ; absolute convergence in the stated disc permits the finite sum to pass through the series. The coefficient of  $z^r$  is  $\alpha_r W_m(q^r)$ , so lemma 4.4 gives (4.15). Division by  $P_b(\xi)$  gives (4.14) exactly on its stated locus.

Finally, analyticity at zero makes the residual series its first nonzero term plus  $O(z^{m+2})$ . Since  $z = q^n \xi^2$ , while  $\alpha_{m+1}(b) > 0$  by proposition 3.4 and  $\xi \neq 0$ , that term is nonzero and yields (4.16).  $\square$

This identity is substantially stronger than the statement that Richardson extrapolation raises the order. Every residual mode and its exact  $q$ -binomial multiplier are known.

#### 4.5 Divided differences and non-asymptotic sign bounds

Let  $x_j = q^j z$ . The quantity  $\sum_j w_j^{(m)} \Phi_b(x_j)$  is the value at zero of the degree- $m$  Lagrange interpolant through the nodes  $x_0, \dots, x_m$ . The standard interpolation remainder therefore yields a second exact form.

**Theorem 4.6** (divided-difference form). *For  $0 \leq z < \pi^2 b^2$ ,*

$$(W_m(T_q)\Phi_b)(z) - 1 = (-1)^m q^{\binom{m+1}{2}} z^{m+1} \Phi_b[0, q^m z, q^{m-1} z, \dots, z], \quad (4.17)$$

where the bracket denotes a divided difference. Since  $\Phi_b$  is absolutely monotone on this interval,

$$q^{\binom{m+1}{2}} \alpha_{m+1}(b) z^{m+1} \leq |(W_m(T_q)\Phi_b)(z) - 1| \quad (4.18)$$

$$\leq \frac{q^{\binom{m+1}{2}} z^{m+1}}{(m+1)!} \Phi_b^{(m+1)}(z). \quad (4.19)$$

For  $0 < z < \pi^2 b^2$  the sign is exactly  $(-1)^m$ ; at  $z = 0$  both sides of (4.17) vanish.

*Proof.* The Lagrange remainder at zero is

$$p_m(0) - \Phi_b(0) = (-1)^m \left( \prod_{j=0}^m x_j \right) \Phi_b[0, x_0, \dots, x_m],$$

and  $\prod_j x_j = q^{\binom{m+1}{2}} z^{m+1}$ . Because every Taylor coefficient  $\alpha_r$  is positive, all derivatives of  $\Phi_b$  are positive and increasing on the stated interval. The Hermite–Genocchi formula bounds the divided difference between  $\Phi_b^{(m+1)}(0)/(m+1)! = \alpha_{m+1}$  and  $\Phi_b^{(m+1)}(z)/(m+1)!$ .  $\square$

**Corollary 4.7** (relative bracketing). *For real  $\xi$  with  $0 < |\xi| < \pi b^{n+1}$  and  $P_b(\xi) \neq 0$ ,*

$$\operatorname{sgn} \left( \frac{\mathcal{R}_{m,n}P_b(\xi)}{P_b(\xi)} - 1 \right) = (-1)^m. \quad (4.20)$$

Thus even and odd extrapolation orders bracket the exact product in relative terms. If  $P_b(\xi) > 0$ , the bracketing is also an ordinary numerical ordering.

## 4.6 Uniform stability of arbitrarily high order

Richardson formulae are often numerically unattractive because their coefficient norms grow rapidly. The geometric nodes here behave unusually well.

**Theorem 4.8** (exact Lebesgue constant). *The amplification factor of the filter is*

$$\Lambda_m(q) = \sum_{j=0}^m |w_j^{(m)}| = \prod_{r=1}^m \frac{1+q^r}{1-q^r} = \frac{(-q; q)_m}{(q; q)_m}. \quad (4.21)$$

For fixed  $q \in (0, 1)$ ,

$$\Lambda_\infty(q) = \prod_{r=1}^{\infty} \frac{1+q^r}{1-q^r} < \infty. \quad (4.22)$$

For  $q = 1/4$ ,

$$\Lambda_\infty(1/4) = 1.969260353668269431947193696967 \dots \quad (4.23)$$

*Proof.* The signs in (4.2) alternate, so

$$\sum_j |w_j^{(m)}| = (-1)^m W_m(-1) = \prod_{r=1}^m \frac{1+q^r}{1-q^r}.$$

The logarithm of the infinite product is  $\sum_r [\log(1+q^r) - \log(1-q^r)] = O(\sum_r q^r)$ , hence it converges.  $\square$

Consequently, absolute input errors bounded by  $\varepsilon$  produce an output error no larger than  $\Lambda_\infty(q)\varepsilon$ , uniformly in the extrapolation order. This is a decisive difference from equally spaced high-order extrapolation.

## 4.7 A rigorous a posteriori certificate

The positive-coefficient error series also compares consecutive truncation levels.

**Theorem 4.9** (successive-level certificate). *Fix  $m$  and a real  $\xi$ . Assume  $|\xi| < \pi b^{n+1}$  and  $P_b(\xi) \neq 0$ . Let  $R_n = \mathcal{R}_{m,n}P_b(\xi)$ . Then*

$$|R_{n+1} - R_n| \leq |R_n - P_b(\xi)| \leq \frac{|R_{n+1} - R_n|}{1 - q^{m+1}}. \quad (4.24)$$

Furthermore,

$$P_b(\xi) - R_n \sim \frac{R_{n+1} - R_n}{1 - q^{m+1}}. \quad (4.25)$$

*Proof.* By (4.14), the relative error is  $(-1)^m A_n$ , where

$$A_n = q^{\binom{m+1}{2}} \sum_{r \geq m+1} \begin{bmatrix} r-1 \\ m \end{bmatrix}_q \alpha_r \xi^{2r} q^{rn} > 0.$$

Every term is multiplied by  $q^r$  when  $n$  increases by one. Hence  $0 < A_{n+1} \leq q^{m+1} A_n$ . Therefore

$$(1 - q^{m+1})A_n \leq A_n - A_{n+1} \leq A_n.$$

Multiplying by  $|P_b(\xi)|$  proves (4.24). The leading term has ratio  $q^{m+1}$ , which proves (4.25).  $\square$

This certificate is practical: it estimates the unknown infinite product from two consecutive accelerated values and uses only the known number  $1 - q^{m+1}$ .

## 5 Physical-space consequences for Rvachev splines

Define the signed extrapolated spline

$$U_{b;m,n}(x) = \sum_{j=0}^m w_j^{(m)} u_{b,n+j}(x). \quad (5.1)$$

It has total mass one and support in  $[-1/(b-1), 1/(b-1)]$ . It need not be nonnegative, but

$$\|U_{b;m,n}\|_{L^1} \leq \Lambda_m(q) < \Lambda_\infty(q). \quad (5.2)$$

Its Fourier transform is  $\mathcal{R}_{m,n}P_b$ .

### 5.1 All-order inverse-differential expansion

**Theorem 5.1** (all-order spline expansion). *For every pair of nonnegative integers  $N, s$ ,*

$$u_{b,n} = \sum_{r=0}^N (-1)^r \alpha_r(b) q^{rn} u_b^{(2r)} + O_{C^s} \left( q^{(N+1)n} \right) \quad (n \rightarrow \infty). \quad (5.3)$$

*Equivalently, as an all-order asymptotic operator identity,*

$$u_{b,n} \sim \exp \left( \sum_{k \geq 1} (-1)^k c_k(b) q^{kn} D^{2k} \right) u_b. \quad (5.4)$$

*Proof.* Fourier transformation turns  $(-1)^r D^{2r} u_b$  into  $\xi^{2r} P_b(\xi)$ , so the displayed finite sum is exactly the first  $N+1$  terms of (3.8).

For completeness, split the inverse Fourier integral at  $|\xi| = b^n$ . On the low-frequency region,  $z = q^n \xi^2 \leq 1$ , and analyticity of  $\Phi_b$  gives a uniform Taylor remainder bounded by  $C q^{(N+1)n} |\xi|^{2N+2} |P_b(\xi)|$ . This remains integrable after multiplication by  $|\xi|^s$ , because  $P_b$  is rapidly decreasing.

On  $|\xi| \geq b^n$ , the first  $n$  factors satisfy

$$|P_{b,n+j}(\xi)| \leq b^{n(n+1)/2} |\xi|^{-n}, \quad j \geq 0,$$

while the remaining factors have modulus at most one. The corresponding weighted Fourier tail is  $O(b^{-n^2/2+O(n)})$ , hence smaller than every fixed geometric rate. The same bound applies to  $P_b$ . Fourier inversion proves the assertion in  $C^s$ .  $\square$

### 5.2 Superconvergence of the extrapolated spline

**Theorem 5.2** (q-binomial  $C^\infty$  expansion). *For fixed  $m, s$  and every  $N \geq m+1$ ,*

$$\begin{aligned} U_{b;m,n} - u_b &= q^{\binom{m+1}{2}} \sum_{r=m+1}^N (-1)^{m+r} \begin{bmatrix} r-1 \\ m \end{bmatrix}_q \alpha_r(b) q^{rn} u_b^{(2r)} \\ &\quad + O_{C^s} \left( q^{(N+1)n} \right). \end{aligned} \quad (5.5)$$

*In particular,*

$$\|U_{b;m,n} - u_b\|_{C^s} \leq C_{b,m,s} q^{(m+1)n}, \quad (5.6)$$

*and the normalized error has the explicit limit*

$$q^{-(m+1)n} (U_{b;m,n} - u_b) \longrightarrow -q^{\binom{m+1}{2}} \alpha_{m+1}(b) u_b^{(2m+2)} \quad \text{in } C^s(\mathbb{R}). \quad (5.7)$$

*Proof.* Apply inverse Fourier transformation to (4.14), using the equivalent angular-frequency identity  $\widehat{(-1)^r u_b^{(2r)} \text{ang}}(\xi) = \xi^{2r} P_b(\xi)$ . The remainder estimate is the same low/high-frequency argument as in theorem 5.1. At  $r = m + 1$ , the sign is  $(-1)^{m+m+1} = -1$ , which gives (5.7).  $\square$

For  $b = 2$ , each additional extrapolation level changes the basic convergence factor from  $4^{-n}$  to  $4^{-(m+1)n}$ , while the filter norm remains below 1.970.

### 5.3 Exact moment matching

The cancellation is stronger than fast convergence: a finite set of moments is already exact.

**Theorem 5.3** (exact moment matching). *For every  $m, n \geq 0$  and every integer  $0 \leq \ell \leq 2m$ ,*

$$\int_{\mathbb{R}} x^\ell U_{b;m,n}(x) dx = \int_{\mathbb{R}} x^\ell u_b(x) dx. \quad (5.8)$$

*All odd moments vanish by symmetry. The first missed even moment is known exactly:*

$$\int_{\mathbb{R}} x^{2m+2} (U_{b;m,n}(x) - u_b(x)) dx = -(2m+2)! q^{\binom{m+1}{2}} \alpha_{m+1}(b) q^{(m+1)n}. \quad (5.9)$$

*Proof.* By (4.14),  $\widehat{U_{b;m,n} \text{ang}} - \widehat{u_b \text{ang}}$  has a zero of order at least  $2m+2$  at  $\xi = 0$ . Since

$$\widehat{f \text{ang}}^{(\ell)}(0) = (-i)^\ell \int x^\ell f(x) dx,$$

all moments through degree  $2m+1$  agree. Symmetry handles every odd degree. The coefficient of  $\xi^{2m+2}$  in the Fourier error is  $(-1)^m q^{\binom{m+1}{2}} \alpha_{m+1} q^{(m+1)n}$ . Converting the derivative at zero back to the  $(2m+2)$ -nd moment contributes  $(-1)^{m+1}$ , leaving the negative sign in (5.9).  $\square$

**Remark 5.4** (quadrature interpretation). For any polynomial  $p$  of degree at most  $2m$ ,

$$\int p(x) U_{b;m,n}(x) dx = \int p(x) u_b(x) dx.$$

Thus  $U_{b;m,n}$  is a signed, compactly supported, uniformly stable density surrogate with exact polynomial expectations through degree  $2m$ .

## 6 Fabius and inverse-Fabius consequences

Under the normalization used throughout the repository, the Fabius function on  $[0, 1]$  is the left half of the translated up-function:

$$F(x) = u_2(x-1), \quad 0 \leq x \leq 1. \quad (6.1)$$

Define the accelerated finite-spline approximation

$$F_{m,n}(x) = U_{2;m,n}(x-1), \quad q = \frac{1}{4}. \quad (6.2)$$

**Corollary 6.1** (accelerated Fabius approximation). *For every  $m, s \geq 0$ ,*

$$\|F_{m,n} - F\|_{C^s([0,1])} = O\left(4^{-(m+1)n}\right), \quad (6.3)$$

and

$$4^{(m+1)n} (F_{m,n} - F) \longrightarrow -4^{-\binom{m+1}{2}} \alpha_{m+1}(2) F^{(2m+2)} \quad \text{in } C^s([0,1]). \quad (6.4)$$

The global statement is possible because all derivatives of the compactly supported up-function vanish at the endpoints. Inversion is more delicate:  $F'(x)$  tends to zero at the ends, and this is precisely where the repository's Lambert– $W$  asymptotics become relevant.

**Theorem 6.2** (interior inverse-Fabius superconvergence). *Let  $K = [a, b] \Subset (0, 1)$ , and put  $J = F(K)$ . Let  $G = F^{-1}$  on  $J$ . For sufficiently large  $n$ , the restriction of  $F_{m,n}$  to a neighborhood of  $K$  is strictly increasing and has an inverse branch  $G_{m,n}$  on  $J$ . Then*

$$\|G_{m,n} - G\|_{C^s(J)} = O\left(4^{-(m+1)n}\right) \quad (6.5)$$

for every fixed  $s$ , and uniformly for  $y \in J$ ,

$$4^{(m+1)n}(G_{m,n}(y) - G(y)) \longrightarrow 4^{-\binom{m+1}{2}} \alpha_{m+1}(2) \frac{F^{(2m+2)}(G(y))}{F'(G(y))}. \quad (6.6)$$

*Proof.* On a slightly larger compact interval  $K' \Subset (0, 1)$ , positivity of the up-function on the interior of its support gives  $F' \geq c > 0$ : by (6.1) and the refinement equation,  $F'(x) = 2u_2(2x - 1)$ , and  $u_2 > 0$  on  $(-1, 1)$ , so  $F' > 0$  on  $(0, 1)$ ; continuity and compactness of  $K'$  supply the constant. The  $C^1$  convergence in corollary 6.1 implies  $F'_{m,n} \geq c/2$  for large  $n$ , so the inverse branch exists. Standard differentiation of inverse functions gives (6.5).

For the leading term, write  $F_{m,n} = F + \varepsilon_n$ , set  $x = G(y)$ , and solve

$$0 = F(x + \delta_n) - F(x) + \varepsilon_n(x + \delta_n) = F'(x)\delta_n + \varepsilon_n(x) + o(4^{-(m+1)n}).$$

Equation (6.4) gives  $\varepsilon_n(x) \sim -4^{-\binom{m+1}{2}} \alpha_{m+1} 4^{-(m+1)n} F^{(2m+2)}(x)$ , which proves (6.6).  $\square$

## 6.1 Relation to Lambert– $W$ endpoint asymptotics

The inverse theorem is intentionally restricted to compact interior intervals. Near  $y = 0$ , the factor  $1/F'(G(y))$  becomes large and fixed-order inversion is no longer uniform. The repository's endpoint theory expresses the dominant scale of  $G(y)$  using a Lambert– $W$  variable together with a periodic dyadic phase. The two results therefore occupy complementary limits:

| regime                                         | small parameter | governing mechanism               |
|------------------------------------------------|-----------------|-----------------------------------|
| $y \downarrow 0$                               | $1/\log(1/y)$   | Lambert– $W$ and dyadic phase     |
| $n \rightarrow \infty, y \in J \Subset (0, 1)$ | $4^{-n}$        | Bernoulli–Bell tail and q-filter. |

A genuinely new two-parameter problem is to choose  $n = n(y)$  so that (6.6) remains uniform while  $y \downarrow 0$ . The exact factor  $F^{(2m+2)}(G(y))/F'(G(y))$ , combined with the repository's all-order endpoint expansion, reduces that problem to a concrete matching calculation rather than a vague stability question; see appendix 10.

## 7 Numerical verification

All computations in this section used 90-decimal-digit arithmetic. The infinite product was continued until a geometric bound for the omitted logarithmic tail fell below  $10^{-70}$ . The explicit weighted formula and the triangular table (4.5) were evaluated independently and agreed to the working precision. Symbolic checks verified (4.12) for a range of orders as an identity in the symbol  $q$ .

Table 8: Absolute errors at  $\xi = 1$ .

| $n$ | $m = 0$                 | $m = 1$                  | $m = 2$                  | $m = 3$                  |
|-----|-------------------------|--------------------------|--------------------------|--------------------------|
| 3   | $8.2128 \times 10^{-4}$ | $1.1050 \times 10^{-7}$  | $3.0877 \times 10^{-12}$ | $1.9989 \times 10^{-17}$ |
| 4   | $2.0524 \times 10^{-4}$ | $6.9036 \times 10^{-9}$  | $4.8225 \times 10^{-14}$ | $7.8050 \times 10^{-20}$ |
| 5   | $5.1304 \times 10^{-5}$ | $4.3143 \times 10^{-10}$ | $7.5345 \times 10^{-16}$ | $3.0485 \times 10^{-22}$ |
| 6   | $1.2826 \times 10^{-5}$ | $2.6964 \times 10^{-11}$ | $1.1772 \times 10^{-17}$ | $1.1908 \times 10^{-24}$ |

Table 9: Absolute errors at  $\xi = 3$ .

| $n$ | $m = 0$                 | $m = 1$                 | $m = 2$                  | $m = 3$                  |
|-----|-------------------------|-------------------------|--------------------------|--------------------------|
| 3   | $4.5982 \times 10^{-3}$ | $5.5691 \times 10^{-6}$ | $1.4003 \times 10^{-9}$  | $8.1582 \times 10^{-14}$ |
| 4   | $1.1454 \times 10^{-3}$ | $3.4676 \times 10^{-7}$ | $2.1800 \times 10^{-11}$ | $3.1753 \times 10^{-16}$ |
| 5   | $2.8608 \times 10^{-4}$ | $2.1652 \times 10^{-8}$ | $3.4031 \times 10^{-13}$ | $1.2392 \times 10^{-18}$ |
| 6   | $7.1504 \times 10^{-5}$ | $1.3529 \times 10^{-9}$ | $5.3162 \times 10^{-15}$ | $4.8397 \times 10^{-21}$ |

### 7.1 Absolute errors

For  $b = 2$ , the reference values are

$$\begin{aligned} P(1) &= 0.9456037838988178752649922474364995 \dots, \\ P(3) &= 0.5857191765350684829125292624510928 \dots, \\ P(10) &= -0.03192757801277845828274435480432725 \dots \end{aligned}$$

The observed reduction on increasing  $n$  is approximately  $4^{-(m+1)}$ , exactly as predicted.

### 7.2 Leading constants

At  $\xi = 3$ , divide the signed error by  $q^{(m+1)n}$ . The predicted limits from (4.16) are

| $m$ | predicted limit                       | computed at $n = 11$                  |
|-----|---------------------------------------|---------------------------------------|
| 0   | 0.29285958826753424146                | 0.29285960991271731281                |
| 1   | -0.022696618090733903713              | -0.022696619832602340475              |
| 2   | 0.000365296265765254499               | 0.000365296293045216575               |
| 3   | $-1.36214818263598388 \times 10^{-6}$ | $-1.36214828256045162 \times 10^{-6}$ |

The alternating signs and limiting constants are resolved without fitting any free parameter.

### 7.3 A posteriori estimates

For example, at  $\xi = 3, n = 4, m = 3$ , the true absolute error is  $3.17531383672 \times 10^{-16}$ . The computable difference between successive accelerated values is  $3.16292144780 \times 10^{-16}$ , and [theorem 4.9](#) gives the upper bound  $3.17532506132 \times 10^{-16}$ . The certificate therefore encloses the true error in an interval of relative width  $q^{m+1} = 4^{-4} = 1/256 \approx 3.9 \times 10^{-3}$ , whose upper endpoint exceeds the true error by only about  $3.5 \times 10^{-6}$  in relative terms, and it uses no reference value of  $P(3)$ .

## 8 A universal geometric-product theorem

The q-filter is not tied to the sine function. Its universality clarifies which parts of the construction are special to Rvachev’s function.



Table 10: Absolute errors at  $\xi = 10$ , beyond the first sign change of the infinite product but still inside the tail disk for the displayed  $n$ .

| $n$ | $m = 0$                 | $m = 1$                 | $m = 2$                  | $m = 3$                  |
|-----|-------------------------|-------------------------|--------------------------|--------------------------|
| 3   | $2.9276 \times 10^{-3}$ | $3.9468 \times 10^{-5}$ | $1.1012 \times 10^{-7}$  | $7.1217 \times 10^{-11}$ |
| 4   | $7.0230 \times 10^{-4}$ | $2.3635 \times 10^{-6}$ | $1.6504 \times 10^{-9}$  | $2.6705 \times 10^{-13}$ |
| 5   | $1.7380 \times 10^{-4}$ | $1.4617 \times 10^{-7}$ | $2.5525 \times 10^{-11}$ | $1.0327 \times 10^{-15}$ |
| 6   | $4.3341 \times 10^{-5}$ | $9.1120 \times 10^{-9}$ | $3.9782 \times 10^{-13}$ | $4.0240 \times 10^{-18}$ |

**Theorem 8.1** (universal analytic-factor acceleration). *Let  $\rho > 0$ , and let  $\varphi$  be even, analytic and zero-free on the disc  $|w| < \rho$ , with  $\varphi(0) = 1$  and*

$$\log \varphi(w) = - \sum_{k=1}^{\infty} a_k w^{2k}. \quad (8.1)$$

For  $b > 1$ , define

$$Q_b(\xi) = \prod_{\nu \geq 1} \varphi(\xi/b^\nu), \quad Q_{b,n}(\xi) = \prod_{\nu=1}^n \varphi(\xi/b^\nu).$$

Then for  $|\xi| < b\rho$ , so that  $z = q^n \xi^2$  lies in the disc  $|z| < b^2 \rho^2$  on which the tail series converges,

$$\frac{Q_{b,n}(\xi)}{Q_b(\xi)} = \exp \left( \sum_{k \geq 1} \frac{a_k}{b^{2k} - 1} \xi^{2k} q^{kn} \right). \quad (8.2)$$

Let its Taylor coefficients in  $z = q^n \xi^2$  be  $\beta_r$ . The same weights (4.2) satisfy

$$\frac{\sum_{j=0}^m w_j^{(m)} Q_{b,n+j}(\xi)}{Q_b(\xi)} - 1 = (-1)^m q^{\binom{m+1}{2}} \sum_{r \geq m+1} \begin{bmatrix} r-1 \\ m \end{bmatrix}_q \beta_r z^r. \quad (8.3)$$

If  $a_k \geq 0$  for every  $k$  and  $a_1 > 0$ , then  $\beta_r > 0$  for every  $r \geq 0$ , by the same recurrence as (3.11), and the sign law, divided-difference bounds, and a posteriori certificate remain valid. Nonnegativity alone is not enough:  $\varphi \equiv 1$  makes every  $\beta_r$  with  $r \geq 1$  vanish, and with it the error and its sign.

*Proof.* For  $|\xi| < b\rho$  every factor  $\varphi(\xi/b^\nu)$  is evaluated inside the disc of analyticity, and  $\limsup_k |a_k|^{1/k} \leq \rho^{-2}$  makes the tail radius at least  $b^2 \rho^2$ ; for  $\varphi = \text{sinc}_{\text{rad}}$  one has  $\rho = \pi$  and this is the disc of [theorem 3.2](#). The proof of [theorem 3.3](#) uses only the geometric sum  $\sum_{\ell \geq 1} b^{-2k\ell} = 1/(b^{2k} - 1)$ . The  $q$ -filter calculation then uses only the Taylor expansion and [lemma 4.4](#).  $\square$

Thus the sinc-specific information is concentrated in the Bernoulli values of  $a_k = \zeta(2k)/(k\pi^{2k})$ . The  $q$ -binomial extrapolation, its stability, and its exact mode cancellation belong to the geometry of the scales rather than to the elementary factor.

## 9 Lean formalization roadmap

The analytic proof naturally separates into formalization-sized modules.

1. **Finite  $q$ -algebra (formalized).** The finite  $q$ -Pochhammer and Gaussian-binomial core is in `FiniteQBinomialCore.lean`; the closed weights are in `GeometricQBinomialLagrange.lean`; and the principal specialization, all positive moments, scaled exact-row formula,

and shifted Lagrange specialization are in `GeometricCompleteHomogeneous.lean` and `GeometricResidualMoments.lean`. The report-facing rational geometric modules cited above, at checkpoint `f77ff7c296b99050107948f5aa2a9488aec01d05`, additionally identify the weight polynomial with  $W_m$ , prove the  $q$ -Pochhammer moment presentation, the sign of every residual moment, the exact finite norm, and dedicated quarter-base corollaries. In particular the cancellations  $1 \leq r \leq m$  and (4.12) have exact Lean counterparts. The companion `QBinomialVandermonde.lean` exhaustively exports `Fabiús.gaussianBinomial_add_vandermonde`, `Fabiús.gaussianBinomial_add_vandermonde'`, `Fabiús.gaussianBinomial_add_central`, `Fabiús.gaussianBinomial_add_central_min`, `Fabiús.gaussianBinomial_two_mul_add_shifted_central`, `Fabiús.gaussianBinomial_two_mul_sub_shifted_central`, and `Fabiús.gaussianBinomial_two_mul_sub_shifted_central_Icc`, together with the total signed forms `Fabiús.gaussianBinomial_two_mul_int_shifted_central` and `Fabiús.gaussianBinomial_two_mul_int_shifted_central_finsum`. All nine are denominator-free commutative-semiring identities. `FiniteQBinomialCore.lean` supplies the reusable integer-index extension `Fabiús.gaussianBinomialInt`, its cast and lower/upper zero laws `Fabiús.gaussianBinomialInt_ofNat`, `Fabiús.gaussianBinomialInt_eq_zero_of_neg`, and `Fabiús.gaussianBinomialInt_eq_zero_of_lt`, and the total reflection law `Fabiús.gaussianBinomialInt_symm`. Consequently the two final Vandermonde theorems state the monograph's single convolution for every  $k \in \mathbb{Z}$ , including  $|k| > N$ , both on the natural support and as a literal finite-support sum over  $\mathbb{Z}$ .

`QPochhammerElementaryIdentities.lean` has exactly thirteen public theorems. Its commutative-ring surface is `Fabiús.finiteQPochhammerIn_base_reversal_units`, `Fabiús.finiteQPochhammerIn_inv_base_reversal_units`, `Fabiús.prod_pow_sub_pow_eq_finiteQPochhammerIn`, `Fabiús.one_sub_mul_gaussianBinomial_one`, `Fabiús.gaussianBinomial_adjacent_mul`, and `Fabiús.gaussianBinomial_row_adjacent_mul`. The first two use unit parameters; the remaining four are root-safe polynomial identities, so the terminating numerator and adjacent cross identities remain valid at roots of unity. Both cross identities are total for all natural  $n, k$ , including on and above the row boundary by zero extension. Its field surface is `Fabiús.finiteQPochhammerIn_base_reversal`, `Fabiús.finiteQPochhammerIn_inv_base_reversal`, `Fabiús.pow_mul_finiteQPochhammerIn_inv_pow_eq`, `Fabiús.finiteQPochhammerIn_inv_pow_eq_self_div`, `Fabiús.finiteQPochhammerIn_inv_pow_eq_zero_of_lt`, `Fabiús.gaussianBinomial_adjacent_div`, and `Fabiús.gaussianBinomial_row_adjacent_div`. The numerical reversal theorems require exactly  $a \neq 0$  and  $q \neq 0$ . For  $k \leq N$ , the cleared terminating formula needs  $q \neq 0$ , while its displayed quotient also needs  $(q; q)_{N-k} \neq 0$ ; the  $k > N$  zero clause needs only  $q \neq 0$ . For  $k < n$ , the first adjacent quotient assumes  $\begin{bmatrix} n \\ k \end{bmatrix}_q \neq 0$  and  $1 - q^{k+1} \neq 0$ , and the second assumes  $\begin{bmatrix} n-1 \\ k \end{bmatrix}_q \neq 0$  and  $1 - q^{n-k} \neq 0$ . Neither adjacent quotient requires  $q \neq 0$ ; unlike the total cross identities, both quotients retain the strict triangular condition.

The finite Cauchy companion is also complete. `QBinomialCauchy.lean` has exactly one public definition and five public theorems, counting the compatibility spelling `Fabiús.finiteQPochhammerIn_mul_eq_sum_gaussianBinomial` of the primary identity. Its primary and reflected identities are

$$\begin{aligned} (uv; q)_n &= \sum_{k=0}^n \begin{bmatrix} n \\ k \end{bmatrix}_q (u; q)_k v^k (v; q)_{n-k}, \\ &= \sum_{k=0}^n \begin{bmatrix} n \\ k \end{bmatrix}_q (v; q)_k v^{n-k} (u; q)_{n-k}, \end{aligned}$$

namely `Fabiús.finite_qCauchy_identity` and `Fabiús.finite_qCauchy_identity_reflected`. The definition `Fabiús.qBernsteinBasis` is  $B_{n,k}^{(q)}(x) = \begin{bmatrix} n \\ k \end{bmatrix}_q x^k (x; q)_{n-k}$ , and

`Fabiús.sum_qBernsteinBasis` proves  $\sum_{k=0}^n B_{n,k}^{(q)}(x) = 1$ . Finally, `Fabiús.finite_qCauchy_second_identity` proves

$$(ac; q)_n (bc; q)_n = \sum_{k=0}^n \begin{bmatrix} n \\ k \end{bmatrix}_q (a; q)_k (b; q)_k c^k (c; q)_{n-k} (abcq^k; q)_{n-k}.$$

All parameters and degrees are arbitrary over every commutative ring. These are polynomial identities at  $q = 0$ , roots of unity, in positive characteristic, and with zero divisors; no division or cancellation is used.

The symmetric-function companion has an equally explicit boundary. `SymmetricFunctionOrthogonality.lean` has exactly one public definition, `Fabiús.elementarySymmetricEval`, and the six public theorems `Fabiús.elementarySymmetricEval_eq_eval_esymm`, `Fabiús.elementarySymmetricEval_zero`, `Fabiús.elementarySymmetricEval_comp_equiv`, `Fabiús.elementarySymmetricEval_option_succ`, `Fabiús.elementarySymmetricEval_fin_succ`, and `Fabiús.sum_elementarySymmetricEval_mul_completeHomogeneousEval`. Human-readably, the first three identify the evaluator with Mathlib’s universal elementary symmetric polynomial, prove  $e_0 = 1$ , and prove reindexing invariance. The next two give the adjoining-variable recurrence

$$e_{r+1}(x, a) = xe_r(a) + e_{r+1}(a)$$

on both an `Option` family and a literal head–tail `Fin` family. Together with the existing `Fabiús.completeHomogeneousEval_option_succ`, which states

$$h_{r+1}(x, a) = xh_r(x, a) + h_{r+1}(a),$$

this proves both weighted-Pascal laws over every commutative semiring. The last theorem proves, over every commutative ring,

$$\sum_{k=0}^n (-1)^k e_k(a) h_{n-k}(a) = \begin{cases} 1, & n = 0, \\ 0, & n > 0, \end{cases}$$

for every finite family, including the empty family and degree zero. It does not require division, a domain, a characteristic restriction, or a nonvanishing assumption.

The formal generating-series companion is now exhaustive as well. `SymmetricFunctionGenerating.lean` has exactly two public definitions, `Fabiús.elementarySymmetricGeneratingSeries` and `Fabiús.completeHomogeneousGeneratingSeries`, and six public theorems: `Fabiús.coeff_elementarySymmetricGeneratingSeries`, `Fabiús.coeff_completeHomogeneousGeneratingSeries`, `Fabiús.elementarySymmetricGeneratingSeries_eq_prod`, `Fabiús.elementarySymmetricGeneratingSeries_neg_mul_completeHomogeneousGeneratingSeries`, `Fabiús.completeHomogeneousGeneratingSeries_eq_invOfUnit_elementarySymmetricGeneratingSeries_neg`, and `Fabiús.prod_one_sub_qPow_X_mul_gaussianBinomialGeneratingSeries`. The definitions, coefficient formulas, and finite product  $E_a(X) = \prod_i (1 + a_i X)$  hold over every commutative semiring. Over every commutative ring the remaining declarations prove

$$E_{-a}(X)H_a(X) = 1, \quad H_a(X) = \text{invOfUnit}(E_{-a}(X), 1),$$

and, for the geometric alphabet  $1, q, \dots, q^n$ ,

$$\left( \prod_{j=0}^n (1 - q^j X) \right) \left( \sum_{k \geq 0} \begin{bmatrix} n+k \\ k \end{bmatrix}_q X^k \right) = 1 \quad \text{in } R[[X]].$$

These are formal power-series identities only. Analytic evaluation and convergence under hypotheses such as  $|a_i z| < 1$  or  $|z| < 1$  remain open.

The general weighted inversion is now exact. `SymmetricFunctionTransform.lean` has exactly four public definitions and five public theorems, exhaustively `Fabijs.completeHomogeneousKernel`, `Fabijs.signedElementaryKernel`, `Fabijs.completeHomogeneousKernel_left_orthogonality`, `Fabijs.completeHomogeneousKernel_right_orthogonality`, `Fabijs.completeHomogeneousTransform`, `Fabijs.signedElementaryTransform`, `Fabijs.signedElementaryTransform_completeHomogeneousTransform`, and `Fabijs.weightedSymmetricFunction_inversion`. For a finite alphabet, the complete-homogeneous kernel and transform already require only a commutative semiring; the signed kernel and transform, both total kernel orthogonalities, both whole-function inverse compositions, and the inversion iff use a commutative ring. Every transform target needs only an additive commutative monoid with its module structure. Both kernels are extended by zero above the diagonal. The proofs reuse the exactly one definition and one theorem of `FiniteTriangularTransform.lean`, `Fabijs.lowerTriangularTransform` and `Fabijs.lowerTriangularTransform_comp`; the latter itself needs only a semiring acting on an additive commutative monoid and a total ordered finite-convolution hypothesis. The finite second Cauchy identity still does not supply a  $q$ -Pfaff–Saalschütz summation.

The finite inversion layer is now formalized without denominators as well. `BitPositionQBinomial.lean` supplies the zero-based subset theorem `Fabijs.sum_pow_sum_powersetCard_eq_gaussianBinomial` and the literal one-based interval theorem `Fabijs.sum_pow_sum_powersetCard_Icc_eq_gaussianBinomial`, both total in the row and subset size. The four definitions and seven theorems of `QBinomialInversion.lean` are, exhaustively, `Fabijs.gaussianBinomialKernel`, `Fabijs.gaussianBinomialInverseKernel`, `Fabijs.scaledGaussianBinomialKernel`, `Fabijs.scaledGaussianBinomialInverseKernel`, `Fabijs.gaussianBinomial_mul`, `Fabijs.sum_gaussianBinomial_alternating_mul_pow`, `Fabijs.sum_gaussianBinomial_alternating`, `Fabijs.gaussianBinomialKernel_left_orthogonality`, `Fabijs.gaussianBinomialKernel_right_orthogonality`, `Fabijs.scaledGaussianBinomialKernel_left_orthogonality`, and `Fabijs.scaledGaussianBinomialKernel_right_orthogonality`. The chain law holds over a commutative semiring; the alternating and orthogonality theorems over a commutative ring. The independent scale is arbitrary and is never inverted.

The four definitions and four theorems of `QBinomialTransform.lean` are, exhaustively, `Fabijs.scaledGaussianBinomialTransform`, `Fabijs.scaledGaussianBinomialInverseTransform`, `Fabijs.gaussianBinomialTransform`, `Fabijs.gaussianBinomialInverseTransform`, `Fabijs.scaledGaussianBinomialInverseTransform_transform`, `Fabijs.scaledGaussianBinomialTransform_inverseTransform`, `Fabijs.scaledGaussianBinomial_inversion`, and `Fabijs.gaussianBinomial_inversion`. The two forward definitions need only  $[Semiring\ R]\ [AddCommMonoid\ M]\ [Module\ R\ M]$ , while the two signed inverse definitions need  $[Ring\ R]$  with the same target assumptions. All four theorems use  $[CommRing\ R]\ [AddCommMonoid\ M]\ [Module\ R\ M]$ ; the two compositions are equalities of whole sequence functions and the triangular relations form an exact iff. Their refactored proofs reuse `Fabijs.lowerTriangularTransform_comp`, and the Gaussian kernels are zero-extended above the row. No topology or convergence is involved.

The new `QDifferenceAnnihilation.lean` surface consists exhaustively of four public theorems: `Fabijs.sum_scaledGaussianBinomialInverseKernel_mul_pow`, `Fabijs.sum_gaussianBinomialInverseKernel_mul_geometric_pow`, `Fabijs.qDifference_sum_eval2_eq_map_coeff_mul`, and `Fabijs.qDifference_sum_eval2_eq_zero_of_degree_lt`.

For

$$K_{q,s}^{-1}(n, k) = (-s)^{n-k} q^{\binom{n-k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q,$$

the scaled characteristic identity is

$$\sum_{k=0}^n K_{q,s}^{-1}(n, k) z^k = \prod_{j=0}^{n-1} (z - sq^j).$$

At  $s = 1$  and  $z = q^d$ , its complete monomial moment is

$$\sum_{k=0}^n K_{q,1}^{-1}(n, k) (q^k)^d = \prod_{j=0}^{n-1} (q^d - q^j),$$

which vanishes for every  $d < n$ . More strongly, if  $A$  is a semiring,  $R$  is a commutative ring,  $\varphi: A \rightarrow R$  is a semiring homomorphism, and  $P \in A[X]$  has degree at most  $n$ , then the same row evaluated on the scalar extension of  $P$  is

$$\varphi([X^n]P) \prod_{j=0}^{n-1} (q^n - q^j).$$

It annihilates  $P$  whenever  $\deg P < n$ . These statements include  $n = 0$  and the zero polynomial; nodes may collide and the displayed surviving product may vanish. No division, distinctness, nonzero or invertible base, domain, characteristic, topology, or convergence assumption is used.

The global bridge `Fabiús.geometricQBinomialWeightNumerator_eq_scaledGaussianBinomialInverseKernel` now belongs to `GeometricQBinomialLagrange.lean`. It identifies the denominator-free geometric numerator with  $K_{q,q}^{-1}(n, k)$  for all natural indices, including above the diagonal, and is precisely the  $s = q$  specialization of the characteristic identity. It does not remove the separate field, finite-node injectivity, and in-range assumptions of normalized Lagrange quotients.

Finally, `QBinomialInversionSpecializations.lean` has the two definitions `Fabiús.qGaussianResidualCoeff` and `Fabiús.qGaussianReconstructionCoeff`, and exactly the four theorems `Fabiús.qGaussianResidualCoeff_eq`, `Fabiús.qGaussianReconstructionCoeff_eq`, `Fabiús.qGaussianReconstructionCoeff_residualCoeff_delta`, and `Fabiús.qGaussianResidualCoeff_reconstructionCoeff_delta`. At base  $q^2$  and independent scale  $-q$ , these are the residual and reconstruction rows

$$(-q)^{n-k} \begin{bmatrix} n \\ k \end{bmatrix}_{q^2}, \quad q^{(n-k)^2} \begin{bmatrix} n \\ k \end{bmatrix}_{q^2},$$

The definitions and these two pointwise equalities require only an arbitrary, possibly noncommutative ring. Exactly the two total finite convolution theorems require a commutative ring, and their values are the Kronecker delta.

**Current finite geometric q-layer API.** The following counts and names are exhaustive for the additional modules used by this part.

- `GeometricCompleteHomogeneous.lean` has six public theorems: `Fabiús.completeHomogeneousEval_geometric`, `Fabiús.completeHomogeneousEval_scaled_geometric`, `Fabiús.completeHomogeneousEvalOn_range_pow_eq_gaussianBinomial`, `Fabiús.completeHomogeneousEvalOn_range_pow_eq_gaussianBinomial_degree`, `Fabiús.gaussianBinomial_add_symm`, and `Fabiús.g`

aussianBinomial\_symm\_via\_completeHomogeneous. They prove both principal-specialization orientations, common scaling, and both generic Gaussian symmetry laws over every commutative semiring, without division, distinctness, or cancellation.

- `GeometricLagrangeCompleteHomogeneous.lean` has five public theorems: `Fabiús.completeHomogeneousEvalOn_geometric_range`, `Fabiús.sum_geometricLagrangeWeight_mul_pow_succ_add_eq_gaussianBinomial`, `Fabiús.geometricLagrangeQMoment_eq_residual_gaussianBinomial`, `Fabiús.completeHomogeneousEvalOn_geometric_range_eq_qBinomial`, and `Fabiús.geometricLagrangeQMoment_eq_residual_qBinomial_via_completeHomogeneous`. The first is semiring-level; the field statement uses finite-node injectivity; the rational quotient bridges retain their explicit nonzero-Pochhammer or  $0 < q < 1$  assumptions.
- `GeometricLagrangeQMoments.lean` has one public definition, `Fabiús.geometricLagrangeQMoment`, and exactly 37 public theorems: `Fabiús.geometricLagrangeQMoment_eq_weightPolynomial_eval`, `Fabiús.geometricLagrangeQMoment_eq_forwardRichardson_eval`, `Fabiús.geometricRootPolynomial_inv_eval_pow_mul_signedPowers`, `Fabiús.geometricRootPolynomial_inv_eval_pow_mul_triangular`, `Fabiús.geometricRootPolynomial_inv_eval_one_mul_triangular`, `Fabiús.geometricLagrangeQMoment_eq_qPochhammer`, `Fabiús.geometricLagrangeQMoment_zero`, `Fabiús.geometricLagrangeQMoment_eq_zero`, `Fabiús.geometricRootPolynomial_inv_eval_pow_eq_qPochhammer_of_le`, `Fabiús.geometricLagrangeQMoment_eq_residual_qPochhammer`, `Fabiús.qPochhammer_self_add`, `Fabiús.qPochhammer_self_pos_of_pos_of_lt_one`, `Fabiús.qBinomial_pos_of_pos_of_lt_one`, `Fabiús.gaussianBinomial_eq_qBinomial_of_pos_of_lt_one`, `Fabiús.qPochhammer_pow_pos_of_pos_of_lt_one`, `Fabiús.qPochhammer_tail_div_self_eq_qBinomial`, `Fabiús.geometricLagrangeQMoment_eq_residual_qBinomial`, `Fabiús.geometricLagrangeQMoment_firstUncancelled`, `Fabiús.negOnePow_mul_geometricLagrangeQMoment_eq_positiveResidual`, `Fabiús.negOnePow_mul_geometricLagrangeQMoment_pos`, `Fabiús.qPochhammer_self_succ`, `Fabiús.qBinomial_succ_succ_of_pos_of_lt_one'`, `Fabiús.qBinomial_succ_succ_of_pos_of_lt_one`, `Fabiús.qBinomial_theorem_of_pos_of_lt_one`, `Fabiús.sum_qBinomial_triangular_succ_eq_neg_qPochhammer`, `Fabiús.abs_geometricLagrangeWeight_eq_qBinomial`, `Fabiús.abs_geometricLagrangeWeight_eq_sign_mul`, `Fabiús.abs_geometricLagrangeWeight_complement_eq_qBinomial`, `Fabiús.sum_abs_geometricLagrangeWeight_eq_qPochhammer_ratio`, `Fabiús.neg_qPochhammer_div_self_eq_prod`, `Fabiús.sum_abs_geometricLagrangeWeight_eq_prod`, `Fabiús.quarterGeometricLagrangeQMoment_eq_qPochhammer`, `Fabiús.quarterGeometricLagrangeQMoment_eq_zero`, `Fabiús.quarterGeometricLagrangeQMoment_eq_residual_qPochhammer`, `Fabiús.quarterGeometricLagrangeQMoment_eq_residual_qBinomial`, `Fabiús.quarterGeometricLagrangeQMoment_firstUncancelled`, and `Fabiús.sum_abs_quarterGeometricLagrangeWeight_eq_qPochhammer_ratio`. These are finite rational identities. Quotient statements retain their nonzero denominators, and positivity, sign, and absolute-value formulas retain  $0 < q < 1$ ; the module makes no analytic convergence or remainder claim.
- `FinitePolynomialFilterExactness.lean` has five public theorems: `Fabiús.polynomialFilter_response_eq`, `Fabiús.polynomialFilter_exact`, `Fabiús.normalizedGeometricRootPolynomial_filter_exact`, `Fabiús.forwardGeometricRichardsonPolynomial_filter_exact`, and `Fabiús.forwardGeometricRichardsonPolynomial_filter_firstUncancelled`. The first two are arbitrary commutative-semiring response and exactness laws; the field-valued geometric

forms retain their nonzero-base and normalization- denominator hypotheses and expose the first surviving mode.

- `QuarterCatalanGerm.lean` has two public definitions, `Fabius.quarterCatalanCoefficient` and `Fabius.quarterCatalanGermSeries`, and thirteen public theorems: `Fabius.quarterCatalanCoefficient_zero`, `Fabius.quarterCatalanCoefficient_succ_eq_report`, `Fabius.quarterCatalanGermSeries_coeff`, `Fabius.quarterCatalanGermSeries_coeff_succ`, `Fabius.quarterCatalanGermSeries_constantCoeff`, `Fabius.quarterCatalanGermSeries_equation`, `Fabius.powerSeries_quadratic_injectiveOn_zeroConstant`, `Fabius.eq_quarterCatalanGermSeries_of_equation`, `Fabius.existsUnique_quarterCatalanGermSeries`, `Fabius.dyadicGermTwo_functionalEquation`, `Fabius.rescale_dyadicGermTwo_eq_quadraticInverse`, `Fabius.dyadicGermTwo_eq_rescale_quadraticInverse`, and `Fabius.coeff_dyadicGermTwo_succ`. This is the unique zero-constant solution of  $D + 4D^2 = (4/9)X$  in  $\mathbb{Q}[[X]]$ , with its complete coefficient formula and exact rescaling bridge to the distinguished dyadic germ and the quadratic inverse. The separate downstream module `FabiusInverseQuarterJet.lean` has exactly two public theorems, `Fabius.iteratedDeriv_centeredFabiusInv_quarter_eq_quadraticInverse` and `Fabius.iteratedDeriv_fabiusInv_five_seventy_two_succ`. They identify the full derivative jet of the actual centered Fabius inverse at  $5/72 = F(1/4)$  with the factorial-scaled coefficients of the inverse of  $X + 4X^2$ , and hence prove

$$G^{(m+1)}(5/72) = (m+1)!(-4)^m C_m.$$

This is an equality of smooth jets only: neither module proves convergence of the formal germ, equality with the inverse on a neighborhood, or analyticity at the quarter anchor.

- `QuarterCatalanRichardson.lean` has three public definitions, `Fabius.finiteRescaleFilter`, `Fabius.geometricRichardsonPowerSeriesFilter`, and `Fabius.quarterCatalanRichardsonFilter`, and 15 public theorems: `Fabius.finiteRescaleFilter_coeff`, `Fabius.geometricRichardsonPowerSeriesFilter_coeff`, `Fabius.geometricRichardsonPowerSeriesFilter_coeff_zero`, `Fabius.geometricRichardsonPowerSeriesFilter_coeff_eq_zero`, `Fabius.geometricRichardsonPowerSeriesFilter_coeff_eq_qPochhammer`, `Fabius.geometricRichardsonPowerSeriesFilter_coeff_eq_qBinomial`, `Fabius.geometricRichardsonPowerSeriesFilter_firstUncancelled_coeff_of_nonzero`, `Fabius.geometricRichardsonPowerSeriesFilter_firstUncancelled_coeff`, `Fabius.quarterCatalanRichardsonFilter_coeff`, `Fabius.quarterCatalanRichardsonFilter_coeff_zero`, `Fabius.quarterCatalanRichardsonFilter_coeff_eq_zero`, `Fabius.quarterCatalanRichardsonFilter_coeff_eq_zero_of_le`, `Fabius.quarterCatalanRichardsonFilter_coeff_eq_qBinomial`, `Fabius.quarterCatalanRichardsonFilter_coeff_succ_eq_qBinomial`, and `Fabius.quarterCatalanRichardsonFilter_firstUncancelled_coeff`. These are coefficientwise formal-power-series identities only: no convergence, real error sign, remainder estimate, or analytic acceleration is asserted.

The additional theorem `Fabius.sum_lagrangeEvalWeight_mul_pow_card_add_zero` in `LagrangeResidualMoments.lean` gives every higher evaluation-at-zero moment of a nonempty distinct field-valued node family as the negative signed nodal product times its complete homogeneous function. Its nonempty premise is exactly what removes the exceptional empty-row  $0^0$  term.

2. **Formal and unconditionally summable series (partially formalized).** The coefficientwise geometric filter and its Gaussian residual are in `GeometricPowerSeriesFilter.lean`;



`AnalyticSeriesFilter.lean` proves the matching exact tail for an abstract unconditionally summable coefficient series, and `RvachevQBinomialFilter.lean` connects the named even-moment series to the actual infinite Rvachev product. What remains here is to derive the report’s Bell recurrence (3.11) from  $\Phi_b = \exp(\sum c_k z^k)$  and connect those coefficients to the *finite-prefix quotient*  $P_{b,n}/P_b$ .

3. **Analytic sinc logarithm (formalized; finite-prefix quotient bridge open).** The complex exponential and real logarithmic forms of (3.4) are proved in `SincZetaSeries.lean`. The dyadic geometric-scale sum, exact finite-prefix factorization, and certified zeta remainder are proved in `SincZetaDyadic.lean` and `SincZetaRemainder.lean`. What remains here is the general- $b$  reciprocal-product coefficient package and its identification with the quotient  $P_{b,n}/P_b$  in the analytic filter interface; this is not supplied by the infinite-product theorem (4.13).
4. **Interpolation and positivity (partially formalized).** The finite Lagrange weights, exact residual sign, and abstract unconditionally summable Gaussian tail are proved, as is its exact entire Rvachev-product specialization. The divided-difference representation for the finite-prefix quotient, its absolute-monotonicity argument, analytic sign, quantitative bounds, and uniform control remain frontier work.
5. **Fourier/measure layer.** Define the uniform box measures and their finite convolutions. Moment matching follows from derivatives of the characteristic function at zero and may be formalized before the full  $C^\infty$  theorem.
6. **Asymptotic layer.** Prove low-frequency Taylor bounds and the high-frequency superexponential estimate. This is the technically heaviest step, but it is independent of  $q$ -binomial algebra.
7. **Inverse-function layer.** Apply a quantitative inverse function theorem on compact intervals where  $F'$  has a positive lower bound, then derive (6.6) from the first-order perturbation identity.

The accompanying Python file supplies exact test vectors for the algebraic modules and high-precision regression tests for the analytic statements.

## 10 Further frontier problems

### 10.1 A matched Lambert–Richardson endpoint theory

Let  $G = F^{-1}$  and let  $G_{m,n}$  be the accelerated inverse branch. Determine the largest endpoint region  $0 < y \leq y_n$  on which

$$G_{m,n}(y) - G(y) \sim 4^{-\binom{m+1}{2}} \alpha_{m+1}(2) 4^{-(m+1)n} \frac{F^{(2m+2)}(G(y))}{F'(G(y))}$$

remains uniform. Substitution of the repository’s Lambert- $W$  expansion for  $G(y)$  should identify a sharp transition curve  $y_n$ , together with a dyadically periodic modulation. This would connect two asymptotic theories that are presently separate.

### 10.2 Optimal order in finite precision

Although  $\Lambda_m(1/4) < 1.970$ , the finite products themselves may suffer relative loss near zeros. A complete finite-precision theory should choose  $m$  and  $n$  adaptively from: (i) the certificate (4.24); (ii) the distance to the shared zero set; and (iii) the working precision. Because the filter norm is uniformly bounded, one expects the optimal order to be governed mainly by product evaluation rather than extrapolation instability.

### 10.3 Positivity restoration

The signed density  $U_{b;m,n}$  has exact low moments but need not be positive. Is there a nonlinear, positivity-preserving correction that retains  $q^{(m+1)n}$  convergence and the moment constraints? One possibility is a small entropy projection onto the convex set of nonnegative densities with the same moments. The exact first missed moment (5.9) provides a natural scale for such a correction.

### 10.4 Dyadic rational evaluation from accelerated splines

At a dyadic point, each finite spline  $u_{2,n+j}$  is exactly computable from integrated Thue–Morse data and q-binomial formulae already present in the repository. Substituting those values into (4.3) should yield explicit rational or q-rational approximants to  $F(x)$  with a certified  $4^{-(m+1)n}$  remainder. This would produce a new bridge between exact dyadic arithmetic and the analytic sinc tail.

### 10.5 Complex domains and pole-aware extrapolation

The nearest tail pole at  $z = \pi^2 b^2$  controls the Taylor expansion. For complex  $\xi$ , one can conformally map the slit  $z$ -plane before applying a geometric interpolation filter. A pole-aware variant may retain exact q-mode cancellation while extending useful acceleration beyond the first omitted sinc zero.

### 10.6 Other refinable and compactly supported functions

Apply theorem 8.1 to elementary factors arising from compactly supported refinable distributions, Bernoulli convolutions, or symmetric B-splines. The main classification question is: which elementary factors have nonnegative logarithmic cumulants, and therefore inherit the exact sign law and successive-level certificate?

## 11 Conclusion

The truncation of the Rvachev sinc product carries more structure than a generic convergent product. Its omitted tail is an analytic function of the single geometric variable  $z = 4^{-n}\xi^2$ . Bernoulli numbers describe the logarithm of that function, Bell polynomials describe the function itself, and Lagrange interpolation on the nodes  $1, q, \dots, q^m$  produces a q-binomial Richardson filter with a complete, sign-definite error series.

The most notable consequences are exact rather than heuristic: the q-moments of the weights are known in closed form; the filter norm stays bounded at arbitrarily high order; consecutive accelerated levels certify the error; and the associated signed box splines match the first  $2m$  moments of the limiting up-density for every  $n$ . Fourier inversion converts the same algebra into  $C^\infty$ -superconvergence with an explicit derivative profile. Translation and local inversion then give a new interior approximation theory for the Fabius function and its inverse, complementary to the repository's Lambert– $W$  endpoint asymptotics.

These results create concrete next steps: connect the compiled infinite-product Gaussian filter to the report's finite-prefix quotient and its Bell coefficients, prove the ordered and uniform analytic error estimates, build a dyadic exact-evaluation implementation, and develop a matched two-parameter theory linking truncation acceleration to the periodic Lambert phase near the endpoint.

## A Exact coefficients and filters at $b = 2$

The next two Bell coefficients, useful for regression tests, are

$$\alpha_6 = \frac{1884491755011469}{1980124051433319792000000},$$

$$\alpha_7 = \frac{151616054172597829}{6278781742186853835936000000}.$$

The order-four filter is

$$\mathcal{R}_{4,n}P = \frac{1}{722925}P_n - \frac{4}{8505}P_{n+1} + \frac{64}{2025}P_{n+2} - \frac{4096}{8505}P_{n+3} + \frac{1048576}{722925}P_{n+4}.$$

The order-five weights are

$$\left( -\frac{1}{739552275}, \frac{4}{2168775}, -\frac{64}{127575}, \frac{4096}{127575}, -\frac{1048576}{2168775}, \frac{1073741824}{739552275} \right).$$

Despite the large integer numerators, the sum of absolute values is below 1.968 at order five.

## B Corpus audit map

This appendix records the source families used to establish the non-duplication boundary. Historical variants were read for content rather than treated as independent discoveries. The family headings below describe the audit snapshot, not current navigation. The non-elementarity material is now consolidated in Chapter 1 of the canonical inverse-Fabius synthesis at `drafts/inverse-and-sampling/Inverse_Fabius_Analyticity_Asymptotics_and_Computability/`; the consolidated Thue–Morse and Fourier-decay sources are respectively `drafts/thue-morse/Thue_Morse_Atlas_and_Frontiers/Thue_Morse_Atlas_and_Frontiers.tex` and `drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay/Rvachev_Up_Fourier_Decay.tex`. The exact Fourier-decay predecessors remain available at immutable commit [2e3567feb149](#).

### Top-level expositions at the audit snapshot

- `Fabius_Function_and_Rvachev_Up/Fabius_Function_and_Rvachev_Up.tex`
- `FabiusFunction_Mathematical_Glossary/FabiusFunction_Mathematical_Glossary.tex`
- `Non_Elementarity_of_the_Fabius_Function/Non_Elementarity_of_the_Fabius_Function.tex`
- `fabius_lean_walkthrough/fabius_lean_walkthrough.tex`
- `semi-formalized-research-frontiers/semi-formalized-research-frontiers.tex`

### Frontier drafts at the audit snapshot

- `drafts/Thue_Morse_Formula_Atlas/Thue_Morse_Formula_Atlas.tex`
- LaTeX source in `Rvachev_Up_Fourier_Decay-2`
- LaTeX source in `Rvachev_Up_Fourier_Decay-3`
- LaTeX source in `Rvachev_Up_Fourier_Decay_Comparative_Audit`
- LaTeX source in `Rvachev_Up_Fourier_Decay_Gentle_Guide`
- LaTeX source in `Rvachev_Up_Fourier_Decay_Second_Wave_Audit`
- LaTeX source in `asymptotic-decay-rate-of-an-infinite-product-of-sinc-functions`
- LaTeX source in `rvachev_up_fourier_decay-1`
- LaTeX sources in `rvachev_up_fourier_decay-4` through `rvachev_up_fourier_decay-8`

## Paper-source directories

- `AIF_2017__67_2_637_0.tex` and its corrected source
- `Lacunary_products.tex` (arXiv:1502.06738v2)
- `09-Function.tex` (arXiv:1702.05442v1)
- `157-Arithmetic-v3.tex` (arXiv:1702.06487v3)
- `document.tex` (arXiv:2211.13570v1)
- `ubiq15-corrected.tex`

## Historical archive families

- `Fabius Asymptotic.tex`
- `K-fold summation over signed Thue-Morse.tex`
- `Non_Elementarity_of_the_Fabius_Function.tex`
- `Small_Argument_Asymptotics.tex`
- `fabius-inverse-dyadic-closed-form.tex`
- Historical primary exposition variants under `Fabius_Function_and_Rvachev_Up`, `Fabius_and_Rvachev_Up-2`, `Fabius_and_Rvachev_up`, and `Fabius_function_and_up_function`
- `Fabius_Function_Missing_Parts` variants
- `Fabius_q_Special_Functions` and `fabius-q-special-functions`
- `Fabius_Function_Lean_Walkthrough` and `fabius_lean_walkthrough` variants
- `FabiusFunction_Mathematical_Glossary` variants
- `Fabius_Dyadic_Asymptotic_Bridge`
- `Fabius_Dyadic_Formulae_and_Alternative_Representations`
- `Fabius_Dyadic_Formulae_to_Asymptotics`
- `Fabius_Dyadic_q_Connections`
- `Fabius_Thue_Morse_Convergence_Rate`
- `Fabius_Thue_Morse_Convergence_Rates`
- `Primary_Exposition_Gap_Register`
- `Repeated_Integration_and_Rvachev_Up`
- `Rvachev_Up_from_Repeated_Integration`
- Earlier consolidated semi-formalized-research-frontiers source
- `Thue_Morse_Formula_Atlas-1`, `Thue_Morse_Formula_Atlas-2`, and `Thue_Morse_Formula_Atlas-3`

The current consolidated frontier source explicitly records the absorption of several former notebooks and their checksums. That provenance was used when checking whether a proposed statement was genuinely absent or merely moved.

## C Reproducibility notes

The verification material for this part is kept in the volume directory under `assets/fabius_frontier_results/`:

- `assets/fabius_frontier_results/verify_frontier_results.py` — exact symbolic and high-precision numerical checks;

- `assets/fabius_frontier_results/numerical_results.txt` and `assets/fabius_frontier_results/numerical_results.csv` — machine-readable output;
- `assets/fabius_frontier_results/README.txt` — the absorbed package’s own run notes.

The absorbed report’s source and PDF are not part of the package: they were deleted when it was merged into this volume, and git history is their archive. The volume itself is rebuilt from its own directory by running `pdflatex on geometric_q_fabius_frontiers.tex` three times, so that the table of contents and every cross-reference settle.

The script checks the q-moment identity symbolically for several arbitrary orders, generates the exact Bernoulli and Bell coefficients, verifies agreement between the explicit and triangular filters, reproduces [tables 8 to 10](#), checks the leading constants, and asserts the two-sided certificate (4.24) at representative points.

## Part III

# Finite Dyadic Sinc Products and Piecewise-Polynomial Approximants to Rvachev’s Up-Function

### Abstract

Let

$$\Phi(\xi) = \widehat{\text{up}}_{2\pi}(\xi) = \prod_{j=0}^{\infty} \text{sinc}_{\text{rad}}\left(\frac{\pi\xi}{2^j}\right), \quad \text{sinc}_{\text{rad}} z = \frac{\sin z}{z},$$

under the Fourier convention  $\widehat{f}_{2\pi}(\xi) = \int_{\mathbb{R}} f(x) e^{-2\pi i x \xi} dx$ . This report studies the inverse transforms  $p_n$  of the first  $n$  factors. They are nonstationary dyadic box splines: compactly supported, nonnegative, even, log-concave, piecewise polynomial of degree at most  $n - 1$ , and  $C^{n-2}$ . Their knots unexpectedly form a uniform dyadic grid, while the jumps of the top derivative carry the signed Thue–Morse sequence. An explicit truncated-power formula, exact cell formulas, recursions, Fourier-zero multiplicities, partition-of-unity identities, cumulants, and sharp derivative geometry are obtained.

The main convergence result is an exact law, not merely an upper bound. For every  $r \geq 0$  and  $n \geq r + 3$ ,

$$\left\| p_n^{(r)} - \text{up}^{(r)} \right\|_{\infty} = \frac{2^{\binom{r+3}{2}-1}}{9 \cdot 4^n}.$$

The extremum occurs at an explicit dyadic point. The associated distribution functions satisfy the exact Kolmogorov law  $d_K = 1/(9 \cdot 4^n)$ . At all orders,  $p_n$  admits a uniform differential expansion whose coefficients are complete Bell polynomials in Bernoulli-number data from  $1/\Phi$ ; geometric Richardson combinations then give signed  $4^{-Mn}$  extrapolation with uniformly bounded coefficient variation.

A second, positive acceleration theory is developed from a new uniform-scale-mixture representation of  $\text{up}$ . The exact tail is a mixture of uniform boxes governed by a probability measure  $\nu$  on  $[0, 1]$ . Positive Gauss, Radau, and Lobatto quadrature for  $\nu$  turns the continuum tail into finite positive mixtures of boxes. A one-signed Peano-kernel argument and the dyadic derivative plateaux give exact  $C^r$  errors for the whole hierarchy. In particular, a single variance-matched extra box yields an exact  $16^{-n}$  law, resolving a positive variance-matching direction left open in the repository’s frontier volume; right-Radau rules preserve the exact support  $[-1, 1]$  and attain arbitrarily high algebraic orders. Reproducible FFT and exact-rational experiments verify the formulas to numerical precision.

# 1 Scope, source audit, and novelty boundary

## 1.1 Documents examined

The source audit covered the living TeX documents under `Analysis/FabiusFunction/docs`, especially the primary synthesis *The Fabius Function and Rvachev’s Up-Function*, the mathematical glossary, the Lean walkthrough, and the consolidated volume *Non-formalized Research Frontiers*. The recursive archive index was also examined to identify the historical TeX families: early notes, standalone studies, primary-exposition drafts, supplements,  $q$ -calculus articles, Lean walkthroughs, glossaries, and per-topic frontier dossiers. The archive explicitly records frozen first-introduced versions and prunes byte-identical copies already present in the live tree, so the living consolidated sources are the appropriate mathematical baseline for avoiding duplication [91, 79, 70].

The existing corpus already develops the probability law of up, the infinite sinc product, exact dyadic arithmetic, Thue–Morse identities, derivative tilings, moments,  $q$ -binomial structure, endpoint Lambert– $W$  asymptotics, and a repeated-integration spline  $f_n$ . In particular, the frontier volume proves a sharp error law for  $f_n$  and explicitly contrasts it with the ordinary prefix density  $p_n$ , whose sharp law is not studied there. It then asks for a positive compactly supported tail replacement with matching variance, predicting a possible  $16^{-n}$  rate but asserting no such scheme [79].

## 1.2 What is new here

The principal additions of this report are:

1. the exact uniform-grid truncated-power representation of the ordinary inverse transform  $p_n$ , with Thue–Morse jump weights;
2. a sharp derivative-plateau theorem for  $p_n$ ;
3. the exact  $C^r$  prefix error law and exact Kolmogorov distance;
4. a uniform all-orders expansion of  $p_n$  in even derivatives of up, with Bell–Bernoulli coefficient formulas;
5. stable  $q = 1/4$  Richardson weights and their closed  $q$ -binomial form;
6. a uniform-scale-mixture representation of the exact tail;
7. a general positive quadrature theorem giving exact error constants for Gauss, Radau, and Lobatto tail surrogates;
8. explicit positive fourth- and sixth-order schemes, including exact-support accelerants.

Every theorem below is accompanied by a mathematical proof. These results have not been added to the repository’s Lean development and should therefore be read as a research contribution awaiting independent review and formalization. Numerical experiments are used as verification, never as a substitute for proof.

# 2 Fourier normalization and the probabilistic head–tail model

## 2.1 The infinite product

We use

$$\widehat{f}_{2\pi}(\xi) = \int_{\mathbb{R}} f(x) e^{-2\pi i x \xi} dx, \quad f(x) = \int_{\mathbb{R}} \widehat{f}_{2\pi}(\xi) e^{2\pi i x \xi} d\xi. \quad (2.1)$$

For  $a > 0$ , define the centered uniform density

$$u_a(x) = \frac{1}{2a} \mathbf{1}_{[-a, a]}(x). \quad (2.2)$$

Then

$$\widehat{u}_{a\,2\pi}(\xi) = \text{sinc}_{\text{rad}}(2\pi a\xi). \quad (2.3)$$

Rvachev's normalized up-function is the even  $C^\infty$  probability density on  $[-1, 1]$  whose transform is

$$\Phi(\xi) = \widehat{\text{up}}_{2\pi}(\xi) = \prod_{j=0}^{\infty} \text{sinc}_{\text{rad}}\left(\frac{\pi\xi}{2^j}\right). \quad (2.4)$$

This normalization agrees with the compactly supported function studied by Rvachev and Arias de Reyna [61, 33].

## 2.2 Partial products and random sums

For  $n \geq 1$ , set

$$\Phi_n(\xi) = \prod_{j=0}^{n-1} \text{sinc}_{\text{rad}}\left(\frac{\pi\xi}{2^j}\right), \quad \widehat{p}_{n\,2\pi} = \Phi_n. \quad (2.5)$$

By equation (2.3),

$$p_n = u_{2^{-1}} * u_{2^{-2}} * \cdots * u_{2^{-n}}. \quad (2.6)$$

Let  $U_0, U_1, \dots$  be independent  $\text{Unif}[-1, 1]$  variables and put

$$S_n = \sum_{j=0}^{n-1} 2^{-j-1} U_j, \quad X = \sum_{j=0}^{\infty} 2^{-j-1} U_j. \quad (2.7)$$

Then  $p_n$  is the density of  $S_n$  and up is the density of  $X$ . Introduce

$$\delta_n = 2^{-n}, \quad A_n = 1 - 2^{-n}. \quad (2.8)$$

The omitted tail is a scaled independent copy of the whole sum:

$$X \stackrel{d}{=} S_n + \delta_n X', \quad X' \stackrel{d}{=} X, \quad X' \text{ independent of } S_n. \quad (2.9)$$

Consequently,

$$\Phi(\xi) = \Phi_n(\xi) \Phi(\delta_n \xi), \quad \text{up} = p_n * \rho_{\delta_n}, \quad \rho_{\delta}(x) = \delta^{-1} \text{up}(x/\delta). \quad (2.10)$$

The identity equation (2.10) is the engine behind both the exact error law and all-order acceleration.

## 3 Exact inverse transforms: dyadic box splines with Thue–Morse knots

### 3.1 Uniform knot formula

Write

$$\tau_k = (-1)^{\text{wt}_2(k)}, \quad (3.1)$$

where  $\text{wt}_2(k)$  is the binary digit sum. For  $n = 1$ , where the exponent in (3.2) is 0, the convention  $y_+^0 = \mathbf{1}_{(0,\infty)}(y)$  is used away from knots.

**Theorem 3.1** (Exact truncated-power formula). *For every  $n \geq 1$ ,*

$$p_n(x) = \frac{2^{n(n-1)/2}}{(n-1)!} \sum_{k=0}^{2^n-1} \tau_k \left( x + A_n - \frac{k}{2^{n-1}} \right)_+^{n-1}. \quad (3.2)$$



The knots are the uniformly spaced points

$$t_{n,k} = -A_n + \frac{k}{2^{n-1}}, \quad 0 \leq k < 2^n. \quad (3.3)$$

In distributions,

$$D^n p_n = 2^{n(n-1)/2} \sum_{k=0}^{2^n-1} \tau_k \delta_{t_{n,k}}. \quad (3.4)$$

*Proof.* For one box,

$$Du_a = \frac{1}{2a}(\delta_{-a} - \delta_a).$$

Convolving the  $n$  derivatives in [equation \(2.6\)](#) gives a signed atomic measure. Choose a binary vector  $b = (b_0, \dots, b_{n-1}) \in \{0, 1\}^n$ ; its atom lies at

$$-A_n + \sum_{j=0}^{n-1} b_j 2^{-j} = -A_n + \frac{k}{2^{n-1}}, \quad k = \sum_{j=0}^{n-1} b_j 2^{n-1-j},$$

and its sign is  $(-1)^{\sum b_j} = \tau_k$ . The normalization is

$$\prod_{j=0}^{n-1} \frac{1}{2 \cdot 2^{-j-1}} = \prod_{j=0}^{n-1} 2^j = 2^{n(n-1)/2}.$$

This proves [equation \(3.4\)](#). Integrating  $n$  times from  $-\infty$ , using compact support, gives [equation \(3.2\)](#).  $\square$

The equal spacing is special to the dyadic weights. The  $2^n$  subset sums of  $1, 1/2, \dots, 2^{1-n}$  fill a complete integer grid after multiplication by  $2^{n-1}$ , while their parity signs are exactly Thue–Morse. Thus a nonstationary convolution of boxes of different widths becomes a uniform-knot spline with highly structured signed top derivative.

### 3.2 Cell formula and exact arithmetic

Set

$$y = 2^{n-1}(x + A_n). \quad (3.5)$$

Inside the support, [equation \(3.2\)](#) becomes

$$p_n(x) = \frac{2^{-(n-1)(n-2)/2}}{(n-1)!} \sum_{0 \leq k \leq \lfloor y \rfloor} \tau_k (y - k)^{n-1}. \quad (3.6)$$

This has several immediate consequences.

**Corollary 3.2** (Piecewise-polynomial and arithmetic structure). *The function  $p_n$  is supported on  $[-A_n, A_n]$ , is polynomial of degree at most  $n-1$  on each of the  $2^n-1$  open knot cells, and belongs to  $C^{n-2}(\mathbb{R})$  for  $n \geq 2$ . At rational arguments it has rational values; at dyadic arguments all computations may be performed with integers after one global power-of-two scaling.*

On the cell  $(t_{n,J}, t_{n,J+1})$ ,

$$p_n^{(n-1)}(x) = 2^{n(n-1)/2} \sum_{k=0}^J \tau_k. \quad (3.7)$$

The elementary Thue–Morse prefix identity

$$\sum_{k=0}^J \tau_k = \begin{cases} \tau_{J/2}, & J \text{ even,} \\ 0, & J \text{ odd} \end{cases} \quad (3.8)$$

therefore shows that every second top-derivative cell is identically zero, while the remaining cells carry a compressed Thue–Morse pattern.

### 3.3 Convolution and scaling recursions

Appending the final box in [equation \(2.6\)](#) gives

$$p_n(x) = 2^{n-1} \int_{x-2^{-n}}^{x+2^{-n}} p_{n-1}(t) dt, \quad (3.9)$$

$$p'_n(x) = 2^{n-1} (p_{n-1}(x + 2^{-n}) - p_{n-1}(x - 2^{-n})). \quad (3.10)$$

The product shell  $\Phi_n(\xi) = \text{sinc}_{\text{rad}}(\pi\xi)\Phi_{n-1}(\xi/2)$  yields the equivalent dilation recursion

$$p_n(x) = \int_{2x-1}^{2x+1} p_{n-1}(s) ds, \quad (3.11)$$

$$p'_n(x) = 2(p_{n-1}(2x+1) - p_{n-1}(2x-1)). \quad (3.12)$$

These formulas permit exact construction of all cell polynomials by repeated primitive and shift operations. They avoid the severe cancellation that can occur in the global truncated-power sum for large  $n$ .

## 4 Structural properties of the finite-product splines

### 4.1 Shape, plateau, and partition of unity

**Proposition 4.1** (Basic geometry). *For every  $n \geq 1$ :*

- (i)  $p_n \geq 0$ ,  $p_n$  is even, and  $\int p_n = 1$ ;
- (ii)  $\text{supp } p_n = [-A_n, A_n]$ ;
- (iii)  $p_n$  is log-concave and therefore unimodal;
- (iv)  $p_n(x) = 1$  for  $|x| \leq \delta_n$ ;
- (v) for  $n \geq 2$ ,

$$\sum_{m \in \mathbb{Z}} p_n(x - m) = 1, \quad p_n(x) + p_n(1 - x) = 1 \quad (0 \leq x \leq 1). \quad (4.1)$$

*Proof.* The first three assertions follow from convolution of centered uniform densities; log-concavity is preserved by convolution. For the plateau, split off the largest box  $u_{1/2}$ . The remaining convolution is supported on  $[-1/2 + \delta_n, 1/2 - \delta_n]$ . If  $|x| \leq \delta_n$ , the interval  $[x - 1/2, x + 1/2]$  contains that support, so convolution with  $u_{1/2}$  integrates the entire remaining density and equals 1.

At every nonzero integer  $k$ , the factor  $\text{sinc}_{\text{rad}}(\pi\xi)$  gives  $\Phi_n(k) = 0$ , while  $\Phi_n(0) = 1$ . Poisson summation therefore gives the first identity in [equation \(4.1\)](#). On  $[0, 1]$ , the support radius is less than 1, so only the translates  $m = 0, 1$  survive; evenness converts  $p_n(x - 1)$  to  $p_n(1 - x)$ .  $\square$

### 4.2 Fourier zeros

Let  $\nu_2(k)$  denote the exponent of 2 in a nonzero integer  $k$ .

**Proposition 4.2** (Exact zero multiplicities). *For  $k \in \mathbb{Z} \setminus \{0\}$ ,*

$$\text{ord}_{\xi=k} \Phi_n = 1 + \min\{\nu_2(k), n - 1\}. \quad (4.2)$$

*For fixed  $k$ , this stabilizes to  $1 + \nu_2(k)$ , the multiplicity of the corresponding zero of  $\Phi$ .*

This 2-adic spectral stratification is the Fourier counterpart of the Thue–Morse jump pattern. It also explains why the splines reproduce constants by integer translation, while higher-order cancellation appears selectively on dyadic frequency sublattices.

### 4.3 Cumulants and the variance defect

Let  $B_{2m}$  be the Bernoulli numbers with  $B_2 = 1/6$  and  $B_4 = -1/30$ . The even cumulants of  $X$  are

$$\kappa_{2m}(X) = \frac{B_{2m}}{2m(1 - 2^{-2m})}, \quad \kappa_{2m+1}(X) = 0. \quad (4.3)$$

Indeed, a uniform  $U \sim \text{Unif}[-1, 1]$  has  $\kappa_{2m}(U) = 2^{2m} B_{2m}/(2m)$ , and the geometric weights sum exactly. From equation (2.9),

$$\kappa_{2m}(S_n) = (1 - \delta_n^{2m})\kappa_{2m}(X). \quad (4.4)$$

In particular,

$$\mathbb{E}X^2 = \frac{1}{9}, \quad \mathbb{E}X^4 = \frac{19}{675}, \quad \mathbb{E}X^6 = \frac{583}{59535}, \quad (4.5)$$

and

$$\text{Var } X - \text{Var } S_n = \frac{\delta_n^2}{9}. \quad (4.6)$$

The raw prefix's leading  $4^{-n}$  error is exactly this missing variance transported through the second derivative.

### 4.4 Sharp derivative plateaux

Define

$$c_m = 1 - 2^{-m} \quad (m \geq 0), \quad c_0 = 0. \quad (4.7)$$

The living primary exposition proves the corresponding sharp derivative norm for up through its exact dyadic derivative tiling [91]. The finite splines possess a stronger local plateau.

**Theorem 4.3** (Finite-prefix derivative geometry). *Fix  $m \geq 0$  and  $n \geq m + 1$ . Interpreting the top derivative at the limiting case  $n = m + 1$  as its canonical almost-everywhere step representative, one has*

$$\|p_n^{(m)}\|_{L^\infty} = 2^{\binom{m+1}{2}}. \quad (4.8)$$

Moreover,

$$p_n^{(m)}(x) = (-1)^m 2^{\binom{m+1}{2}} \quad \text{for almost every } x \in [c_m - \delta_n, c_m + \delta_n]. \quad (4.9)$$

For  $n \geq m + 2$  the derivative is classical and the equality holds pointwise on the closed interval. By evenness, the reflected plateau has the corresponding parity sign.

*Proof.* Factor

$$p_n = p_{m+1} * q_{m+1,n},$$

where  $q_{m+1,n}$  is the convolution of the remaining boxes and is a probability density supported on

$$[-2^{-m-1} + \delta_n, 2^{-m-1} - \delta_n].$$

By equations (3.7) and (3.8), the step function  $p_{m+1}^{(m)}$  takes only the values 0 and  $\pm 2^{\binom{m+1}{2}}$ . Convolution with a probability measure cannot increase its supremum norm, proving the upper bound.

The point  $c_m$  is the midpoint of the rightmost nonzero cell of  $p_{m+1}^{(m)}$ , whose half-width is  $2^{-m-1}$ . Translating the support of  $q_{m+1,n}$  across any  $x \in [c_m - \delta_n, c_m + \delta_n]$  remains inside that cell. Hence the convolution averages a constant and gives equation (4.9), which also proves sharpness.  $\square$

## 5 Sharp convergence of the ordinary partial products

### 5.1 Smooth-test and transport estimates

The coupling [equation \(2.9\)](#) immediately gives, for  $1 \leq p < \infty$ ,

$$W_p(\mathcal{L}(S_n), \mathcal{L}(X)) \leq \delta_n (\mathbb{E}|X|^p)^{1/p} \leq \delta_n. \quad (5.1)$$

The mean-zero tail improves smooth tests by one full power. If  $\varphi \in C^2(\mathbb{R})$  has bounded second derivative, then

$$|\mathbb{E}\varphi(X) - \mathbb{E}\varphi(S_n)| \leq \frac{\delta_n^2}{18} \|\varphi''\|_\infty. \quad (5.2)$$

For convex  $\varphi$ , the left side has the nonnegative sign  $\mathbb{E}\varphi(X) - \mathbb{E}\varphi(S_n)$  by conditional Jensen. The constant is sharp for  $\varphi(x) = x^2$ .

### 5.2 Exact $C^r$ error law

**Theorem 5.1** (Sharp ordinary-prefix law). *For every integer  $r \geq 0$  and every  $n \geq r + 3$ ,*

$$\left\| p_n^{(r)} - \text{up}^{(r)} \right\|_\infty = \frac{2^{\binom{r+3}{2}-1}}{9 \cdot 4^n}. \quad (5.3)$$

*The extremum is attained, among other points, at*

$$x = c_{r+2} = 1 - 2^{-(r+2)}, \quad (5.4)$$

*and the signed value is*

$$p_n^{(r)}(c_{r+2}) - \text{up}^{(r)}(c_{r+2}) = (-1)^{r+1} \frac{2^{\binom{r+3}{2}-1}}{9 \cdot 4^n}. \quad (5.5)$$

*If  $\|f\|_{C^r} = \max_{0 \leq j \leq r} \|f^{(j)}\|_\infty$ , then the same right-hand side is exactly  $\|p_n - \text{up}\|_{C^r}$ .*

*Proof.* Let  $g = p_n^{(r)}$ . From [equation \(2.10\)](#),

$$\text{up}^{(r)}(x) = \mathbb{E}g(x - \delta_n X).$$

The function  $g$  has a bounded second weak derivative when  $n \geq r + 3$ . Taylor's formula with integral remainder gives

$$g(x - h) = g(x) - hg'(x) + h^2 \int_0^1 (1-t)g''(x - th) dt.$$

Set  $h = \delta_n X$  and take expectations. Since  $\mathbb{E}X = 0$  and  $\mathbb{E}X^2 = 1/9$ ,

$$\begin{aligned} \left| \text{up}^{(r)}(x) - p_n^{(r)}(x) \right| &\leq \frac{\delta_n^2}{18} \left\| p_n^{(r+2)} \right\|_\infty \\ &= \frac{\delta_n^2}{18} 2^{\binom{r+3}{2}}, \end{aligned}$$

where [theorem 4.3](#) was used in the second line. This is the claimed upper bound.

At  $x = c_{r+2}$ , the entire interval  $[x - \delta_n, x + \delta_n]$  lies in the plateau from [equation \(4.9\)](#) for  $p_n^{(r+2)}$ . The Taylor remainder therefore averages a constant, and every inequality above is an equality. Its sign is obtained from the plateau sign  $(-1)^{r+2}$ . Finally, the constants in [equation \(5.3\)](#) increase strictly with the derivative order, proving the  $C^r$  statement.  $\square$

For  $r = 0$ ,

$$\|p_n - \text{up}\|_\infty = \frac{4}{9 \cdot 4^n} \quad (n \geq 3). \quad (5.6)$$

This is exactly one half of the repeated-integration error  $8/(9 \cdot 4^n)$  in the frontier volume: omitting the tail has variance defect  $-\delta_n^2/9$ , whereas replacing it by a full-radius uniform box has variance defect  $+2\delta_n^2/9$ , in the units of [\(4.6\)](#).

### 5.3 Exact Kolmogorov distance

Let

$$P_n(x) = \int_{-\infty}^x p_n(t) dt, \quad P(x) = \int_{-\infty}^x \text{up}(t) dt. \quad (5.7)$$

**Corollary 5.2** (Exact CDF convergence). *For every  $n \geq 2$ ,*

$$\|P_n - P\|_{\infty} = \frac{1}{9 \cdot 4^n}. \quad (5.8)$$

The maximum is attained at  $x = 1/2$ , where  $P_n(x) - P(x) = 1/(9 \cdot 4^n)$ .

*Proof.* Apply the proof of [theorem 5.1](#) to  $g = P_n$ . Now  $g'' = p'_n$ , whose supremum is 2 by [theorem 4.3](#). The derivative  $p'_n$  is identically  $-2$  on  $[1/2 - \delta_n, 1/2 + \delta_n]$ , so equality occurs at  $1/2$ .  $\square$

### 5.4 $L^p$ consequences and endpoint flatness

For  $1 \leq p < \infty$  and  $n \geq r + 3$ ,

$$\|p_n^{(r)} - \text{up}^{(r)}\|_{L^p} \leq 2^{1/p} \frac{2^{\binom{r+3}{2}-1}}{9 \cdot 4^n}. \quad (5.9)$$

The all-orders expansion in [appendix 6](#) sharpens this to

$$4^n \|p_n^{(r)} - \text{up}^{(r)}\|_{L^p} \longrightarrow \frac{1}{18} \|\text{up}^{(r+2)}\|_{L^p}, \quad 1 \leq p \leq \infty. \quad (5.10)$$

Although  $p_n$  vanishes on the thin endpoint layers  $A_n < |x| \leq 1$ , this causes no  $O(\delta_n)$  uniform loss. Since every derivative of  $\text{up}$  vanishes at  $\pm 1$ , for every  $M$  and fixed  $r$ ,

$$\sup_{A_n \leq |x| \leq 1} |\text{up}^{(r)}(x)| = O_{r,M}(\delta_n^M). \quad (5.11)$$

The endpoint discrepancy is beyond every algebraic order; the  $4^{-n}$  error is an interior moment effect.

## 6 All-orders differential expansion

### 6.1 The reciprocal product and Bell–Bernoulli coefficients

Near the origin, define

$$\frac{1}{\Phi(z)} = \exp\left(\sum_{j=1}^{\infty} c_j(2)(2\pi z)^{2j}\right) = \sum_{m=0}^{\infty} a_m(2\pi z)^{2m}. \quad (6.1)$$

This is the base-two case of the volume’s single reciprocal-product coefficient layer, [theorem 3.3](#) and [proposition 3.4](#), where it is proved: under the substitution  $z_{\text{II}} = (2\pi z)^2$  one has  $a_m = \alpha_m(2)$ , and the Bernoulli cumulants  $c_k(b)$  of [\(3.7\)](#) specialize to

$$c_j(2) = \frac{\zeta(2j)}{j\pi^{2j}(4^j - 1)} = \frac{|B_{2j}|}{2j(2j)!(1 - 2^{-2j})}. \quad (6.2)$$

The Bell recurrence [\(3.11\)](#) then gives

$$\begin{aligned} a_0 &= 1, & a_1 &= \frac{1}{18}, & a_2 &= \frac{31}{16200}, \\ a_3 &= \frac{2347}{42865200}, & a_4 &= \frac{1904369}{1311675120000}, & a_5 &= \frac{3309760579}{88561680752160000}. \end{aligned} \quad (6.3)$$

## 6.2 Uniform expansion theorem

**Theorem 6.1** (All-orders prefix expansion). *Fix integers  $r \geq 0$  and  $M \geq 1$ . As  $n \rightarrow \infty$ ,*

$$p_n^{(r)}(x) = \sum_{m=0}^{M-1} (-1)^m a_m \delta_n^{2m} \text{up}^{(r+2m)}(x) + O_{r,M}(\delta_n^{2M}) \quad (6.4)$$

uniformly for  $x \in \mathbb{R}$ .

*Proof.* By equation (2.10),

$$\Phi_n(\xi) = \Phi(\xi) \frac{1}{\Phi(\delta_n \xi)}.$$

Choose  $0 < c < 1$ . On  $|\xi| \leq c/\delta_n$ , the reciprocal is analytic and its Taylor remainder after  $M - 1$  terms is bounded by  $C_M \delta_n^{2M} |\xi|^{2M}$ . Multiplication by  $|\xi|^r |\Phi(\xi)|$  is integrable because  $\Phi$  is Schwartz, so Fourier inversion bounds the low-frequency remainder by  $O_{r,M}(\delta_n^{2M})$ .

For the high-frequency tail, retain any fixed  $K$  factors of  $\Phi_n$ :

$$|\Phi_n(\xi)| \leq C_K |\xi|^{-K}.$$

Choosing  $K > r + 2M + 1$  makes the integral over  $|\xi| > c/\delta_n$  smaller than  $O(\delta_n^{2M})$ . The high-frequency tails of the finitely many  $\Phi(\xi) \xi^{2m}$  terms are superalgebraically small. Finally,

$$(-1)^m \widehat{\text{up}^{(2m)}}_{2\pi}(\xi) = (2\pi\xi)^{2m} \Phi(\xi),$$

which yields equation (6.4) after differentiating  $r$  times.  $\square$

In particular,

$$p_n = \text{up} - \frac{\delta_n^2}{18} \text{up}'' + \frac{31\delta_n^4}{16200} \text{up}^{(4)} - \frac{2347\delta_n^6}{42865200} \text{up}^{(6)} + O(\delta_n^8). \quad (6.5)$$

The leading profile is therefore

$$4^n (p_n^{(r)} - \text{up}^{(r)}) \longrightarrow -\frac{1}{18} \text{up}^{(r+2)} \quad \text{uniformly.} \quad (6.6)$$

The sharp derivative norm  $\|\text{up}^{(r+2)}\|_\infty = 2^{\binom{r+3}{2}}$  is exactly  $18 \cdot 4^n$  times the constant in equation (5.3): the asymptotic profile  $-\delta_n^2 \text{up}^{(r+2)} / 18$  already attains its supremum at the dyadic extremizer for every admissible  $n$ .

## 6.3 Signed geometric Richardson acceleration

Let  $q = 1/4$  and, for  $M \geq 1$ , define

$$R_{n,M} = \sum_{j=0}^{M-1} w_{M,j} p_{n+j}, \quad (6.7)$$

where

$$\begin{aligned} w_{M,j} &= (-1)^{M-1-j} \frac{q^{\binom{M-j}{2}}}{(q; q)_j (q; q)_{M-1-j}} \\ &= \frac{(-1)^{M-1-j} q^{\binom{M-j}{2}}}{(q; q)_{M-1}} \begin{bmatrix} M-1 \\ j \end{bmatrix}_q. \end{aligned} \quad (6.8)$$

Here  $(q; q)_k = \prod_{\ell=1}^k (1 - q^\ell)$ . These are the Lagrange weights that extrapolate from the geometric nodes  $q^j$ ,  $0 \leq j < M$ , to the origin:

$$\sum_{j=0}^{M-1} w_{M,j} q^{jm} = \delta_{m0} \quad (0 \leq m < M). \quad (6.9)$$

Consequently,

$$R_{n,M}^{(r)} = \text{up}^{(r)} - a_M q^{\binom{M}{2}} \delta_n^{2M} \text{up}^{(r+2M)} + O_{r,M}(\delta_n^{2M+2}). \quad (6.10)$$

The first weight vectors are

$$\begin{aligned} M = 2 : & \quad \left( -\frac{1}{3}, \frac{4}{3} \right), \\ M = 3 : & \quad \left( \frac{1}{45}, -\frac{4}{9}, \frac{64}{45} \right), \\ M = 4 : & \quad \left( -\frac{1}{2835}, \frac{4}{135}, -\frac{64}{135}, \frac{4096}{2835} \right). \end{aligned}$$

Despite alternating signs, the coefficient variation stays uniformly bounded:

$$\sum_{j=0}^{M-1} |w_{M,j}| = \frac{(-q; q)_{M-1}}{(q; q)_{M-1}} < \frac{(-q; q)_\infty}{(q; q)_\infty} = 1.969260353668 \dots \quad (6.11)$$

This supplies a stable signed arbitrary-order method. At a dyadic point  $x$  of depth  $s$  (the least  $s$  with  $2^s x \in \mathbb{Z}$ ) the combination is moreover *exact* rather than asymptotic: for  $n \geq s + 1$  the finite spline  $p_n$  equals  $\text{up}(x)$  plus exactly  $\lfloor s/2 \rfloor$  geometric modes of ratio  $4^{-1}, \dots, 4^{-\lfloor s/2 \rfloor}$  with no remainder, so  $R_{n,M}$  with  $M = \lfloor s/2 \rfloor + 1$  returns  $\text{up}(x)$  exactly from  $M$  consecutive rational samples; see the consolidated dyadic-extraction volume in `inverse-and-sampling/dyadic-up-extraction/` (its corollary due:cor:sharp-richardson), whose  $\text{up}_n$  is this volume's  $p_{n+1}$ . The next section constructs an entirely positive alternative.

## 7 A positive quadrature hierarchy for the exact tail

### 7.1 Up as a scale mixture of uniform boxes

The function  $\text{up}$  is even, nonnegative, and decreasing on  $[0, 1]$ . Define a measure  $\eta$  on  $[0, 1]$  by

$$\mathrm{d}\eta(r) = -2r \text{up}'(r) \mathrm{d}r. \quad (7.1)$$

Integration by parts gives  $\eta([0, 1]) = 1$ .

**Theorem 7.1** (Uniform scale mixture). *Let  $R \sim \eta$  and  $U \sim \text{Unif}[-1, 1]$  be independent. Then*

$$X \stackrel{d}{=} RU, \quad (7.2)$$

and

$$\text{up}(x) = \int_0^1 u_r(x) \mathrm{d}\eta(r), \quad (7.3)$$

$$\Phi(\xi) = \int_0^1 \text{sinc}_{\text{rad}}(2\pi r \xi) \mathrm{d}\eta(r). \quad (7.4)$$

If  $Z = R^2$  and  $\nu$  is its law, then

$$\mathrm{d}\nu(z) = -\text{up}'(\sqrt{z}) \mathrm{d}z, \quad 0 < z < 1, \quad (7.5)$$

and

$$m_k := \int_0^1 z^k \mathrm{d}\nu(z) = \mathbb{E}R^{2k} = (2k + 1)\mathbb{E}X^{2k}. \quad (7.6)$$



*Proof.* For  $x \geq 0$ ,

$$\int_0^1 u_r(x) d\eta(r) = \int_x^1 \frac{1}{2r} (-2r \operatorname{up}'(r)) dr = \operatorname{up}(x) - \operatorname{up}(1) = \operatorname{up}(x).$$

Evenness handles  $x < 0$ , proving the density identity and hence [equation \(7.2\)](#). Fourier transformation gives [equation \(7.4\)](#). The change of variables  $z = r^2$  gives [equation \(7.5\)](#). Finally,

$$\mathbb{E}R^{2k} = \int_0^1 -2r^{2k+1} \operatorname{up}'(r) dr = 2(2k+1) \int_0^1 r^{2k} \operatorname{up}(r) dr = (2k+1)\mathbb{E}X^{2k}.$$

□

The first moments of  $\nu$  are

$$m_0 = 1, \quad m_1 = \frac{1}{3}, \quad m_2 = \frac{19}{135}, \quad m_3 = \frac{583}{8505}. \quad (7.7)$$

All  $m_k$  are rational. Therefore every monic orthogonal polynomial generated from  $\nu$  has rational coefficients, and all Gaussian or Radau nodes are algebraic numbers obtainable by exact arithmetic before root isolation.

## 7.2 Quadrature tail surrogates

Let

$$Q = \sum_{j=1}^J w_j \delta_{z_j}, \quad 0 \leq z_j \leq 1, \quad w_j > 0, \quad \sum_j w_j = 1, \quad (7.8)$$

be a positive quadrature rule for  $\nu$ . In spatial convolution formulas we use the measure convention  $u_0 = \delta_0$ . Let  $Z_Q$  have law  $Q$ , let  $U$  be an independent uniform variable, and define

$$Y_Q = \sqrt{Z_Q} U. \quad (7.9)$$

The associated positive approximant is

$$g_{n,Q} = p_n * \mathcal{L}(\delta_n Y_Q) = \sum_{j=1}^J w_j (p_n * u_{\delta_n \sqrt{z_j}}), \quad (7.10)$$

$$\widehat{g_{n,Q}}_{2\pi}(\xi) = \Phi_n(\xi) \sum_{j=1}^J w_j \operatorname{sinc}_{\text{rad}}(2\pi \delta_n \sqrt{z_j} \xi). \quad (7.11)$$

Thus every rule produces a finite positive mixture of explicit piecewise-polynomial box splines.

For a smooth function  $h$ , introduce

$$A_h(z) = \frac{1}{2} \int_{-1}^1 h(\sqrt{z} u) du. \quad (7.12)$$

Repeated integration by parts gives the key transfer identity

$$A_h^{(\ell)}(z) = \frac{1}{2^{2\ell+1} \ell!} \int_{-1}^1 (1-u^2)^\ell h^{(2\ell)}(\sqrt{z} u) du. \quad (7.13)$$

Since

$$\int_{-1}^1 (1-u^2)^\ell du = \frac{2^{2\ell+1} (\ell!)^2}{(2\ell+1)!}, \quad (7.14)$$

we obtain

$$\|A_h^{(\ell)}\|_\infty \leq \frac{\ell!}{(2\ell+1)!} \|h^{(2\ell)}\|_\infty. \quad (7.15)$$

This is the bridge by which matching a moment of the squared scale  $Z$  cancels two spatial derivatives.

### 7.3 General exact quadrature error theorem

Assume  $Q$  is exact for polynomials of degree at most  $\ell - 1$ , and its Peano kernel of order  $\ell$  has one sign. This includes Gaussian, one-endpoint Radau, and Lobatto rules for a positive measure on an interval [50, 95]. Set

$$\varepsilon_{Q,\ell} = Q(z^\ell) - m_\ell, \quad \Gamma_{Q,\ell} = \frac{|\varepsilon_{Q,\ell}|}{(2\ell + 1)!}. \quad (7.16)$$

**Theorem 7.2** (Exact positive quadrature law). *Suppose  $\varepsilon_{Q,\ell} \neq 0$ . For every  $r \geq 0$  and  $n \geq r + 2\ell + 1$ ,*

$$\left\| g_{n,Q}^{(r)} - \text{up}^{(r)} \right\|_\infty = \Gamma_{Q,\ell} 2^{\binom{r+2\ell+1}{2}} 2^{-2\ell n}. \quad (7.17)$$

The extremum is attained, among other points, at  $x = c_{r+2\ell}$ , where it has sign  $\text{sgn}(\varepsilon_{Q,\ell})(-1)^r$ .

*Proof.* Put  $f = p_n^{(r)}$  and, for fixed  $x$ , let  $h_x(v) = f(x - \delta_n v)$ . Then

$$g_{n,Q}^{(r)}(x) - \text{up}^{(r)}(x) = Q(A_{h_x}) - \nu(A_{h_x}).$$

A one-signed Peano kernel and exactness through degree  $\ell - 1$  imply

$$|Q(A) - \nu(A)| \leq \frac{|Q(z^\ell) - \nu(z^\ell)|}{\ell!} \|A^{(\ell)}\|_\infty.$$

Using equation (7.15) and  $\|h_x^{(2\ell)}\|_\infty = \delta_n^{2\ell} \|p_n^{(r+2\ell)}\|_\infty$  gives

$$|g_{n,Q}^{(r)}(x) - \text{up}^{(r)}(x)| \leq \Gamma_{Q,\ell} \delta_n^{2\ell} 2^{\binom{r+2\ell+1}{2}}.$$

At  $x = c_{r+2\ell}$ , every argument  $x - \delta_n \sqrt{z} u$  lies in the constant plateau of  $p_n^{(r+2\ell)}$  from theorem 4.3. Hence  $A_{h_x}^{(\ell)}$  is constant, the Peano inequality is an equality, and the asserted sign follows.  $\square$

Two familiar approximations fit the same theorem. Ordinary truncation is  $Q = \delta_0$ ,  $\ell = 1$ , with  $\Gamma = 1/18$ . The repeated-integration spline is  $Q = \delta_1$ ,  $\ell = 1$ , with  $\Gamma = 1/9$ . Their factor-of-two error difference is therefore a one-line quadrature moment calculation.

### 7.4 Explicit positive accelerated schemes

**Example 7.3** (One-node Gauss: a single variance-matched box). The one-node Gauss rule for  $\nu$  is  $Q = \delta_{1/3}$ . It matches  $m_0, m_1$  and has

$$\varepsilon_{Q,2} = \frac{1}{9} - \frac{19}{135} = -\frac{4}{135}, \quad \Gamma_{Q,2} = \frac{1}{4050}.$$

Thus

$$G_n = p_n * u_{\delta_n/\sqrt{3}} \quad (7.18)$$

is a single positive box spline, and for  $n \geq r + 5$ ,

$$\left\| G_n^{(r)} - \text{up}^{(r)} \right\|_\infty = \frac{2^{\binom{r+5}{2}}}{4050 \cdot 16^n}. \quad (7.19)$$

Its support radius is  $1 - \delta_n + \delta_n/\sqrt{3} < 1$ . Endpoint flatness makes the support deficit superalgebraically small, while the interior error is exactly fourth order.

**Example 7.4** (Two-node Lobatto: exact support and fourth order). The endpoint rule matching  $m_1 = 1/3$  is

$$Q = \frac{2}{3}\delta_0 + \frac{1}{3}\delta_1. \quad (7.20)$$

It gives

$$H_n = \frac{2}{3}p_n + \frac{1}{3}(p_n * u_{\delta_n}), \quad (7.21)$$

which is nonnegative, integrates to one, and has exact support  $[-1, 1]$ . Since

$$\varepsilon_{Q,2} = \frac{1}{3} - \frac{19}{135} = \frac{26}{135}, \quad \Gamma_{Q,2} = \frac{13}{8100},$$

$$\left\| H_n^{(r)} - \text{up}^{(r)} \right\|_{\infty} = \frac{13}{8100} \frac{2^{\binom{r+5}{2}}}{16^n} \quad (n \geq r+5). \quad (7.22)$$

This explicitly realizes the positive, compactly supported, variance-matched  $16^{-n}$  direction proposed but not constructed in the frontier document.

**Example 7.5** (Two-node right Radau: exact support and sixth order). The two-node right-Radau rule, constrained to contain  $z = 1$  and exact through  $z^2$ , is

$$Q = \frac{15}{16}\delta_{13/45} + \frac{1}{16}\delta_1. \quad (7.23)$$

Its first missed moment is

$$\varepsilon_{Q,3} = \frac{704}{42525}, \quad \Gamma_{Q,3} = \frac{44}{13395375}. \quad (7.24)$$

Hence

$$J_n = \frac{15}{16}(p_n * u_{\delta_n \sqrt{13/45}}) + \frac{1}{16}(p_n * u_{\delta_n}) \quad (7.25)$$

is positive with exact support  $[-1, 1]$ , and

$$\left\| J_n^{(r)} - \text{up}^{(r)} \right\|_{\infty} = \frac{44}{13395375} \frac{2^{\binom{r+7}{2}}}{64^n} \quad (n \geq r+7). \quad (7.26)$$

## 7.5 Arbitrary order with positivity

Standard quadrature immediately produces a hierarchy.

| Rule with $M$ nodes               | Exact degree | $\ell$   | Rate           | Support behavior                        |
|-----------------------------------|--------------|----------|----------------|-----------------------------------------|
| Gauss                             | $2M - 1$     | $2M$     | $2^{-4Mn}$     | Positive; strictly inside $[-1, 1]$     |
| Right Radau (node 1)              | $2M - 2$     | $2M - 1$ | $2^{-(4M-2)n}$ | Positive; exact support $[-1, 1]$       |
| Lobatto (nodes 0, 1, $M \geq 2$ ) | $2M - 3$     | $2M - 2$ | $2^{-(4M-4)n}$ | Positive; exact support, includes $p_n$ |

The constants are not merely asymptotic: once  $n \geq r + 2\ell + 1$ , they are exactly

$$\frac{|Q(z^\ell) - m_\ell|}{(2\ell + 1)!} 2^{\binom{r+2\ell+1}{2}}. \quad (7.27)$$

Thus positive arbitrary-order acceleration reduces to the classical algebra of orthogonal polynomials for the explicitly defined measure  $d\nu(z) = -\text{up}'(\sqrt{z}) dz$ .

## 8 Numerical and exact-arithmetic experiments

### 8.1 Method

The accompanying script uses the exact product samples

$$\Phi_n(k/L), \quad \Phi(k/L),$$

on a periodic box of length  $L = 4$  and applies an inverse FFT with  $2^{17}$  modes. Every function considered is supported in  $[-1, 1]$ , so its periodization has no overlap. Derivatives are obtained by multiplying the transform by  $(2\pi i\xi)^r$ . The infinite product is truncated after 48 factors, beyond double-precision relevance on the active frequency range. Separate routines evaluate [equation \(3.2\)](#) with exact rational arithmetic, generate the  $a_m$ , and generate the Richardson weights.

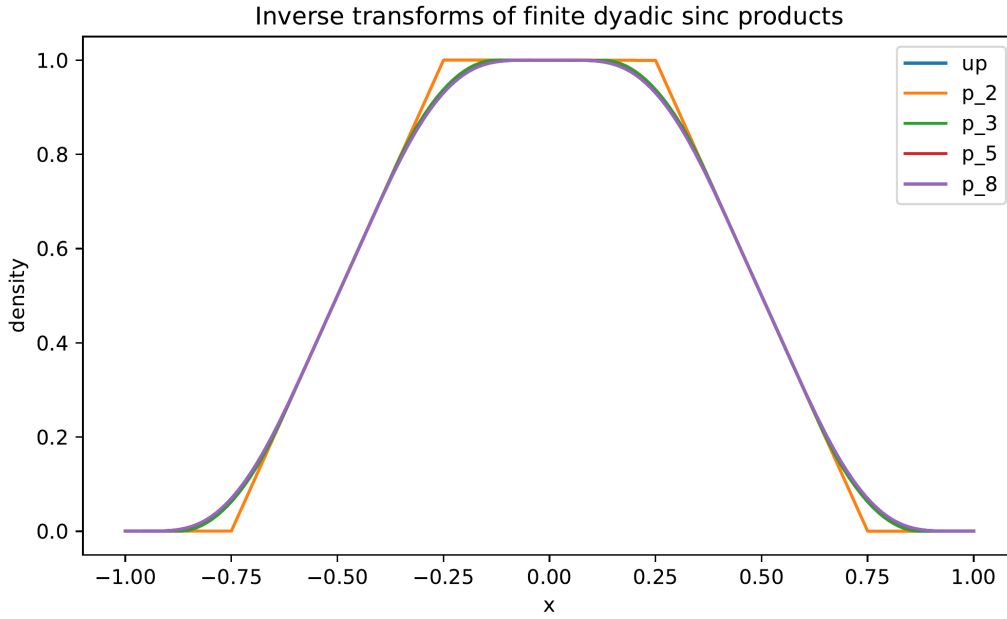


Figure 3: Inverse transforms of the first few finite sinc products. The support expands to  $[-1, 1]$ , the polynomial degree and smoothness increase, and the central plateau shrinks dyadically.

### 8.2 Verification of exact constants

Selected FFT ratios of measured supremum error to the exact formula are shown below.

| Scheme     | $r$ | $n$ | measured error                  | measured/exact |
|------------|-----|-----|---------------------------------|----------------|
| Raw prefix | 0   | 3   | $6.94444444448 \times 10^{-3}$  | 1.000000000006 |
| Raw prefix | 2   | 5   | $5.55555555771 \times 10^{-2}$  | 1.00000000387  |
| Raw prefix | 4   | 7   | 7.1111128754                    | 1.00000024810  |
| Gauss-1    | 0   | 5   | $2.41126543254 \times 10^{-7}$  | 1.00000000183  |
| Lobatto-2  | 0   | 5   | $1.56732253097 \times 10^{-6}$  | 1.00000000069  |
| Radau-2    | 0   | 7   | $1.56639950187 \times 10^{-12}$ | 1.000079775969 |

The raw and fourth-order laws agree essentially to machine precision. The Radau error is already  $1.6 \times 10^{-12}$  at its first admissible  $n$ , so Fourier truncation and roundoff visibly affect the last digits. Full tables are included in `assets/finite_sinc_products_report/sharp_error_verification.csv` and `assets/finite_sinc_products_report/positive_acceleration_verification.csv`.

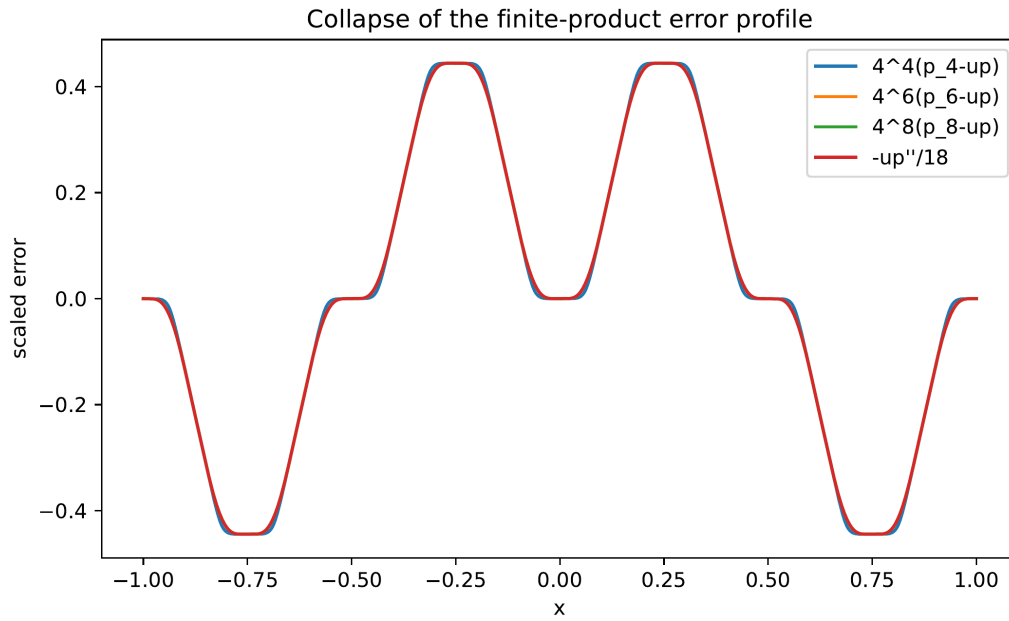


Figure 4: The profiles  $4^n(p_n - \text{up})$  collapse onto  $-\text{up}''/18$ , in agreement with [equation \(6.6\)](#).

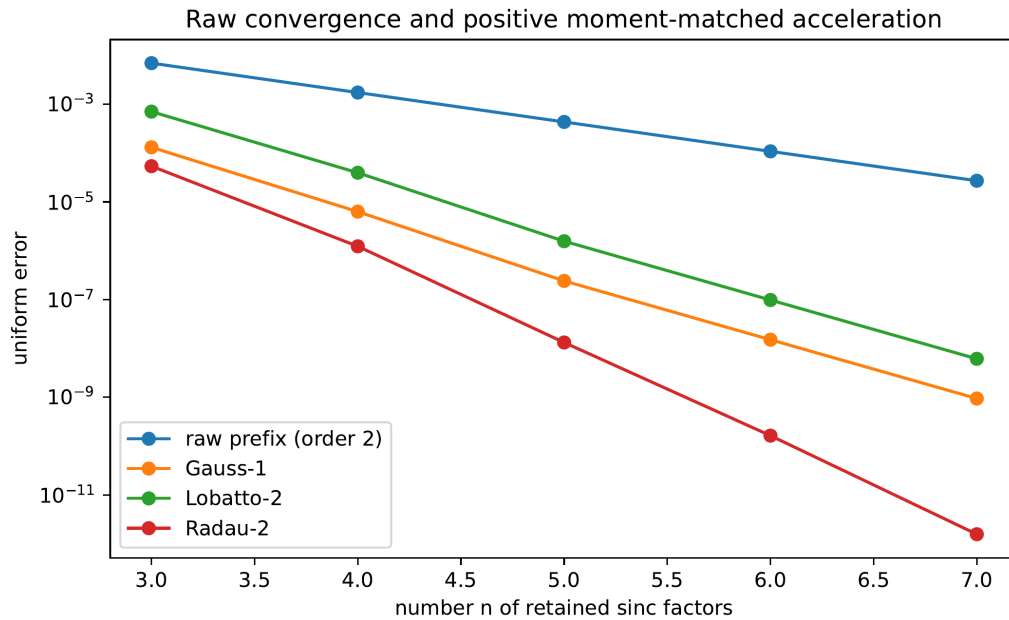


Figure 5: Raw second-order convergence and positive Gauss, Lobatto, and Radau tail acceleration. The highest-order Radau values approach the double-precision floor.

## 9 Algorithms and implementation considerations

### 9.1 Exact cell construction

There are three complementary exact representations.

**Global truncated powers.** Equation (3.2) evaluates a point in  $O(2^n)$  signed rational operations and is ideal for proof and low-order checks. Only knots left of  $x$  need be included.

**Primitive recursion.** If a piecewise-polynomial primitive of  $p_{n-1}$  is stored cellwise, then equation (3.9) evaluates  $p_n$  by two shifted primitive values. This constructs the complete spline with exact coefficients while reusing adjacent-cell data.

**Top-down integration of the Thue–Morse step.** Start from the top derivative values in equations (3.7) and (3.8), integrate cell by cell, and determine constants by continuity and vanishing at the left support endpoint. This uses only additions, multiplications by the mesh, and division by small integers. It is particularly attractive for formal verification.

### 9.2 Positive quadrature construction

The moment sequence  $m_k = (2k+1)\mathbb{E}X^{2k}$  can be generated exactly from the cumulants equation (4.3). A practical  $M$ -node Gauss construction is:

1. compute  $m_0, \dots, m_{2M}$  as rational numbers;
2. perform exact Gram–Schmidt on  $1, z, \dots, z^M$ , or solve the Hankel system for the monic orthogonal polynomial  $P_M$ ;
3. isolate the  $M$  simple roots of  $P_M$  in  $(0, 1)$ ;
4. solve the Vandermonde moment equations for positive weights;
5. insert the nodes and weights in equation (7.11) or equation (7.10).

Right-Radau rules are obtained by orthogonalizing against the modified measure  $(1 - z) d\nu(z)$  and then adjoining  $z = 1$ ; Lobatto rules use  $z(1 - z) d\nu(z)$  and adjoin both endpoints. Exact rational recurrence coefficients make certified algebraic output feasible.

### 9.3 Numerical stability

The global formula equation (3.2) contains large alternating terms and should not be evaluated naively in floating point at high  $n$ . Stable options are:

- exact rational arithmetic followed by one final conversion;
- cellwise Horner evaluation after primitive recursion;
- FFT inversion when values on a dense uniform grid are needed;
- convolution on adaptive polynomial pieces for positive quadrature mixtures.

The Richardson coefficients have uniformly bounded  $\ell^1$  norm by equation (6.11); their cancellation is therefore much milder than a generic high-order extrapolation. Positive quadrature schemes avoid cancellation entirely.

## 10 Further deductions and research directions

### 10.1 Results that appear immediately accessible

**Open problem 10.1** (Lean formalization of the prefix law). *Formalize [theorems 3.1, 4.3 and 5.1](#). The proof ingredients are already close to the repository’s APIs: finite uniform convolutions, Thue–Morse prefix sums, the exact tail decomposition, bounded weak second derivatives, and the formally proved sharp derivative tiling of up.*

**Open problem 10.2** (Certified quadrature generator). *Build an exact-arithmetic pipeline that returns the rational orthogonal polynomial, certified algebraic nodes, and positive weights for Gauss and right-Radau rules of  $\nu$ . The output would be a library of positive compactly supported approximants with machine-checkable error constants.*

**Open problem 10.3** (Norms beyond  $L^\infty$ ). *The expansion gives sharp leading constants in every  $L^p$  norm. The exact derivative distribution identities in the primary exposition suggest that some  $L^p$  constants, rearrangement-invariant norms, or signed error distributions may admit closed forms.*

### 10.2 Conjectures

**Conjecture 10.4** ( $q$ -special orthogonal structure). *The monic orthogonal polynomials of  $d\nu(z) = -\text{up}'(\sqrt{z})dz$  have recurrence coefficients admitting closed  $q$ -product or  $q$ -continued-fraction expressions at  $q = 1/4$ . Their rationality is automatic from the rational moments; the conjecture is that the rational numbers have a simple structural form involving  $(q; q)_m$ , Gaussian binomials, or the reciprocal-product coefficients  $a_m$ .*

**Conjecture 10.5** (Lambert edge scaling of quadrature nodes). *The extreme Gauss and Radau nodes have refined endpoint asymptotics controlled by the same lower-Lambert phase that governs the flat endpoint of the Fabius/Rvachev function. The leading zero distribution may be classical, but the edge correction should retain the repository’s log-periodic dyadic phase.*

**Conjecture 10.6** (Positive minimax tail rule). *Among positive  $M$ -component uniform scale mixtures supported in  $[-1, 1]$  and required to include the endpoint scale 1, the right-Radau rule minimizes the first nonzero worst-case  $C^r$  error constant simultaneously for all fixed  $r$ . The exact plateau formula reduces this to an extremal moment problem on  $[0, 1]$ .*

**Conjecture 10.7** (Shape preservation). *For every right-Radau rule of  $\nu$  and every sufficiently large  $n$ , the accelerated density  $g_{n,Q}$  is strictly decreasing on  $(0, 1)$  and has no spurious interior oscillation. Positivity is automatic, but monotonicity of a finite mixture is not; a proof may follow from total positivity of the uniform-box kernel.*

### 10.3 Broader extensions

**Other geometric bases.** For  $0 < q < 1$ , infinite products generated by widths  $(1 - q)q^j$  have the same exact head–tail scaling. The Bell–Bernoulli expansion and positive tail quadrature survive. The dyadic case  $q = 1/2$  is exceptional because subset sums form a uniform grid and the top derivative is exactly Thue–Morse. For the bases  $q = 1/a$  with  $a \geq 2$  the sharp leading  $C^r$  constants are already available in closed form ([corollary 18.3](#)); what remains open here is the overlapping range  $1/2 < q < 1$ , together with the plateau and extremal-set structure, which is special to  $q = 1/2$ .

**Multivariate box splines.** Products of directional sinc factors invert to multivariate box splines. A geometric infinite product produces a compactly supported  $C^\infty$  atomic function, and finite prefixes again provide polynomial approximants. Positive quadrature of the tail scale may extend with radial or matrix-valued moments.



**Inverse Fabius propagation.** On compact subintervals where  $\text{up}$  is bounded away from zero, the exact error expansion can be transported through inverse-function calculus to obtain high-order approximations of the inverse Fabius function. Near the flat endpoint, the algebraic prefix expansion must be matched with the Lambert– $W$  asymptotic already developed in the repository.

**Spectral and sampling questions.** The zero multiplicities [equation \(4.2\)](#) suggest dyadic Strang–Fix-type conditions, multiresolution reconstruction formulas, and exact cancellation identities for sampled derivatives. The finite splines could serve as compactly supported preconditioners whose approximation order is enhanced by the positive tail rules.

## 11 Conclusion

The inverse Fourier transforms of finite dyadic sinc products are much more rigid than generic finite convolutions. Binary subset sums force a uniform knot grid; Thue–Morse signs govern the top derivative; and a precise derivative plateau survives every later convolution. Combined with the exact self-similar tail of the Rvachev distribution, this local geometry turns a Taylor bound into an equality and yields the sharp law

$$\left\| p_n^{(r)} - \text{up}^{(r)} \right\|_{\infty} = \frac{2^{\binom{r+3}{2}-1}}{9 \cdot 4^n}.$$

The same mechanism converts one-signed quadrature errors into exact functional errors. The resulting Gauss–Radau–Lobatto hierarchy supplies positive piecewise-polynomial approximants of arbitrary order, with explicit constants and exact support whenever a right endpoint node is included. This closes the repository’s variance-matched  $16^{-n}$  frontier and opens a tractable new interface among dyadic harmonic analysis, Thue–Morse combinatorics, Bell–Bernoulli series, orthogonal polynomials, and certified numerical approximation.

## A Proof of the scale-derivative identity

For completeness, we prove [equation \(7.13\)](#). The case  $\ell = 0$  is the definition. Assume it holds at order  $\ell$ . Differentiating under the integral and using  $\partial_z h^{(2\ell)}(\sqrt{z}u) = u h^{(2\ell+1)}(\sqrt{z}u)/(2\sqrt{z})$  gives

$$A_h^{(\ell+1)}(z) = \frac{1}{2^{2\ell+2}\ell!\sqrt{z}} \int_{-1}^1 u(1-u^2)^{\ell} h^{(2\ell+1)}(\sqrt{z}u) \, du.$$

Since

$$\frac{d}{du}(1-u^2)^{\ell+1} = -2(\ell+1)u(1-u^2)^{\ell},$$

integration by parts has no boundary term and yields

$$A_h^{(\ell+1)}(z) = \frac{1}{2^{2\ell+3}(\ell+1)!} \int_{-1}^1 (1-u^2)^{\ell+1} h^{(2\ell+2)}(\sqrt{z}u) \, du.$$

This is the desired induction step. The formula extends continuously to  $z = 0$  by Taylor expansion.

## B Reproducibility manifest

The verification material for this part is kept in the volume directory under `assets/finite_sinc_products_report/`, each figure being present both as the `.png` file this part loads and as a `.pdf` variant:

- `assets/finite_sinc_products_report/finite_sinc_experiments.py` – documented numerical and exact code;

- `assets/finite_sinc_products_report/finite_sinc_approximants.png` – generated approximation figure;
- `assets/finite_sinc_products_report/scaled_error_profiles.png` – generated error-profile figure;
- `assets/finite_sinc_products_report/convergence_comparison.png` – generated convergence figure;
- `assets/finite_sinc_products_report/sharp_error_verification.csv` – raw-prefix numerical table;
- `assets/finite_sinc_products_report/positive_acceleration_verification.csv` – accelerated numerical table;
- `assets/finite_sinc_products_report/exact_coefficients.txt` – exact series and Richardson coefficients;
- `assets/finite_sinc_products_report/exact_rational_samples.csv` – exact rational spline samples.

The absorbed report’s own source and PDF were deleted when it was merged; git history is their archive. Run, from that directory,

```
python finite_sinc_experiments.py --output-dir . --fft-power 17
```

to regenerate every figure and table. The volume itself is rebuilt from its own directory by running `pdflatex` on `geometric_q_fabiús_frontiers.tex` three times, so that the table of contents and every cross-reference settle.

## Part IV

# Fourier Images of the Repeated-Integration Approximants

### Abstract

The repeated-integration approximants  $f_n$  to Rvachev’s up-function differ from the limit, on the Fourier side, by an exact dilation of a single universal meromorphic transfer function  $A(z) = \text{sinc}_{\text{rad}} z / \Phi(z) = \prod_{m \geq 1} (1 - z^2 / (\pi^2 m^2))^{1 - \nu_2(m)}$ :  $\widehat{f_n}_{\text{ang}} = \Phi \cdot A(2^{-n}t)$ . The valuation-weighted divisor of  $A$  yields digit-sum zero counts, a Thue–Morse sign law, the exact Taylor radius  $4\pi$  with dominant-pole coefficient asymptotics and an arithmetic Darboux hierarchy, the complete finite/limit zero-multiplicity filtration, a sharp  $o(2^n)$  relative-convergence window, forward and inverse conditioning thresholds at  $4\pi 2^n$  and  $\pi 2^n$ , and the impossibility of any globally stable convolutional correction. All-orders expansions hold in every polynomially weighted Fourier  $L^p$  space and every Sobolev space, with explicit leading norm constants; the algebraic Fourier tails have exact mean-square constants and the sharp threshold  $f_n \in H^s$  iff  $s < n + \frac{1}{2}$ . Interpreting the construction as a tail closure and matching more moments of the true self-similar tail produces positive piecewise-polynomial approximants of arbitrary order — atomic, dyadic-atomic, and polynomial-density menus with exact constants — complementing the box-mixture hierarchy of Part III.

## 1 Scope, sources, and relation to the prefix theory

This part is the editorial consolidation of two ninth-wave reports written independently on the same day and on the same subject — the global Fourier geometry of the *repeated-integration* approximants

to Rvachev’s up-function: *Fourier Images of the Piecewise-Polynomial Approximants to Rvachev’s Up-Function* (meromorphic tail transfer, weighted-norm asymptotics, moment-matched acceleration) and *Fourier Images of Recursive Piecewise-Polynomial Approximations to Rvachev’s Up Function* (2-adic divisors, critical spectral layers, exact Fourier tails, accelerated closures). Their shared core — the transfer quotient, its cotangent product and divisor, the Taylor radius  $4\pi$  and dominant-pole asymptotics, the finite zero filtration, and the cumulant defects — is stated once below with the stronger of the two treatments; each report’s unique layers (respectively: the weighted- $L^p$  master theorem with its norm constants, the arithmetic Darboux hierarchy, and the Gauss/Lobatto/polynomial closure menu; the valuation-weighted canonical product, the digital divisor laws and Thue–Morse sign, the sharp critical window, the two conditioning thresholds, the deconvolution obstruction, the exact mean-square tail constants with the sharp Sobolev threshold, and the dyadic-node closure menu) are kept in full. Every shared constant was verified identical between the sources before merging; the two independently derived zero-order laws, stated in different groupings, agree in all four valuation regimes.

Where Part III (appendix 3) studies the ordinary prefix densities  $p_n$  in physical space, the present part studies the twin family  $f_n$  (the prefix with its smallest box doubled) on the Fourier side. The two theories meet in the closure framework of appendix 11, where the three families of accelerated tail surrogates — the box mixtures of appendix 7, and the atomic and polynomial closures below — are compared as quadratures of three different laws.

*Convention.* This part uses radian frequency,

$$\widehat{g}_{\text{ang}}(t) = \int_{\mathbb{R}} g(x) e^{-itx} dx, \quad \text{sinc}_{\text{rad}} z = \begin{cases} \sin z/z, & z \neq 0, \\ 1, & z = 0, \end{cases} \quad (1.1)$$

so that

$$\Phi(t) := \widehat{\text{up}}_{\text{ang}}(t) = \prod_{k=1}^{\infty} \text{sinc}_{\text{rad}}\left(\frac{t}{2^k}\right), \quad \Phi(0) = 1, \quad (1.2)$$

with up the even  $C^\infty$  probability density on  $[-1, 1]$ . The frequency-normalized product used elsewhere in this volume is  $\Phi(2\pi\xi)$ ; in particular the zeros of (1.2) sit at  $2\pi m$ ,  $m \in \mathbb{Z} \setminus \{0\}$ , with order  $1 + \nu_2(m)$ , and  $\Phi(\pi) = 0.5537712758888100953442174163 \dots$  equals the frequency-normalized product at one half.

## 2 The repeated-integration family and its Fourier products

### 2.1 Recursion, box splines, and the head–tail model

With  $f_0 = \frac{1}{2}\mathbf{1}_{(-1,1)}$ , the recursive integration scheme

$$f_n(x) = 2 \int_{-1}^x (f_{n-1}(2t+1) - f_{n-1}(2t-1)) dt = \int_{2x-1}^{2x+1} f_{n-1}(t) dt \quad (2.1)$$

produces compactly supported piecewise-polynomial probability densities converging to up. Writing  $\beta_a = \frac{1}{2a}\mathbf{1}_{[-a,a]}$  for the centered uniform density and unrolling the recursion,

$$f_n = \beta_{2^{-1}} * \beta_{2^{-2}} * \dots * \beta_{2^{-n}} * \beta_{2^{-n}} = p_n * \beta_h, \quad h := 2^{-n}, \quad (2.2)$$

where  $p_n = \beta_{2^{-1}} * \dots * \beta_{2^{-n}}$  is the ordinary prefix of appendix 3: a univariate box spline of degree  $n-1$  supported on  $[-1+h, 1-h]$ , each box occurring once. It is  $f_n$ , not  $p_n$ , that is the degree- $n$  spline on  $[-1, 1]$  obtained by repeating the smallest box. Probabilistically, if  $X_n$  has density  $f_n$ ,  $X$  has density up, and  $S_n = \sum_{k=1}^n 2^{-k} U_k$  with  $U_k$  independent uniform on  $[-1, 1]$ , then

$$X_n \stackrel{d}{=} S_n + hU, \quad X \stackrel{d}{=} S_n + hX', \quad X' \stackrel{d}{=} X, \quad (2.3)$$

with all variables independent: the finite construction replaces the true self-similar tail  $hX'$  by a single uniform variable  $hU$ . This closure point of view drives the entire part.

## 2.2 Finite products and the master factorization

With  $P_n(t) = \prod_{k=1}^n \text{sinc}_{\text{rad}}(t/2^k) = \widehat{p}_{n \text{ ang}}(t)$ ,

$$F_n(t) := \widehat{f}_{n \text{ ang}}(t) = P_n(t) \text{sinc}_{\text{rad}}(ht), \quad \Phi(t) = P_n(t) \Phi(ht). \quad (2.4)$$

**Definition 2.1** (Tail-transfer quotient). The *tail-transfer function* (or correction quotient) is the even meromorphic function

$$A(z) = \frac{\text{sinc}_{\text{rad}} z}{\Phi(z)}, \quad (2.5)$$

interpreted at common zeros by meromorphic continuation.

**Theorem 2.2** (Master factorization). *For every  $n \geq 1$ ,*

$$F_n(t) = \Phi(t) A(2^{-n}t), \quad (2.6)$$

*as an identity of entire functions after the implicit cancellations, and in the cancellation-free form  $F_n(t) - \Phi(t) = P_n(t)(\text{sinc}_{\text{rad}}(ht) - \Phi(ht))$ .*

*Proof.* Divide the two identities in (2.4). □

All stage dependence is thus an exact dilation of one universal meromorphic profile: the finite-stage problem separates into the rapidly decaying global carrier  $\Phi(t)$  and the universal microscopic transfer  $A(2^{-n}t)$ . The identity is stronger than an asymptotic statement – in the renormalized coordinate  $z = 2^{-n}t$  the profile  $\mathcal{R}_n(z) = F_n(2^n z)/\Phi(2^n z) = A(z)$  is independent of  $n$ , so the apparent convergence of the finite products is entirely the critical window moving to infinity, a boundary layer with an exact rather than asymptotic inner solution.

## 3 Exact forms and the canonical valuation product

### 3.1 Functional equations and renormalization products

The scaling identity  $\Phi(2z) = \text{sinc}_{\text{rad}} z \Phi(z)$  gives three equivalent descriptions of the transfer quotient and a dyadic functional equation:

$$A(z) = \frac{\Phi(2z)}{\Phi(z)^2} = \frac{\cos(z/2)}{\Phi(z/2)}, \quad (3.1)$$

$$\frac{A(z)}{A(z/2)} = \frac{\text{sinc}_{\text{rad}} z}{\text{sinc}_{\text{rad}}^2(z/2)} = \frac{z}{2} \cot \frac{z}{2}, \quad (3.2)$$

the second form of (3.1) by  $\text{sinc}_{\text{rad}} z = \text{sinc}_{\text{rad}}(z/2) \cos(z/2)$  and one cancellation. Iterating (3.2) from  $A(0) = 1$  yields the cotangent renormalization product

$$A(z) = \prod_{j=1}^{\infty} \left( \frac{z}{2^j} \cot \frac{z}{2^j} \right), \quad (3.3)$$

normally convergent on compact pole-free sets since  $u \cot u = 1 - u^2/3 + \mathcal{O}(u^4)$ ; collecting equal cosine scales in  $\text{sinc}_{\text{rad}} z = \prod_{j \geq 1} \cos(z/2^j)$  gives the companion form  $A(z) = \cos(z/2) \prod_{j \geq 3} \cos(z/2^j)^{-(j-2)}$ , which makes the dyadic pole hierarchy visually transparent.

### 3.2 The valuation-weighted canonical product

The arithmetic structure is clearest in a canonical product obtained by regrouping Euler's  $\text{sinc}_{\text{rad}} z = \prod_{r \geq 1} (1 - z^2/(\pi^2 r^2))$  across the dyadic scales: a fixed factor  $1 - z^2/(\pi^2 m^2)$  occurs in  $\text{sinc}_{\text{rad}}(z/2^k)$  once for every  $k \geq 1$  with  $2^k \mid m$ , i.e.  $\nu_2(m)$  times in  $\Phi$ .

**Theorem 3.1** (Canonical products). *As an entire function,*

$$\Phi(z) = \prod_{m=1}^{\infty} \left(1 - \frac{z^2}{\pi^2 m^2}\right)^{\nu_2(m)}, \quad (3.4)$$

*and as a meromorphic function,*

$$A(z) = \prod_{m=1}^{\infty} \left(1 - \frac{z^2}{\pi^2 m^2}\right)^{1-\nu_2(m)}, \quad (3.5)$$

*both locally uniformly in their natural domains; the exponent sum is legitimate because  $\sum_m \nu_2(m)m^{-2} < \infty$  (count the multiples of each  $2^j$ ).*

### 3.3 The complete divisor and local principal parts

**Theorem 3.2** (Divisor, removable values, and leading jets). *For every nonzero integer  $m$ ,*

$$\text{ord}_{z=\pi m} A = 1 - \nu_2(|m|). \quad (3.6)$$

*Explicitly:  $A$  has a simple zero at every odd multiple of  $\pi$ ; the points  $2\pi q$  with  $q$  odd are removable with the nonzero values*

$$A(2\pi q) = -\frac{1}{\Phi(\pi q)}; \quad (3.7)$$

*and at  $z_0 = \pi 2^r q > 0$  with  $q$  odd and  $r \geq 2$  there is a pole of order  $r - 1$  with leading jet*

$$A(z) = -\frac{z_0^{r-1}}{\Phi(\pi q)} (z - z_0)^{1-r} (1 + \mathcal{O}(z - z_0)). \quad (3.8)$$

*Equivalently, parametrizing the poles as  $z = 4\pi q'$  with  $q' \in \mathbb{Z} \setminus \{0\}$ , the pole order is  $1 + \nu_2(q')$ . In particular the nearest singularities are the simple poles at  $\pm 4\pi$ , with*

$$\text{Res}_{z=4\pi} A = -\frac{4\pi}{\Phi(\pi)} = -22.6923481977102997779\dots, \quad \text{Res}_{z=-4\pi} A = +\frac{4\pi}{\Phi(\pi)}. \quad (3.9)$$

*Proof.* The order formula is immediate from (3.5). For the local jets use  $A(z) = \cos(z/2)/\Phi(z/2)$  and put  $w = z/2$ ,  $w_0 = z_0/2$ . Iterating the head splitting  $r - 1$  times,

$$\Phi(w) = \left( \prod_{k=1}^{r-1} \text{sinc}_{\text{rad}}(w/2^k) \right) \Phi(w/2^{r-1});$$

at  $w = w_0$  every displayed factor vanishes linearly, all the associated integer multiples are even except the last, so the product of their sign slopes is  $-1$  and  $\Phi(w) = -((w - w_0)/w_0)^{r-1} \Phi(\pi q)(1 + \mathcal{O}(w - w_0))$ . For  $r \geq 2$ ,  $\cos(z_0/2) = 1$ , giving (3.8); for  $r = 1$ ,  $\cos(\pi q) = -1$  gives (3.7). The residue at  $4\pi$  is the case  $r = 2$ ,  $q = 1$  of (3.8).  $\square$

The leading-jet formula (3.8) evaluates, at every pole, the top Laurent coefficient in finite terms of  $\Phi$  at odd multiples of  $\pi$  — the first layer of the closed-jet program posed in [appendix 12](#).

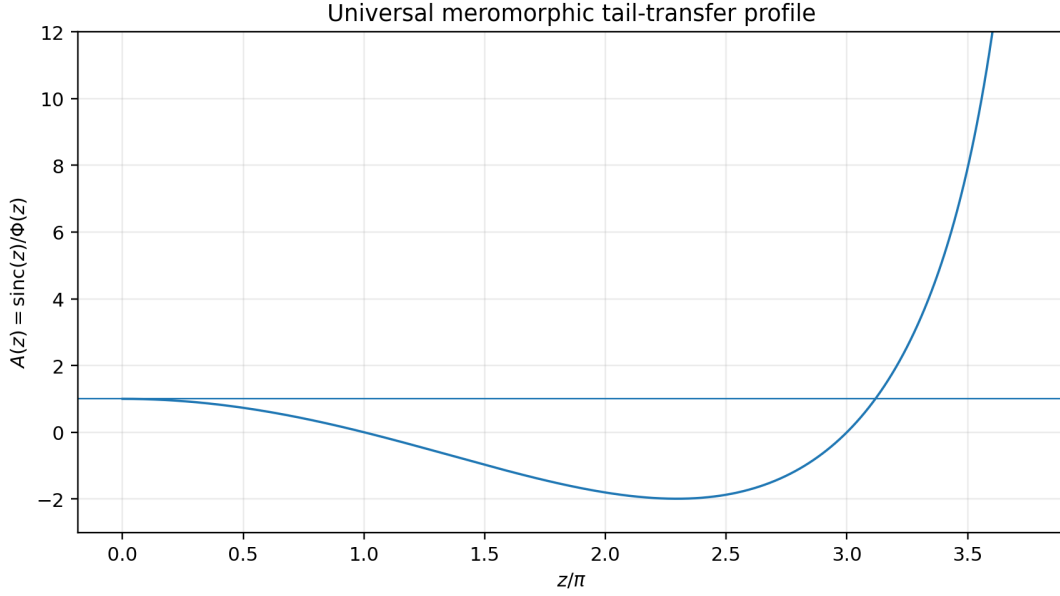


Figure 6: The universal transfer function on the first positive meromorphic cell: simple zeros at  $\pi$  and  $3\pi$ , a removable point at  $2\pi$  with value  $-1/\Phi(\pi)$ , and the first pole at  $4\pi$ .

#### 4 Digital divisor laws and the Thue–Morse sign

The signed divisor sequence  $1 - \nu_2(m)$  has three exact arithmetic transforms, and its cumulative parity is a digit sum.

**Theorem 4.1** (Digital divisor laws). *Let  $\text{wt}_2(M)$  be the binary digit sum. Then*

$$\sum_{m=1}^M \text{ord}_{\pi m} A = \sum_{m=1}^M (1 - \nu_2(m)) = M - \nu_2(M!) = \text{wt}_2(M), \quad (4.1)$$

*the total pole multiplicity on  $\{\pi, \dots, M\pi\}$  is  $\lceil M/2 \rceil - \text{wt}_2(M)$ , and the divisor has the Lambert and Dirichlet series*

$$\begin{aligned} \sum_{m \geq 1} (1 - \nu_2(m)) q^m &= \frac{q}{1-q} - \sum_{j \geq 1} \frac{q^{2^j}}{1-q^{2^j}} \quad (|q| < 1), \\ \sum_{m \geq 1} \frac{1 - \nu_2(m)}{m^s} &= \zeta(s) \frac{2^s - 2}{2^s - 1} \quad (\Re s > 1). \end{aligned} \quad (4.2)$$

*Proof.* Legendre’s formula  $\nu_2(M!) = M - \text{wt}_2(M)$  gives (4.1); the pole count is  $\sum_{m \leq M} \max\{\nu_2(m) - 1, 0\} = \sum_{j \geq 2} \lfloor M/2^j \rfloor = \nu_2(M!) - \lfloor M/2 \rfloor$ . Since  $\nu_2(m) = \sum_{j \geq 1} \mathbf{1}_{2^j | m}$ , summation in  $m$  gives both series, using  $\sum_m \nu_2(m) m^{-s} = \zeta(s)/(2^s - 1)$ .  $\square$

**Corollary 4.2** (Thue–Morse sign law). *For every integer  $M \geq 0$  and every real  $x \in (\pi M, \pi(M+1))$ ,*

$$\text{sgn } A(x) = (-1)^{\text{wt}_2(M)}. \quad (4.3)$$

*Proof.*  $A > 0$  near 0; crossing  $\pi m$  flips the sign according to the parity of the signed order  $1 - \nu_2(m)$  (poles flip parity exactly as zeros do), and the cumulative parity through  $M$  is  $\text{wt}_2(M)$  by (4.1).  $\square$

The Thue–Morse sequence thus appears twice in the theory of the same splines, by different mechanisms: as subset-sum parity in the spatial truncated-power coefficients (equation (3.2) for the prefix family), and as a cumulative valuation divisor in the signs of the universal Fourier correction.

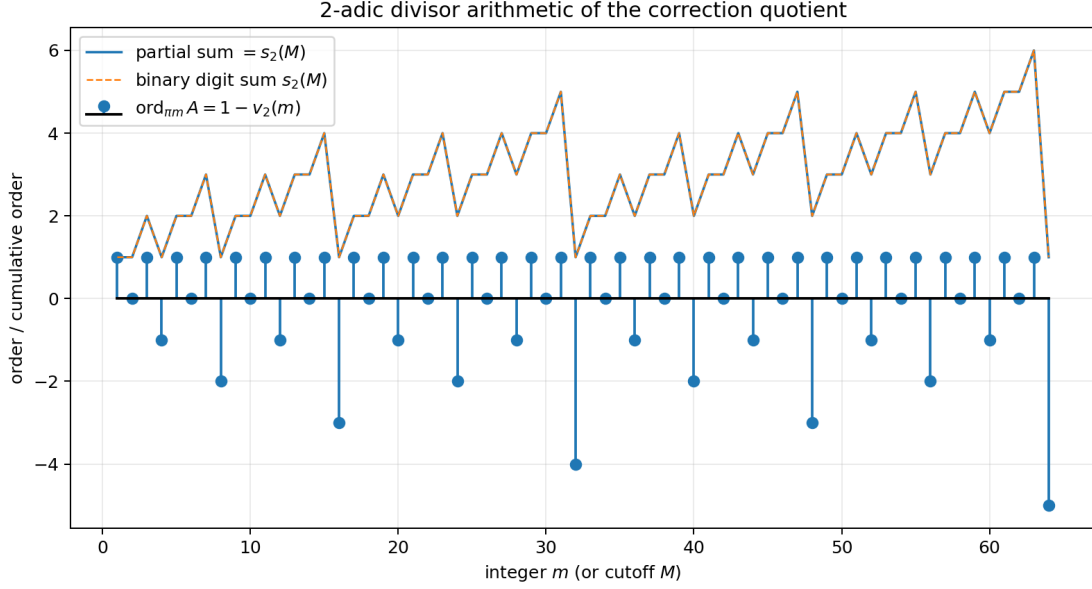


Figure 7: The signed divisor order  $1 - \nu_2(m)$  and its cumulative sum, which coincides exactly with the binary digit sum  $\text{wt}_2(M)$ .

## 5 Taylor coefficients: Bernoulli series, radius, and dominant poles

### 5.1 The local logarithm and its arithmetic explanation

For  $|z| < \pi$ , the zero-free branch of  $\log A$  at the origin is

$$\log A(z) = \sum_{r=1}^{\infty} \ell_r z^{2r}, \quad \ell_r = -\frac{2^{2r-1} |B_{2r}|}{r (2r)!} \frac{4^r - 2}{4^r - 1}, \quad (5.1)$$

the local series already recorded with the repeated-integration corpus. The canonical product explains its shape arithmetically: logarithmic expansion of (3.5) gives

$$\log A(z) = -\sum_{r=1}^{\infty} \frac{z^{2r}}{r\pi^{2r}} \sum_{m \geq 1} \frac{1 - \nu_2(m)}{m^{2r}} = -\sum_{r=1}^{\infty} \frac{\zeta(2r)}{r\pi^{2r}} \frac{2^{2r} - 2}{2^{2r} - 1} z^{2r}, \quad (5.2)$$

by the divisor Dirichlet series (4.2); Euler's evaluation of  $\zeta(2r)$  recovers (5.1). The coefficient factor  $(4^r - 2)/(4^r - 1)$  is thus the divisor zeta quotient  $\zeta(2r)^{-1} \sum_m (1 - \nu_2(m)) m^{-2r}$ , and simultaneously the exact  $2r$ th cumulant defect ratio of appendix 9.

Writing  $A(z) = \sum_{r \geq 0} \rho_r z^{2r}$ , differentiation of  $A = \exp \sum \ell_r w^r$  in  $w = z^2$  gives the exact rational recurrence  $r\rho_r = \sum_{k=1}^r k\ell_k \rho_{r-k}$ , whence

$$\begin{aligned} A(z) = 1 - \frac{z^2}{9} + \frac{2z^4}{2025} + \frac{z^6}{2679075} + \frac{58z^8}{10247461875} \\ + \frac{12622z^{10}}{345944065438125} + \frac{449289436z^{12}}{1933714893977851359375} + \cdots \end{aligned} \quad (5.3)$$

### 5.2 Radius and dominant-pole asymptotics

**Theorem 5.1** (Taylor radius and dominant-pole law). *The Maclaurin series of  $A$  has radius of convergence exactly  $4\pi$ : the zeros at  $\pm\pi, \pm3\pi$  do not obstruct analyticity, the points  $\pm2\pi$  are removable, and the first*



*genuine singularities are the simple poles at  $\pm 4\pi$ . Moreover there is a function  $H$  holomorphic in  $|z| < 8\pi$  with*

$$A(z) = \frac{2/\Phi(\pi)}{1 - z^2/(4\pi)^2} + H(z), \quad (5.4)$$

*so for every fixed  $\rho < 8\pi$ ,*

$$\begin{aligned} \rho_r &= \frac{2}{\Phi(\pi)(4\pi)^{2r}} + \mathcal{O}_\rho(\rho^{-2r}), \\ \lim_{r \rightarrow \infty} \rho_r(4\pi)^{2r} &= \frac{2}{\Phi(\pi)} = 3.6115993860280564486832331013\dots, \end{aligned} \quad (5.5)$$

*and  $\rho_{r+1}/\rho_r \rightarrow (16\pi^2)^{-1}$ . In particular  $\rho_r > 0$  for all sufficiently large  $r$ .*

*Proof.* The paired principal parts at  $\pm 4\pi$  determined by (3.9) sum to the rational model in (5.4); subtraction removes both first poles, the next poles are the double poles at  $\pm 8\pi$ , and Cauchy's estimate on  $|z| = \rho < 8\pi$  bounds the remainder coefficients.  $\square$

Exponentiating the convergent logarithmic series is justified only for  $|z| < \pi$ , but the resulting formal recurrence computes the Taylor series of the meromorphic continuation, convergent throughout  $|z| < 4\pi$  — the ordinary series reaches four times farther than the logarithm.

The dominant-pole approximation is accurate remarkably early: exact rational computation gives

| $r$                 | 3       | 5       | 10       | 20           | 30                | 50                    |
|---------------------|---------|---------|----------|--------------|-------------------|-----------------------|
| $\rho_r(4\pi)^{2r}$ | 1.46985 | 3.58279 | 3.611538 | 3.6115993859 | 3.611599386028056 | 3.6115993860280564487 |

with  $\rho_r > 0$  verified for  $2 \leq r \leq 50$ .

**Proposition 5.2** (Arithmetic Darboux hierarchy). *For  $q \geq 1$  put  $R_q = 4\pi q$  and  $d_q = 1 + \nu_2(q)$ , and write the principal part of  $A$  at  $R_q$  as  $\sum_{\ell=1}^{d_q} c_{q,\ell}(z - R_q)^{-\ell}$ . Then for each  $Q \geq 1$  and  $R_Q < \rho < R_{Q+1}$ ,*

$$\rho_r = \sum_{q=1}^Q \sum_{\ell=1}^{d_q} 2(-1)^\ell c_{q,\ell} R_q^{-2r-\ell} \binom{2r+\ell-1}{\ell-1} + \mathcal{O}_{Q,\rho}(\rho^{-2r}) : \quad (5.6)$$

*each positive integer  $q$  contributes the exponential scale  $(4\pi q)^{-2r}$  times a polynomial in  $r$  of degree  $\nu_2(q)$ , so the 2-adic hierarchy of pole orders becomes an arithmetic hierarchy of coefficient corrections*

$$\rho_r \sim \frac{2}{\Phi(\pi)(4\pi)^{2r}} + \frac{\alpha_1 r + \alpha_0}{(8\pi)^{2r}} + \frac{\beta_0}{(12\pi)^{2r}} + \frac{\gamma_2 r^2 + \gamma_1 r + \gamma_0}{(16\pi)^{2r}} + \dots, \quad (5.7)$$

*with the top constant of each block determined by the leading jet (3.8).*

*Proof.* Evenness pairs the principal part at  $-R_q$  with signs  $(-1)^\ell$ ; the coefficient of  $z^{2r}$  in a paired principal part is  $2(-1)^\ell c_{q,\ell} R_q^{-2r-\ell} \binom{2r+\ell-1}{\ell-1}$ . Subtract all paired principal parts with  $q \leq Q$  and apply Cauchy's estimate in  $|z| < R_{Q+1}$ .  $\square$

**Conjecture 5.3** (Global coefficient positivity). *Apart from  $\rho_1 = -1/9$ , every Taylor coefficient of  $A$  is positive.*

The dominant-pole law proves this for all large  $r$ , and exact arithmetic verifies it through  $r = 50$ , but all nonconstant coefficients of  $\log A$  are negative, so exponentiation produces alternating combinatorial sums and the finite gap is genuinely open. Candidate routes: an explicit positive representation of  $A(z) - 1 + z^2/9$  in  $z^2$ ; a majorization of the negative logarithmic cumulants after exponentiation; a pole-subtracted Mittag-Leffler expansion with positivity-preserving remainder; or a sign-regular interpretation of the tail comparison.

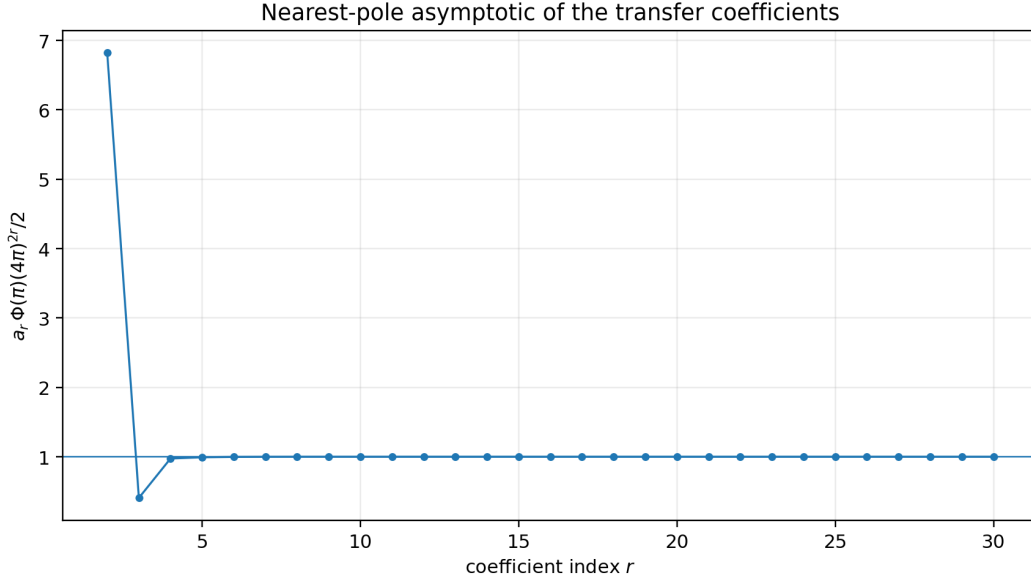


Figure 8: Normalized exact rational coefficients  $\rho_r \Phi(\pi)(4\pi)^{2r}/2$  against the nearest-pole prediction 1; the asymptotic is already highly accurate from  $r = 4$  onward. The nearly linear semilogarithmic error profile is consistent with the double poles at  $\pm 8\pi$ , whose contribution carries a linear polynomial factor in  $r$  per equation (5.7).

## 6 The finite zero filtration and its divisor realization

Both  $F_n$  and  $\Phi$  vanish exactly on  $2\pi\mathbb{Z} \setminus \{0\}$ , but with different multiplicities — a distinction invisible in magnitude plots and precisely responsible for the zeros and poles of the quotient.

**Theorem 6.1** (Exact finite multiplicities). *Let  $m \in \mathbb{Z} \setminus \{0\}$  and  $v = \nu_2(|m|)$ . Then  $\text{ord}_{2\pi m} \Phi = v + 1$ , while*

$$\text{ord}_{2\pi m} F_n = \min(n, v+1) + \mathbf{1}_{v \geq n-1} = \min(v+1, n-1) + 2 \cdot \mathbf{1}_{v \geq n-1} = \begin{cases} v+1, & v \leq n-2, \\ n+1, & v \geq n-1. \end{cases} \quad (6.1)$$

(The two groupings were derived independently in the two sources, from the factorizations  $F_n = P_n \cdot \text{sinc}_{\text{rad}}(t/2^n)$  and  $F_n = (\prod_{k < n} \text{sinc}_{\text{rad}}(t/2^k)) \text{sinc}_{\text{rad}}^2(t/2^n)$  respectively, and agree in all regimes.) Consequently the defect  $\Delta_n(m) = \text{ord}_{2\pi m} F_n - \text{ord}_{2\pi m} \Phi$  is

$$\Delta_n(m) = \begin{cases} 0, & v \leq n-2 \quad (\text{exact jet resolution}), \\ 1, & v = n-1 \quad (\text{over-resolved by one order}), \\ 0, & v = n \quad (\text{exact again}), \\ n-v, & v \geq n+1 \quad (\text{unresolved high dyadic jet}). \end{cases} \quad (6.2)$$

*Proof.*  $\text{sinc}_{\text{rad}}(t/2^k)$  vanishes at  $t = 2\pi m$  iff  $2^{k-1} \mid m$ , i.e.  $k \leq v+1$ ; among  $k = 1, \dots, n$  this contributes  $\min(n, v+1)$ , and the duplicated smallest box one more exactly when  $v \geq n-1$ .  $\square$

**Corollary 6.2** (Divisor-level realization). *For every nonzero integer  $M$ ,  $\text{ord}_{t=2^n\pi M} (F_n/\Phi) = 1 - \nu_2(|M|)$ : at the critical scaling  $t = 2^n z$  the three layers  $v = n-1, v = n, v \geq n+1$  of (6.2) are exactly the divisor of  $A$  — the over-resolved layer  $v = n-1$  is the simple zero of  $A$  at odd multiples of  $\pi$ , the re-agreement layer  $v = n$  is the removable point at  $2\pi \cdot \text{odd}$ , and each unresolved layer  $v \geq n+1$  is the pole of order  $v - n$ .*

## 7 The critical spectral boundary layer

### 7.1 The sharp relative-convergence window

Absolute convergence  $F_n \rightarrow \Phi$  is strong on the real axis because both sides decay; relative convergence has an exact scale.

**Theorem 7.1** (Sharp window criterion). *Let  $T_n > 0$ . With the quotient understood meromorphically,*

$$\sup_{|t| \leq T_n} \left| \frac{F_n(t)}{\Phi(t)} - 1 \right| \longrightarrow 0 \quad \Longleftrightarrow \quad \frac{T_n}{2^n} \longrightarrow 0, \quad (7.1)$$

and in the convergent regime

$$\sup_{|t| \leq T_n} \left| \frac{F_n(t)}{\Phi(t)} - 1 \right| = \frac{T_n^2}{9 \cdot 4^n} (1 + \mathcal{O}((T_n/2^n)^2)). \quad (7.2)$$

For fixed  $0 < c < 4\pi$  the critical supremum  $\sup_{|t| \leq c2^n} |F_n/\Phi - 1| = \max_{|z| \leq c} |A(z) - 1|$  is independent of  $n$  and nonzero; more generally, truncating the Taylor series of  $A$  after  $J - 1$  terms leaves a relative error  $\mathcal{O}(4^{-J\varepsilon^n})$  on the subcritical band  $|t| \leq 2^{(1-\varepsilon)n}$ .

*Proof.* By theorem 2.2,  $F_n/\Phi = A(t/2^n)$ . If  $T_n/2^n \rightarrow 0$ , the local expansion  $A(z) = 1 - z^2/9 + \mathcal{O}(z^4)$  gives both convergence and (7.2). Otherwise a subsequence has  $T_n/2^n \geq \varepsilon > 0$ , and since  $A$  is nonconstant analytic,  $\max_{|z| \leq \varepsilon} |A(z) - 1| > 0$  (infinite if the window reaches a pole). The critical identity is a change of variables.  $\square$

Hence  $2^n$  is not merely a convenient scale but the exact boundary between uniform relative convergence and a nontrivial limiting profile.

### 7.2 Two conditioning thresholds and the deconvolution obstruction

Bandlimited to  $|t| \leq c2^n$ , the exact relation between the finite and limiting transforms is the Fourier multiplier identity  $\Pi f_n = A(2^{-n}D) \Pi$  up with symbol  $A(2^{-n}t)$ . Forward application is governed by the first pole, inversion by the first zero.

**Theorem 7.2** (Forward and inverse thresholds). *Let  $M_+(c) = \sup_{|z| \leq c} |A(z)|$  and  $M_-(c) = \sup_{|z| \leq c} |A(z)|^{-1}$ . Then  $M_+(c) < \infty$  iff  $c < 4\pi$ , and  $M_-(c) < \infty$  iff  $c < \pi$ , with the blowup laws*

$$M_+(c) \sim \frac{4\pi}{\Phi(\pi)(4\pi - c)} \quad (c \uparrow 4\pi), \quad M_-(c) \sim \frac{\pi \Phi(\pi)}{\pi - c} \quad (c \uparrow \pi), \quad (7.3)$$

the second from the local factorization  $A(z) = \frac{\pi - z}{\pi \Phi(\pi)} (1 + \mathcal{O}(\pi - z))$  near the first zero (using  $\Phi(\pi/2) = \frac{\pi}{2}\Phi(\pi)$ ).

**Theorem 7.3** (No global stable convolutional correction). *There is no compactly supported distribution, finite measure, or  $L^1$  kernel  $K_n$  with  $f_n = \text{up} * K_n$ , and none  $L_n$  with  $\text{up} = f_n * L_n$ .*

*Proof.* The first identity forces  $\widehat{K_n \text{ ang}}(t) = A(2^{-n}t)$  everywhere by meromorphic continuation; the transform of a compactly supported distribution is entire and that of a finite measure is bounded on  $\mathbb{R}$ , while  $A(2^{-n}t)$  has a real pole at  $4\pi 2^n$ . The reverse identity requires the symbol  $1/A(2^{-n}t)$ , with a real pole already at  $\pi 2^n$ .  $\square$

The obstruction is stronger than ill-conditioning: the global translation-invariant correction kernel does not exist in any of the usual stable classes. Away from the divisor one still has the exact shell transfer

$$\log |F_n(2^n z)| - \log |\Phi(2^n z)| = \log |A(z)|, \quad (7.4)$$

so every asymptotic decay law for the limiting product on the sequence  $t = 2^n z$  transfers to the finite products with an exact  $O(1)$  profile correction — a direct bridge from the Fourier-decay corpus (which studies the envelope and transfer-operator behaviour of  $\Phi$ ) to the finite piecewise-polynomial approximants, to be supplemented at divisor points by the multiplicity filtration (6.1).

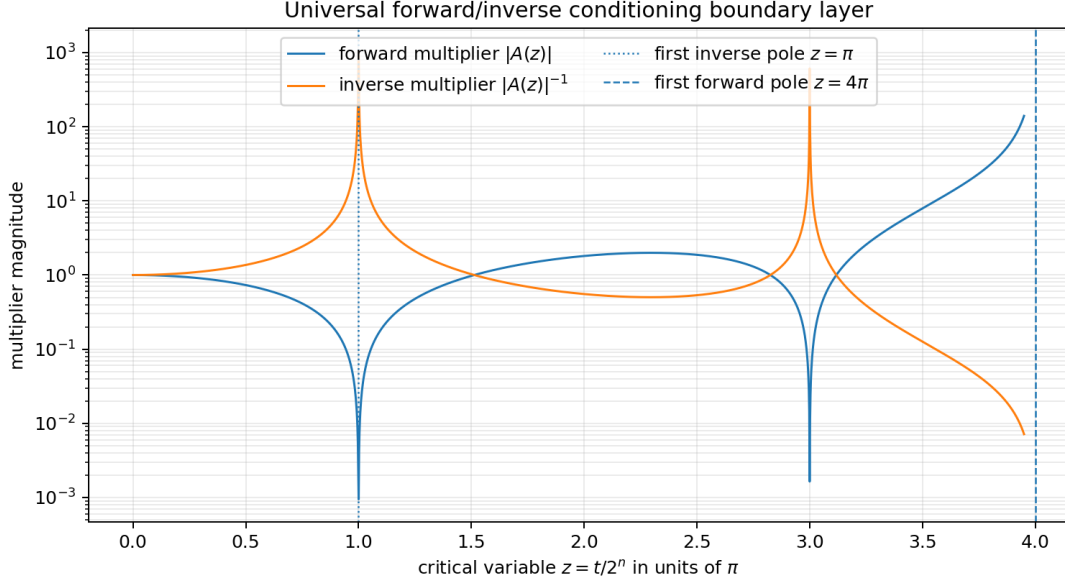


Figure 9: Forward and inverse multiplier magnitudes on the critical scale  $z = t/2^n$ : inverse instability begins at the first zero  $z = \pi$ , forward instability at the first pole  $z = 4\pi$ , with further inverse peaks at the odd multiples of  $\pi$ .

## 8 All-orders norm asymptotics

### 8.1 The weighted- $L^p$ master theorem

The local Taylor series of  $A$  promotes to a global norm theorem in every polynomially weighted Fourier space, because the head product supplies arbitrarily many powers of algebraic decay: for every fixed  $N$  and  $n \geq N$ ,  $|P_n(t)| \leq \prod_{k=1}^N \min(1, 2^k/|t|) \leq C_N(1 + |t|)^{-N}$ . For  $s \geq 0$ ,  $1 \leq p \leq \infty$ , write  $\|g\|_{L_s^p} = \|(1 + |t|)^s g\|_{L^p(\mathbb{R})}$ .

**Theorem 8.1** (Weighted Fourier expansion). *Let  $J \geq 1$ ,  $s \geq 0$ ,  $1 \leq p \leq \infty$ . As  $n \rightarrow \infty$ ,*

$$4^{Jn} \left[ F_n(t) - \Phi(t) \sum_{j=0}^{J-1} \rho_j 4^{-jn} t^{2j} \right] \longrightarrow \rho_J t^{2J} \Phi(t) \quad \text{in } L_s^p(\mathbb{R}), \quad (8.1)$$

and hence  $\left\| F_n - \Phi \sum_{j < J} \rho_j 4^{-jn} t^{2j} \right\|_{L_s^p} = |\rho_J| 4^{-Jn} \|t^{2J} \Phi\|_{L_s^p} + o(4^{-Jn})$ .

*Proof.* Let  $D_J(z) = \text{sinc}_{\text{rad}} z - \Phi(z) \sum_{j < J} \rho_j z^{2j}$ , an entire function with  $D_J(z) = \rho_J z^{2J} + \mathcal{O}(z^{2J+2})$ , and let  $H_J(z) = D_J(z)/z^{2J}$  (value  $\rho_J$  at 0), continuous and bounded on  $\mathbb{R}$  since  $|D_J(z)| \leq C_J |z|^{2J}$  for  $|z| \geq 1$ . The two identities (2.4) give exactly

$$4^{Jn} \left[ F_n(t) - \Phi(t) \sum_{j < J} \rho_j 4^{-jn} t^{2j} \right] = t^{2J} P_n(t) H_J(2^{-n} t).$$

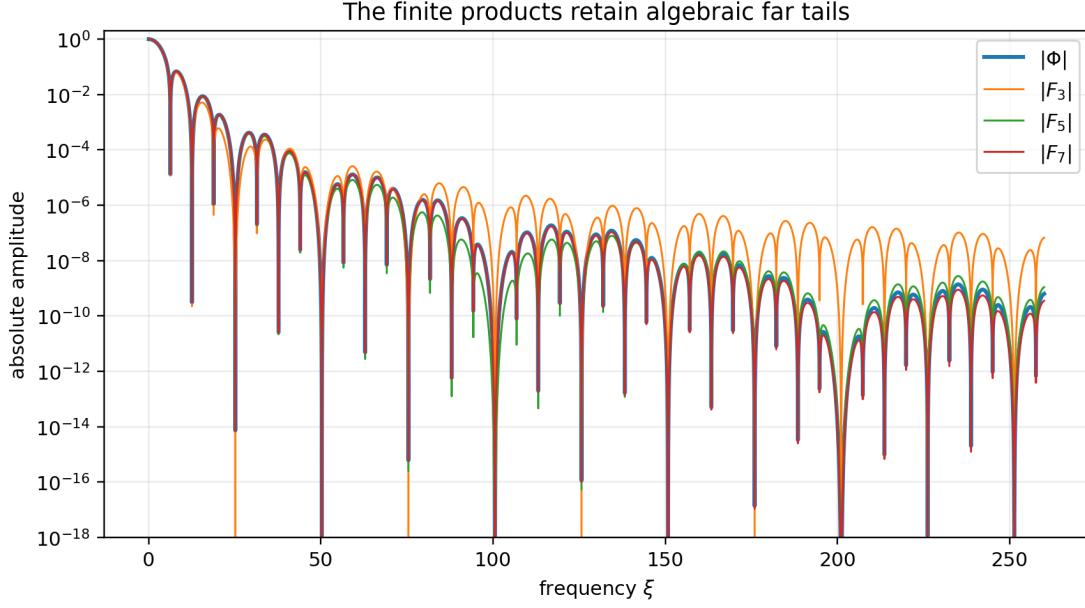


Figure 10: Absolute finite and limiting spectra on a logarithmic scale: each fixed finite product has only algebraic far-field decay (appendix 10), while  $\Phi$  is rapidly decreasing; increasing  $n$  pushes the discrepancy outward to the boundary layer  $|t| \asymp 2^n$ .

Pointwise,  $P_n \rightarrow \Phi$  and  $H_J(2^{-n}t) \rightarrow \rho_J$ ; choosing  $N > s + 2J + 1/p$  in the head-decay bound gives an  $n$ -independent  $L^p$  majorant after multiplication by  $(1 + |t|)^s$ , and dominated convergence concludes.  $\square$

**Corollary 8.2** (Universal leading profile and norm constants).  $4^n(F_n - \Phi) \rightarrow -\frac{t^2}{9}\Phi$  in every  $L_s^p$ , and  $4^n \|F_n - \Phi\|_{L^p} \rightarrow \frac{1}{9} \|t^2\Phi\|_{L^p}$ . The unweighted constants, by lobe-wise quadrature, are

$$C_\infty = \frac{1}{9} \sup_t t^2 |\Phi(t)| = 0.650851567393992 \dots \quad (\text{maximizer } t_* = 3.750573330779642 \dots), \quad (8.2)$$

$C_1 = \frac{1}{9} \int t^2 |\Phi| dt = 11.0486184433431 \dots$  and  $C_2 = \frac{1}{9} (\int t^4 |\Phi|^2 dt)^{1/2} = 2.00390775107366 \dots$  (reproducible numerical values, not certified enclosures); the measured scaled norms at  $n = 8$  agree with all three limits to five significant digits.

## 8.2 Sobolev expansions, energies, and monotonicity

Since  $\widehat{\text{up}^{(2j)}_{\text{ang}}} = (-1)^j t^{2j} \Phi$ , the  $p = 2$  case transfers to physical space: for every fixed real  $s \geq 0$  and  $J \geq 1$ , with  $a_j = (-1)^j \rho_j$ ,

$$4^{Jn} \left[ f_n - \sum_{j=0}^{J-1} a_j 4^{-jn} \text{up}^{(2j)} \right] \longrightarrow a_J \text{up}^{(2J)} \quad \text{in } H^s(\mathbb{R}) \quad (8.3)$$

(also in every  $C^m$  via  $p = \infty$ ) – the known physical-space expansion

$$f_n = \text{up} + \frac{4^{-n}}{9} \text{up}'' + \frac{2 \cdot 16^{-n}}{2025} \text{up}^{(4)} - \frac{64^{-n}}{2679075} \text{up}^{(6)} + \dots$$

is simultaneously an asymptotic expansion in every fixed Sobolev topology. On the energy level, with the convolution coefficients  $d_q = \sum_{j=0}^q \rho_j \rho_{q-j}$  and  $c_q = \sum_{j=1}^{q-1} \rho_j \rho_{q-j}$ , expanding  $|F_n|^2 = |\Phi|^2 A(ht)^2$

gives, for every fixed  $s$  and  $M$ ,

$$\begin{aligned} \|f_n\|_{\dot{H}^s}^2 &= \sum_{q=0}^{M-1} d_q 4^{-qn} \|\text{up}\|_{\dot{H}^{s+q}}^2 + \mathcal{O}_{s,M}(4^{-Mn}) \\ &= \|\text{up}\|_{\dot{H}^s}^2 - \frac{2 \cdot 4^{-n}}{9} \|\text{up}\|_{\dot{H}^{s+1}}^2 + \frac{29 \cdot 16^{-n}}{2025} \|\text{up}\|_{\dot{H}^{s+2}}^2 + \cdots, \end{aligned} \quad (8.4)$$

$$\begin{aligned} \|f_n - \text{up}\|_{\dot{H}^s}^2 &= \sum_{q=2}^{M-1} c_q 4^{-qn} \|\text{up}\|_{\dot{H}^{s+q}}^2 + \mathcal{O}_{s,M}(4^{-Mn}) \\ &= \frac{16^{-n}}{81} \|\text{up}\|_{\dot{H}^{s+2}}^2 - \frac{4 \cdot 64^{-n}}{18225} \|\text{up}\|_{\dot{H}^{s+3}}^2 + \cdots. \end{aligned} \quad (8.5)$$

Two consequences: the error-norm ratio  $\|f_{n+1} - \text{up}\|_{\dot{H}^s} / \|f_n - \text{up}\|_{\dot{H}^s} \rightarrow \frac{1}{4}$ , and  $\|f_n\|_{\dot{H}^s}$  is eventually strictly increasing to  $\|\text{up}\|_{\dot{H}^s}$  (from the negative  $4^{-n}$  term in (8.4)).

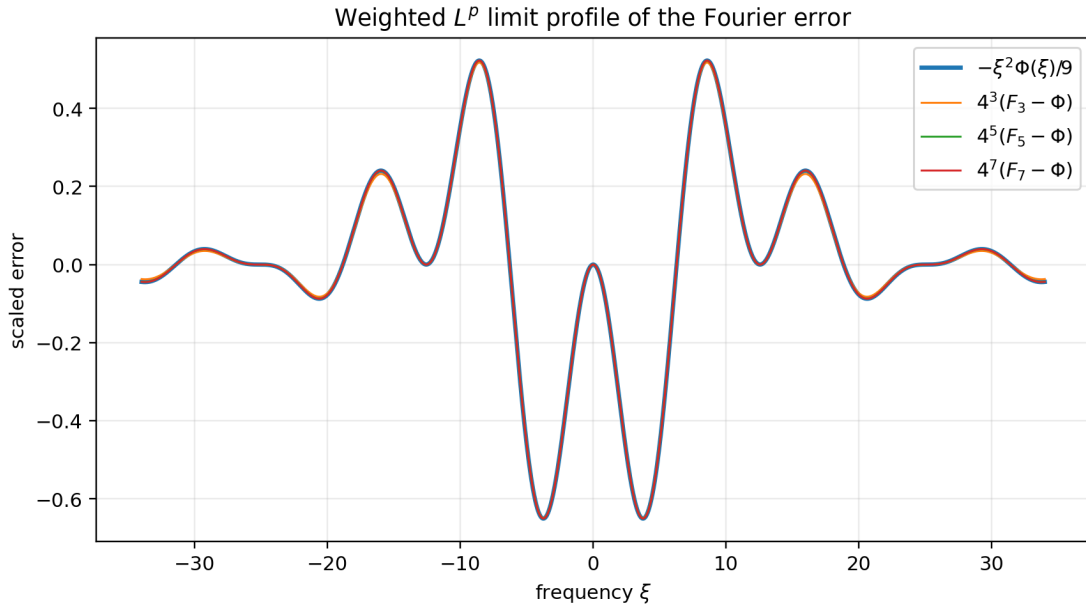


Figure 11: The scaled errors  $4^n(F_n - \Phi)$  converge globally to  $-t^2\Phi/9$ , not only near the origin; [theorem 8.1](#) makes the convergence valid in every polynomially weighted  $L^p$ .

## 9 Cumulant defects and finite moment polynomials

The transfer function is the quotient of the characteristic function of a uniform tail by that of the true self-similar tail, so cumulants diagonalize the comparison: comparing  $\log A(ht)$  with the cumulant series of (2.3) gives, for every  $r \geq 1$ ,

$$\kappa_{2r}(X) = \frac{2^{2r} B_{2r}}{2r(4^r - 1)}, \quad \kappa_{2r}(X_n) - \kappa_{2r}(X) = \frac{2^{2r} B_{2r}}{2r} \cdot \frac{4^r - 2}{4^r - 1} \cdot 4^{-rn}, \quad (9.1)$$

with all odd cumulants zero: the  $r$ th term of the Bernoulli series (5.1), evaluated at  $ht$ , is the  $2r$ th cumulant term of  $X_n$  minus that of  $X$ , so that  $\kappa_{2r}(X_n) - \kappa_{2r}(X) = (-1)^r (2r)! \ell_r 4^{-rn}$ , and the leading  $4^{-n}$  rate is exactly the variance mismatch  $\text{Var}(X_n) - \text{Var}(X) = \frac{2}{9} 4^{-n}$  between the uniform tail (variance  $\frac{1}{3}$ ) and the true tail (variance  $\frac{1}{9}$ ). Bell-polynomial conversion gives finite-stage moment

polynomials in  $q = 4^{-n}$  of degree at most the half-order,

$$\mathbb{E}X_n^2 = \frac{1}{9} + \frac{2}{9}q, \quad \mathbb{E}X_n^4 = \frac{19}{675} + \frac{4}{27}q + \frac{16}{675}q^2, \quad \mathbb{E}X_n^6 = \frac{583}{59535} + \frac{38}{405}q + \frac{16}{405}q^2 - \frac{16}{59535}q^3, \quad (9.2)$$

against the limiting moments  $\mathbb{E}X^2 = \frac{1}{9}$ ,  $\mathbb{E}X^4 = \frac{19}{675}$ ,  $\mathbb{E}X^6 = \frac{583}{59535}$ ,  $\mathbb{E}X^8 = \frac{132809}{32531625}$  (cumulants  $\frac{1}{9}$ ,  $-\frac{2}{225}$ ,  $\frac{16}{3969}$ ,  $-\frac{16}{3825}$ ) that determine the accelerated rules of [appendix 11](#).

## 10 Exact algebraic Fourier tails and the Sobolev threshold

For fixed  $n$  the finite spline has only finite smoothness, visible in the exact algebraic factorization

$$F_n(t) = 2^{n(n+3)/2} t^{-(n+1)} \left( \prod_{k=1}^{n-1} \sin(t/2^k) \right) \sin^2(t/2^n). \quad (10.1)$$

The oscillatory numerator prevents a pointwise constant, but its mean square is exactly computable through a dyadic trigonometric integral.

**Theorem 10.1** (Exact mean-square tail). *Let  $I_n = \frac{1}{\pi} \int_0^\pi \sin^4 u \prod_{j=1}^{n-1} \sin^2(2^j u) du$ . Then*

$$I_n = \frac{2^{n+2} + (-1)^{n+1}}{3 \cdot 2^{2n+1}}, \quad I_1 = \frac{3}{8}, \quad I_2 = \frac{5}{32}, \quad I_3 = \frac{11}{128}, \quad I_4 = \frac{21}{512}, \quad I_5 = \frac{43}{2048}, \quad \dots, \quad (10.2)$$

and consequently

$$\begin{aligned} \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T t^{2n+2} |F_n(t)|^2 dt &= 2^{n(n+3)} I_n, \\ \int_{|t| > T} |F_n|^2 dt &= \frac{2^{n(n+3)+1} I_n}{2n+1} T^{-2n-1} + \mathcal{O}_n(T^{-2n-2}), \end{aligned} \quad (10.3)$$

with the ideal low-pass error  $\|f_n - \Pi_T f_n\|_{H^s}^2 = \frac{2^{n(n+3)} I_n}{\pi(2n+1-2s)} T^{-(2n+1-2s)} + \mathcal{O}_n(T^{-(2n+2-2s)})$  for  $s < n + \frac{1}{2}$ , and the sharp threshold

$$f_n \in H^s(\mathbb{R}) \iff s < n + \frac{1}{2}. \quad (10.4)$$

*Proof.* Let  $Q_n(u) = \prod_{j=1}^{n-1} \sin^2(2^j u)$ , with constant Fourier coefficient  $2^{1-n}$  and  $\cos 4u$ -coefficient  $c_n$ . Since  $\sin^4 u = (3 - 4 \cos 2u + \cos 4u)/8$  and all frequencies of  $Q_n$  are multiples of 4,  $I_n = \frac{3}{8} 2^{1-n} + \frac{c_n}{16}$ . The highest-frequency coefficient of  $Q_n$  is  $(-1)^{n-1} 2^{3-2n}$  at frequency  $2^{n+1} - 4$ ; multiplying by  $\sin^2(2^n u) = (1 - \cos(2^{n+1} u))/2$  gives the recurrence  $c_{n+1} = \frac{c_n}{2} - \frac{(-1)^{n-1} 2^{3-2n}}{4}$  with  $c_2 = -\frac{1}{2}$ , solved by  $c_n = -\frac{2^{2-n}}{3} - \frac{(-1)^{n-1} 2^{3-2n}}{3}$ ; substitution gives (10.2). The tail laws follow by substituting  $u = t/2^n$  in (10.1), averaging the bounded periodic numerator against  $t^{-\alpha}$  by integration by parts, and applying Plancherel; positivity of  $I_n$  makes the weighted integral diverge at  $s = n + \frac{1}{2}$ , proving sharpness.  $\square$

The threshold is the Fourier counterpart of the spline's final piecewise-constant derivative stage; (10.3) adds the leading constants, not just the exponents.

## 11 Positive moment-matched closures

### 11.1 The closure framework and the rate theorem

The construction (2.3) replaces the exact scaled tail law by a tractable one — a *closure*. For a symmetric probability measure  $\nu$  on  $[-1, 1]$  with characteristic function  $\psi_\nu(z) = \int e^{-izx} d\nu(x)$ , define

$$g_{n,\nu} = p_n * (D_h \nu), \quad \widehat{g_{n,\nu}}_{\text{ang}}(t) = P_n(t) \psi_\nu(2^{-n}t), \quad (11.1)$$



where  $D_h$  scales by  $h = 2^{-n}$ . The original spline is  $\nu = \text{Unif}[-1, 1]$ ; every finitely supported  $\nu$  gives a positive finite sum of shifted head splines  $g_{n,\nu} = \sum_j w_j p_n(\cdot - 2^{-n}t_j)$ , a piecewise-polynomial density of degree  $n - 1$ ; a  $\nu$  with piecewise-polynomial density of degree  $d$  gives degree at most  $n + d$ ; and the support is exactly  $[-1, 1]$  iff  $\text{supp } \nu$  reaches both endpoints (since  $\text{supp } p_n = [-1 + h, 1 - h]$ ).

**Theorem 11.1** (Moment matching determines the spectral order). *If  $\int x^k d\nu = \mathbb{E}X^k$  for  $0 \leq k \leq 2M - 1$  and  $c_M = \frac{(-1)^M}{(2M)!} (\int x^{2M} d\nu - \mathbb{E}X^{2M}) \neq 0$ , then for every  $s \geq 0$ ,  $1 \leq p \leq \infty$ ,*

$$4^{Mn} (\widehat{g_{n,\nu}}_{\text{ang}} - \Phi) \longrightarrow c_M t^{2M} \Phi(t) \quad \text{in } L_s^p(\mathbb{R}), \quad (11.2)$$

and correspondingly in every  $C^m$  and  $\dot{H}^s$  in physical space.

*Proof.*  $\psi_\nu(z) - \Phi(z) = c_M z^{2M} + \mathcal{O}(z^{2M+2})$  by symmetry and Taylor expansion at 0; the exact difference is  $P_n(t)(\psi_\nu(ht) - \Phi(ht))$ , and the bounded-remainder plus head-decay argument of [theorem 8.1](#) applies verbatim.  $\square$

**Theorem 11.2** (Arbitrary-order positive support-filling closures). *For every  $r \geq 0$  there is a symmetric, finitely supported probability measure  $\nu_r$  on  $[-1, 1]$  with positive mass at both endpoints matching the up moments through degree  $2r + 1$ ; consequently  $\|g_{n,\nu_r} - \text{up}\|_{C^m} = \mathcal{O}_{m,r}(4^{-(r+1)n})$  and likewise in every  $\dot{H}^s$ , with  $g_{n,\nu_r}$  nonnegative, of total mass one, and of support exactly  $[-1, 1]$ .*

*Proof.* Push the up law forward under  $x \mapsto x^2$  to a measure on  $[0, 1]$ ; a positive Gauss–Radau rule containing the endpoint 1 and exact through degree  $r$  exists by standard quadrature theory; lifting each node  $y_j$  to the symmetric pair  $\pm\sqrt{y_j}$  with half weights (a zero node to 0) gives  $\nu_r$ . The rates follow from [theorem 11.1](#).  $\square$

## 11.2 The explicit menu

Solving the moment equations with the rational moments of [appendix 9](#) yields two complementary explicit families: rules with minimal atom counts (irrational nodes allowed), and rules with dyadic rational nodes. Both sources’ overlapping entries agree exactly; the variance-matched three-point rule below was derived independently in both.

| closure          | tail measure $\nu$                                                                                                                                               | $M$ | $c_M$                    | structure                                              |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|--------------------------|--------------------------------------------------------|
| original uniform | $\text{Unif}[-1, 1]$                                                                                                                                             | 1   | $-1/9$                   | exact support; the spline $f_n$ itself                 |
| variance box     | $\text{Unif}[-1/\sqrt{3}, 1/\sqrt{3}]$                                                                                                                           | 2   | $-1/4050$                | narrower support; one rescaled box                     |
| Lobatto-3        | $\frac{8}{9}\delta_0 + \frac{1}{18}(\delta_{-1} + \delta_1)$                                                                                                     | 2   | $7/2025$                 | exact support; three shifts                            |
| Gaussian-3       | $\frac{32}{57}\delta_0 + \frac{25}{114}(\delta_{-a} + \delta_a)$ , $a = \sqrt{19/75}$                                                                            | 3   | $1238/334884375$         | narrower support; three shifts                         |
| dyadic-5         | weights $\frac{376}{675}, \frac{896}{2025}, \frac{1}{2025}$ at $0, \pm\frac{1}{2}, \pm 1$ (pair totals)                                                          | 3   | $71/21432600$            | exact support; five dyadic shifts                      |
| Lobatto-5        | $\frac{78976}{153675}\delta_0 + \frac{2470629}{10262075}(\delta_{-b} + \delta_b) + \frac{619}{270450}(\delta_{-1} + \delta_1)$ , $b = \sqrt{683/3087}$           | 4   | $1006207/24604155961875$ | exact support; five shifts                             |
| dyadic-7         | pair totals $\frac{10411}{2679075}, \frac{1003648}{2679075}, \frac{581632}{2679075}$ at $\pm 1, \pm\frac{1}{2}, \pm\frac{1}{4}$ , center $\frac{120376}{297675}$ | 4   | $330593/10493400960000$  | exact support; seven dyadic shifts                     |
| polynomial $q_3$ | density $\frac{35}{32}(1 - x^2)^3$                                                                                                                               | 2   | $2/22275$                | exact support; one $C^{n+2}$ density of degree $n + 6$ |

The convergence scales are  $4^{-Mn}$ : rate  $16^{-n}$  for the three  $M = 2$  rules,  $64^{-n}$  for Gaussian–3 and dyadic–5,  $256^{-n}$  for Lobatto–5 and dyadic–7. Rules including  $\pm 1$  preserve the exact support; Gaussian rules maximize order per atom; the polynomial density achieves  $16^{-n}$  with a single continuous piecewise-polynomial representation and the smallest leading constant of the  $M = 2$  entries ( $C'_{q3,\infty} = 0.0507\dots$  against  $C_{L3,\infty} = 1.952\dots$  and  $C_{VB,\infty} = 0.1394\dots$ , computed against  $\sup |t|^4 |\Phi| = 564.7796\dots$  at  $t = 16.3025\dots$ ). A fixed dyadic ladder is not guaranteed to retain positivity indefinitely, so [theorem 11.2](#) is the robust existence result while the dyadic menu gives convenient low-order formulas. Cancellation makes direct double-precision verification of the  $64^{-n}$  and  $256^{-n}$  rules unreliable; the verification packages evaluate the scaled point errors at 120 decimal digits and reproduce the predicted leading profiles (Gaussian–3 at  $n = 11$ : 3.21182 against the limit 3.21168; Lobatto–5 at  $n = 14$ : 873.5187 against 873.5103).

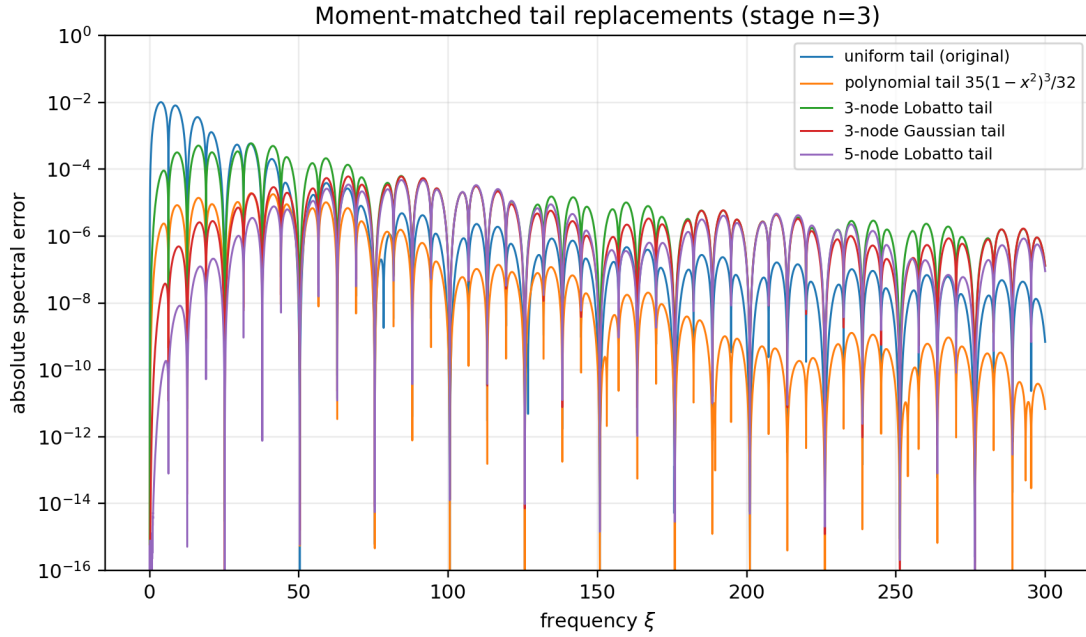


Figure 12: Absolute Fourier errors of the closure menu at stage  $n = 3$ ; the faster rates are visible over most of the frequency range, with leading profiles attaining maxima on progressively more distant lobes.

### 11.3 Three closure families and the prefix quadratures

Part III and this part together produce three structurally different positive acceleration families over the *same* head spline  $p_n$ :

1. *box mixtures* ([equation \(7.10\)](#)): quadrature of the squared-scale-mixture law ( $X \stackrel{d}{=} RU$  with  $Z = R^2 \sim \nu$ ) – each node contributes a box  $p_n * u_{\delta\sqrt{z_j}}$ , giving mixtures of box splines of full degree;
2. *atomic closures* (this part): quadrature of the up law itself (equivalently, Gauss–Radau on the pushforward under  $x \mapsto x^2$  lifted symmetrically) – each node contributes a translate  $p_n(\cdot - ht_j)$ , giving sums of shifted head splines of degree  $n - 1$ ;
3. *polynomial closures*: a continuous tail density, giving one smooth piecewise-polynomial density of slightly higher degree.

The variance-matched members of the first two families have the same leading constant  $|c_2| = 1/4050$  – the box mixture  $p_n * u_{2^{-n}/\sqrt{3}}$  and the variance box above are the same approximant seen from the

two quadrature variables — while the endpoint-preserving members differ genuinely (the box family’s Lobatto rule keeps  $p_n$  itself as a component; the atomic family’s endpoint atoms shift the head to  $\pm 1$ ). Moment order controls the global scale  $4^{-Mn}$  in every family; what distinguishes them is support exactness, smoothness, atom count, and the divisor of the critical transfer  $B_\nu = \psi_\nu/\Phi$ , which for generic rules has poles on the full dyadic zero lattice — the original uniform tail is special in sharing many zeros with  $\Phi$ , which is what produced the removable/pole hierarchy of [theorem 3.2](#). Moment order and critical zero matching are independent design objectives ([appendix 12](#)).

## 12 Directions, conjectures, and formalization targets

1. *Positivity* ([conjecture 5.3](#)) and the pole-resolved transseries (5.7): compute the full Laurent jets at  $4\pi q$  from finite products of  $\Phi$ -values at odd multiples of  $\pi$  (the leading jets are (3.8)), reorganize into an optimally truncated arithmetic transseries, and decide eventual monotonicity of the normalized ratio.
2. *Hybrid tail design*: for prescribed  $M, K$ , find positive  $\nu$  matching moments through  $2M - 1$  while  $\psi_\nu$  shares prescribed zero jets with  $\Phi$  at  $2\pi, \dots, 2\pi K$  — a finite-dimensional positive trigonometric moment problem (semidefinite search, then exact certificates); minimize atoms or polynomial degree. Rational or Padé spectral multipliers encoding the first poles offer an alternative postprocessing route around the kernel obstruction of [theorem 7.3](#), with evenness, Thue–Morse sign intervals, and no spurious real poles below a chosen band as natural constraints.
3. *Optimal closures*: minimize the number of atoms at fixed moment order with endpoint masses; minimize the first unmatched moment; determine the largest order retaining positivity on a fixed dyadic node ladder; minimize the forward multiplier norm on a target band (quadrature meets minimax filter design).
4. *Modified local recursions*: derive finite vector-valued refinement systems whose  $n$ th iterate is exactly a Gaussian, Lobatto, or polynomial-tail closure, preserving the one-line locality of (2.1) at higher order.
5. *Base- $q$  analogues*: for the base- $q$  sinc product  $\Pi_q(z) = \prod_{k \geq 1} \text{sinc}_{\text{rad}}(z/q^k)$  — not the centered characteristic function  $\varphi_q$  of (0.3.5), whose arguments carry the factor  $(1 - q)q^j/2$  — and  $A_q = \text{sinc}_{\text{rad}} z/\Pi_q$ , the Euler regrouping suggests  $\text{ord}_{\pi m} A_q = 1 - \nu_q(m)$  with divisor Dirichlet series  $\zeta(s)(q^s - 2)/(q^s - 1)$ ; determine which spectral and sign phenomena survive and which are genuinely binary (for  $q = 2$  the cumulative divisor collapses to the digit sum and the functional equation to the cotangent product).
6. *Two-parameter asymptotics*: allow the Fourier weight  $s_n$  to grow with  $n$  and locate the transition between the fixed-weight theorem and the exact boundary layer; on the physical side this quantifies how much spectral regularity a finite spline possesses at a given stage, refining (10.4).
7. *Transfer-operator bridge*: insert the shell identity (7.4) into the Fourier-decay corpus to obtain a joint envelope/profile theorem for finite products, with exceptional arithmetic points from (6.1) and uniformity near the first zero and pole.
8. *Lean targets*: the finite multiplicity formula (6.1); the divisor count and digit-sum identity (4.1); the Taylor radius and first residues; the  $I_n$  recurrence and closed form (10.2); the cumulant defect (9.1); exact moment verification, positivity, and support of the explicit closures; and the shift-sum identity behind (11.1). The weighted- $L^p$  theorem is modular: a bounded remainder, a finite-product decay lemma, and dominated convergence.

## 13 Verification and reproducibility

The two absorbed packages provide independent verification pipelines, retained verbatim under `assets/Rvachev_Piecewise_Approximation_Fourier_Images/` (exact SymPy coefficients through order 30, 120-digit mpmath evaluation of  $\Phi(\pi)$ , the residue, and the cancellation-sensitive accelerated errors, lobe-wise maximization, composite Gauss–Legendre norms on 700 lobes, symbolic verification of every quadrature rule; figures and CSV audit artifacts) and `assets/rvachev_fourier_frontier_report/` (exact coefficients through order 50, functional-identity checks, divisor partial sums against  $wt_2$ , finite multiplicity tables, 100-digit quadrature of  $I_n$  against the closed form, rational verification of the dyadic closures; figures, coefficient CSV, and text summary). Both scripts regenerate every figure and table from scratch (`fourier_piecewise_experiments.py`, `rvachev_fourier_experiments.py`). All shared constants agree between the two packages; the numerical work is ancillary throughout — every theorem above has a complete proof independent of the computations.

## Part V

# Finite Dyadic Sinc Products and Exact Transport Geometry of Rvachev Spline Approximants

### Abstract

Let

$$\Phi_N(\xi) = \prod_{j=1}^N \text{sinc}_{\text{rad}}\left(\frac{\xi}{2^j}\right), \quad \Phi(\xi) = \prod_{j=1}^{\infty} \text{sinc}_{\text{rad}}\left(\frac{\xi}{2^j}\right), \quad \text{sinc}_{\text{rad}} z = \frac{\sin z}{z},$$

and let  $u_N$  and  $u = \text{up}$  be their inverse Fourier transforms in the angular-frequency convention. The established Fabius–Rvachev corpus identifies  $u_N$  as a degree- $N - 1$  compactly supported spline with a Thue–Morse truncated-power formula and proves strong convergence, an all-orders  $4^{-N}$  deconvolution expansion, and signed  $q$ -Richardson acceleration. Those facts are used here as input rather than relabelled as new.

The main new result is an exact convergence-geometry package. Writing  $\mu_N = u_N \, dx$  and  $\mu = u \, dx$ , we prove that  $\mu_N$  increases to  $\mu$  in convex order while remaining more peaked about the origin. The density error changes sign at the fixed points  $\pm \frac{1}{2}$ , independently of  $N$ . Consequently  $W_1(\mu_N, \mu) = 4^{-N}/9$  for every  $N \geq 1$ , and for every  $N \geq 2$

$$W_1(\mu_N, \mu) = d_K(\mu_N, \mu) = \frac{4^{-N}}{9}, \quad d_{\text{TV}}(\mu_N, \mu) = \frac{2 \cdot 4^{-N}}{9},$$

while the stop-loss and order-two Zolotarev distances both equal  $4^{-N}/18$  for every  $N \geq 1$ . We also prove  $W_{\infty}(\mu_N, \mu) = 2^{-N}$ , derive an exact Thue–Morse spline formula for the call and put transforms of  $\mu_N$ , and show that the call-potential gap  $K_N = C - C_N$  has maximum and total area both equal to  $4^{-N}/18$ . These identities imply a positivity obstruction: no convex combination of consecutive finite-product laws can cancel the leading  $4^{-N}$  error in any of the preceding finite metrics; signed weights are structurally necessary for Richardson acceleration.

The report further derives the all-orders stop-loss profile, smooth-test and interior-quantile expansions, and a finite- $p$  versus  $p = \infty$  transport crossover. Fully reproducible FFT experiments verify the exact constants and give strong evidence for new entropy, forward-relative-entropy, Fisher-information, and fixed- $p$  Wasserstein asymptotics. The latter are explicitly marked conjectural where a weighted endpoint remainder theorem is still required.

## 1 Audit boundary, status discipline, and contribution map

1.1 Pinned corpus and recursive reading strategy

The source boundary is the repository commit

21758aba2f8e200342be2a7b321247dac80efc00

from 28 August 2026. A recursive inventory of Analysis/FabiusFunction/docs returned 95 paths with extension .tex. The tree contains the canonical exposition, imported papers, historical archive copies, a living consolidated frontier volume, and current thematic draft consolidations. Reading every historical revision as an independent mathematical source would overcount the same theorem many times. Accordingly, the audit followed the repository’s own consolidation policy:

1. inventory every .tex path recursively;
2. read the canonical exposition and the consolidated frontier volume as the current mathematical spine;
3. read the live draft manifest and every topic-specific source capable of overlapping finite sinc products, splines,  $q$ -Richardson acceleration, transport, entropy, or multiresolution analysis;
4. normalize archived duplicates against their live successor and use imported papers as literature baselines rather than repository discoveries;
5. run formula-level searches for the proposed new invariants: convex order, peakedness, single crossing, stop-loss, Zolotarev, exact Wasserstein/Kolmogorov/TV constants, and positive-mixture obstructions.

This is the same cumulative-audit methodology used by the recent frontier reports. The phrases “proved here” and “new” below mean that no explicit counterpart was located in the pinned corpus after the preceding source- and formula-level audit. They do not assert exhaustive worldwide priority.

1.2 Established material deliberately excluded from novelty claims

| Layer            | Audited prior material used as input                                                                                                                                                     |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Function theory  | Existence, uniqueness, support, symmetry, positivity, flatness, refinement equations, and the relation between the bounded Fabius CDF and the even up-density.                           |
| Probability      | Weighted sums of independent uniform variables, finite atomic and finite spline laws, exact tail coupling, reflection, and weak/strong convergence.                                      |
| Finite splines   | Thue–Morse truncated-power formulas, exact dyadic knots, repeated-integration formulas, support bounds, and exact rational evaluation at dyadic points.                                  |
| Fourier analysis | Infinite and finite sinc products, local uniform convergence, Fourier inversion, rapid decay, zeros and multiplicities, Poisson summation, and cosine sampling formulas.                 |
| Asymptotics      | The logarithmic tail expansion in powers of $4^{-N}$ , Bell–Bernoulli coefficient algebra, all-orders physical-space expansion, and zero-preserving signed $q$ -Richardson acceleration. |
| Endpoint theory  | Correct lower-Lambert coordinate, nonconstant periodic Gamma–zeta phase, and all-orders endpoint and inverse-endpoint expansions.                                                        |

| Layer           | Audited prior material used as input                                                                                  |
|-----------------|-----------------------------------------------------------------------------------------------------------------------|
| Multiresolution | Haar and Faber–Schauder expansions, exact energy tails, Bernstein/Kummer representations, and weighted-cube geometry. |

The finite-spline construction and the  $q$ -Richardson machinery are therefore restated only compactly enough to make the new deductions self-contained.

### 1.3 New theorem package

The proved contribution is the following linked package.

1. [proved here](#): a convex-order chain  $\mu_N \preceq_{\text{cx}} \mu_{N+1} \preceq_{\text{cx}} \mu$  together with a peakedness chain for all centered intervals;
2. [proved here](#): the exact moment identity
$$\mathbb{E} |X_N| = \frac{5}{18} - \frac{4^{-N}}{9};$$
3. [proved here](#): a fixed single-crossing law for  $u_N - u$ , with crossings at 0 and  $\pm \frac{1}{2}$  and no  $N$ -dependent drift;
4. [proved here](#): exact  $W_1$ , Kolmogorov, total-variation, stop-loss, Zolotarev, and  $W_\infty$  errors, including the sole first-stage exception;
5. [proved here](#): an exact piecewise-polynomial call/put potential and an all-orders profile whose leading term is  $u/18$ ;
6. [proved here](#): an exact no-go theorem for positive mixture acceleration in all the preceding finite metrics;
7. [proved here](#): monotone entropy and Fisher-information facts, including the exact criterion  $I(u_N) < \infty \iff N \geq 3$  and the exact divergence  $D(u||u_N) = \infty$ ;
8. [conjectural](#): weighted endpoint expansions for entropy, forward KL, Fisher information, and fixed finite- $p$  Wasserstein distance, supported by high-resolution computations.

## 2 Normalization and the common probability space

### 2.1 Fourier convention

Throughout the report,

$$\widehat{f}_{\text{ang}}(\xi) = \int_{\mathbb{R}} f(x) e^{-i\xi x} dx, \quad f(x) = \frac{1}{2\pi} \int_{\mathbb{R}} \widehat{f}_{\text{ang}}(\xi) e^{i\xi x} d\xi. \quad (2.1)$$

The sinc function is entire after filling the removable singularity:

$$\text{sinc}_{\text{rad}} z = \begin{cases} \sin z/z, & z \neq 0, \\ 1, & z = 0. \end{cases}$$

Define

$$\Phi_N(\xi) = \prod_{j=1}^N \text{sinc}_{\text{rad}}\left(\frac{\xi}{2^j}\right), \quad \Phi(\xi) = \prod_{j=1}^{\infty} \text{sinc}_{\text{rad}}\left(\frac{\xi}{2^j}\right). \quad (2.2)$$

The infinite product converges locally uniformly on  $\mathbb{C}$  and rapidly decreases on the real axis. Its inverse Fourier transform is the normalized Rvachev function  $u = \text{up}$ , supported on  $[-1, 1]$  with  $u(0) = 1$  and  $\int u = 1$ .

## 2.2 Uniform coordinates and exact tail self-similarity

Let  $V_1, V_2, \dots$  be independent  $\text{Unif}[-1, 1]$  variables, and put

$$X_N = \sum_{j=1}^N 2^{-j} V_j, \quad X = \sum_{j=1}^{\infty} 2^{-j} V_j. \quad (2.3)$$

Then  $\Phi_N$  and  $\Phi$  are the characteristic functions of  $X_N$  and  $X$ , and their densities are  $u_N$  and  $u$ . Set

$$\varepsilon_N = 2^{-N}, \quad a_N = 1 - \varepsilon_N. \quad (2.4)$$

Removing the first  $N$  coordinates and relabelling them gives the exact identity

$$X = X_N + \varepsilon_N X', \quad (2.5)$$

where  $X'$  is an independent copy of  $X$ . Equivalently,

$$\Phi(\xi) = \Phi_N(\xi) \Phi(\varepsilon_N \xi), \quad u = u_N * k_N, \quad k_N(x) = \varepsilon_N^{-1} u(x/\varepsilon_N). \quad (2.6)$$

This identity is the source of both the elementary coupling rate  $\varepsilon_N$  and the much sharper cancellation rate  $\varepsilon_N^2 = 4^{-N}$ .

Let  $H_N$  and  $H$  denote the CDFs of  $X_N$  and  $X$  and write

$$d_N(x) = H_N(x) - H(x). \quad (2.7)$$

All laws are symmetric and have mean zero.

## 3 Exact finite inverse transforms: dyadic box splines

### 3.1 Convolution and support

The density of  $2^{-j} V_j$  is

$$b_j(x) = 2^{j-1} \mathbf{1}_{[-2^{-j}, 2^{-j}]}(x), \quad (3.1)$$

so

$$u_N = b_1 * b_2 * \dots * b_N. \quad (3.2)$$

This immediately implies positivity, symmetry, unit mass, and

$$\text{supp } u_N = [-a_N, a_N]. \quad (3.3)$$

Each  $b_j$  is log-concave; convolution preserves log-concavity, hence  $u_N$  is even, log-concave, and nonincreasing on  $[0, \infty)$ . Because the first box has half-width  $1/2$  while the remaining boxes have total half-width  $1/2 - \varepsilon_N$ , there is an exact central plateau:

$$u_N(x) = 1 \quad (|x| \leq \varepsilon_N). \quad (3.4)$$

The limiting function is flat but not locally constant at the origin:  $u(0) = 1$  and  $u^{(r)}(0) = 0$  for every  $r \geq 1$ .



### 3.2 Thue–Morse knots and truncated powers

Put

$$\kappa_{N,m} = -1 + \frac{2m+1}{2^N}, \quad 0 \leq m < 2^N, \quad (3.5)$$

and let  $\tau(m) = \varepsilon_m$  be the signed Thue–Morse sequence.

**Theorem 3.1** (Finite Thue–Morse spline; [baseline](#)). *For  $N \geq 1$ ,*

$$u_N(x) = \frac{2^{\binom{N}{2}}}{(N-1)!} \sum_{m=0}^{2^N-1} \tau(m) (x - \kappa_{N,m})_+^{N-1}. \quad (3.6)$$

*Thus  $u_N$  is piecewise polynomial of degree  $N-1$ , belongs to  $C^{N-2}(\mathbb{R})$  for  $N \geq 2$ , and its distributional  $N$ th derivative is*

$$D^N u_N = 2^{\binom{N}{2}} \sum_{m=0}^{2^N-1} \tau(m) \delta_{\kappa_{N,m}}. \quad (3.7)$$

*Equivalently,*

$$[u_N^{(N-1)}]_{\kappa_{N,m}} = 2^{\binom{N}{2}} \tau(m). \quad (3.8)$$

*Proof.* For independent uniforms on intervals of lengths  $\ell_1, \dots, \ell_N$ , inclusion–exclusion gives the density

$$\frac{1}{(N-1)! \prod_j \ell_j} \sum_{S \subseteq \{1, \dots, N\}} (-1)^{|S|} \left( x - x_{\min} - \sum_{j \in S} \ell_j \right)_+^{N-1}.$$

Here  $\ell_j = 2^{1-j}$ ,  $x_{\min} = -a_N$ , and  $\prod_j \ell_j = 2^{-\binom{N}{2}}$ . The subset sum has a unique binary encoding  $m$ , its sign is  $\tau(m)$ , and its translated knot is precisely (3.5). Distributional differentiation proves (3.7) and (3.8).  $\square$

*Remark 3.2.* The right side of (3.6) is a severely cancelling sum of  $2^N$  large truncated powers. Its positivity and compact support are nearly invisible algebraically and immediate probabilistically. Direct floating-point evaluation is therefore ill-conditioned for large  $N$ ; recursive convolution, B-spline evaluation, or the Fourier-series method of [appendix 11](#) is preferable.

### 3.3 Refinement recurrences

Convolving by the  $(N+1)$ st box gives

$$u_{N+1}(x) = 2^N \int_{x-2^{-(N+1)}}^{x+2^{-(N+1)}} u_N(t) dt, \quad (3.9)$$

and differentiation gives

$$u'_{N+1}(x) = 2^N \left[ u_N(x + 2^{-(N+1)}) - u_N(x - 2^{-(N+1)}) \right]. \quad (3.10)$$

Separating the first coordinate in (2.3), for  $\varepsilon_N < x < a_N$ ,

$$u_N(x) = 1 - H_{N-1}(2x-1), \quad u'_N(x) = -2u_{N-1}(2x-1), \quad (3.11)$$

while

$$u(x) = 1 - H(2x-1), \quad u'(x) = -2u(2x-1), \quad 0 < x < 1. \quad (3.12)$$

These elementary identities will turn the CDF ordering into a complete sign diagram for the density error.

## 4 Established all-orders deconvolution expansion

This section records the minimum asymptotic machinery needed later. It is part of the audited baseline.

### 4.1 Reciprocal product and Bell coefficients

From (2.6),

$$\Phi_N(\xi) = \frac{\Phi(\xi)}{\Phi(\varepsilon_N \xi)}. \quad (4.1)$$

Near zero,

$$-\log \Phi(z) = \sum_{r=1}^{\infty} c_r z^{2r}, \quad c_r = \frac{\zeta(2r)}{r\pi^{2r}(4^r - 1)}. \quad (4.2)$$

Define  $b_m$  by

$$\frac{1}{\Phi(z)} = \exp\left(\sum_{r \geq 1} c_r z^{2r}\right) = \sum_{m=0}^{\infty} b_m z^{2m}. \quad (4.3)$$

These are the base-two values of the volume's single reciprocal-product coefficient layer: in the notation of [theorem 3.3](#) and [proposition 3.4](#) one has  $c_r = c_r(2)$  and  $b_m = \alpha_m(2)$ , so  $b_m$  is the complete Bell polynomial (3.10) in the  $c_r$  and obeys the recurrence (3.11); the same numbers are Part III's  $a_m$  in (6.1). Explicitly,

$$b_0 = 1, \quad b_1 = \frac{1}{18}, \quad b_2 = \frac{31}{16200}, \quad b_3 = \frac{2347}{42865200}, \quad b_4 = \frac{1904369}{1311675120000}. \quad (4.4)$$

### 4.2 Physical-space expansion

**Theorem 4.1** (All-orders finite-product expansion; [baseline](#)). *For fixed integers  $M, r \geq 0$ ,*

$$u_N = \sum_{m=0}^M (-1)^m b_m \varepsilon_N^{2m} u^{(2m)} + \mathcal{O}_{C^r}(\varepsilon_N^{2M+2}). \quad (4.5)$$

*In particular,*

$$4^N(u_N - u) \longrightarrow -\frac{1}{18}u'' \quad \text{in every } C^r(\mathbb{R}). \quad (4.6)$$

*Proof architecture.* Insert (4.3) into (4.1). Under the Fourier convention (2.1), multiplication by  $\xi^{2m}$  corresponds to  $(-1)^m D^{2m}$ . A compact-frequency Taylor expansion plus the rapid real-axis decay of  $\Phi$ , followed by a growing-frequency cutoff, controls every derivative of the remainder. The repository contains the corresponding certified logarithmic tail and  $q$ -Richardson forms.  $\square$

The leading coefficient has an immediate variance interpretation. Since

$$\text{Var } X = \frac{1}{3} \sum_{j \geq 1} 4^{-j} = \frac{1}{9}, \quad (4.7)$$

small convolution by the tail  $\varepsilon_N X'$  has variance  $\varepsilon_N^2/9$ , and the inverse heat-like correction has coefficient one half of that variance, namely  $\varepsilon_N^2/18$ .

## 5 Convex order and peakedness

### 5.1 Two orderings

For integrable probability measures with equal means,  $\nu \preceq_{\text{cx}} \mu$  means

$$\int \varphi \, d\nu \leq \int \varphi \, d\mu$$

for every convex  $\varphi$  for which the expectations exist. For symmetric laws on  $\mathbb{R}$ , say that  $\nu$  is *more peaked* than  $\mu$  if

$$\nu([-r, r]) \geq \mu([-r, r]) \quad (r \geq 0). \quad (5.1)$$

Convex order compares spread through convex tests; peakedness compares every centered interval. Neither implication holds for arbitrary symmetric laws, but both arise here from the same convolution chain.

**Lemma 5.1** (Centered intervals are maximal). *Let  $f$  be an even density nonincreasing on  $[0, \infty)$ . Then for every  $r \geq 0$  the map*

$$y \mapsto \int_{-r}^r f(x - y) \, dx$$

*is even and nonincreasing for  $y \geq 0$ . In particular, every interval of length  $2r$  has at most as much  $f$ -mass as  $[-r, r]$ .*

*Proof.* For  $y > 0$ , differentiation at regular points gives

$$\frac{d}{dy} \int_{-r}^r f(x - y) \, dx = f(-r - y) - f(r - y) = f(r + y) - f(|r - y|) \leq 0.$$

Approximation handles the finitely nonsmooth spline case. This is the one-dimensional content of Anderson's symmetric-unimodal inequality [60].  $\square$

**Theorem 5.2** (Convex-order and peakedness chain; [proved here](#)). *For every  $N \geq 1$ ,*

$$\mu_N \preceq_{\text{cx}} \mu_{N+1} \preceq_{\text{cx}} \mu, \quad (5.2)$$

*and*

$$\mathbb{P}(|X_N| \leq r) \geq \mathbb{P}(|X_{N+1}| \leq r) \geq \mathbb{P}(|X| \leq r) \quad (r \geq 0). \quad (5.3)$$

*Consequently,*

$$d_N(x) \geq 0 \quad (x \geq 0), \quad d_N(x) \leq 0 \quad (x \leq 0). \quad (5.4)$$

*Proof.* Write  $X_{N+1} = X_N + Z_{N+1}$  with  $Z_{N+1}$  independent and mean zero. Conditional Jensen gives

$$\mathbb{E}\varphi(X_N + Z_{N+1}) \geq \mathbb{E}\varphi(X_N),$$

and iteration, followed by dominated convergence, proves (5.2).

The density  $u_N$  is even and unimodal. Conditional on the independent tail  $T$ , the event  $|X_N + T| \leq r$  asks for the  $u_N$ -mass of a translate of  $[-r, r]$ . The preceding lemma bounds it by the centered mass, proving (5.3). For a symmetric continuous law,

$$H(x) = \frac{1}{2} + \frac{1}{2}\mathbb{P}(|X| \leq x), \quad x \geq 0,$$

which gives (5.4).  $\square$

## 6 Exact moments and the single-crossing law

### 6.1 A quadratic absolute-value identity

The exact metric constants all originate in one elementary calculation.

**Lemma 6.1.** *If  $U \sim \text{Unif}[-\frac{1}{2}, \frac{1}{2}]$  and  $|r| \leq \frac{1}{2}$ , then*

$$\mathbb{E}|U + r| = r^2 + \frac{1}{4}. \quad (6.1)$$

*Proof.* The zero of  $v + r$  lies inside the integration interval, and direct integration gives

$$\int_{-1/2}^{1/2} |v + r| dv = \frac{1}{2} \left( \frac{1}{2} - r \right)^2 + \frac{1}{2} \left( \frac{1}{2} + r \right)^2 = r^2 + \frac{1}{4}.$$

□

**Theorem 6.2** (Exact first absolute moment; [proved here](#)). *For every  $N \geq 1$ ,*

$$\boxed{\mathbb{E}|X_N| = \frac{5}{18} - \frac{4^{-N}}{9}, \quad \mathbb{E}|X| = \frac{5}{18}.} \quad (6.2)$$

*Proof.* Write  $X_N = U + R_N$ , where  $U = V_1/2$  is uniform on  $[-1/2, 1/2]$  and

$$R_N = \sum_{j=2}^N 2^{-j} V_j, \quad |R_N| \leq \frac{1}{2}.$$

Conditioning and using (6.1),

$$\mathbb{E}|X_N| = \frac{1}{4} + \mathbb{E}R_N^2.$$

Independence gives

$$\mathbb{E}R_N^2 = \frac{1}{3} \sum_{j=2}^N 4^{-j} = \frac{1}{36} (1 - 4^{1-N}),$$

which proves the finite formula. Letting  $N \rightarrow \infty$  proves the limit. □

### 6.2 The density error crosses at a fixed location

**Theorem 6.3** (Single-crossing theorem; [proved here](#)). *For every  $N \geq 1$ , up to the irrelevant values at spline knots,*

$$u_N(x) - u(x) \begin{cases} > 0, & 0 < |x| < \frac{1}{2}, \\ = 0, & x = 0 \text{ or } |x| = \frac{1}{2}, \\ < 0, & \frac{1}{2} < |x| < 1. \end{cases} \quad (6.3)$$

*Thus the positive half-axis crossing is exactly  $x = 1/2$  for every stage; it does not merely converge there.*

*Proof.* For  $0 \leq x \leq \varepsilon_N$ , (3.4) gives  $u_N(x) = 1 \geq u(x)$ , with equality only at  $x = 0$ . For  $\varepsilon_N < x < a_N$ , subtract (3.12) from (3.11):

$$u_N(x) - u(x) = -d_{N-1}(2x - 1). \quad (6.4)$$

By (5.4), the right side is positive for  $x < 1/2$ , zero at  $1/2$ , and negative for  $x > 1/2$ . For  $a_N \leq x < 1$ ,  $u_N(x) = 0 < u(x)$ . Evenness completes the proof. For  $N = 1$ , the same conclusion follows directly from  $u_1 = \mathbf{1}_{[-1/2, 1/2]}$ . □

**Corollary 6.4.** *For  $x \geq 0$ ,  $d_N$  is nondecreasing on  $[0, 1/2]$  and nonincreasing on  $[1/2, \infty)$ . Hence*

$$\|H_N - H\|_\infty = d_N(1/2). \quad (6.5)$$

## 7 Exact convergence metrics

### 7.1 Definitions

For laws  $\nu, \mu$  on  $\mathbb{R}$ , define

$$W_1(\nu, \mu) = \inf \mathbb{E} |Y - Z|, \quad (7.1)$$

$$d_K(\nu, \mu) = \sup_x |F_\nu(x) - F_\mu(x)|, \quad (7.2)$$

$$d_{TV}(\nu, \mu) = \frac{1}{2} \int |f_\nu - f_\mu|, \quad (7.3)$$

$$W_\infty(\nu, \mu) = \inf \text{ess sup} |Y - Z|, \quad (7.4)$$

where the infima range over couplings. In one dimension,

$$W_1(\nu, \mu) = \int_{\mathbb{R}} |F_\nu(x) - F_\mu(x)| dx. \quad (7.5)$$

**Theorem 7.1** (Exact metric collapse; [proved here](#)). *For every  $N \geq 1$ ,*

$$\boxed{W_1(\mu_N, \mu) = \frac{4^{-N}}{9}}, \quad (7.6)$$

*and for every  $N \geq 2$  this is also the Kolmogorov distance,*

$$\boxed{W_1(\mu_N, \mu) = d_K(\mu_N, \mu) = \frac{4^{-N}}{9}}, \quad (7.7)$$

$$\boxed{\|u_N - u\|_1 = \frac{4^{1-N}}{9}, \quad d_{TV}(\mu_N, \mu) = \frac{2 \cdot 4^{-N}}{9}}. \quad (7.8)$$

*At the first stage the remaining three constants are exceptional, while  $W_1$  keeps its general value  $4^{-1}/9 = 1/36$ :*

$$W_1(\mu_1, \mu) = \frac{1}{36}, \quad d_K(\mu_1, \mu) = \frac{5}{72}, \quad \|u_1 - u\|_1 = \frac{5}{18}, \quad d_{TV}(\mu_1, \mu) = \frac{5}{36}. \quad (7.9)$$

*Proof.* By (5.4),

$$W_1(\mu_N, \mu) = 2 \int_0^\infty d_N(x) dx.$$

For any symmetric integrable law,

$$\mathbb{E} |X| = 2 \int_0^\infty (1 - H(x)) dx.$$

Subtracting the finite and limiting versions and applying (6.2) gives

$$W_1(\mu_N, \mu) = \mathbb{E} |X| - \mathbb{E} |X_N| = \frac{4^{-N}}{9}.$$

This step uses only (5.4) and (6.2), so it is valid for every  $N \geq 1$ .

For  $N \geq 2$ , write  $X_N = U + R_N$  with  $R_N = \frac{1}{2} X'_{N-1}$ . Conditioning on  $R_N$  at the threshold  $1/2$  gives

$$H_N(1/2) = 1 - \mathbb{E}(R_N)_+ = 1 - \frac{1}{4} \mathbb{E} |X_{N-1}|.$$

The limiting identity is  $H(1/2) = 1 - \frac{1}{4}\mathbb{E}|X|$ , hence

$$d_N(1/2) = \frac{1}{4}(\mathbb{E}|X| - \mathbb{E}|X_{N-1}|) = \frac{4^{-N}}{9}.$$

By (6.5), this is the Kolmogorov distance.

The single-crossing theorem shows that the positive density error is supported on  $(-1/2, 1/2)$ . Its mass is

$$\int_{-1/2}^{1/2} (u_N - u) dx = 2d_N(1/2).$$

Positive and negative masses are equal, so the  $L^1$  norm is twice this quantity, namely  $4d_N(1/2)$ . This proves (7.8).

For  $N = 1$ ,  $H_1(1/2) = 1$  and

$$H(1/2) = 1 - \frac{1}{4}\mathbb{E}|X|,$$

so  $d_1(1/2) = 5/72$ . The same crossing argument gives the exceptional  $L^1$  and TV constants, while the  $W_1$  formula remains  $4^{-1}/9 = 1/36$ .  $\square$

*Remark 7.2* (Why the constants coincide). The equality  $W_1 = d_K$  is not dimensionally universal; it is a rigid consequence of the chosen support normalization, symmetry, and the exact first-coordinate quadratic identity. The equality persists at every  $N \geq 2$ , not merely asymptotically.

## 7.2 Exact essential-supremum transport and the nonoptimal truncation coupling

**Theorem 7.3** (Endpoint-controlled transport; [proved here](#)). *For every  $N \geq 1$ ,*

$$\boxed{W_\infty(\mu_N, \mu) = 2^{-N}}. \quad (7.10)$$

*The synchronous truncation coupling (2.5) has exact  $p$ -cost*

$$(\mathbb{E}|X - X_N|^p)^{1/p} = 2^{-N} (\mathbb{E}|X|^p)^{1/p}, \quad 1 \leq p < \infty, \quad (7.11)$$

*and is optimal for  $p = \infty$  but dramatically nonoptimal for  $p = 1$ .*

*Proof.* The coupling (2.5) satisfies  $|X - X_N| \leq 2^{-N}$ , proving the upper bound. The support of  $\mu_N$  ends at  $a_N = 1 - 2^{-N}$  while every interval  $(1 - \delta, 1)$  has positive  $\mu$ -mass. Any coupling must therefore move points arbitrarily close to 1 by at least  $1 - a_N = 2^{-N}$  in essential supremum. This proves equality. Equation (7.11) follows directly from  $X - X_N = 2^{-N}X'$ .  $\square$

For  $p = 1$ , the synchronous cost is  $5 \cdot 2^{-N}/18$ , whereas the optimum is  $4^{-N}/9$ . Their ratio is

$$\frac{(5/18)2^{-N}}{4^{-N}/9} = 5 \cdot 2^{N-1}. \quad (7.12)$$

Thus the obvious independent-tail coupling loses an exponentially growing factor in  $W_1$ , even though it is exactly optimal in  $W_\infty$ .

## 8 Stop-loss potentials, call splines, and Zolotarev distance

### 8.1 Call and put transforms

Define

$$C_N(t) = \mathbb{E}(X_N - t)_+, \quad C(t) = \mathbb{E}(X - t)_+, \quad K_N(t) = C(t) - C_N(t). \quad (8.1)$$

Convex order gives  $K_N \geq 0$ . Because both laws are symmetric with mean zero,

$$K_N(-t) = K_N(t). \quad (8.2)$$

At regular points,

$$K'_N(t) = H(t) - H_N(t) = -d_N(t), \quad K''_N(t) = u(t) - u_N(t). \quad (8.3)$$

**Theorem 8.1** (Exact finite call spline; [proved here](#)). *Let  $\kappa_{N,m}$  be (3.5). Then*

$$P_N(t) := \mathbb{E}(t - X_N)_+ = \frac{2^{\binom{N}{2}}}{(N+1)!} \sum_{m=0}^{2^N-1} \tau(m) (t - \kappa_{N,m})_+^{N+1}, \quad (8.4)$$

and

$$C_N(t) = P_N(t) - t. \quad (8.5)$$

Thus the call and put transforms are exact degree- $N+1$  piecewise-polynomial splines with the same Thue–Morse knot set.

*Proof.* The second distributional derivative of the right side of (8.4) is exactly (3.6), and both the function and its first derivative vanish to the left of the support. Therefore it equals  $\mathbb{E}(t - X_N)_+$ . Finally,

$$P_N(t) - C_N(t) = \mathbb{E}[(t - X_N)_+ - (X_N - t)_+] = \mathbb{E}(t - X_N) = t.$$

□

## 8.2 Exact height, area, and order-two ideal metric

**Theorem 8.2** (Stop-loss geometry; [proved here](#)). *For every  $N \geq 1$ ,  $K_N$  is even, nonnegative, nondecreasing on  $(-\infty, 0]$ , and nonincreasing on  $[0, \infty)$ . Moreover,*

$$\|K_N\|_\infty = K_N(0) = \frac{4^{-N}}{18}, \quad (8.6)$$

$$\int_{\mathbb{R}} K_N(t) dt = \frac{4^{-N}}{18}. \quad (8.7)$$

*Proof.* Equation (5.4) and  $K'_N = -d_N$  show monotonicity and that the maximum occurs at zero. Symmetry gives

$$K_N(0) = \mathbb{E}(X)_+ - \mathbb{E}(X_N)_+ = \frac{1}{2} (\mathbb{E}|X| - \mathbb{E}|X_N|) = \frac{4^{-N}}{18}.$$

For the area, use the standard call representation of twice differentiable convex tests, or simply test with  $\varphi(x) = x^2/2$ :

$$\int K_N(t) dt = \frac{1}{2} (\mathbb{E}X^2 - \mathbb{E}X_N^2).$$

Since

$$\text{Var } X - \text{Var } X_N = \frac{1}{3} \sum_{j>N} 4^{-j} = \frac{4^{-N}}{9},$$

(8.7) follows. □

The equality between height and area is another normalization-specific rigidity. It says that the entire positive potential has effective width one at every scale even though its detailed profile changes with  $N$ .



**Definition 8.3** (Order-two Zolotarev distance). For equal-mean laws with finite second moments, define

$$\zeta_2(\nu, \mu) = \sup_{\|f''\|_\infty \leq 1} \left| \int f d\nu - \int f d\mu \right|, \quad (8.8)$$

where  $f'$  is locally absolutely continuous. Affine terms are immaterial because the masses and means agree.

**Corollary 8.4** (Exact stop-loss and Zolotarev metrics; [proved here](#)). For every  $N \geq 1$ ,

$$d_{\text{SL}}(\mu_N, \mu) := \sup_t |C_N(t) - C(t)| = \zeta_2(\mu_N, \mu) = \frac{4^{-N}}{18}. \quad (8.9)$$

*Proof.* The stop-loss equality is (8.6). For admissible  $f$ ,

$$\mathbb{E}f(X) - \mathbb{E}f(X_N) = \int_{\mathbb{R}} K_N(t) f''(t) dt,$$

so the absolute value is at most  $\int K_N = 4^{-N}/18$ . Equality is attained by  $f(x) = x^2/2$ , whose second derivative is one.  $\square$

### 8.3 All-orders stop-loss profile

Integrating (4.5) twice, using flatness at the support endpoints, gives a direct analogue of the density expansion.

**Theorem 8.5** (All-orders potential expansion; [proved herefrom baseline](#)). For fixed  $M, r \geq 0$ ,

$$K_N(t) = \sum_{m=1}^M (-1)^{m+1} b_m \varepsilon_N^{2m} u^{(2m-2)}(t) + \mathcal{O}_{C^r}(\varepsilon_N^{2M+2}). \quad (8.10)$$

In particular,

$$4^N K_N \longrightarrow \frac{u}{18} \quad \text{in every } C^r(\mathbb{R}). \quad (8.11)$$

At  $t = 0$ , all terms after the first vanish because  $u(0) = 1$  and  $u^{(k)}(0) = 0$  for  $k \geq 1$ , consistently with the exact value  $K_N(0) = 4^{-N}/18$  established in (8.6).

*Proof.* By (8.3),  $K_N'' = u - u_N$ . Substitute (4.5). The unique twice-integrated solution vanishing with its derivative outside the support is (8.10). The leading term uses  $b_1 = 1/18$ .  $\square$

## 9 A positivity obstruction to acceleration

Signed  $q$ -Richardson combinations can cancel successive powers of  $4^{-N}$ , but they cease to be probability measures. The exact crossing theorem proves that this loss of positivity is not accidental.

Let  $J \geq 0$ , let  $\alpha_j \geq 0$  with  $\sum_{j=0}^J \alpha_j = 1$ , and define the mixture

$$\nu_{N,\alpha} = \sum_{j=0}^J \alpha_j \mu_{N+j}, \quad A(\alpha) = \sum_{j=0}^J \alpha_j 4^{-j}. \quad (9.1)$$

**Theorem 9.1** (Exact positive-mixture errors; [proved here](#)). For  $N \geq 2$ ,

$$W_1(\nu_{N,\alpha}, \mu) = d_K(\nu_{N,\alpha}, \mu) = \frac{4^{-N}}{9} A(\alpha), \quad (9.2)$$

$$d_{\text{TV}}(\nu_{N,\alpha}, \mu) = \frac{2 \cdot 4^{-N}}{9} A(\alpha), \quad (9.3)$$

$$d_{\text{SL}}(\nu_{N,\alpha}, \mu) = \zeta_2(\nu_{N,\alpha}, \mu) = \frac{4^{-N}}{18} A(\alpha). \quad (9.4)$$

The mixture density exceeds  $u$  on  $(-1/2, 0) \cup (0, 1/2)$ , equals it at 0 and  $\pm 1/2$  (up to knot conventions), and is below  $u$  for  $1/2 < |x| < 1$ .

*Proof.* CDF, density, and call errors are linear under mixing. Every  $d_{N+j}$  has the same sign on each half-axis and the same maximizer  $1/2$ ; every  $u_{N+j} - u$  has the same sign on each side of  $|x| = 1/2$ ; and every call gap is nonnegative with its maximum at zero. Therefore no absolute-value cancellation occurs in any defining integral or supremum. Substituting the exact stagewise constants and summing proves all formulas.  $\square$

**Corollary 9.2** (No positive Richardson acceleration). *Among mixtures supported on the window  $N, N+1, \dots, N+J$ , every metric in (9.2)–(9.4) is minimized by the trivial choice  $\alpha_J = 1$ . In particular, no nontrivial convex combination cancels the  $4^{-N}$  term or improves the geometric order relative to its finest constituent.*

*Proof.* Since  $4^{-j}$  is strictly decreasing, the convex average  $A(\alpha)$  is minimized uniquely at its smallest entry  $4^{-J}$ .  $\square$

---

**Structural interpretation.** Richardson acceleration needs negative weights because the entire positive cone generated by the finite-product laws points to the same side of the target in convex order, CDF order on each half-axis, density single-crossing order, and call-potential order. The obstruction is simultaneous, exact, and metric-independent within this family.

---

## 10 Smooth tests, CDFs, quantiles, and moment hierarchies

### 10.1 Smooth-test expansion

Let  $\varphi$  be smooth with derivatives of at most polynomial growth. Pairing (4.5) with  $\varphi$  and integrating by parts gives

$$\mathbb{E}\varphi(X) - \mathbb{E}\varphi(X_N) = \sum_{m=1}^M (-1)^{m+1} b_m \varepsilon_N^{2m} \mathbb{E}\varphi^{(2m)}(X) + \mathcal{O}(\varepsilon_N^{2M+2}). \quad (10.1)$$

In particular,

$$\mathbb{E}\varphi(X) - \mathbb{E}\varphi(X_N) = \frac{4^{-N}}{18} \mathbb{E}\varphi''(X) + \mathcal{O}(16^{-N}). \quad (10.2)$$

For nonsmooth convex  $\varphi$ , the exact distributional form is

$$\mathbb{E}\varphi(X) - \mathbb{E}\varphi(X_N) = \int_{\mathbb{R}} K_N(t) \varphi''(dt), \quad (10.3)$$

where  $\varphi''$  is the nonnegative second-derivative measure. Formula (10.3) is the precise bridge between convex order and the potential expansion.

### 10.2 CDF and interior quantile expansion

Integrating (4.5) once gives

$$H_N(x) = H(x) + \sum_{m=1}^M (-1)^m b_m \varepsilon_N^{2m} u^{(2m-1)}(x) + \mathcal{O}_{C^r}(\varepsilon_N^{2M+2}). \quad (10.4)$$

Thus

$$d_N(x) = -\frac{4^{-N}}{18} u'(x) + \mathcal{O}(16^{-N}). \quad (10.5)$$

This leading term has the exact sign from (5.4), because  $u'$  is positive on  $(-1, 0)$  and negative on  $(0, 1)$ .

Let  $Q = H^{-1}$  and  $Q_N = H_N^{-1}$ . On every compact probability interval  $[\delta, 1 - \delta]$ ,

$$Q_N(s) - Q(s) = \frac{4^{-N}}{18} \frac{u'(Q(s))}{u(Q(s))} + \mathcal{O}_\delta(16^{-N}). \quad (10.6)$$

The logarithmic derivative  $u'/u$  is the location score. It is unbounded at the flat endpoints, which is why (10.6) is not uniform in  $s$  and why  $W_\infty$  has the slower scale  $2^{-N}$ .

### 10.3 Exact cumulant tail

Since  $X = X_N + \varepsilon_N X'$  with independent terms, every cumulant satisfies

$$\kappa_r(X) - \kappa_r(X_N) = \varepsilon_N^r \kappa_r(X'). \quad (10.7)$$

Odd cumulants vanish. Thus the variance error is exactly  $4^{-N}/9$ , while the  $2r$ th cumulant error is of order  $4^{-rN}$ . The all-orders density expansion packages all these distinct moment scales into one differential series.

## 11 Numerical verification and reproducibility

### 11.1 Stable period-two Fourier reconstruction

Because  $u_N$  and  $u$  are supported inside one period of length two, their periodizations agree with them on  $[-1, 1]$ . Poisson summation gives

$$u_N(x) = \frac{1}{2} + \sum_{n=1}^{\infty} \Phi_N(\pi n) \cos(\pi n x), \quad (11.1)$$

$$u(x) = \frac{1}{2} + \sum_{n=1}^{\infty} \Phi(\pi n) \cos(\pi n x). \quad (11.2)$$

For every  $N \geq 1$  the even coefficients with  $n \neq 0$  vanish, since the  $j = 1$  factor  $\text{sinc}_{\text{rad}}(\pi n/2)$  does; for the limit the odd coefficients decay superalgebraically as well, whereas at finite  $N$  one has only  $\Phi_N(\pi n) = \mathcal{O}(n^{-N})$ . On the grid

$$x_k = -1 + \frac{2k}{M}, \quad M = 2^{18},$$

these series are inverse FFTs after the phase correction  $(-1)^n$ . Derivatives are obtained by multiplying the  $n$ th coefficient by  $(i\pi n)^r$ . CDFs and call potentials are then cumulative trapezoidal integrals. This route avoids the catastrophic cancellation of (3.6).

The script `assets/fabius_finite_products_frontier/finite_sinc_experiments.py`, kept in the volume directory, is fully commented. It has no network dependency and writes all figures and tables from scratch.

### 11.2 Approximants and leading profiles

#### 11.3 Exact metric checks

The largest relative error in the tabulated  $W_1$  values is  $2.2 \times 10^{-7}$ , occurring at the smallest absolute value  $N = 10$ ; the largest stop-loss relative error is  $1.1 \times 10^{-8}$ . The same run numerically returns

$$\int u = 1, \quad \int |u'| = 2, \quad \int |u''| = 8 \quad (11.3)$$

to double precision. The last identity also follows analytically from  $u'(1/2) = -2u(0) = -2$  and the sign pattern of  $u''$ .

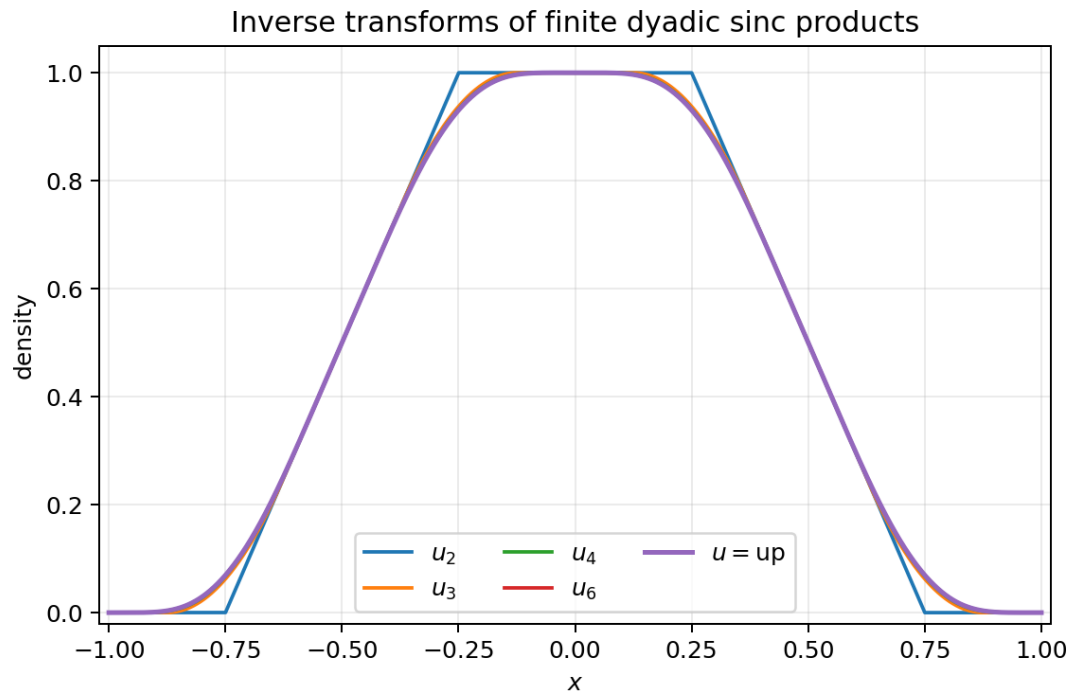


Figure 13: The inverse transforms  $u_N$  are nonnegative degree- $N - 1$  splines with support  $[-1 + 2^{-N}, 1 - 2^{-N}]$  and plateau  $[-2^{-N}, 2^{-N}]$ . Their geometric convergence is already visually rapid.

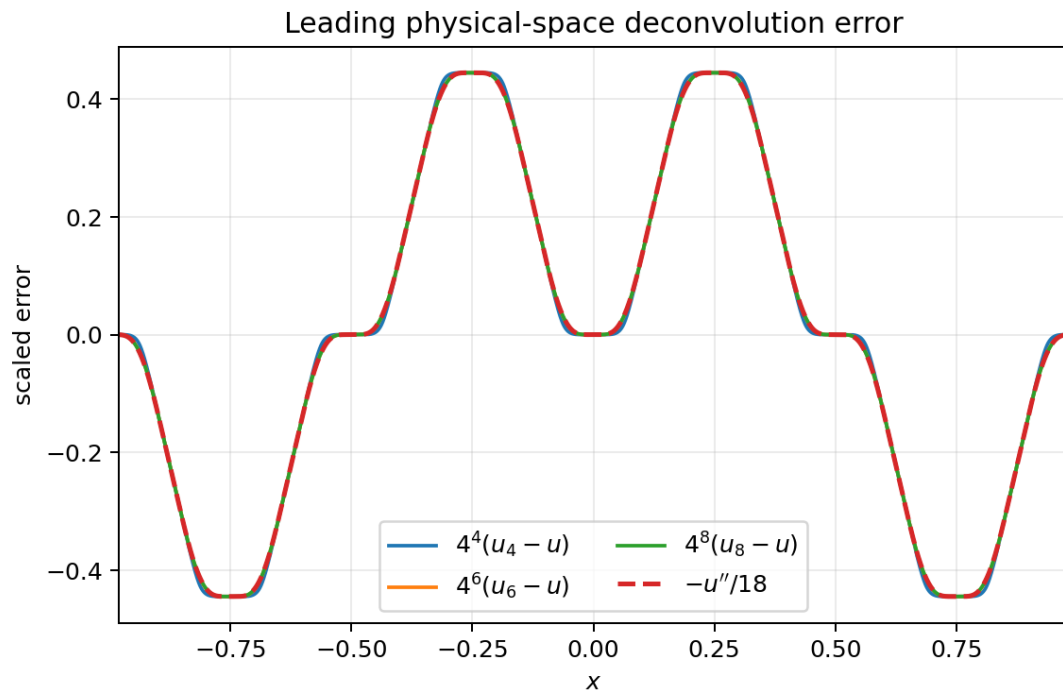


Figure 14: The scaled errors  $4^N(u_N - u)$  collapse onto the baseline limit  $-u''/18$ . The zero pattern at 0 and  $\pm 1/2$  is exact at every stage.

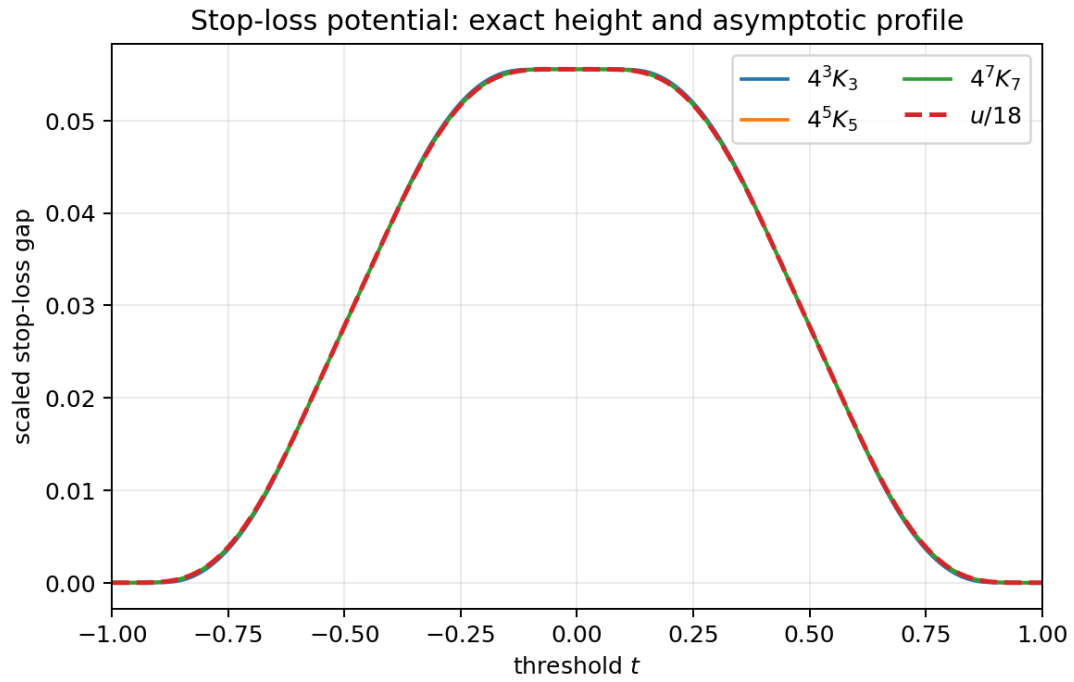


Figure 15: The exact call-potential gaps satisfy  $K_N(0) = 4^{-N}/18$  and converge after scaling to  $u/18$ . The central flatness is inherited from the up-function.

Table 12: FFT verification of two representative exact identities. The table uses  $M = 2^{18}$  period-two samples.

| $N$ | $W_1$ numeric             | $W_1$ exact               | rel. error | $\max K_N$ numeric        | exact                     | rel. error |
|-----|---------------------------|---------------------------|------------|---------------------------|---------------------------|------------|
| 2   | $6.944444 \times 10^{-3}$ | $6.944444 \times 10^{-3}$ | $2.82e-10$ | $3.472222 \times 10^{-3}$ | $3.472222 \times 10^{-3}$ | $1.86e-10$ |
| 3   | $1.736111 \times 10^{-3}$ | $1.736111 \times 10^{-3}$ | $2.85e-11$ | $8.680556 \times 10^{-4}$ | $8.680556 \times 10^{-4}$ | $1.63e-12$ |
| 4   | $4.340278 \times 10^{-4}$ | $4.340278 \times 10^{-4}$ | $5.68e-10$ | $2.170139 \times 10^{-4}$ | $2.170139 \times 10^{-4}$ | $8.65e-12$ |
| 5   | $1.085069 \times 10^{-4}$ | $1.085069 \times 10^{-4}$ | $1.83e-09$ | $5.425347 \times 10^{-5}$ | $5.425347 \times 10^{-5}$ | $9.21e-11$ |
| 6   | $2.712674 \times 10^{-5}$ | $2.712674 \times 10^{-5}$ | $1.82e-09$ | $1.356337 \times 10^{-5}$ | $1.356337 \times 10^{-5}$ | $1.04e-10$ |
| 7   | $6.781684 \times 10^{-6}$ | $6.781684 \times 10^{-6}$ | $1.42e-10$ | $3.390842 \times 10^{-6}$ | $3.390842 \times 10^{-6}$ | $3.21e-10$ |
| 8   | $1.695421 \times 10^{-6}$ | $1.695421 \times 10^{-6}$ | $5.80e-08$ | $8.477105 \times 10^{-7}$ | $8.477105 \times 10^{-7}$ | $5.12e-09$ |
| 9   | $4.238552 \times 10^{-7}$ | $4.238553 \times 10^{-7}$ | $1.15e-07$ | $2.119276 \times 10^{-7}$ | $2.119276 \times 10^{-7}$ | $2.77e-09$ |
| 10  | $1.059638 \times 10^{-7}$ | $1.059638 \times 10^{-7}$ | $2.17e-07$ | $5.298191 \times 10^{-8}$ | $5.298191 \times 10^{-8}$ | $1.10e-08$ |

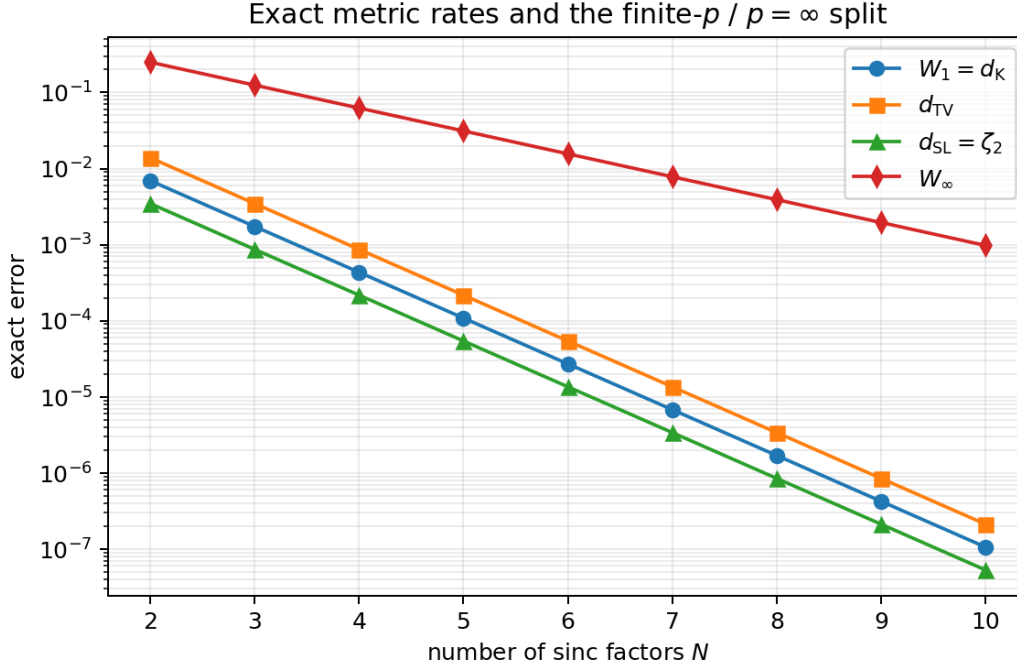


Figure 16: The finite metrics have exact  $4^{-N}$  rates, whereas  $W_\infty$  has exact rate  $2^{-N}$ . This is the first indication of the finite- $p/p = \infty$  crossover.

## 12 Information geometry and finite-order transport frontiers

This section separates rigorous monotonicity statements from asymptotic conjectures requiring endpoint-weighted remainder control.

### 12.1 Entropy monotonicity and convergence

For a density  $f$ , write

$$h(f) = - \int f \log f, \quad I(f) = \int \frac{(f')^2}{f},$$

with the usual conventions.

**Theorem 12.1** (Entropy chain; [proved here](#)). *The sequence  $h(u_N)$  is strictly increasing and*

$$h(u_N) \uparrow h(u). \quad (12.1)$$

*Proof.* Since  $X_{N+1} = X_N + 2^{-(N+1)} V_{N+1}$  with an independent nondegenerate summand, the entropy-power inequality gives  $h(u_{N+1}) > h(u_N)$ . The densities converge in  $L^1$ , are supported in  $[-1, 1]$ , and satisfy  $0 \leq u_N \leq 1$  because  $u_N$  is a convolution of the unit-height first box with a probability density. The continuous function  $-x \log x$  on  $[0, 1]$  is uniformly bounded, so convergence in measure plus dominated convergence yields  $h(u_N) \rightarrow h(u)$ .  $\square$

**Proposition 12.2** (Asymmetric relative entropy; [proved here](#)). *For every finite  $N$ ,*

$$D(u \| u_N) = +\infty, \quad D(u_N \| u) < \infty. \quad (12.2)$$

*Proof.* The limiting density is strictly positive on  $(-1, 1)$ , while  $u_N$  vanishes on the two strips  $a_N < |x| < 1$  of positive  $u$ -mass. Hence the reverse logarithmic ratio is infinite there. In the forward direction,  $u_N$  is bounded and supported on the compact subinterval  $[-a_N, a_N]$  of  $(-1, 1)$ , on which the continuous  $u$  is bounded below by a positive constant; hence  $u_N \log(u_N/u)$  is bounded and integrable.  $\square$

## 12.2 Fisher information

Near a support endpoint, the spline formula gives

$$u_N(a_N - y) \asymp y^{N-1}, \quad u'_N(a_N - y) \asymp y^{N-2}. \quad (12.3)$$

Therefore  $(u'_N)^2/u_N \asymp y^{N-3}$ .

**Theorem 12.3** (Finite-stage Fisher criterion and monotonicity; [proved here](#)).

$$I(u_N) < \infty \iff N \geq 3. \quad (12.4)$$

For  $N \geq 3$ ,

$$I(u_{N+1}) \leq I(u_N). \quad (12.5)$$

Moreover  $I(u) \leq \lim_{N \rightarrow \infty} I(u_N)$ .

*Proof.* The integrability criterion follows from (12.3). For monotonicity, the score of a convolution is the conditional expectation of the score of the first summand given the sum. Conditional Jensen therefore decreases its squared  $L^2$  norm. The final inequality is lower semicontinuity of Fisher information under weak convergence.  $\square$

Equality of the limit with  $I(u)$  is strongly supported numerically and is included in the conjecture below.

## 12.3 Weighted information asymptotics

Set

$$I = I(u) = \int_{-1}^1 \frac{u'(x)^2}{u(x)} dx, \quad J = \int_{-1}^1 \frac{u''(x)^2}{u(x)} dx, \quad (12.6)$$

and

$$L = \int_{-1}^1 u(x) ((\log u(x))'')^2 dx. \quad (12.7)$$

The FFT computation gives

$$\begin{aligned} h(u) &\approx 0.297039912826071, & I &\approx 11.8007105434191, \\ J &\approx 1619.745753585, & L &\approx 485.548634253. \end{aligned} \quad (12.8)$$

**Conjecture 12.4** (Information expansion). As  $N \rightarrow \infty$ ,

$$h(u) - h(u_N) = \frac{4^{-N}}{18} I + \mathcal{O}(16^{-N}), \quad (12.9)$$

$$D(u_N \| u) = \frac{16^{-N}}{648} J + \mathcal{O}(64^{-N}), \quad (12.10)$$

$$I(u_N) = I(u) + \frac{4^{-N}}{9} L + \mathcal{O}(16^{-N}). \quad (12.11)$$

In particular,  $I(u_N) \downarrow I(u)$ .

*Proof architecture.* Formally insert

$$u_N = u - \frac{\varepsilon_N^2}{18} u'' + \mathcal{O}(\varepsilon_N^4)$$



into the entropy, relative-entropy, and Fisher functionals. The first variations give (12.9); the quadratic term of KL gives (12.10). For Fisher information, one integration-by-parts reduction turns the first variation into

$$\frac{\varepsilon_N^2}{9} \int u((\log u)'')^2.$$

The missing rigorous step is a version of (4.5) in weighted spaces containing inverse powers of  $u$ . Uniform  $C^r$  control alone is insufficient because  $u$  is flat at  $\pm 1$ . The repository’s lower-Lambert endpoint expansion appears strong enough to supply the needed split-domain domination.  $\square$

## 12.4 Fixed-order Wasserstein asymptotics

For  $1 \leq p < \infty$ , define the score moment

$$J_p = \int_{-1}^1 |u'(x)|^p u(x)^{1-p} dx = \int_0^1 |(\log u)'(Q(s))|^p ds. \quad (12.12)$$

Flatness makes  $J_p$  finite for each fixed  $p$ , although  $J_p^{1/p} \rightarrow \infty$  as  $p \rightarrow \infty$ .

**Conjecture 12.5** (Finite- $p$  transport law). *For every fixed  $1 \leq p < \infty$ ,*

$$\boxed{W_p(\mu_N, \mu) \sim \frac{4^{-N}}{18} J_p^{1/p}.} \quad (12.13)$$

For  $p = 1$ , this is the proved exact formula because  $J_1 = \int |u'| = 2$ . For  $p = 2$ ,

$$W_2(\mu_N, \mu) \sim \frac{4^{-N}}{18} \sqrt{I(u)}. \quad (12.14)$$

*Proof architecture.* The interior quantile expansion (10.6) gives

$$|Q_N(s) - Q(s)|^p \sim \left( \frac{4^{-N}}{18} \right)^p |(\log u)'(Q(s))|^p.$$

Integrating over  $s$  gives (12.13). The endpoint layers must again be controlled with the lower-Lambert expansion; fixed  $p$  should make their contribution negligible at the  $4^{-Np}$  scale.  $\square$

Table 13: Numerical evidence for the information and  $W_2$  conjectures. “Ratio” means measured value divided by the displayed leading asymptotic term.

| $N$ | entropy gap               | ratio      | $D(u_N \  u)$              | ratio      | $W_2$                     | ratio      |
|-----|---------------------------|------------|----------------------------|------------|---------------------------|------------|
| 3   | $1.057601 \times 10^{-2}$ | 1.03244285 | $5.896794 \times 10^{-4}$  | 0.96628234 | $3.018666 \times 10^{-3}$ | 1.01230978 |
| 4   | $2.580642 \times 10^{-3}$ | 1.00770183 | $3.762085 \times 10^{-5}$  | 0.98636290 | $7.477592 \times 10^{-4}$ | 1.00304444 |
| 5   | $6.414394 \times 10^{-4}$ | 1.00188982 | $2.373649 \times 10^{-6}$  | 0.99573677 | $1.865139 \times 10^{-4}$ | 1.00075897 |
| 6   | $1.601326 \times 10^{-4}$ | 1.00046967 | $1.488143 \times 10^{-7}$  | 0.99883278 | $4.660192 \times 10^{-5}$ | 1.00018926 |
| 7   | $4.001904 \times 10^{-5}$ | 1.00011722 | $9.308961 \times 10^{-9}$  | 0.99969898 | $1.164881 \times 10^{-5}$ | 1.00004596 |
| 8   | $1.000388 \times 10^{-5}$ | 1.00002929 | $5.819411 \times 10^{-10}$ | 0.99992405 | $2.912086 \times 10^{-6}$ | 1.00000594 |
| 9   | $2.500915 \times 10^{-6}$ | 1.00000732 | $3.637339 \times 10^{-11}$ | 0.99998105 | $7.280062 \times 10^{-7}$ | 0.99998492 |
| 10  | $6.252253 \times 10^{-7}$ | 1.00000183 | $2.273371 \times 10^{-12}$ | 0.99999627 | $1.820097 \times 10^{-7}$ | 1.00002973 |

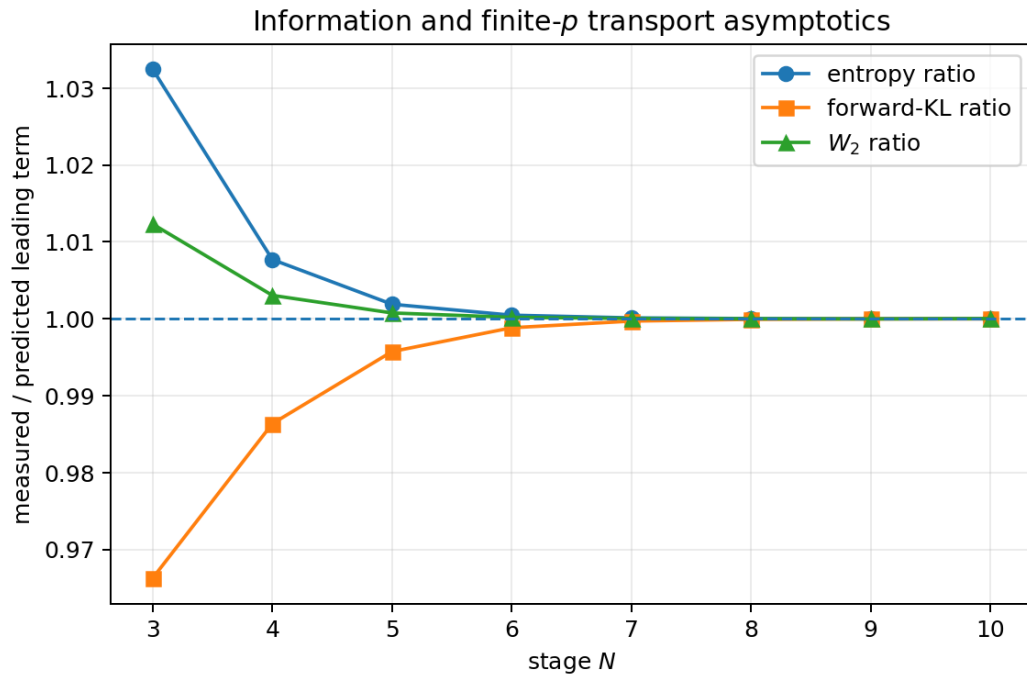


Figure 17: Measured values divided by the predicted leading terms in (12.9), (12.10), and (12.14). All ratios approach one rapidly.

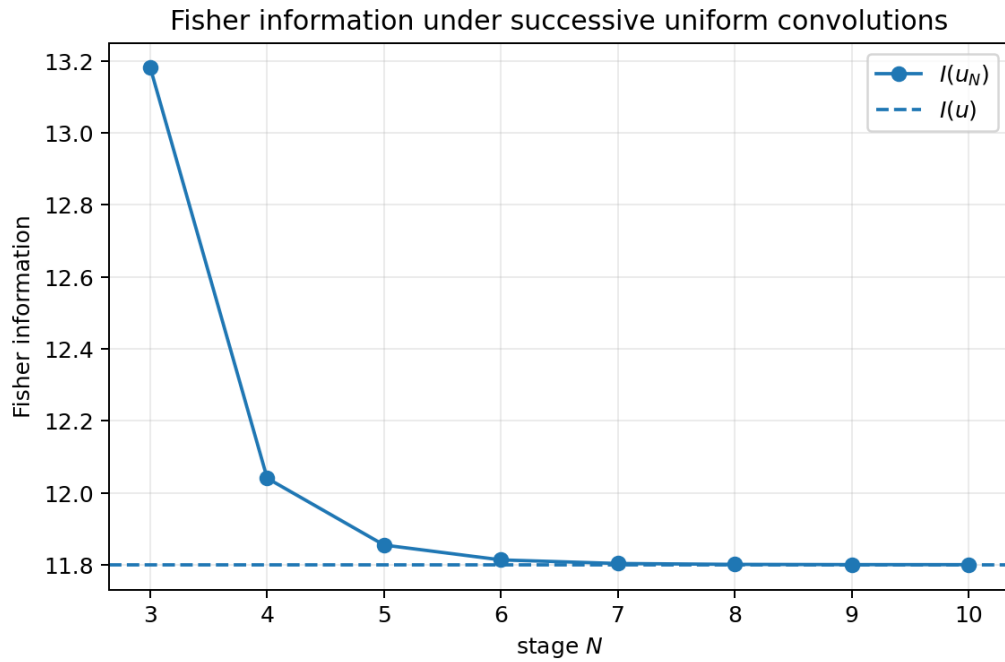


Figure 18: Numerical monotone convergence of  $I(u_N)$  to  $I(u) \approx 11.8007105434191$ . The measured first correction agrees with  $4^{-N}L/9$ .

## 12.5 The transport-order crossover near twice the truncation stage

The exact endpoint result  $W_\infty = 2^{-N}$  cannot be obtained by sending  $p \rightarrow \infty$  after taking the fixed- $p$  asymptotic. The lower-Lambert endpoint law suggests

$$J_p^{1/p} = 2^{p/2} p^{\theta+o(1)} \times (\text{bounded log-periodic factor}) \quad (12.15)$$

for a computable exponent  $\theta$ . Balancing

$$4^{-N} J_p^{1/p} \asymp 2^{-N}$$

then places the transition near

$$p = 2N + \mathcal{O}(\log N). \quad (12.16)$$

**Conjecture 12.6** (Double-scaling transport crossover). *There is a nontrivial profile  $\mathcal{W}(\lambda, \vartheta)$  such that, with  $p = 2N + \lambda \log N + \mathcal{O}(1)$  and with  $\vartheta$  the dyadic lower-Lambert phase,*

$$2^N W_p(\mu_N, \mu) \longrightarrow \mathcal{W}(\lambda, \vartheta) \quad (12.17)$$

*along phase-locked subsequences. The profile interpolates between the score-moment regime and the endpoint value one.*

The periodic phase is not cosmetic: the repository's endpoint theory proves that it is nonconstant. Any crossover theorem that averages it away prematurely is likely to miss subsequential structure.

## 13 Further deductions and research directions

### 13.1 Weighted endpoint remainder theorem

The most immediate target is a theorem of the schematic form

$$\int_{-1}^1 \frac{|u_N - u + \varepsilon_N^2 u''/18|^q}{u^{q-1}} dx = \mathcal{O}(\varepsilon_N^{4q}) \quad (13.1)$$

for suitable  $q$ , together with derivative analogues. A proof should split the interval in the exact lower-Lambert coordinate, use the periodic endpoint transseries in the boundary layer, and use the ordinary  $C^r$  expansion in the bulk. This single technical result would rigorously settle (12.9), (12.10), and fixed- $p$  cases of (12.13).

### 13.2 Exact Wasserstein values beyond first order

The one-dimensional optimal coupling is monotone, so

$$W_p^p(\mu_N, \mu) = \int_0^1 |Q_N(s) - Q(s)|^p ds. \quad (13.2)$$

The exact dyadic arithmetic of  $H_N$  and the inverse-Fabius infrastructure suggest several possibilities:

1. exact recurrences or rapidly convergent series for  $W_2^2$ ;
2. arithmetic information about quadrature approximants to (13.2);
3. bounds obtained from the exact call potential and one-dimensional Sobolev inequalities;
4. phase-locked asymptotics for  $p$  growing with  $N$ .

### 13.3 Total positivity and variation diminution

Every finite approximant is a convolution of boxes, hence belongs to the classical Pólya-frequency/spline setting of Curry and Schoenberg [26]. The fixed density crossing suggests a stronger variation-diminishing statement for the deconvolution chain:

**Conjecture 13.1** (Higher derivative crossing hierarchy). *For each fixed  $r$ , the error  $u_N^{(r)} - u^{(r)}$  has, for all sufficiently large  $N$ , the same number and ordering of sign changes as  $-u^{(r+2)}$ , and the zeros converge with complete  $4^{-N}$  expansions. At special dyadic zeros forced by flatness, some crossings are exact at every stage.*

Such a theorem would turn the all-orders  $C^r$  expansion into exact shape information and could yield closed formulas for additional  $L^1$  derivative norms.

### 13.4 Nonlinear positive acceleration

The positive-mixture theorem rules out linear convex acceleration, but not nonlinear transformations that preserve nonnegativity and mass. Promising candidates include:

1. exponentiating a Richardson combination of logarithmic characteristic-function tails and then proving positive definiteness;
2. monotone transport extrapolation at the quantile level;
3. entropy projection of a signed Richardson density onto the convex set of probability densities with prescribed moments;
4. call-potential extrapolation followed by projection onto the cone of nonnegative convex-order gaps.

Any such method must respect the exact support deficit, the fixed crossing at  $1/2$ , and the zero divisor of the finite Fourier product.

### 13.5 Formalization roadmap

The most Lean-friendly new results are elementary once the probability model is available:

1. the conditional formula (6.1);
2. the exact absolute moment (6.2);
3. peakedness from one-dimensional interval rearrangement;
4. the error identity (6.4) and single crossing;
5. exact  $W_1$ , Kolmogorov, and TV constants;
6. the call spline (8.4);
7. exact potential height/area and the positive-mixture no-go theorem.

The information conjectures should wait until the weighted endpoint calculus is formalized. The exact metric package, by contrast, uses only finite sums, conditioning, symmetry, monotonicity, and elementary integration.

## 14 Compact theorem and formula index

| Object                | Formula or property                                                                        |
|-----------------------|--------------------------------------------------------------------------------------------|
| Finite product        | $\Phi_N(\xi) = \prod_{j=1}^N \text{sinc}_{\text{rad}}(\xi/2^j)$ .                          |
| Random variable       | $X_N = \sum_{j=1}^N 2^{-j} V_j$ , $V_j \sim \text{Unif}[-1, 1]$ .                          |
| Tail                  | $X = X_N + 2^{-N} X'$ , $\Phi = \Phi_N \Phi(2^{-N} \cdot)$ .                               |
| Support/plateau       | $\text{supp } u_N = [-1 + 2^{-N}, 1 - 2^{-N}]$ , $u_N = 1$ on $[-2^{-N}, 2^{-N}]$ .        |
| Knots                 | $\kappa_{N,m} = -1 + (2m + 1)/2^N$ .                                                       |
| Spline                | Equation (3.6); degree $N - 1$ , smoothness $C^{N-2}$ .                                    |
| Leading density error | $4^N(u_N - u) \rightarrow -u''/18$ .                                                       |
| Convex order          | $\mu_N \preceq_{\text{cx}} \mu_{N+1} \preceq_{\text{cx}} \mu$ .                            |
| Absolute moment       | $\mathbb{E} X_N  = 5/18 - 4^{-N}/9$ .                                                      |
| Single crossing       | $u_N - u > 0$ for $0 <  x  < 1/2$ , negative for $1/2 <  x  < 1$ .                         |
| $W_1$                 | $4^{-N}/9$ for all $N \geq 1$ .                                                            |
| Kolmogorov            | $4^{-N}/9$ for $N \geq 2$ ; $5/72$ for $N = 1$ .                                           |
| Total variation       | $2 \cdot 4^{-N}/9$ for $N \geq 2$ ; $5/36$ for $N = 1$ .                                   |
| $W_\infty$            | $2^{-N}$ .                                                                                 |
| Call potential        | $K_N = C - C_N \geq 0$ , exact finite spline from (8.4).                                   |
| Stop-loss/Zolotarev   | Both equal $4^{-N}/18$ .                                                                   |
| Potential profile     | $4^N K_N \rightarrow u/18$ in every $C^r$ .                                                |
| Positive mixture      | Every finite metric is multiplied by $A(\alpha) = \sum \alpha_j 4^{-j}$ ; no cancellation. |
| Entropy               | $h(u_N) \uparrow h(u)$ .                                                                   |
| Reverse KL            | $D(u  u_N) = \infty$ .                                                                     |
| Fisher                | Finite iff $N \geq 3$ ; nonincreasing thereafter.                                          |
| Conjectured $W_p$     | $W_p \sim 4^{-N} J_p^{1/p}/18$ for fixed finite $p$ .                                      |

## A The first-stage exception in detail

The first approximant is

$$u_1 = \mathbf{1}_{[-1/2, 1/2]}.$$

The CDF difference still has one positive maximum at  $1/2$ , but the reduction through  $X_{N-1}$  would require the degenerate stage  $X_0 = 0$ . Directly,

$$d_1(1/2) = H_1(1/2) - H(1/2) = 1 - \left(1 - \frac{1}{4} \mathbb{E}|X|\right) = \frac{5}{72}.$$

The density crossing then gives

$$\|u_1 - u\|_1 = 4d_1(1/2) = \frac{5}{18}.$$

By contrast,  $W_1$  depends on the absolute-moment difference and remains on the universal formula

$$W_1 = \mathbb{E}|X| - \mathbb{E}|X_1| = \frac{5}{18} - \frac{1}{4} = \frac{1}{36}.$$

The stop-loss value also remains universal:

$$K_1(0) = \frac{1}{2} W_1 = \frac{1}{72}.$$

## B Derivation of the Fisher first variation

Let  $f_\eta = u + \eta v$  with  $\int v = 0$ . The first variation of Fisher information is

$$\left. \frac{d}{d\eta} I(f_\eta) \right|_{\eta=0} = \int \left( \frac{2u'v'}{u} - \frac{u'^2 v}{u^2} \right) dx. \quad (\text{B.1})$$

For the finite-product deconvolution,  $v = -u''/18$ . Put  $s = (\log u)'$ . Since

$$\frac{u''}{u} = s' + s^2, \quad \frac{u'''}{u} = s'' + 3ss' + s^3,$$

the bracket before the factor  $-1/18$  becomes

$$u(2ss'' + 5s^2s' + s^4).$$

Integration by parts gives

$$\int 2uss'' = -2 \int u((s')^2 + s^2s'),$$

and

$$3 \int us^2s' = - \int us^4.$$

The quartic terms cancel, leaving

$$\int \left( \frac{2u'u'''}{u} - \frac{u'^2 u''}{u^2} \right) dx = \int u(2ss'' + 5s^2s' + s^4) dx = -2 \int u(s')^2 dx = -2L.$$

With the deconvolution perturbation  $v = -\varepsilon_N^2 u''/18$ , the first variation (B.1) is  $-\varepsilon_N^2/18$  times this bracket, that is  $+\varepsilon_N^2 L/9$ , the conjectured first correction in (12.11).

## C Reproducibility ledger

### C.1 Archive contents

This part's verification material is kept in the volume directory under `assets/fabiús_finite_products_frontier/`. The paths below are relative to the volume root; the absorbed report's own source and PDF were deleted when it was merged, and git history is their archive.

| Path                                                                               | Purpose                                                     |
|------------------------------------------------------------------------------------|-------------------------------------------------------------|
| <code>assets/fabiús_finite_products_frontier/finite_sinc_experiments.py</code>     | Standalone, fully commented numerical experiment.           |
| <code>assets/fabiús_finite_products_frontier/figures/</code>                       | Vector figures included in this part, with raster previews. |
| <code>assets/fabiús_finite_products_frontier/data/metric_verification.csv</code>   | Numerical and exact metric values.                          |
| <code>assets/fabiús_finite_products_frontier/data/information_transport.csv</code> | Entropy, KL, Fisher, and $W_2$ data.                        |
| <code>assets/fabiús_finite_products_frontier/data/reference_constants.csv</code>   | Computed normalization and information constants.           |
| <code>assets/fabiús_finite_products_frontier/data/verification_log.txt</code>      | Human-readable diagnostics and maximum errors.              |

---

```
assets/fabius_finite_products_frontier/R  The absorbed package's own run notes.
EADME.md
assets/fabius_finite_products_frontier/r  Python dependencies.
equirements.txt
```

---

## C.2 Audited source hashes

The four directly materialized overlap-critical sources had the following SHA-256 hashes at audit time:

| Source                       | SHA-256                                  |
|------------------------------|------------------------------------------|
| Fabius_Function_and_Rvachev_ | c2f203778b1889bff22e4839f369c16e50e9ef45 |
| Up.tex                       | 799717f6f1357dc1df3f40d1                 |
| fabius-frontier-results.tex  | 32953767492749ce7668a28896e76075ab3ab1f9 |
|                              | c42f370c4e31e35638f21857                 |
| Fabius_Rvachev_Multiresoluti | 832c790e8bd247ea1569c7ef1ac09017a9c083d7 |
| on_Representations.tex       | c8f5ed17d57035e2aa82175c                 |
| Fabius_Dyadic_q_Connections. | 81a0a911ac7ab28e12c6b87ccaf76ffccaf32e48 |
| tex                          | f873abff18fb7a2d01bcf3e5                 |

## C.3 Numerical command

From `assets/fabius_finite_products_frontier/`, the complete numerical layer is regenerated by

```
python finite_sinc_experiments.py --output . --grid-power 18 --max-stage 10
```

The volume itself is rebuilt from its own directory by running `pdflatex` on `geometric_q_fabius_frontiers.tex` three times, so that the table of contents and every cross-reference settle.

## Part VI

# Atomic Sinc-Product Splines Beyond the Binary Point

### Abstract

The attached Russian chapter develops *atomic functions*: compactly supported solutions of differential–functional equations, with Rvachev’s function up as the principal example. This report supplies a mathematically edited English translation of the chapter’s surviving material and expands it in the notation of the current Analysis/FabiusFunction documentation. The central family is

$$\widehat{h_{a\,2\pi}}(\xi) = \prod_{j \geq 1} \text{sinc}_{\text{rad}}(2\pi a^{-j} \xi), \quad a > 1,$$

where  $\text{sinc}_{\text{rad}} z = \sin z/z$ . It interpolates between the critical binary atom  $h_2 = \text{up}$  and, for  $a > 2$ , a class of  $C^\infty$  functions that are locally polynomial off a Cantor-type exceptional set.

The translation is followed by a repository-wide crosswalk and several extensions. We derive the complete Bernoulli–cumulant and Bell-polynomial moment calculus for  $h_a$ ; a weighted Prouhet identity giving the entire derivative hierarchy; exact derivative norms for every  $a \geq 2$ ; a finite truncated-power formula and a self-similar polynomial-gap atlas; a symbolic localization formula for derivatives on the exceptional set; a rational-power Strang–Fix theorem that reproduces arbitrary polynomials by one uniform lattice of shifts whenever some positive power of  $a$  is



rational; an elementary log-Gaussian Fourier envelope; and an all-orders asymptotic expansion for the ordinary finite sinc-product splines. The leading correction is

$$a^{2N}(h_{a,N} - h_a) \longrightarrow -\frac{h_a''}{6(a^2 - 1)} \quad \text{in every fixed } C^r \text{ norm.}$$

A deterministic Python program checks the exact moment and Prouhet identities, computes Fourier inversions without Monte Carlo sampling, and reproduces all figures and tables in the package. Claims not yet formalized in Lean are kept outside the repository’s trusted theorem boundary and are labeled accordingly.

This part also absorbs two further independent reconstructions of the same chapter (fourteenth and fifteenth waves). Their distinctive layers are folded in below. From the fourteenth wave: the complete fractal-string geometry of the exceptional set for  $a > 2$  (geometric zeta function, lattice of complex dimensions, an exact tube formula with nonconstant logarithmic oscillation, and Minkowski non-measurability); the local polynomial degree as a geometric random variable, with an exponential scaling limit as  $a \downarrow 2$ ; quantitative Gaussian ( $a \downarrow 1$ ) and uniform ( $a \rightarrow \infty$ ) parameter limits; an exact arbitrary-base negative-Laplace decomposition whose one-periodic correction has Gamma–zeta Fourier modes  $-\Gamma(-\chi_k)\zeta(1-\chi_k)/\log a$ ; and a canonical Fup ladder extending the source’s surviving Fup<sub>1</sub> construction. From the fifteenth wave: signed closed forms for the leading coefficient on every polynomial gap; a derivative equimeasurability theorem giving the complete value distribution and all  $L^p$  norms of every derivative for  $a > 2$ ; the classical Fup<sub>n</sub> hierarchy with an exact triangular dyadic averaging reconstruction of up and a quantitative central-limit regime; and the edge pantograph equations that make the Lambert- $W$  structure of the endpoint visible already at the level of the differential–functional equation. Two revised fourteenth-wave editions add the spectral Stieltjes–Wigert bridge (the Fourier energy of every  $h_a$  realizes the scaled Stieltjes–Wigert moments  $a^{n(n+2)}/2^n$ , a non-lognormal representing measure with closed Hankel determinants and orthogonal polynomials), the Mellin law of the distance to  $K_a$  with its shifted complex-dimension poles, and — for provenance — the Russian source scan and OCR themselves, archived under `assets/`.

**Status legend and novelty convention.** [S] denotes material translated or reconstructed from the attached Russian source; [R] denotes facts already present in the living repository documentation or its formalization crosswalk; [N] denotes a derivation that appears new relative to the audited repository corpus, not a claim of worldwide priority; and [C] denotes a conjecture, heuristic, or future research program. Numerical agreement is evidence only, never a substitute for proof.

**Source condition.** The supplied `rvachev.tex` is a 1,267-line OCR transcription containing the full Chapter 3 through equation (3.100), where Section 5 stops in midstream. The supplied PDF contains only eleven printed pages, 116–126, beginning in the middle of Section 2. The OCR has predictable defects: lost superscripts, confused Latin and Cyrillic glyphs, corrupted binomial coefficients, equation numbers such as “(8.85)”, and occasional reversed Fourier signs. Whenever the PDF image resolves a formula, it is preferred. Otherwise the reconstruction is checked against adjacent identities, normalization, and the source’s own proof. No text is silently completed beyond the surviving endpoint of Section 5.

The fourteenth- and fifteenth-wave packages reconstructed the same OCR and page images independently and reached the same readings on every equation the editions transcribe in common, including the damaged ranges (3.51)–(3.58) and (3.73)–(3.75), for which all editions record the invariant content (finite moment reduction at cylinder endpoints; the bit-recursive evaluator with remainder at most  $\text{up}(-1 + 2^{-N})$ ) rather than a fabricated literal transcription. The fifteenth wave additionally repairs the endpoint indexing of (3.51)–(3.58) into an explicit binary-encoded form. The packages’ OCR-reconstruction ledgers are merged into the concordance appendix of this part; the absorbed source documents are archived in git history, with SHA-256 hashes in the volume’s provenance list.

Two revised editions of the fourteenth-wave report arrived subsequently and are merged below

(their unique layers: the spectral Stieltjes–Wigert bridge, the rearrangement form of equimeasurability, and the distance–Mellin law). Decisively for provenance, the first revised edition shipped the surviving *Russian source itself*: the eleven-page scan source\_rvachev\_scan.pdf, against which the translation layers of this part were checked. Both raw source files — the scan and the OCR transcription — were deleted from the working tree once their recoverable content had been merged: three independent reconstructions agree on every recoverable equation, the merged repair ledger in the concordance appendix records every correction, and each file’s SHA-256 remains in the provenance list, with git history archiving the files themselves. A revised edition of the *fifteenth-wave* report is merged as well; its post-merge layers are the  $q$ -Gaussian Gram geometry of the derivative tower, the log-Weibull jet-intermittency law, and the proof of the  $\text{Fup}_n$  Edgeworth program. A second, *expanded* edition of that report followed and is likewise merged; it is audit-aware (it explicitly marks the previously merged layers as inherited baseline), and its genuinely new layer is the closed orthogonalization of the derivative tower — the Gaussian-binomial Gram–Schmidt theorem with its Rogers–Szegő identification, the wrapped-heat-kernel circle model, and the MacMahon determinant constant with triple-product Riesz bounds — together with the overlap-regime theta conjecture in the register. Two expanded *fourteenth-wave* editions arrived last (both audit-aware, both re-shipping byte-identical copies of the recorded source scan and OCR, which were again not retained); their disjoint new layers are merged below: the physical-space Stieltjes–Wigert differential ladder with its derivative-jet Gram determinants and the autocorrelation germ (first edition), and the derivative-energy factorization with its Bernoulli-convolution limit and exact entropy laws (second edition), plus two register conjectures. The merge also identified the spectral ladder with the fifteenth-wave Gram–Schmidt vectors ([proposition 13.13](#)) — and, in checking that identification, caught and repaired a sign-convention slip in the first printing of [theorem 13.3](#) (the closed coefficients orthogonalize the *raw* tower; the sign-corrected basis takes alternating coefficients). Two expanded *fifteenth-wave* editions then closed the round (both audit-aware; the first re-shipped byte-identical copies of the scan, the OCR, and the two previous editions of its own lineage — all with SHA-256 already recorded, none retained): their disjoint new layers are the nodal-polynomial reading of the Gram–Schmidt residuals with the exact inverse transform and positive Cholesky, the minimum-phase theta whitening filter, the Schur-minor strict total positivity of the kernel, the two-term jet tail with its sharp exponential–Orlicz threshold, and the highest-jet partial-theta law with the joint jet–distance transform — plus five register conjectures. Their independently written statements of the closed orthogonalization agree with the repaired sign convention.

## 1 Executive summary of the mathematical contribution

The report has two intertwined purposes. First, it makes the attached chapter usable in the current English-language Fabius/Rvachev corpus. Second, it treats the chapter’s parameter  $a$  as a genuine deformation parameter rather than as merely a route to  $a = 2$ . The main results are as follows.

1. [S] The source construction is normalized to the repository’s Fourier convention and translated from the general atomic equation through the  $h_a$  family, the critical atom up, polynomial reproduction, the uniqueness theorem, and the surviving definition of  $\text{Fup}_1$ .
2. [R] The random-series law is placed directly over the repository’s geometric- $q$  probability layer. With  $q = a^{-1}$  and  $f_q$  denoting the density of the weighted uniform sum on  $[0, 1]$ ,

$$h_a(x) = \frac{a-1}{2} f_{1/a} \left( \frac{1+(a-1)x}{2} \right).$$

At  $a = 2$ , this is precisely the formalized bridge  $f_{1/2}(u) = 2 \text{up}(2u - 1)$ .

3. [N] Every even cumulant has the closed form

$$\kappa_{2m}(X_a) = \frac{2^{2m} B_{2m}}{2m(a^{2m} - 1)}, \quad \kappa_{2m+1}(X_a) = 0,$$

and all moments follow from complete Bell polynomials or a triangular recurrence. This unifies the source's coefficient estimates with the repository's Bernoulli–Bell machinery.

4. [N] For  $a \geq 2$  the derivative supports separate at every scale, which upgrades the source's binary norm identity to

$$\|h_a^{(n)}\|_\infty = \frac{a^{n(n+3)/2}}{2^n} \|h_a\|_\infty = \frac{a^{n(n+3)/2+1}}{2^{n+1}}.$$

The second equality uses  $\|h_a\|_\infty = a/2$ ; the companion exact identity is  $\|h_a^{(n)}\|_1 = a^{n(n+1)/2}$ .

5. [N] For  $a > 2$ , the exceptional set is the self-similar set  $K_a = \{\sum_{j \geq 1} \varepsilon_j a^{-j} : \varepsilon_j = \pm 1\}$ , with  $\dim_H K_a = \log 2 / \log a$ . Its complement is an explicit disjoint union of polynomial gaps. On  $K_a$ , one branch of the  $2^n$ -term derivative formula survives, yielding an exact symbolic localization formula.

6. [N] If  $a^d = p/q \in \mathbb{Q}_{>1}$  in lowest terms, then for every degree  $n$  the lattice spacing

$$\delta_{a,n} = \frac{2}{ap^n}$$

reproduces every polynomial of degree at most  $n$  by shifts of  $h_a$ . The coefficients are constructive:

$$P(x) = \sum_{k \in \mathbb{Z}} \delta_{a,n} M_a(\partial_x)^{-1} P|_{x=k\delta_{a,n}} h_a(x - k\delta_{a,n}).$$

For  $a = 2$  this reduces exactly to the dyadic spacing  $2^{-n}$  in the source.

7. [N] The ordinary prefix spline  $h_{a,N}$ , obtained by stopping the sinc product after  $N$  factors, has a full deconvolution expansion governed by  $R_a(z) = 1/M_a(z)$ . The first uniform correction and its exact norm are

$$a^{2N}(h_{a,N} - h_a) \rightarrow -\frac{h_a''}{6(a^2 - 1)}, \quad a^{2N}\|h_{a,N} - h_a\|_\infty \rightarrow \frac{a^6}{48(a^2 - 1)}$$

for  $a \geq 2$ .

8. [N] (Fourteenth wave.) The complementary gaps of  $K_a$  form a self-similar fractal string with geometric zeta function  $\ell_0^s/(1 - 2a^{-s})$ , complex dimensions  $D_a + 2\pi i k / \log a$  on a vertical lattice, and an exact tube formula whose normalized profile is continuous, one-periodic in  $\log_a(1/\varepsilon)$ , and nonconstant; hence  $K_a$  is not Minkowski measurable. A Lebesgue-typical point of the support lies in a gap whose exact polynomial degree is geometric with parameter  $(a - 2)/a$ , and  $\frac{a-2}{2} N_a \rightarrow \text{Exp}(1)$  as  $a \downarrow 2$ .
9. [N] (Fourteenth wave.) Quantitative parameter limits at both ends:  $X_a/\sigma_a \rightarrow N(0, 1)$  as  $a \downarrow 1$  with explicit standardized-cumulant scaling, and  $aX_a \rightarrow \text{Unif}[-1, 1]$  as  $a \rightarrow \infty$  with exact  $W_\infty$  and total-variation rates and an exactly uniform expanding core.
10. [N] (Fourteenth wave.) The endpoint saddle structure is now exact at the transform level in every base: the log-Laplace sum admits the closed decomposition of [theorem 21.1](#), whose one-periodic term is real analytic with Fourier modes  $-\Gamma(-\chi_k)\zeta(1 - \chi_k)/\log a$ ,  $\chi_k = 2\pi i k / \log a$ . [C] The density-level all-orders saddle expansion in the lower-Lambert coordinate remains a research target.

11. [N] (Fifteenth wave.) For  $a > 2$  the derivative formula is a complete distributional identity: every  $h_a^{(n)}$  is equimeasurable with an amplified, fairly signed copy of  $h_a$  living on a fraction  $(2/a)^n$  of the support, giving all  $L^p$  norms at once,  $\|h_a^{(n)}\|_p = \frac{a^{n(n+3)/2}}{2^n} (2/a)^{n/p} \|h_a\|_p$ , and the polynomial on a depth- $r$  gap has signed leading coefficient  $(-1)^{N_+(\omega)} a^{(r+1)(r+2)/2} / (2^{r+1} r!)$ .
12. [N] (Fifteenth wave.) The source's  $\widehat{\text{Fup}}_1$  identity begins the classical hierarchy  $\widehat{\text{Fup}}_{n\ 2\pi}(t/2\pi) = \text{sinc}_{\text{rad}}(t2^{-n-1})^n \widehat{\text{up}}_{2\pi}(t2^{-n}/2\pi)$ : up is reconstructed from  $\text{Fup}_n$  by an exact triangular mask of  $n(n+1)/2$  dyadic averaging steps, every cumulant of  $\text{Fup}_n$  is closed, and the standardized hierarchy is asymptotically Gaussian with Berry–Esseen rate  $O(n^{-1/2})$  and all standardized cumulants  $O(n^{1-m})$ . The edge pantograph equation  $F'_a(x) = \frac{a^2}{2(a-1)} F_a(ax)$  generalizes the Fabius relation  $F' = 2F(2 \cdot)$  to every base.
13. [N] (Revised fourteenth-wave editions.) The  $p = 2$  thinning law turns the Fourier energy of every  $h_a$ ,  $a \geq 2$ , into a concrete representing measure for the scaled Stieltjes–Wigert moment problem  $m_n = a^{n(n+2)}/2^n$  – an explicit non-lognormal moment twin of the lognormal  $N(2 \log a - \log 2, 2 \log a)$ , with closed  $q$ -Pearson equations, Hankel determinants, and monic orthogonal polynomials. The distance from a Lebesgue-typical point to  $K_a$  factorizes as  $\Delta \stackrel{d}{=} \frac{1}{2} \ell_0 a^{-N_a} V$  and has an explicit Mellin transform whose poles are the complex dimensions shifted by  $-1$ ; its distribution function is the exact tube formula. The Russian source scan and OCR were verified against the merged text and then deleted; their SHA-256 hashes stand in the provenance list, and git history archives both.
14. [N] (Revised fifteenth-wave edition.) The normalized derivative tower splits into two orthogonal  $q$ -Gaussian Riesz sequences with exact kernel  $q^{(j-k)^2}$ ,  $q = 1/a$ , Gram determinants  $\prod_r (q^2; q^2)_r$ , Gram–Schmidt pivots  $(q^2; q^2)_N$ , and sharp Riesz constants  $\vartheta_4(0, q)$ ,  $\vartheta_3(0, q)$ . The leading jet amplitude at a uniform point has an exact geometric staircase tail, a log-Weibull law, and no finite positive moments. The  $\text{Fup}_n$  Edgeworth program is proved to all orders, uniformly on  $\mathbb{R}$  and after any fixed number of differentiations ([theorem 8.5](#)).
15. [N] (Expanded fifteenth-wave edition.) The Gram–Schmidt process of that geometry is closed in Gaussian binomials:  $\psi_n^* = \sum_j q^{n-j} \begin{bmatrix} n \\ j \end{bmatrix}_{q^2} e_j$  orthogonalizes each parity tower with norms  $\|\psi_n^*\|_2^2 = (q^2; q^2)_n$ , giving an explicit Cholesky factorization and inverse Gram matrix, and identifying the orthogonal polynomials as Rogers–Szegő polynomials at base  $a^{-2}$ . Each parity tower is unitarily a monomial sequence on the circle against the wrapped heat kernel at time  $\log a$ ; the determinant constant is MacMahon's plane-partition product  $\mathfrak{M}(a^{-2})$ ; and the sharp Riesz constants close up under the Jacobi triple product. The register gains the overlap-regime theta conjecture ([conjecture 23.12](#)).
16. [N] (Expanded fourteenth-wave editions.) The spectral orthogonal polynomials act on  $\mathcal{L} = -d^2/dx^2$  to give a closed, compactly supported orthogonal ladder  $\Upsilon_{a,n} = P_{a,n}(\mathcal{L})h_a$  with norms  $c^{2n} q^{-n(2n+1)} (q; q)_n \|h_a\|_2^2$ , an explicit  $q$ -binomial expansion in even derivatives, and a three-term operator recurrence – and  $\Upsilon_{a,n}$  coincides with the fifteenth wave's Gram–Schmidt vector  $(-1)^n \|h_a^{(2n)}\|_2 \psi_n^*$  ([proposition 13.13](#)). Both unnormalized parity jet Grams have closed determinants ([theorem 13.14](#)). The autocorrelation  $h_a * h_a$  has exact normalized germ  $a^{n(n+2)}/2^n$  at the origin with zero Taylor radius, and the derivative ladder is provably incomplete in its cyclic subspace (Riesz density criterion). Separately, every normalized  $p$ -energy measure of every derivative factors exactly as a level- $n$  Bernoulli address sum plus an  $a^{-n}$ -compressed base profile ([theorem 11.7](#)):  $W_\infty$ -convergence at rate  $2a^{-n}/(a-1)$  to the symmetric Bernoulli convolution  $\nu_a$  – uniform at  $a = 2$ , the equal-weight Cantor measure on  $K_a$  for  $a > 2$  – with exact Hausdorff support rate  $b_1 a^{-n}$  and exact Rényi/Shannon entropy law  $H_\beta(n) = H_\beta(0) + n \log(2/a)$ , so the derivative hierarchy carries the information dimension  $D_a$  of its singular locus. The register gains [conjectures 23.13](#) and [23.14](#).

17. [N] (Expanded fifteenth-wave editions.) The closed orthogonalization acquires four closures: the residuals are nodal polynomials,  $P_n(z) = q^{n^2} \prod_{j < n} (z - q^{-2j})$ , so orthogonality is interpolation at geometric nodes and the pivot is the value at the next node; the triangular transform has the explicit inverse  $\eta_n = \sum_k q^{(n-k)^2} \begin{bmatrix} n \\ k \end{bmatrix}_{q^2} r_k$ , giving the entrywise positive Cholesky factorization of the positive kernel (theorem 13.4); the innovation filters converge in  $\ell^1$  to the causal minimum-phase Szegő factor  $A_q(z) = 1/(-qz; q^2)_\infty$  with the exact whitening identity  $|A_q|^2 \vartheta_3 = (q^2; q^2)_\infty$  (theorem 13.9); and every mixed minor of the kernel is a positive Schur-polynomial expression, so the kernel is strictly totally positive, with oscillation and checkerboard-inverse consequences (theorem 13.10). The jet-intermittency law gains its two-term tail  $-\gamma_a \sqrt{\log x} - (\log \frac{a}{2} / (2 \log a)) \log \log x + \mathcal{O}(1)$  and the sharp Orlicz threshold  $\mathbb{E} \exp(\theta \sqrt{\log A}) < \infty \iff \theta < \gamma_a$  (theorem 15.7), and the highest local jet  $\mathcal{J}_a = |h_a^{(N_a)}(Y)|$  obeys an exact staircase law whose reciprocal moments are partial theta functions, jointly with the distance to  $K_a$ : the fractal zeta denominator of the distance-Mellin law deforms, for  $s > 0$ , into an entire partial theta series (theorem 15.8). The register gains conjectures 23.15 to 23.19, and the algebraic-breakpoint conjecture acquires its transcendental-dichotomy sharpening.

## 2 Corpus audit, normalization, and editorial method

### 2.1 Repository corpus used for non-duplication

The living documents treated as canonical were the following recursively located `.tex` sources under `Analysis/FabiusFunction/docs`:

- `Fabius_Function_and_Rvachev_Up/Fabius_Function_and_Rvachev_Up.tex`, the primary mathematical exposition;
- `FabiusFunction_Mathematical_Glossary/FabiusFunction_Mathematical_Glossary.tex`, which fixes notation and terminology;
- `Non_Elementarity_of_the_Fabius_Function/Non_Elementarity_of_the_Fabius_Function.tex`;
- `fabius_lean_walkthrough/fabius_lean_walkthrough.tex`;
- `semi-formalized-research-frontiers/semi-formalized-research-frontiers.tex`.

The recursive archive was also inventoried by its content groups—early notes, glossary, Lean walk-through, primary expositions and supplements,  $q$ -calculus, research frontiers, and standalone studies. The archive README explicitly marks these as frozen historical versions and prunes byte-identical duplicates. Accordingly, archived copies were used for provenance and gap checking, not counted as independent mathematical claims. The frontier volume was treated as the canonical quarantine boundary: a numerical match or plausible derivation does not move a claim into the formalized primary exposition.

A machine-readable clone was not available in the execution container, so the audit used the public rendered tree and raw canonical sources retrieved from GitHub. This limitation affects file-system statistics, not the mathematical cross-comparison stated in this report. The package includes `corpus_audit.txt`, which records the scope and the exact URLs used.

### 2.2 Fourier, probability, and function conventions

We follow the repository convention

$$\widehat{f}_{2\pi}(\xi) = \int_{\mathbb{R}} f(x) e^{-2\pi i x \xi} dx, \quad f(x) = \int_{\mathbb{R}} \widehat{f}_{2\pi}(\xi) e^{2\pi i x \xi} d\xi, \quad (2.1)$$



whenever both integrals are classical. Also

$$\text{sinc}_{\text{rad}} z = \begin{cases} \sin z/z, & z \neq 0, \\ 1, & z = 0. \end{cases}$$

The bounded Fabius function is  $F_{\text{bd}}$  – written  $F$  in the source and in this part’s running prose where no confusion with a distribution function is possible – the even Rvachev atom is up, and the signed Thue–Morse sign is

$$\tau(k) = \varepsilon_k.$$

We write  $B_n$  for Bernoulli numbers with  $B_1 = -1/2$ , as fixed in (0.4.1),  $M_a$  for a moment generating function, and  $K_a$  for the self-similar exceptional set. This part’s parameter is the base  $a > 1$ , related to the canonical parameter of definition 0.2.1 by  $q = a^{-1}$ ; its  $X_a$  is the  $[-1, 1]$ -supported variable (5.6), so  $X_a = b_a \cdot 2X_q^\circ$  with  $b_a = (a - 1)^{-1}$ , and the same letter  $X_q$  elsewhere in the volume denotes the  $[0, 1]$ -supported canonical variable. See table 1.

### 2.3 Editorial reconstruction rules

The translation is not a facsimile. It is a mathematical edition governed by four rules.

1. Fourier signs and  $2\pi$  factors are converted consistently to (2.1).
2. A source formula is corrected only when the adjacent derivation, the PDF image, or a normalization identity makes the correction forced.
3. Theorem hypotheses that are visibly damaged are marked as reconstructed. The most important case is the uniqueness theorem in Section 4: its derivative growth denominator is determined by equation (3.87) and by the source’s counterexample  $\text{up} + \alpha \text{up}'$ .
4. Material beyond equation (3.100) is not supplied as though translated. Later  $\text{Fup}_n$  remarks are explicitly presented as extensions, not recovered source text.

## 3 English translation I: why atomic functions arise

### [S] Source translation or source reconstruction

**Translated opening argument.** Algebraic and trigonometric polynomials are universal approximation tools, but they are not finite: changing one coefficient affects the approximant globally. For numerical work one often prefers local functions with supports of small diameter. Their matrices are sparse or banded, so a fixed amount of memory can support higher-dimensional approximation spaces. Classical finite splines, including Schoenberg’s  $B$ -splines [28], are local, but a spline of fixed degree has fixed smoothness and therefore does not remain optimal as the target smoothness increases. Classical polynomials are universal but nonlocal; ordinary splines are local but not universal. The natural objective is a system that is both local and universal and is still computationally tractable.

A degree- $n$  compact spline is an  $(n+1)$ -fold convolution of interval indicators. The source’s decisive step is to continue this construction indefinitely. Let  $u_\alpha$  denote the probability density  $(2\alpha)^{-1} \mathbf{1}_{[-\alpha, \alpha]}$ . For positive numbers  $\alpha_k \downarrow 0$  with  $\sum_k \alpha_k < \infty$ , the infinite convolution

$$\varphi_\alpha = u_{\alpha_1} * u_{\alpha_2} * \cdots$$

has Fourier transform

$$\widehat{\varphi}_\alpha(2\pi)(\xi) = \prod_{k=1}^{\infty} \text{sinc}_{\text{rad}}(2\pi\alpha_k\xi). \quad (3.1)$$

The summability of the half-widths gives compact support in  $[-\sum_k \alpha_k, \sum_k \alpha_k]$ ; sufficiently rapid decay of the product gives  $C^\infty$  smoothness. The simplest self-similar choice is  $\alpha_k = 2^{-k}$ , and the source denotes the resulting atom by  $\widehat{\text{up}}$ :

$$\widehat{\text{up}}_{2\pi}(\xi) = \prod_{k=1}^{\infty} \text{sinc}_{\text{rad}}(2\pi 2^{-k} \xi). \quad (3.2)$$

The historical motivation described in the source is geometric. A compactly supported differentiable “hump” has a derivative consisting of a positive hump and a negative well. Rvachev asked in 1967 whether these two pieces could be scaled copies of the original hump. With support normalized to  $[-1, 1]$ , this becomes

$$y'(x) = 2y(2x + 1) - 2y(2x - 1). \quad (3.3)$$

The normalized nonzero compactly supported solution is  $\widehat{\text{up}}$ . Unlike a classical polynomial, which is characterized by a homogeneous constant-coefficient differential equation, an atomic function is characterized by a differential operator balanced against contracted and shifted copies of itself.

## 4 English translation II: the general atomic equation

### 4.1 Definition and Fourier reduction

**Definition 4.1** (Atomic function, source normalization). Let  $a > 1$ , let  $L = P(D_F)$  be a constant-coefficient ordinary differential operator, and let

$$(Ry)(x) = \sum_{k=1}^M c_k y(ax - b_k).$$

A nonzero compactly supported solution of

$$P(D_F)y(x) = \lambda \sum_{k=1}^M c_k y(ax - b_k) \quad (4.1)$$

is called an *atomic function* associated with  $(P, R, \lambda)$ . Here we take  $D = (2\pi i)^{-1} d/dx$ , so that  $\widehat{Df}_{2\pi}(\xi) = \xi \widehat{f}_{2\pi}(\xi)$ .

Put

$$\mathcal{E}(\xi) = \frac{1}{a} \sum_{k=1}^M c_k e^{-2\pi i b_k \xi / a}. \quad (4.2)$$

Then Fourier transformation of (4.1) gives

$$P(\xi)Y(\xi) = \lambda \mathcal{E}(\xi)Y(\xi/a). \quad (4.3)$$

Iteration yields

$$Y(\xi) \prod_{j=0}^N P(\xi a^{-j}) = \lambda^{N+1} \prod_{j=0}^N \mathcal{E}(\xi a^{-j}) Y(\xi a^{-N-1}). \quad (4.4)$$

If  $Y(0) = 1$ , the formal solution is

$$Y(\xi) = \prod_{j=0}^{\infty} \frac{\lambda \mathcal{E}(\xi a^{-j})}{P(\xi a^{-j})}, \quad (4.5)$$

with all apparent poles required to be removable.



## 4.2 The source zero-matching criterion

Let  $A$  be the multiset of zeros of  $P$  and  $B$  the multiset of zeros of  $\mathcal{E}$ , each counted with multiplicity. A denominator zero produced by  $P(\xi a^{-j})$  at  $\xi = a^j \alpha$  must be canceled by a numerator zero from a later scale. The source encodes precisely this requirement.

**Theorem 4.2** (Source Theorem 1, edited). *Assume the zero at the origin has first been normalized so that the ratio in (4.5) tends to one as  $\xi \rightarrow 0$ . A nonzero compactly supported solution of (4.1) with integral one exists if and only if there is a multiplicity-preserving injection*

$$\iota : A \longrightarrow B$$

such that, for every  $\alpha \in A$ , one has

$$\iota(\alpha) = a^{-m(\alpha)} \alpha$$

for some integer  $m(\alpha) \geq 0$ . Under this condition the product (4.5) is entire of exponential type, rapidly decreasing on the real axis, and its inverse Fourier transform is the unique normalized compactly supported solution.

*Translation of the proof architecture.* Necessity follows from (4.4). For fixed  $\xi$ ,  $Y(\xi a^{-N-1})$  is eventually nonzero because it tends to  $Y(0) = 1$ . Hence every zero generated by a factor  $P(\xi a^{-j})$  must occur in the zero multiset of a numerator factor  $\mathcal{E}(\xi a^{-k})$ . Following each zero along its  $a$ -adic orbit produces the stated injection; the source devotes several pages to removing collisions and preserving multiplicity.

For sufficiency, use the injection to cancel every denominator zero in the infinite product. On a fixed disk, all sufficiently late factors are analytic and equal to  $1 + \mathcal{O}(a^{-j})$ , so the tail converges locally uniformly. The finite uncanceled prefactor is entire after the prescribed removals. The exponential-polynomial numerator controls the exponential type, while repeated use of (4.3) gives arbitrary real-axis power decay. The Paley–Wiener theorem then gives compact support, and direct substitution verifies (4.1).  $\square$

*Remark 4.3.* The criterion is best understood as a divisibility statement in an  $a$ -adic orbit category: the symbol  $P$  creates potential poles at geometrically expanded zeros, and the exponential polynomial  $\mathcal{E}$  must supply a disjoint chain of zeros at later contraction levels. This point of view connects the chapter to modern refinement equations and subdivision masks.

## 4.3 Solutions with zero integral

The source next treats  $Y(0) = 0$ . Let  $r$  be the order of vanishing and write  $Y(\xi) = \xi^r G(\xi)$  with  $G(0) \neq 0$ . Since  $Y(\xi/a) = a^{-r} \xi^r G(\xi/a)$ , the reduced transform  $G$  satisfies the same functional equation with the eigenvalue scaled by  $a^{-r}$ . In physical space the solution is a derivative of the normalized atom.

**Theorem 4.4** (Source Theorem 2, edited). *Under the zero-matching condition of theorem 4.2, every compactly supported solution whose Fourier transform has a zero of order  $r$  at the origin is a scalar multiple of the  $r$ th derivative of a normalized atomic function. The corresponding spectral parameter is multiplied by  $a^r$  relative to the integral-one solution. Different derivative orders are linearly independent.*

This elementary observation is already a seed of the derivative hierarchies that later produce Thue–Morse signs and exact norm growth.

# 5 English translation III: the first-order family $h_a$

## 5.1 Reduction to the symmetric equation

The source specializes to a first-order operator and two shifts. After a translation of  $x$  and an affine rescaling, the equation becomes

$$y'(x) + \alpha y(x) = \lambda(y(ax - 1) + c y(ax + 1)). \quad (5.1)$$

### 5.1.1 The nonzero-drift family

Before the symmetric reduction, the source keeps arbitrary shifts  $b_1 \neq b_2$  and coefficients  $c_1, c_2$ . When  $\alpha \neq 0$ , normalization at frequency zero gives

$$\lambda = \frac{a\alpha}{c_1 + c_2}. \quad (5.2)$$

The zero-matching condition is satisfied precisely along a countable geometric family: after setting  $c_1 = 1$ , for some integer  $m \geq 0$  one must have

$$c_2 = -\exp((b_2 - b_1)\alpha a^{-m-1}). \quad (5.3)$$

For complex data this statement is read on the corresponding logarithm branch. Thus, for fixed  $b_1, b_2$  and  $\alpha \neq 0$ , the source obtains countably many admissible two-shift masks. This part of the chapter is structurally important: the compact atom is selected by a geometric collision between the zero of the differential symbol and a zero of the exponential mask.

### 5.1.2 Zero drift and the symmetric atom

For  $\alpha = 0$ , normalization forces  $c = -1$  and  $\lambda = -a^2/2$  in the source's order of terms. Equivalently,

$$h'_a(x) = \frac{a^2}{2}(h_a(ax+1) - h_a(ax-1)). \quad (5.4)$$

The integral-one solution is

$$\widehat{h_{a\,2\pi}}(\xi) = \Phi_a(\xi) := \prod_{j=1}^{\infty} \text{sinc}_{\text{rad}}(2\pi a^{-j}\xi). \quad (5.5)$$

The critical value is  $h_2 = \text{up}$ .

## 5.2 Probability model and support

Let  $U_1, U_2, \dots$  be independent uniform random variables on  $[-1, 1]$  and set

$$X_a = \sum_{j=1}^{\infty} a^{-j} U_j. \quad (5.6)$$

Then  $\Phi_a$  is the characteristic function of  $X_a$ , and  $h_a$  is its density. Since the tail is absolutely bounded,

$$\text{supp } h_a = [-b_a, b_a], \quad b_a = \sum_{j \geq 1} a^{-j} = \frac{1}{a-1}. \quad (5.7)$$

Separating the first summand in (5.6) gives the distributional fixed point

$$X_a \stackrel{d}{=} \frac{U + X'_a}{a},$$

where  $X'_a$  is an independent copy. Therefore

$$h_a(x) = \frac{a}{2} \int_{ax-1}^{ax+1} h_a(u) \, du, \quad (5.8)$$

and differentiation recovers (5.4).

**[S] Source translation or source reconstruction**

The probability interpretation in the attached chapter is not decorative. It immediately explains support, positivity, evenness, normalization, and the functional equation. It also places the construction in the classical theory of infinite convolutions developed by Jessen and Wintner [8], while the geometric weights make the law exactly self-similar.

**5.3 Maclaurin coefficients and source decay estimates**

Write, in the source's angular variable  $t$ ,

$$\Phi_a(t/2\pi) = \sum_{k \geq 0} c_{2k}(a) t^{2k}, \quad c_{2k+1}(a) = 0.$$

The one-step identity

$$\Phi_a(t/2\pi) = \frac{\sin(t/a)}{t/a} \Phi_a(t/(2\pi a))$$

gives the triangular recurrence

$$c_{2k}(a) = \frac{1}{a^{2k} - 1} \sum_{\ell=0}^{k-1} \frac{(-1)^{k-\ell} c_{2\ell}(a)}{(2k - 2\ell + 1)!}. \quad (5.9)$$

The source iterates this recurrence to obtain a decay estimate. The sharp statement, and the one used below, is

$$|c_{2k}(a)| = \frac{\mathbb{E} X_a^{2k}}{(2k)!} \leq \frac{b_a^{2k}}{(2k)!} = \frac{1}{(a-1)^{2k} (2k)!}, \quad b_a = \frac{1}{a-1}, \quad (5.10)$$

where  $X_a$  is the random variable (5.6) and  $b_a$  its support endpoint (5.7). Indeed  $G(t) := \Phi_a(t/2\pi) = \mathbb{E} e^{itX_a}$  is entire, its Maclaurin coefficients are  $c_{2k} = (-1)^k \mathbb{E} X_a^{2k} / (2k)!$  and  $c_{2k+1} = 0$  by symmetry, and  $|X_a| \leq b_a$  almost surely; equivalently,  $G$  has exponential type  $b_a$  because  $|\operatorname{sinc}_{\text{rad}} w| \leq \sinh |w| / |w| \leq e^{|w|}$  makes  $|G(t)| \leq e^{|t| \sum_j a^{-j}}$ .

**Warning 5.1** (A superfactorial bound would be false here). A decay estimate of the shape  $|c_{2k}(a)| \leq C_a a^{-k(k+1)} (\prod_{j \leq k} j!)^{-1}$  cannot hold for any constant  $C_a$ : the superfactorial  $\prod_{j \leq k} j!$  grows like  $k^{k^2/2}$  and so outgrows  $(2k)!$ , while  $|c_{2k}|$  is of exact order  $(2k)!^{-1}$  by (5.10) and the fact that  $G$  has positive type. Solving the recurrence (5.9) in exact rational arithmetic makes the failure explicit: at  $a = 2$  the ratio of  $|c_{2k}|$  to that expression with  $C_a = 1$  is  $6.7 \times 10^{-1}$  at  $k = 3$  but already  $4.2 \times 10^{38}$  at  $k = 10$  and  $9.4 \times 10^{62}$  at  $k = 12$ , and it grows without bound; the same happens at  $a = 3$  and  $a = 3/2$ . The genuine quadratic-in- $n$  scale in this theory is not in the Maclaurin coefficients but in the derivative ladder on the real axis,  $\|h_a^{(n)}\|_\infty = a^{n(n+3)/2+1} / 2^{n+1}$ , which is what places  $h_a$  in a quadratic-exponential Denjoy–Carleman class.

Since

$$\Phi_a^{(2k)}(0) = (-1)^k (2\pi)^{2k} \int x^{2k} h_a(x) \, dx,$$

(5.10) is exactly a moment bound. In later sections the coefficient recurrence is replaced by a closed Bernoulli–Bell calculus.

**6 English translation IV: the separated regime  $a > 2$**

## 6.1 The central plateau and the polynomial-gap cascade

When  $a > 2$ , the two intervals on which the terms in (5.4) can be nonzero are separated near the origin. Put

$$b_1 = b_a - \frac{2b_a}{a} = \frac{a-2}{a(a-1)}. \quad (6.1)$$

For  $|x| < b_1$ , both  $ax - 1$  and  $ax + 1$  lie outside the support of one derivative branch in such a way that  $h'_a(x) = 0$ . The normalization in (5.8) then gives

$$h_a(x) = \frac{a}{2}, \quad |x| \leq b_1. \quad (6.2)$$

At the next scale there are two intervals on which  $h_a$  is linear; at the next there are four intervals on which it is quadratic; and so on. The source sums the lengths of these intervals and obtains the entire support measure. Thus  $h_a$  is locally a polynomial at almost every point, while a nowhere-dense Cantor-type set remains exceptional.

### [S] Source translation or source reconstruction

The attached source calls the functions  $h_a$ ,  $a > 2$ , “splines of class  $C^\infty$ .” This phrase is intentionally broader than the usual finite-knot spline: the knot set is fractal, the local polynomial degrees are unbounded, and the function is globally  $C^\infty$  because every transition is flat to the required order.

The source computes the first pieces explicitly. On the central plateau, (6.2) holds. Integrating the differential equation from the left endpoint gives values at the first gap endpoints, and a second integration gives the adjacent linear pieces. Repeating the operation recursively expresses every local polynomial coefficient rationally in  $a$ . The printed formulas are somewhat cumbersome, but their conceptual content is simple: a word in the two inverse branches determines the interval; iterating (5.4) determines the first nonzero derivative there; and integration from a flat boundary determines the polynomial.

## 6.2 Taylor series and the exceptional set in the source

At a point lying strictly inside a polynomial gap, the Taylor series is exactly the local polynomial. At an endpoint of such a gap, the Taylor series may still terminate, but it need not equal the function on any full neighborhood. At the remaining exceptional points the source asserts zero Taylor radius. This gives a striking class of compactly supported  $C^\infty$  functions for which the Taylor series is a polynomial almost everywhere, even though the function is neither globally polynomial nor analytic.

The attached argument encodes the exceptional points by signed radix- $a$  expansions and differentiates repeatedly. Its signs satisfy

$$\delta_{2k-1} = \delta_k, \quad \delta_{2k} = -\delta_k, \quad \delta_1 = 1,$$

which is exactly the shifted Thue–Morse rule. The OCR obscures several indices, so a clean symbolic form is developed in [appendix 14](#).

## 6.3 Source evaluation formulae

For a depth- $n$  endpoint, the source applies Taylor’s integral remainder from the left support boundary and inserts the  $n$ th derivative formula. Only moments of  $h_a$  remain. Consequently every value and derivative at every finite signed radix- $a$  endpoint is a finite linear combination of the even moments

$$m_{2r}(a) = \int_{-b_a}^{b_a} x^{2r} h_a(x) \, dx.$$

This is an exact algorithm. The Bernoulli–Bell formulas in [appendix 10](#) turn those moment inputs into explicit rational functions of  $a$  whenever  $a$  is rational or algebraic.

## 7 English translation V: the critical atom $\text{up}$

### 7.1 Definition, support, and differential equation

At  $a = 2$  the central gap collapses. The source defines

$$\text{up} = h_2, \quad \widehat{\text{up}}_{2\pi}(\xi) = \prod_{j \geq 1} \text{sinc}_{\text{rad}}(2\pi 2^{-j} \xi), \quad (7.1)$$

with

$$\text{up}'(x) = 2 \text{up}(2x + 1) - 2 \text{up}(2x - 1), \quad \text{supp up} = [-1, 1]. \quad (7.2)$$

Repeated differentiation produces the finite signed sum

$$\text{up}^{(n)}(x) = 2^{\binom{n+1}{2}} \sum_{k=1}^{2^n} \tau(k-1) \text{up}(2^n x + 2^n + 1 - 2k). \quad (7.3)$$

The source records the exact norm

$$\|\text{up}^{(n)}\|_{C[-1,1]} = 2^{\binom{n+1}{2}}. \quad (7.4)$$

### 7.2 Nowhere analyticity and dyadic Taylor polynomials

At a dyadic rational  $x = k2^{-n}$  in  $[-1, 1]$ , the Taylor series terminates after finitely many terms. At every nondyadic point of the support, the source states that the Taylor radius is zero. Since a terminating Taylor series at a dyadic point does not agree with  $\text{up}$  on any neighborhood,  $\text{up}$  is analytic nowhere on its support. This is the critical contrast with  $a > 2$ : separated cylinders give local polynomial intervals of full measure, while at  $a = 2$  the cylinders only touch and the exceptional geometry fills the support.

### 7.3 Moments and exact dyadic values

Let

$$\mu_{2r} = \int_{-1}^1 x^{2r} \text{up}(x) \, dx.$$

The source relates these moments to the even Maclaurin coefficients of the sinc product and introduces half-moments used in exact dyadic evaluation. Taylor's integral formula, followed by (7.3), yields a finite expression for

$$\text{up}(-1 + k2^{-n}) \quad (7.5)$$

in terms of  $\tau$ , binomial coefficients, and the moments. For the first point in a dyadic cell the expression simplifies to a half-moment divided by  $2^{\binom{n}{2}}(n-1)!$ . This is one of the bridges to the exact arithmetic results later developed by Arias de Reyna and Haugland [31, 37], and by the repository.

### 7.4 Recursive evaluation on arbitrary points

On the interval

$$[-1 + 2^{-n}, -1 + 2^{-n+1}],$$

the source considers

$$\varphi_n(x) = \text{up}(x) + \text{up}(x - 2^{-n}).$$

Its derivative of order  $n+1$  vanishes there, so  $\varphi_n$  is a polynomial of degree at most  $n$ . Therefore

$$\text{up}(x) = \sum_{\ell=0}^n \frac{\text{up}^{(\ell)}(-1 + 2^{-n})}{\ell!} (x + 1 - 2^{-n})^\ell - \text{up}(x - 2^{-n}). \quad (7.6)$$

Iterating across the binary expansion of  $x$  reduces arbitrary evaluation to values exponentially close to the flat endpoint. The source combines this with its coefficient bounds to obtain a rapidly convergent series indexed by the binary digits. In modern language this is a bit-recursive exact evaluator with a superexponentially small tail.

## 8 English translation VI: polynomial reproduction and uniqueness

### 8.1 The source difference equation

Consider a locally finite shift sum on the dyadic lattice

$$\varphi(x) = \sum_{k \in \mathbb{Z}} c_k \operatorname{up}(x - k2^{-n}). \quad (8.1)$$

Differentiating  $n + 1$  times and using (7.3), the source obtains the constant-coefficient difference equation

$$\sum_{j=1}^{2^{n+1}} \delta_j c_{k+j} = 0, \quad k \in \mathbb{Z}, \quad (8.2)$$

where  $\delta_j = \tau(j - 1)$ . Its characteristic polynomial is

$$Q_n(z) = \sum_{j=1}^{2^{n+1}} \delta_j z^{2^{n+1}-j} = \prod_{r=0}^n (z^{2^r} - 1). \quad (8.3)$$

Thus every coefficient sequence producing a polynomial of degree at most  $n$  is a finite sum of polynomially weighted roots of unity. In particular, the root  $z = 1$  has multiplicity  $n + 1$ , so any polynomial sequence  $c_k$  of degree at most  $n$  produces a polynomial in  $x$  of degree at most  $n$ .

A cleaner cyclotomic decomposition, useful for comparing with the source's 2-adic formula, is

$$Q_n(z) = \prod_{s=0}^n \Phi_{2^s}(z)^{n-s+1}, \quad (8.4)$$

where  $\Phi_1(z) = z - 1$ . Hence a root of exact order  $2^s$  has multiplicity  $n - s + 1$ . The cosine–sine expression in the source is precisely the real form of the standard solution of this recurrence.

### 8.2 Reconstructed source uniqueness theorem

The OCR destroys the denominator in hypothesis 2 of the uniqueness theorem. Equations (3.87) and the final counterexample force the following reading.

**Theorem 8.1** (Source uniqueness theorem, reconstructed). *Let  $\varphi \in C^\infty(\mathbb{R})$  be supported in  $[-1, 1]$  and suppose*

$$1. \int_{-1}^1 \varphi(x) \, dx = 1;$$

2.

$$\lim_{n \rightarrow \infty} \frac{\|\varphi^{(n)}\|_\infty}{2^{\binom{n+2}{2}}} = 0; \quad (8.5)$$

3. *for each  $n \in \mathbb{N}_0$  there are coefficients  $c_j^{(n)}$  such that the locally finite sum*

$$\sum_{j \in \mathbb{Z}} c_j^{(n)} \varphi(x - j2^{-n}) = x^n$$

*holds for every  $x \in \mathbb{R}$ .*

Then  $\varphi = \text{up}$ .

*Source proof, edited.* Reproduction of constants implies a partition of unity by integer translates. From this, the source constructs another compactly supported normalized function  $\varphi_1$  satisfying

$$\varphi'(x) = 2\varphi_1(2x+1) - 2\varphi_1(2x-1).$$

Repeating the argument produces  $\varphi_m$  with

$$\varphi'_m(x) = 2\varphi_{m+1}(2x+1) - 2\varphi_{m+1}(2x-1). \quad (8.6)$$

The growth condition implies  $2^{-m}\|\varphi_{m+1}\|_\infty \rightarrow 0$  along a subsequence. Put  $\gamma_m = \text{up} - \varphi_m$ . The same integral refinement equation holds for  $\gamma_m$ , and the symmetry relation on adjacent unit intervals gives a one-step norm contraction. Iteration yields

$$\|\gamma_0\|_\infty \leq 2^{-m}\|\gamma_{m+1}\|_\infty.$$

Taking the subsequence from the growth condition gives  $\gamma_0 = 0$ .  $\square$

The source observes that the growth scale is sharp in its method. Indeed,  $\varphi = \text{up} + \alpha \text{up}'$  still integrates to one and has the same broad reproduction behavior, but

$$\lim_{n \rightarrow \infty} \frac{\|\varphi^{(n)}\|_\infty}{2^{\binom{n+2}{2}}} = |\alpha|,$$

so the theorem cannot identify it with  $\text{up}$ .

### 8.3 The surviving beginning of the $\text{Fup}_n$ hierarchy

The source's Section 5 ends after defining only  $\text{Fup}_1$ . Using

$$\text{sinc}_{\text{rad}}(t/2) = \cos(t/4) \text{sinc}_{\text{rad}}(t/4),$$

it rewrites

$$\prod_{j \geq 1} \text{sinc}_{\text{rad}}(t2^{-j}) = \cos(t/4) (\text{sinc}_{\text{rad}}(t/4))^2 \prod_{j \geq 3} \text{sinc}_{\text{rad}}(t2^{-j}).$$

Define  $\text{Fup}_1$  by the Fourier transform

$$\widehat{\text{Fup}_1}_{2\pi}(t/2\pi) = (\text{sinc}_{\text{rad}}(t/4))^2 \prod_{j \geq 3} \text{sinc}_{\text{rad}}(t2^{-j}). \quad (8.7)$$

Then multiplication by  $\cos(t/4)$  gives the exact averaging identity

$$\text{up}(x) = \frac{1}{2} (\text{Fup}_1(x - 1/4) + \text{Fup}_1(x + 1/4)). \quad (8.8)$$

Nothing beyond this identity survives in the supplied files.

### 8.4 A canonical $\text{Fup}$ ladder

The following continuation is new material (fourteenth wave), not translated source text. The same halving identity that produced  $\text{Fup}_1$  forces a unique one-parameter ladder. In the source's angular variable, put

$$H(t) = \prod_{j \geq 2} \text{sinc}_{\text{rad}}(t2^{-j}), \quad (8.9)$$

and define  $G_n$  for  $n \geq 0$  by

$$\widehat{G_n}_{2\pi}(t/2\pi) = H(t) \text{sinc}_{\text{rad}}(t2^{-(n+1)}). \quad (8.10)$$

Then  $G_0 = \text{up}$  and  $G_1 = \text{Fup}_1$ .



**Proposition 8.2** (Dyadic Fup ladder). *For  $n \geq 1$ ,*

$$G_{n-1}(x) = \frac{1}{2} \left( G_n(x - 2^{-(n+1)}) + G_n(x + 2^{-(n+1)}) \right), \quad (8.11)$$

*and the support radius of  $G_n$  is  $\frac{1}{2} + 2^{-(n+1)}$ . Moreover,*

$$G_n \longrightarrow 2 \operatorname{up}(2 \cdot) \quad \text{in } C^m(\mathbb{R}) \text{ for every fixed } m. \quad (8.12)$$

*Proof.* The halving identity  $\operatorname{sinc}_{\text{rad}}(t2^{-n}) = \cos(t2^{-(n+1)}) \operatorname{sinc}_{\text{rad}}(t2^{-(n+1)})$  gives  $\widehat{G_{n-1} 2\pi}(t/2\pi) = \cos(t2^{-(n+1)}) \widehat{G_n 2\pi}(t/2\pi)$ , which is (8.11). Pointwise  $\widehat{G_n 2\pi}(t/2\pi) \rightarrow H(t)$ , and  $H$  is the transform of the density of  $\sum_{j \geq 2} 2^{-j} U_j = X_2/2$ , namely  $2 \operatorname{up}(2x)$ . Since  $|t|^m H(t)$  is integrable and the products are dominated by finitely many common factors, dominated Fourier inversion gives  $C^m$  convergence. The support radius is the sum of the uniform half-widths in (8.10).  $\square$

Iterating (8.11) writes up as an average of  $2^n$  translates of  $G_n$ ; in the limit, the translate locations distribute as the centered dyadic Bernoulli sum, whose law is uniform on  $[-1/2, 1/2]$ , and the first-step convolution identity for up is recovered. The Fup construction is thus a finite Rademacher factorization of the uniform smoothing mask.

### 8.5 The classical Fup<sub>n</sub> hierarchy and its Gaussian regime

A second canonical continuation (fifteenth wave) is the historically standard hierarchy, which piles the redistributed multiplicity onto a single scale instead of sliding one factor. In the source's angular variable, define

$$\widehat{\operatorname{Fup}_n 2\pi}(t/2\pi) = \operatorname{sinc}_{\text{rad}}(t2^{-n-1})^n \widehat{\operatorname{up} 2\pi}(t2^{-n}/2\pi), \quad \operatorname{Fup}_0 = \operatorname{up}, \quad (8.13)$$

which reduces to (8.7) at  $n = 1$ . Probabilistically, Fup<sub>n</sub> is the density of

$$Z_n = 2^{-n} X_2 + 2^{-n-1} \sum_{r=1}^n U_r, \quad (8.14)$$

with  $X_2$  the binary random series and  $U_r$  independent uniforms on  $[-1, 1]$ , so

$$\operatorname{supp} \operatorname{Fup}_n = \left[ -\frac{n+2}{2^{n+1}}, \frac{n+2}{2^{n+1}} \right]. \quad (8.15)$$

**Theorem 8.3** (Triangular reconstruction at every level). *For every  $n \geq 0$ ,*

$$\widehat{\operatorname{up} 2\pi}(t/2\pi) = \widehat{\operatorname{Fup}_n 2\pi}(t/2\pi) \prod_{r=2}^{n+1} \cos(t2^{-r})^{r-1}. \quad (8.16)$$

*Equivalently, with  $N_n = n(n+1)/2$ , the function up is the average of the  $2^{N_n}$  translates of Fup<sub>n</sub> by the signed sums  $\sum_{r=2}^{n+1} \sum_{s=1}^{r-1} \varepsilon_{r,s} 2^{-r}$ ,  $\varepsilon_{r,s} \in \{\pm 1\}$ . Letting  $n \rightarrow \infty$  recovers the infinite cosine-multiplicity factorization*

$$\widehat{\operatorname{up} 2\pi}(t/2\pi) = \prod_{r=2}^{\infty} \cos(t2^{-r})^{r-1}. \quad (8.17)$$

*Proof.* By (8.13),  $\widehat{\operatorname{up} 2\pi}/\widehat{\operatorname{Fup}_n 2\pi} = \prod_{j=1}^n \operatorname{sinc}_{\text{rad}}(t2^{-j})/\operatorname{sinc}_{\text{rad}}(t2^{-n-1})$  at matched arguments. Iterating the halving identity  $\operatorname{sinc}_{\text{rad}}(t2^{-j}) = [\prod_{r=j+1}^{n+1} \cos(t2^{-r})] \operatorname{sinc}_{\text{rad}}(t2^{-n-1})$  and counting the pairs  $j < r$  gives each cosine the multiplicity  $r-1$ , which proves (8.16); expanding every cosine as a two-point averaging multiplier gives the translate form. The infinite product (8.17) follows because  $\widehat{\operatorname{Fup}_n 2\pi}(t/2\pi) \rightarrow 1$  locally uniformly.  $\square$

**Theorem 8.4** (Exact cumulants and central-limit regime). *All odd cumulants of  $Z_n$  vanish, and for  $m \geq 1$ ,*

$$\kappa_{2m}(Z_n) = 2^{-2mn} \frac{2^{2m-1} B_{2m}}{m} \left( \frac{1}{2^{2m}-1} + \frac{n}{2^{2m}} \right), \quad \sigma_n^2 := \kappa_2(Z_n) = 4^{-n} \frac{3n+4}{36}. \quad (8.18)$$

As  $n \rightarrow \infty$ , the standardized variable  $W_n = Z_n/\sigma_n$  satisfies  $W_n \Rightarrow N(0, 1)$  with Kolmogorov distance  $O(n^{-1/2})$ , and  $\kappa_{2m}(W_n) = O(n^{1-m})$  for every  $m \geq 2$ ; the first non-Gaussian correction is of order  $1/n$  and is determined by  $B_4 = -\frac{1}{30}$ .

*Proof.* Cumulants add along (8.14):  $\kappa_{2m}(Z_n) = 2^{-2mn} \kappa_{2m}(X_2) + n 2^{-2m(n+1)} \kappa_{2m}(U)$ , and inserting  $\kappa_{2m}(U) = 2^{2m-1} B_{2m}/m$  and the  $a = 2$  case of (10.3) gives (8.18);  $m = 1$  gives  $\sigma_n^2$ . After multiplication by  $2^{n+1}$ ,  $Z_n$  is a sum of  $n$  independent uniforms plus the bounded perturbation  $2X_2$ , so the classical central limit theorem and the Berry–Esseen bound apply; dividing (8.18) by  $\sigma_n^{2m} \asymp n^m 2^{-2mn}$  gives  $\kappa_{2m}(W_n) = O(n^{1-m})$ .  $\square$

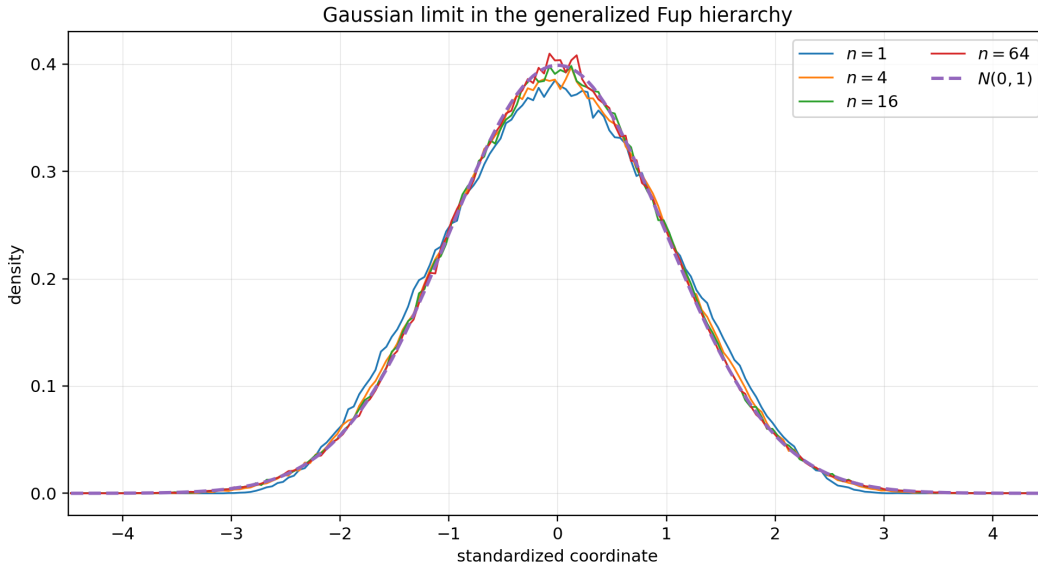


Figure 19: Standardized  $\text{Fup}_n$  densities for  $n = 1, 4, 16, 64$  against the standard Gaussian. The exact fourth standardized cumulants are approximately  $-0.456, -0.214, -0.0682, -0.0183$  (theorem 8.4).

## 8.6 The Edgeworth program is a theorem

The all-orders Edgeworth expansion, conjectured by the fifteenth wave and recorded in the register below, is proved by its revised edition. Let  $g_n$  be the density of  $W_n = Z_n/\sigma_n$ , whose exact standardized cumulants are, by (8.18),

$$\lambda_{2m,n} := \kappa_{2m}(W_n) = \frac{2^{2m-1} B_{2m}}{m} \left( n + \frac{2^{2m}}{2^{2m}-1} \right) \left( \frac{9}{3n+4} \right)^m, \quad \lambda_{2,n} = 1, \quad (8.19)$$

and for  $R \geq 1$  define the exact-cumulant Edgeworth polynomial

$$\mathcal{E}_{R,n}(x) = \sum_{\substack{k_2, \dots, k_{R+1} \geq 0 \\ \sum_m (m-1) k_m \leq R}} H_{2 \sum_m m k_m}(x) \prod_{m=2}^{R+1} \frac{1}{k_m!} \left( \frac{\lambda_{2m,n}}{(2m)!} \right)^{k_m}, \quad (8.20)$$

with  $H_k$  the probabilists' Hermite polynomials and  $\phi(x) = e^{-x^2/2}/\sqrt{2\pi}$ .

**Theorem 8.5** (Uniform differentiated all-orders Edgeworth expansion). *For all fixed integers  $R, r \geq 0$  there is  $C_{R,r}$  with*

$$\sup_{x \in \mathbb{R}} \left| g_n^{(r)}(x) - \frac{d^r}{dx^r} (\phi(x) \mathcal{E}_{R,n}(x)) \right| \leq C_{R,r} n^{-R-1} \quad (8.21)$$

for all large  $n$ : the expansion is uniform on the whole real line, to arbitrary algebraic order, after any fixed number of differentiations.

*Proof (condensed from the revised fifteenth wave).* Rescale to  $V_n = 2^{n+1}Z_n = 2X_2 + U_1 + \cdots + U_n$  and  $s_n^2 = (3n+4)/9$ . Then  $W_n = V_n/s_n$  has characteristic function

$$\chi_n(t) = \widehat{\text{up}}_{2\pi}(2t/(2\pi s_n)) \text{sinc}_{\text{rad}}(t/s_n)^n$$

in the volume's convention. Fix  $0 < \delta < \pi/2$ . On  $|t| \leq \delta s_n$  the cumulant series converges,  $|\lambda_{2m,n}| \leq C_m n^{1-m}$ , and  $|\text{sinc}_{\text{rad}} u| \leq e^{-c_\delta u^2}$  for  $|u| \leq \delta$ , so truncating the exponential of the cumulant series at total weight  $\sum (m-1)k_m \leq R$  leaves an error  $\leq C_R n^{-R-1}(1 + |t|^{M_R})e^{-ct^2}$ , integrable against every fixed  $|t|^r$ . On  $|t| > \delta s_n$ , substituting  $u = t/s_n$  and using  $|\widehat{\text{up}}_{2\pi}| \leq 1$ ,  $\max_{\delta \leq |u| \leq A} |\text{sinc}_{\text{rad}} u| < 1$ , and  $|\text{sinc}_{\text{rad}} u| \leq |u|^{-1}$  beyond a fixed  $A > 1$  makes  $\int_{|t| > \delta s_n} |t|^r |\chi_n(t)|$  exponentially small in  $n$ , as is the tail of the Gaussian–polynomial approximant. Hence  $\int |t|^r |\chi_n(t) - e^{-t^2/2} Q_{R,n}(t)| dt = \mathcal{O}(n^{-R-1})$ , and Fourier inversion  $-(it)^k e^{-t^2/2} \mapsto H_k(x)\phi(x)$  for even  $k$  – identifies the approximant with  $\phi \mathcal{E}_{R,n}$  and its derivatives.  $\square$

The first two nontrivial orders, uniformly in  $x$ :

$$g_n = \phi \left[ 1 + \frac{\lambda_{4,n}}{24} H_4 \right] + \mathcal{O}(n^{-2}), \quad g_n = \phi \left[ 1 + \frac{\lambda_{4,n}}{24} H_4 + \frac{\lambda_{6,n}}{720} H_6 + \frac{\lambda_{8,n}^2}{1152} H_8 \right] + \mathcal{O}(n^{-3}), \quad (8.22)$$

$$\lambda_{4,n} = -\frac{18(15n+16)}{25(3n+4)^2}, \quad \lambda_{6,n} = \frac{144(63n+64)}{49(3n+4)^3}.$$

The exact-cumulant convention absorbs the finite- $n$  corrections into the displayed coefficients instead of re-expanding them in powers of  $1/n$ .

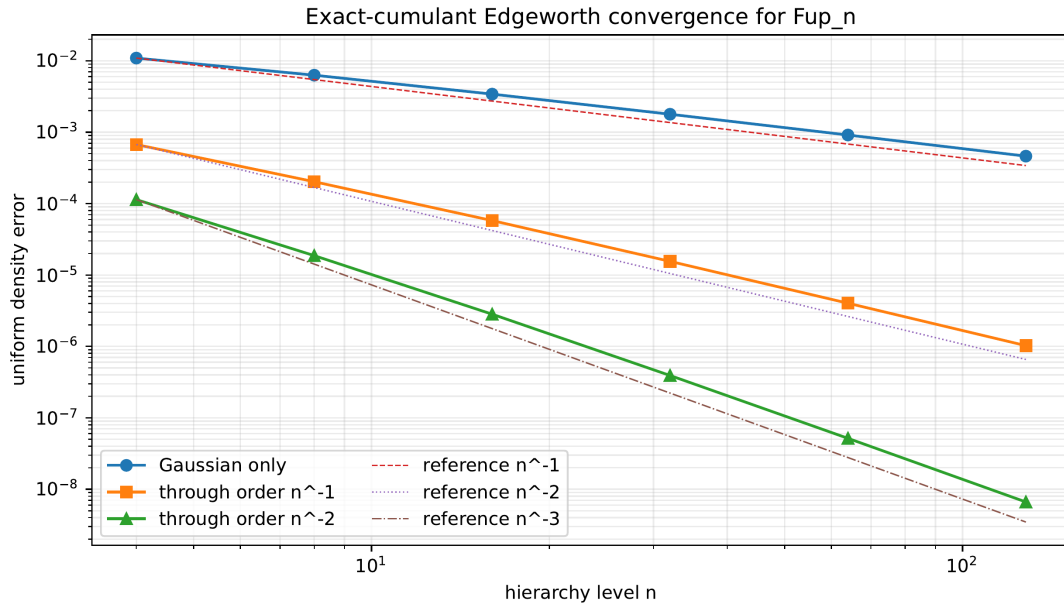


Figure 20: Deterministic FFT inversion confirming the uniform error scales of theorem 8.5: Gaussian  $= n^{-1}$ , with  $H_4 = n^{-2}$ , with  $H_4, H_6, H_8 = n^{-3}$ .

**[N] Repository-relative new derivation**

The two continuations of the source's  $F_{up_1}$  complement each other. The ladder  $G_n$  of [proposition 8.2](#) keeps the support radius essentially fixed and converges to the rescaled atom  $2 \text{ up}(2 \cdot)$ ; the classical hierarchy  $F_{up_n}$  shrinks geometrically and, after standardization, converges to the Gaussian. Both arise from redistributing cosine factors of the same binary sinc product, and both admit exact finite reconstructions of  $\text{up}$ : a  $2^n$ -translate average for the ladder, a  $2^{n(n+1)/2}$ -translate triangular mask for the hierarchy. The general multiplicity-vector program of [appendix 23](#) now possesses two explicit one-parameter solution families.

## 9 Repository crosswalk: $F_{bd}$ , $\text{up}$ , and the geometric- $q$ law

### 9.1 The three normalizations

**[R] Repository baseline or crosswalk**

The living repository carefully distinguishes three related objects: the bounded CDF-style Fabius function  $F_{bd}$ , its signed global extension  $F_{gl}$ , and the even compactly supported atom  $\text{up}$ . On the basic interval their relation is

$$\text{up}(x) = F_{bd}(1 - |x|), \quad |x| \leq 1, \quad (9.1)$$

so in particular

$$F_{bd}(u) = \text{up}(u - 1), \quad 0 \leq u \leq 1.$$

Another useful form is

$$F'_{bd}(u) = 2 \text{ up}(2u - 1), \quad 0 < u < 1.$$

The exact endpoint, dyadic, Fourier, and inverse-Fabius statements in the repository use these normalizations explicitly; they should not be conflated.

The attached chapter is written almost entirely in the  $\text{up}$  normalization. Equation (9.1) transfers its dyadic formulas and endpoint flatness to  $F_{bd}$ . In particular, the small- $u$  behavior of  $F_{bd}(u)$  is the left endpoint behavior of  $\text{up}(-1 + u)$ , and the inverse Fabius problem is an inversion of the endpoint tail of the  $\text{up}$  density.

### 9.2 Affine identification with the repository's geometric- $q$ family

Let  $0 < q < 1$ , let  $B_0, B_1, \dots$  be independent uniform random variables on  $[0, 1]$ , and define

$$S_q = (1 - q) \sum_{n=0}^{\infty} q^n B_n. \quad (9.2)$$

Write  $f_q$  for its density. This is the geometric weighted-uniform law already used in the repository. The parameterized source family is exactly its centered and rescaled version.

**Proposition 9.1** (Affine  $q$ -crosswalk). *For  $a > 1$ , put  $q = a^{-1}$  and  $b_a = (a - 1)^{-1}$ . Then*

$$X_a \stackrel{d}{=} b_a(2S_q - 1), \quad (9.3)$$

and hence

$$h_a(x) = \frac{a - 1}{2} f_{1/a} \left( \frac{1 + (a - 1)x}{2} \right). \quad (9.4)$$

*Proof.* Put  $U_n = 2B_n - 1$ . Since the weights in (9.2) sum to one,

$$2S_q - 1 = (1 - q) \sum_{n \geq 0} q^n U_n.$$

For  $q = a^{-1}$ , multiplication by  $b_a$  changes the weight of  $U_n$  to  $a^{-(n+1)}$ , which is precisely (5.6). The density transformation formula gives (9.4).  $\square$

At  $a = 2$ , (9.4) becomes the repository bridge

$$f_{1/2}(u) = 2 \operatorname{up}(2u - 1).$$

For general  $a$ , the source's first-order equation therefore supplies a density- level refinement equation and a compactly supported  $C^\infty$  package for the entire geometric- $q$  family. The current formalization contains the weighted uniform law, support, smoothness, and the binary bridge, and it now closes the uncentered general- $q$  Fourier product. For  $q, z \in \mathbb{C}$  with  $\|q\| < 1$ , writing  $S_q(z) = \prod_{n \geq 0} \operatorname{sinc}_{\text{rad}}(\pi q^n z)$ , the theorems `hasProdLocallyUniformly_geometricSincProduct`, `geometricSincProductFactors_multipliable`, `hasProd_geometricSincProduct`, and `geometricSincProduct_differentiable` give locally uniform product convergence, genuine pointwise `Multipliable` and `HasProd` witnesses, and entire-ness. For real  $|q| < 1$ ,  $t \in \mathbb{R}$ , and  $z_q(t) = (1 - q)t/(2\pi)$ , including  $q = 0$  and negative contractions, `charFun_geometricUniformDistribution_eq_phase_mul_geometricSincProduct` and `charFun_geometricUniformDistribution_eq_phase_mul_geometricReciprocalGamma` give

$$\begin{aligned} \phi_q(t) &= e^{it/2} S_q(z_q(t)) \\ &= e^{it/2} G_q(z_q(t)) G_q(-z_q(t)), \\ G_q(z) &= \operatorname{geometricReciprocalGamma}(q, z). \end{aligned}$$

The two further declarations `tendstoLocallyUniformly_prefix_sinc_charFun` and `tendstoUniformlyOn_prefix_sinc_charFun` prove that the full phase-bearing finite prefixes converge locally uniformly on the real frequency line, and uniformly on every compact subset of  $\mathbb{R}$ , to  $\phi_q$ . Their hypothesis is again exactly  $|q| < 1$ , so this includes  $q = 0$  and negative contractions. A separately named centered and  $h_a$ -scaled Fourier wrapper, the derivative hierarchy, plateau geometry, cumulants, and uniqueness remain the frontier obligations relevant here. The next sections address several of those obligations on paper.

### 9.3 Log-concavity and the overlapping regime

**Proposition 9.2** (Global log-concavity). *For every  $a > 1$ , the density  $h_a$  is even and log-concave on its support. Consequently it is unimodal and attains its maximum at the origin.*

*Proof.* Every finite density  $h_{a,N}$  is a convolution of uniform densities on intervals. Interval indicators are log-concave, convolution preserves log-concavity by the Prékopa–Leindler theorem, and weak limits of log-concave probability measures remain log-concave. The limiting measure has the continuous density  $h_a$ , so the density itself is log-concave. Evenness follows from the symmetric summands.  $\square$

This places the atomic family in convex-geometric probability as well as in refinement theory. For  $a > 2$ , log-concavity coexists with the exact central plateau and the fractal hierarchy of higher-degree gaps. At  $a = 2$  the plateau collapses to one point. In the genuinely overlapping regime  $1 < a < 2$ , the two branches in the derivative equation overlap at every scale, so the exact norm and one-branch localization arguments below no longer apply, although smoothness, log-concavity, cumulants, and rational-power polynomial reproduction all remain valid.

## 10 Bernoulli cumulants, Bell moments, and Lambert-series structure

### 10.1 Moment generating function

The moment generating function of  $X_a$  is

$$M_a(z) = \mathbb{E}e^{zX_a} = \prod_{j=1}^{\infty} \frac{\sinh(a^{-j}z)}{a^{-j}z}. \quad (10.1)$$

The classical Bernoulli expansion

$$\log \frac{\sinh z}{z} = \sum_{m=1}^{\infty} \frac{2^{2m-1} B_{2m}}{m(2m)!} z^{2m}, \quad |z| < \pi, \quad (10.2)$$

can be summed over the geometric scales exactly.

**Theorem 10.1** (Complete cumulant formula). *All odd cumulants of  $X_a$  vanish, and for  $m \geq 1$ ,*

$$\kappa_{2m}(X_a) = \frac{2^{2m} B_{2m}}{2m(a^{2m} - 1)}. \quad (10.3)$$

Equivalently,

$$\log M_a(z) = \sum_{m=1}^{\infty} \frac{2^{2m-1} B_{2m}}{m(2m)!(a^{2m} - 1)} z^{2m}. \quad (10.4)$$

*Proof.* Insert  $a^{-j}z$  in (10.2), sum over  $j \geq 1$ , and use

$$\sum_{j=1}^{\infty} a^{-2mj} = \frac{1}{a^{2m} - 1}.$$

The coefficient of  $z^{2m}/(2m)!$  in the logarithm of a moment generating function is the cumulant  $\kappa_{2m}$ . □

#### [N] Repository-relative new derivation

Formula (10.3) is the shortest exact description of the source's moment sequence for general  $a$ . Bernoulli numbers encode the uniform summands; the denominator  $a^{2m} - 1$  is the geometric scale sum. The binary formulas in the repository are recovered by  $a = 2$ , but the closed parameterized result and its consequences below are not duplicated there.

### 10.2 Bell-polynomial moments

Let

$$m_n(a) = \mathbb{E}X_a^n = \int x^n h_a(x) \, dx.$$

Then

$$m_n(a) = Y_n(\kappa_1, \dots, \kappa_n), \quad (10.5)$$

where  $Y_n$  is the complete exponential Bell polynomial. Since the law is symmetric, it is more efficient to use the recurrence

$$m_0(a) = 1, \quad m_{2n}(a) = \sum_{j=1}^n \binom{2n-1}{2j-1} \kappa_{2j}(a) m_{2n-2j}(a). \quad (10.6)$$

The first moments simplify to

$$m_2(a) = \frac{1}{3(a^2 - 1)}, \quad (10.7)$$

$$m_4(a) = \frac{3a^2 + 7}{15(a^2 - 1)^2(a^2 + 1)}, \quad (10.8)$$

$$m_6(a) = \frac{3a^6 + 18a^4 + 18a^2 + 31}{21(a^2 - 1)^3(a^2 + 1)(a^4 + a^2 + 1)}. \quad (10.9)$$

For  $a = 2$  this gives

$$m_2 = \frac{1}{9}, \quad m_4 = \frac{19}{675}, \quad m_6 = \frac{583}{59535}, \quad m_8 = \frac{132809}{32531625}. \quad (10.10)$$

These exact values agree with the repository's binary moment machinery and with the attached source's recurrence.

### 10.3 A Lambert series in the scale variable

Put  $q = a^{-2}$ . Then (10.4) becomes

$$\log M_a(z) = \sum_{m=1}^{\infty} \frac{2^{2m-1} B_{2m}}{m(2m)!} \frac{q^m}{1 - q^m} z^{2m}. \quad (10.11)$$

Thus the cumulant generating function is a Bernoulli-weighted Lambert series. This connects the source family simultaneously to the repository's Lambert series,  $q$ -binomial, and Bell-polynomial chapters. The roles are cleanly separated:

- Bernoulli numbers come from a single uniform interval;
- $q^m/(1 - q^m)$  sums that cumulant over geometric scales;
- Bell polynomials exponentiate cumulants back to moments.

Expanding each Lambert factor as  $\sum_{r \geq 1} q^{mr}$  and collecting powers of  $q$  gives the equivalent divisor-polynomial form (fourteenth wave)

$$\log M_a(z) = \sum_{n \geq 1} q^n \sum_{m|n} \frac{2^{2m-1} B_{2m}}{m(2m)!} z^{2m}, \quad q = a^{-2}. \quad (10.12)$$

Each coefficient of  $q^n$  is a polynomial over the divisors of  $n$ , the exact analogue for this family of the divisor sums that organize the repository's Lambert-series chapters.

### 10.4 $q$ -binomial organization of finite products

For the  $N$ -factor product, the exponential numerator can be organized by the finite  $q$ -binomial theorem

$$\prod_{j=1}^N (1 - zq^j) = \sum_{k=0}^N (-1)^k q^{k(k+1)/2} \begin{bmatrix} N \\ k \end{bmatrix}_q z^k. \quad (10.13)$$

Taking  $q = a^{-1}$  and separating the elementary phase in each sine factor gives finite exponential sums for  $\Phi_{a,N}$ . In physical space the same subset expansion appears as the signed knot formula in [theorem 14.2](#). For  $a = 2$ , the subset sums are dyadic and the signs reduce to Thue–Morse parity. This is a connection, not a new binary identity: the repository already develops the  $q$ -binomial theorem extensively.



## 11 Weighted Prouhet identities and the full derivative hierarchy

### 11.1 The signed exponential product

For a sign vector  $\varepsilon = (\varepsilon_0, \dots, \varepsilon_{n-1}) \in \{\pm 1\}^n$ , define

$$s_a(\varepsilon) = \sum_{j=0}^{n-1} \varepsilon_j a^j, \quad w(\varepsilon) = \prod_{j=0}^{n-1} \varepsilon_j. \quad (11.1)$$

Then

$$\sum_{\varepsilon \in \{\pm 1\}^n} w(\varepsilon) e^{z s_a(\varepsilon)} = \prod_{j=0}^{n-1} (e^{a^j z} - e^{-a^j z}). \quad (11.2)$$

The right side has a zero of exact order  $n$  at  $z = 0$ .

**Theorem 11.1** (Weighted Prouhet identity). *For  $0 \leq m < n$ ,*

$$\sum_{\varepsilon} w(\varepsilon) s_a(\varepsilon)^m = 0, \quad (11.3)$$

*and the first nonzero signed power sum is*

$$\sum_{\varepsilon} w(\varepsilon) s_a(\varepsilon)^n = 2^n n! a^{n(n-1)/2}. \quad (11.4)$$

*Proof.* Differentiate (11.2)  $m$  times at zero. Every factor on the right is  $2a^j z + \mathcal{O}(z^3)$ , so derivatives below order  $n$  vanish, while the coefficient of  $z^n$  is  $2^n a^{0+1+\dots+(n-1)}$ .  $\square$

For  $a = 2$ , the locations are consecutive odd integers and the weights, when the locations are sorted, are the Thue–Morse signs. Thus [theorem 11.1](#) is the scale-deformed version of the classical Prouhet cancellation behind the source’s derivative formula.

### 11.2 All derivatives of $h_a$

**Theorem 11.2** (Derivative hierarchy). *For every  $a > 1$  and  $n \geq 0$ ,*

$$h_a^{(n)}(x) = \left(\frac{a^2}{2}\right)^n a^{n(n-1)/2} \sum_{\varepsilon \in \{\pm 1\}^n} w(\varepsilon) h_a(a^n x + s_a(\varepsilon)). \quad (11.5)$$

*Proof.* The case  $n = 1$  is (5.4). Differentiating the  $n$ th formula multiplies the prefactor by  $a$  and replaces each term by the two branches from (5.4). Relabeling the new sign as the coefficient of  $a^n$  produces the  $(n + 1)$ st formula.  $\square$

At  $a = 2$ , (11.5) becomes (7.3). The moment cancellations in [theorem 11.1](#) also explain why polynomial coefficient sequences of degree  $n$  survive exactly through the source’s difference operator.

### 11.3 Exact derivative norms for $a \geq 2$

The source states the binary norm identity only. Separation of the support cylinders gives a parameterized extension.

**Theorem 11.3** (Exact derivative norms). *For every  $a \geq 2$  and  $n \geq 0$ ,*

$$\|h_a^{(n)}\|_{\infty} = \frac{a^{n(n+3)/2}}{2^n} \|h_a\|_{\infty}. \quad (11.6)$$

Moreover,

$$\|h_a\|_\infty = \frac{a}{2}, \quad (11.7)$$

so

$$\|h_a^{(n)}\|_\infty = \frac{a^{n(n+3)/2+1}}{2^{n+1}}. \quad (11.8)$$

The exact total-variation companion is

$$\|h_a^{(n)}\|_1 = a^{n(n+1)/2}. \quad (11.9)$$

*Proof.* For a fixed sign vector, the corresponding term in (11.5) is supported on

$$I_\varepsilon = \{x : |a^n x + s_a(\varepsilon)| \leq b_a\}.$$

For  $a > 2$  these  $2^n$  intervals are disjoint; for  $a = 2$  adjacent intervals only touch, and  $h_a$  vanishes at the touching endpoints. Hence at every  $x$  at most one summand is nonzero. Each affine map  $x \mapsto a^n x + s_a(\varepsilon)$  maps its support interval onto the full support  $[-b_a, b_a]$ , so the upper bound from one surviving summand is attained. The prefactor is

$$\left(\frac{a^2}{2}\right)^n a^{n(n-1)/2} = \frac{a^{n(n+3)/2}}{2^n}.$$

For  $a > 2$ , the central plateau (6.2) has height  $a/2$ . At  $a = 2$ ,  $\text{up}(0) = 1 = a/2$  by the partition-of-unity symmetry, proving (11.7). For the  $L^1$  norm, each of the  $2^n$  disjoint affine copies has integral  $a^{-n}$ , while the common prefactor is  $a^{n(n+3)/2}/2^n$ . Their absolute integrals therefore sum to  $a^{n(n+1)/2}$ .  $\square$

**Corollary 11.4** (Critical binary identity). *At  $a = 2$ ,*

$$\|\text{up}^{(n)}\|_\infty = \|\text{up}^{(n)}\|_1 = 2^{\binom{n+1}{2}},$$

*exactly as in the attached source.*

#### [N] Repository-relative new derivation

The exact norm formula is useful in three different directions: it quantifies the zero Taylor radius on recurrent symbolic orbits; it gives closed constants in the finite-product asymptotics; and it provides a sharply normalized growth condition for future uniqueness theorems at general  $a$ .

### 11.4 Equimeasurability and the full $L^p$ ladder

For  $a > 2$  the separation of the  $2^n$  summand supports upgrades the norm identities to a complete distributional statement (fifteenth wave). Write  $C_{a,n} = a^{n(n+3)/2}/2^n$  for the prefactor of (11.5).

**Theorem 11.5** (Derivative equimeasurability). *Let  $a \geq 2$ ,  $n \geq 1$ , and let  $\Phi : \mathbb{R} \rightarrow \mathbb{R}$  be Borel measurable, whenever the integrals exist. (At  $a = 2$  the cylinders of the proof tile the support rather than being separated, meeting only in a null set; the change of variables is unaffected, and the atom at 0 has weight  $1 - (2/a)^n = 0$ .) Then*

$$\begin{aligned} \frac{1}{2b_a} \int_{-b_a}^{b_a} \Phi(h_a^{(n)}(x)) \, dx &= \left[1 - \left(\frac{2}{a}\right)^n\right] \Phi(0) \\ &+ \frac{1}{2} \left(\frac{2}{a}\right)^n \frac{1}{2b_a} \int_{-b_a}^{b_a} [\Phi(C_{a,n} h_a(t)) + \Phi(-C_{a,n} h_a(t))] \, dt. \end{aligned} \quad (11.10)$$

In particular, for every  $1 \leq p < \infty$ ,

$$\|h_a^{(n)}\|_p = C_{a,n} \left(\frac{2}{a}\right)^{n/p} \|h_a\|_p, \quad (11.11)$$

interpolating the exact  $L^1$  and  $L^\infty$  identities of [theorem 11.3](#), and the set  $\{h_a^{(n)} \neq 0\}$  has relative measure exactly  $(2/a)^n$  in the support.

*Proof.* Each of the  $2^n$  summands of (11.5) is supported on its own cylinder interval  $I_\varepsilon$  of length  $2b_a a^{-n}$ ; the cylinders are pairwise disjoint, and off their union the derivative vanishes, which accounts for the atom at 0 with weight  $1 - (2/a)^n$ . On one cylinder the substitution  $t = a^n x + s_a(\varepsilon)$  maps onto  $(-b_a, b_a)$  with  $dx = a^{-n} dt$  and turns the summand into  $w(\varepsilon) C_{a,n} h_a(t)$ ; exactly half of the  $2^n$  weights  $w(\varepsilon)$  are  $+1$  and half are  $-1$ . Summing the cylinders gives (11.10), and  $\Phi(u) = |u|^p$  gives (11.11).  $\square$

**Corollary 11.6** (Exact derivative-value law). *If  $Y$  is uniform on  $(-b_a, b_a)$ , then*

$$h_a^{(n)}(Y) \stackrel{d}{=} \begin{cases} 0 & \text{with probability } 1 - (2/a)^n, \\ \pm C_{a,n} h_a(Y') & \text{each with probability } \frac{1}{2}(2/a)^n, \end{cases}$$

where  $Y'$  is again uniform on  $(-b_a, b_a)$  and the fair sign is independent of  $Y'$ . Thus every derivative is an amplified, fairly signed, sparsely supported copy of  $h_a$  itself.

At  $a = 2$  the cylinders tile the support up to a null set, so the same change of variables gives the all- $p$  identity  $\|\text{up}^{(n)}\|_p = 2^{n(n+1)/2} \|\text{up}\|_p$ , with no atom at zero.

The revised fourteenth-wave editions derive the same law independently and record two refinements worth keeping: the change of variables is valid for *all*  $a \geq 2$  and all  $0 < p < \infty$  at once (at  $a = 2$  the cylinders meet only in a null set), and at the level of rearrangements it is an exact thinning of the decreasing rearrangement,

$$(h_a^{(n)})^*(t) = C_{a,n} h_a^*\left(\frac{t}{(2/a)^n}\right) \quad \left(0 < t < \frac{2}{a-1}(2/a)^n\right), \quad (11.12)$$

zero afterward; equivalently  $\int \Theta(|h_a^{(n)}|/C_{a,n}) = (2/a)^n \int \Theta(h_a)$  for every nonnegative Borel  $\Theta$  with  $\Theta(0) = 0$ , both sides possibly infinite.

## 11.5 Derivative energies, Bernoulli convolutions, and the entropy of the singular locus

The same cylinder decomposition answers a different question (second expanded fourteenth-wave edition): not how large the derivatives are, but *where* their energy sits. For  $a \geq 2$ ,  $p > 0$ , and  $n \geq 0$  let

$$d\mu_{a,n,p}(x) = \frac{|h_a^{(n)}(x)|^p}{\|h_a^{(n)}\|_p^p} dx, \quad (11.13)$$

the normalized  $p$ -energy measure of the  $n$ th derivative, and let  $Y_{a,p}$  have the base density  $h_a^p/\|h_a\|_p^p$  on  $[-b_a, b_a]$ . Let  $\varepsilon_1, \varepsilon_2, \dots$  be independent fair signs, independent of  $Y_{a,p}$ , and

$$S_{a,n} = \sum_{j=1}^n \varepsilon_j a^{-j}, \quad \nu_a = \text{Law}\left(\sum_{j=1}^{\infty} \varepsilon_j a^{-j}\right),$$

the symmetric Bernoulli convolution with contraction ratio  $1/a$  [96].

**Theorem 11.7** (Exact derivative-energy factorization and universal limit). *For every  $a \geq 2$ ,  $p > 0$ , and  $n \geq 0$ ,*

$$\mu_{a,n,p} = \text{Law}(S_{a,n} + a^{-n}Y_{a,p}), \quad (11.14)$$

*so the characteristic function factors exactly,*

$$\widehat{\mu_{a,n,p}}_{\text{ang}}(t) = \widehat{\mu_{a,0,p}}_{\text{ang}}(t/a^n) \prod_{j=1}^n \cos(t/a^j), \quad (11.15)$$

*and the digit coupling gives*

$$W_{\infty}(\mu_{a,n,p}, \nu_a) \leq \frac{2a^{-n}}{a-1}. \quad (11.16)$$

*The limit measure is therefore independent of  $p$  and of everything about the base profile. For  $a > 2$  the essential support of  $h_a^{(n)}$  is the level- $n$  Hutchinson approximant of the attractor  $K_a$  of (14.2), at exact Hausdorff distance*

$$d_H(\text{supp } \mu_{a,n,p}, K_a) = b_1 a^{-n} = \frac{a-2}{a^{n+1}(a-1)}, \quad (11.17)$$

*and  $\nu_a$  is the equal-weight self-similar measure on  $K_a$ , of Hausdorff and information dimension  $D_a = \log 2 / \log a$ . At  $a = 2$  the measure  $\nu_a$  is the uniform law on  $[-1, 1]$ .*

*Proof.* On each level- $n$  cylinder of (11.5) exactly one affine copy of  $h_a$  survives almost everywhere, and the equimeasurability substitution shows the energy density there is  $(a/2)^n \cdot h_a^p(a^n x + s_a(\varepsilon)) / \|h_a\|_p^p$ : each of the  $2^n$  cylinders carries energy exactly  $2^{-n}$ , its center is the address sum  $S_{a,n}$  after relabeling signs, and its internal coordinate is  $a^{-n}Y_{a,p}$ . This is (11.14), and independence of the digits gives (11.15). Coupling  $\mu_{a,n,p}$  and  $\nu_a$  through the same first  $n$  signs leaves a difference  $|a^{-n}Y_{a,p} - \sum_{j>n} \varepsilon_j a^{-j}| \leq 2b_a a^{-n}$ , which is (11.16). For  $a > 2$  the deepest point of a cylinder relative to  $K_a$  is the image of the central-gap midpoint, at distance  $b_1 a^{-n}$ , giving (11.17); the dimension statements follow from the open-set condition with equal ratios, and at  $a = 2$  the address sum is a random binary expansion of a uniform point.  $\square$

**Corollary 11.8** (Exact Rényi and Shannon entropy laws). *Write  $\rho_{a,n,p}$  for the density of  $\mu_{a,n,p}$  and, for  $\beta > 0$ ,  $\beta \neq 1$ , let  $H_{\beta}(\rho) = (1-\beta)^{-1} \log \int \rho^{\beta}$  be the differential Rényi entropy, with  $H_1 = H$  the Shannon entropy. Then for all admissible parameters*

$$H_{\beta}(\rho_{a,n,p}) = H_{\beta}(\rho_{a,0,p}) + n \log \frac{2}{a} \quad (0 < \beta \leq \infty). \quad (11.18)$$

*Proof.* The measure is a disjoint equal-weight mixture of  $2^n$  affine copies of the base density scaled by  $a^{-n}$ , so  $\int \rho_{a,n,p}^{\beta} = (a/2)^{n(\beta-1)} \int \rho_{a,0,p}^{\beta}$ ; the Rényi formula, its  $\beta \rightarrow 1$  limit, and the  $\beta = \infty$  sup-norm reading all give (11.18).  $\square$

**Remark 11.9** (A critical entropy discontinuity, and the dimension). For  $a > 2$  every differential entropy decreases linearly at the exact rate  $n \log(a/2)$  while  $\mu_{a,n,p}$  converges weakly to the singular measure  $\nu_a$  (differential entropy  $-\infty$ ): the loss rate is the derivative hierarchy's information-theoretic reading of singularity. At  $a = 2$  the entropies are constant in  $n$ , yet the weak limit is uniform on  $[-1, 1]$  – so unless the base energy profile is already uniform, differential entropy does not converge to the entropy of the limit. The discrete address entropy is exactly  $n \log 2$  against the geometric scale  $n \log a$ , and their ratio recovers the dimension  $D_a = \log 2 / \log a$  of the nonanalytic locus: a compactly supported  $C^{\infty}$  function carries the information dimension of its singular set in the exact entropy budget of its derivative energies. At criticality,  $\widehat{\mu_{2,n,p}}_{\text{ang}}(t) \rightarrow \text{sinc}_{\text{rad}} t$  by the classical cosine product: the same binary product that turns Thue–Morse differences into derivatives turns every normalized derivative energy into the uniform law.

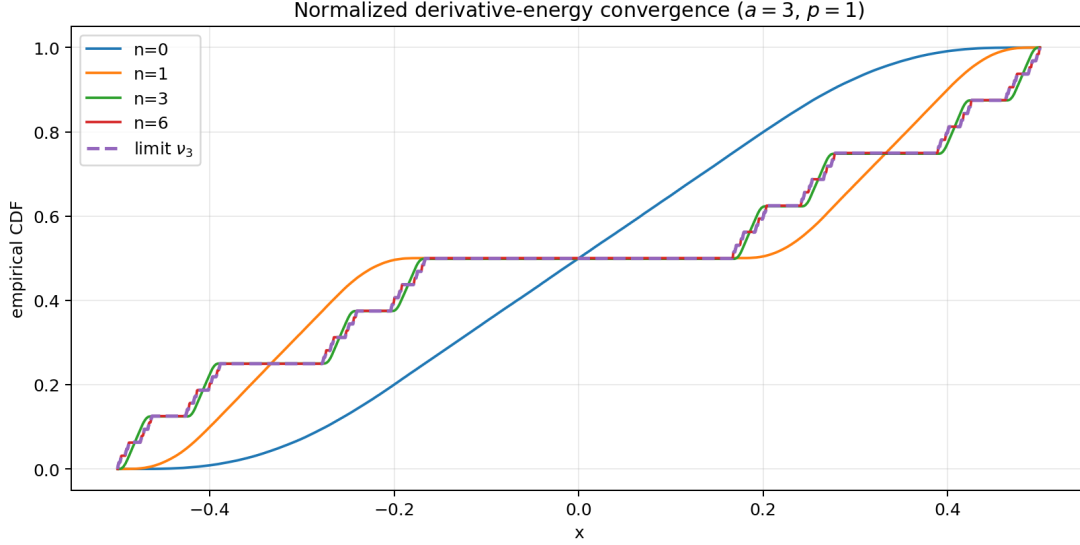


Figure 21: Distribution functions of the normalized  $p = 1$  derivative-energy measures  $\mu_{3,n,1}$  for increasing  $n$ , against the Bernoulli-convolution limit  $\nu_3$  (theorem 11.7). The exact coupling error (11.16) is at most  $3^{-n}$ .

## 12 Spectral moments and the Stieltjes–Wigert bridge

The  $p = 2$  rung of the ladder has a moment-theoretic consequence (revised fourteenth wave): the Fourier energy of every  $h_a$ ,  $a \geq 2$ , is an explicit representing measure for a scaled Stieltjes–Wigert moment problem.

### 12.1 The spectral probability law and its moments

In the source’s angular variable, let  $T_a$  have the symmetric probability density  $|\widehat{h_a}_{2\pi}(t/2\pi)|^2/I_a$  with  $I_a = \int_{\mathbb{R}} |\widehat{h_a}_{2\pi}(t/2\pi)|^2 dt = 2\pi \|h_a\|_2^2$ , and put  $Z_a = T_a^2$ .

**Theorem 12.1** (General-base spectral moments). *For every  $a \geq 2$  and  $n \geq 0$ ,*

$$\mathbb{E} Z_a^n = \frac{\|h_a^{(n)}\|_2^2}{\|h_a\|_2^2} = \frac{a^{n(n+2)}}{2^n}. \quad (12.1)$$

*Proof.* Parseval converts frequency moments into derivative energies,  $\int t^{2n} |\widehat{h_a}_{2\pi}(t/2\pi)|^2 dt = 2\pi \|h_a^{(n)}\|_2^2$ , and the  $p = 2$  case of theorem 11.5 gives  $\|h_a^{(n)}\|_2^2 / \|h_a\|_2^2 = C_{a,n}^2 (2/a)^n = a^{n(n+2)} / 2^n$ .  $\square$

**Corollary 12.2** (A continuum of non-lognormal moment twins). *Let  $Z_a^{\text{ln}}$  be lognormal with  $\log Z_a^{\text{ln}} \sim N(2 \log a - \log 2, 2 \log a)$ . Then  $\mathbb{E}(Z_a^{\text{ln}})^n = a^{n(n+2)} / 2^n = \mathbb{E} Z_a^n$  for all  $n$ , but the two laws are distinct: the spectral density behaves like  $I_a^{-1} z^{-1/2}$  as  $z \downarrow 0$ , whereas the lognormal density vanishes faster than every power. Hence this Stieltjes moment sequence is indeterminate (Heyde’s classical lognormal indeterminacy [13]), and the sinc product supplies, for every base  $a \geq 2$ , a concrete second representing density.*

### 12.2 The scaled Stieltjes–Wigert package

With  $q = a^{-2}$  and  $c = a/2$ , the moments read  $m_n = c^n q^{-n(n+1)/2}$ : the Stieltjes–Wigert sequence after the dilation  $z \mapsto cz$ . The refinement equation gives the two moment twins distinct exact  $q$ -Pearson

scale equations,

$$w_{\text{sp}}(a^2 z) = \frac{1}{a} \text{sinc}_{\text{rad}}^2(\sqrt{z}) w_{\text{sp}}(z), \quad w_{\text{ln}}(a^2 z) = \frac{1}{2az} w_{\text{ln}}(z), \quad (12.2)$$

(the spectral one from  $\widehat{h_{a,2\pi}}$ 's one-step refinement, written in the angular variable), and the shared moment functional has closed Hankel determinants and monic orthogonal polynomials:

$$\Delta_N = \det[m_{i+j}]_{i,j=0}^N = c^{N(N+1)} q^{-N(N+1)(4N+5)/6} \prod_{k=1}^N (q; q)_k, \quad (12.3)$$

$$P_{a,n}(z) = \sum_{k=0}^n (-1)^{n-k} c^{n-k} q^{k^2-n^2} \begin{bmatrix} n \\ k \end{bmatrix}_q z^k, \quad \int_0^\infty P_{a,n}^2 d\rho = c^{2n} q^{-n(2n+1)} (q; q)_n, \quad (12.4)$$

for either representing measure  $\rho$  – the dilated monic Stieltjes–Wigert system [44]. (Hankel: factor  $c^{i+j} q^{-i(i+1)/2-j(j+1)/2-ij}$  from rows and columns and evaluate the Vandermonde determinant in  $1, q^{-1}, \dots, q^{-N}$ ; the squared norm is  $\Delta_n/\Delta_{n-1}$ .) The repository's  $q$ -moment algebra thus acquires, from the derivative thinning law alone, a one-parameter family of concrete sinc-product representatives of the scaled Stieltjes–Wigert moments. The open Nevanlinna–Pick question for this family is recorded in [appendix 23](#).

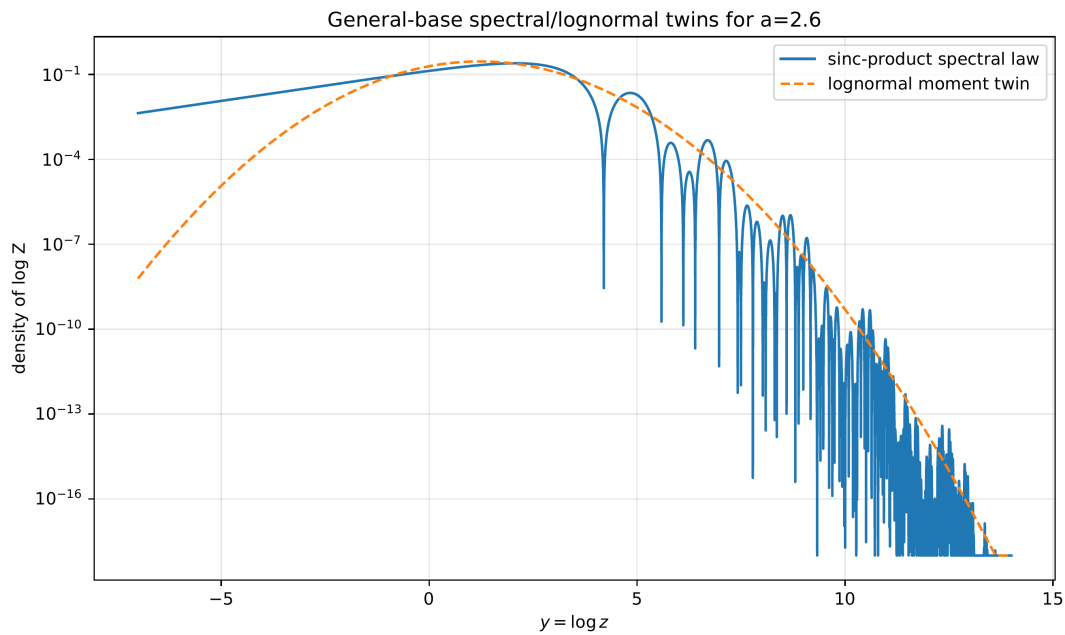


Figure 22: The spectral density of  $Z_a$  against its lognormal moment twin for representative bases. The two densities share every moment (12.1) yet differ on an interval ([corollary 12.2](#)).

### 13 The derivative tower as a $q$ -Gaussian Riesz sequence

The  $L^2$  ladder also fixes every *mutual* inner product of the derivatives (revised and expanded editions of both reconstructions), and the resulting Hilbert-space geometry is exactly solvable – down to a closed Gram–Schmidt process, an explicit unitary circle model, and a compactly supported Stieltjes–Wigert ladder in physical space.

**Theorem 13.1** (Exact derivative Gram kernel). *Let  $a > 2$  and  $\psi_n = h_a^{(n)} / \|h_a^{(n)}\|_2$ . Then  $\langle \psi_n, \psi_m \rangle = 0$  for  $n - m$  odd and*

$$\langle \psi_n, \psi_m \rangle = (-1)^{(n-m)/2} a^{-(n-m)^2/4} \quad (n - m \text{ even}). \quad (13.1)$$

Consequently, for each parity  $\epsilon \in \{0, 1\}$  the sign-corrected tower  $\eta_j = (-1)^j \psi_{2j+\epsilon}$  has the stationary  $q$ -Gaussian kernel

$$\langle \eta_j, \eta_k \rangle = q^{(j-k)^2}, \quad q = a^{-1}, \quad (13.2)$$

and the even and odd towers are mutually orthogonal.

*Proof.* Parity of  $h_a^{(n)}$  is  $(-1)^n$ , killing odd differences. For  $n = m + 2r$ ,  $r$  integrations by parts give  $\langle h_a^{(m+2r)}, h_a^{(m)} \rangle = (-1)^r \|h_a^{(m+r)}\|_2^2$ . Normalizing by  $\|h_a^{(s)}\|_2^2 = a^{s(s+2)} 2^{-s} \|h_a\|_2^2$  (theorem 12.1) at  $s = (n+m)/2$  cancels every power of 2, and the  $a$ -exponent is  $s(s+2) - \frac{1}{2}[n(n+2) + m(m+2)] = -(n-m)^2/4$ . The sign correction  $(-1)^{j+k}$  absorbs  $(-1)^{(n-m)/2} = (-1)^{j-k}$ .  $\square$

**Theorem 13.2** (Gram determinants, pivots, and sharp Riesz bounds). For  $0 < q < 1$  and  $G_N(q) = [q^{(i-j)^2}]_{i,j=0}^N$ ,

$$\det G_N(q) = \prod_{d=1}^N (1 - q^{2d})^{N+1-d} = \prod_{r=1}^N (q^2; q^2)_r, \quad (13.3)$$

so the squared Gram–Schmidt pivot of  $\eta_N$  against its predecessors is  $\det G_N / \det G_{N-1} = (q^2; q^2)_N$ . Moreover, for every finitely supported  $(c_j)$ ,

$$\vartheta_4(0, q) \sum_j |c_j|^2 \leq \left\| \sum_j c_j \eta_j \right\|_2^2 \leq \vartheta_3(0, q) \sum_j |c_j|^2, \quad (13.4)$$

with both Jacobi–theta constants optimal: each parity tower is a Riesz sequence in  $L^2(\mathbb{R})$ . Since the parity blocks are orthogonal and carry the same kernel, the complete normalized sequence  $(\psi_n)_{n \geq 0}$  is a Riesz sequence with the same optimal constants, and the full Gram determinants factor by parity,  $D_{2N} = (\det G_{N-1})^2$  and  $D_{2N+1} = \det G_N \det G_{N-1}$ .

*Proof.* Factoring  $q^{i^2}$  from row  $i$  and  $q^{j^2}$  from column  $j$  leaves the Vandermonde matrix  $((q^{-2i})^j)$ ; the extracted powers cancel against the  $q^{-2\sum j^2}$  from the Vandermonde product, and each difference  $d = j - i$  occurs  $N + 1 - d$  times. For the bounds, the quadratic form has Toeplitz symbol  $\Theta_q(\theta) = \sum_k q^{k^2} e^{ik\theta} = \vartheta_3(\theta/2, q)$ , whose Jacobi triple product  $\prod_r (1 - q^{2r})(1 + 2q^{2r-1} \cos \theta + q^{4r-2})$  is factorwise increasing in  $\cos \theta$ : the extremes are  $\vartheta_4(0, q)$  at  $\theta = \pi$  and  $\vartheta_3(0, q)$  at 0, attained asymptotically by concentrated Fejér sequences.  $\square$

The determinants converge to a classical constant (expanded fifteenth wave): rearranging (13.3) as  $\det G_N = (q^2; q^2)_N^{N+1} \prod_{d=1}^N (1 - q^{2d})^{-d}$  gives exactly

$$\det G_N(q) = (q^2; q^2)_\infty^{N+1} \mathfrak{M}(q^2) (1 + \mathcal{O}(Nq^{2N})), \quad \mathfrak{M}(x) = \prod_{d \geq 1} (1 - x^d)^{-d}, \quad (13.5)$$

with  $\mathfrak{M}$  MacMahon’s plane-partition generating function [21]: the per-vector entropy of the tower is  $\log(q^2; q^2)_\infty$  and the excess constant counts plane partitions at  $x = a^{-2}$ . The sharp Riesz constants themselves close up under the Jacobi triple product,

$$A_a = \vartheta_4(0, q) = (q^2; q^2)_\infty (q; q^2)_\infty^2, \quad B_a = \vartheta_3(0, q) = (q^2; q^2)_\infty (-q; q^2)_\infty^2, \quad (13.6)$$

with values

| $a$ | $A_a$        | $B_a$        | $B_a/A_a$ |
|-----|--------------|--------------|-----------|
| 2   | 0.1211242080 | 2.1289368272 | 17.5765   |
| 3   | 0.3579231273 | 1.6914596817 | 4.7258    |
| 4   | 0.5078048711 | 1.5078201299 | 2.9693    |
| 10  | 0.8001999980 | 1.2002000020 | 1.4999    |



The theorem ties the Rvachev derivative tower to Gaussian Toeplitz kernels, Jacobi theta functions, and finite  $q$ -Pochhammer products at once, and shows that — in stark contrast to the explosive unnormalized norms — the normalized towers stay uniformly well conditioned no matter how many derivatives are taken: even at the critical base  $a = 2$  the condition number of the full infinite tower is  $B_2/A_2 \approx 17.58$ , and it decays to 1 like  $4q$  as  $a \rightarrow \infty$ .

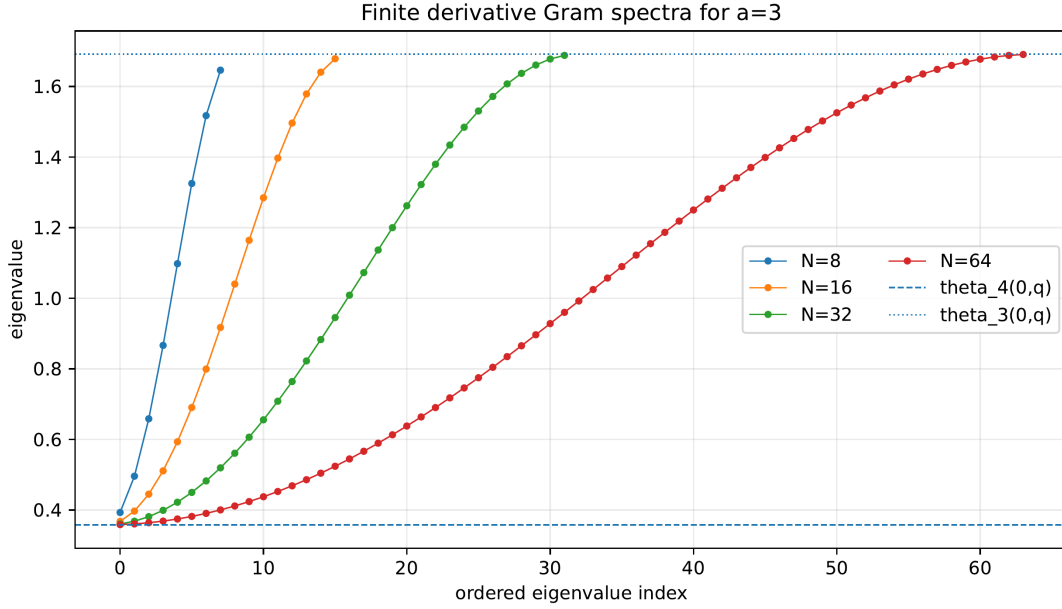


Figure 23: Eigenvalues of the finite  $q$ -Gaussian derivative Gram matrices at  $a = 3$  against the sharp infinite Riesz bounds  $\vartheta_4(0, \frac{1}{3})$  and  $\vartheta_3(0, \frac{1}{3})$  (theorem 13.2).

### 13.1 Closed orthogonalization: Gaussian binomials and Rogers–Szegő innovations

The Gram–Schmidt process for the stationary kernel (13.2) is itself exactly solvable (expanded fifteenth-wave editions): the orthogonal directions, the Cholesky factorization, and the inverse Gram matrix are all single Gaussian-binomial sums, the residuals are nodal polynomials at geometric nodes, and the limiting innovation filter is an explicit theta spectral factor.

**Theorem 13.3** (Closed  $q$ -Gram–Schmidt). *Fix a parity  $\epsilon$  and abbreviate  $e_j = \psi_{2j+\epsilon}$ , the raw normalized derivatives, whose Gram matrix is the signed kernel  $\tilde{G}_N = [(-1)^{i-j} q^{(i-j)^2}]_{i,j=0}^N$  of theorem 13.1. The vectors*

$$\psi_n^* = \sum_{j=0}^n q^{n-j} \begin{bmatrix} n \\ j \end{bmatrix}_{q^2} e_j \quad (13.7)$$

satisfy  $\langle \psi_n^*, e_k \rangle = 0$  for  $0 \leq k < n$  and  $\langle \psi_n^*, e_n \rangle = \|\psi_n^*\|_2^2 = (q^2; q^2)_n$ : up to normalization,  $\psi_n^*$  is the  $n$ -th Gram–Schmidt vector of the tower. Equivalently, the unit lower-triangular matrix  $C = [q^{n-j} \begin{bmatrix} n \\ j \end{bmatrix}_{q^2}]_{n \geq j \geq 0}$  factors the raw kernel and its inverse explicitly:

$$C \tilde{G}_N C^T = \text{diag}((q^2; q^2)_0, \dots, (q^2; q^2)_N), \quad (\tilde{G}_N^{-1})_{jk} = \sum_{n=\max(j,k)}^N \frac{q^{2n-j-k}}{(q^2; q^2)_n} \begin{bmatrix} n \\ j \end{bmatrix}_{q^2} \begin{bmatrix} n \\ k \end{bmatrix}_{q^2},$$

re-deriving the determinant (13.3) and its pivots. In the sign-corrected basis  $\eta_j = (-1)^j e_j$  of (13.2) the coefficients simply alternate:  $(-1)^n \psi_n^* = \sum_j (-q)^{n-j} \begin{bmatrix} n \\ j \end{bmatrix}_{q^2} \eta_j$  is the  $n$ -th Gram–Schmidt vector there, and the positive kernel  $G_N = [q^{(i-j)^2}]$  of theorem 13.2 has  $(G_N^{-1})_{jk} = (-1)^{j-k} (\tilde{G}_N^{-1})_{jk}$ ; determinants, pivots, and distances are unchanged, since the two kernels are conjugate by  $\text{diag}((-1)^i)$ .

*Proof.* The  $q$ -binomial theorem at base  $q^2$  reads  $\sum_{j=0}^n \begin{bmatrix} n \\ j \end{bmatrix}_{q^2} q^{j(j-1)} x^j = \prod_{r=0}^{n-1} (1 + q^{2r} x)$ . Since  $q^{n-j} (-1)^{j-k} q^{(j-k)^2} = (-1)^k q^{n+k^2} \cdot q^{j(j-1)} (-q^{-2k})^j$ ,

$$\langle \psi_n^*, e_k \rangle = \sum_{j=0}^n q^{n-j} \begin{bmatrix} n \\ j \end{bmatrix}_{q^2} (-1)^{j-k} q^{(j-k)^2} = (-1)^k q^{n+k^2} \prod_{r=0}^{n-1} (1 - q^{2(r-k)}).$$

For  $0 \leq k < n$  the factor  $r = k$  vanishes. At  $k = n$  the product telescopes to  $(-1)^n q^{-n(n+1)} (q^2; q^2)_n$  and the prefactor  $(-1)^n q^{n+n^2}$  cancels sign and power exactly. Expanding  $\|\psi_n^*\|_2^2$  against the defining sum (13.7) then keeps only the  $j = n$  term. The two matrix identities are the same computation read as  $C \tilde{G}_N C^\top = D$  and  $\tilde{G}_N^{-1} = C^\top D^{-1} C$ ; conjugating by  $\text{diag}((-1)^i)$  transports both to the sign-corrected basis.  $\square$

The expanded fifteenth-wave editions add two closures: the residuals are nodal polynomials at geometric nodes, and the triangular transform has an equally explicit inverse.

**Theorem 13.4** (Nodal form and the exact inverse transform). *Write  $Q = q^2$  and  $r_n = (-1)^n \psi_n^* = \sum_k (-q)^{n-k} \begin{bmatrix} n \\ k \end{bmatrix}_Q \eta_k$  for the monic Gram–Schmidt residuals of the sign-corrected tower.*

1. *The generating polynomial  $P_n(z) = \sum_k (-1)^{n-k} q^{n-k} \begin{bmatrix} n \\ k \end{bmatrix}_Q q^{k^2} z^k$  factors as the nodal product*

$$P_n(z) = q^{n^2} \prod_{j=0}^{n-1} (z - q^{-2j}), \quad \langle r_n, \eta_j \rangle = q^{j^2} P_n(q^{-2j}) : \quad (13.8)$$

*orthogonality is interpolation at the geometric nodes  $1, Q^{-1}, \dots, Q^{-(n-1)}$ , and the pivot is the evaluation at the next node,  $\|r_n\|_2^2 = q^{n^2} P_n(q^{-2n}) = (Q; Q)_n$ .*

2. *The transform inverts in closed form:*

$$\boxed{\eta_n = \sum_{k=0}^n q^{(n-k)^2} \begin{bmatrix} n \\ k \end{bmatrix}_Q r_k,} \quad (13.9)$$

*that is,  $C'_{nk} = (-q)^{n-k} \begin{bmatrix} n \\ k \end{bmatrix}_Q$  and  $B_{nk} = q^{(n-k)^2} \begin{bmatrix} n \\ k \end{bmatrix}_Q$  satisfy  $C'_N B_N = B_N C'_N = I$ , and the positive kernel factors as  $G_N = B_N D_N B_N^\top$  with  $D_N = \text{diag}((Q; Q)_0, \dots, (Q; Q)_N)$ : the Cholesky factor of the strictly totally positive kernel is itself entrywise positive.*

*Proof.* Since  $q^{n-k} q^{k^2} = q^n Q^{\binom{k}{2}}$ , Gauss's identity  $\sum_k (-1)^k Q^{\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_Q z^k = (z; Q)_n$  gives  $P_n(z) = (-1)^n q^n (z; Q)_n = q^{n^2} \prod_j (z - q^{-2j})$ , and  $(k-j)^2 = k^2 - 2kj + j^2$  turns  $\langle r_n, \eta_j \rangle$  into  $q^{j^2} P_n(q^{-2j})$ . The nodes  $j < n$  annihilate; at  $j = n$ , extracting  $q^{-2n}$  from each factor of  $P_n(q^{-2n})$  leaves  $(Q; Q)_n q^{-2n^2}$ , and the prefactors  $q^{n^2} \cdot q^{n^2}$  cancel exactly. For the inverse, put  $d = n - m$  and  $j = k - m$ ; the trinomial identity  $\begin{bmatrix} n \\ k \end{bmatrix}_Q \begin{bmatrix} k \\ m \end{bmatrix}_Q = \begin{bmatrix} n \\ m \end{bmatrix}_Q \begin{bmatrix} d \\ j \end{bmatrix}_Q$  gives

$$\sum_{k=m}^n B_{nk} C'_{km} = \begin{bmatrix} n \\ m \end{bmatrix}_Q \sum_{j=0}^d (-1)^j q^{(d-j)^2 + j^2} \begin{bmatrix} d \\ j \end{bmatrix}_Q = \begin{bmatrix} n \\ m \end{bmatrix}_Q (-1)^d q^d \sum_{i=0}^d (-1)^i Q^{\binom{i}{2}} \begin{bmatrix} d \\ i \end{bmatrix}_Q,$$

substituting  $i = d - j$  and  $q^{i^2 - i} = Q^{\binom{i}{2}}$ ; the inner sum is  $(1; Q)_d$  by Gauss, which vanishes for  $d \geq 1$  and is 1 at  $d = 0$ . Triangularity makes the one-sided inverse two-sided, and  $C' G_N C'^\top = D_N$  rearranges to  $G_N = B_N D_N B_N^\top$ .  $\square$

**Remark 13.5** (Exact formal boundary for the  $q$ -Gaussian inverse). Only the coefficient-inversion component of the preceding two theorems is currently exact in Lean. In `QBinomialInversionSp`

specializations. In `lean`, at Gaussian base  $Q = q^2$  and independent scale  $s = -q$ , the definitions `Fabiús.qGaussianResidualCoeff` and `Fabiús.qGaussianReconstructionCoeff` have the closed forms

$$C'_{n,k} = (-q)^{n-k} \begin{bmatrix} n \\ k \end{bmatrix}_{q^2}, \quad B_{n,k} = q^{(n-k)^2} \begin{bmatrix} n \\ k \end{bmatrix}_{q^2}$$

by `Fabiús.qGaussianResidualCoeff_eq` and `Fabiús.qGaussianReconstructionCoeff_eq`. These two definitions and pointwise theorems require only an arbitrary ring. The two declarations `Fabiús.qGaussianReconstructionCoeff_residualCoeff_delta` and `Fabiús.qGaussianResidualCoeff_reconstructionCoeff_delta` prove both total finite convolutions over a commutative ring, hence  $C'_N B_N = B_N C'_N = I$  for every  $N$ . Together with `Fabiús.scaledGaussianBinomialInverseTransform_transform` and `Fabiús.scaledGaussianBinomialTransform_inverseTransform` from `QBinomialTransform.lean`, this proves the module-valued reconstruction (13.9) whenever  $r = C'\eta$ . It needs only a commutative ring acting on an additive commutative monoid;  $q$  need not be nonzero or invertible.

This does *not* mark all of [theorem 13.3](#) or [theorem 13.4](#) formalized. No current declaration in this tranche proves the displayed Hilbert inner products or norms, the Gram or inverse-Gram formulas, the nodal product and its evaluations, the pivots, or  $G_N = B_N D_N B_N^T$ . Those analytic and Gram-factorization components remain open; only (13.9) and the two-sided coefficient matrix inverse are exact here.

*Remark 13.6* (Rogers–Szegő identification). Substituting  $e_j \mapsto z^j$  turns (13.7) into  $q^n H_n(z/q; q^2)$ , where  $H_n(z; q) = \sum_j \begin{bmatrix} n \\ j \end{bmatrix}_q z^j$  is the Rogers–Szegő polynomial: the derivative tower of  $h_a$  is orthogonalized by the classical  $q$ -Hermite circle polynomials at base  $a^{-2}$  [44]. This is forced by [proposition 13.8](#) below, because the Rogers–Szegő polynomials are precisely the orthogonal polynomials of the theta weight on the circle.

**Corollary 13.7** (Uniform innovation). *For every  $n \geq 1$ ,*

$$\text{dist}_{L^2}(e_n, \text{span}(e_0, \dots, e_{n-1}))^2 = (q^2; q^2)_n \downarrow (q^2; q^2)_\infty > 0 :$$

*no matter how many derivatives have been recorded, the next one retains an innovation of norm at least  $(q^2; q^2)_\infty^{1/2}$  — new derivatives never become asymptotically redundant, in sharp contrast with the explosive growth of the unnormalized norms.*

*Proof.* The squared distance is the Gram–Schmidt pivot  $\det G_n / \det G_{n-1} = (q^2; q^2)_n$  of [theorem 13.2](#), decreasing to the convergent infinite product.  $\square$

**Proposition 13.8** (Circle model: the wrapped heat kernel at time  $\log a$ ). *Fix a parity  $\epsilon$  and let*

$$\Theta_a(\theta) = \sum_k (-1)^k q^{k^2} e^{ik\theta} = \vartheta_4\left(\frac{\theta}{2}, q\right) = \sqrt{\frac{\pi}{\log a}} \sum_k \exp\left(-\frac{(\theta - \pi - 2\pi k)^2}{4 \log a}\right), \quad q = a^{-1},$$

*a strictly positive analytic weight. Then  $\psi_{2j+\epsilon} \mapsto z^j$  extends to a unitary equivalence between  $\overline{\text{span}}\{\psi_{2j+\epsilon}\}_{j \geq 0} \subset L^2(\mathbb{R})$  and the closure of the polynomials in  $L^2(\mathbb{T}, \Theta_a \frac{d\theta}{2\pi})$ : each parity tower of normalized derivatives is the monomial sequence of a weighted Hardy space whose weight is the wrapped heat kernel at time  $\log a$ , centred at the antipode  $\theta = \pi$ . The sign correction  $\eta_j = (-1)^j e_j$  of (13.2) rotates the weight to  $\vartheta_3(\theta/2, q)$ , centred at  $\theta = 0$ .*

*Proof.* The monomial Gram matrix of the weight is its normalized  $2\pi$ -periodic Fourier coefficient sequence  $\widehat{\Theta}_{a \text{ FS}; 2\pi}[j - k] := (2\pi)^{-1} \int_0^{2\pi} \Theta_a(\theta) e^{-i(j-k)\theta} d\theta = (-1)^{j-k} q^{(j-k)^2}$ , which by [theorem 13.1](#) is the Gram matrix of the raw tower; a stationary positive definite kernel determines its closed span up to canonical unitary equivalence, and the equivalence matches the two distinguished total sequences. The theta identity is the Jacobi triple product; the Gaussian resummation is Poisson summation applied to  $(-1)^k q^{k^2} = e^{-k^2 \log a + \pi i k}$ .  $\square$

Two readings follow. First, taking two more derivatives of  $h_a$  is, in this model, plain multiplication by the coordinate  $z$  — the most rigid shift there is; every spectral statement about high derivatives becomes a statement about a Toeplitz operator with an explicit theta symbol. Second, the base  $a$  enters only as the heat time  $t = \log a$ : the family  $a \mapsto \Theta_a$  solves the heat equation on the circle, flattening to the uniform weight (an asymptotically orthonormal tower) as  $a \rightarrow \infty$ . The separated regime only ever sees times  $t > \log 2$ , safely away from the short-time Dirac degeneration — which is why the towers stay uniformly conditioned for all  $a > 2$ , with the table above quantifying the approach to orthonormality.

The innovation filters themselves stabilize (second expanded fifteenth-wave edition): far from the left boundary, Gram–Schmidt becomes an explicit stationary minimum-phase filter.

**Theorem 13.9** (Minimum-phase theta whitening). *Write the monic residuals of [theorem 13.4](#) in lag form,  $r_n = \sum_{r=0}^n a_{n,r} \eta_{n-r}$  with  $a_{n,r} = (-1)^r q^r \begin{bmatrix} n \\ r \end{bmatrix}_Q$ ,  $Q = q^2$ . Then  $(a_{n,r})_{r \geq 0} \rightarrow (a_r)_{r \geq 0}$  in  $\ell^1$  as  $n \rightarrow \infty$ , where*

$$a_r = \frac{(-1)^r q^r}{(Q; Q)_r}, \quad A_q(z) = \sum_{r \geq 0} a_r z^r = \frac{1}{(-qz; Q)_\infty} \quad (|z| \leq 1),$$

whose denominator zeros  $z = -q^{-2m-1}$ ,  $m \geq 0$ , all lie outside the closed unit disk. Against the positive Toeplitz symbol  $\vartheta_3(\theta/2, q) = \sum_k q^{k^2} e^{ik\theta}$  of [theorem 13.2](#), Jacobi’s triple product gives the exact whitening identity

$$\boxed{|A_q(e^{i\theta})|^2 \vartheta_3\left(\frac{\theta}{2}, q\right) = (Q; Q)_\infty \quad (\theta \in \mathbb{R}),} \quad (13.10)$$

and  $\|r_n\|_2^2 = (Q; Q)_n \downarrow (Q; Q)_\infty$  is the stationary innovation variance of [corollary 13.7](#).

*Proof.* For fixed  $r$ ,  $\begin{bmatrix} n \\ r \end{bmatrix}_Q = (Q^{n-r+1}; Q)_r / (Q; Q)_r \rightarrow 1/(Q; Q)_r$ , with summable majorant  $|a_{n,r}| \leq q^r / (Q; Q)_\infty$ , so dominated convergence gives the  $\ell^1$  limit; Euler’s theorem  $\sum_r x^r / (Q; Q)_r = 1/(x; Q)_\infty$  evaluates the generating function at  $x = -qz$ . The triple product  $\vartheta_3(\theta/2, q) = (Q; Q)_\infty (-qe^{i\theta}; Q)_\infty (-qe^{-i\theta}; Q)_\infty$  — the same factorization used for [theorem 13.2](#) — divides out to (13.10).  $\square$

In prediction-theory language: the finite Gaussian-binomial rows are exact finite-past innovation filters, and their limit  $A_q$  is the causal minimum-phase Szegő spectral factor of the bi-infinite  $q$ -Gaussian covariance. One proof of the closed orthogonalization thus lives at geometric interpolation nodes ([theorem 13.4](#)), the other on the unit circle. The rate at which the finite rows approach the stationary filter is posed in the register ([conjecture 23.16](#)).

Beyond consecutive sections, every minor of the kernel is closed, and positive: the answer is a Schur polynomial.

**Theorem 13.10** (Schur minors and strict total positivity). *Let  $0 < q < 1$  and let  $0 \leq r_0 < \dots < r_N$  and  $0 \leq c_0 < \dots < c_N$  be row and column indices. With  $x_i = q^{-2r_i}$  and the partition  $\lambda = (c_N - N, \dots, c_1 - 1, c_0)$ ,*

$$\boxed{\det[q^{(r_i - c_j)^2}]_{i,j=0}^N = q^{\sum_i r_i^2 + \sum_j c_j^2} \prod_{0 \leq i < j \leq N} (x_j - x_i) s_\lambda(x_0, \dots, x_N) > 0.} \quad (13.11)$$

Hence the infinite matrix  $[q^{(i-j)^2}]_{i,j \geq 0}$  is strictly totally positive. In particular a sparse parity subfamily  $\eta_{m_0}, \dots, \eta_{m_N}$  has Gram determinant

$$q^{2 \sum_i m_i^2} \prod_{i < j} (q^{-2m_j} - q^{-2m_i}) s_{(m_N - N, \dots, m_0)}(q^{-2m_0}, \dots, q^{-2m_N}),$$

recovering (13.3) at  $m_i = i$ , where the partition is empty and the Schur factor is 1.

*Proof.* Extract  $q^{r_i^2}$  from row  $i$  and  $q^{c_j^2}$  from column  $j$ ; the remaining matrix is the generalized Vandermonde  $[x_i^{c_j}]$ , and the bialternant definition of Schur polynomials gives  $\det[x_i^{c_j}] = \prod_{i < j} (x_j - x_i) s_\lambda(x)$ . Since the  $r_i$  increase and  $q < 1$ ,  $0 < x_0 < \dots < x_N$ : the Vandermonde factor is positive, and a Schur polynomial has nonnegative coefficients and is strictly positive at positive arguments.  $\square$

**Corollary 13.11** (Oscillation and the checkerboard inverse). *Every finite section of the positive kernel is an oscillation matrix in Karlin’s sense [58]: its Gram maps are variation diminishing, its eigenvalues are simple and positive, and its inverse has checkerboard signs — exactly the  $(-1)^{j-k}$  pattern displayed in closed form by theorem 13.3. The Schur factor quantifies how a sparse choice of derivative orders changes the Euclidean volume of the normalized system.*

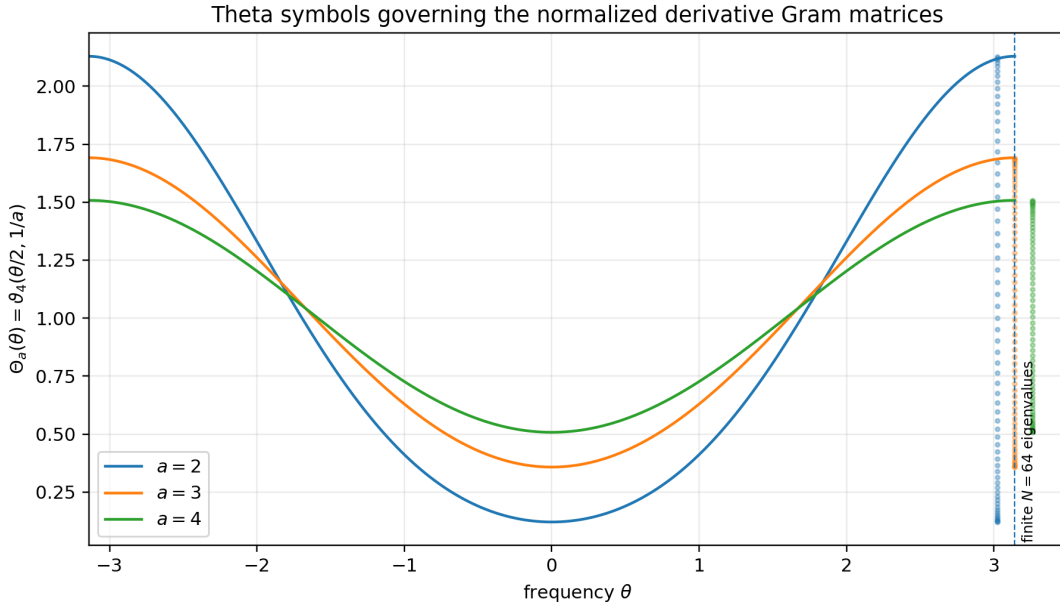


Figure 24: The exact Toeplitz symbols  $\Theta_a$  for  $a = 2, 3, 4$  (proposition 13.8). The vertical point clouds at the right are the eigenvalues of the corresponding  $64 \times 64$  Gram sections; they lie inside, and increasingly fill, the optimal theta intervals  $[A_a, B_a]$  of (13.6).

### 13.2 The physical-space Stieltjes–Wigert ladder and the autocorrelation germ

The spectral orthogonal polynomials (12.4) pull back through the Fourier transform (expanded fourteenth-wave edition): applied to the positive operator  $\mathcal{L} = -d^2/dx^2$ , they orthogonalize the even-derivative span of  $h_a$  inside  $L^2(\mathbb{R})$  by the same closed  $q$ -binomial formulas, with no Gram–Schmidt recursion. Here  $q = a^{-2}$  and  $c = a/2$  as in appendix 12, and  $d\rho_{\text{sp}} = w_{\text{sp}}(z) dz$  denotes the spectral law of  $Z_a$  from theorem 12.1.

**Theorem 13.12** (Stieltjes–Wigert differential ladder). *Let  $a \geq 2$  and  $\Upsilon_{a,n} = P_{a,n}(\mathcal{L})h_a$  with  $P_{a,n}$  the monic polynomials (12.4). Then:*

1.  $\widehat{\Upsilon_{a,n} \text{ ang}}(t) = P_{a,n}(t^2) \widehat{h_{a \text{ ang}}}(t)$ , and each  $\Upsilon_{a,n}$  is  $C^\infty$  with support in  $[-b_a, b_a]$ ;
2. the family  $\{\Upsilon_{a,n}\}_{n \geq 0}$  is orthogonal in  $L^2(\mathbb{R})$ , with

$$\|\Upsilon_{a,n}\|_2^2 = \|h_a\|_2^2 c^{2n} q^{-n(2n+1)}(q; q)_n; \quad (13.12)$$

3. explicitly, in derivatives,

$$\Upsilon_{a,n} = (-1)^n \sum_{k=0}^n c^{n-k} q^{k^2-n^2} \begin{bmatrix} n \\ k \end{bmatrix}_q h_a^{(2k)}, \quad (13.13)$$

so  $\text{span}\{\Upsilon_{a,0}, \dots, \Upsilon_{a,N}\} = \text{span}\{h_a, h_a'', \dots, h_a^{(2N)}\}$  for every  $N$ ;

4. with  $\Upsilon_{a,-1} = 0$  and  $\Upsilon_{a,0} = h_a$ , the ladder obeys the three-term operator recurrence

$$\Upsilon_{a,n+1} = (\mathcal{L} - \alpha_n)\Upsilon_{a,n} - \beta_n\Upsilon_{a,n-1}, \quad \begin{aligned} \alpha_n &= cq^{-2n-1}(1+q-q^{n+1}), \\ \beta_n &= c^2q^{-(4n-1)}(1-q^n). \end{aligned} \quad (13.14)$$

*Proof.*  $\mathcal{L}$  acts on the Fourier side as multiplication by  $t^2$ , so (i) is polynomial functional calculus, and Parseval turns ladder inner products into integrals of  $P_{a,n}P_{a,m}$  against the spectral law of [theorem 12.1](#): orthogonality and (13.12) are exactly the statements recorded in (12.4). Expanding  $P_{a,n}(\mathcal{L})$  with  $\mathcal{L}^k = (-1)^k \partial^{2k}$  collapses the signs  $(-1)^{n-k}(-1)^k$  to the global  $(-1)^n$ , giving (iii); the change of basis is triangular with invertible diagonal. For (iv), the monic three-term recurrence  $zP_{a,n} = P_{a,n+1} + \alpha_n P_{a,n} + \beta_n P_{a,n-1}$  has  $\beta_n = \|P_{a,n}\|^2 / \|P_{a,n-1}\|^2$ , which (12.4) evaluates to  $c^2 q^{-(4n-1)}(1-q^n)$ , while comparing subleading coefficients gives  $\alpha_n = p_{n,n-1} - p_{n+1,n} = cq^{-2n-1}(1+q-q^{n+1})$ , valid also at  $n=0$ ; substituting  $z = \mathcal{L}$  finishes.  $\square$

An indeterminate Stieltjes moment problem in frequency thus produces an explicit orthogonal system of compactly supported smooth functions in physical space. Moreover, the two closed orthogonalizations of the derivative tower — this spectral ladder and the  $q$ -Gaussian Gram–Schmidt of [theorem 13.3](#) — are one and the same:

**Proposition 13.13** (The two ladders coincide). *For every  $a \geq 2$  and  $n \geq 0$ , the spectral ladder is the even-parity closed Gram–Schmidt vector, rescaled by the raw derivative norm:*

$$\Upsilon_{a,n} = (-1)^n \|h_a^{(2n)}\|_2 \psi_n^* \quad (\epsilon = 0). \quad (13.15)$$

Likewise  $(-1)^n \|h_a^{(2n+1)}\|_2 \psi_n^*$  at odd parity equals  $\tilde{P}_{a,n}(\mathcal{L}) h'_a$ , where  $\tilde{P}_{a,n}$  are the monic orthogonal polynomials of the once-shifted spectral measure  $z \, d\rho_{\text{sp}}(z)$ .

*Proof.* Both sides of (13.15) are orthogonal to  $h_a, h_a'', \dots, h_a^{(2n-2)}$  and lie in the span through  $h_a^{(2n)}$  with the same top coefficient  $(-1)^n$  (on the right because  $e_n = h_a^{(2n)} / \|h_a^{(2n)}\|_2$  carries coefficient 1 in  $\psi_n^*$ ); Gram–Schmidt vectors of a flag are unique given the top coefficient. Directly: with  $q_* = a^{-1}$ , so that  $\begin{bmatrix} n \\ k \end{bmatrix}_{q_*} = \begin{bmatrix} n \\ k \end{bmatrix}_q$ , the coefficient of  $h_a^{(2k)}$  on the right is  $(-1)^n \|h_a^{(2n)}\|_2 q_*^{n-k} \begin{bmatrix} n \\ k \end{bmatrix}_q / \|h_a^{(2k)}\|_2$ , and the norm ladder (12.1) gives  $\|h_a^{(2n)}\|_2 q_*^{n-k} / \|h_a^{(2k)}\|_2 = c^{n-k} q^{k^2-n^2}$ , matching (13.13) term by term; the norms agree because  $\|h_a^{(2n)}\|_2^2(q; q)_n = \|h_a\|_2^2 c^{2n} q^{-n(2n+1)}(q; q)_n$ . The odd case is the same argument run on the shifted moments  $m_{s+1}$ , whose Parseval realization is  $\langle \tilde{P}_n(\mathcal{L})h'_a, \tilde{P}_m(\mathcal{L})h'_a \rangle = \|h_a\|_2^2 \int \tilde{P}_n \tilde{P}_m z \, d\rho_{\text{sp}}$ .  $\square$

The unnormalized parity Gram matrices then close in one line each.

**Theorem 13.14** (Derivative-jet Gram determinants). *For  $a \geq 2$  and  $N \geq 0$ , put  $E_N^{\text{jet}} = [\langle h_a^{(2i)}, h_a^{(2j)} \rangle]_{i,j=0}^N$  and  $O_N^{\text{jet}} = [\langle h_a^{(2i+1)}, h_a^{(2j+1)} \rangle]_{i,j=0}^N$  (mixed parities are orthogonal). Then*

$$\det E_N^{\text{jet}} = \|h_a\|_2^{2(N+1)} c^{N(N+1)} q^{-N(N+1)(4N+5)/6} \prod_{k=1}^N (q; q)_k, \quad (13.16)$$

$$\det O_N^{\text{jet}} = \|h_a\|_2^{2(N+1)} c^{(N+1)^2} q^{-(N+1)(N+2)(4N+3)/6} \prod_{k=1}^N (q; q)_k. \quad (13.17)$$



In particular every finite family of even, and every finite family of odd, derivatives of  $h_a$  is linearly independent in  $L^2(\mathbb{R})$ .

*Proof.* Pull out the norms:  $\det E_N^{\text{jet}} = \prod_{i=0}^N \|h_a^{(2i)}\|_2^2 \cdot \det[\langle \psi_{2i}, \psi_{2j} \rangle]$ , and the normalized determinant is  $\prod_{r=1}^N (q; q)_r$  by [theorem 13.2](#) (the raw kernel's alternating signs cancel under conjugation by  $\text{diag}((-1)^i)$ ). By (12.1),  $\prod_{i=0}^N \|h_a^{(2i)}\|_2^2 = \|h_a\|_2^{2(N+1)} c^{N(N+1)} q^{-N(N+1)(4N+5)/6}$  from  $\sum_{i=0}^N i(2i+1) = N(N+1)(4N+5)/6$ ; the odd case uses  $\sum_{i=0}^N (2i+1)(i+1) = (N+1)(N+2)(4N+3)/6$ . The even determinant equivalently restates the moment Hankel (12.3):  $\det E_N^{\text{jet}} = \|h_a\|_2^{2(N+1)} \Delta_N$ .  $\square$

Finally, the second-order statistics of  $X_a$  close at the origin. Let  $\text{acf}_a = h_a * h_a$ , the density of the sum of two independent copies of  $X_a$ ; since  $h_a$  is even, it is also the autocorrelation of  $h_a$ , with  $\widehat{\text{acf}_a}_{\text{ang}} = \widehat{h_a}_{\text{ang}}^2$ .

**Theorem 13.15** (Autocorrelation germ). *For  $a \geq 2$  and  $n \geq 0$ , all odd derivatives of  $\text{acf}_a$  vanish at 0, and*

$$\frac{(-1)^n \text{acf}_a^{(2n)}(0)}{\text{acf}_a(0)} = \frac{\|h_a^{(n)}\|_2^2}{\|h_a\|_2^2} = \frac{a^{n(n+2)}}{2^n}. \quad (13.18)$$

The Taylor series of  $\text{acf}_a$  at the origin has radius of convergence zero.

*Proof.* Fourier inversion gives  $\text{acf}_a^{(2n)}(0) = (-1)^n (2\pi)^{-1} \int t^{2n} |\widehat{h_a}_{\text{ang}}(t)|^2 dt = (-1)^n \|h_a^{(n)}\|_2^2$  and  $\text{acf}_a(0) = \|h_a\|_2^2$ ; the moment value is [theorem 12.1](#). The  $2n$ th Taylor coefficient thus grows like  $a^{n^2}$  against  $(2n)! = e^{\mathcal{O}(n \log n)}$ , so its  $2n$ th root diverges.  $\square$

**Remark 13.16** (Plateau versus germ, and incompleteness of the ladder). For  $a > 2$  the density  $h_a$  is exactly constant on its central plateau, while its autocorrelation is not analytic at the origin in the strongest possible sense: the spatial plateau and the frequency-moment explosion coexist on sharply different analytic levels. The germ also shows the ladder is provably *incomplete*: the spectral law  $\rho_{\text{sp}}$  is the spectral measure of  $\mathcal{L}$  in the state  $h_a$ , so the cyclic subspace of  $h_a$  under  $\mathcal{L}$  is unitarily  $L^2(\rho_{\text{sp}})$  – but by the Riesz density criterion for indeterminate moment problems, polynomials are dense in  $L^2(\rho)$  only for  $N$ -extremal (hence discrete) representing measures [44], and  $\rho_{\text{sp}}$  is absolutely continuous. Hence some vectors  $g(\mathcal{L})h_a$  of the cyclic subspace are orthogonal to every even derivative of  $h_a$ ; exhibiting an explicit bounded one is the subject of [conjecture 23.14](#).

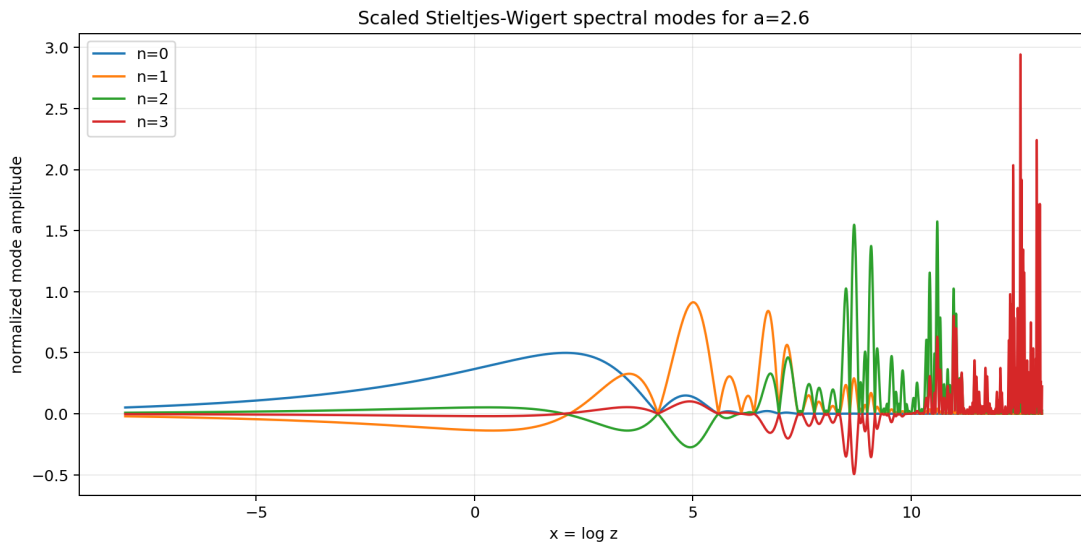


Figure 25: The first four normalized spectral modes  $P_{a,n}(t^2) \widehat{h_a}_{\text{ang}}(t)$  of the differential ladder at  $a = 2.6$ , displayed against  $\log(t^2)$  ([theorem 13.12](#)). Their numerical Gram matrix is the identity to the precision recorded in the package's `orthogonal_ladder_validation_a_2_6.csv`.



## 14 The fractal spline atlas for $a > 2$

### 14.1 The exceptional self-similar set

Define the two contractions

$$T_{\pm}(x) = \frac{x \pm 1}{a} \quad (14.1)$$

and their attractor

$$K_a = T_{-}(K_a) \cup T_{+}(K_a) = \left\{ \sum_{j=1}^{\infty} \varepsilon_j a^{-j} : \varepsilon_j \in \{\pm 1\} \right\}. \quad (14.2)$$

For  $a > 2$  the two first-level pieces are disjoint. The open central gap is

$$G_{\emptyset} = (-b_1, b_1), \quad b_1 = \frac{a-2}{a(a-1)}.$$

For a finite word  $\omega \in \{-, +\}^r$ , let  $T_{\omega}$  be the corresponding composition and put

$$G_{\omega} = T_{\omega}(G_{\emptyset}).$$

**Theorem 14.1** (Gap decomposition and dimension). *For  $a > 2$ ,*

$$(-b_a, b_a) \setminus K_a = \bigsqcup_{r=0}^{\infty} \bigsqcup_{|\omega|=r} G_{\omega}. \quad (14.3)$$

*At depth  $r$  there are  $2^r$  gaps, each of length  $2b_1 a^{-r}$ , and*

$$\sum_{r=0}^{\infty} 2^r (2b_1 a^{-r}) = 2b_a. \quad (14.4)$$

*Consequently  $K_a$  has Lebesgue measure zero and*

$$\dim_{\text{H}} K_a = \frac{\log 2}{\log a}. \quad (14.5)$$

*Proof.* This is the standard open-set-condition analysis of the two-map iterated function system. Removing the first central gap leaves exactly the two scaled support copies; iterating gives (14.3). The length calculation is geometric, and (14.4) follows after substituting  $b_1 = b_a(1 - 2/a)$ . The similarity dimension solves  $2a^{-s} = 1$ .  $\square$

### 14.2 Finite spline formula

Let  $h_{a,N}$  be the inverse Fourier transform of the finite product

$$\Phi_{a,N}(\xi) = \prod_{j=1}^N \text{sinc}_{\text{rad}}(2\pi a^{-j} \xi). \quad (14.6)$$

It is the density of  $\sum_{j=1}^N a^{-j} U_j$ , hence a compact spline of degree  $N - 1$ .

**Theorem 14.2** (Explicit finite atomic spline). *For  $N \geq 1$ ,*

$$h_{a,N}(x) = \frac{a^{N(N+1)/2}}{2^N (N-1)!} \sum_{\varepsilon \in \{\pm 1\}^N} w(\varepsilon) \left( x + \sum_{j=1}^N \varepsilon_j a^{-j} \right)_+^{N-1}. \quad (14.7)$$

*Moreover,*

$$h_a = h_{a,N} * [a^N h_a(a^N \cdot)]. \quad (14.8)$$

*Proof.* The density of  $a^{-j}U_j$  is  $(a^j/2)\mathbf{1}_{[-a^{-j}, a^{-j}]}$ . Repeated integration of the signed endpoint representation of an interval indicator gives (14.7). Splitting  $X_a = \sum_{j=1}^N a^{-j}U_j + a^{-N}X'_a$  gives (14.8).  $\square$

The formula makes the  $q$ -binomial connection concrete: its knots are signed geometric subset sums, and its coefficients are parity signs. It also supplies an exact symbolic implementation independent of FFT inversion, although direct floating-point evaluation at large  $N$  suffers catastrophic cancellation.

### 14.3 Local polynomial degree

**Theorem 14.3** (Exact degree and derivative amplitude on every gap). *If  $|\omega| = r$ , then  $h_a$  is a polynomial of degree exactly  $r$  on  $G_\omega$ . More precisely, throughout that gap,*

$$\left| h_a^{(r)}(x) \right| = \frac{a^{r(r+3)/2+1}}{2^{r+1}}, \quad h_a^{(r+1)}(x) = 0. \quad (14.9)$$

*Proof.* On the empty-word gap,  $h_a = a/2$ , so the assertion holds for  $r = 0$ . Assume a depth- $r$  gap  $G$  has constant nonzero  $r$ th derivative of magnitude  $C_r$  and vanishing  $(r+1)$ st derivative. If  $x = T_+(y) = (y+1)/a$  with  $y \in G$ , then only the branch  $h_a(ax-1) = h_a(y)$  survives in a neighborhood, whereas for  $x = T_-(y) = (y-1)/a$  only  $h_a(ax+1) = h_a(y)$  survives. Differentiating the one-branch identity  $r$  additional times gives

$$|h_a^{(r+1)}(x)| = \frac{a^{r+2}}{2} C_r, \quad h_a^{(r+2)}(x) = 0.$$

Thus  $C_{r+1} = a^{r+2}C_r/2$  with  $C_0 = a/2$ . Solving this recursion yields (14.9), which is nonzero and proves exact degree.  $\square$

This removes the possibility of an exceptional algebraic cancellation: every word of length  $r$  carries a genuinely degree- $r$  piece, with a leading derivative whose magnitude is the global  $r$ th-derivative norm.

**Corollary 14.4** (Signed leading coefficient). *Let  $|\omega| = r$  and let  $N_+(\omega)$  be the number of  $+$  symbols in  $\omega$ . On  $G_\omega$  the polynomial piece of  $h_a$  has leading coefficient*

$$L_\omega = (-1)^{N_+(\omega)} \frac{a^{(r+1)(r+2)/2}}{2^{r+1} r!}. \quad (14.10)$$

*Proof.* The magnitude is (14.9) divided by  $r!$ , since  $a^{r(r+3)/2+1} = a^{(r+1)(r+2)/2}$ . For the sign, follow the one-branch induction in the proof of theorem 14.3: on the image  $T_+(G)$  of a gap  $G$  only the branch  $-h_a(ax-1)$  of (5.4) survives after differentiation, reversing the sign of the leading derivative, while on  $T_-(G)$  the surviving branch  $+h_a(ax+1)$  preserves it. Starting from the positive plateau value  $a/2$  on  $G_\emptyset$ , each  $+$  letter contributes one sign reversal (fifteenth wave).  $\square$

The first magnitudes at  $a = 3$  are  $\frac{3}{2}$ ,  $\frac{27}{4}$ ,  $\frac{3^6}{16}$ , and  $\frac{3^{10}}{96}$ ; the fifteenth-wave package verifies these by polynomial fits on all gaps through generation three.

### 14.4 Derivative localization on $K_a$

For  $x \in K_a$ , the separated expansion

$$x = \sum_{r=1}^{\infty} \varepsilon_r a^{-r} \quad (14.11)$$

is unique. Define the left shift

$$\mathcal{T}_a^n x = \sum_{r=n+1}^{\infty} \varepsilon_r a^{n-r}. \quad (14.12)$$

**Theorem 14.5** (One-branch symbolic localization). *Let  $a > 2$ ,  $x \in K_a$ , and let  $(\varepsilon_r)$  be its unique digit sequence. Then*

$$h_a^{(n)}(x) = (-1)^n \left( \prod_{r=1}^n \varepsilon_r \right) \frac{a^{n(n+3)/2}}{2^n} h_a(\mathcal{T}_a^n x). \quad (14.13)$$

*Proof.* In (11.5), choose the sign attached to the power  $a^{n-r}$  to be  $-\varepsilon_r$ . Its large prefix cancels  $a^n x$ , leaving exactly  $\mathcal{T}_a^n x \in [-b_a, b_a]$ . Any other sign choice has a first mismatched digit. Because  $a > 2$ , that leading discrepancy exceeds the largest possible remaining tail, so the resulting argument lies outside  $[-b_a, b_a]$ . Thus exactly one term survives. Its weight is  $(-1)^n \prod_{r=1}^n \varepsilon_r$ , and the common prefactor is the one in (11.6).  $\square$

**Lemma 14.6** (Strict positivity in the open support). *For every  $a > 1$ ,*

$$h_a(x) > 0 \quad (|x| < b_a).$$

*Proof.* Fix  $|x| < b_a$  and choose  $N$  so large that  $x$  lies in the interior of the finite-sum support  $[-\sum_{j=1}^N a^{-j}, \sum_{j=1}^N a^{-j}]$ . The finite convolution  $h_{a,N}$  is strictly positive in that interior. Writing  $X_a = X_{a,N} + Z_N$ , the tail  $Z_N$  assigns positive probability to every neighborhood of zero (choose finitely many uniform coordinates in small intervals and bound the remaining geometric tail). Hence  $h_a(x) = \mathbb{E}[h_{a,N}(x - Z_N)] > 0$ .  $\square$

**Corollary 14.7** (Terminating Taylor germs). *If the digit sequence of  $x \in K_a$  is eventually constant, then  $h_a^{(n)}(x) = 0$  for every sufficiently large  $n$ . Hence the Taylor series at  $x$  is a polynomial. Nevertheless  $x$  is a gap endpoint, so this polynomial need not agree with  $h_a$  on a neighborhood.*

**Corollary 14.8** (A zero-radius criterion). *Suppose there are a compact set  $J \subset K_a \cap (-b_a, b_a)$ , a constant  $c_J > 0$ , and infinitely many  $n$  such that  $\mathcal{T}_a^n x \in J$  and  $h_a \geq c_J$  on  $J$ . Then the Taylor series of  $h_a$  at  $x$  has radius zero.*

*Proof.* Along the stated subsequence,

$$|h_a^{(n)}(x)| \geq c_J \frac{a^{n(n+3)/2}}{2^n}.$$

The  $n$ th root of  $|h_a^{(n)}(x)|/n!$  tends to infinity along that subsequence.  $\square$

**Theorem 14.9** (Complete Taylor-germ classification). *Let  $a > 2$ .*

1. *If  $x \notin K_a$ , then  $x$  lies in a unique polynomial gap and the Taylor series of  $h_a$  agrees locally with the exact polynomial on that gap.*
2. *If  $x \in K_a$  has an eventually constant digit sequence, then its Taylor series terminates but does not agree with  $h_a$  on a full neighborhood.*
3. *If  $x \in K_a$  is not eventually constant, then its Taylor series has radius zero.*

*Proof.* The first two statements follow from the gap theorem and the terminating-germ corollary. For the third, a non-eventually-constant sign sequence has infinitely many digit changes. Let  $J_a \subset K_a$  be the union of the two compact second-level cylinders whose first two digits are opposite. This set is disjoint from the support endpoints, so by lemma 14.6  $c_a := \min_{J_a} h_a > 0$ . Whenever  $\varepsilon_{n+1} \neq \varepsilon_{n+2}$ , the shifted point  $\mathcal{T}_a^n x$  belongs to  $J_a$ . Thus the zero-radius criterion applies along infinitely many  $n$ .  $\square$

The source states this dichotomy in a longer signed-radix calculation. The one-branch formula reduces it to the elementary observation that every non-eventually-constant binary word contains infinitely many changes.

*Remark 14.10* (Endpoint jet reduction and the exact engine). At a generation- $n$  cylinder endpoint  $x$  the digit sequence is eventually constant, and the shifted point  $\mathcal{T}_a^\ell x$  is again a cylinder endpoint, of generation  $n - \ell$ . [Theorem 14.5](#) then reads as an exact jet reduction: for  $0 \leq \ell < n$ ,

$$h_a^{(\ell)}(x) = \pm \frac{a^{\ell(\ell+3)/2}}{2^\ell} h_a(\mathcal{T}_a^\ell x), \quad (14.14)$$

with the sign  $(-1)^\ell \prod_{r \leq \ell} \varepsilon_r$  from the digits consumed. The fifteenth-wave package writes (14.14) in a fully binary-indexed form with the Thue–Morse sign spelled out, repairing the damaged source indices (3.51)–(3.58), and combines it with the Taylor remainder into an explicit chain

Bernoulli numbers  $\rightarrow$  cumulants  $\rightarrow$  moments  
 $\rightarrow$  endpoint values  $\rightarrow$  endpoint jets  $\rightarrow$  every gap polynomial,

an exact symbolic engine for  $h_a$  over  $\mathbb{Q}(a)$  at any algebraic base. With [corollary 14.4](#) certifying the top coefficient independently, the engine is self-checking.

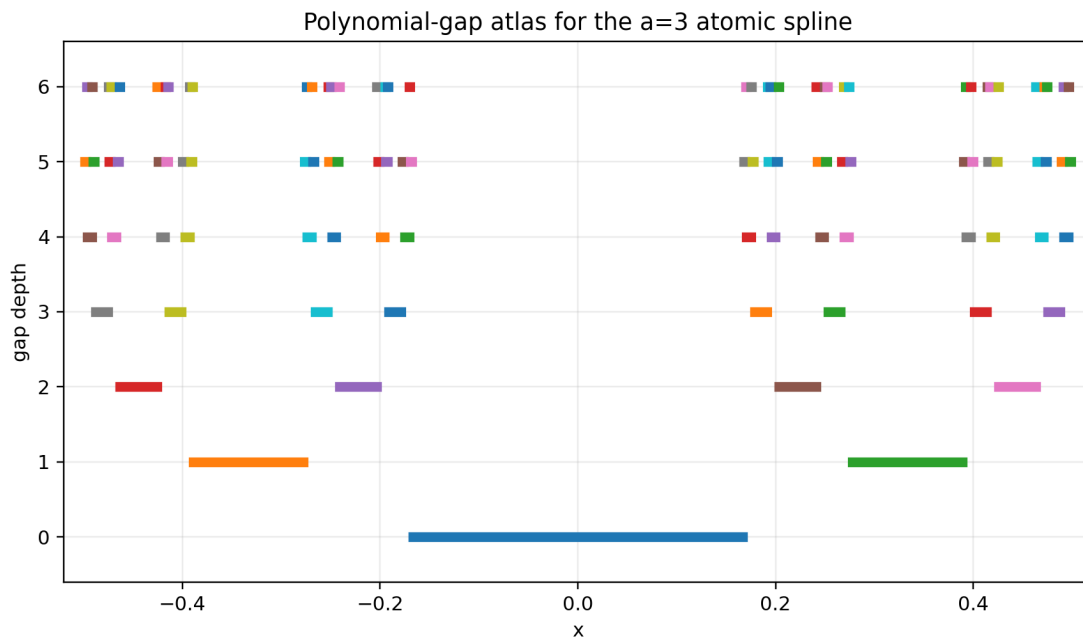


Figure 26: The first seven levels of the gap decomposition for  $a = 3$ . The depth  $r$  gaps are the intervals on which the local polynomial degree is exactly  $r$ . Their union has full measure in the support  $[-1/2, 1/2]$ .

## 15 Fractal strings, complex dimensions, and the exact tube formula

This section (fourteenth wave) upgrades the qualitative dimension count of [theorem 14.1](#) to a complete quantitative geometry of the exceptional set: every classical statistic of  $K_a$ —its geometric zeta function, its complex dimensions, its tube volume, and the distribution of the local polynomial degree—has a closed form.

### 15.1 The gap string and its complex dimensions

By [theorem 14.1](#), the bounded complementary intervals of  $K_a$  form the self-similar fractal string

$$\mathcal{L}_a : \quad \ell_r = \ell_0 a^{-r} \text{ with multiplicity } 2^r \ (r \geq 0), \quad \ell_0 = 2b_1 = \frac{2(a-2)}{a(a-1)}. \quad (15.1)$$

**Theorem 15.1** (Dimension, geometric zeta function, and complex dimensions). *For  $a > 2$ , the Hausdorff and Minkowski dimensions of  $K_a$  agree,*

$$\dim_{\mathrm{H}} K_a = \dim_{\mathrm{M}} K_a = D_a := \frac{\log 2}{\log a}, \quad (15.2)$$

*and the geometric zeta function of the string (15.1),*

$$\zeta_{\mathcal{L}_a}(s) = \sum_{r \geq 0} 2^r \ell_r^s = \frac{\ell_0^s}{1 - 2a^{-s}}, \quad \Re s > D_a, \quad (15.3)$$

*extends meromorphically to  $\mathbb{C}$  with simple poles exactly at the vertical lattice*

$$s_k = D_a + \frac{2\pi i k}{\log a}, \quad k \in \mathbb{Z}, \quad \operatorname{res}_{s=s_k} \zeta_{\mathcal{L}_a}(s) = \frac{\ell_0^{s_k}}{\log a}. \quad (15.4)$$

*Proof.* The two-map system (14.1) satisfies the open-set condition for  $a > 2$ , so Hutchinson's theorem [35] gives  $\dim_{\mathrm{H}} K_a = D_a$  from the similarity equation  $2a^{-D} = 1$ ; the tube formula below shows that  $\varepsilon^{D_a-1} V_a(\varepsilon)$  is bounded above and below by positive constants, which pins the Minkowski dimension at  $D_a$  as well. The gap sum (15.3) is geometric. Solving  $2a^{-s} = 1$  gives the pole lattice, and differentiating the denominator gives  $d(1 - 2a^{-s})/ds = 2a^{-s} \log a = \log a$  at every  $s_k$ , hence the residue.  $\square$

The vertical arithmetic progression (15.4) is the canonical lattice pattern of a self-similar fractal string in the sense of Lapidus and van Frankenhuysen [45]. What is distinctive here is that it enters the analysis of a  $C^\infty$  function: the singular set of the smooth spline  $h_a$  carries the same complex dimensions as its complementary gap string.

## 15.2 An exact tube formula

For  $\varepsilon > 0$ , let

$$V_a(\varepsilon) = |\{x \in (-b_a, b_a) \setminus K_a : \operatorname{dist}(x, K_a) < \varepsilon\}| \quad (15.5)$$

be the inner tube volume. Each gap of length  $\ell$  contributes  $\min(2\varepsilon, \ell)$ .

**Theorem 15.2** (Exact tube formula and Minkowski non-measurability). *Let  $a > 2$  and  $0 < 2\varepsilon < \ell_0$ , and put  $N(\varepsilon) = \lfloor \log_a(\ell_0/(2\varepsilon)) \rfloor$ . Then*

$$V_a(\varepsilon) = 2\varepsilon(2^{N(\varepsilon)+1} - 1) + \frac{\ell_0(2/a)^{N(\varepsilon)+1}}{1 - 2/a}. \quad (15.6)$$

*With  $x = \log_a(\ell_0/(2\varepsilon))$  and  $\theta = \{x\}$  its fractional part,*

$$\varepsilon^{D_a-1} V_a(\varepsilon) = \mathcal{M}_a(\theta) - 2\varepsilon^{D_a}, \quad (15.7)$$

*where the one-periodic profile*

$$\mathcal{M}_a(\theta) = \ell_0^{D_a} \left[ 2^{2-D_a-\theta} + \frac{2^{1-D_a} a^{(D_a-1)(1-\theta)}}{1 - 2/a} \right], \quad 0 \leq \theta < 1, \quad (15.8)$$

*is continuous under periodic identification and nonconstant. Hence  $K_a$  is not Minkowski measurable, although its upper and lower Minkowski contents are finite and positive. The logarithmic average of the profile is*

$$\int_0^1 \mathcal{M}_a(\theta) \, d\theta = \ell_0^{D_a} 2^{1-D_a} \left( \frac{1}{\log 2} + \frac{1}{\log(a/2)} \right). \quad (15.9)$$

*Proof.* The depth- $r$  gaps with  $r \leq N(\varepsilon)$  are wider than  $2\varepsilon$  and contribute  $2\varepsilon$  each; the remaining gaps are covered entirely. Summing the finite geometric count and the infinite geometric length tail gives (15.6). Substituting  $\varepsilon = (\ell_0/2)a^{-x}$  and using  $a^{D_a} = 2$  turns the two terms into (15.7)–(15.8); matching the one-sided limits  $\theta \rightarrow 1^-$  and  $\theta = 0$  reduces, after dividing by  $2^{1-D_a}$ , to the identity  $1 + \frac{1}{1-2/a} = 2 + \frac{2/a}{1-2/a}$ , so the profile is continuous on the circle. It is nonconstant because  $2^{-\theta}$  and  $a^{(D_a-1)(1-\theta)}$  are nonconstant exponentials with positive coefficients that are linearly independent from the constants. Integrating each exponential over one period gives (15.9). Boundedness of the positive continuous profile between its minimum and maximum gives finite positive upper and lower Minkowski contents, and nonconstancy denies Minkowski measurability.  $\square$

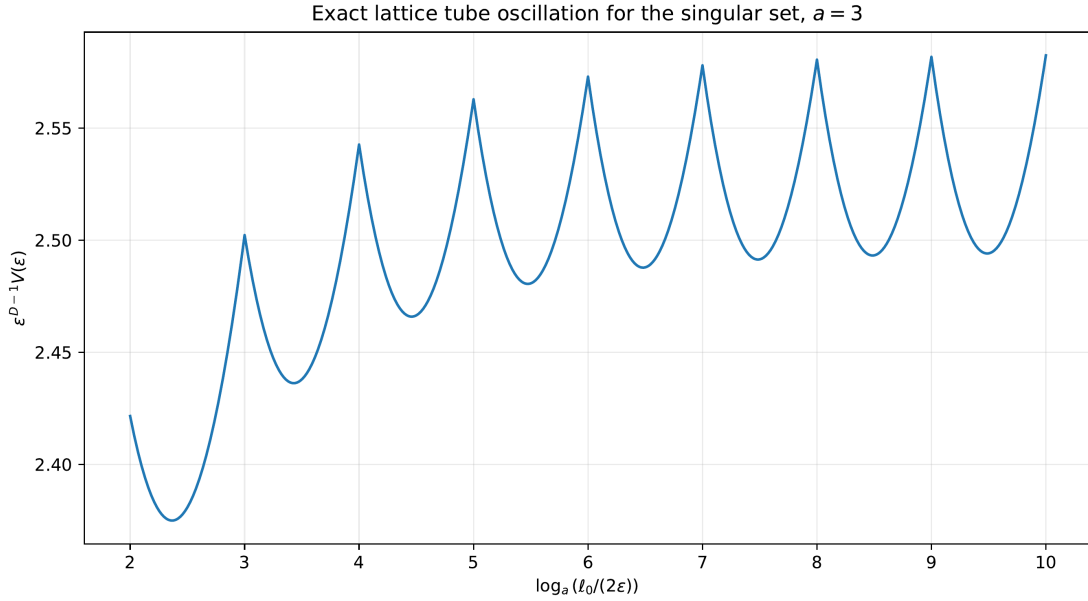


Figure 27: The exact normalized tube volume for  $a = 3$ . The approach to a nonconstant one-periodic profile in  $\log_a(1/\varepsilon)$  verifies the lattice oscillation of [theorem 15.2](#).

### 15.3 The local polynomial degree as a random variable

Choose a point uniformly with respect to normalized Lebesgue measure on  $(-b_a, b_a)$ . Since  $K_a$  is null, the point almost surely lies in a unique gap; let  $N_a$  be the exact polynomial degree of  $h_a$  on that gap, which by [theorem 14.3](#) equals the gap depth.

**Theorem 15.3** (Geometric degree law and critical exponential limit). *For  $a > 2$ ,*

$$\mathbb{P}(N_a = r) = \frac{a-2}{a} \left(\frac{2}{a}\right)^r, \quad r = 0, 1, 2, \dots, \quad (15.10)$$

so that

$$\mathbb{E}N_a = \frac{2}{a-2}, \quad \text{Var}(N_a) = \frac{2a}{(a-2)^2}. \quad (15.11)$$

With  $p_a = (a-2)/a$ , as  $a \downarrow 2$ ,

$$p_a N_a \xrightarrow{d} \text{Exp}(1), \quad \text{equivalently} \quad \frac{a-2}{2} N_a \xrightarrow{d} \text{Exp}(1). \quad (15.12)$$

*Proof.* The total length of the depth- $r$  gaps is  $2^r \ell_0 a^{-r}$ ; dividing by  $2b_a = 2/(a-1)$  gives (15.10), a geometric law on  $\mathbb{N}_0$  with success parameter  $p_a$ , whence (15.11). For the limit,  $\mathbb{P}(p_a N_a > x) = (1 - p_a)^{\lfloor x/p_a \rfloor + 1} \rightarrow e^{-x}$  for every  $x > 0$ , and  $p_a \sim (a-2)/2$ .  $\square$

This makes the phrase “polynomial almost everywhere” quantitative: near the critical base, a typical polynomial piece has degree of exact order  $(a - 2)^{-1}$ . It also identifies the first marginal of the critical double-scaling program of [appendix 23](#): the depth variable, rescaled by  $(a - 2)/2$ , is exponential in the limit, and what remains open is the joint limit of the word address, the gap geometry, and the polynomial coefficients.

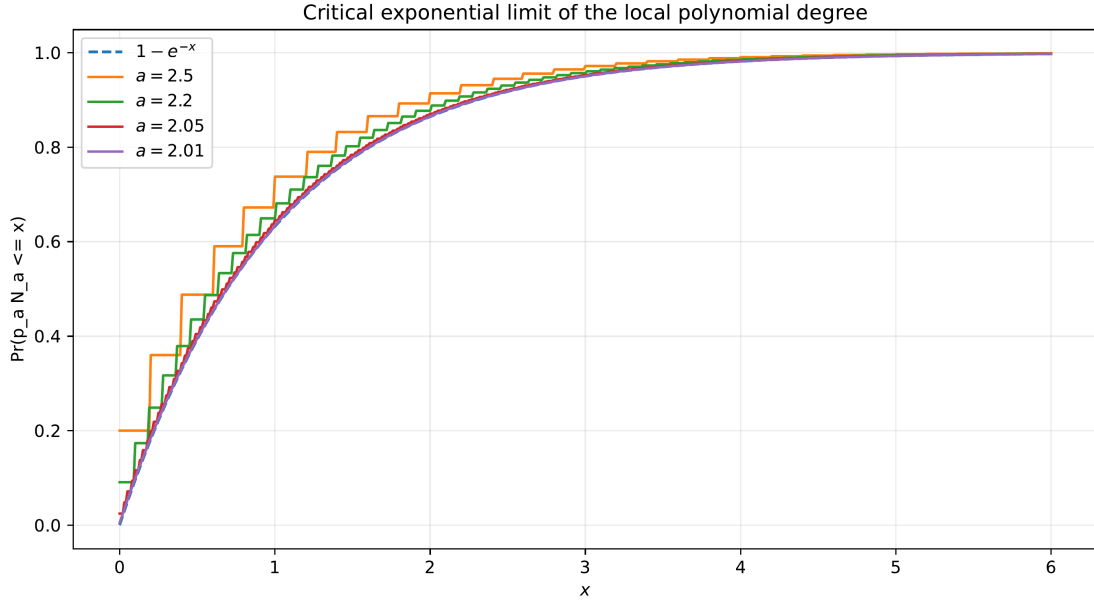


Figure 28: Distribution functions of  $p_a N_a$  approaching  $1 - e^{-x}$  as  $a \downarrow 2$  ([theorem 15.3](#)).

### 15.4 Distance to the nonanalytic set and the shifted zeta law

The revised fourteenth-wave edition refines the degree law into a complete description of the local geometry seen from a typical point. Let  $Q$  be uniform on  $(-b_a, b_a)$ , let  $\mathcal{G}$  be the length of the gap containing it, and let  $\Delta = \text{dist}(Q, K_a)$ .

**Theorem 15.4** (Mellin law of the distance to  $K_a$ ). *For  $a > 2$  there is  $V \sim \text{Unif}[0, 1]$ , independent of  $N_a$ , with*

$$\mathcal{G} = \ell_0 a^{-N_a}, \quad \Delta \stackrel{d}{=} \frac{\mathcal{G}}{2} V. \quad (15.13)$$

Consequently, for  $\Re s > D_a - 1$ ,

$$\mathbb{E} \mathcal{G}^s = \frac{a-2}{a} \frac{\ell_0^s}{1 - 2a^{-(s+1)}}, \quad \mathbb{E} \Delta^s = \frac{a-2}{a} \frac{(\ell_0/2)^s}{(s+1)(1 - 2a^{-(s+1)})}, \quad (15.14)$$

with meromorphic continuations whose poles form the shifted lattice  $s = D_a - 1 + 2\pi i k / \log a$  (plus the elementary pole  $s = -1$  for  $\Delta$ ): the complex dimensions (15.4) reappear, shifted by  $-1$ , in the statistics of a Lebesgue-typical point. Moreover, for  $0 < 2r < \ell_0$ ,

$$\mathbb{P}(\Delta > r) = 1 - \frac{V_a(r)}{2b_a} : \quad (15.15)$$

the exact tube formula (15.6) is also the exact distribution function of the distance.

*Proof.* Conditional on  $N_a = r$ , the point is uniform in one gap of length  $\ell_0 a^{-r}$ , so its distance to the nearer endpoint is uniform on  $[0, \ell_0 a^{-r}/2]$ , giving (15.13) with  $V$  independent of the depth. Combining with (15.10) and summing the geometric series gives (15.14);  $\mathbb{E} V^s = 1/(s+1)$  supplies the extra factor. The event  $\{\Delta \leq r\}$  is the inner  $r$ -tube of  $K_a$  up to the null set  $K_a$  itself.  $\square$



**Corollary 15.5** (Critical logarithmic gap scale). *As  $a \downarrow 2$ , with  $p_a = (a - 2)/a$ ,*

$$p_a \log_a \frac{\ell_0}{G} \xrightarrow{d} \text{Exp}(1), \quad p_a \log_a \frac{\ell_0}{2\Delta} \xrightarrow{d} \text{Exp}(1),$$

*since the first variable equals  $p_a N_a$  and  $p_a \log_a V \rightarrow 0$  in probability ( $\mathbb{E}(-\log V) = 1$ ); Slutsky's theorem completes the second.*

Replacing Lebesgue sampling by  $h_a(x) dx$  would couple the gap geometry to the values of the atomic function itself; the resulting weighted distance law is no longer a geometric series and is left as a further direction.

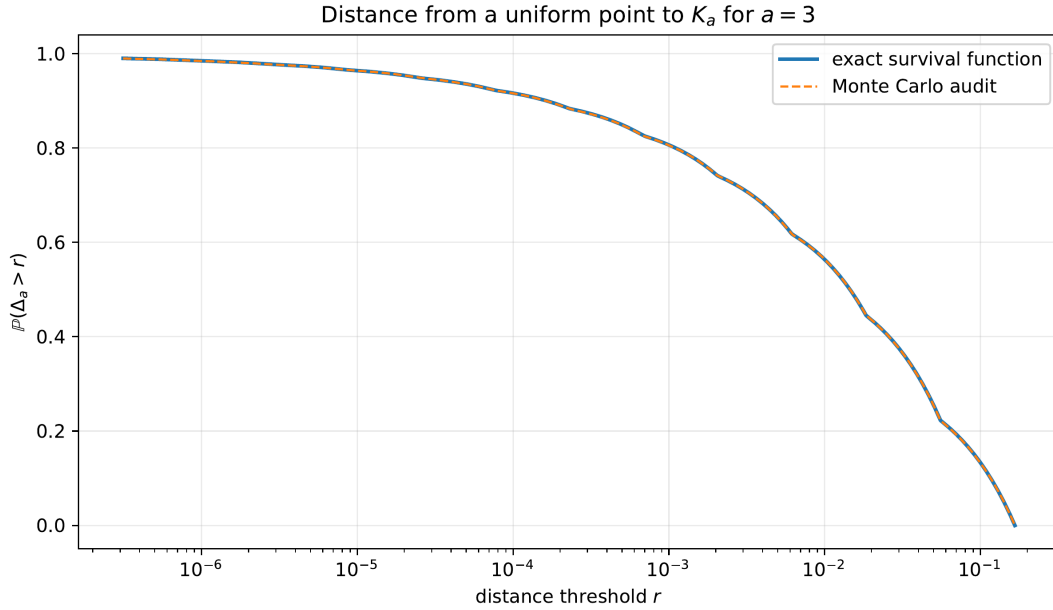


Figure 29: The distance  $\Delta$  to the nonanalytic set at  $a = 3$ : empirical law of a uniform sample against the exact tube-formula distribution (15.15).

### 15.5 Log-Weibull intermittency of the leading jets

Combining the degree law with the signed leading coefficients quantifies how violently the highest jets grow (revised fifteenth wave). Let  $A = |L_\omega|$  be the leading-coefficient magnitude of the gap polynomial at a uniform point, i.e.  $A = L_{N_a}$  with  $L_r = a^{(r+1)(r+2)/2} / (2^{r+1} r!)$  by corollary 14.4.

**Theorem 15.6** (Exact step tail and log-Weibull law). *For  $a > 2$  the levels  $L_r$  are strictly increasing ( $L_{r+1}/L_r = a^{r+2}/(2(r+1)) > 1$ ), the tail is the exact geometric staircase*

$$\mathbb{P}(A \geq L_r) = \mathbb{P}(N_a \geq r) = (2/a)^r, \quad (15.16)$$

and

$$\lim_{x \rightarrow \infty} \frac{\log \mathbb{P}(A > x)}{\sqrt{\log x}} = -\log\left(\frac{a}{2}\right) \sqrt{\frac{2}{\log a}}. \quad (15.17)$$

*Every positive moment of  $A$  diverges, while  $\mathbb{E}[\log(1+A)]^p < \infty$  for all  $p > 0$ ; conditional on  $N_a = r \geq 1$ , the sign of the leading coefficient is a fair Rademacher variable.*

*Proof.* Stirling gives  $\log L_r = \frac{\log a}{2} r^2 - r \log r + \mathcal{O}(r)$ , so the crossing index of level  $x$  is  $r(x) \sim \sqrt{2 \log x / \log a}$ ; the tail is constant between consecutive levels and equals  $(2/a)^{r(x)}$  up to one index, giving (15.17). For  $s > 0$  the summands of  $\mathbb{E} A^s$  have logarithms  $\frac{s \log a}{2} r^2 + \mathcal{O}(r \log r) \rightarrow \infty$ ;  $\log(1 + L_r) = \mathcal{O}(r^2)$  against a geometric law has all polynomial moments. Exactly half of the  $2^r$  depth- $r$  words have even  $N_+(\omega)$  for  $r \geq 1$ .  $\square$

A random point almost surely lies on a single finite polynomial piece, yet increasingly rare pieces carry superexponentially large top coefficients: spatial sparsity is exponential while amplitude growth is quadratic-exponential. This *jet intermittency* warns against judging numerical conditioning from the almost-everywhere polynomial description alone, and complements the uniformly well-conditioned *normalized derivative towers* of [appendix 13](#).

The leading limit (15.17) loses the first unbounded correction; Stirling inversion recovers it, together with the exact exponential-Orlicz threshold of the jet law (expanded fifteenth-wave editions).

**Theorem 15.7** (Two-term jet tail and sharp exponential integrability). *Let  $a > 2$  and  $\gamma_a = \log(a/2)\sqrt{2/\log a}$ . As  $x \rightarrow \infty$ ,*

$$\log \mathbb{P}(A > x) = -\gamma_a \sqrt{\log x} - \frac{\log(a/2)}{2 \log a} \log \log x + \mathcal{O}(1), \quad (15.18)$$

*so the tail is comparable, above some  $x_0$ , to  $e^{-\gamma_a \sqrt{\log x}} (\log x)^{-\log(a/2)/(2 \log a)}$ . Moreover*

$$\mathbb{E} \exp(\theta \sqrt{\log A}) < \infty \iff \theta < \gamma_a, \quad (15.19)$$

*with divergence at the boundary  $\theta = \gamma_a$ .*

*Proof.* Refining Stirling,  $\log L_r = \frac{\log a}{2} r^2 - r \log r + \mathcal{O}(r)$ , so the crossing index of level  $x$  satisfies  $r(x) = \sqrt{2 \log x / \log a} + \log \log x / (2 \log a) + \mathcal{O}(1)$ : substituting  $r = \sqrt{2y / \log a} + \delta$  into the quadratic term and matching the  $-r \log r \sim -\frac{1}{2} \sqrt{2y / \log a} \log y$  correction pins  $\delta$ . Multiplying by  $-\log(a/2)$  in the staircase  $\mathbb{P}(A > x) = (2/a)^{r(x)}$  gives (15.18). For the threshold,  $\sqrt{\log L_r} = cr - \log r / \sqrt{2 \log a} + \mathcal{O}(1)$  with  $c = \sqrt{\log a / 2}$ , so the  $r$ th summand of  $\mathbb{E} \exp(\theta \sqrt{\log A})$  has logarithm  $(\theta c - \log \frac{a}{2})r - \theta \log r / \sqrt{2 \log a} + \mathcal{O}(1)$ : it decays geometrically iff  $\theta < \log(\frac{a}{2})/c = \gamma_a$ , and at equality it is comparable to  $r^{-\log(a/2)/\log a}$  with exponent in  $(0, 1)$ , a divergent series.  $\square$

The bounded term in (15.18) cannot be replaced by a constant without resolving the staircase phase – crossing one level multiplies the tail by  $2/a$ ; the structure of the remainder’s limit set is posed in the register ([conjecture 23.17](#)).

## 15.6 The highest local jet: partial theta functions and the joint distance law

A more intrinsic observable than the leading *coefficient* is the highest nonvanishing *derivative* itself (second expanded fifteenth-wave edition). On a depth- $r$  gap,  $h_a^{(r)}$  is the constant  $r! |L_\omega|$ , so with  $Y$  uniform on  $(-b_a, b_a)$ ,

$$\mathcal{J}_a = |h_a^{(N_a)}(Y)|, \quad \mathcal{J}_a \in \{\mathcal{J}_r\}_{r \geq 0}, \quad \mathcal{J}_r = r! L_r = \frac{a^{(r+1)(r+2)/2}}{2^{r+1}},$$

and  $\log \mathcal{J}_r = \alpha r^2 + \beta r + \gamma$  is exactly quadratic, with  $\alpha = \frac{1}{2} \log a$ ,  $\beta = \frac{3}{2} \log a - \log 2$ ,  $\gamma = \log a - \log 2$ . Write  $\vartheta_p(z; \rho) = \sum_{n \geq 0} \rho^{n^2} z^n$  for the partial theta function.

**Theorem 15.8** (Staircase law, partial-theta reciprocal moments, and the joint distance transform). *For  $a > 2$ , with  $p_a = 1 - 2/a$ :*

1.  $\mathbb{P}(\mathcal{J}_a = \mathcal{J}_r) = p_a (2/a)^r$  and  $\mathbb{P}(\mathcal{J}_a \geq \mathcal{J}_r) = (2/a)^r$ ; *inverting the quadratic gives the exact staircase tail  $\mathbb{P}(\mathcal{J}_a \geq x) = (2/a)^{\lceil \nu_a(x) \rceil}$  with  $\nu_a(x) = (-\beta + \sqrt{\beta^2 + 4\alpha(\log x - \gamma)})/(2\alpha)$ , hence the same log-Weibull envelope (15.17) and the same moment dichotomy as  $A$ .*
2. *All reciprocal moments are finite and partial-theta:*

$$\mathbb{E} \mathcal{J}_a^{-s} = p_a 2^s a^{-s} \vartheta_p(2^{s+1} a^{-1-3s/2}; a^{-s/2}) \quad (s > 0). \quad (15.20)$$

3. Jointly with the distance  $\Delta$  of (15.14), for  $s > 0$  and  $\Re t > -1$ ,

$$\mathbb{E}[\mathcal{J}_a^{-s} \Delta^t] = \frac{p_a}{t+1} \left(\frac{\ell_0}{2}\right)^t 2^s a^{-s} \vartheta_p(2^{s+1} a^{-1-3s/2-t}; a^{-s/2}). \quad (15.21)$$

At  $s = 0$  the partial theta degenerates to the geometric series  $1/(1 - 2a^{-1-t})$  and (15.21) reduces to the distance-Mellin law with its pole lattice  $t = D_a - 1 + 2\pi i k / \log a$ ; for every  $s > 0$  the quadratic factor makes the transform entire in  $t$  apart from the endpoint pole  $t = -1$ .

*Proof.* (i) combines the degree law (theorem 15.3) with the signed leading coefficients; the levels increase strictly. (ii): the sum factors exactly,

$$p_a \sum_r \left(\frac{2}{a}\right)^r 2^{s(r+1)} a^{-s(r+1)(r+2)/2} = p_a 2^s a^{-s} \sum_r a^{-sr^2/2} (2^{s+1} a^{-1-3s/2})^r.$$

(iii): conditional on  $N_a = r$  the point is uniform in its gap, so  $\mathbb{E}[\Delta^t \mid N_a = r] = (\ell_0/2)^t a^{-rt}/(t+1)$ , which multiplies the  $z$ -argument by  $a^{-t}$ ; at  $s = 0$  the series is geometric with ratio  $2a^{-1-t}$ , matching theorem 15.4.  $\square$

The genuinely new point of (iii) is the coupling: reciprocal highest jets replace the fractal zeta denominator  $1 - 2a^{-1-t}$  by an entire partial theta series, and the limit  $s \downarrow 0$  is a singular perturbation in which an entire function must recreate an infinite lattice of complex poles; the conjectured saddle mechanism is posed in the register (conjecture 23.18).

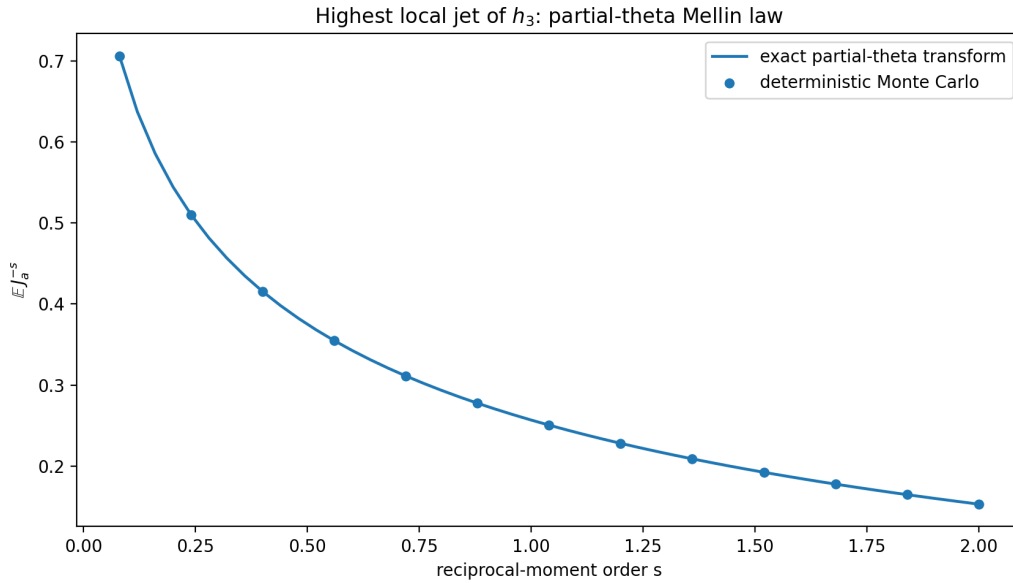


Figure 30: The exact reciprocal-moment curve (15.20) at  $a = 3$  against a fixed-seed simulation of  $1.2 \times 10^6$  uniform points.

#### [N] Repository-relative new derivation

Two periodic phenomena with the same vertical spacing  $2\pi/\log a$  occur in this part: the complex dimensions (15.4) of the gap string, producing tube oscillations in  $\log_a(1/\varepsilon)$ , and the Gamma-zeta Fourier modes of the endpoint Laplace correction in appendix 21, producing endpoint oscillations in  $\log_a u$ . They have different amplitudes and arise from different objects—one from the geometry of  $K_a$ , one from the law of  $X_a$  near its endpoint—but both are consequences of the same exact multiplicative lattice. Whether a natural spectral operator carries one profile into the other is posed in appendix 23.

## 16 Polynomial reproduction beyond the binary point

### 16.1 Zeros of the Fourier product

The zeros of  $\Phi_a$  are

$$\xi = \frac{ka^j}{2}, \quad k \in \mathbb{Z} \setminus \{0\}, \quad j \geq 1. \quad (16.1)$$

A uniform lattice of translates with spacing  $\delta$  reproduces polynomials of degree  $n$  when  $\Phi_a$  and its first  $n$  derivatives vanish at every nonzero frequency  $m/\delta$ . This is the one-dimensional Strang–Fix mechanism [25, 14]. The binary source exploits it in difference-equation form. For general  $a$ , zero collisions are controlled by rational powers of  $a$ .

**Theorem 16.1** (Rational-power Strang–Fix theorem). *Assume that  $a > 1$  and that the least positive integer  $d$  for which  $a^d$  is rational exists. Write*

$$a^d = \rho = \frac{p}{q}, \quad p, q \in \mathbb{N}_0, \quad p > q, \quad (p, q) = 1. \quad (16.2)$$

For a target degree  $n \geq 0$ , define

$$\xi_{a,n} = \frac{ap^n}{2}, \quad \delta_{a,n} = \xi_{a,n}^{-1} = \frac{2}{ap^n}. \quad (16.3)$$

Then  $\Phi_a$  has a zero of multiplicity at least  $n + 1$  at every nonzero frequency  $m/\delta_{a,n}$ . Consequently every polynomial  $P$  of degree at most  $n$  has an exact locally finite representation

$$P(x) = \sum_{k \in \mathbb{Z}} c_k(P) h_a(x - k\delta_{a,n}), \quad (16.4)$$

where

$$c_k(P) = [\delta_{a,n} M_a(\partial_x)^{-1} P]_{x=k\delta_{a,n}}. \quad (16.5)$$

Because  $P$  has finite degree and  $M_a(0) = 1$ , the inverse operator in (16.5) is a finite triangular differential operator on  $P$ .

*Proof.* For  $\ell = 0, 1, \dots, n$ , inspect the product factor with index  $j = 1 + \ell d$ . At  $\xi = \xi_{a,n}$ ,

$$2\xi_{a,n} a^{-j} = p^n a^{-\ell d} = p^{n-\ell} q^\ell \in \mathbb{N}_0.$$

Hence each of these  $n + 1$  distinct sinc factors vanishes. The same is true at  $m\xi_{a,n}$  for every nonzero integer  $m$ , proving the multiplicity claim.

Let  $r \leq n$  and periodize  $x^r h_a(x)$  with period  $\delta = \delta_{a,n}$ . Poisson summation gives Fourier coefficients proportional to derivatives of  $\Phi_a$  at  $m/\delta$ . Every nonzero Fourier coefficient vanishes by the multiplicity just proved, while the zero coefficient is the moment  $m_r(a)/\delta$ . Thus

$$\sum_{k \in \mathbb{Z}} (x - k\delta)^r h_a(x - k\delta) = \frac{m_r(a)}{\delta}, \quad 0 \leq r \leq n. \quad (16.6)$$

For a polynomial  $p$  of degree at most  $n$ , expand  $p(k\delta) = p(x - (x - k\delta))$  and use (16.6):

$$\begin{aligned} \sum_{k \in \mathbb{Z}} p(k\delta) h_a(x - k\delta) &= \frac{1}{\delta} \sum_{r=0}^n \frac{(-1)^r m_r(a)}{r!} p^{(r)}(x) \\ &= \frac{1}{\delta} M_a(-\partial_x) p(x). \end{aligned}$$

Here  $\partial_x = d/dx$  is the ordinary derivative (not the Fourier-normalized symbol  $D_F$  used in equation (4.1)). The law is symmetric, so  $M_a(-\partial_x) = M_a(\partial_x)$ . Choosing  $p = \delta M_a(\partial_x)^{-1} P$  proves the representation.  $\square$

**[N] Repository-relative new derivation**

The theorem extends the source's polynomial reproduction from the single binary point to every parameter whose multiplicative orbit meets the rationals. The result is not merely existential: the coefficient polynomial is obtained by applying the reciprocal moment-generating operator, whose coefficients are the same Bernoulli–Bell quantities derived in [appendix 10](#).

**16.2 Explicit low-degree coefficient sequences**

Let  $\delta = \delta_{a,n}$ . Since the law is centered,

$$M_a(\partial_x) = 1 + \frac{m_2(a)}{2} \partial_x^2 + \frac{m_4(a)}{24} \partial_x^4 + \dots$$

Thus the first reproducing sequences are

$$P(x) = 1 : \quad c_k = \delta, \quad (16.7)$$

$$P(x) = x : \quad c_k = \delta(k\delta), \quad (16.8)$$

$$P(x) = x^2 : \quad c_k = \delta((k\delta)^2 - m_2(a)), \quad (16.9)$$

$$P(x) = x^3 : \quad c_k = \delta((k\delta)^3 - 3m_2(a)k\delta). \quad (16.10)$$

For degree four,

$$c_k = \delta((k\delta)^4 - 6m_2(a)(k\delta)^2 + 6m_2(a)^2 - m_4(a)). \quad (16.11)$$

These are Appell-type inverse-moment polynomials.

**Example 16.2** (Integer parameters). If  $a = A \in \mathbb{N}_0$ , then  $d = 1$ ,  $p = A$ , and

$$\delta_{A,n} = \frac{2}{A^{n+1}}. \quad (16.12)$$

At  $A = 2$ , this is exactly the source lattice  $2^{-n}$ . At  $A = 3$ , quadratic reproduction uses spacing  $2/27$ , while the compact support has width one; each  $x$  therefore sees only finitely many of the coefficient terms.

**Example 16.3** (A nonrational parameter with a rational power). For  $a = \sqrt{2}$ , take  $d = 2$ ,  $p = 2$ ,  $q = 1$ . Degree  $n$  reproduction is available on the lattice

$$\delta_{\sqrt{2},n} = \frac{2}{\sqrt{2} 2^n} = 2^{1/2-n}.$$

The repeated Fourier zeros come from the factors with odd indices  $1, 3, 5, \dots, 2n + 1$ .

**16.3 What fails without rational power relations**

If no positive power of  $a$  is rational, two distinct sinc factors cannot share a nonzero zero: equality  $ka^j/2 = \ell a^r/2$  would force  $a^{j-r} = \ell/k \in \mathbb{Q}$ . Hence every nonzero zero of  $\Phi_a$  is simple.

**[C] Conjectural or programmatic statement**

**Uniform-lattice obstruction.** Suppose  $a > 1$  has no positive rational power. Under the usual local stability or tempered-growth hypotheses on the coefficient sequence, no scalar uniform lattice of shifts of  $h_a$  reproduces every affine polynomial. Equivalently, the absence of repeated Fourier zeros is not only a failure of the standard Strang–Fix construction but a genuine obstruction to all stable one-generator uniform-lattice reproduction.

A proof should formulate the minimal necessity theorem appropriate to compactly supported  $C^\infty$  generators and locally finite, polynomial-growth coefficients. Multi-generator schemes may evade the obstruction and form a separate research direction.

## 17 Fourier decay and smoothness

### 17.1 An elementary log-Gaussian envelope

Let  $t = 2\pi|\xi|$  and, for  $t \geq a$ , put

$$N = \lfloor \log_a t \rfloor.$$

For  $1 \leq j \leq N$ , the elementary estimate  $|\operatorname{sinc}_{\text{rad}}(ta^{-j})| \leq a^j/t$  gives

$$|\Phi_a(\xi)| \leq a^{N(N+1)/2} t^{-N}. \quad (17.1)$$

Consequently,

$$\log |\Phi_a(\xi)| \leq -\frac{(\log t)^2}{2 \log a} + \mathcal{O}(\log t). \quad (17.2)$$

The bound is faster than every negative power of  $|\xi|$ , so the compactly supported density is  $C^\infty$ .

*Remark 17.1.* For  $a = 2$ , the repository contains substantially sharper binary estimates and periodic refinements. The point of (17.2) is its uniformly transparent dependence on  $a$ : it already displays the quadratic logarithmic exponent and is sufficient for all fixed-order Fourier inversions used below.

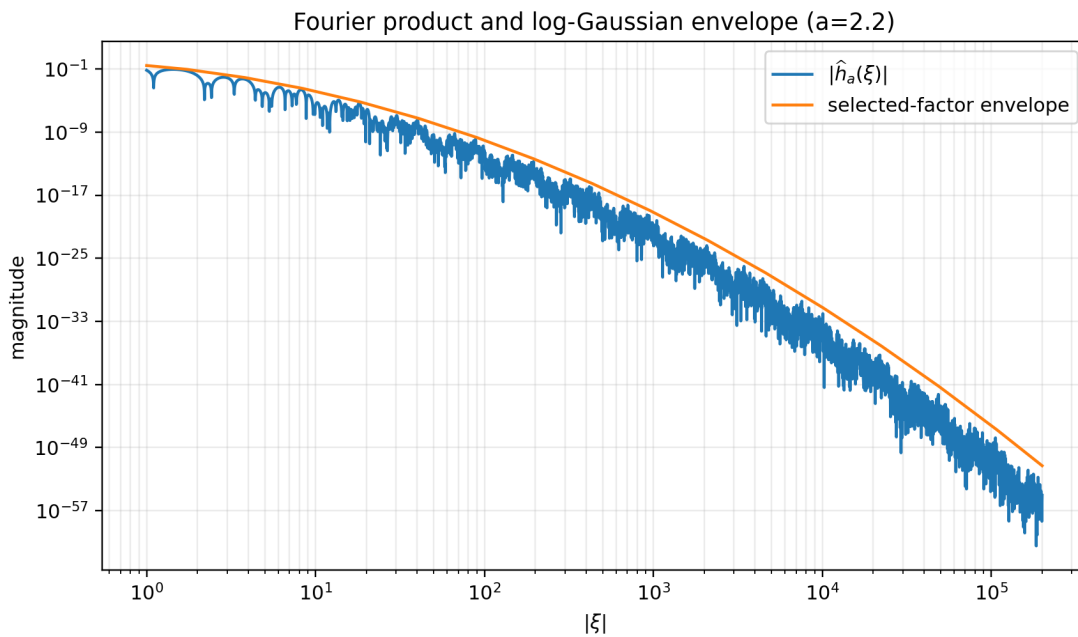


Figure 31: The Fourier product for  $a = 2.2$  and the selected-factor bound (17.1). The oscillatory zeros sit well below the envelope, while the quadratic decay in  $\log |\xi|$  is clearly visible.

### 17.2 Derivative growth from Fourier decay and from self-similarity

Fourier inversion and (17.2) imply a rough Gevrey-like estimate of the form

$$\|h_a^{(r)}\|_\infty \leq \exp\left(\frac{\log a}{2} r^2 + \mathcal{O}_a(r \log r)\right).$$

For  $a \geq 2$ , theorem 11.3 is exact and stronger. The two proofs illuminate different structures: the Fourier argument survives for all  $a > 1$ , including overlapping cylinders, whereas the exact norm depends on separation at  $a \geq 2$ .

## 18 Ordinary finite sinc products: an all-orders convergence expansion

### 18.1 Prefix splines and exact tail scaling

The finite spline  $h_{a,N}$  from (14.6) is the most literal piecewise-polynomial approximation to  $h_a$ . Its tail is exactly a scaled copy of the full law:

$$X_a = X_{a,N} + a^{-N} X'_a,$$

which gave the convolution identity (14.8). On the Fourier side,

$$\Phi_{a,N}(\xi) = \frac{\Phi_a(\xi)}{\Phi_a(a^{-N}\xi)}. \quad (18.1)$$

This identity turns approximation of  $h_a$  into a small-argument expansion of the reciprocal moment generating function.

Define

$$R_a(z) = \frac{1}{M_a(z)} = \sum_{m=0}^{\infty} r_m(a) z^{2m}. \quad (18.2)$$

By (10.4),

$$\log R_a(z) = - \sum_{m=1}^{\infty} \frac{2^{2m-1} B_{2m}}{m(2m)! (a^{2m} - 1)} z^{2m}. \quad (18.3)$$

In particular,

$$r_0(a) = 1, \quad (18.4)$$

$$r_1(a) = -\frac{1}{6(a^2 - 1)}, \quad (18.5)$$

$$r_2(a) = \frac{1}{72(a^2 - 1)^2} + \frac{1}{180(a^4 - 1)} = \frac{7a^2 + 3}{360(a^2 - 1)^2(a^2 + 1)}. \quad (18.6)$$

**Theorem 18.1** (All-orders prefix expansion). *Fix  $a > 1$ , derivative order  $r \geq 0$ , and truncation order  $M \geq 1$ . Then, as  $N \rightarrow \infty$ ,*

$$h_{a,N}^{(r)} = \sum_{m=0}^{M-1} r_m(a) a^{-2mN} h_a^{(r+2m)} + \mathcal{O}_{a,r,M}(a^{-2MN}) \quad (18.7)$$

uniformly on  $\mathbb{R}$ .

*Proof.* By evenness,  $\Phi_a(\eta) = M_a(2\pi i\eta)$ . Thus (18.1) is

$$\Phi_{a,N}(\xi) = \Phi_a(\xi) R_a(2\pi i a^{-N} \xi).$$

On  $|\xi| \leq ca^N$ , Taylor's theorem for the analytic function  $R_a$  gives a remainder bounded by  $C a^{-2MN} |\xi|^{2M}$ . Multiplication by  $(2\pi i \xi)^r \Phi_a(\xi)$  is integrable by (17.2). On the complementary frequencies, both the full product and the  $N$ -factor product have a bound of order  $\exp(-c_a N^2)$  after choosing  $c$  below the first reciprocal zero. This is smaller than  $a^{-2MN}$  for fixed  $M$ . Fourier inversion proves the uniform expansion.  $\square$

**Corollary 18.2** (First correction). *For every fixed  $r \geq 0$ ,*

$$a^{2N} (h_{a,N}^{(r)} - h_a^{(r)}) \longrightarrow -\frac{h_a^{(r+2)}}{6(a^2 - 1)} \quad (18.8)$$

uniformly on  $\mathbb{R}$ .



**Corollary 18.3** (Exact leading norm for  $a \geq 2$ ). For  $a \geq 2$ ,

$$a^{2N} \|h_{a,N} - h_a\|_\infty \longrightarrow \frac{a^6}{48(a^2 - 1)}. \quad (18.9)$$

More generally,

$$a^{2N} \|h_{a,N}^{(r)} - h_a^{(r)}\|_\infty \longrightarrow \frac{1}{6(a^2 - 1)} \frac{a^{(r+2)(r+5)/2+1}}{2^{r+3}}. \quad (18.10)$$

*Proof.* Combine [corollary 18.2](#) with the exact derivative norm [\(11.8\)](#).  $\square$

#### [N] Repository-relative new derivation

The sign of the first correction is important. The ordinary Fourier-prefix spline has leading profile  $-h_a''/[6(a^2 - 1)]$ . The current repository also studies a different finite approximation produced by repeated integration and support correction; its leading binary coefficient is positive. The two schemes should not be identified: they invert different finite operators and therefore have different asymptotic signatures.

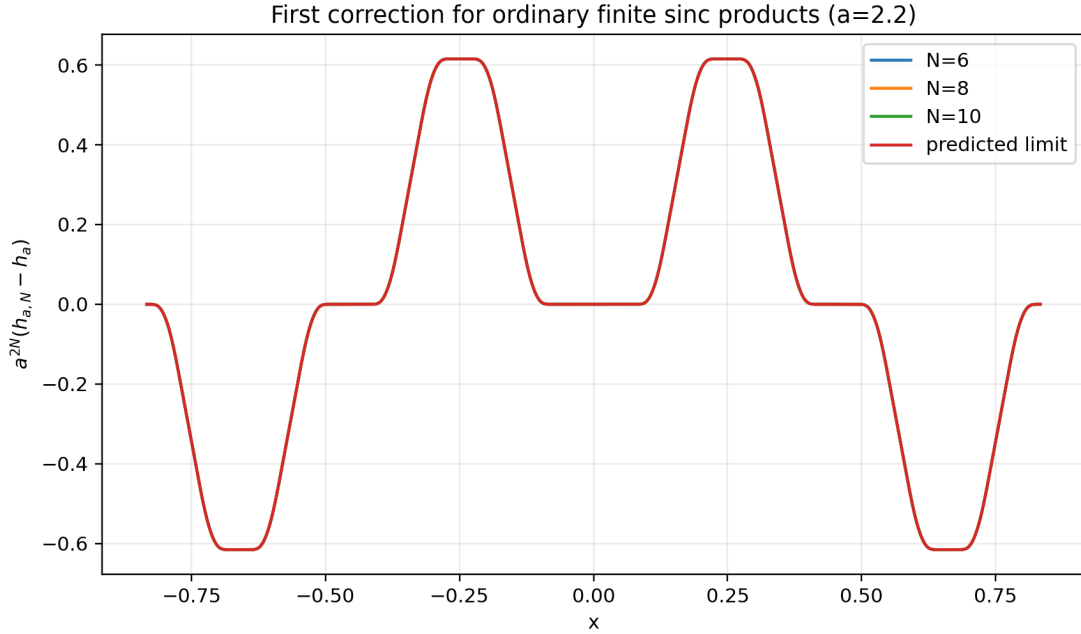


Figure 32: For  $a = 2.2$ , the scaled errors  $a^{2N}(h_{a,N} - h_a)$  for  $N = 6, 8, 10$  are visually indistinguishable from  $-h_a''/[6(a^2 - 1)]$ . The computation uses FFT inversion of the explicit products, not random sampling.

## 18.2 Interpretation as an inverse approximate identity

The convolution relation  $h_a = h_{a,N} * g_N$ , where  $g_N(x) = a^N h_a(a^N x)$ , is an approximate-identity statement in the direction  $h_{a,N} \mapsto h_a$ . Recovering  $h_{a,N}$  from  $h_a$  is therefore a formal deconvolution. The reciprocal series  $R_a$  is exactly the inverse of the tail moment operator. This explains why the coefficients are not raw moments but inverse Bell polynomials in the cumulants.

## 19 The critical deformation $a \downarrow 2$

## 19.1 A deterministic coupling bound

Use the same uniform variables in the definitions of  $X_a$  and  $X_2$ . For  $a > 2$ ,

$$|X_a - X_2| \leq \sum_{j=1}^{\infty} (2^{-j} - a^{-j}) = 1 - \frac{1}{a-1} = \frac{a-2}{a-1}. \quad (19.1)$$

Thus the  $L^\infty$  Wasserstein distance satisfies

$$W_\infty(\mathcal{L}(X_a), \mathcal{L}(X_2)) \leq \frac{a-2}{a-1}. \quad (19.2)$$

If  $H_a$  denotes the CDF of  $X_a$ , the coupling immediately gives the bracket

$$H_2(x - \eta_a) \leq H_a(x) \leq H_2(x + \eta_a), \quad \eta_a = \frac{a-2}{a-1}. \quad (19.3)$$

**Proposition 19.1** ( $C^r$  convergence to up). *For every fixed  $r \geq 0$ ,*

$$h_a \longrightarrow \text{up} \quad \text{in } C^r(\mathbb{R}) \quad (a \downarrow 2). \quad (19.4)$$

*Proof.* The Fourier products converge pointwise to the binary product. For  $a \in [2, 2 + \varepsilon]$ , the log-Gaussian bound (17.2) may be chosen with one integrable majorant after multiplication by any fixed power  $|\xi|^r$ . Dominated convergence in Fourier inversion gives the result.  $\square$

## 19.2 Collapse of the fractal spline geometry

As  $a = 2 + \varepsilon \downarrow 2$ ,

$$b_1 = \frac{a-2}{a(a-1)} = \frac{\varepsilon}{2} + \mathcal{O}(\varepsilon^2), \quad (19.5)$$

$$\dim_{\text{H}} K_a = \frac{\log 2}{\log(2 + \varepsilon)} = 1 - \frac{\varepsilon}{2 \log 2} + \mathcal{O}(\varepsilon^2). \quad (19.6)$$

The full-measure polynomial-gap set therefore degenerates in a singular way: individual gaps shrink, their exceptional complement increases in dimension to one, and the limiting up is nowhere analytic.

### [C] Conjectural or programmatic statement

**Critical boundary-layer scaling.** There is a nontrivial double-scaling limit in which  $a = 2 + \varepsilon$ , the gap depth  $r$  tends to infinity, and  $r\varepsilon$  remains bounded. After centering at an endpoint of a depth- $r$  gap and rescaling by its width  $a^{-r}b_1$ , the polynomial pieces should converge to universal profiles governed by the binary up derivative hierarchy.

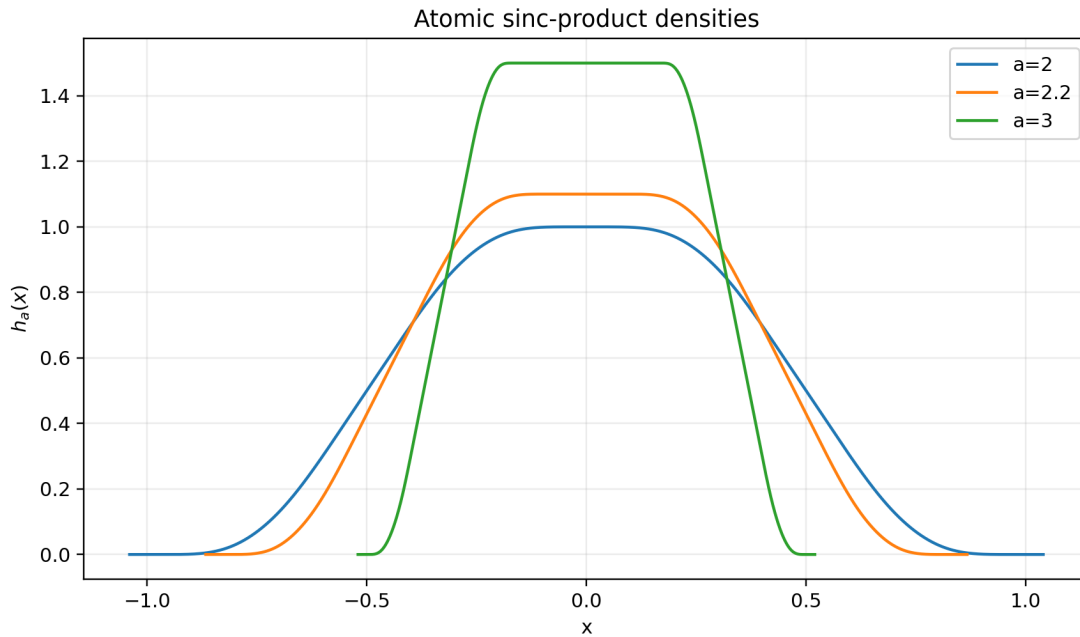


Figure 33: Profiles of  $h_a$  at the critical point  $a = 2$  and in the separated regime. The central plateau has height  $a/2$  and half-width  $(a - 2)/(a(a - 1))$  for  $a > 2$ ; it collapses to the single maximum of up at  $a = 2$ .

## 20 Parameter limits: the Gaussian and uniform ends of the family

The previous section treats the critical deformation  $a \downarrow 2$ . This section (fourteenth wave) settles the two extreme ends of the parameter range, completing the limit atlas of the family: normalized Gaussian behavior as  $a \downarrow 1$ , exact uniform behavior as  $a \rightarrow \infty$ , and quantitative rates at both ends.

### 20.1 The many-small-summands limit $a \downarrow 1$

By (10.7), the standard deviation is  $\sigma_a = (3(a^2 - 1))^{-1/2}$ ; let  $Z_a = X_a/\sigma_a$ .

**Theorem 20.1** (Gaussian limit and first characteristic correction). *As  $a \downarrow 1$ ,*

$$Z_a \xrightarrow{d} N(0, 1). \quad (20.1)$$

*The standardized cumulants are, for  $m \geq 2$ ,*

$$\lambda_{2m}(a) = \frac{\kappa_{2m}(X_a)}{\sigma_a^{2m}} = \frac{3^m 2^{2m} B_{2m} (a^2 - 1)^m}{2m (a^{2m} - 1)} = \mathcal{O}((a - 1)^{m-1}), \quad (20.2)$$

*with in particular*

$$\lambda_4(a) = -\frac{6}{5} \frac{a^2 - 1}{a^2 + 1}. \quad (20.3)$$

*Consequently, for every fixed  $T > 0$ , uniformly for real  $|t| \leq T$ ,*

$$\mathbb{E} e^{itZ_a} = e^{-t^2/2} \left[ 1 - \frac{a^2 - 1}{20(a^2 + 1)} t^4 + \mathcal{O}_T((a - 1)^2) \right]. \quad (20.4)$$

*Proof.* The triangular array  $Z_a = \sum_j \sqrt{3(a^2 - 1)} a^{-j} U_j$  has total variance one and maximal coefficient tending to zero, so the Lindeberg theorem gives (20.1). Formula (20.2) is (10.3) divided by  $\sigma_a^{2m}$ ; writing

$a = 1 + \eta$ , the ratio  $(a^2 - 1)^m / (a^{2m} - 1) \sim 2^{m-1} \eta^{m-1} / m$  gives the order. Since the log-characteristic function of  $Z_a$  is  $-t^2/2 + \lambda_4(a)t^4/4! + \mathcal{O}_T((a-1)^2)$  on  $|t| \leq T$ , and  $\lambda_4/4! = -\frac{a^2-1}{20(a^2+1)}$ , exponentiation gives (20.4).  $\square$

### [C] Conjectural or programmatic statement

**Uniform local Edgeworth expansion.** The characteristic expansion (20.4) and the superpolynomial Fourier decay of equation (17.2) make an all-orders local Edgeworth expansion for the normalized densities  $\sigma_a h_a(\sigma_a \cdot)$  and their derivatives plausible. The open difficulty is uniform control of the large-frequency range as  $a \downarrow 1$ : in normalized units the envelope (17.2) becomes effective only beyond frequencies of order  $\sigma_a \asymp (a-1)^{-1/2}$ , so the central and tail regimes must be split at a threshold that itself diverges in the joint limit.

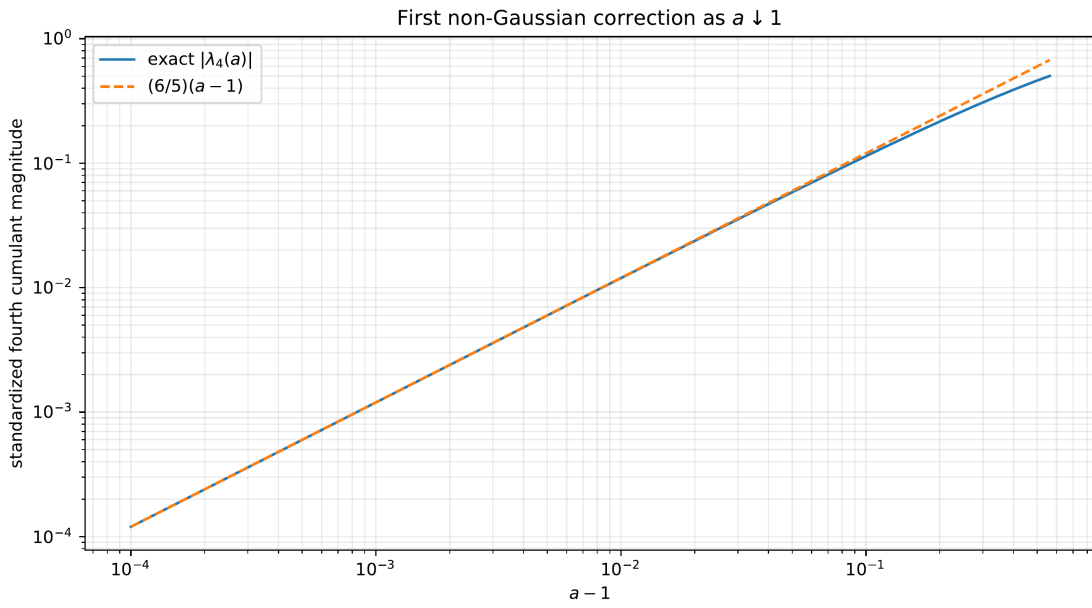


Figure 34: The exact standardized fourth cumulant (20.3) and its first asymptotic term as  $a \downarrow 1$ .

## 20.2 The single-dominant-summand limit $a \rightarrow \infty$

Rescale by the leading weight: put  $Y_a = aX_a = U_1 + T_a$  with  $T_a = \sum_{j \geq 2} a^{-(j-1)} U_j$ , so  $|T_a| \leq 1/(a-1)$  almost surely.

**Theorem 20.2** (Uniform limit with exact core). *Let  $U \sim \text{Unif}[-1, 1]$  and let  $g_a(y) = h_a(y/a)/a$  denote the density of  $Y_a$ . For every  $a > 2$ ,*

$$W_\infty(\mathcal{L}(Y_a), \mathcal{L}(U)) \leq \frac{1}{a-1}, \quad d_{\text{TV}}(\mathcal{L}(Y_a), \mathcal{L}(U)) \leq \frac{1}{2(a-1)}, \quad (20.5)$$

and the convergence is exact on an expanding central core:

$$g_a(y) = \frac{1}{2} \quad \text{for} \quad |y| \leq \frac{a-2}{a-1}. \quad (20.6)$$

All discrepancy is confined to the two boundary layers  $\frac{a-2}{a-1} < |y| \leq \frac{a}{a-1}$ , each of width  $2/(a-1)$ .

*Proof.* The coupling  $Y_a = U_1 + T_a$  gives the  $W_\infty$  bound. Conditional on  $T_a = r$ , the law of  $Y_a$  is uniform on  $[r-1, r+1]$ , whose total variation distance from  $\text{Unif}[-1, 1]$  is  $|r|/2$ ; averaging over  $r$  and

convexity of total variation give the second bound. Finally  $g_a(y) = h_a(y/a)/a$ , and the plateau (6.2) of half-width  $b_1$  becomes (20.6) because  $ab_1 = (a-2)/(a-1)$ , with plateau height  $(a/2)/a = 1/2$ .  $\square$

Together with [proposition 19.1](#), the family therefore interpolates between three classical laws with explicit rates: Gaussian after normalization at  $a \downarrow 1$ , the nowhere-analytic critical atom at  $a = 2$ , and the single uniform digit after rescaling at  $a \rightarrow \infty$ .

## 21 Endpoint tails, inverse Fabius behavior, and Lambert $W$

### 21.1 The endpoint random series

Let

$$Y_a = b_a - X_a. \quad (21.1)$$

Writing  $V_j = (1 - U_j)/2$ , with  $V_j$  uniform on  $[0, 1]$ , gives

$$Y_a = 2 \sum_{j=1}^{\infty} a^{-j} V_j. \quad (21.2)$$

Its Laplace transform is

$$L_a(s) = \mathbb{E} e^{-sY_a} = \prod_{j=1}^{\infty} \frac{1 - e^{-2sa^{-j}}}{2sa^{-j}}. \quad (21.3)$$

For large  $s$ , roughly the first  $\log_a s$  factors contribute  $a^j/(2s)$  and the remaining factors are close to one. Therefore

$$\log L_a(s) = -\frac{(\log s)^2}{2 \log a} + \mathcal{O}(\log s), \quad (21.4)$$

with an unavoidable bounded oscillation in the fractional part of  $\log_a s$ . This is the same discrete-scale mechanism that produces the periodic correction in the repository's binary endpoint asymptotics. The heuristic (21.4) is made exact, in every base, in [theorem 21.1](#) below: in the variable  $u = 2s$ , the full log-Laplace transform is a quadratic-plus-linear polynomial in  $\log u$ , plus a real-analytic one-periodic function of  $\log_a u$ , plus an explicit exponentially small tail, with no error term.

### 21.2 Why the lower Lambert branch appears

A Bromwich or Laplace saddle for the small- $y$  density balances  $sy$  against  $-\log L_a(s)$ . Differentiating the leading term gives

$$sy \asymp \frac{\log s}{\log a}. \quad (21.5)$$

If  $u = \log s$ , then  $ye^u \asymp u/\log a$ , so

$$u \asymp -W_{-1}(-y \log a). \quad (21.6)$$

The lower branch is required because  $u \rightarrow +\infty$  as  $y \downarrow 0$ ; see [\[52\]](#) for the branch structure of Lambert  $W$ . For  $a = 2$ , this is exactly the coordinate used in the repository's inverse and small-argument expansions.

### 21.3 The edge pantograph equation

The same structure is visible without any transform (fifteenth wave). Let  $g_a(u) = h_a(-b_a + u)$  for  $0 \leq u \leq 2b_a$ . For  $0 \leq u \leq 2/a$  the branch  $h_a(a(-b_a + u) - 1)$  of (5.4) lies outside the support, while  $a(-b_a + u) + 1 = -b_a + au$ , so the edge profile solves the pantograph equation

$$g'_a(u) = \frac{a^2}{2} g_a(au), \quad 0 \leq u \leq \frac{2}{a}, \quad (21.7)$$

and iterating along the flat edge gives the exact identity

$$g_a^{(n)}(u) = \frac{a^{n(n+3)/2}}{2^n} g_a(a^n u), \quad 0 \leq u \leq 2a^{-n}, \quad (21.8)$$

the endpoint case of the one-branch localization (14.13). Normalizing by  $F_a(x) = \frac{2}{a} h_a(-b_a + b_a x)$  produces a generalized Fabius profile with  $F_2 = F$  on  $[0, 1]$  and

$$F'_a(x) = \frac{a^2}{2(a-1)} F_a(ax), \quad 0 \leq x \leq \frac{2(a-1)}{a}, \quad (21.9)$$

with  $h_a$  extended by zero beyond its support; for  $a \geq \frac{3}{2}$  this range contains  $[0, 1/a]$ , on which the recursion is closed in the unit normalization, and at  $a = 2$  it specializes to the repository identity  $F'(x) = 2F(2x)$ . Pantograph equations of this type sit alongside de Rham's functional equations [20], and (21.8) already forces the quadratic-in-depth exponents whose inversion is Lambert- $W$ ; the following subsections make that heuristic exact at the transform level.

### 21.4 The exact periodic decomposition in every base

The heuristics above are now made exact (fourteenth wave). For  $u > 0$ , put

$$\varkappa(u) = \log \frac{1 - e^{-u}}{u}, \quad \Lambda_a(u) = \sum_{j=1}^{\infty} \varkappa(ua^{-j}), \quad (21.10)$$

so that  $\log L_a(s) = \Lambda_a(2s)$  by (21.3). The kernel  $\varkappa$  is exactly the kernel of the binary Mellin reflection theory of Part I (theorem 10.1); here it is summed over a single geometric orbit in an arbitrary base. Termwise,

$$\Lambda_a(au) = \Lambda_a(u) + \varkappa(u). \quad (21.11)$$

Define the exponentially convergent, nonnegative tail sum

$$E_a(u) = \sum_{n=0}^{\infty} -\log(1 - e^{-ua^n}) \geq 0, \quad u > 0, \quad (21.12)$$

and put

$$P_a(x) = \Lambda_a(a^x) + \frac{\log a}{2} x^2 - \frac{\log a}{2} x - E_a(a^x). \quad (21.13)$$

**Theorem 21.1** (General-base exact periodic decomposition). *For every  $a > 1$ , the function  $P_a$  is one-periodic, and for every  $u > 0$ ,*

$$\Lambda_a(u) = -\frac{(\log u)^2}{2 \log a} + \frac{\log u}{2} + P_a(\log_a u) + E_a(u), \quad (21.14)$$

with

$$0 \leq E_a(u) \leq \frac{e^{-u}}{(1 - e^{-u})(1 - e^{-(a-1)u})} = \mathcal{O}_a(e^{-u}) \quad (u \rightarrow \infty). \quad (21.15)$$

*Proof.* Dropping the  $n = 0$  term in (21.12) gives  $E_a(a^{x+1}) = E_a(a^x) + \log(1 - e^{-a^x})$ , while  $\varkappa(a^x) = \log(1 - e^{-a^x}) - x \log a$ . With the dilation identity (21.11),

$$P_a(x+1) - P_a(x) = \varkappa(a^x) + x \log a - (E_a(a^{x+1}) - E_a(a^x)) = 0,$$

so  $P_a$  is one-periodic. Rearranging (21.13) at  $x = \log_a u$  gives (21.14). For the tail bound, use  $-\log(1 - z) \leq z/(1 - z)$  for  $0 < z < 1$  and  $a^n \geq 1 + n(a - 1)$ , then sum the resulting geometric series.  $\square$

At  $a = 2$ , the bound (21.15) becomes  $E_2(u) \leq e^{-u}/(1 - e^{-u})^2$ , exactly the tail form in the repository's primary binary exposition; theorem 21.1 is thus a base- $a$  renormalization of the same mechanism, not an analogy.

## 21.5 Gamma–zeta Fourier modes of the periodic term

Let

$$\overline{P}_a = \int_0^1 P_a(x) \, dx, \quad \Psi_a = P_a - \overline{P}_a, \quad \chi_k = \frac{2\pi i k}{\log a}. \quad (21.16)$$

**Theorem 21.2** (General-base Gamma–zeta modes). *For every  $a > 1$  and every  $k \neq 0$ ,*

$$\widehat{\Psi}_a \text{FS}[k] = \int_0^1 \Psi_a(x) e^{-2\pi i k x} \, dx = -\frac{\Gamma(-\chi_k) \zeta(1 - \chi_k)}{\log a}. \quad (21.17)$$

*The coefficients decay exponentially in  $|k|$ , so  $P_a$  is real analytic; and every nonzero mode is nonzero, so  $P_a$  is nonconstant.*

*Proof sketch.* On the strip  $-1 < \Re s < 0$ , the kernel transform is  $\int_0^\infty \varkappa(u) u^{s-1} \, du = -\Gamma(s) \zeta(s+1)$  (theorem 10.1, there for the same kernel). The harmonic sum  $\Lambda_a(u) = \sum_{j \geq 1} \varkappa(ua^{-j})$  therefore has Mellin transform  $-\Gamma(s) \zeta(s+1) a^s / (1 - a^s)$  on the same strip, and Mellin inversion with the contour pushed across  $\Re s = 0$  produces: the quadratic and linear terms of (21.14) from the triple pole at  $s = 0$ ; the lattice modes from the simple poles of  $1/(1 - a^s)$  at  $s = \chi_k$ , whose residues sample the kernel transform at  $s = -\chi_k$  after pairing  $k \rightarrow -k$ , giving (21.17); and a remainder absorbed by  $E_a$  [49]. Stirling's bound  $|\Gamma(\sigma + it)| \asymp e^{-\pi|t|/2} |t|^{\sigma-1/2}$  gives exponential decay in  $|k|$ . Since  $\Gamma$  has no zeros and  $\zeta(1 + it) \neq 0$  for  $t \neq 0$ , every mode (21.17) is nonzero.  $\square$



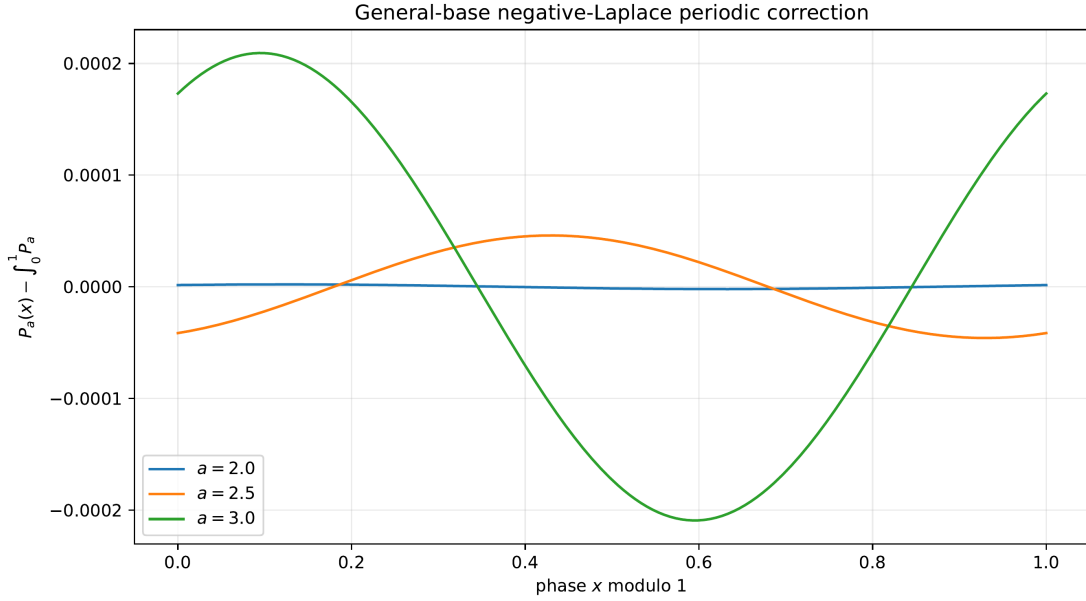


Figure 35: Centered periodic corrections  $\Psi_a = P_a - \bar{P}_a$  for several bases. The dyadic oscillation has amplitude of order  $10^{-6}$ ; the amplitude grows substantially with the base, following the mode formula (21.17).

The mean of the profile, which (21.16) introduces but does not evaluate, is the  $k = 0$  member of the same computation.

**Corollary 21.3** (Mean of the periodic profile). *For every  $a > 1$ ,*

$$\bar{P}_a = -\frac{\log a}{12} + \frac{\gamma_1 + \frac{\gamma^2}{2} - \frac{\pi^2}{12}}{\log a}, \quad (21.18)$$

where  $\gamma$  is Euler's constant and  $\gamma_1$  the first Stieltjes constant in the convention  $\zeta(1+s) = s^{-1} + \gamma - \gamma_1 s + \mathcal{O}(s^2)$ .

*Proof.* The Fourier coefficients of  $\Psi_a$  come from the simple poles of  $1/(1-a^s)$  in the proof of [theorem 21.2](#); the mean  $\bar{P}_a$  is what the triple pole at  $s = 0$  contributes beyond the quadratic and linear terms already displayed in (21.14). Expanding the two factors of the Mellin integrand at  $s = 0$ ,

$$\Gamma(s)\zeta(s+1) = \frac{1}{s^2} - \left(\gamma_1 + \frac{\gamma^2}{2} - \frac{\pi^2}{12}\right) + \mathcal{O}(s), \quad \frac{a^s}{1-a^s} = -\frac{1}{s \log a} - \frac{1}{2} - \frac{s \log a}{12} + \mathcal{O}(s^2),$$

the constant term of the product  $-\Gamma(s)\zeta(s+1)a^s/(1-a^s)$  is  $-\log a/12 + (\gamma_1 + \gamma^2/2 - \pi^2/12)/\log a$ , which is (21.18). Note that the  $1/s$  coefficient of  $\Gamma(s)\zeta(s+1)$  vanishes, which is why no  $\gamma$ -linear term survives.  $\square$

**Remark 21.4** (The same Mellin constant, two different constants). The combination  $\gamma_1 + \gamma^2/2 - \pi^2/12$  in (21.18) is the same Laurent constant of  $-\Gamma(s)\zeta(s+1)$  at  $s = 0$  that governs the  $q \uparrow 1$  large-deviation rate of Parts X and XI, where it appears as (8.2). The two constants are nevertheless *not* equal, and a reader who meets the same Greek letters twice should not assume one theorem. Here

$$(\log a) \bar{P}_a = \left(\gamma_1 + \frac{\gamma^2}{2} - \frac{\pi^2}{12}\right) - \frac{(\log a)^2}{12},$$

whereas the endpoint constant of Part X carries  $-(\log 2)^2/2$  in place of  $-(\log a)^2/12$ . The objects differ in kind as well as in coefficient: (21.18) is the mean of a periodic profile at a *fixed* base, in the

spatial variable near a support endpoint, while Part X's constant is a *parameter-limit* rate as  $q \uparrow 1$ . What they share is the kernel, not the theorem. Numerically, (21.18) agrees with direct quadrature of  $\int_0^1 P_a$  to twenty-six digits at  $a = 3/2, 2, 3, 5$ ; at  $a = 2$ ,  $\bar{P}_2 = -1.10904536543418668\dots$

#### [R] Repository baseline or crosswalk

At  $a = 2$ , the mode lattice  $\chi_k = 2\pi i k / \log 2$  is precisely the vertical lattice of the repository's binary endpoint theory, and Part I's reflection formula (10.6) is the composite-scale refinement in which an additional spectral factor  $A(2^w)$  multiplies the same  $\Gamma$ - $\zeta$  kernel transform. The theorem above is the single-scale, arbitrary-base member of that family. Note also the second appearance of the lattice  $2\pi i k / \log a$  in this part: the complex dimensions (15.4) of the gap string share the vertical spacing but not the amplitudes.

### 21.6 The exact saddle and the general-base lower-Lambert coordinate

The decomposition also pins the saddle coordinate. The exact saddle equation for the endpoint density at depth  $y$  is

$$y + 2\Lambda'_a(2s_a(y)) = 0. \quad (21.19)$$

Differentiating (21.14),

$$y s = \log_a(2s) - \frac{1}{2} - \frac{P'_a(\log_a 2s)}{\log a} - 2s E'_a(2s),$$

so, ignoring the bounded periodic derivative and the exponentially small tail, the variable  $\rho = \log_a(2s/\sqrt{a})$  satisfies

$$\rho a^{-\rho} = \frac{y\sqrt{a}}{2}, \quad (21.20)$$

whose large solution is the general-base lower-Lambert coordinate

$$r_a(y) = -\frac{1}{\log a} W_{-1}\left(-\frac{y\sqrt{a} \log a}{2}\right) \rightarrow \infty \quad (y \downarrow 0). \quad (21.21)$$

The half-unit shift hidden in  $\sqrt{a}$  is exactly the linear term  $+\frac{1}{2} \log u$  of the decomposition; at  $a = 2$ , this normalization matches the repository's binary inverse coordinate.

#### [C] Conjectural or programmatic statement

**All-orders endpoint expansion for  $h_a$  (refined).** Let  $\delta \downarrow 0$  and  $U_a(\delta) = -W_{-1}(-c_a \delta)$  for a normalization constant  $c_a > 0$ . There exist smooth 1-periodic coefficient functions  $\Psi_{a,j}$  such that

$$\log h_a(-b_a + \delta) \sim -\frac{U_a(\delta)^2}{2 \log a} + A_a U_a(\delta) + B_a \log U_a(\delta) + \sum_{j \geq 0} \frac{\Psi_{a,j}(\{U_a(\delta)/\log a\})}{U_a(\delta)^j}. \quad (21.22)$$

The periodic phase records the exact position of the saddle between neighboring geometric scales. An analogous expansion should hold for the CDF and for the inverse of the affine  $q$ -CDF. At  $a = 2$ , after normalization, it must reduce to the existing binary expansion in the repository.

Of the tasks formerly open here, the identification of the normalization  $c_a = \sqrt{a} \log a / 2$  and of the leading-order constants, and the construction of the periodic phase, are settled at the transform level by (21.21) and theorems 21.1 and 21.2: the order-one oscillation of the conjectured expansion must be  $\Psi_a$  evaluated at the saddle phase, with the saddle jets generated by the derivatives of  $P_a$ . What remains open is the density level: existence and uniqueness of the exact saddle (21.19) for small  $y$ , the steepest-descent contour with a uniform remainder, and uniformity of all constants for  $a$  in compact subsets of  $(1, \infty)$ .

## 22 Deterministic numerical experiments

### 22.1 Reproducibility design

The thirteenth-wave package is preserved in the volume directory under `assets/atomic_sinc_splines_report_package/`; the `data/` and `figures/` names used in the thirteenth-wave subsections below are relative to that directory. Its program `numerics.py` uses three independent modes of verification.

1. **Exact rational arithmetic.** SymPy evaluates Bernoulli numbers, cumulants, and moments as exact fractions. The weighted Prouhet sums are computed as Python integers.
2. **Direct Fourier inversion.** Densities and derivatives are obtained from the explicit sinc products on a zero-padded FFT grid. The computational box is wider than the compact support, so periodic copies do not overlap.
3. **Profile-level asymptotics.** The scaled function  $a^{2N}(h_{a,N} - h_a)$  is compared pointwise with  $-h_a''/[6(a^2 - 1)]$ , not merely through a single fitted scalar.

No Monte Carlo simulation is used. The exact identities remain valid independently of the numerical section.

### 22.2 Exact moment table

The following sample is generated from (10.6); the full CSV includes even moments through order twelve for  $a = 2, 11/5$ , and 3.

| $a$  | order | exact moment    |
|------|-------|-----------------|
| 2    | 2     | 1/9             |
| 2    | 4     | 19/675          |
| 2    | 6     | 583/59535       |
| 2    | 8     | 132809/32531625 |
| 11/5 | 2     | 25/288          |
| 11/5 | 4     | 33625/2018304   |
| 3    | 2     | 1/24            |
| 3    | 4     | 17/4800         |

Table 17: Selected exact moments from `data/moments.csv`.

### 22.3 Weighted Prouhet checks

For  $a = 2, 3$  and  $1 \leq n \leq 8$ , the program enumerates all  $2^n$  sign vectors and checks

$$\sum_{\epsilon} w(\epsilon) s_a(\epsilon)^m = \begin{cases} 0, & m < n, \\ 2^n n! a^{n(n-1)/2}, & m = n. \end{cases}$$

Every row in `data/prouhet_checks.csv` is marked True. For example, the largest check is

$$\sum_{\epsilon \in \{\pm 1\}^8} w(\epsilon) s_3(\epsilon)^8 = 236132421576711045120.$$

### 22.4 Prefix-spline convergence

The leading norm constants predicted by (18.9) are

$$\frac{4}{9} \quad (a = 2), \quad 0.615125347222 \dots \quad (a = 2.2), \quad \frac{243}{128} = 1.8984375 \quad (a = 3). \quad (22.1)$$

The FFT calculations give the following scaled errors.

| $a$ | $N$ | $a^{2N}\ h_{a,N} - h_a\ _\infty$ | predicted limit  |
|-----|-----|----------------------------------|------------------|
| 2   | 4   | 0.44444444444445                 | 0.44444444444444 |
| 2   | 8   | 0.444444444444598                | 0.44444444444444 |
| 2.2 | 4   | 0.6151253472224                  | 0.6151253472222  |
| 2.2 | 8   | 0.6151253473366                  | 0.6151253472222  |
| 3   | 4   | 1.898437500006                   | 1.898437500000   |
| 3   | 8   | 1.898437535656                   | 1.898437500000   |

Table 18: Selected rows from `data/prefix_errors.csv`. At very large  $N$ , double precision eventually loses relative accuracy because the unscaled error approaches machine epsilon; the moderate- $N$  rows are the meaningful validation range.

The profile residual

$$\left\| a^{2N}(h_{a,N} - h_a) + \frac{h_a''}{6(a^2 - 1)} \right\|_\infty$$

falls from approximately  $1.67 \times 10^{-2}$  at  $N = 4$  to  $1.69 \times 10^{-5}$  at  $N = 8$  for  $a = 2.2$ . This verifies the function-level statement rather than just the limiting norm.

## 22.5 Fourteenth-wave verification package

The second package (`assets/Atomic_Functions_Beyond_Dyadic_Report/`, script `experiments.py`, fixed seed 20260828) verifies the layers merged from the fourteenth wave.

- **Periodicity and decomposition.** On a grid, the residual in  $P_a(x+1) - P_a(x)$  stays below  $3 \times 10^{-14}$  for  $a \in \{1.2, 1.5, 2, 2.6, 3, 5\}$  (`data/periodicity_residuals.csv`).
- **Gamma-zeta modes.** Numerical Fourier coefficients of  $\Psi_a$  agree with (21.17) to between  $10^{-12}$  and  $10^{-16}$  (`data/periodic_fourier_validation.csv`); the larger dyadic first-mode error reflects resolving a mode of amplitude about  $10^{-6}$  with double-precision quadrature, not a failure of the identity. During the merge, the identity was re-derived independently at 30-digit precision, confirming the sign and value of (21.17) at  $a \in \{2, 3\}$ ,  $k \in \{1, 2\}$  to between  $10^{-13}$  and  $10^{-28}$ , and the exact decomposition (21.14) to  $10^{-31}$ .
- **Tube formula and degree law.** The exact tube volume (15.6) matches the direct gap sum, the normalized profile matches (15.8), and the empirical laws of  $p_a N_a$  approach  $1 - e^{-x}$  (figures 27 and 28; `data/geometry_and_norms.csv`).
- **Bell moments.** For  $a = 2.6$ , a fixed-seed simulation of 250,000 random sums confirms the exact Bell moments (10.6) through order ten at the expected Monte Carlo accuracy (`data/moment_validation_a_2_6.csv`).

## 22.6 Fifteenth-wave verification package

The third package is preserved under `assets/Rvachev_Atomic_Functions_Report/`. Its deterministic script `numerical_experiments.py` verifies the layers merged from the fifteenth wave by iterating the exact fixed-point operator  $T_a f(x) = \frac{a}{2} \int_{ax-1}^{ax+1} f$  on a support grid.

- **Fixed-point anatomy of  $h_3$ .** Mass one to  $2.3 \times 10^{-16}$ , reflection symmetry to  $1.4 \times 10^{-13}$ , and the exact plateau  $h_3 = \frac{3}{2}$  on  $[-\frac{1}{6}, \frac{1}{6}]$  (`ha_a3_diagnostics.csv`).
- **Signed leading coefficients.** Polynomial fits on every gap through generation three recover (14.10), including the magnitudes  $\frac{3}{2}$ ,  $\frac{27}{4}$ ,  $\frac{3^6}{16}$ ,  $\frac{3^{10}}{96}$  (`polynomial_gap_fit_a3.csv`, `gap_length_check_a3.csv`).

- **Degree law.** A 400,000-point uniform sample at  $a = 3$  (`local_degree_monte_carlo.txt`) gives empirical mean 2.003075 and variance 6.02305 against the exact values 2 and 6 of [theorem 15.3](#).
- **Fup<sub>n</sub> Gaussian regime.** Samples drawn directly from (8.14) match the exact standardized cumulants of [theorem 8.4](#) (`fup_clt_cumulants.csv`; [figure 19](#)).

The degree-law samples here and the Bell-moment samples above are the only Monte Carlo items in the three packages; every identity they touch is also verified exactly.

## 23 Conjectures and future research directions

### 23.1 A focused conjecture register

The following problems are sufficiently precise to support proof attempts or formalization.

**Conjecture 23.1** (No stable scalar lattice without rational powers). *If  $a^d \notin \mathbb{Q}$  for every  $d \geq 1$ , then no compactly supported scalar shift-invariant system generated by  $h_a$  on one uniform lattice reproduces all affine polynomials with polynomial-growth coefficients.*

**Conjecture 23.2** (Strict log-concavity in the overlapping regime). *For  $1 < a \leq 2$ , the function  $\log h_a$  is strictly concave on  $(-b_a, b_a)$ . Equivalently, the only flat interval in the full family occurs for  $a > 2$ , where it is the central support-separation plateau.*

**Conjecture 23.3** (Periodic Lambert expansion). *The endpoint density, CDF, and inverse CDF admit full lower-Lambert expansions of the form (21.22), with coefficient functions analytic in the phase and smooth in  $a$  on compact subintervals of  $(1, \infty)$ . The transform-level structure — the exact quadratic–linear–periodic decomposition, the Gamma–zeta modes of the phase, and the normalization  $c_a = \sqrt{a} \log a/2$  of the Lambert coordinate — is now proved ([theorems 21.1](#) and [21.2](#)); the conjecture retains the density-level saddle inversion and its uniformity in  $a$ .*

**Conjecture 23.4** (Critical double scaling). *The  $a \downarrow 2$  polynomial-gap boundary layer has a universal limit after simultaneously sending the depth to infinity and keeping  $(a - 2)r$  bounded. The first marginal is now known: by [theorem 15.3](#), the depth of a Lebesgue-typical gap, rescaled by  $(a - 2)/2$ , converges to a standard exponential law. The remaining problem is the joint limit of word address, rescaled gap geometry, and polynomial coefficients, normalized by the exact derivative amplitudes of [theorem 14.3](#).*

**Conjecture 23.5** (Overlap-regime nowhere analyticity). *For every  $1 < a \leq 2$ , the density  $h_a$  is nonanalytic at every point of the open interior of its support. The case  $a = 2$  is the source theorem. For  $1 < a < 2$  the two refinement branches overlap, so the gap-degree and one-branch localization arguments disappear. Global nonanalyticity on the full support is easy: the Fourier transform decays only quadratic-logarithmically, which no function analytic on a neighborhood of its compact support can match. The open problem is pointwise: a proof may require a localized Paley–Wiener criterion stable under the overlap refinement operator, or a transfer-operator argument, and the cases of Pisot, Salem, algebraic non-Pisot, and transcendental  $a$  may need different orbit estimates (fifteenth wave).*

**Conjecture 23.6** (Arithmetic of algebraic breakpoints). *For algebraic  $a > 2$ , the exact values of  $h_a$  and its derivatives at cylinder endpoints lie in the field generated by  $a$  and the even moments, with denominator growth controlled by the ideals generated by  $a^{2^m} - 1$ . The cumulant formula (10.3) suggests a systematic algebraic-denominator theory generalizing the dyadic rational arithmetic of Arias de Reyna and Haugland [31, 37]. The expanded editions sharpen the expected dichotomy: for algebraic  $a$  the denominators should be controlled by the algebraic norms of  $a^{2^m} - 1$ , while transcendental parameters generically admit no comparable finite-denominator normalization — the dyadic case  $a = 2$  would be the first member of a genuinely arithmetic hierarchy.*

**Conjecture 23.7** (A bridge between the two periodic profiles). *There is a natural spectral operator, built from the smoothing transform or its transfer operator, relating the tube profile (15.8) of the gap string to the negative-Laplace profile  $P_a$  of (21.13). The two profiles share the vertical lattice  $2\pi i k / \log a$  but have very different amplitudes; even a rigorous obstruction to a simple relation would clarify how the geometry of  $K_a$  and the endpoint probability of  $X_a$  decouple. The distance-Mellin law (15.14) adds a third appearance of the lattice, shifted by  $-1$ , in typical-point statistics.*

**Conjecture 23.8** (Nevanlinna–Pick coordinate of the spectral density). *For each  $a \geq 2$ , the representing measure of theorem 12.1 has a Nevanlinna–Pick parameter in the indeterminate scaled Stieltjes–Wigert problem admitting an explicit Mellin-product representation in terms of  $\widehat{h}_{a \text{ ang}}$ ; after the scaling  $z \mapsto 2z/a$  it should depend real analytically on  $a$  away from the critical endpoint, with a controlled limit as  $a \downarrow 2$ . A first step is to compute the Cauchy transform of the spectral density at negative arguments and compare it with the canonical entire functions of the Stieltjes–Wigert problem; the exact  $q$ -Pearson equation (12.2) should turn this into a multiplicative Riemann–Hilbert problem (revised fourteenth wave).*

**Conjecture 23.9** (Uniform all-orders Edgeworth expansion for  $\text{Fup}_n$ ). *The standardized density of  $\text{Fup}_n$  admits, uniformly on  $\mathbb{R}$ , an Edgeworth expansion to arbitrary order, with coefficients that are explicit polynomials in Bernoulli numbers and Hermite polynomials generated by the exact cumulants (8.18), and a remainder one order smaller than the last retained term. This is now theorem 8.5, proved uniformly on  $\mathbb{R}$  and after any fixed number of differentiations, exactly by the expected ingredients. The continuous-parameter Edgeworth program for  $a \downarrow 1$  in appendix 20 remains open, and the discrete side advances to the beyond-all-orders layer:*

**Conjecture 23.10** (Exponentially improved  $\text{Fup}_n$  asymptotics). *After optimal truncation of the algebraic expansion of theorem 8.5, the remaining error admits a transseries governed by the nearest nonreal zeros of  $\text{sinc}_{\text{rad}}(t/s_n)^n$ , with exponentially small sectors uniform through the central and moderate-deviation regimes: saddlepoint continuation, zero-induced Stokes transitions, and a uniform bridge from the central-limit scale to the compact-support edge (revised fifteenth wave).*

**Conjecture 23.11** (Graph-directed atomic splines). *For an  $m$ -branch first-order atomic equation whose affine images satisfy the open-set condition, the nonanalyticity set is the graph-directed attractor of the branch system, the complementary components carry polynomials of exact finite degree, and the dimension of the nonanalyticity set is the zero of the associated pressure determinant. The two-branch theory of appendices 14 and 15 is the simplest case; unequal contraction ratios and matrix-valued masks should produce graph-directed smooth fractal splines with the same four-layer structure (gap atlas, degree law, fractal string, tube formula).*

**Conjecture 23.12** (Asymptotic theta geometry in the overlap regime). *Let  $1 < a < 2$  and  $e_n = h_a^{(n)} / \|h_a^{(n)}\|_2$ . For each fixed pair  $(j, k)$  and parity  $\epsilon$ , the translated correlations  $\langle e_{2r+2j+\epsilon}, e_{2r+2k+\epsilon} \rangle$  possess, as  $r \rightarrow \infty$ , either a limit or a finite-dimensional log-periodic limit cycle; in the absence of a nonconstant cycle the limit is the same theta kernel  $(-1)^{j-k} a^{-(j-k)^2}$  as in the separated regime (theorem 13.1). Integration by parts shows the correlations are determined entirely by second finite differences of  $\log \|h_a^{(n)}\|_2$ , so the conjecture reduces to a sharp two-term derivative-norm asymptotic in the overlapping regime, where the closed ladder of theorem 12.1 is unavailable. Discrete dilation makes a log-periodic obstruction plausible; proving its absence or computing its phase would connect the Fourier-decay transfer operator directly to Hilbert-space conditioning (expanded fifteenth wave).*

**Conjecture 23.13** (Overlap-regime derivative-energy limit). *For  $1 < a < 2$  and every  $p > 0$ , the normalized derivative-energy measures  $\mu_{a,n,p}$  of (11.13) converge weakly to the same Bernoulli convolution  $\nu_a$ , after a suitable treatment of the overlapping derivative copies; whenever  $\nu_a$  is absolutely continuous the convergence should be quantitative in a negative Sobolev norm, but the exact  $p$ -independence and the clean entropy law (11.18) of the separated regime should acquire an overlap-pressure correction. The factorization (11.14) uses disjoint cylinders and does not extend formally below  $a = 2$ ; a transfer-operator*



formulation may identify the correct replacement and expose exceptional algebraic parameters. This is the energy-measure companion of [conjecture 23.12](#): both ask how the exactly solvable separated-regime derivative geometry degrades across the overlap threshold (second expanded fourteenth-wave edition).

**Conjecture 23.14** (Explicit spectral null modes). *For each  $a \geq 2$ , the orthogonal complement of the polynomials in  $L^2(d\rho_{\text{sp}})$  — nontrivial by the Riesz density criterion, since the spectral law is absolutely continuous and the moment problem indeterminate ([theorem 13.15](#)) — contains a nonzero bounded function with an explicit multiplicative-harmonic expansion in  $\log_a z$ : an explicit  $G_a \neq 0$  with  $\int_0^\infty z^n G_a(z) d\rho_{\text{sp}}(z) = 0$  for all  $n \geq 0$ . The  $q$ -Pearson scale equation ([12.2](#)) is the natural route; transported through the Fourier multiplier  $\widehat{h}_{a \text{ ang}}$ , such a mode would describe explicitly what the derivative tower misses inside its cyclic subspace (expanded fourteenth-wave edition).*

**Conjecture 23.15** (Infinite dual derivative tower). *For each parity tower, the canonical biorthogonal dual system guaranteed by the sharp Riesz bounds of [theorem 13.2](#) has exponentially decaying connection coefficients whose generating function is  $1/\vartheta_3(\theta/2, q)$ , and these coefficients admit a closed bilateral basic-hypergeometric representation inheriting a second-order  $q$ -difference equation from the Rogers–Szegő recurrence. Existence is not in question; what is missing is a coefficient formula comparable in simplicity to the innovation filter of [theorem 13.9](#), whose Jacobi-triple-product structure makes a bilateral answer plausible (expanded fifteenth wave).*

**Conjecture 23.16** (Finite-section theta boundary layer). *For fixed lag  $r$ , the difference between the finite innovation coefficient  $(-1)^r q^r \begin{bmatrix} n \\ r \end{bmatrix}_{q^2}$  and its stationary limit  $(-1)^r q^r / (q^2; q^2)_r$  admits a complete expansion in powers of  $q^{2(n-r+1)}$ , uniformly for  $r \leq \rho n$  for each fixed  $0 < \rho < 1$ , with a distinct error-function transition in the corner regime  $n - r = \mathcal{O}(1)$ . A single uniform formula crossing from the stationary bulk to the triangular boundary would give sharp finite-section inverse and preconditioner errors for the derivative Gram matrices (second expanded fifteenth wave).*

**Conjecture 23.17** (Centered staircase limit set). *The centered remainder  $\log \mathbb{P}(A > x) + \gamma_a \sqrt{\log x} + (\log \frac{a}{2} / (2 \log a)) \log \log x$  of [theorem 15.7](#) has a compact interval of limit points whose endpoints are explicit in the adjacent-level ratios  $L_{r+1}/L_r$ , and after logarithmic Cesàro smoothing the remainder converges to a genuine constant (expanded fifteenth wave).*

**Conjecture 23.18** (Partial-theta recreation of the complex dimensions). *In the double scaling  $s \downarrow 0$ ,  $t - (D_a - 1) = \mathcal{O}(s^{1/2})$ , the joint transform ([15.21](#)) has a universal saddle transition whose outer expansion recreates the full pole lattice  $D_a - 1 + 2\pi i k / \log a$  of the distance-Mellin law, and whose exponentially small corrections encode the neighboring complex dimensions. This is a concrete partial-theta analogue of the Lambert– $W$  endpoint program: a quadratic cutoff regularizes a geometric pole, and the lattice must re-emerge nonuniformly as the cutoff disappears (second expanded fifteenth wave).*

**Conjecture 23.19** (Graph-directed Gaussian-binomial prediction). *For separated multi-branch atomic splines, derivative towers decomposed into irreducible branch characters have block-Toeplitz Gram kernels whose finite innovations are controlled by multivariate Gaussian-binomial or Macdonald-type interpolation polynomials, and whose limiting spectral factors are matrix theta products. The two-branch parity split of [theorem 13.9](#) is the scalar rank-one case; a proof would connect the graph-directed program of the register with matrix subdivision masks and multivariate  $q$ -orthogonal polynomial theory (second expanded fifteenth wave).*

## 23.2 Promising proof programs

### 23.2.1 Symbolic dynamics and quantitative Taylor growth

The exact formula ([14.13](#)) and [theorem 14.9](#) settle the qualitative Taylor-radius classification. The next problem is quantitative: determine the limsup and liminf of  $\log |h_a^{(n)}(x)|/n^2$  from recurrence statistics of the digit shift, and classify points whose derivatives exhibit prescribed subleading growth. This



links multifractal symbolic dynamics to Denjoy–Carleman-type local regularity. The fifteenth wave sharpens the target: the Taylor-growth exponent

$$\tau(x) = \limsup_{n \rightarrow \infty} \frac{\log(|h_a^{(n)}(x)|/n!)}{n^2}$$

should have level sets whose dimensions are given by a pressure function for the two-shift coding of  $K_a$  — the one-branch formula expresses the active term as a cocycle over the full shift, so a subadditive thermodynamic formalism with bounded distortion is the natural machine.

### 23.2.2 Strang–Fix necessity and multi-generator extensions

The rational-power theorem supplies a sharp sufficient arithmetic condition. The next step is a necessity theorem under the exact coefficient class used by the source. If scalar reproduction fails, one should ask whether a finite vector of generators

$$h_a, h'_a, \dots, h_a^{(r)}$$

or several phase-shifted copies can create the missing Fourier multiplicity. This becomes a matrix-valued zero-interpolation problem.

### 23.2.3 Factor redistribution and $\text{Fup}_n$

Equation (8.8) is the first instance of moving a cosine factor out of the binary sinc product and increasing the multiplicity of a smaller-scale sinc. A systematic hierarchy should index multiplicity vectors  $\mathbf{m} = (m_1, m_2, \dots)$  with finite deviation from  $(1, 1, \dots)$ :

$$\widehat{H_{\mathbf{m}}}_{2\pi}(\xi) = \prod_{j \geq 1} \text{sinc}_{\text{rad}}(2\pi 2^{-j} \xi)^{m_j}.$$

Cosine identities then relate neighboring vectors by finite shift averages. The one-parameter subfamily obtained by sliding a single doubled factor is now constructed in closed form: the ladder  $G_n$  of [proposition 8.2](#) satisfies an exact two-translate recursion and converges in every  $C^m$  to  $2 \text{up}(2 \cdot)$ . The open questions concern the general multiplicity vectors: support, smoothness, exact dyadic values, polynomial reproduction order, and whether the resulting family forms a useful multiscale basis. A complete historical comparison also awaits a full copy of the source’s  $\text{Fup}_n$  section, which breaks off at equation (3.100).

### 23.2.4 Mellin analysis of the endpoint product

This program is now carried out at the transform level: [theorem 21.1](#) isolates the periodic phase exactly, and [theorem 21.2](#) computes its Fourier modes by sampling the kernel Mellin transform of Part I on the lattice  $-\chi_k$ . The remaining work is the saddle inversion at the density level with a uniform remainder in  $a$ , for which the binary repository’s lower-Lambert and periodic-jet technology is the model.

### 23.2.5 Formalization roadmap

A Lean development can be staged so that every analytic obligation is small and reusable.

1. Define  $X_a$  and prove the affine bridge to the existing weighted-uniform law.
2. Prove the Fourier product and the integral refinement equation.
3. Formalize the Bernoulli cumulants as an identity of formal power series, then derive the moment recurrence.

4. Prove the finite derivative formula by induction over sign vectors and identify the binary signs with `thueMorseSign`.
5. For  $a \geq 2$ , prove support-cylinder separation, the exact  $L^\infty$  and  $L^1$  derivative norms, the exact polynomial-gap degree, and the Taylor-germ classification.
6. For rational-power  $a$ , formalize the repeated-zero lemma; first prove periodized moment identities for compactly supported smooth functions, then obtain polynomial reproduction.
7. Formalize the fourteenth-wave layers with elementary proofs: the geometric zeta function and tube formula are finite/geometric-series computations over the gap string; the degree law is a length count; the uniform limit is an explicit coupling; and the exact periodic decomposition [theorem 21.1](#) is a telescoping identity of convergent sums, independent of any Mellin theory. Only the Gamma–zeta mode formula requires the strip-extended kernel transform already listed as an obligation in Part I.
8. Likewise the fifteenth-wave layers: the signed leading coefficient is the same one-branch induction that gives the degree; equimeasurability is a change of variables over disjoint cylinders; the  $\text{Fup}_n$  cumulants are finite cumulant additivity; and the triangular reconstruction is an iterated halving identity. Only the Berry–Esseen constant in the  $\text{Fup}_n$  central limit imports nonelementary input.
9. Keep the endpoint saddle, overlapping-regime strict log-concavity, and critical double-scaling claims in the frontier volume until their exact hypotheses and remainders are checked.

## 24 Conclusion

The attached chapter is more than an early account of one unusual bump function. It contains a general spectral criterion for compactly supported solutions of differential–functional equations, a probability model for an entire geometric family, a first view of fractal  $C^\infty$  splines, an exact Thue–Morse derivative calculus, and a polynomial reproduction theorem. In the current repository these themes are distributed among probability,  $q$ -calculus, exact dyadic arithmetic, Fourier decay, Bell polynomials, and Lambert- $W$  asymptotics. The parameterized treatment reunifies them.

The most robust new organizing principle is that the same geometric scale controls four different structures:

- $a^{-j}$  is the convolution width in probability;
- $a^j/2$  is the Fourier zero scale;
- signed sums of  $a^j$  are the derivative/Prouhet locations;
- signed sums of  $a^{-j}$  are the fractal gap endpoints.

Rational relations among powers of  $a$  create repeated Fourier zeros and hence polynomial reproduction. Separation at  $a > 2$  creates local polynomials and exact one-branch derivative localization. At the critical point  $a = 2$ , the gaps collapse, the signs become the classical Thue–Morse sequence, and the Fabius/Rvachev theory in the repository is recovered.

The merged fourteenth-wave layers add a fifth manifestation of the same scale: the vertical lattice  $2\pi i k / \log a$  appears twice, as the complex dimensions of the gap string governing tube oscillations and as the Gamma–zeta Fourier modes governing endpoint oscillations, while the parameter limits  $a \downarrow 1$ ,  $a \downarrow 2$ , and  $a \rightarrow \infty$  bracket the family between the Gaussian, the nowhere-analytic critical atom, and the uniform law with explicit rates at every end.

## A Concordance with the attached source

| Source range   | Subject                                                                              | Normalized location in this report                                         |
|----------------|--------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| (3.1)–(3.8)    | General atomic equation, Fourier iteration, zero matching, infinite product          | <a href="#">Equations (4.1) and (4.5)</a> and <a href="#">theorem 4.2</a>  |
| (3.9)–(3.12)   | Zero integral and derivative tower                                                   | <a href="#">Theorem 4.4</a>                                                |
| (3.13)–(3.23)  | First-order two-shift equation and normalization                                     | <a href="#">Equations (5.1) and (5.4)</a>                                  |
| (3.24)–(3.28)  | Infinite sinc product, probability law, support, self-similarity                     | <a href="#">Equations (5.5) to (5.8)</a>                                   |
| (3.29)–(3.38)  | Maclaurin recurrence and coefficient/moment decay                                    | <a href="#">Equations (5.9) and (5.10)</a>                                 |
| (3.39)–(3.59)  | Full-measure local polynomial set, Cantor-type exceptional set, endpoint evaluation  | <a href="#">Theorems 14.1, 14.3 and 14.5</a>                               |
| (3.60)–(3.64)  | up Fourier product, equation, support, derivatives, exact $L^\infty$ and $L^1$ norms | <a href="#">Equations (7.1) and (7.3)</a> and <a href="#">theorem 11.3</a> |
| (3.65)–(3.75)  | Moments, dyadic values, recursive series evaluator                                   | <a href="#">Equations (7.5) and (7.6)</a>                                  |
| (3.76)–(3.83)  | Shift sums, difference equation, characteristic roots                                | <a href="#">Equations (8.1), (8.3) and (8.4)</a>                           |
| (3.84)–(3.97)  | Uniqueness of the representing function                                              | <a href="#">Theorem 8.1</a>                                                |
| (3.98)–(3.100) | First factor redistribution and $\text{Fup}_1$                                       | <a href="#">Equations (8.7) and (8.8)</a>                                  |

The fourteenth-wave package reconstructed the same equation ranges independently under the labels (S3.1)–(S3.100) and reached identical readings, as did the fifteenth-wave package with its own repairs ledger and crosswalk table. Their OCR-reconstruction ledgers are merged and deduplicated below; the absorbed source documents themselves are archived in git history, with SHA-256 hashes in the volume’s provenance list.

### Merged OCR-reconstruction ledger

| Source location                                    | OCR/transcription issue                                                                                                    | Reconstruction adopted                                                                                                                                             |
|----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Fourier convention, (3.2)                          | mixed signs in the exponential and in $D$                                                                                  | one convention fixed globally; every mask sign and inverse transform adjusted consistently.                                                                        |
| (3.5), (3.8)                                       | the mask read variously as $g, \xi, \varepsilon, \mathcal{E}$                                                              | uniformly $\mathcal{E}$ , with the origin normalization fixing $\lambda$ .                                                                                         |
| t a - j, a - i<br>(3.31)–(3.35)                    | geometric scales flattened indices alternating among $j, \ell, k$ ; factorial bars read as absolute values (1 1 for $l!$ ) | $ta^{-j}, a^{-j}$ , from the printed superscripts. the coefficient recurrence re-derived from the one-step refinement; the source’s superfactorial bound retained. |
| C_n^2 in exponents<br>up, mixed Cyrillic<br>glyphs | damaged binomials<br>name of the atom                                                                                      | $\binom{n}{2}$ (triangular exponents).<br>up.                                                                                                                      |
| Pj(k)                                              | digit notation                                                                                                             | $p_j(k)$ , the $j$ th binary digit.                                                                                                                                |
| $\delta$ indices shifted by<br>one                 | sign sequence                                                                                                              | $\delta_k = \varepsilon_{k-1}$ , matching the printed recursion $\delta_{2k-1} = \delta_k, \delta_{2k} = -\delta_k$ .                                              |

| Source location       | OCR/transcription issue                                           | Reconstruction adopted                                                                                                                                                                                         |
|-----------------------|-------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (3.48)                | damaged parentheses and powers of $a$                             | superseded by the general derivative induction.                                                                                                                                                                |
| (3.51)–(3.58)         | binary subscripts, powers, and summation limits heavily corrupted | invariant content kept (finite moment reduction at cylinder endpoints); the fifteenth wave rebuilds the indexing in binary-encoded form ( <a href="#">remark 14.10</a> ); no literal transcription fabricated. |
| (3.59)                | denominator $!!$ read as 11                                       | restored in the gap Taylor polynomial.                                                                                                                                                                         |
| (3.63)–(3.64)         | visually ambiguous derivative exponent                            | $2^{\binom{n+1}{2}}$ by direct induction, consistent with the norm identity.                                                                                                                                   |
| (3.66)                | index mismatch in the half-moment definition                      | a single half-moment sequence with the recurrence from integration by parts.                                                                                                                                   |
| (3.73)–(3.75)         | binary-tail variables unrecoverable                               | the recursive identity and the remainder bound $\leq \text{up}(-1 + 2^{-N})$ stated instead.                                                                                                                   |
| (8.85)                | chapter-number typo                                               | read as (3.85).                                                                                                                                                                                                |
| Uniqueness hypothesis | growth denominator split and malformed                            | $2^{\binom{n+2}{2}}$ , confirmed by the counterexample $\text{up} + \alpha \text{up}'$ .                                                                                                                       |
| (3.98)                | inconsistent factor index                                         | re-derived from $\text{sinc}_{\text{rad}} z = \cos(z/2) \text{sinc}_{\text{rad}}(z/2)$ .                                                                                                                       |

## B Reproducibility and package contents

The thirteenth-wave verification material is kept in the volume directory under `assets/atomic_sinc_splines_report_package/`:

- `assets/atomic_sinc_splines_report_package/numerics.py`, with detailed comments and deterministic algorithms;
- `assets/atomic_sinc_splines_report_package/figures/`, containing all four PNG figures used in this part;
- `assets/atomic_sinc_splines_report_package/data/moments.csv`, `assets/atomic_sinc_splines_report_package/data/prouhet_checks.csv`, and `assets/atomic_sinc_splines_report_package/data/prefix_errors.csv`;
- `assets/atomic_sinc_splines_report_package/requirements.txt`, `assets/atomic_sinc_splines_report_package/README.md`, and `assets/atomic_sinc_splines_report_package/corpus_audit.txt`.

The absorbed report’s own source and PDF were deleted when it was merged; git history is their archive. From that directory, reproduce the numerical outputs with

```
python numerics.py
```

The volume itself is rebuilt from its own directory by running `pdflatex` on `geometric_q_fabius_frontiers.tex` three times, so that the table of contents and every cross-reference settle. The computations require no network access.

The fourteenth-wave archive (`assets/Atomic_Functions_Beyond_Dyadic_Report/`) contains the script `experiments.py` (NumPy, Matplotlib, mpmath; fixed seed 20260828), the five figures in vector form, the four CSV tables cited in the numerics section, `README.md`, and `requirements.txt`. From that directory,

```
python experiments.py --output .
```

regenerates every figure and table deterministically.

The fifteenth-wave archive (`assets/Rvachev_Atomic_Functions_Report/`) contains the deterministic script `numerical_experiments.py` (Python 3.10+, NumPy, SciPy, Matplotlib), the three figures, six CSV/TXT diagnostic tables, `README.md`, and `requirements.txt`; running

```
python numerical_experiments.py
```

regenerates all of them. In every archive under `assets/`, the absorbed report source itself is deleted after merging; git history and the provenance hashes are the archive of record.

## Part VII

# Signed and Reciprocal $q$ -Fabius Frontiers

### Abstract

The geometric-uniform family

$$Y_q = (1 - q) \sum_{j \geq 0} q^j U_j, \quad U_j \stackrel{\text{iid}}{\sim} \text{Unif}[0, 1],$$

contains the classical Fabius distribution at  $q = 1/2$  and, through the affine crosswalk of Part VI, the entire atomic family  $h_a$  at  $q = 1/a$ . This part studies the full signed and reciprocal parameter orbit  $\{q, -q, q^{-1}, -q^{-1}\}$ , merging two independently written sixteenth- and seventeenth-wave reports whose shared core agreed identity for identity. Three exact symmetries organize everything:  $q \mapsto -q$  is an affine sign conjugacy of centered uniform digits, so negative parameters create no new normalized shapes;  $q \mapsto q^m$  is residue-class decimation, which exhibits the Fabius law as an explicit convolution of two quarter-base laws and is the probabilistic face of  $q$ -Pochhammer dissection; and  $q \mapsto q^{-1}$  is digit reversal at the finite level and exact convolution inversion  $M_q(t)M_{1/q}(-t) = 1$  at the germ level. The centered transform is a product of  $q^2$ -Pochhammer symbols over the sine zeros, packaging the spectral zeta function and the Bernoulli cumulant dictionary  $\kappa_n = \frac{B_n}{n}(1 - q)^n/(1 - q^n)$  for  $n \geq 2$ , the mean  $\kappa_1 = \frac{1}{2}$  standing apart under the convention  $B_1 = -\frac{1}{2}$ .

The reciprocal germs at  $q = \pm 2, \pm 4$  are not probability laws on  $\mathbb{R}$ , but they are exactly the Fourier transforms of infinite convolutions of Laplace laws: their moments are moments of a positive measure on the vertical line  $\Re z = 1/2$ , their Hankel determinants have sign  $(-1)^{\binom{n}{2}}$ , and their orthogonal polynomials have all zeros on that line. The inverse of geometric averaging is a  $q$ -Fabius–Bernoulli Appell family interpolating Bernoulli polynomials ( $q = 0$ ) and centered monomials ( $q \rightarrow 1^-$ ). A moment polynomial  $\mathcal{P}_n(q) = \frac{(q; q)_n}{(1 - q)^n} \frac{m_n(q)}{n!}$  clears all poles, and the exact inverse-geometric endpoint lattice  $G_q(q^n) = q^{\binom{n+1}{2}} \mathcal{P}_n(q)/(q; q)_n$  generalizes the classical  $F(2^{-n})$  formula, with all derivative jets and a uniform two-term endpoint asymptotic  $\log G_q(x) = -\ell^2/(2L) - \ell \log \ell/L + \mathcal{O}(\ell)$  and its square-root/log-log inversion. A two-nome binary-cube partition function unifies Gaussian  $q$ -binomial coefficients and Thue–Morse Prouhet cancellation as perpendicular faces of one finite product; its renormalized maximal Prouhet jet is exactly the  $q$ -Fabius moment generating function, and its position-energy layers carry the Grassmannian/Hermitian finite-geometry square of Fu, Reiner, Stanton, and Thiem. Finite signed derivative combs interpolate between the full dyadic Thue–Morse grid at  $q = 1/2$  and a signed Cantor skeleton of Hausdorff dimension  $1/2$  at  $q = 1/4$ . The general- $q$  endpoint cocycle conjectured by the sixteenth wave is proved here by specializing Part VI’s exact periodic Laplace decomposition; the density-level all-orders Lambert- $W_{-1}$  expansion remains the principal open program.

## 1 Audit boundary, twin provenance, and the merge

## 1.1 The two sources and their agreement

This part merges two same-topic reports written independently against same-day snapshots of the repository: the sixteenth wave (*Beyond the Dyadic Fabius Web*: negative- $q$  conjugacy,  $q$ -power decimation, sinc–Pochhammer factorization, Thue–Morse desingularization, reciprocal-radix limits) and the seventeenth wave (*Signed and Reciprocal  $q$ -Fabius Frontiers*: the reciprocal moment germ and its Laplace dual, two-nome Pochhammer–Prouhet partition functions, moment polynomials, and inverse-geometric endpoint lattices). On their common core the two reports proved the same theorems by the same mechanisms — the affine sign conjugacy with dilation  $(1+q)/(1-q)$ , the residue-class multisection with weights  $q^r/[m]_q$ , the Bernoulli cumulant dictionary, and the spectral  $q^2$ -Pochhammer factorization — with every shared constant verified identical before merging. Their unique layers are complementary: the sixteenth wave contributes the finite reciprocal-radix duality by digit reversal, the Thue–Morse desingularization of finite laws, the digit-position master product with its position-parity companion, reciprocal  $q$ -Lagrange row reversal, the exact plateau phase, and general-radix Bell–Bernoulli formulas; the seventeenth wave contributes the reciprocal meromorphic germ and its convolution inversion, the Laplace/vertical-line spectral dual with Hankel signatures and orthogonal polynomials, the  $q$ -Fabius–Bernoulli Appell family, the moment polynomial  $\mathcal{P}_n$ , the two-nome partition function, the finite-geometry square, the box-spline comb geometry, and the exact endpoint lattice with its uniform asymptotics.

---

**Novelty and proof-status convention.** Novelty markers in this part are relative to the two audited repository snapshots named in the provenance list, not claims of worldwide priority. Statements carry the volume’s usual flags: **proved here** for results proved in the merged sources, **derived from audited corpus theorems** for deductions from audited corpus theorems, and **conjectural/open** for conjectures and programs. Numerical evidence never substitutes for proof; both packages are deterministic and their exact checks are summarized in the numerics section.

---

## 1.2 Roles of the letter $q$

Following the seventeenth wave, four roles of the letter  $q$  are kept separate whenever two of them coexist: the geometric ratio of the random series ( $Y_q, M_q$ ); the Gaussian-binomial or position-energy nome ( $\rho$  in  $(u; \rho)_N, \left[ \begin{smallmatrix} N \\ k \end{smallmatrix} \right]_\rho$ ); the geometric digit-projection scale ( $\sigma$  in  $A_{\sigma, N}$ ); and the finite-field cardinality ( $Q$ ). The spectral Pochhammer nome is always  $q^2$ : centering makes every factor even, and evenness forgets the sign of the geometric scale. The classical Fabius case has all of these equal to powers of  $1/2$ , which is contingent, not structural.

## 1.3 The two Klein-four orbits

The maps  $s(q) = -q$  and  $r(q) = q^{-1}$  commute and generate a Klein four-group away from  $q = 0$ . The parameter values studied throughout this part form two orbits,

$$\mathcal{O}_2 = \left\{ \frac{1}{2}, -\frac{1}{2}, 2, -2 \right\}, \quad \mathcal{O}_4 = \left\{ \frac{1}{4}, -\frac{1}{4}, 4, -4 \right\},$$

with means and (formal) variances



| $q$    | status                              | mean  | variance/formal second cumulant |
|--------|-------------------------------------|-------|---------------------------------|
| $1/2$  | Fabius probability law              | $1/2$ | $1/36$                          |
| $-1/2$ | affine Fabius law on $[-1, 2]$      | $1/2$ | $1/4$                           |
| $2$    | reciprocal moment germ              | $1/2$ | $-1/36$                         |
| $-2$   | reciprocal signed germ              | $1/2$ | $-1/4$                          |
| $1/4$  | quartic geometric probability law   | $1/2$ | $1/20$                          |
| $-1/4$ | affine quartic law on $[-1/3, 4/3]$ | $1/2$ | $5/36$                          |
| $4$    | reciprocal quartic germ             | $1/2$ | $-1/20$                         |
| $-4$   | reciprocal signed quartic germ      | $1/2$ | $-5/36$                         |

The four exterior entries are formal second cumulants, not variances of laws on  $\mathbb{R}$ ; each is exactly the negative of the variance of the reciprocal-base limit law at  $1/q$ . [Remark 3.4](#), at the end of [appendix 3](#), separates the two objects that live at  $q = \pm 2, \pm 4$  and pins down the sign that relates them.

## 2 The geometric-uniform family and its reciprocal germ

### 2.1 Definition, support, and uniqueness

Let  $(U_j)_{j \geq 0}$  be independent  $\text{Unif}[0, 1]$  variables and, for real  $|q| < 1$ ,

$$Y_q = (1 - q) \sum_{j=0}^{\infty} q^j U_j, \quad Y_q - \frac{1}{2} = (1 - q) \sum_{j=0}^{\infty} q^j V_j, \quad V_j = U_j - \frac{1}{2}. \quad (2.1)$$

The weights sum to one, so  $\mathbb{E}Y_q = 1/2$  for every real  $|q| < 1$ , and the symmetry of the  $V_j$  makes the law of  $Y_q$  symmetric about  $1/2$ , including at negative  $q$ . For  $0 \leq q < 1$  the support is  $[0, 1]$ ; at  $q = 1/2$  the CDF is the classical bounded Fabius function  $F$ . For  $q = -a$  with  $0 < a < 1$ , summing the positive and negative coefficient parts gives

$$\text{supp } Y_{-a} = \left[ -\frac{a}{1-a}, \frac{1}{1-a} \right], \quad (2.2)$$

an interval (not a Cantor set), with midpoint  $1/2$ . Separating the first digit gives the affine fixed point

$$Y_q \stackrel{d}{=} (1 - q)U + qY'_q, \quad (2.3)$$

and (sixteenth wave) the map  $\nu \mapsto \text{Law}((1 - q)U + qZ)$ ,  $Z \sim \nu$ , is a  $W_1$ -contraction with ratio  $|q|$  on laws with finite first moment, so (2.3) has  $\text{Law}(Y_q)$  as its unique finite-mean solution. We write  $G_q(x) = \mathbb{P}(Y_q \leq x)$  and  $g_q = G'_q$  in the probability regime, and  $m_n(q) = \mathbb{E}Y_q^n$ ,  $a_n(q) = m_n(q)/n!$ .

### 2.2 The universal Mahler equation and the reciprocal germ

Put  $E(z) = (e^z - 1)/z$ ,  $E(0) = 1$ . For  $|q| < 1$  the moment generating function is entire,

$$M_q(t) = \mathbb{E}e^{tY_q} = \prod_{j=0}^{\infty} E((1 - q)q^j t), \quad (2.4)$$

and splitting the factor  $j = 0$  gives the normalized Mahler equation

$$\boxed{M_q(t) = E((1 - q)t) M_q(qt)}. \quad (2.5)$$

For  $|q| > 1$  the series diverges, but (seventeenth wave) there is a canonical *reciprocal germ*

$$M_q(t) = \left[ \prod_{j=1}^{\infty} E((1 - q)q^{-j} t) \right]^{-1}, \quad (2.6)$$



meromorphic on  $\mathbb{C}$  and analytic at the origin; reindexing shows it satisfies the same equation (2.5). The unit circle, not the sign of  $q$ , separates the probability and reciprocal regimes.

**Theorem 2.1** (Uniqueness of the normalized formal solution). *Let  $q \in \mathbb{C}$  with  $q^n \neq 1$  for all  $n \geq 1$ . There is a unique formal power series  $M_q \in \mathbb{C}[[t]]$  with  $M_q(0) = 1$  satisfying (2.5); it is realized by (2.4) for  $|q| < 1$  and by (2.6) for  $|q| > 1$ . Equivalently, the exponential moment coefficients obey*

$$(1 - q^n) a_n(q) = \sum_{r=1}^n \frac{(1 - q)^r q^{n-r}}{(r+1)!} a_{n-r}(q), \quad a_0 = 1, \quad (2.7)$$

which at  $q = 1/2$  is the classical Fabius half-moment recurrence.

*Proof.* Comparing the coefficient of  $t^n$  in (2.5) gives (2.7), and  $1 - q^n \neq 0$  determines  $a_n$  recursively. The convergent products satisfy the same equation and normalization.  $\square$

Since  $E(-z) = e^{-z} E(z)$  and the weights sum to one,  $M_q(-t) = e^{-t} M_q(t)$  in every regime. The centered germ

$$C_q(t) = e^{-t/2} M_q(t), \quad C_q(-t) = C_q(t), \quad (2.8)$$

is even, so all centered odd cumulants vanish – ordinary reflection about  $1/2$  inside the disk, an exact algebraic symmetry outside it.

### 3 The Klein-four sign and inversion calculus

#### 3.1 Negative parameters are affine positive laws

**Theorem 3.1** (Affine sign conjugacy; both waves independently). *For  $0 < a < 1$ , put*

$$\lambda_a = \frac{1+a}{1-a}, \quad \beta_a = \frac{a}{1-a}.$$

Then

$$\boxed{Y_{-a} \stackrel{d}{=} \lambda_a Y_a - \beta_a, \quad Y_{-a} - \frac{1}{2} \stackrel{d}{=} \lambda_a \left(Y_a - \frac{1}{2}\right),} \quad (3.1)$$

equivalently  $M_{-a}(t) = e^{-\beta_a t} M_a(\lambda_a t)$ ,  $C_{-a}(t) = C_a(\lambda_a t)$ , and, for the centered characteristic functions (4.4),  $\psi_{-a}(t) = \psi_a(\lambda_a t)$ .

*Proof.* More is true: the equality holds almost surely under a digitwise coupling. With  $V_j = U_j - \frac{1}{2}$ , define the reflected digits

$$V_j^- = (-1)^j V_j, \quad U_j^- = \frac{1}{2} + V_j^- = \begin{cases} U_j, & j \text{ even}, \\ 1 - U_j, & j \text{ odd}, \end{cases}$$

so that  $(U_j^-)_{j \geq 0}$  is again an iid  $\text{Unif}[0, 1]$  sequence. Building  $Y_{-a}$  from it gives, almost surely,

$$Y_{-a} - \frac{1}{2} = (1+a) \sum_{j \geq 0} (-a)^j V_j^- = (1+a) \sum_{j \geq 0} a^j V_j = \lambda_a \left( (1-a) \sum_{j \geq 0} a^j V_j \right) = \lambda_a \left( Y_a - \frac{1}{2} \right),$$

the rearrangement being justified by absolute convergence. Forgetting the coupling gives the distributional identity, and the transform forms follow by applying  $\mathbb{E}e^{t \cdot}$  and  $\mathbb{E}e^{it \cdot}$ .  $\square$

**Corollary 3.2** (Complete transport). *For  $0 < a < 1$ ,*

$$G_{-a}(x) = G_a\left(\frac{x + \beta_a}{\lambda_a}\right), \quad g_{-a}(x) = \frac{1}{\lambda_a} g_a\left(\frac{x + \beta_a}{\lambda_a}\right), \quad (3.2)$$

$$g_{-a}^{(k)}(x) = \lambda_a^{-(k+1)} g_a^{(k)}\left(\frac{x + \beta_a}{\lambda_a}\right), \quad k \geq 0, \quad (3.3)$$

$$G_{-a}^{-1}(u) = \lambda_a G_a^{-1}(u) - \beta_a, \quad 0 < u < 1. \quad (3.4)$$

Consequently  $g_{-a} \in C_c^\infty(\mathbb{R})$ , with support (2.2); the  $L^p$ , Sobolev and entropy consequences of (3.3) are [corollary 3.2](#), and its Thue–Morse consequence at  $a = \frac{1}{2}$  is [corollary 4.2](#). Every interior or endpoint expansion at  $+a$  transfers exactly to  $-a$ ; there is no independent negative-parameter asymptotic problem. In particular

$$Y_{-1/2} \stackrel{d}{=} 3Y_{1/2} - 1, \quad Y_{-1/4} \stackrel{d}{=} \frac{5}{3}Y_{1/4} - \frac{1}{3} : \quad (3.5)$$

the negative half-base law is the Fabius law stretched by 3 onto  $[-1, 2]$ , with  $G_{-1/2}(x) = F\left(\frac{x+1}{3}\right)$ , quantile  $G_{-1/2}^{-1}(u) = 3F^{-1}(u) - 1$ , and density  $\frac{2}{3} \text{up}\left(\frac{2x-1}{3}\right)$  in Rvachev normalization: every exact formula for  $F^{-1}$  is an exact formula for the negative dyadic quantile. The repository’s selected total function at this parameter is the negative of that density ([corollary 4.1](#)).

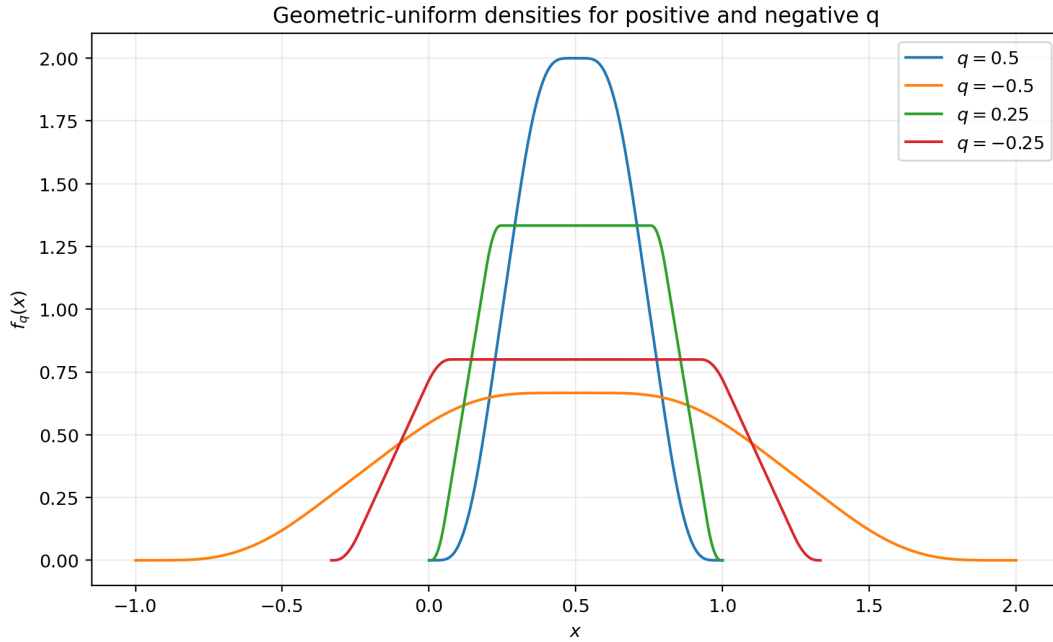


Figure 36: Physical densities for  $q = \pm\frac{1}{2}, \pm\frac{1}{4}$  (sixteenth-wave package). The negative curves are exact affine stretches of their positive partners; the quarter-base profiles carry positive-length plateaus, while the half-base plateau is critical and collapses to the center ([theorem 4.2](#)).

Standardizing removes the sign entirely (sixteenth wave): since  $\sigma_{-a} = \lambda_a \sigma_a$  by (0.4.5), the variance-normalized laws satisfy

$$\boxed{Z_{-a} \stackrel{d}{=} Z_a}, \quad (3.6)$$

and the support-normalized density profile

$$\mathcal{U}_q(x) = R_q g_q\left(\frac{1}{2} + R_q x\right), \quad R_q = \frac{1-q}{2(1-|q|)}, \quad -1 \leq x \leq 1, \quad (3.7)$$

obeys  $\mathcal{U}_{-a} = \mathcal{U}_a$ , with  $\mathcal{U}_{1/2} = \text{up}$ . All normalized shape statistics are even functions of  $q$ .

### 3.2 Reciprocal inversion and the Cayley coordinate

**Theorem 3.3** (Reciprocal convolution inversion; seventeenth wave). *For every  $q$  off the unit circle,*

$$\boxed{M_q(t) M_{1/q}(-t) = 1, \quad C_q(t) C_{1/q}(t) = 1.} \quad (3.8)$$

*At the coefficient level, the reciprocal moment row is the signed binomial-convolution inverse of the original row:  $\sum_{k=0}^n \binom{n}{k} m_k(q) (-1)^{n-k} m_{n-k}(1/q) = \delta_{n0}$ .*

*Proof.* For  $|q| < 1$ ,  $M_{1/q}(-t)^{-1} = \prod_{j \geq 1} E((1 - q^{-1})q^j(-t)) = \prod_{j \geq 1} E((1 - q)q^{j-1}t) = M_q(t)$  by (2.6); interchange  $q \leftrightarrow 1/q$  for the other case, and combine with reflection for the centered form.  $\square$

In the Cayley coordinate  $c(q) = (1 + q)/(1 - q)$  one has  $c(-q) = c(q)^{-1}$  and  $c(1/q) = -c(q)$ , and the whole calculus is the pair

$$C_{-q}(t) = C_q(c(q)t), \quad C_{1/q}(t) = C_q(t)^{-1}. \quad (3.9)$$

For the two orbits, with  $M = M_{1/2}$  and  $N = M_{1/4}$ :

$$M_{-1/2}(t) = e^{-t} M(3t), \quad M_2(t) = e^t / M(t), \quad M_{-2}(t) = e^{2t} / M(3t), \quad (3.10)$$

$$M_{-1/4}(t) = e^{-t/3} N(5t/3), \quad M_4(t) = e^t / N(t), \quad M_{-4}(t) = e^{4t/3} / N(5t/3). \quad (3.11)$$

Even centered cumulants transform by the two laws (4.2),  $\kappa_{2m}(-q) = c(q)^{2m} \kappa_{2m}(q)$  and  $\kappa_{2m}(1/q) = -\kappa_{2m}(q)$ .

### 3.3 Finite reciprocal duality by digit reversal

The germ identity has an exact finite ancestor (sixteenth wave). For  $q \notin \{0, 1\}$  with  $q^m \neq 1$ , the normalized finite law

$$Y_q^{[m]} = \alpha_{m,q} \sum_{j=0}^{m-1} q^j U_j, \quad \alpha_{m,q} = \frac{1 - q}{1 - q^m}, \quad (3.12)$$

makes sense for *every* such real  $q$ , and reversing the iid digit list proves

$$\boxed{Y_q^{[m]} \stackrel{d}{=} Y_{q^{-1}}^{[m]}} \quad \text{and} \quad Y_q^{[m]} \implies Y_{q^{-1}} \quad (m \rightarrow \infty, |q| > 1), \quad (3.13)$$

both being proved in Part VIII: the identity in the sharper pathwise form as [theorem 7.1](#), from  $[m]_{q^{-1}} = q^{1-m} [m]_q$ , and the weak limit with its four orbit specializations as [theorem 7.2](#). Thus the exterior parameters 2,  $-2$ , 4,  $-4$ , read through normalized finite sums, converge exactly to the interior laws at  $1/2$ ,  $-1/2$ ,  $1/4$ ,  $-1/4$ : values  $|q| > 1$  must not be inserted into the infinite series, but they have exact finite and limiting meanings after normalization and digit reversal. The germ-level inversion (3.8) and the finite digit reversal are the analytic and combinatorial faces of the same duality.

*Remark 3.4* (The two meanings of  $q = \pm 2, \pm 4$ ). Two different objects live at a parameter outside the unit circle, and this volume asserts something true about each. The first is the *reciprocal germ*  $M_q$  – the unique normalized solution of the Mahler equation ([theorem 2.1](#)), realized outside the disc by the inverse product (2.6) – whose cumulants are the values at  $q$  of the rational functions  $\kappa_n$  of [theorem 0.4.2](#), continued past the unit circle by [theorem 4.1](#). For  $|q| > 1$  these have  $\kappa_2(q) < 0$ , so the germ is the moment generating function of no probability law on  $\mathbb{R}$ ; it is instead the characteristic function of a positive measure carried by the vertical line  $\Re z = \frac{1}{2}$  ([theorem 7.1](#)). The second is the *reciprocal-base limit law*: the weak limit of the normalized finite laws  $Y_q^{[N]} = [N]_q^{-1} \sum_{n < N} q^n U_n$  of (3.12) (written  $X_{q,N}$  in Part VIII), which by digit reversal equals  $Y_{q^{-1}}^{[N]}$  exactly and converges to the honest compactly supported law  $Y_{1/q}$  ([theorems 7.1 and 7.2](#)). The bridge between the two is one sign,

and it is not a defect of either statement: the reciprocal-invariant finite ratio of [theorem 7.4](#) has two different limits, according to which side of the unit circle  $q$  lies on,

$$\lim_{N \rightarrow \infty} \frac{[N]_{q^s}}{[N]_q^s} = \begin{cases} \frac{(1-q)^s}{1-q^s}, & |q| < 1, \\ (-1)^{s+1} \frac{(1-q)^s}{1-q^s}, & |q| > 1, \end{cases}$$

whence

$$\kappa_{2m}(Y_{1/q}) = \lim_{N \rightarrow \infty} \kappa_{2m}(Y_q^{[N]}) = -\kappa_{2m}(q) \quad (|q| > 1).$$

So the inversion law  $\kappa_{2m}(1/q) = -\kappa_{2m}(q)$  of (4.2) and the reciprocal invariance of every finite ratio ([theorem 7.4](#)) are the *same* statement, read at  $N = \infty$  and at finite  $N$ . Concretely, the four formal second cumulants  $-\frac{1}{36}, -\frac{1}{4}, -\frac{1}{20}, -\frac{5}{36}$  of the orbit table at  $q = 2, -2, 4, -4$  are exactly  $-\text{Var}(Y_{1/2}), -\text{Var}(Y_{-1/2}), -\text{Var}(Y_{1/4}), -\text{Var}(Y_{-1/4})$ . An independent corroboration of the same sign is the Laplace dual: with  $r = 1/q$ ,  $C_q(t) = \mathbb{E}e^{itL_r}$  places the germ's moments on  $\Re z = \frac{1}{2}$ , and  $\kappa_{2m}(iL) = (-1)^m \kappa_{2m}(L)$  is precisely the  $i^{2m}$  of that vertical embedding ([theorem 7.1](#)); at  $m = 1$ ,  $\kappa_2(L_r) = \text{Var}(Y_{1/q}) = -\kappa_2(q)$ .

## 4 Cumulants, spectral zeta, and shape

### 4.1 The Bernoulli cumulant dictionary

**Theorem 4.1** (Signed/reciprocal cumulant dictionary; both waves). *The cumulant dictionary of [theorem 0.4.2](#), proved there for real  $|q| < 1$ , extends verbatim to every  $q$  with  $q^n \neq 1$  — in particular to the reciprocal regime  $|q| > 1$  — as an identity between the coefficients of the germ of [theorem 2.1](#) and rational functions of  $q$ :*

$$\kappa_1(q) = \frac{1}{2}, \quad \kappa_n(q) = \frac{B_n}{n} \frac{(1-q)^n}{1-q^n} = \frac{B_n}{n} \frac{(1-q)^{n-1}}{[n]_q} \quad (n \geq 2), \quad (4.1)$$

with  $[n]_q = 1 + q + \dots + q^{n-1}$ ; all odd cumulants above the mean vanish, and  $\kappa_2(q) = \frac{1}{12}(1-q)/(1+q)$ , which on  $|q| < 1$  is the variance (0.4.5). Along the Klein-four orbit the even cumulants obey the two symmetry laws

$$\kappa_{2m}(1/q) = -\kappa_{2m}(q), \quad \kappa_{2m}(-q) = c(q)^{2m} \kappa_{2m}(q). \quad (4.2)$$

Moments are complete Bell polynomials,  $m_n(q) = \mathcal{B}_n^{\exp}(\frac{1}{2}, \kappa_2, 0, \kappa_4, 0, \dots)$ , hence exact rationals at every rational  $q$ .

*Proof.* For real  $|q| < 1$  the dictionary is [theorem 0.4.2](#). Both sides of (4.1) are rational in  $q$  order by order and satisfy the unique recurrence (2.7) of [theorem 2.1](#), so the identity of rational functions persists wherever that recurrence determines the germ, in particular on  $|q| > 1$ , where the germ is (2.6). The two symmetries (4.2) are read off the Möbius factor  $(1-q)/(1+q)$  and the Cayley coordinate  $c(q) = (1+q)/(1-q)$ : inversion flips every even centered cumulant, sign change scales it by  $c(q)^{2m}$ .  $\square$

The standardized cumulants  $\lambda_{2m}(q) = \kappa_{2m}(Z_q)$  of (0.4.6) are even in  $q$ , which is what makes the normalized sign collapse (3.6) algebraically transparent. Outside the unit circle there is no  $Z_q$  to standardize by, since  $\sigma_q^2 = \kappa_2(q) < 0$ ; there  $\lambda_{2m}$  is read as the rational function, which by (4.2) satisfies the formal inversion law

$$\lambda_{2m}(1/q) = (-1)^{m+1} \lambda_{2m}(q), \quad (4.3)$$

so that  $\lambda_4(1/q) = -\lambda_4(q)$  while  $\lambda_2 \equiv 1$ .

---

**Crosswalk to Part VI.** Under the affine identification of [proposition 9.1](#) ( $X_a \stackrel{d}{=} b_a(2Y_q - 1)$  at  $q = 1/a$ ), the dictionary (4.1) is equivalent to the cumulant formula (10.3), and the standardized kurtosis in (0.4.6) is the  $\lambda_4$  of Part VI's Gaussian limit. Likewise the Lindeberg limit  $Z_q \Rightarrow N(0, 1)$  as  $|q| \rightarrow 1^-$ , the uniform limit  $Y_q \rightarrow U_0$  in  $L^2$  as  $q \rightarrow 0$ , the log-Gaussian Fourier envelope, and the exact plateau below are the  $Y_q$ -coordinate faces of [appendix 20, equation \(17.2\)](#), and the plateau (6.2); they are stated here once in the  $q$  normalization and not re-proved.

---

## 4.2 Spectral zeta and Fourier decay

The centered characteristic function is the sinc product

$$\psi_q(t) = \mathbb{E} e^{it(Y_q - 1/2)} = \prod_{n=0}^{\infty} \text{sinc}_{\text{rad}}\left(\frac{(1-q)q^n t}{2}\right), \quad \psi_q(t) = \text{sinc}_{\text{rad}}\left(\frac{(1-q)t}{2}\right) \psi_q(qt), \quad (4.4)$$

with positive zeros  $\tau_{n,k} = 2\pi k / ((1-q)|q|^n)$  for  $q \neq 0$ ; in the degenerate case  $q = 0$  only the scale  $n = 0$  survives, the zero list collapses to  $\tau_{0,k} = 2\pi k$ , and the closed form below — like the formalized statements of the crosswalk boxes — still holds verbatim. The indexed spectral zeta function evaluates in closed form (sixteenth wave, specializing the exponent-sequence machinery of Part I):

$$\mathcal{Z}_q(s) = \sum_{n \geq 0} \sum_{k \geq 1} \tau_{n,k}^{-s} = \left(\frac{1-q}{2\pi}\right)^s \frac{\zeta(s)}{1-|q|^s}, \quad \log \psi_q(t) = - \sum_{m \geq 1} \frac{\mathcal{Z}_q(2m)}{m} t^{2m}, \quad (4.5)$$

and  $\kappa_{2m}(q) = (-1)^{m+1} 2(2m-1)! \mathcal{Z}_q(2m)$ . The elementary cutoff argument gives the superpolynomial envelope  $\log |\psi_q(t)| \leq -(\log |t|)^2 / (2\lambda) + \mathcal{O}_q(\log |t|)$  with  $\lambda = -\log |q|$ , so for every real  $0 < |q| < 1$  the law has a compactly supported  $C^\infty$  density, all of whose derivatives vanish at both support endpoints. The excluded case  $q = 0$  (the uniform law) marks a sharp regularity transition.

## 4.3 Log-concavity and the exact plateau phase

**Theorem 4.2** (Shape of the whole real family; sixteenth wave). *For every real  $|q| < 1$  the law of  $Y_q$  is log-concave. For  $0 < |q| < 1$  its density  $g_q$  is symmetric about  $1/2$ , positive on the interior of its support, log-concave, and unimodal; at  $q = 0$  the density is  $\mathbf{1}_{[0,1]}$ , log-concave in the extended sense. For  $0 < a \leq \frac{1}{2}$ ,*

$$g_a(x) = \frac{1}{1-a} \quad \text{on } [a, 1-a], \quad g_{-a}(x) = \frac{1}{1+a} \quad \text{on } \left[\frac{a^2}{1-a}, 1 - \frac{a^2}{1-a}\right], \quad (4.6)$$

*these being the exact global maxima; for  $a < \frac{1}{2}$  the plateaus have positive length, and at  $|q| = \frac{1}{2}$  they collapse to the central maximizer. In either law the plateau carries mass  $(1-2a)/(1-a)$ .*

*Proof.* Log-concavity: the partial sums  $(1-q) \sum_{n \leq N} q^n U_n$  are convolutions of interval uniforms — a negative coefficient only reflects an interval — and so are log-concave by the Prékopa–Leindler theorem. For  $q \neq 0$  and  $N \geq 2$  the truncated transforms  $\prod_{n \leq N} \text{sinc}_{\text{rad}}((1-q)q^n t/2)$  are bounded in modulus by their first three factors, hence by the  $N$ -independent integrable envelope  $\min\{1, C_q |t|^{-3}\}$  with  $C_q = 8(1-q)^{-3} |q|^{-3}$ , and they converge pointwise to (4.4); Fourier inversion turns this into uniform convergence of the truncated densities to  $g_q$ , and a locally uniform limit of log-concave functions is log-concave. At  $q = 0$  the law is  $\text{Unif}[0, 1]$ . Plateau: the fixed point (2.3) writes  $Y_a = (1-a)U + aY'_a$  with  $aY'_a$  supported in  $[0, a]$ , so [lemma 5.2](#) applies with  $\alpha = 0$ ,  $\beta = a$ , and  $a$  there equal to  $1-a$  here, its hypothesis  $1-a \geq a$  being exactly  $a \leq \frac{1}{2}$ ; the height is  $1/(1-a)$  on  $[a, 1-a]$ . Convolution with a uniform kernel gives the matching upper bound. The negative case is the affine image under [corollary 3.2](#):  $\lambda_a a - \beta_a = a^2/(1-a)$  and the height is  $\lambda_a^{-1}(1-a)^{-1} = 1/(1+a)$ . Height times length is  $(1-2a)/(1-a)$  in both cases.  $\square$

In Part VI coordinates the positive plateau is exactly the central plateau of  $h_a$  at  $a = 1/q$ ; the phase boundary  $|q| = \frac{1}{2}$  is the critical base  $a = 2$ . The conjecture that  $\mathcal{U}_q$  is *strictly* log-concave for  $\frac{1}{2} < |q| < 1$ , and that below the transition (4.6) is the *maximal* flat interval with strictly log-concave shoulders, is the register entry already recorded in Part VI's future directions ("strict log-concavity in the overlapping regime") and is not repeated here; by [corollary 3.2](#) the negative cases are affine images of the positive ones.

## 5 Scale multisection and the half-quarter bridge

**Theorem 5.1** (Geometric multisection; both waves). *Let  $m \geq 2$  and  $[m]_q \neq 0$ ,  $q^n \neq 1$ . As analytic germs,*

$$M_q(t) = \prod_{r=0}^{m-1} M_{q^m} \left( \frac{q^r}{[m]_q} t \right), \quad (5.1)$$

and for  $|q| < 1$  this is the distributional identity

$$Y_q \stackrel{d}{=} \sum_{r=0}^{m-1} \frac{q^r}{[m]_q} Y_{q^m}^{(r)} \quad (5.2)$$

with independent summands. On the frequency axis and on cumulants this reads

$$\psi_q(t) = \prod_{r=0}^{m-1} \psi_{q^m} \left( \frac{q^r t}{[m]_q} \right), \quad \kappa_s(Y_q) = \frac{[m]_{q^s}}{[m]_q^s} \kappa_s(Y_{q^m}) \quad (s \geq 1), \quad (5.3)$$

the second being the cumulant compatibility check, i.e. the  $q$ -integer identity  $\sum_{r < m} (q^r / [m]_q)^n = [m]_{q^n} / [m]_q^n$ .

*Proof.* Split the digit index as  $j = mk + r$ ; the inner sums are independent copies of  $Y_{q^m}$  with weights  $(1 - q)q^r / (1 - q^m)$ . The reciprocal regime follows from the product definition (2.6) by the same reindexing.  $\square$

The specializations at  $m = 2$  organize the two orbits into one system:

$$Y_{1/2} \stackrel{d}{=} \frac{2}{3} Y_{1/4}^{(0)} + \frac{1}{3} Y_{1/4}^{(1)}, \quad (5.4)$$

$$Y_{-1/2} \stackrel{d}{=} 2Y_{1/4}^{(0)} - Y_{1/4}^{(1)}, \quad M_2(t) = M_4(t/3) M_4(2t/3). \quad (5.5)$$

On the frequency axis the same two lines are the sinc splittings

$$\psi_{-a}(t) = \psi_a(\lambda_a t) = \psi_{a^2} \left( \frac{t}{1-a} \right) \psi_{a^2} \left( \frac{at}{1-a} \right), \quad \psi_{1/2}(3t) = \psi_{1/4}(2t) \psi_{1/4}(t), \quad (5.6)$$

the first obtained from (5.1) at  $q = -a$ ,  $m = 2$ , where  $[2]_{-a} = 1 - a$  and  $\psi_{a^2}$  is even, combined with the sign law (3.1); the second is the case  $q = \frac{1}{2}$ ,  $m = 2$  after  $t \mapsto 3t$ , and is exactly the interlacing of the even and odd factors of the dyadic sinc product into two quarter-base products.

**Lean crosswalk: exact fixed half-quarter law.** The fixed first line [equation \(5.4\)](#) is now exact in `GeometricUniformMultisection.lean`. Its two definitions `evenCoordinates` and `oddCoordinates` split a coordinate sequence by parity. The theorem `geometricUniformSeries_one_half_multisection` proves pointwise that

$$Y_{1/2}(\omega) = \frac{2}{3}Y_{1/4}(\omega^{(0)}) + \frac{1}{3}Y_{1/4}(\omega^{(1)}).$$

The theorem `geometricUniformDistribution_one_half_multisection` proves that the two quarter-ratio processes are independent copies under the product-uniform law and identifies the corresponding product pushforward, while `geometricUniformDistribution_one_half_conv_one_quarter` gives the exact convolution of the  $2/3$ - and  $1/3$ -scaled quarter laws. These fixed identities need no user hypothesis.

The compound result in [theorem 5.1](#) remains partial: the formal API does not yet supply general  $q$ ,  $m$ , [equation \(5.1\)](#), cumulant compatibility, signed reciprocal or all- $m$  decimation, the centered-series or density formula [equation \(5.7\)](#), or its Fourier–Pochhammer,  $Z$ , and comb counterparts. In particular, a convolution of scaled measures alone does not establish the displayed density identity.

More generally  $Y_{1/2} \stackrel{d}{=} (2^m - 1)^{-1} \sum_{j < m} 2^{m-1-j} Y_{2^{-m}}^{(j)}$ : the Fabius law is a convolutional barycenter of the  $q = 2^{-m}$  family. In the unnormalized centered coordinate  $R_q = \sum_{j \geq 0} q^j V_j$  the identity is cleaner still (seventeenth wave),

$$R_q \stackrel{d}{=} \sum_{r=0}^{m-1} q^r R_{q^m}^{(r)}, \quad \text{so at } q = \frac{1}{2}, m = 2 : \quad u_{1/2} = u_{1/4} * (2u_{1/4}(2 \cdot)), \quad (5.7)$$

where  $u_q$  denotes the density of  $R_q$  ( $u_{1/2}$  is the Rvachev bump in its standard support normalization); the sixteenth wave states the same fact as  $\text{up} = \mathcal{D}_{2/3} \mathcal{U}_{1/4} * \mathcal{D}_{1/3} \mathcal{U}_{1/4}$  with  $\mathcal{D}_c$  the  $L^1$ -dilation by  $c$ . Multisection is simultaneously: an independent-sum decomposition of random digits, a convolution of densities, a factorization of Fourier products, and the elementary Pochhammer dissection  $(a; q)_\infty = \prod_{r < m} (aq^r; q^m)_\infty$  – the same identity in four coordinate systems, not four analogies.

## 6 Spectral $q^2$ -Pochhammer products and zero divisors

### 6.1 The factorization inside and outside the disk

**Theorem 6.1** (Spectral Pochhammer factorization; both waves). *For real  $|q| < 1$ ,*

$$\boxed{\psi_q(t) = \prod_{k=1}^{\infty} \left( \frac{(1-q)^2 t^2}{4\pi^2 k^2}; q^2 \right)_{\infty}}, \quad (6.1)$$

*locally uniformly in  $t \in \mathbb{C}$ ; equivalently  $M_q(t) = e^{t/2} \prod_k \left( -\frac{(1-q)^2 t^2}{4\pi^2 k^2}; q^2 \right)_{\infty}$ . For  $|q| > 1$  (seventeenth wave),*

$$M_q(t) = e^{t/2} \prod_{k=1}^{\infty} \left( -\frac{(1-q^{-1})^2 t^2}{4\pi^2 k^2}; q^{-2} \right)_{\infty}^{-1} : \quad (6.2)$$

*inversion is literally a zero–pole exchange in the same contracting nome.*

*Proof.* Insert Euler’s product  $\text{sinc}_{\text{rad}} z = \prod_k (1 - z^2/\pi^2 k^2)$  into every factor of [\(4.4\)](#); absolute convergence of the double perturbation series permits reordering, and the product over the scale index is a  $q^2$ -Pochhammer symbol. Outside the disk, apply this to  $1/C_{1/q}$ .  $\square$



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**Lean crosswalk: general complex sinc–Pochhammer factorization.** The current formal API does not prove the whole compound statement [theorem 6.1](#). The module `RvachevPochhammerFactorization.lean` contains exactly one definition, `complexQPochhammerInf`, and exactly the ten theorems `complexQPochhammerInf_eq_tprod`, `complexQPochhammerInf_eq_qPochhammerInfIn`, `multipliable_one_sub_mul_pow_complex`, `hasProd_complexQPochhammerInf`, `tendsto_finiteQPochhammerIn_complex`, `summable_norm_sineTerm_qpow_pair`, `geometricSincProduct_eq_tprod_pair`, `geometricSincProduct_eq_tprod_complexQPochhammerInf`, `rvachevFourierProduct_eq_tprod_complexQPochhammerInf`, and `rvachevFourier_eq_tprod_complexQPochhammerInf`. The first two equalities are unconditional; in particular, the second identifies the historical complex symbol with `qPochhammerInfIn`. The following three convergence results allow arbitrary complex  $a$  and require exactly  $\|q\| < 1$ . The generic `QPochhammerDissection.lean` proves the exact finite residue-class formulas through `finiteQPochhammerIn_dissection` and `finiteQPochhammerIn_dissection_remainder`. The companion `QPochhammerInfinite.lean`, with one definition `qPochhammerInfIn` and 29 theorems, proves finite-prefix convergence, finite-shift and factor-removal identities, infinite dissection, and the exact zero locus for a fixed contracting nome. Its declarations `hasProdLocallyUniformly_qPochhammerInfIn`, `differentiable_qPochhammerInfIn`, `qPochhammerInfIn_eq_zero_iff_exists_inv_pow`, `hasDerivAt_qPochhammerInfIn_inv_pow`, and `deriv_qPochhammerInfIn_inv_pow_ne_zero` give fixed- $q$  local uniformity in  $a$ , complex entire-ness, the zeros  $q^{-j}$ , and their explicit nonzero derivatives under their stated field hypotheses. The division-free theorem `deriv_qPochhammerInfIn_ne_zero_of_mul_pow_eq_one` gives actual derivative nonvanishing whenever  $aq^j = 1$ , including  $q = 0$ , and `analyticOrderAt_qPochhammerInfIn_of_eq_zero` states that every zero has analytic order exactly one, also including  $q = 0$ . The zero-definition companion `QPochhammerEntire.lean` exposes the four legacy compatibility wrappers together with the analytic-order theorem, for exactly five public theorems: `hasProdLocallyUniformly_complexQPochhammerInf`, `complexQPochhammerInf_differentiable`, `complexQPochhammerInf_eq_zero_iff`, `complexQPochhammerInf_eq_zero_iff_eq_inv_pow`, and `analyticOrderAt_complexQPochhammerInf_of_eq_zero`. For each fixed complex strict contraction  $q$ , they transfer the generic `qPochhammerInfIn` results to the older `complexQPochhammerInf` names, without duplicate analytic proofs. They expose locally uniform convergence on the whole complex  $a$ -plane and entire-ness in  $a$ , identify the zero set in the division-free form  $\exists j, 1 - aq^j = 0$ , including  $q = 0$ , and give analytic order one at every zero; for  $q \neq 0$  the fourth theorem rewrites that set as the literal reciprocal-power lattice. These symbol-level APIs do not assert joint holomorphy in  $q$  or explicit uniform-in- $q$  tail or derivative-kernel bounds. The separate outer-product leaf described below supplies the missing local-uniform convergence over the spectral index.

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**Lean crosswalk: spectral-product consequences and remaining boundary.** The next three theorems of `RvachevPochhammerFactorization.lean` give absolute summability of the paired Euler perturbations, the paired scale-zero tprod, and, for every complex  $q, z$  with  $\|q\| < 1$ , the global identity

$$S_q(z) = \prod_{k \geq 0} \left( \frac{z^2}{(k+1)^2}; q^2 \right)_{\infty}.$$

They include  $q = 0$ , negative and nonreal contractions, and all individual zero factors. The final two spectral theorems specialize this result to the dyadic scale and nome  $1/4$ ; the Fourier-transform form additionally assumes a bounded Fabius witness satisfying `IsFabius`.

After the substitution  $z = (1 - q)t/(2\pi)$ , the general product theorem gives the Pochhammer-product part of the inside formula (6.1). A separate final sinc-product tranche proves the general uncentered characteristic-function bridge. For  $q, z \in \mathbb{C}$ ,  $\|q\| < 1$ , the four theorems `hasProdLocallyUniformly_geometricSincProduct`, `geometricSincProductFactors_multiplicable`, `hasProd_geometricSincProduct`, and `geometricSincProduct_differentiable` give locally uniform convergence, pointwise `Multiplicable/HasProd`, and entire-ness for  $S_q(z) = \prod_{n \geq 0} \text{sinc}_{\text{rad}}(\pi q^n z)$ . For real  $|q| < 1$ ,  $t \in \mathbb{R}$ , and  $z_q(t) = (1 - q)t/(2\pi)$ , the declarations `charFun_geometricUniformDistribution_eq_phase_mul_geometricSincProduct` and `charFun_geometricUniformDistribution_eq_phase_mul_geometricReciprocalGamma` give

$$\begin{aligned} \phi_q(t) &= e^{it/2} S_q(z_q(t)) \\ &= e^{it/2} G_q(z_q(t)) G_q(-z_q(t)), \\ G_q(z) &= \text{geometricReciprocalGamma}(q, z), \end{aligned}$$

including  $q = 0$  and negative contractions. Theorems `tendstoLocallyUniformly_prefix_sinc_charFun` and `tendstoUniformlyOn_prefix_sinc_charFun` additionally prove local uniform convergence of the full phase-bearing prefixes on  $\mathbb{R}$  and its explicit uniform-on-every-compact-set form.

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**Lean crosswalk: the fixed-symbol and outer-product layers.** The source-only module `QPochhammerEntire.lean` now closes the exact fixed-symbol analytic layer without adding a second definition: its API has zero public definitions and five public theorems. For arbitrary  $a, q \in \mathbb{C}$  with  $\|q\| < 1$ , `hasProdLocallyUniformly_complexQPochhammerInf` proves locally uniform convergence on the whole  $a$ -plane, and `complexQPochhammerInf_differentiable` proves that  $a \mapsto (a; q)_\infty$  is entire. The division-free statement `complexQPochhammerInf_eq_zero_iff` identifies its zeros exactly with the displayed factor zeros

$$(a; q)_\infty = 0 \iff \exists j \in \mathbb{N}_0, 1 - aq^j = 0.$$

This spelling includes the degenerate nome  $q = 0$ . Under the necessary additional hypothesis  $q \neq 0$ , `complexQPochhammerInf_eq_zero_iff_eq_inv_pow` gives the literal reciprocal-power lattice  $a = (q^j)^{-1}$ ; and `analyticOrderAt_complexQPochhammerInf_of_eq_zero` proves analytic order one at every factor zero, including the sole zero  $a = 1$  when  $q = 0$ .

The zero-definition, three-theorem leaf `GeometricPochhammerNormalConvergence.lean` now closes the outer-product analytic subclaim. Its general declaration `hasProdLocallyUniformly_geometricSincProduct_complexQPochhammerInf` proves, for every complex  $q$  with  $\|q\| < 1$ , that the product over  $k$  of the displayed nome- $q^2$  symbols converges locally uniformly on the whole  $z$ -plane to  $S_q(z)$ . This includes the degenerate contraction  $q = 0$  without a division or nonzero hypothesis. The two specializations `hasProdLocallyUniformly_rvachevFourierProduct_complexQPochhammerInf` and `hasProdLocallyUniformly_rvachevFourier_complexQPochhammerInf` give the nome- $1/4$  Rvachev product and then the Fourier transform of every bounded Fabius witness satisfying `IsFabius`. After the linear substitution  $z = (1 - q)t/(2\pi)$ , the general theorem is exactly the outer locally-uniform (normal-convergence) part of the inside-disk formula (6.1).

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**Lean crosswalk: what remains a frontier obligation.** There is still no single declaration with the manuscript’s named centered  $\psi_q$  or MGF wrapper, and there is no outside- $|q| > 1$  reciprocal formula, pole divisor, or zero–pole exchange. Consequently the compound label [theorem 6.1](#) remains only partially formalized: its inside-disk pointwise factorization and outer normal convergence are exact, while the centered/MGF packaging and outside-disk reciprocal layer remain frontier obligations.

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**Lean crosswalk: latest q-calculus and theta tranche.** The final source union adds eight facade-reachable modules with five public definitions and fifty-three public theorems. Their ownership is exact: `GaussianBinomialPalindromic.lean` has zero definitions and twelve theorems, `reflect_add_of_natDegree_le`, `reflect_one'`, `gaussianBinomial_natDegree_le`, `gaussianBinomial_zero_left`, `gaussianBinomial_diag'`, `reflect_gaussianBinomial`, `coeff_gaussianBinomial_reflect`, `coeff_gaussianBinomial_zero`, `coeff_gaussianBinomial_top`, `gaussianBinomial_natDegree`, `gaussianBinomial_mononic`, and `two_mul_derivative_gaussianBinomial_eval_one`. Over a commutative semiring these give the degree bound, reflection and coefficient symmetry, constant and top coefficients, and the division-free first-moment identity; exact degree and monicity additionally require nontriviality.

`QExponential.lean` has the three definitions `qDeriv`, `qExp`, and `qExpBig`, and eight theorems: `qFactorial_mul_one_sub_pow`, `qFactorial_ne_zero`, `qDeriv_mul`, `hasSum_qExp`, `hasSum_qExpBig`, `qExp_mul_qExpBig_neg`, `qDeriv_qExp`, and `qDeriv_qExpBig`. The product rule is unconditional; the analytic statements retain the displayed strict-nome, argument-disc, and nonzero-point hypotheses. `JacksonIntegral.lean` has the definition `jacksonIntegral` and seven theorems: `qDeriv_jacksonIntegral`, `one_sub_mul_pow_mul_qDeriv`, `tendsto_jackson_sum_qDeriv`, `jacksonIntegral_qDeriv`, `tendsto_jackson_sum_parts`, `jackson_parts_of_tendsto`, and `jacksonIntegral_mul_qDeriv`. They separate partial-sum telescoping from the summable fundamental-theorem and integration-by-parts forms; no unconditional convergence is inferred.

`ThetaQuasiPeriodicity.lean` has the definition `bilateralTheta` and six theorems: `thetaExponent_add_one`, `pow_thetaExponent_add_one`, `hasSum_bilateralTheta`, `bilateralTheta_eq_prod`, `bilateralTheta_mul_left`, and `bilateralTheta_eq_zero_iff`. The last two retain  $q \neq 0$  and  $z \neq 0$ , and the zero set is exactly  $z = -q^m$  for  $m \in \mathbb{Z}$ . The zero-definition, two-theorem `JacobiCubic.lean` surface is `two_mul_add_one_le_three_pow` and `hasSum_jacobi_cubic`; the second is Jacobi's cubic identity over  $\mathbb{C}$  under exactly  $\|q\| < 1$ .

`QPochhammerLogDerivative.lean` has zero definitions and ten theorems: `one_sub_le_norm_one_sub_mul_pow`, `summable_pow_div_one_sub_mul_pow`, `summable_log_one_sub_mul_pow`, `one_sub_mul_pow_ne_zero`, `qPochhammerInfIn_eq_cexp_tsum_log`, `hasDerivAt_tsum_log_one_sub_mul_pow`, `hasDerivAt_qPochhammerInfIn`, `hasDerivAt_lambert_series`, `tsum_neg_pow_div_one_sub_mul_pow_eq`, and `hasDerivAt_qPochhammerInfIn_lambert`. Its two logarithmic-derivative forms require complex  $\|q\| < 1$  and  $\|a\| < 1$ . `QPochhammerOrderDerivative.lean` has zero definitions and three theorems, `hasDerivAt_const_cpow'`, `hasDerivAt_qPochhammerInfIn_mul_cpow`, and `hasDerivAt_qPochhammerC`; the order derivative keeps  $q \neq 0$ ,  $\|q\| < 1$ , and  $\|aq^\alpha\| < 1$ .

Finally, `GaussianBinomialPolynomialStructure.lean` remains the distinct zero-definition, five-theorem  $\mathbb{N}_0[X]$  compatibility surface: `natDegree_gaussianBinomial_universal`, `gaussianBinomial_universal_mononic`, `coeff_zero_gaussianBinomial_universal`, `gaussianBinomial_universal_reflect`, and `coeff_gaussianBinomial_universal_symm`. At this historical tranche the facade census was exactly 657 modules and 8,756 public declarations. The 238-page PDF from that stage remains a historical artifact. Current render-validation receipts are maintained externally; no package-wide checksum ledger is used.

The nome is  $q^2$ , not  $q$ : centering makes every factor even, and evenness forgets the sign of each geometric scale. Thus  $q = \pm \frac{1}{2}$  share nome  $\frac{1}{4}$  and  $q = \pm \frac{1}{4}$  share nome  $\frac{1}{16}$ , with  $\psi_{-1/2}(t) = \psi_{1/2}(3t)$  and  $\psi_{-1/4}(t) = \psi_{1/4}(5t/3)$ , while their reciprocals use the same nomes with inverse factors. Multisection of the random digits and multisection of the Pochhammer product are the same identity (sixteenth wave): dissecting (6.1) by  $(a; Q)_\infty = \prod_{j < m} (aQ^j; Q^m)_\infty$  reproduces [theorem 5.1](#) factor by factor.

## 6.2 Integer zero divisors at reciprocal-integer ratio

For the base- $b$  member it is convenient to renormalize the centered transform as

$$S_q(z) = \prod_{j=0}^{\infty} \text{sinc}_{\text{rad}}(\pi q^j z) \quad (|q| < 1), \quad (6.3)$$

so that at  $q = 1/b$  the factor  $j = 0$  vanishes exactly at the nonzero integers. This is the geometric `cSincProduct` of the crosswalk boxes above, and is written  $S_q$  throughout: the letter  $\Phi$  is reserved in this volume for the generalized Rvachev transform and, in Part XII, for a characteristic function (table 1).

**Proposition 6.2** (Base- $b$  zero divisor; seventeenth wave). *Let  $q = 1/b$  with integer  $b \geq 2$ . At every nonzero integer  $n$ ,*

$$\text{ord}_{z=n} S_{1/b}(z) = 1 + \nu_b(n), \quad \nu_b(n) = \max\{j \geq 0 : b^j \mid n\}. \quad (6.4)$$

*In particular  $\text{ord}_{z=n} S_{1/4}(z) = 1 + \lfloor \nu_2(n)/2 \rfloor$ : the quartic transform sees binary valuation levels in pairs, the spectral counterpart of even/odd scale decimation.*

*Proof.* The factor  $\text{sinc}_{\text{rad}}(\pi z/b^j)$  vanishes simply at  $z = n \neq 0$  exactly when  $b^j \mid n$ , that is, for  $0 \leq j \leq \nu_b(n)$ ; all other factors are nonzero there.  $\square$

At  $b = 2$  this is the classical multiplicity  $1 + \nu_2(n)$  of the dyadic corpus; the general- $b$  statement is the single-scale case of Part I's exponent-sequence zero calculus.

## 7 Reciprocal Laplace duality and vertical moment geometry

The formal variance  $-\kappa_2(1/q)$  of a reciprocal germ is negative, so  $M_q$  with  $q > 1$  is the MGF of no real probability law. The seventeenth wave resolves this positively: the obstruction disappears on a vertical line.

**Theorem 7.1** (Positive Laplace representation; seventeenth wave). *Let  $|q| > 1$ ,  $r = 1/q$ , and let  $Z_{j,k}$  ( $j \geq 0, k \geq 1$ ) be independent standard symmetric Laplace variables (density  $\frac{1}{2}e^{-|x|}$ ). With scales  $b_{j,k}(r) = |1 - r| |r|^j / (2\pi k)$ , the double series*

$$L_r = \sum_{j \geq 0} \sum_{k \geq 1} b_{j,k}(r) Z_{j,k} \quad (7.1)$$

*converges almost surely and in  $L^2$ , and*

$$\boxed{C_q(t) = \mathbb{E} e^{itL_r}, \quad m_n(q) = \mathbb{E} \left( \frac{1}{2} + iL_r \right)^n.} \quad (7.2)$$

*The reciprocal moments are ordinary polynomial moments of a positive measure on the vertical line  $\Re z = \frac{1}{2}$ .*

*Proof.* By (6.1) and reciprocity,  $C_q(t) = 1/C_r(t) = \prod_{j,k} (1 + b_{j,k}(r)^2 t^2)^{-1}$ , the characteristic function of the displayed Laplace convolution;  $\sum b_{j,k}^2 = (1 - r)^2 \zeta(2) / (4\pi^2(1 - r^2)) < \infty$  gives Kolmogorov convergence, and  $M_q(t) = e^{t/2} C_q(t) = \mathbb{E} \exp(t(\frac{1}{2} + iL_r))$  near 0.  $\square$

The Laplace cumulants close the circle through Euler's evaluation: a Laplace variable of scale  $b$  has even cumulants  $(2m)! b^{2m}/m$ , so  $\kappa_{2m}(L_r)$  is a  $\zeta(2m)$ -sum over the double lattice, and  $B_{2m} = (-1)^{m+1} 2(2m)! \zeta(2m) / (2\pi)^{2m}$  turns it into  $\kappa_{2m}(q) = (-1)^m \kappa_{2m}(L_r)$  — exactly the  $i^{2m}$  of the vertical embedding.

**Theorem 7.2** (Hankel signature; seventeenth wave). *For real  $|q| > 1$  and every  $n \geq 1$ , the moment Hankel determinant  $\Delta_n(q) = \det(m_{i+j}(q))_{0 \leq i,j < n}$  satisfies*

$$\boxed{\operatorname{sgn} \Delta_n(q) = (-1)^{\binom{n}{2}},} \quad (7.3)$$

and no  $\Delta_n(q)$  vanishes.

*Proof.* The real Hankel determinant  $H_n$  of  $L_r$  is strictly positive (its law has full support, so  $1, x, \dots, x^{n-1}$  are independent in  $L^2$ ). Translating a moment sequence is a unit-triangular change of basis, and scaling by  $a$  multiplies the  $n$ th determinant by  $a^{n(n-1)}$ ; applying both to  $\frac{1}{2} + iL_r$  gives  $\Delta_n(q) = i^{n(n-1)} H_n = (-1)^{\binom{n}{2}} H_n$ .  $\square$

Consequently the reciprocal moment functional is quasi-definite: its monic orthogonal polynomials exist, satisfy  $P_{n+1}(z) = (z - \frac{1}{2})P_n(z) + \beta_n^{(r)} P_{n-1}(z)$  with  $\beta_n^{(r)} > 0$  inherited from the symmetric Jacobi recurrence of  $L_r$ , and have all zeros on  $\Re z = \frac{1}{2}$ . Reciprocal  $q$  does not destroy positivity; it rotates it onto a vertical line. Determining the Jacobi coefficients  $\beta_n^{(r)}$ , their asymptotics, and their relation to the double geometric/harmonic scale lattice is posed in the frontier section.

## 8 $q$ -Fabius–Bernoulli Appell polynomials

On polynomials, geometric averaging  $(\mathcal{A}_r f)(x) = \mathbb{E}f(x + Y_r)$  is the multiplier  $M_r(D)$ , so reciprocity identifies its inverse with no triangular solve (seventeenth wave):

$$\boxed{\mathcal{A}_r^{-1} = M_{1/r}(-D).} \quad (8.1)$$

The reciprocal germ is the deconvolution symbol for geometric-uniform smoothing. Define the Appell family

$$\sum_{n \geq 0} \mathfrak{B}_n^{(r)}(x) \frac{t^n}{n!} = \frac{e^{xt}}{M_r(t)}, \quad (8.2)$$

so that  $\frac{d}{dx} \mathfrak{B}_n^{(r)} = n \mathfrak{B}_{n-1}^{(r)}$  and

$$\boxed{\mathbb{E} \mathfrak{B}_n^{(r)}(x + Y_r) = x^n.} \quad (8.3)$$

The family interpolates two classical boundaries:

$$\mathfrak{B}_n^{(0)}(x) = B_n(x) \quad (\text{ordinary Bernoulli polynomials}), \quad \mathfrak{B}_n^{(r)}(x) \longrightarrow \left(x - \frac{1}{2}\right)^n \quad (r \rightarrow 1^-), \quad (8.4)$$

and from outside the disk,  $M_q(t) \rightarrow te^t/(e^t - 1) = \sum B_n(1)t^n/n!$  as  $|q| \rightarrow \infty$ : Bernoulli polynomials sit at both ends of the sign/reciprocal diagram. The Mahler equation induces the refinement

$$\mathfrak{B}_n^{(r)}(x) = \sum_{k=0}^n \binom{n}{k} B_k (1-r)^k r^{n-k} \mathfrak{B}_{n-k}^{(r)}\left(\frac{x}{r}\right) \quad (8.5)$$

(with  $B_1 = -\frac{1}{2}$ ), a Bernoulli-polynomial analogue of the Fabius Mahler equation; and the Bell form  $\mathfrak{B}_n^{(r)}(x) = \mathcal{B}_n^{\exp}(x - \frac{1}{2}, -\kappa_2(r), 0, -\kappa_4(r), 0, \dots)$  gives

$$\mathfrak{B}_2^{(r)} = z^2 - v_r, \quad \mathfrak{B}_3^{(r)} = z^3 - 3v_r z, \quad \mathfrak{B}_4^{(r)} = z^4 - 6v_r z^2 + 3v_r^2 - \kappa_4(r), \quad z = x - \frac{1}{2}, \quad v_r = \kappa_2(r),$$

with the reflection  $\mathfrak{B}_n^{(r)}(1-x) = (-1)^n \mathfrak{B}_n^{(r)}(x)$ . Their arithmetic (von Staudt–Clausen analogues, multiplication theorems, root geometry) is a fresh research program.

## 9 The Fabiús–Pochhammer moment polynomial

The exponential moments  $a_n(q)$  have poles at roots of unity. Clearing them with the finite Pochhammer symbol produces a polynomial invariant (seventeenth wave):

$$\mathcal{P}_n(q) = \frac{(q; q)_n}{(1 - q)^n} \frac{m_n(q)}{n!}. \quad (9.1)$$

**Theorem 9.1** (Polynomial recurrence and boundaries).  $\mathcal{P}_n \in \mathbb{Q}[q]$  with  $\deg \mathcal{P}_n \leq \binom{n}{2}$ ,  $\mathcal{P}_0 = 1$ , and

$$\mathcal{P}_n(q) = \sum_{k=0}^{n-1} \frac{q^k}{(n - k + 1)!} \frac{(q; q)_{n-1}}{(q; q)_k} \mathcal{P}_k(q), \quad (9.2)$$

with boundary data  $\mathcal{P}_n(0) = \frac{1}{(n+1)!}$  and leading coefficient  $[q^{\binom{n}{2}}] \mathcal{P}_n = B_n(1)/n!$  (so odd  $n > 1$  drop strictly below the degree bound).

*Proof.* Multiply (2.7) by  $(q; q)_{n-1}/(1 - q)^n$  and substitute  $a_k = (1 - q)^k \mathcal{P}_k/(q; q)_k$ ; the quotient  $(q; q)_{n-1}/(q; q)_k$  is a polynomial and the degree bound follows by induction. At  $q = 0$  only  $k = 0$  survives; as  $q \rightarrow \infty$ ,  $m_n(q)/n! \rightarrow B_n(1)/n!$  while  $(q; q)_n/(1 - q)^n \sim q^{\binom{n}{2}}$ .  $\square$

The first values are

$$\begin{aligned} \mathcal{P}_1 &= \frac{1}{2}, & \mathcal{P}_2 &= \frac{q+2}{12}, & \mathcal{P}_3 &= \frac{q^2+q+1}{24}, \\ \mathcal{P}_4 &= -\frac{(q^2+q+1)(q^4-2q^3-5q^2-3q-6)}{720}, \end{aligned}$$

and the exact factorizations continue with

$$\begin{aligned} \mathcal{P}_5 &= -\frac{[3]_q[5]_q(q^3-2q^2-2)}{1440}, \\ \mathcal{P}_7 &= \frac{[3]_q[5]_q[7]_q(2q^8-6q^7+4q^6-3q^5+6q^4-4q^3+7q^2+3)}{120960}, \end{aligned} \quad (9.3)$$

with an analogous  $[3]_q[5]_q$  factor in  $\mathcal{P}_6$ .

**Conjecture 9.2** (Odd  $q$ -integer divisor; seventeenth wave). *For every  $n \geq 3$ ,  $\prod_{3 \leq d \leq n, d \text{ odd}} [d]_q$  divides  $\mathcal{P}_n(q)$  in  $\mathbb{Q}[q]$ . Checked symbolically through  $n = 16$ . Since  $\prod_d [d]_q = \prod_e \Phi_e(q)^{\lfloor (n/e+1)/2 \rfloor}$  over odd cyclotomic indices  $e$ , repeated cyclotomic jets — not mere vanishing at primitive odd roots of unity — must be controlled; the cancellation-order calculus of the two-nome partition function below is a candidate mechanism.*



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**Post-snapshot Lean status: Exact equation and theorem.** The source-only module `GeometricUniformMomentPolynomial.lean` has one public definition, `geometricUniformMomentPolynomial`, and exactly eight public theorems: `geometricUniformMomentPolynomial_zero`, `geometricUniformMomentPolynomial_succ`, `geometricUniformMomentPolynomial_natDegree_le`, `geometricUniformMomentPolynomial_eval_zero`, `geometricUniformMomentPolynomial_one`, `geometricUniformMomentPolynomial_two`, `geometricUniformMomentPolynomial_three`, and `geometricUniformMomentPolynomial_four`. It supplies the recursively defined rational polynomial, the residual-product recurrence (9.2), the triangular degree bound, the value at zero, and the displayed cases  $P_0, \dots, P_4$ . The companion `GeometricUniformMomentPolynomialBridge.lean` has no public definitions and exactly one public theorem, `geometricUniformMomentPolynomial_eval2_eq_mgf_taylorCoefficient`. For every real  $q$  with  $|q| < 1$  and every  $n$ , it identifies that polynomial with the finite- $q$ -Pochhammer normalization of the  $n$ -th Taylor coefficient of the genuine geometric-uniform MGF. Hence (9.1) is Exact in the real contraction regime, including signed  $q$  and  $q = 0$ . The subsequent source-only `GeometricUniformComplexMomentProduct.lean` leaf has one public definition, `geometricUniformComplexMomentProduct`, and exactly three public theorems, `hasProdLocallyUniformly_geometricUniformComplexMomentProduct`, `differentiable_geometricUniformComplexMomentProduct`, and `geometricUniformMomentPolynomial_eval2_eq_complexMomentProduct_taylorCoefficient`. For every complex  $q$  with  $\|q\| < 1$ , it constructs the actual infinite product, proves locally uniform convergence and complex differentiability on the whole plane, and identifies its normalized Taylor coefficient with the same polynomial. This is a complex analytic analogue, not a complex probability-moment reading of (9.1); that equation is Exact only in the real contraction regime stated above. The subsequent source-only `GeometricUniformExteriorComplexMomentGerm.lean` leaf also has one public definition, `geometricUniformExteriorComplexMomentGerm`, and exactly two public theorems, `analyticAt_geometricUniformExteriorComplexMomentGerm` and `geometricUniformMomentPolynomial_eval2_eq_exteriorComplexMomentGerm_taylorCoefficient`. For every complex  $q$  with  $1 < \|q\|$ , it defines the actual reciprocal germ  $(\mathcal{A}_{q^{-1}}(-z))^{-1}$ , proves its analyticity at the origin, and identifies its normalized Taylor coefficient with the same polynomial. The canonical  $q$ -monograph compound label `thm:qF-moment-polynomial` is completed by the final source-only `GeometricUniformMomentRatFunc.lean` leaf. Its exhaustive one-definition/ four-theorem API is `geometricUniformMomentRatFunc`, `qFactorial_mul_geometricUniformMomentRatFunc`, `eval_geometricUniformMomentRatFunc_eq_complexMomentProduct_taylorCoefficient`, `eval_geometricUniformMomentRatFunc_eq_exteriorComplexMomentGerm_taylorCoefficient`, and `eval_geometricUniformMomentRatFunc_one`. It packages the single rational function  $a_n = d_n/n! = \mathcal{P}_n/[n]_q!$ , proves the global  $q$ -factorial pole-clearing identity, specializes it safely to the inner and exterior analytic coefficients, and handles  $q = 1$  through  $[n]_1! = n!$  rather than totalized  $0/0$ . No `RatFunc.eval` identity is asserted at a genuine reduced-denominator zero, and no general unit-circle analytic-germ claim is made.

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**Post-snapshot Lean status: moment reciprocity.** The subsequent source-only `GeometricUniformMomentReciprocity.lean` leaf has one public definition, `geometricUniformComplexMomentGerm`, and exactly five public theorems: `geometricUniformComplexMomentGerm_of_norm_lt_one`, `geometricUniformComplexMomentGerm_of_one_lt_norm`, `analyticAt_geometricUniformComplexMomentGerm`, `geometricUniformComplexMomentGerm_reciprocity`, and `geometricUniformComplexMomentGerm_moment_convolution`. The first two identify the strict inner and exterior branches, the third proves analyticity at the origin off the unit circle, and the last two prove, for  $q \neq 0$  and  $\|q\| \neq 1$ ,

$$M_q(z)M_{q^{-1}}(-z) = 1$$

as a local germ equality and give its exact all-order binomial derivative convolution. The germ formulation is essential: at a genuine zero of the inner product, Lean’s total inverse does not yield a global pointwise reciprocal identity. Thus the q-monograph label `thm:qF-reciprocity` is `Exact`.

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**Post-snapshot Lean status: sharp degree and follow-up leaves.** The final source-only `GeometricUniformMomentPolynomialDegree.lean` leaf has no public definitions and exactly three public theorems: `coeff_geometricUniformMomentPolynomial_choose_two`, `coeff_geometricUniformMomentPolynomial_choose_two_sub_one`, and `geometricUniformMomentPolynomial_natDegree_eq`. Writing  $T_n = \binom{n}{2}$ , they prove

$$[q^{T_n}]P_n = \frac{B_n(1)}{n!}, \quad [q^{T_n-1}]P_n = -\frac{B_n(1)}{n!} + \frac{B_{n-1}(1)}{2(n-1)!} \quad (n \geq 2),$$

and exact degree  $T_n$  when  $n = 1$  or  $n$  is even, versus  $T_n - 1$  otherwise. Lean’s `bernoulli`’s  $n$  is precisely  $B_n(1) = (-1)^n B_n$ , with Mathlib using  $B_1 = -1/2$ . Consequently [theorem 9.1](#) is `Exact`: its algebraic recurrence, bound, zero and small cases, its inner strict-disk and exterior analytic normalizations, and its leading/odd-degree clause are all supplied. The sharper q-monograph label `prop:qF-P-degree-sharp` is `Exact` as well. Neither promotion fills any missing clause of the broader geometric-uniform MGF theorem, but the `RatFunc` leaf above makes `thm:qF-moment-polynomial` `Exact`.

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**Post-snapshot Lean status (continued): Census and crosswalk.** The  $1 + 8$  algebraic leaf raised the historical 903/11,448 checkpoint to the historical 904 modules and 11,457 public declarations; the exhaustive  $0 + 1$  real bridge raised the next historical checkpoint to 905 modules and 11,458 public declarations; and the exhaustive  $1 + 2$  complex leaf raised the next incoming-branch checkpoint to 906 modules and 11,461 public declarations; in the merged chronology it formed the historical 918/11,568 checkpoint. The exhaustive zero-definition/one-theorem `HalfQBinomialRootSimplicity.lean` leaf then gives the merged-main 919/11,569 checkpoint. Its theorem `halfQBinomial_sum_rootMultiplicity_two_pow`, combined with the existing complete rational root classification and half-base Gaussian-coefficient bridge, makes `cor:halfbase-root-locus` `Exact`. It makes no arbitrary-base or arbitrary-field simplicity claim, and `cor:qbinom-inversion-law` remains `Partial`.

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**Post-snapshot Lean status: later census and scope.** The exhaustive  $1 + 2$  exterior leaf then gives 920/11,572, and the exhaustive  $0 + 3$  sharp-degree leaf gives the historical 921/11,575 checkpoint. The subsequent `RvachevLaurentLeading.lean` ( $1 + 6$ ) leaf gives 922/11,582, and `FinitePrefixAppellRecovery.lean` ( $11 + 17$ ) gives the historical 923/11,610 checkpoint. The final `RatFunc` ( $1 + 4$ ) leaf gives the historical 924/11,615 checkpoint. The existing `ProbabilityLaplaceMoments.lean` module then adds exactly the two public theorems `weightedSumDistribution_real_Ici_eq_rvachevUp_of_nonneg` and `integral_pow_weightedSumDistribution_eq_mul_intervalIntegral_rvachevUp`. `Atomlessness` makes the first the exact closed-tail convention, while the second is the full-law power-moment formula for every natural degree at least one; together with the existing global up/Fabiús identities, they make `prop:up-tail` and `cor:up-moments` `Exact`. This zero-definition/two-theorem increment gives the intermediate 924/11,617 checkpoint. The unrelated one-definition/one-theorem `RvachevLegendreBiorthogonality.lean` leaf then gives the historical 925/11,619 repository checkpoint. Subsequent source-only tranches, including the reciprocity leaf above, give the historical reciprocity checkpoint 931/11,685. The subsequently merged upstream `DyadicBoundaryIdentity.lean` and `FinitePrefixThueMorseCollapse.lean` modules add two modules and ten public declarations, making the historical dyadic/finite-prefix census 933/11,695. The incoming union adds one module and fourteen public declarations: the new zero-definition/six-theorem `ProuhetBaseTwoBridge.lean` module, one theorem added to `DyadicBoundaryIdentity.lean`, and seven theorems added to `ThueMorseNewmanSelfSimilarity.lean`. That tranche formed the historical 934/11,709 checkpoint. At the earlier frozen-merge checkpoint, the audited upstream census was 957/11,920 before the retained unconditional public `complexQPochhammerInf_eq_qPochhammerInfIn` bridge; the historical semantic-union checkpoint is 957/11,921. The residual-existence certificate `exists_eq_in_residual_interval` is included in the current union computed by `scripts/doc_audit.py` and pinned in `docs/doc_audit_baseline.json`. The `q` forward ledger is 181 `Exact` / 79 `Partial` / 14 `None` / 8 interface rows, the relevant Dyadic Gaussian–Thue–Morse chapter is 13/43/0/0, and the five-publication source concordance is 103 Lean-proved / 375 human-proved frontier / 60 not-applicable / 9 conjecture rows. The Laurent and finite-prefix leaves are included here for the census and exhaustive API counts; their `Exact` inverse-series status rows belong to the inverse-series crosswalk. These crosswalks were initially published as source-only additions. The earlier PDF is a historical receipt; current render-validation receipts are maintained externally.

The polynomial is not decorative: the exact endpoint lattice of [appendix 14](#) reads  $G_q(q^n) = q^{\binom{n+1}{2}} \mathcal{P}_n(q) / (q; q)_n$ , so  $(q; q)_n$  is precisely the pole-clearing factor of the moment recurrence surfacing as the endpoint denominator — Mersenne factors at  $q = \frac{1}{2}$ , factors  $4^j - 1$  at  $q = \frac{1}{4}$ .

## 10 Two master products on the binary cube

The Thue–Morse product and the  $q$ -Pochhammer symbol are unified twice over by the two waves, through two complementary four-parameter products. For  $\epsilon \in \{0, 1\}^N$  write  $|\epsilon| = \sum_j \epsilon_j$  (Hamming weight),  $\Omega(\epsilon) = \sum_j j \epsilon_j$  (position energy), and  $A_{\sigma, N}(\epsilon) = \sum_j \epsilon_j \sigma^j$  (geometric projection).

### 10.1 The two-nome partition function

**Definition 10.1** (Seventeenth wave). For parameters  $u, t, \rho, \sigma$  set

$$\mathcal{Z}_N(u, t; \rho, \sigma) = \prod_{j=0}^{N-1} (1 - u \rho^j e^{t \sigma^j}) = \sum_{\epsilon \in \{0, 1\}^N} (-u)^{|\epsilon|} \rho^{\Omega(\epsilon)} e^{t A_{\sigma, N}(\epsilon)}. \quad (10.1)$$

Its  $t = 0$  face is finite Pochhammer/Gaussian calculus,  $\mathcal{Z}_N(u, 0; \rho, \sigma) = (u; \rho)_N = \sum_k (-u)^k \rho^{\binom{k}{2}} \begin{bmatrix} N \\ k \end{bmatrix}_\rho$ ; its  $(u, \rho) = (1, 1)$  face is the geometric Prouhet generator  $\prod_{j < N} (1 - e^{t\sigma^j})$ . The  $q$ -Pochhammer and Thue–Morse products are perpendicular faces of one object, and the two nomes must not be identified prematurely. Splitting  $j = mk + r$  gives simultaneous multisection in both nomes,

$$\mathcal{Z}_{mN}(u, t; \rho, \sigma) = \prod_{r=0}^{m-1} \mathcal{Z}_N(u\rho^r, t\sigma^r; \rho^m, \sigma^m), \quad (10.2)$$

whose faces are finite Pochhammer dissection and the comb convolution law of [appendix 12](#), and whose normalized Prouhet jet recovers [theorem 5.1](#). The mixed coefficients  $\mathcal{M}_{N,k,m} = \sum_{|\epsilon|=k} \rho^{\Omega(\epsilon)} A_{\sigma,N}(\epsilon)^m = (-1)^k m! [u^k t^m] \mathcal{Z}_N$  refine each Gaussian row by geometric power moments, and the order of vanishing at  $t = 0$  is exactly  $\#\{j < N : u\rho^j = 1\}$  – classical Prouhet cancellation is the maximal case  $u = \rho = 1$ , while root-of-unity nomes select arithmetic progressions of digit positions and give exactly controlled *partial* Prouhet cancellations.

## 10.2 The digit-position product and its position-parity companion

The sixteenth wave couples the position energy to the *encoded integer* instead of the projected moment. With  $b_j(n)$  the binary digits of  $n < 2^m$ ,  $\text{wt}_2(n) = \sum_j b_j(n)$ , and  $\pi_2(n) = \sum_j j b_j(n)$ ,

$$\Xi_m(u, z; q) = \prod_{j=0}^{m-1} (1 - uq^j z^{2^j}) = \sum_{n=0}^{2^m-1} (-u)^{\text{wt}_2(n)} q^{\pi_2(n)} z^n, \quad (10.3)$$

which simultaneously tracks the integer, its Hamming weight, and its occupied bit positions. Setting  $z = 1$  gives  $(u; q)_m = \sum_{n < 2^m} (-u)^{\text{wt}_2(n)} q^{\pi_2(n)}$ , and at fixed weight

$$\sum_{\substack{n < 2^m \\ \text{wt}_2(n)=k}} q^{\pi_2(n)} = q^{\binom{k}{2}} \begin{bmatrix} m \\ k \end{bmatrix}_q, \quad (10.4)$$

a specialization of the  $q$ -digital binomial theorem of Mansour and Nguyen [59]. The shifted variant  $\mathcal{T}_m(z; q) = \prod_{j < m} (1 - q^{j+1} z^{2^j}) = \sum_n \varepsilon_n q^{\rho_2(n)} z^n$  with  $\rho_2 = \pi_2 + \text{wt}_2$  satisfies  $\mathcal{T}_m(1; q) = (q; q)_m$ ; at negative nome  $q = -r$  its coefficients become  $\chi(n) r^{\rho_2(n)}$  with the *position parity*  $\chi(n) = (-1)^{\pi_2(n)}$ . The pair  $(\varepsilon_n, \chi(n))$  is 2-automatic with the four-state recurrences

$$\chi(2n) = \chi(2n+1) = \chi(n) \varepsilon_n, \quad (10.5)$$

since doubling shifts every occupied position by one. The infinite product satisfies the Mahler equation  $\Xi_\infty(u, z; q) = (1 - uz) \Xi_\infty(uq, z^2; q)$  and the finite reciprocity  $\Xi_m(u, z; q) = (-u)^m q^{\binom{m}{2}} z^{2^m-1} \Xi_m(u^{-1}, z^{-1}; q^{-1})$  – the digit-polynomial counterpart of the reciprocal law and the Lagrange row reversal below.

## 10.3 Prouhet moment transfer, desingularization, and Bell–Bernoulli

The Prouhet face carries an exact moment calculus, stated by the two waves in two equivalent dresses and merged here.

**Theorem 10.2** ( $q$ -Prouhet moment transfer; both waves). *Let  $q \neq 0$  be complex,  $N \geq 1$ , and let  $S_{N,m}(q) = \sum_\epsilon (-1)^{|\epsilon|} A_{q,N}(\epsilon)^m$  be the signed power sums of the geometric subset sums. With  $T_{q,N} = \sum_{j < N} q^j U_j$ ,*

$$S_{N,m}(q) = 0 \quad (0 \leq m < N), \quad \boxed{S_{N,N+\ell}(q) = (-1)^N \frac{(N+\ell)!}{\ell!} q^{\binom{N}{2}} \mathbb{E} T_{q,N}^\ell} \quad (\ell \geq 0). \quad (10.6)$$

*Equivalently (general-radix Bell–Bernoulli form), the same subset sums obey*

$$S_{N,N+\ell}(q) = (-1)^N q^{\binom{N}{2}} \frac{(N+\ell)!}{\ell!} \mathcal{B}_\ell^{\text{exp}}(\gamma_1, \dots, \gamma_\ell), \quad (10.7)$$

$$\gamma_1 = \frac{1}{2}[N]_q, \quad \gamma_{2r} = \frac{B_{2r}}{2r} [N]_{q^{2r}}, \quad \gamma_{2r+1} = 0 \quad (r \geq 1),$$

*and the finite product desingularizes into a normalized moment transform: with  $\alpha_{N,q} = (1-q)/(1-q^N)$ ,*

$$\mathbb{E} e^{t(Y_q^{[N]} - 1/2)} = \frac{(-1)^N e^{-t/2} \prod_{j < N} (1 - e^{\alpha_{N,q} q^j t})}{\alpha_{N,q}^N q^{\binom{N}{2}} t^N}, \quad (10.8)$$

*which for  $|q| > 1$  converges locally uniformly to the centered transform of  $Y_{1/q}$ . Normalizing by  $[N]_q^s$  turns the unnormalized cumulants  $\gamma_s$  of  $T_{q,N}$  into the reciprocal-invariant ratio  $\kappa_s(Y_q^{[N]}) = \gamma_s/[N]_q^s$  of theorem 7.4.*

*Proof.* Factor  $1 - e^z = -zE(z)$  in each term of the Prouhet face:  $\prod_{j < N} (1 - e^{q^j t}) = (-1)^N q^{\binom{N}{2}} t^N \prod_{j < N} E(q^j t)$ . The explicit  $t^N$  proves the vanishings; comparing the coefficient of  $t^{N+\ell}$  gives (10.6), and expanding  $\log \prod E(q^j t)$  by the uniform cumulant series gives (10.7) (the two right sides are the same number:  $\mathbb{E} T_{q,N}^\ell = \ell! [t^\ell] \prod_{j < N} E(q^j t)$ ). Dividing by the maximal Prouhet zero and centering gives (10.8), and (3.13) identifies the  $|q| > 1$  limit.  $\square$

The first surviving moment is  $S_{N,N}(q) = (-1)^N N! q^{\binom{N}{2}}$ , and every  $q$ -Fabius moment is a normalized limit of first nonvanishing Thue–Morse jets:  $m_\ell(q) = (1-q)^\ell \lim_N (-1)^N q^{-\binom{N}{2}} \frac{\ell!}{(N+\ell)!} S_{N,N+\ell}(q)$ . In particular  $b = 2, -2, 4, -4$  produce, through renormalized classical, negabinary, base-four, and negative base-four Thue–Morse exponential sums, exactly the laws at  $q = \frac{1}{2}, -\frac{1}{2}, \frac{1}{4}, -\frac{1}{4}$ . The Fabius MGF is the normalized residual jet of the maximal Prouhet zero (equation (10.8) at  $N \rightarrow \infty$ ; the same statement in the partition-function coordinate is  $M_q(t) = \lim_N \mathcal{Z}_N(1, (1-q)t; 1, q) \cdot [(-1)^N (1-q)^N q^{\binom{N}{2}} t^N]^{-1}$ ).

Two parity remarks connect the layers (sixteenth wave). Centering the finite transform turns each factor into a hyperbolic sine; since  $\sinh$  is odd, the radix sign change accumulates a global parity,

$$e^{-\frac{1}{2}[N]-at} \prod_{j < N} (1 - e^{(-a)^j t}) = (-1)^{\binom{N}{2}} e^{-\frac{1}{2}[N]at} \prod_{j < N} (1 - e^{a^j t}), \quad (10.9)$$

whereas the continuous centered factors are even sines and absorb signs invisibly. At the measure level (seventeenth wave), the signed comb  $\Delta_{q,N} = \sum_\epsilon (-1)^{|\epsilon|} \delta_{A_{q,N}(\epsilon)}$  transports by

$$\Delta_{-a,N} = (-1)^{\lfloor N/2 \rfloor} \tau_{-c_{a,N}} \Delta_{a,N}, \quad \Delta_{q^{-1},N} = (D_{q^{-(N-1)}})_* \Delta_{q,N}, \quad (10.10)$$

( $\tau_c$  translation,  $D_c$  dilation,  $c_{a,N} = a(1 - a^{2\lfloor N/2 \rfloor})/(1 - a^2)$ , by complementing odd-position digits and by digit reversal respectively): sign is translation up to a global parity, inversion is dilation. The Klein-four geometry acts identically on the infinite germ and on every finite comb.

## 11 Position-energy layers and the finite-geometry square

At fixed Hamming weight the position-energy layer is Gaussian,  $H_{N,k}(\rho) = \sum_{|\epsilon|=k} \rho^{\Omega(\epsilon)} = \rho^{\binom{k}{2}} \begin{bmatrix} N \\ k \end{bmatrix}_\rho$ , with reciprocity  $H_{N,k}(\rho) = \rho^{k(N-1)} H_{N,k}(\rho^{-1})$  and  $\begin{bmatrix} N \\ k \end{bmatrix}_\rho = \rho^{k(N-k)} \begin{bmatrix} N \\ k \end{bmatrix}_{\rho^{-1}}$ . At negative nome the signs are governed by two quadratic parities (seventeenth wave):

$$\text{sgn } H_{N,k}(-a) = (-1)^{\binom{k}{2}} \quad (0 < a < 1), \quad \text{sgn } H_{N,k}(-a) = (-1)^{\binom{k}{2} + k(N-k)} \quad (a > 1), \quad (11.1)$$

and pairing factors splits the negative row into two positive  $a^2$ -rows,  $(z; -a)_{2m} = (z; a^2)_m (-az; a^2)_m$  — the finite counterpart of sign conjugacy and decimation. Interpreting the nome as a prime power connects the two orbits to finite geometry:  $\begin{bmatrix} N \\ k \end{bmatrix}_Q$  counts  $k$ -planes in  $\mathbb{F}_Q^N$ ,  $\begin{bmatrix} N \\ k \end{bmatrix}_{1/Q} = Q^{-k(N-k)} \# \text{Gr}(k, \mathbb{F}_Q^N)$ , and by Fu, Reiner, Stanton, and Thiem [4] the negative nome counts nondegenerate subspaces of Hermitian spaces:  $\begin{bmatrix} N \\ k \end{bmatrix}_{-Q} = (-Q)^{-k(N-k)} U_Q(N, k)$  and  $\begin{bmatrix} N \\ k \end{bmatrix}_{-1/Q} = Q^{-2k(N-k)} U_Q(N, k)$  over  $\mathbb{F}_{Q^2}$ . The two orbits therefore carry a finite-geometry square:  $\pm 2, \pm \frac{1}{2}$  correspond to Grassmannians over  $\mathbb{F}_2$  and Hermitian geometry over  $\mathbb{F}_4$ ;  $\pm 4, \pm \frac{1}{4}$  to  $\mathbb{F}_4$  and  $\mathbb{F}_{16}$ . (For example  $\# \text{Gr}(2, \mathbb{F}_2^4) = 35$  and  $U_2(4, 2) = 240$  give  $\begin{bmatrix} 4 \\ 2 \end{bmatrix}_{1/2} = 35/16$  and  $\begin{bmatrix} 4 \\ 2 \end{bmatrix}_{-1/2} = 15/16$ .)

## 12 Box-spline combs and the quartic Cantor skeleton

The centered finite law  $R_{q,N} = \sum_{j < N} q^j V_j$  is a nonuniform box spline: with radius  $A_{q,N}^{\text{rad}} = \frac{1}{2} \sum_{j < N} q^j$ ,

$$B_{q,N}(x) = \frac{1}{(N-1)! q^{\binom{N}{2}}} \sum_{\epsilon \in \{0,1\}^N} (-1)^{|\epsilon|} \left( x + A_{q,N}^{\text{rad}} - A_{q,N}(\epsilon) \right)_+^{N-1}, \quad (12.1)$$

$$D^N B_{q,N} = q^{-\binom{N}{2}} \tau_{-A_{q,N}^{\text{rad}}} \Delta_{q,N},$$

so the  $q$ -Prouhet cancellation is precisely the vanishing of the first  $N-1$  distributional moments of the derivative comb, and  $N$ -fold integration converts the signed atoms into a positive spline. At  $q = \frac{1}{2}$  the knots  $A_{1/2,N}(\epsilon)$  fill a complete dyadic grid and the signs are the Thue–Morse sequence in binary order; at  $q = \frac{1}{4}$  the knots use only the digits  $\{0, 1\}$  in base four, and their infinite closure is the self-similar set

$$K_4^{\text{raw}} = \left\{ \sum_{j \geq 0} \epsilon_j 4^{-j} : \epsilon_j \in \{0, 1\} \right\}, \quad \dim_{\text{H}} K_4^{\text{raw}} = \frac{\log 2}{\log 4} = \frac{1}{2}. \quad (12.2)$$

The finite derivative comb is a signed Cantor skeleton even though the limiting density is  $C^\infty$  with full interval support: integration through arbitrarily many uniform scales erases the gaps. The comb multisection

$$\Delta_{q,2N} = \Delta_{q^2,N} * (D_q)_* \Delta_{q^2,N} \quad (12.3)$$

factors the full dyadic Thue–Morse comb into a convolution of two quartic Cantor combs — the singular, derivative-level ancestor of the smooth convolution (5.4). (This comb calculus complements the dyadic-comb quadrature volume of the corpus, which studies unsigned equal-weight combs against  $F$ ; here the combs are the signed Thue–Morse derivative skeletons of the family itself.)

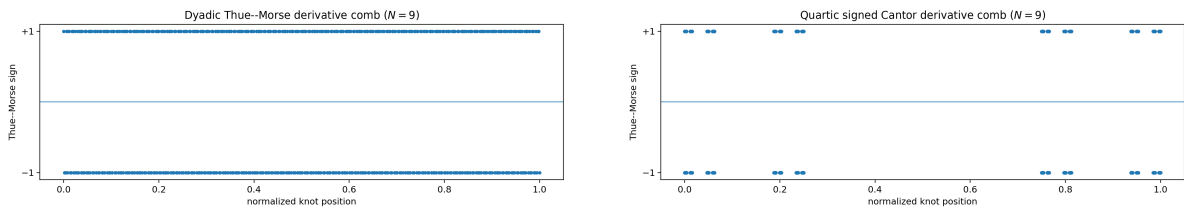


Figure 37: Signed derivative combs at level  $N = 9$  (ordinate: Thue–Morse sign). Left,  $q = 1/2$ : a full dyadic grid. Right,  $q = 1/4$ : base-four digits  $\{0, 1\}$  already displaying the dimension- $1/2$  Cantor geometry of (12.2).



### 13 Reciprocal $q$ -Lagrange rows

For evaluation at zero from geometric nodes  $1, q, \dots, q^n$ , the Lagrange weights have the closed form  $\lambda_{n,k}^{(q)} = (-1)^{n-k} q^{\binom{n-k+1}{2}} [n]_q / (q; q)_n$ , and digit – here node – reversal gives the exact row reversal (sixteenth wave)

$$\lambda_{n,k}^{(q)} = \lambda_{n,n-k}^{(q^{-1})} : \quad (13.1)$$

the rows for  $q = 2, -2, 4, -4$  are exact reversals of the rows for  $\frac{1}{2}, -\frac{1}{2}, \frac{1}{4}, -\frac{1}{4}$  (the geometric node list for  $q^{-1}$ , read backwards and rescaled by  $q^n$ , is the node list for  $q$ , and weights for evaluation at zero are scaling-invariant). For example, at  $n = 5$ ,  $q = \frac{1}{2}$  the exact row is  $(-\frac{1}{9765}, \frac{2}{315}, -\frac{8}{63}, \frac{64}{63}, -\frac{1024}{315}, \frac{32768}{9765})$ , and the  $q = 2$  row is its reversal. This is the interpolation-theoretic face of the same reciprocity that inverts the germ and dilates the comb, and it connects directly to the geometric-Lagrange and  $q$ -Richardson machinery of Parts I–II.

### 14 Exact inverse-geometric endpoint lattices

#### 14.1 The stop-loss simplex formula

**Theorem 14.1** (Stop-loss endpoint formula; seventeenth wave). *Let  $0 < q < 1$ ,  $n \geq 1$ , and  $0 \leq c \leq (1 - q)/q$ . Then*

$$G_q(q^n c) = \frac{q^{n(n+1)/2}}{(1 - q)^n n!} \mathbb{E} (c - Y_q)_+^n. \quad (14.1)$$

*Proof.* Split  $Y_q = (1 - q) \sum_{j < n} q^j U_j + q^n Y'_q$  and condition on  $Y'_q = y$ . The event becomes  $\sum_{j < n} (1 - q) q^j U_j \leq q^n (c - y)$ ; the threshold is at most  $q^n c \leq (1 - q) q^{n-1}$ , the smallest coordinate width, so the upper cube constraints are inactive and the conditional probability is the full simplex volume  $q^{n^2} (c - y)_+^n / (n! \prod_{j < n} (1 - q) q^j) = q^{n(n+1)/2} (c - y)_+^n / ((1 - q)^n n!)$ . Average over  $y$ .  $\square$

#### 14.2 The lattice, its jets, and negative transport

For  $0 < q \leq \frac{1}{2}$  the value  $c = 1$  is admissible, and reflection  $1 - Y_q \stackrel{d}{=} Y_q$  turns the stop-loss moment into a raw moment.

**Corollary 14.2** (Inverse-geometric endpoint identity). *For  $0 < q \leq \frac{1}{2}$  and  $n \geq 0$ ,*

$$G_q(q^n) = \frac{q^{\binom{n+1}{2}}}{(1 - q)^n} \frac{m_n(q)}{n!} = \frac{q^{\binom{n+1}{2}}}{(q; q)_n} \mathcal{P}_n(q). \quad (14.2)$$

*At  $q = \frac{1}{2}$  this is the classical dyadic formula  $F(2^{-n}) = 2^{-\binom{n}{2}} m_n(\frac{1}{2})/n!$ ; at  $q = \frac{1}{4}$  it gives a new exact inverse-quartic lattice with first values*

$$G_{1/4}(\frac{1}{4}) = \frac{1}{6}, \quad G_{1/4}(\frac{1}{16}) = \frac{1}{240}, \quad G_{1/4}(\frac{1}{64}) = \frac{1}{51840}, \quad G_{1/4}(\frac{1}{256}) = \frac{121}{6768230400}. \quad (14.3)$$

**Theorem 14.3** (Endpoint derivative jets; seventeenth wave). *For  $0 < q \leq \frac{1}{2}$ ,  $n \geq 0$ , and  $0 \leq r \leq n$ ,*

$$G_q^{(r)}(q^n) = (1 - q)^{-r} q^{-\binom{r}{2}} G_q(q^{n-r}) = \frac{q^{\binom{n-r+1}{2} - \binom{r}{2}}}{(1 - q)^n} \frac{m_{n-r}(q)}{(n - r)!}. \quad (14.4)$$

*Proof.* Conditioning on the first digit gives  $G'_q(x) = \frac{1}{1-q} [G_q(x/q) - G_q((x-1+q)/q)]$ ; on the low interval  $0 \leq x \leq 1 - q$  the second term vanishes, so  $G'_q(x) = (1 - q)^{-1} G_q(x/q)$  – the CDF face of



the edge pantograph equation (21.7). Iterating  $r$  times multiplies by  $(1 - q)^{-r} q^{-(0+1+\dots+(r-1))}$  and dilates the argument to  $x/q^r$ ; at  $x = q^n$  every intermediate argument stays in the low interval because  $q \leq \frac{1}{2}$  (at  $q = \frac{1}{2}$ , by flat matching of the two smooth branches).  $\square$

In particular  $g_q(q^n) = q^{n(n-1)/2} (1 - q)^{-n} m_{n-1}(q) / (n - 1)!$ , and at  $r = n$  the jet collapses to the elementary value  $(1 - q)^{-n} q^{-\binom{n}{2}}$ .

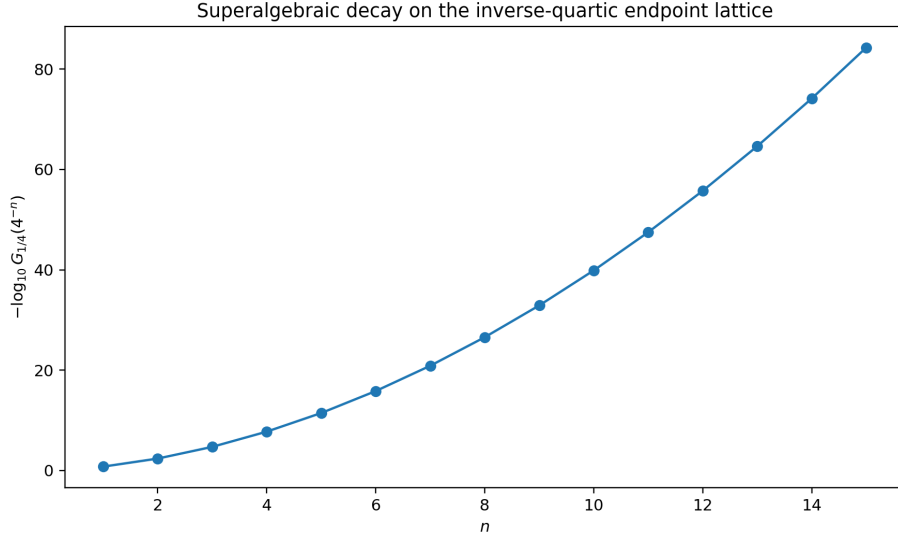


Figure 38: Exact inverse-quartic lattice values (14.3) on a logarithmic ordinate. The quadratic growth of  $-\log G_{1/4}(4^{-n})$  is the discrete shadow of the two-term endpoint law of theorem 15.1.

By the affine transport (3.2), the negative laws inherit the whole lattice at the shifted nodes  $\lambda_a a^n - \beta_a$  (accumulating at the left endpoint  $-\beta_a$ ), with jets scaled by  $\lambda_a^{-r}$ ; there is no independent negative-parameter endpoint problem. Successive lattice ratios obey exactly  $G_q(q^{n+1})/G_q(q^n) = \frac{q^{n+1}}{(1-q)(n+1)} \cdot \frac{m_{n+1}(q)}{m_n(q)}$ , hence  $G_q(q^{n+1})/G_q(q^n) \sim q^{n+1}/((1-q)(n+1))$  since  $m_{n+1}(q)/m_n(q) \rightarrow 1$  by the right-endpoint flatness of  $G_q$  at 1 – faster than any geometric rate, the first visible sign of log-squared decay.

## 15 Endpoint asymptotics and the general- $q$ Lambert program

### 15.1 A uniform two-term law and its inversion

**Theorem 15.1** (Uniform two-term endpoint asymptotic; seventeenth wave). *Fix  $0 < q \leq \frac{1}{2}$  and put  $L = \log(1/q)$ ,  $\ell = \log(1/x)$ . Then, as  $x \downarrow 0$ ,*

$$\log G_q(x) = -\frac{\ell^2}{2L} - \frac{\ell \log \ell}{L} + \mathcal{O}_q(\ell). \quad (15.1)$$

*Proof.* At lattice points, (14.2) gives  $\log G_q(q^n) = -\frac{L}{2}n(n+1) - n \log(1-q) - \log n! + \log m_n(q)$ . Reflection gives the crude bounds  $2^{-(n+1)} \leq m_n(q) \leq 1$ , so  $\log m_n = \mathcal{O}(n)$ ; Stirling and  $n = \ell/L$  give the display, and monotone bracketing between consecutive lattice points extends it to all  $x$ .  $\square$

**Corollary 15.2** (Inverse endpoint asymptotic). *For fixed  $0 < q \leq \frac{1}{2}$ , as  $y \downarrow 0$ ,*

$$\log \frac{1}{G_q^{-1}(y)} = \sqrt{2L \log(1/y)} - \frac{1}{2} \log \log(1/y) + \mathcal{O}_q(1). \quad (15.2)$$

The dyadic and quartic cases read  $\log F(x) = -\frac{\log^2(1/x)}{2\log 2} - \frac{\log(1/x)\log\log(1/x)}{\log 2} + \mathcal{O}(\log(1/x))$  and  $\log G_{1/4}(x) = -\frac{\log^2(1/x)}{4\log 2} - \frac{\log(1/x)\log\log(1/x)}{2\log 2} + \mathcal{O}(\log(1/x))$ : the quartic law is less flat at a common  $x$  because its scale ladder descends faster. These are deliberately weaker but uniform statements; the all-orders periodically corrected dyadic expansion lives in the inverse-endpoint volume of the corpus.

## 15.2 The periodic cocycle is now exact

The sixteenth wave formulated the Laplace-side structure as a conjecture: in the additive coordinate  $y = \log((1-r)s)$ ,  $\lambda = \log(1/r)$ , the function  $G_r^{\text{coc}}(y) = \log \mathbb{E} e^{-sY_r}$  satisfies the cocycle equation  $G_r^{\text{coc}}(y) - G_r^{\text{coc}}(y - \lambda) = \log((1 - e^{-e^y})/e^y)$ , and the conjecture asked for the normal form  $-y^2/(2\lambda) - y/2 + C_r + P_r(y/\lambda) + o(1)$  with a real-analytic one-periodic  $P_r$  whose Fourier coefficients come from a Mellin transform.

**Theorem 15.3** (Resolution by the atomic decomposition). *The conjectured normal form holds exactly, with an explicit exponentially small remainder and explicit Fourier coefficients. Namely, with  $u = (1-r)s$  and  $a = 1/r$ ,*

$$\log \mathbb{E} e^{-sY_r} = \varkappa(u) + \Lambda_a(u) = -\frac{(\log u)^2}{2\lambda} - \frac{\log u}{2} + \left[ \varkappa(u) + \log u + P_a(\log_a u) \right] + E_a(u), \quad (15.3)$$

where  $\varkappa, \Lambda_a, P_a, E_a$  are the objects of [theorem 21.1](#):  $P_a$  is one-periodic and real analytic with Fourier modes  $-\Gamma(-\chi_k)\zeta(1-\chi_k)/\lambda$ ,  $\chi_k = 2\pi i k/\lambda$  ([theorem 21.2](#)), the bracket tends to  $P_a(\log_a u)$  up to  $\mathcal{O}(e^{-u})$  since  $\varkappa(u) = -\log u + \mathcal{O}(e^{-u})$ , and  $0 \leq E_a(u) \leq e^{-u}/((1 - e^{-u})(1 - e^{-(a-1)u}))$ .

*Proof.*  $\mathbb{E} e^{-sY_r} = \prod_{j \geq 0} E(-(1-r)r^j s)$  and  $\log E(-v) = \varkappa(v)$ , so the log-Laplace transform is  $\sum_{j \geq 0} \varkappa(ur^j) = \varkappa(u) + \Lambda_{1/r}(u)$ . Apply the exact decomposition ([21.14](#)) at base  $a = 1/r$ .  $\square$

Thus the transform-level half of the general- $q$  endpoint program is closed — not only the existence of the periodic correction but its exact Gamma-zeta Fourier coefficients, uniformly in the base. What remains open is the density level, exactly as in Part VI:

**Conjecture 15.4** (All-orders general- $q$  endpoint expansion; both waves). *For each  $0 < q < 1$ , the density  $g_q$ , the CDF  $G_q$ , and the inverse  $G_q^{-1}$  admit, near the left endpoint, complete transseries in inverse powers of the lower-Lambert coordinate  $w_q(x) = -W_{-1}(-C_q x)$  (normalization  $C_q$  fixed by the saddle of ([15.3](#))), with coefficient functions real analytic and one-periodic in  $w_q(x)/\log(1/q)$ , generated by Bell polynomials in the derivatives of  $P_{1/q}$ , reducing at  $q = \frac{1}{2}$  to the corpus's dyadic lower-Lambert hierarchy and matching [theorem 15.1](#) to two orders. By [corollary 3.2](#), the negative-parameter cases require no separate analysis. This is the same open program as Part VI's refined endpoint conjecture, now with the quartic case singled out: decimation ([5.4](#)) couples the  $q = \frac{1}{4}$  periodic phase to the dyadic one at two interlaced scales, suggesting that the quartic profile can be reconstructed from the dyadic profile through a convolution saddle equation rather than computed from scratch.*

## 16 Numerical experiments and reproducibility

Both packages are deterministic (no random sampling) and preserved verbatim under assets/.

**Sixteenth wave.** The package is preserved under assets/Fabius\_Q\_Connections\_Report/; its script `q_fabius_experiments.py` uses exact rational arithmetic for the finite  $q$ -binomial, Lagrange, and digit-product identities, 80-digit arithmetic for the analytic products, and a padded  $2^{16}$ -point FFT for density inversion. Selected residuals: sign conjugacy at  $r = \frac{1}{2}$  exact; decimation at  $q = \frac{1}{2}, m = 3$  to  $6.7 \cdot 10^{-16}$  (FFT); the sinc-Pochhammer logarithm at  $q = -\frac{1}{2}$  to  $3.7 \cdot 10^{-72}$ ; Lagrange reciprocal reversal exact (zero rational mismatches) for all eight orbit values at  $n = 3, 5, 7$ ; the digit-position product exact at  $m = 8$ ; the standardized sign collapse at  $r = \frac{3}{4}$  to  $4.8 \cdot 10^{-16}$ ; the

quarter-base plateau height to  $6.7 \cdot 10^{-16}$ . Figures: physical densities of the four interior laws (the negative curves exact affine stretches) and support-normalized profiles.

**Seventeenth wave.** The package is preserved under `assets/Signed_Reciprocal_q_Fabiús_Frontiers/`; its script `numerical_experiments.py` uses exact `Fraction` arithmetic for the moment recurrence (order  $\leq 10$  at all eight values), affine conjugacy, reciprocal binomial-convolution inversion, decimation, position-energy layers ( $N \leq 8$ ), the  $q$ -Prouhet transfer ( $N \leq 7$ ,  $\ell \leq 4$ ; all residuals identically zero), the  $\mathcal{P}_n$  recurrence with polynomial long division for the divisor conjecture ( $n \leq 16$ ), inverse-geometric endpoints ( $n \leq 12$  at  $q = \frac{1}{2}, \frac{1}{4}$ ), the Grassmannian/unitary square ( $Q = 2, 4$ ,  $N \leq 7$ ), and the reciprocal Hankel signature through order nine at  $q = \pm 2, \pm 4$ ; plus 90-digit checks of the Mahler, sign, reciprocal, and decimation product identities (largest residual  $1.3 \cdot 10^{-40}$ , from a deliberately finite 300-factor truncation under an argument dilation of five). Its exact computations were also used as a theorem-discovery filter: several plausible reciprocity formulas for  $\mathcal{P}_n$  were rejected before statement — in this subject,  $q \leftrightarrow 1/q$  introduces triangular powers that can make a wrong formula look valid in low degree.

The exact moment table for the eight orbit values (through  $m_6$ ; e.g.  $m_4(\frac{1}{2}) = \frac{143}{1350}$ ,  $m_6(2) = -\frac{16}{59535}$ ,  $m_4(\frac{1}{4}) = \frac{121}{850}$ ) is in `data/q_orbit_moments.csv` of the seventeenth-wave package, whose `data/` directory also carries the Prouhet, divisibility, endpoint, finite-geometry, and Hankel tables named in the sections above.

## 17 Conjectures, problems, and formalization roadmap

### 17.1 Frontier problems

Beyond [conjectures 9.2](#) and [15.4](#), the merged register is:

**Conjecture 17.1** (Generic natural boundary; seventeenth wave). *For generic fixed  $t \neq 0$ , the unit circle  $|q| = 1$  is a natural boundary for  $q \mapsto C_q(t)$ , after removal of isolated removable degeneracies: the cumulant coefficients  $(1 - q)^{2m}/(1 - q^{2m})$  have poles at all roots of unity, and no single-valued meromorphic continuation should cross a nontrivial arc. A proof strategy is to isolate a primitive root-of-unity singularity in one cumulant order that lower orders cannot cancel; the partition function  $\mathcal{Z}_N$  supplies finite approximants with explicit root-of-unity zero orders.*

*Remark 17.2* (Proved on an explicit interval; the merge closed part of it). [corollary 5.5](#) proves the statement outright for every real  $t$  with  $0 < |t| < \pi$  — for all such  $t$ , not merely generic ones, and with no removable degeneracies to discard. The mechanism is not the cumulant-cancellation strategy sketched above but Part IX’s radial action:  $C_q(t) = \Phi(q, -it/2)$  places the germ on the purely imaginary frequency slice of the cyclotomic sinc product, where the spectral dilogarithm  $\mathcal{A}_\omega$  of Part IX is real and of one sign for every root of unity, forcing super-polynomial blow-up or decay along every radius. What remains open is the range  $|t| \geq \pi$ , where the dilogarithm arguments leave the unit disc and cancellation between spectral terms is possible.

**Open problem 17.3** (Reciprocal orthogonal polynomials). *Determine the Jacobi coefficients  $\beta_n^{(r)}$  of the Laplace dual law (7.1) exactly or asymptotically; find the  $q$ -difference structure induced by adding one geometric layer; determine the limiting zero distribution; compare  $r = \frac{1}{2}$  and  $r = \frac{1}{4}$  through multisection; and decide membership in a tractable subclass of generalized gamma convolutions (the signed Laplace differences are not themselves a positive convolution on  $\mathbb{R}_+$ ).*

**Open problem 17.4** (Arithmetic of  $\mathcal{P}_n$  and of the Appell family). *Find the least  $D_n$  with  $D_n \mathcal{P}_n \in \mathbb{Z}[q]$  and compare its prime valuations with von Staudt–Clausen denominators and the Mersenne/Fermat factors of  $(q; q)_n$  at rational  $q$ ; describe the root geometry of  $\mathcal{P}_n$  (forced cyclotomic zeros versus limiting curves); develop multiplication theorems and integrality for  $\mathfrak{B}_n^{(r)}$ . Through decimation, investigate whether the dyadic value arithmetic of Arias de Reyna and Haugland decomposes into bilinear expressions in quarter-base values — negative-parameter value questions reduce affinely and should never be duplicated.*

**Open problem 17.5** (Two-nome phase diagram and root-of-unity renormalization). *Carry out the  $N \rightarrow \infty$  saddle analysis of the mixed coefficients of  $\mathcal{Z}_N(u, t; \rho, \sigma)$  in the regimes  $|\rho|, |\sigma| \leq 1$  and on the unit circles, where partial Prouhet cancellation and cyclotomic resonance occur; construct canonical renormalized limits of the finite laws, Lagrange rows, and digit products as  $q$  approaches a root of unity; and derive the matrix Mahler system of the four-state pair  $(\varepsilon_n, \chi(n))$  with its Dirichlet series. Replacing the lacunary exponent  $2^j$  by  $b^j$  in  $\Xi$  raises the corresponding joint natural-boundary question on multiplicative-dependence loci.*

**Open problem 17.6** (Quartic Cantor renormalization). *Quantify how the dimension- $\frac{1}{2}$  knot geometry of (12.2) controls finite-level derivative norms and uniform/Sobolev convergence rates of the quartic box splines; build a multiresolution basis adapted to base-four digits  $\{0, 1\}$ ; and track the zero divisor  $1 + \lfloor \nu_2(n)/2 \rfloor$  against the dyadic  $1 + \nu_2(n)$  through the convolution (5.7).*

## 17.2 Lean formalization roadmap

The existing formal development (weighted-uniform series, geometric law, affine self-similarity, reflection, support, absolute continuity, the Fabius bridge) now includes the exact fixed half-quarter pointwise split, product-law identity, and scaled-measure convolution in `GeometricUniformMultisection.lean`. It supports an incremental plan, merged from both waves. The remaining low-risk algebraic tranche: negative affine conjugacy (coordinatewise sign flip and affine pushforward), general geometric multisection (reindex  $j = mk + r$ ), the moment recurrence and formal uniqueness away from roots of unity, the reciprocity  $M_q M_{1/q}(-\cdot) = 1$ , the position-energy layer over the finite  $q$ -binomial core, the two-nome partition function with its multisection, and the  $q$ -Prouhet transfer — finite sums and products throughout, suitable for the repository’s single-worker build. The endpoint tranche: the simplex-volume stop-loss formula (upper cube constraints inactive under (14.1)’s threshold), the lattice identity, the derivative jets from the CDF pantograph equation, and the two-term asymptotic (Stirling, crude moment bounds, monotone bracketing) — far lighter than the existing all-orders dyadic saddle machinery. The reciprocal analytic tranche: the inverse product germ and the outside-disk reciprocal Euler rearrangement (the general inside strict-complex-contraction rearrangement is now formalized in `RvachevPochhammerFactorization.lean`), the  $L^2$  Laplace series, Hankel positivity and affine transport of determinants, and the quasi-definite orthogonal recurrence. Three further targets are absorbed from Part VIII’s list: the sign-corrected density  $g_q = \text{sgn}(q) \tilde{f}_q$  of theorem 3.1, which removes the negative-parameter density ambiguity outright; the finite reciprocal duality (3.13) by coefficient extensionality, together with  $q = \pm 2, \pm 4$  wrappers; and the exact plateau (4.6) from the CDF tails. Suggested module names: `GeometricUniformSign`, `GeometricProuhet`, `TwoNomePartition`, `GeometricEndpoint`, `GeometricEndpointJets`, `GeometricEndpointAsymptotic`. The conjectures of this part remain in frontier quarantine until proved.

## 18 Conclusion

“Other  $q$  values” do not replace  $\frac{1}{2}$  by another number in old formulas; three transformations act on three layers. Sign change is an affine map of the outer normalization — the centered law is blind to it, the centered finite transform picks up only the parity  $(-1)^{\binom{N}{2}}$ , and the finite comb translates. Inversion turns smoothing into deconvolution, zeros into poles, real moments into vertical-line moments, Lagrange rows into their reversals, and finite digit lists into their reversals. Decimation splits scales into congruence classes, factoring the Fabius law over the quartic family, the dyadic Thue–Morse comb over quartic Cantor combs, and every Pochhammer product over its dissection. The finite Gaussian nome and the geometric projection scale are independent coordinates of one binary-cube partition function whose renormalized maximal Prouhet jet is the  $q$ -Fabius moment germ, and whose position-energy layers carry Grassmannian and Hermitian finite geometry. Each positive contraction owns an exact inverse-geometric endpoint lattice with all its jets; each negative law inherits everything affinely; each

reciprocal germ is positive on a vertical line. The transform-level endpoint structure is now exact in every base through Part VI's periodic decomposition; the density-level all-orders Lambert- $W_{-1}$  program, the  $\mathcal{P}_n$  arithmetic, and the vertical orthogonal polynomials are the principal open frontiers.

## Part VIII

# Negative Parameters, Reciprocal Bases, and the Gaussian Boundary

## Abstract

The current ProveIt/Analysis/FabiusFunction/docs corpus develops the Fabius and Rvachev functions from probability, functional equations, exact dyadic arithmetic,  $q$ -calculus, infinite sinc products, Thue–Morse cancellation, Legendre expansions, Lambert– $W$  endpoint asymptotics, and formalized analysis. When this report was written, its integrated frontier volume explicitly left open, for the geometric-uniform family, the density interpretation at negative parameters, centered affine transport, log-concavity and central plateaux, explicit Bernoulli cumulants and Bell moments in the generic parameter, and Gaussian-limit statements.

This report gives paper-level solutions or substantial advances for those gaps. For  $0 < r < 1$ , if

$$X_q = (1 - q) \sum_{n \geq 0} q^n U_n, \quad U_n \stackrel{\text{iid}}{\sim} \text{Unif}(0, 1),$$

then the central identity is

$$X_{-r} - \frac{1}{2} \stackrel{d}{=} \frac{1+r}{1-r} \left( X_r - \frac{1}{2} \right).$$

It gives the exact negative-parameter CDF, quantile, support, smooth density, entropy scaling, sinc product, and all derivative transports. It also diagnoses a sign issue that was deliberately left uninterpreted in the repository: the previously selected total function at  $q < 0$  is the *negative* of the Radon–Nikodym density. At  $q = -1/2$  one obtains

$$F_{-1/2}(x) = F\left(\frac{x+1}{3}\right), \quad g_{-1/2}(x) = \frac{2}{3} \text{up}\left(\frac{2x-1}{3}\right), \quad Q_{-1/2}(u) = 3F^{-1}(u) - 1,$$

which transports the complete Lambert– $W_{-1}$ , Legendre, and Thue–Morse derivative structures to the support  $[-1, 2]$ .

A second exact duality concerns normalized finite geometric sums: reversal of the digits proves invariance under  $q \mapsto q^{-1}$ . Consequently the bases 2,  $-2$ , 4,  $-4$ , and all real  $|q| > 1$ , have canonical weak limits given by the reciprocal-base laws. A residue-class decomposition yields a multiplication formula for the laws and their infinite sinc products. The same framework proves log-concavity for every real  $|q| < 1$ , exact central plateaux when  $|q| < 1/2$ , and a Gaussian boundary as  $|q| \uparrow 1$ . The latter comes with an explicit Berry–Esseen bound, closed standardized Bernoulli cumulants, Bell-polynomial moment corrections, and a quartic characteristic-function expansion. Reproducible numerical experiments accompany the proofs.

All claims of novelty in this report mean *novel relative to the inspected repository corpus*; no claim of global publication priority is made.

# 1 Scope, corpus audit, and novelty boundary

## 1.1 The inspected corpus

The review used the current integrated exposition `Fabijs_Function_and_Rvachev_Up.tex`, the mathematical glossary, the non-elementarity article, the Lean walkthrough, the consolidated semi-formalized-research-frontiers.tex, the archive index and its historical families, and the TeX sources stored under papers. The archive is important for priority control: many early notes are frozen historical stages, while the two large integrated sources are the canonical current synthesis. The principal public entry points are [92, 86, 74, 77, 75].

The repository’s own frontier box is unusually specific. It records that the law  $\mu_q$  is absolutely continuous for every  $|q| < 1$ , but that the chosen function called  $f_q$  has only been identified with the Radon–Nikodym density for  $0 < q < 1$ . It separately lists centered affine transport, log-concavity, strictness, central plateaux, explicit Bernoulli cumulants, Bell moments, and Gaussian limits as unpromoted research layers [86]. These declarations make it possible to state a sharp non-duplication target.

## 1.2 What is taken as established

The absorbed report imported seven foundational layers from the repository with a single blanket citation. In the merged volume six of them are proved, and each is now cited by its in-volume number; only the last remains a corpus input.

- (i) existence and almost-sure convergence of the weighted uniform series: [proposition 0.2.2](#);
- (ii) the affine independent-copy equation and uniqueness of its probability-law fixed point for every real  $|q| < 1$ , including negative  $q$ : [proposition 0.2.3](#) among laws with a finite first moment, and [lemma 2.1](#) below among all probability laws;
- (iii) reflection symmetry of the CDF: [proposition 0.2.2](#); its continuity: the law has a density ([theorem 3.1](#) for  $q \neq 0$ , the uniform law at  $q = 0$ );
- (iv) exact support for the geometric law: [\(0.2.3\)](#);
- (v) the density formula for  $0 < q < 1$ : [theorem 3.1](#), which covers every  $0 < |q| < 1$ ; the  $C^\infty$  regularity of  $g_q$  for  $q \neq 0$ : the Fourier envelope recorded at [\(4.5\)](#);
- (vi) the half-base bridge  $F_{1/2} = F$  and  $g_{1/2}(x) = 2 \operatorname{up}(2x - 1)$ : [proposition 0.2.5](#);
- (vii) the Bernoulli/Bell,  $q$ -binomial, sinc-product, Legendre, and Lambert– $W$  toolkits developed there ([existing repository input](#)). The one corpus theorem used inside a proof below is the Thue–Morse derivative-peak identity [\(4.2\)](#) with its higher-order flatness, consumed by [corollary 4.2](#); it is not reproved in this volume.

The report nevertheless restates enough notation to make every new proof self-contained.

## 1.3 New results relative to the corpus

| Frontier item              | Result obtained here                                                                                                                                                                               |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Negative-parameter density | Exact affine duality with the positive parameter; explicit CDF, quantile, support and smooth density. The repository’s selected total function is proved to equal minus the density when $q < 0$ . |
| Centered transport         | Complete coupling, characteristic-function, entropy, $L^p$ , Sobolev, Legendre and endpoint-asymptotic transport.                                                                                  |



| Frontier item                   | Result obtained here                                                                                                                                         |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dyadic negative base            | Exact formulas in terms of $F$ , $F^{-1}$ , and up, plus transferred Thue–Morse derivative peaks and Lambert– $W_{-1}$ phase.                                |
| Shape                           | Log-concavity for all real $ q  < 1$ ; exact central plateau for $0 \leq  q  < 1/2$ , with negative-base interval and height in closed form.                 |
| Multiplication structure        | Residue-class decomposition into independent $q^m$ -laws and the resulting multiplication law for cumulants and infinite sinc products.                      |
| Bases outside the unit interval | Exact finite $q \leftrightarrow q^{-1}$ reversal duality, canonical reciprocal-base limits, convergence bounds, and Bell/ $q$ -bracket reciprocity.          |
| Gaussian boundary               | Standardized sign-invariance, a Berry–Esseen rate, all standardized Bernoulli cumulants, Bell moment corrections, and a quartic mod-Gaussian-type expansion. |

## 2 Geometric-uniform preliminaries

### 2.1 The law, its centered form, and its transform

Let  $(U_n)_{n \geq 0}$  be independent  $\text{Unif}(0, 1)$  random variables. For real  $q$  with  $|q| < 1$ , define

$$X_q = (1 - q) \sum_{n=0}^{\infty} q^n U_n. \quad (2.1)$$

The series converges absolutely almost surely. Put

$$V_n = U_n - \frac{1}{2}, \quad Y_q = X_q - \frac{1}{2} = (1 - q) \sum_{n=0}^{\infty} q^n V_n. \quad (2.2)$$

The centering follows from  $(1 - q) \sum_{n \geq 0} q^n = 1$ . Every  $V_n$  is symmetric, which is the source of the negative-parameter duality. Note the normalization shift across the part boundary recorded in [table 1](#): Part VII's  $Y_q$  is this part's *uncentered*  $X_q$ , and this part's  $Y_q$  is its centering.

Write

$$F_q(x) = \mathbb{P}(X_q \leq x), \quad \psi_q(t) = \mathbb{E} e^{itY_q}, \quad \text{sinc}_{\text{rad}} z = \begin{cases} \sin z/z, & z \neq 0, \\ 1, & z = 0. \end{cases}$$

Independence gives the centered infinite sinc product  $\psi_q(t) = \prod_{n \geq 0} \text{sinc}_{\text{rad}}((1 - q)q^n t/2)$ , which is (0.3.5); (4.4) records it together with the functional equation  $\psi_q(t) = \text{sinc}_{\text{rad}}(\frac{(1-q)t}{2}) \psi_q(qt)$  and the zero list. The signs of the individual arguments disappear because  $\text{sinc}_{\text{rad}}$  is even, but the factor  $1 - q$  remains and will encode the affine stretch between  $q = r$  and  $q = -r$ .

### 2.2 The affine fixed point

If  $X'_q$  is an independent copy of  $X_q$ , then splitting off the first digit yields

$$X_q \stackrel{d}{=} (1 - q)U_0 + qX'_q. \quad (2.3)$$

[Proposition 0.2.3](#) proves that this equation characterizes the law uniquely among probability measures with a finite first moment, for every real  $|q| < 1$  and without any support or density assumption. The moment hypothesis, which that contraction argument needs for its complete metric, is not needed for the conclusion.

**Lemma 2.1** (Uniqueness among all probability laws). *Let  $|q| < 1$  be real and let  $\nu$  be any probability measure on  $\mathbb{R}$  such that  $Z \sim \nu$  satisfies  $Z \stackrel{d}{=} (1 - q)U + qZ'$  with  $U$  uniform on  $(0, 1)$ ,  $Z' \sim \nu$ , and  $U, Z'$  independent. Then  $\nu$  is the law of  $X_q$ .*

*Proof.* Iterating the equation  $N$  times gives, for every  $N \geq 1$ ,

$$Z \stackrel{d}{=} S_N + q^N Z_N, \quad S_N = (1 - q) \sum_{n < N} q^n U_n,$$

with  $Z_N \sim \nu$  independent of  $(U_n)_{n < N}$ . For every  $\varepsilon > 0$ ,

$$\mathbb{P}(|q^N Z_N| > \varepsilon) = \nu(\{z : |z| > \varepsilon |q|^{-N}\}) \longrightarrow 0 \quad (N \rightarrow \infty),$$

because  $\nu$  is a finite measure: no moment is used. Since  $S_N$  converges almost surely to a variable with the law of  $X_q$  (proposition 0.2.2), the sum  $S_N + q^N Z_N$  converges in probability to that variable, hence in law to the law of  $X_q$ ; as its law is  $\nu$  for every  $N$ ,  $\nu$  is that limit.  $\square$

In the positive range,  $X_r$  is supported on  $[0, 1]$ . At  $r = 1/2$ ,

$$F_{1/2} = F, \quad g_{1/2}(x) = 2 \operatorname{up}(2x - 1), \quad (2.4)$$

where  $F$  is the bounded Fabius function and  $\operatorname{up}$  is the Rvachev up-function.

### 2.3 $q$ -brackets, Bernoulli cumulants, and Bell polynomials

For an integer  $m \geq 1$ , set

$$[m]_q = 1 + q + \cdots + q^{m-1} = \frac{1 - q^m}{1 - q} \quad (2.5)$$

when  $q \neq 1$ , with the continuous value  $[m]_1 = m$ . We use the Bernoulli convention  $B_2 = 1/6$ ,  $B_4 = -1/30$ ,  $B_6 = 1/42$ . A centered uniform  $V \sim \operatorname{Unif}(-1/2, 1/2)$  has  $\kappa_{2m}(V) = B_{2m}/2m$  and  $\kappa_{2m+1}(V) = 0$ ; this and the resulting cumulants of  $X_q$  are theorem 0.4.2. The complete exponential Bell polynomial  $\mathcal{B}_n$  converts cumulants to moments:

$$\mathbb{E}Z^n = \mathcal{B}_n(\kappa_1(Z), \dots, \kappa_n(Z)). \quad (2.6)$$

## 3 Negative parameters: an exact affine duality

### 3.1 The coupling, and the notation it fixes

The affine sign conjugacy, with its digitwise coupling, is theorem 3.1: for  $0 < r < 1$ ,

$$X_{-r} - \frac{1}{2} \stackrel{d}{=} c_r \left( X_r - \frac{1}{2} \right), \quad c_r = \frac{1 + r}{1 - r}, \quad (3.1)$$

with almost-sure equality under the reflected digits  $U_n^- = \frac{1}{2} + (-1)^n(U_n - \frac{1}{2})$ , which are again i.i.d. uniform. Support, CDF and quantile transport are corollary 3.2 and (2.2); writing  $Q_q$  for the generalized inverse of  $F_q$ , we fix for the rest of this part the notation

$$\ell_r = -\frac{r}{1 - r}, \quad u_r = \frac{1}{1 - r}, \quad T_r(y) = \frac{1}{2} + c_r \left( y - \frac{1}{2} \right),$$

so that  $T_r$  is the increasing affine bijection of  $[0, 1]$  onto  $[\ell_r, u_r] = \operatorname{supp}(X_{-r})$ , and  $F_{-r} = F_r \circ T_r^{-1}$ ,  $Q_{-r} = T_r \circ Q_r$ .

### 3.2 The correct negative-parameter density and the sign diagnosis

The integrated source defines, as a total symbolic function,

$$\tilde{f}_q(x) = \frac{F_q(x/q) - F_q((x - (1 - q))/q)}{1 - q}. \quad (3.2)$$

For  $q > 0$  it proves that this is the density. At  $q < 0$ , division by  $q$  reverses the CDF interval. The next result resolves the open interpretation exactly.

**Theorem 3.1** (Radon–Nikodym density for every nonzero parameter). *For  $0 < |q| < 1$ , the law of  $X_q$  has density*

$$g_q(x) = \frac{1}{1 - q} \mathbb{P}(x - (1 - q) \leq qX'_q \leq x), \quad (3.3)$$

where  $X'_q$  is an independent copy. Equivalently,

$$\boxed{g_q(x) = \operatorname{sgn}(q) \tilde{f}_q(x)}. \quad (3.4)$$

Thus for  $-1 < q < 0$ , the repository's selected total function is not a probability density but its negative. In particular it integrates to  $-1$ , while  $g_q \geq 0$  and integrates to 1.

*Proof.* Condition on  $X'_q = y$  in equation (2.3). The conditional law is uniform on the interval  $[qy, qy + 1 - q]$ , of length  $1 - q > 0$ . Averaging the conditional density therefore gives equation (3.3). If  $q > 0$ , the event in that formula is

$$\frac{x - (1 - q)}{q} \leq X'_q \leq \frac{x}{q},$$

which gives  $g_q = \tilde{f}_q$ . If  $q < 0$ , the inequalities reverse and

$$g_q(x) = \frac{F_q((x - (1 - q))/q) - F_q(x/q)}{1 - q} = -\tilde{f}_q(x).$$

The CDF is continuous, so endpoint conventions do not matter.  $\square$

The density and all of its derivatives transport affinely,  $g_{-r}(x) = c_r^{-1} g_r(T_r^{-1}(x))$  and  $g_{-r}^{(k)}(x) = c_r^{-(k+1)} g_r^{(k)}(T_r^{-1}(x))$  for every  $k \geq 0$ , so  $g_{-r} \in C_c^\infty(\mathbb{R})$  with support  $[\ell_r, u_r]$ . This is corollary 3.2 and (3.3).

**Corollary 3.2** (Norm and entropy transport). *For  $1 \leq p \leq \infty$  and  $k \geq 0$ ,*

$$\left\| g_{-r}^{(k)} \right\|_p = c_r^{1/p-k-1} \left\| g_r^{(k)} \right\|_p, \quad (3.5)$$

with  $1/\infty = 0$ . The differential entropy  $h(X_r) = -\int g_r \log g_r$  is finite, with

$$\log(1 - r) \leq h(X_r) \leq 0, \quad (3.6)$$

and satisfies

$$h(X_{-r}) = h(X_r) + \log c_r. \quad (3.7)$$

For every order  $\alpha \in (0, 1) \cup (1, \infty)$  the Rényi entropy  $h_\alpha(X) = (1 - \alpha)^{-1} \log \int f_X^\alpha$  obeys the same law,  $h_\alpha(X_{-r}) = h_\alpha(X_r) + \log c_r$ .

*Proof.* Use (3.3) and the substitution  $y = 1/2 + (x - 1/2)/c_r$ . For (3.6): (3.3) gives  $0 \leq g_r \leq 1/(1 - r)$ , so  $-\int g_r \log g_r \geq -\log \|g_r\|_\infty \geq \log(1 - r)$ , while Jensen's inequality on the support  $[0, 1]$ , of length one, gives  $h(X_r) \leq \log 1 = 0$ . The two entropy identities are the Jacobian formula for  $aX + b$  with  $a = c_r > 0$ : if  $f_X$  is the density of  $X$ , then  $aX + b$  has density  $a^{-1} f_X((\cdot - b)/a)$ , so  $\int (a^{-1} f_X((x - b)/a))^\alpha dx = a^{1-\alpha} \int f_X^\alpha$ , which is  $h_\alpha(aX + b) = h_\alpha(X) + \log a$ , and  $\alpha \rightarrow 1$  gives the Shannon case.  $\square$

### 3.3 Characteristic functions and sinc products

Applying characteristic functions to the sign conjugacy gives the centered transform scaling  $\psi_{-r}(t) = \psi_r(c_r t)$ , the last clause of (3.1). On products it is the identity

$$\prod_{n \geq 0} \text{sinc}_{\text{rad}}\left(\frac{(1+r)r^n t}{2}\right) = \prod_{n \geq 0} \text{sinc}_{\text{rad}}\left(\frac{(1-r)r^n c_r t}{2}\right),$$

immediate from  $(1-r)c_r = 1+r$ .

## 4 The exact negative dyadic law

### 4.1 Fabius, inverse Fabius, and Rvachev formulas

Set  $r = 1/2$ . Then  $c_r = 3$ ,  $\ell_r = -1$ , and  $u_r = 2$ .

The CDF, density and quantile of the negative dyadic law are [corollary 3.2](#) at  $a = \frac{1}{2}$  combined with the half-base bridge (2.4):

$$F_{-1/2}(x) = F\left(\frac{x+1}{3}\right), \quad g_{-1/2}(x) = \frac{2}{3} \text{up}\left(\frac{2x-1}{3}\right), \quad Q_{-1/2}(u) = 3F^{-1}(u) - 1,$$

on the support  $[-1, 2]$ . What the sign diagnosis adds is the value of the source-selected total function there.

**Corollary 4.1** (Sign of the selected function at the negative dyadic base). *For every real  $x$ ,*

$$\tilde{f}_{-1/2}(x) = -\frac{2}{3} \text{up}\left(\frac{2x-1}{3}\right). \quad (4.1)$$

*Proof.* Combine [equation \(3.4\)](#) with the density displayed above.  $\square$

This is a direct and unexpectedly economical link between a negative geometric base and the inverse Fabius function. It also means that every exact formula for  $F^{-1}$  or its asymptotics is automatically an exact formula for the negative dyadic quantile.

### 4.2 Thue–Morse derivative peaks on a stretched dyadic grid

The frontier corpus proves the sharp Rvachev derivative identities

$$\left\| \text{up}^{(m)} \right\|_{\infty} = 2^{\binom{m+1}{2}}, \quad \text{up}^{(m)}(c_{m,j}) = \varepsilon_j 2^{\binom{m+1}{2}}, \quad (4.2)$$

where

$$c_{m,j} = -1 + 2^{-m} + j2^{1-m}, \quad 0 \leq j < 2^m. \quad (4.3)$$

It also proves higher-order flatness at this grid:  $\text{up}^{(k)}(c_{m,j}) = 0$  for  $k > m$ . Transport gives a new negative-base atlas.

**Corollary 4.2** (Negative dyadic Thue–Morse atlas). *Define*

$$x_{m,j} = \frac{1 + 3c_{m,j}}{2} = -1 + \frac{3}{2^{m+1}} + \frac{3j}{2^m}. \quad (4.4)$$

*Then*

$$g_{-1/2}^{(m)}(x_{m,j}) = \varepsilon_j \left(\frac{2}{3}\right)^{m+1} 2^{\binom{m+1}{2}}, \quad (4.5)$$

$$\left\| g_{-1/2}^{(m)} \right\|_{\infty} = \frac{2^{\binom{m+1}{2} + m + 1}}{3^{m+1}}, \quad (4.6)$$

$$g_{-1/2}^{(k)}(x_{m,j}) = 0 \quad (k > m). \quad (4.7)$$

*Thus the same Thue–Morse signs occur on a grid whose spacing is stretched from  $2^{1-m}$  to  $3 \cdot 2^{-m}$ .*

*Proof.* From the negative dyadic density of [corollary 3.2](#),

$$g_{-1/2}^{(m)}(x) = \left(\frac{2}{3}\right)^{m+1} \text{up}^{(m)}\left(\frac{2x-1}{3}\right).$$

Set  $(2x-1)/3 = c_{m,j}$  and use [equation \(4.2\)](#) and the higher-flatness result.  $\square$

### 4.3 Legendre-series transport

The absorbed report assumed the Legendre expansion of  $\text{up}$  in  $L^2$  and asserted its transport pointwise. The assumption is not needed: smoothness on the closed interval already forces absolute and uniform convergence, so the identities below hold at every point.

**Proposition 4.3** (Legendre transport). *Let  $P_k$  be the Legendre polynomials on  $[-1, 1]$  and put*

$$b_n = \frac{4n+1}{2} \int_{-1}^1 \text{up}(t) P_{2n}(t) dt, \quad a_n^{(r)} = (4n+1) \int_0^1 g_r(y) P_{2n}(2y-1) dy \quad (0 < r < 1).$$

*Then  $b_n = \mathcal{O}(n^{-k})$  and  $a_n^{(r)} = \mathcal{O}_r(n^{-k})$  for every  $k$ , the even Legendre expansion*

$$\text{up}(t) = \sum_{n=0}^{\infty} b_n P_{2n}(t), \quad -1 \leq t \leq 1, \quad (4.8)$$

*converges absolutely and uniformly, and so does  $g_r(y) = \sum_{n \geq 0} a_n^{(r)} P_{2n}(2y-1)$  on  $[0, 1]$ . Consequently the negative dyadic density has the pointwise expansion*

$$g_{-1/2}(x) = \frac{2}{3} \sum_{n=0}^{\infty} b_n P_{2n}\left(\frac{2x-1}{3}\right), \quad -1 \leq x \leq 2, \quad (4.9)$$

*and, for every  $0 < r < 1$  and  $\ell_r \leq x \leq u_r$ ,*

$$g_{-r}(x) = \frac{1}{c_r} \sum_{n \geq 0} a_n^{(r)} P_{2n}\left(\frac{2x-1}{c_r}\right). \quad (4.10)$$

*No coefficients need to be recomputed: only the Jacobian and the affine Legendre coordinate change.*

*Proof.* Let  $f \in C^\infty[-1, 1]$  be even; the two cases are  $f = \text{up}$  restricted to  $[-1, 1]$  and  $f(t) = g_r((t+1)/2)$ , both smooth on the closed interval because  $\text{up}$  and  $g_r$  are  $C^\infty$  on  $\mathbb{R}$  ([corollary 3.2](#) and (4.5)) and both even because the two densities are symmetric about their centers. The Legendre operator  $\mathcal{L}f = ((1-t^2)f')'$  satisfies  $\mathcal{L}P_k = -k(k+1)P_k$ , and it is symmetric on  $C^2[-1, 1]$  with no boundary condition, since the weight  $1-t^2$  vanishes at  $\pm 1$ :  $\int_{-1}^1 (\mathcal{L}f)g = -\int_{-1}^1 (1-t^2)f'g' = \int_{-1}^1 f\mathcal{L}g$ . Hence for every  $k \geq 1$  and  $n \geq 1$ ,

$$\int_{-1}^1 f P_{2n} = \frac{(-1)^k}{(2n(2n+1))^k} \int_{-1}^1 (\mathcal{L}^k f) P_{2n}, \quad \text{so} \quad \left| \int_{-1}^1 f P_{2n} \right| \leq \frac{2 \|\mathcal{L}^k f\|_\infty}{(2n(2n+1))^k},$$

using  $\sup_{[-1,1]} |P_{2n}| = 1$ . Multiplying by the normalizing factor  $(4n+1)/2$  gives coefficients of order  $n^{1-2k}$  for every  $k$ , which is the stated decay, and  $\sum_n |b_n| < \infty$ . Since  $|P_{2n}| \leq 1$  on  $[-1, 1]$ , the series  $\sum_n b_n P_{2n}$  converges absolutely and uniformly there to a continuous function; by completeness of the Legendre polynomials in  $L^2(-1, 1)$  its  $L^2$  limit is  $f$  (the odd coefficients vanish by evenness), so the continuous limit agrees with the continuous  $f$  almost everywhere, hence everywhere. The same argument gives the expansion of  $g_r$ . Finally  $g_{-1/2}(x) = \frac{2}{3} \text{up}((2x-1)/3)$  and  $g_{-r}(x) = c_r^{-1} g_r(T_r^{-1}(x))$  with  $2T_r^{-1}(x) - 1 = (2x-1)/c_r$  ([corollary 3.2](#)); substituting the uniformly convergent expansions gives (4.9) and (4.10) pointwise.  $\square$

This supplies a direct bridge from the positive- $q$  Legendre theory to every negative parameter.

#### 4.4 Exact transport of the Lambert- $W_{-1}$ endpoint phase

Let  $L = \log 2$ . The function  $\lambda \mapsto \lambda 2^{-\lambda} = \lambda e^{-L\lambda}$  increases on  $(0, 1/L)$  to its maximum  $1/(eL)$  and decreases to 0 afterwards, so for  $0 < z < 1/(eL)$  the equation

$$z = \lambda 2^{-\lambda} \quad (4.11)$$

has exactly two positive solutions;  $\lambda(z)$  denotes the larger one,  $\lambda(z) > 1/L$ . Writing the equation as  $(-L\lambda)e^{-L\lambda} = -Lz$  with  $-Lz \in (-1/e, 0)$ , its two real solutions are  $-L\lambda = W_0(-Lz) \in (-1, 0)$  and  $-L\lambda = W_{-1}(-Lz) < -1$  [52], and the larger  $\lambda$  is the second:

$$\lambda(z) = -\frac{1}{L}W_{-1}(-Lz), \quad 0 < z < \frac{1}{eL}. \quad (4.12)$$

The corpus's all-orders endpoint expansion of the Fabius function is written in this coordinate, and it emphasizes that its periodic coefficient functions must be evaluated at the *exact* phase  $\lambda(z)$  rather than at a phase with merely bounded error. That expansion is imported here as **existing repository input**: it is neither restated nor reproved, and nothing below depends on its form. What this part proves is the transport of the coordinate; the absorbed report presented the boxed formula below without a statement or proof, and it is one change of variables.

**Proposition 4.4** (Exact transport of the endpoint coordinate). *Write  $x = -1 + \xi$  near the left endpoint of the support of  $X_{-1/2}$ . Then for every real  $\xi$  and every  $k \geq 0$ ,*

$$F_{-1/2}(-1 + \xi) = F(\xi/3), \quad g_{-1/2}^{(k)}(-1 + \xi) = 3^{-(k+1)}g_{1/2}^{(k)}(\xi/3), \quad (4.13)$$

and for  $0 < \xi < 3/(eL)$  the transported phase is

$$\boxed{\lambda_{-}(\xi) = \lambda(\xi/3) = -\frac{1}{L}W_{-1}\left(-\frac{L\xi}{3}\right).} \quad (4.14)$$

Consequently, if  $F(z) = \mathcal{E}(z, \lambda(z)) + \rho(z)$  as  $z \downarrow 0$  for some expression  $\mathcal{E}$  and remainder  $\rho$  — in particular for the corpus's all-orders expansion with its one-periodic coefficient functions of  $\lambda$  — then  $F_{-1/2}(-1 + \xi) = \mathcal{E}(\xi/3, \lambda_{-}(\xi)) + \rho(\xi/3)$ : the coefficient functions are evaluated at  $\lambda_{-}$ , the remainder bound transfers verbatim, and each derivative acquires the factor  $3^{-k}$ . The right endpoint follows from the symmetry of both laws about  $1/2$ , and for the quantile

$$Q_{-1/2}(u) + 1 = 3F^{-1}(u), \quad 0 < u < 1. \quad (4.15)$$

*Proof.*  $F_{-1/2}(x) = F((x+1)/3)$ ,  $Q_{-1/2}(u) = 3F^{-1}(u) - 1$  and  $g_{-1/2}^{(k)}(x) = 3^{-(k+1)}g_{1/2}^{(k)}((x+1)/3)$  are **corollary 3.2** and **(3.3)** at  $a = \frac{1}{2}$ , where  $\lambda_{1/2} = 3$  and  $\beta_{1/2} = 1$ ; put  $x = -1 + \xi$ . The phase line is the definition of  $\lambda$  at the argument  $\xi/3$ , admissible exactly when  $\xi/3 < 1/(eL)$ . The transfer statement is the substitution  $z = \xi/3$  in the assumed expansion, and the derivative factors are those of (4.13).  $\square$

## 5 Shape theory: log-concavity and exact plateaux

### 5.1 Log-concavity for all real parameters

**Definition 5.1.** A probability measure  $\mu$  on  $\mathbb{R}$  is log-concave if for nonempty compact sets  $A, B$  and  $0 < \theta < 1$ ,

$$\mu(\theta A + (1 - \theta)B) \geq \mu(A)^\theta \mu(B)^{1-\theta}.$$

An integrable function is log-concave when its logarithm is concave on its positivity set, with the usual extended-value convention.

In this sense the law of  $X_q$  is log-concave for every real  $|q| < 1$ , including  $q = 0$ , where the density is  $\mathbf{1}_{[0,1]}$ ; for  $q \neq 0$  the smooth density  $g_q$  is log-concave on  $\mathbb{R}$ . This is [theorem 4.2](#): there the finite partial sums  $(1 - q) \sum_{n \leq N} q^n U_n$  are convolutions of uniform laws on compact intervals, a negative coefficient merely reflects an interval, and affine images and convolutions preserve log-concavity by the Prékopa theorem [5]. Its proof there then passes to the limit locally uniformly through the sinc-product tail bound. This supplies the global shape theorem that the repository’s frontier list recorded as open when the absorbed report was written; it implies unimodality, and with reflection symmetry it places the mode at  $1/2$ . It does *not* by itself imply strict log-concavity: the same theorem exhibits a whole parameter region where strictness is impossible.

## 5.2 A dominant-digit plateau lemma

**Lemma 5.2** (Uniform convolution plateau). *Let  $U \sim \text{Unif}(0, 1)$ , independent of a random variable  $Y$  supported on  $[\alpha, \beta]$ . Let  $a > 0$  and  $X = aU + Y$ . If  $a \geq \beta - \alpha$ , then the density of  $X$  is exactly  $1/a$  on*

$$[\beta, \alpha + a]. \quad (5.1)$$

*When  $a > \beta - \alpha$ , this is a nondegenerate interval.*

*Proof.* Conditioning on  $Y$ ,

$$f_X(x) = \frac{1}{a} \mathbb{P}(x - a \leq Y \leq x).$$

For  $x \in [\beta, \alpha + a]$ , one has  $x \geq \beta$  and  $x - a \leq \alpha$ , so the event contains the whole support of  $Y$  and has probability one.  $\square$

Applied to the fixed-point equation  $X_r = (1 - r)U + rX'_r$  with  $\alpha = 0$ ,  $\beta = r$  and  $a = 1 - r$ , the lemma gives the exact central plateaux of [theorem 4.2](#): for  $0 < r \leq 1/2$  the positive density equals  $1/(1 - r)$  on  $[r, 1 - r]$ , the negative density equals  $1/(1 + r)$  on  $[r^2/(1 - r), 1 - r^2/(1 - r)]$  – the affine image of the first under [corollary 3.2](#) – each interval collapsing to the point  $x = 1/2$  at  $r = 1/2$ , and in either law the plateau carries mass  $(1 - 2r)/(1 - r)$ .

Strictness above the transition, and the maximality of the plateau interval, are Part VI’s conjecture on strict log-concavity in the overlapping regime, recorded once in the discussion following [theorem 4.2](#); by [theorem 3.1](#) the negative cases follow from the positive ones. The critical value  $r = 1/2$  is structurally natural: it is exactly where the first uniform digit and the entire tail have equal support length. Proving strictness above the transition likely requires an equality-case analysis for repeated Prékopa convolution or a direct argument from the refinement equation.

## 6 Residue-class decomposition and multiplication formulas

### 6.1 Decomposition into independent $q^m$ laws

Splitting the defining series into residue classes  $n = mk + j$  writes  $X_q$  as the weighted independent sum  $X_q \stackrel{\text{d}}{=} [m]_q^{-1} \sum_{j < m} q^j X_{q^m}^{(j)}$ , with the same weighted identity for the centered variables, the transform law  $\psi_q(t) = \prod_{j < m} \psi_{q^m}(q^j t / [m]_q)$  and the cumulant law  $\kappa_s(X_q) = ([m]_{q^s} / [m]_q^s) \kappa_s(X_{q^m})$ . All four are [theorem 5.1](#) and (5.3), proved there as identities of analytic germs valid outside the unit disc as well, and identified with the Pochhammer dissection  $(a; q)_\infty = \prod_{r < m} (aq^r; q^m)_\infty$ .



## 6.2 Even–odd splitting

For  $m = 2$ , (5.2) becomes

$$X_r \stackrel{d}{=} \frac{X_{r^2}^{(0)} + rX_{r^2}^{(1)}}{1+r}, \quad (6.1)$$

$$X_{-r} \stackrel{d}{=} \frac{X_{r^2}^{(0)} - rX_{r^2}^{(1)}}{1-r}. \quad (6.2)$$

Thus a negative-base law is a normalized difference of two independent positive-base laws. Combining this with theorem 3.1 gives the distribution identity

$$\frac{X_{r^2}^{(0)} - rX_{r^2}^{(1)}}{1-r} - \frac{1}{2} \stackrel{d}{=} c_r \left( \frac{X_{r^2}^{(0)} + rX_{r^2}^{(1)}}{1+r} - \frac{1}{2} \right). \quad (6.3)$$

For centered transforms,  $\psi_{q^2}$  is even, so

$$\psi_{-r}(t) = \psi_{r^2}\left(\frac{t}{1-r}\right) \psi_{r^2}\left(\frac{rt}{1-r}\right). \quad (6.4)$$

Together with  $\psi_{-r}(t) = \psi_r(c_r t)$  this is the sinc splitting (5.6), whose  $r = 1/2$  case  $\psi_{1/2}(3t) = \psi_{1/4}(2t)\psi_{1/4}(t)$  is precisely the interlacing of the even and odd factors of the dyadic infinite sinc product into two quarter-base products.

## 6.3 Computational significance

The residue formula provides a multiscale density algorithm. Choose  $m$  so that  $|q|^m$  is small. The  $q^m$ -law then has a broad exact plateau and short refinement depth; convolving its independently scaled copies reconstructs  $g_q$ . This can outperform direct evaluation of a long sinc product when  $|q|$  is close to one. For negative  $q$ , taking even  $m$  reduces every building block to a positive parameter.

# 7 Reciprocal bases and finite geometric normalization

## 7.1 Exact $q \leftrightarrow q^{-1}$ reversal

For an integer  $N \geq 1$ , define the normalized finite law

$$X_{q,N} = \frac{1}{[N]_q} \sum_{n=0}^{N-1} q^n U_n, \quad (7.1)$$

whenever  $[N]_q \neq 0$ . The coefficients sum to one, so  $\mathbb{E}X_{q,N} = 1/2$ . At  $q = 1$ , the continuous extension is the arithmetic mean  $N^{-1} \sum U_n$ .

**Theorem 7.1** (Finite reciprocal-base duality). *Let  $q \neq 0$  and suppose both normalizations are defined. Then*

$$\boxed{X_{q,N} \stackrel{d}{=} X_{1/q,N}}. \quad (7.2)$$

*More strongly, if the digits on the reciprocal side are reversed, the two random variables are equal almost surely.*

*Proof.* The  $q$ -bracket reciprocity identity is

$$[N]_{1/q} = q^{1-N} [N]_q. \quad (7.3)$$

Therefore

$$\frac{(1/q)^{N-1-n}}{[N]_{1/q}} = \frac{q^{-N+1+n}}{q^{1-N}[N]_q} = \frac{q^n}{[N]_q}.$$

The coefficient of  $U_n$  at base  $q$  is exactly the coefficient of the reversed digit  $U_{N-1-n}$  at base  $1/q$ . Since the digit vector is exchangeable, the laws agree; using the reversed vector gives pathwise equality.  $\square$

## 7.2 Canonical limits for $|q| > 1$

**Theorem 7.2** (Reciprocal-base compactification). *Let  $q$  be real with  $|q| > 1$  and put  $p = 1/q$ , so that  $0 < |p| < 1$ ; both normalizations  $[N]_q$  and  $[N]_p$  are then nonzero for every  $N \geq 1$ . Then, for every  $N$ ,  $X_{q,N} \stackrel{d}{=} X_{p,N}$ , and*

$$X_{q,N} \Longrightarrow X_p \quad (N \rightarrow \infty). \quad (7.4)$$

*Thus the normalized finite laws at every real base outside the unit interval have a canonical weak limit. Explicitly,*

$$\begin{aligned} q = 2 : \quad & X_{q,N} \Longrightarrow X_{1/2}, \\ q = -2 : \quad & X_{q,N} \Longrightarrow X_{-1/2} \stackrel{d}{=} 3X_{1/2} - 1, \\ q = 4 : \quad & X_{q,N} \Longrightarrow X_{1/4}, \\ q = -4 : \quad & X_{q,N} \Longrightarrow X_{-1/4} \stackrel{d}{=} \frac{1}{2} + \frac{5}{3} \left( X_{1/4} - \frac{1}{2} \right). \end{aligned}$$

*Proof.* For real  $|q| > 1$  one has  $q^N \neq 1$ , so  $[N]_q \neq 0$ , and  $|p| < 1$  gives  $[N]_p \neq 0$ ; the identity in law is therefore [theorem 7.1](#). Since  $|p| < 1$ ,  $[N]_p \rightarrow (1-p)^{-1}$  and  $\sum_{n < N} p^n U_n \rightarrow \sum_{n \geq 0} p^n U_n$  almost surely ([proposition 0.2.2](#)), so  $X_{p,N} = [N]_p^{-1} \sum_{n < N} p^n U_n \rightarrow (1-p) \sum_{n \geq 0} p^n U_n = X_p$  almost surely. Almost-sure convergence implies convergence in law, and the identity  $X_{q,N} \stackrel{d}{=} X_{p,N}$  transfers the weak limit to  $X_{q,N}$ . The special cases use [theorem 3.1](#).  $\square$

**Proposition 7.3** (Exponential coupling rate). *For  $0 \leq p < 1$ , there is a common-digit coupling such that*

$$|X_{p,N} - X_p| \leq p^N \quad \text{almost surely.} \quad (7.5)$$

*For  $p = -r$ ,  $0 < r < 1$ , and even  $N$ , there is a coupling such that*

$$|X_{-r,N} - X_{-r}| \leq c_r r^N. \quad (7.6)$$

*Hence the same bounds hold for the  $W_\infty$  Wasserstein distance.*

*Proof.* For  $p \geq 0$ , write

$$A_N = (1-p) \sum_{n < N} p^n U_n, \quad T_N = (1-p) \sum_{n \geq N} p^n U_n.$$

Then  $0 \leq A_N \leq 1 - p^N$ ,  $0 \leq T_N \leq p^N$ ,  $X_{p,N} = A_N / (1 - p^N)$ , and  $X_p = A_N + T_N$ . Hence

$$X_{p,N} - X_p = \frac{p^N}{1 - p^N} A_N - T_N$$

is the difference of two numbers in  $[0, p^N]$ .

For negative  $p = -r$ , finite digit reflection ([theorem 3.1](#)) gives

$$X_{-r,N} - \frac{1}{2} \stackrel{d}{=} c_{r,N} \left( X_{r,N} - \frac{1}{2} \right), \quad c_{r,N} = c_r \frac{1 - r^N}{1 - (-r)^N}.$$

When  $N$  is even,  $c_{r,N} = c_r$ . Couple both finite and infinite negative laws by the same affine images of the positive coupling, and apply [equation \(7.5\)](#).  $\square$

### 7.3 Bell/ $q$ -bracket reciprocity

Let  $u_1 = 1/2$ , and for  $s \geq 2$  let  $u_s$  be the  $s$ -th cumulant of a uniform  $U$ , so  $u_s = B_s/s$  for even  $s$  and  $u_s = 0$  for odd  $s > 1$ .

**Theorem 7.4** (Finite cumulants and moments). *For every admissible  $q, N$ ,*

$$\kappa_s(X_{q,N}) = u_s \frac{[N]_{q^s}}{[N]_q^s}. \quad (7.7)$$

Consequently

$$\mathbb{E}X_{q,N}^k = \mathcal{B}_k\left(u_1, u_2 \frac{[N]_{q^2}}{[N]_q^2}, \dots, u_k \frac{[N]_{q^k}}{[N]_q^k}\right). \quad (7.8)$$

Every ratio in [equation \(7.7\)](#) is invariant under  $q \mapsto q^{-1}$ , giving an algebraic Bell-polynomial proof of the moment part of [theorem 7.1](#).

*Proof.* Cumulants add over independent variables and scale homogeneously, so

$$\kappa_s(X_{q,N}) = u_s \sum_{n=0}^{N-1} \left(\frac{q^n}{[N]_q}\right)^s = u_s \frac{[N]_{q^s}}{[N]_q^s}.$$

Apply the moment–cumulant Bell formula. Finally,

$$[N]_{q^{-s}} = q^{s(1-N)}[N]_{q^s}$$

and  $[N]_{q^{-1}}^s = q^{s(1-N)}[N]_q^s$ , so the ratio is reciprocal-invariant.  $\square$

This reciprocity is the probability-law analogue of the involution underlying Gaussian binomial reciprocity. It suggests a systematic dictionary between finite geometric uniform sums, symmetric functions of the coefficient multiset, and denominator-free  $q$ -identities. It is invariant at every finite  $N$ , but its  $N \rightarrow \infty$  limit is not one rational function of  $q$ : the limit acquires a sign on  $|q| > 1$ , and that sign is exactly what separates the reciprocal germ from the reciprocal-base limit law ([remark 3.4](#)).

## 8 The Gaussian boundary $|q| \uparrow 1$

### 8.1 Sign-invariant standardization

Cumulants and variance are [theorem 0.4.2](#) and (0.4.5): for  $-1 < q < 1$ ,  $\sigma_q^2 = \text{Var}(X_q) = (1 - q)/(12(1 + q))$ . Define

$$Z_q = \frac{X_q - 1/2}{\sigma_q}. \quad (8.1)$$

Since  $\sigma_{-r} = c_r \sigma_r$ , the sign collapses under standardization,  $Z_{-r} \stackrel{d}{=} Z_r$ , so the standardized family depends only on  $r = |q|$ ; this is (3.6), whose support-normalized refinement (3.7) says that every normalized shape statistic is an even function of  $q$ .

The centered characteristic function of  $Z_r$  has the useful normalized product

$$\Psi_r(t) = \mathbb{E}e^{itZ_r} = \prod_{n=0}^{\infty} \text{sinc}_{\text{rad}}\left(\sqrt{3(1-r^2)} r^n t\right). \quad (8.2)$$

## 8.2 A quantitative central limit theorem

**Theorem 8.1** (Berry–Esseen Gaussian boundary). *Let  $C_{\text{BE}}$  be any universal constant valid in the classical Berry–Esseen inequality for independent, non-identically distributed summands. Then for  $0 < r < 1$ ,*

$$\sup_{x \in \mathbb{R}} |\mathbb{P}(Z_r \leq x) - \Phi(x)| \leq C_{\text{BE}} \frac{3\sqrt{3}}{4} \frac{(1-r^2)^{3/2}}{1-r^3}. \quad (8.3)$$

The same bound holds for  $Z_{-r}$ . In particular,

$$Z_{\pm r} \implies N(0, 1) \quad (r \uparrow 1), \quad (8.4)$$

and the right side of [equation \(8.3\)](#) is

$$C_{\text{BE}} \sqrt{\frac{3}{2}} \sqrt{1-r} (1 + \mathcal{O}(1-r)). \quad (8.5)$$

*Proof.* Write  $Y_q = \sum_{n \geq 0} a_{n,q} V_n$  with  $a_{n,q} = (1-q)q^n$  and  $r = |q|$ . Since

$$\mathbb{E}|V_n|^3 = \frac{1}{32}, \quad \text{Var}(V_n) = \frac{1}{12},$$

we have

$$\begin{aligned} \sum_{n \geq 0} |a_{n,q}|^3 &= \frac{|1-q|^3}{1-r^3}, \\ \sum_{n \geq 0} a_{n,q}^2 &= \frac{(1-q)^2}{1-r^2}. \end{aligned}$$

The Berry–Esseen Lyapunov ratio is therefore

$$\begin{aligned} \frac{\sum_n \mathbb{E}|a_{n,q} V_n|^3}{(\sum_n \text{Var}(a_{n,q} V_n))^{3/2}} &= \frac{(1/32)}{(1/12)^{3/2}} \frac{(1-r^2)^{3/2}}{1-r^3} \\ &= \frac{3\sqrt{3}}{4} \frac{(1-r^2)^{3/2}}{1-r^3}. \end{aligned}$$

The Berry–Esseen inequality for independent, non-identically distributed summands [[1](#), [18](#), [94](#)] is a finite-sum statement, so it is applied to the truncations and the limit is taken afterwards. Put

$$S_N = \sum_{n < N} a_{n,q} V_n, \quad s_N^2 = \text{Var}(S_N) = \frac{1}{12} \sum_{n < N} a_{n,q}^2, \quad Z^{(N)} = \frac{S_N}{s_N}.$$

For each  $N \geq 1$  the inequality gives  $\sup_x |\mathbb{P}(Z^{(N)} \leq x) - \Phi(x)| \leq C_{\text{BE}} L_N$  with the truncated Lyapunov ratio

$$L_N = \frac{\sum_{n < N} \mathbb{E}|a_{n,q} V_n|^3}{s_N^3} = \frac{3\sqrt{3}}{4} \frac{(1-r^{3N})(1-r^2)^{3/2}}{(1-r^3)(1-r^{2N})^{3/2}}.$$

As  $N \rightarrow \infty$ ,  $L_N \rightarrow L_\infty = \frac{3\sqrt{3}}{4} (1-r^2)^{3/2} / (1-r^3)$ , the ratio  $L_N/L_\infty = (1-r^{3N})/(1-r^{2N})^{3/2}$  exceeding 1 for every finite  $N$ , so that no single truncation yields the boxed constant. Now  $S_N \rightarrow Y_q$  almost surely ([proposition 0.2.2](#)) and  $s_N \rightarrow \sigma_q$ , so  $Z^{(N)} \rightarrow Z_r$  almost surely, hence in law; and  $Z_r$  has a density ([theorem 3.1](#)), so its distribution function is continuous and  $\mathbb{P}(Z^{(N)} \leq x) \rightarrow \mathbb{P}(Z_r \leq x)$  for every  $x$ . Therefore, for each fixed  $x$ ,

$$|\mathbb{P}(Z_r \leq x) - \Phi(x)| = \lim_{N \rightarrow \infty} |\mathbb{P}(Z^{(N)} \leq x) - \Phi(x)| \leq C_{\text{BE}} \lim_{N \rightarrow \infty} L_N = C_{\text{BE}} L_\infty,$$

and the supremum over  $x$  is [equation \(8.3\)](#). The sign collapse [\(3.6\)](#) handles  $-r$ . Expanding  $1-r^2 \sim 2(1-r)$  and  $1-r^3 \sim 3(1-r)$  gives [equation \(8.5\)](#).  $\square$

### 8.3 Closed standardized Bernoulli cumulants

**Corollary 8.2** (Low standardized cumulants and the order estimate). *The even standardized cumulants  $\lambda_{2m}(r) = \kappa_{2m}(Z_r) = \kappa_{2m}(Z_{-r})$  are (0.4.6), and all odd cumulants vanish. In particular,*

$$\lambda_4(r) = -\frac{6}{5} \frac{1-r^2}{1+r^2}, \quad (8.6)$$

$$\lambda_6(r) = \frac{48}{7} \frac{(1-r^2)^2}{1+r^2+r^4}. \quad (8.7)$$

For fixed  $m \geq 2$ ,  $\lambda_{2m}(r) = \mathcal{O}((1-r)^{m-1})$ .

*Proof.* Put  $m = 2$  and  $m = 3$  in (0.4.6) with  $B_4 = -1/30$  and  $B_6 = 1/42$ . Odd cumulants vanish by symmetry, and the negative law has the same standardized distribution by (3.6). Factoring  $1 - r^{2m} = (1 - r^2)(1 + r^2 + \dots + r^{2m-2})$  gives the order estimate.  $\square$

### 8.4 Bell-polynomial moment correction

**Corollary 8.3** (First correction to Gaussian moments). *For every fixed integer  $m \geq 2$ ,*

$$\mathbb{E}Z_r^{2m} = (2m-1)!! + \binom{2m}{4}(2m-5)!! \lambda_4(r) + \mathcal{O}((1-r)^2) \quad (8.8)$$

$$= (2m-1)!! - \frac{6}{5} \binom{2m}{4}(2m-5)!! \frac{1-r^2}{1+r^2} + \mathcal{O}((1-r)^2). \quad (8.9)$$

All odd moments vanish.

*Proof.* Use the moment–cumulant Bell expansion:  $\mathbb{E}Z_r^{2m}$  is the sum over set partitions of  $\{1, \dots, 2m\}$  of the product of the cumulants  $\lambda_{|B|}$  over the blocks  $B$ . Blocks of odd size contribute 0, since all odd cumulants vanish. The Gaussian term consists entirely of pair blocks ( $\lambda_2 = 1$ ) and equals  $(2m-1)!!$ . Since  $\lambda_{2k}(r) = \mathcal{O}((1-r)^{k-1})$  by corollary 8.2, a partition with blocks of sizes  $2k_1, \dots, 2k_j$  contributes  $\mathcal{O}((1-r)^{\sum_i(k_i-1)})$ ; the only partitions of order exactly  $1-r$  have one block of size four and  $m-2$  pairs, and their number is

$$\binom{2m}{4}(2m-5)!!,$$

with the convention  $(-1)!! = 1$  at  $m = 2$ , where the formula  $\mathbb{E}Z_r^4 = 3 + \lambda_4(r)$  is exact. Every block of size at least six, and every product of two fourth cumulants, contributes  $\mathcal{O}((1-r)^2)$ ; the implied constant depends on  $m$ , through the number of such partitions. Insert equation (8.6).  $\square$

### 8.5 Quartic characteristic correction

**Theorem 8.4** (First mod-Gaussian-type correction). *For  $T > 0$  put  $r_0(T) = \sqrt{\max\{0, 1 - 1/(3T^2)\}}$ . For  $r_0(T) < r < 1$  and  $|t| \leq T$  every factor of (8.2) is positive,  $\log \Psi_r(t)$  is the absolutely convergent sum of the logarithms of the factors, and*

$$\log \Psi_r(t) = -\frac{t^2}{2} - \frac{1-r^2}{20(1+r^2)}t^4 + \mathcal{O}_T((1-r)^2), \quad (8.10)$$

where the remainder term, written  $\varrho_r(t)$ , is explicitly bounded by

$$|\varrho_r(t)| \leq \frac{27}{2600} |t|^6 \frac{(1-r^2)^2}{1+r^2+r^4} \leq \frac{T^6(1-r)^2}{24}. \quad (8.11)$$

The restriction on  $r$  cannot be dropped:  $\Psi_r$  vanishes at  $t = \pi/\sqrt{3(1-r^2)}$ , which lies in  $[-T, T]$  as soon as  $1-r^2 \geq \pi^2/(3T^2)$  (at  $r = 0$ ,  $\Psi_0(t) = \text{sinc}_{\text{rad}}(\sqrt{3}t)$  vanishes at  $t = \pi/\sqrt{3} \approx 1.81$ ), and there  $\log \Psi_r(t)$  is undefined. Consequently, for every fixed real  $t$ ,

$$\lim_{r \uparrow 1} \frac{\log \Psi_r(t) + t^2/2}{1-r} = -\frac{t^4}{20}. \quad (8.12)$$

*Proof.* Put  $z_n = \sqrt{3(1-r^2)} r^n t$ . For  $r > r_0(T)$  and  $|t| \leq T$  one has  $3(1-r^2)T^2 < 1$ , hence  $|z_n| \leq |z_0| < 1 < \pi$  for every  $n$ : all factors of (8.2) are positive, and since  $\sum_n z_n^2 < \infty$  the logarithm of the product is the absolutely convergent sum of the logarithms. For  $|z| < \pi$ , Euler's product  $\text{sinc}_{\text{rad}} z = \prod_{k \geq 1} (1 - z^2/(\pi^2 k^2))$  gives the exact series

$$\begin{aligned} \log \text{sinc}_{\text{rad}} z &= - \sum_{m \geq 1} \frac{\zeta(2m)}{m \pi^{2m}} z^{2m} = -\frac{z^2}{6} - \frac{z^4}{180} + R(z), \\ |R(z)| &\leq z^6 \sum_{m \geq 3} \frac{\zeta(2m)}{m \pi^{2m}} \leq \frac{z^6}{2600} \quad (|z| \leq 1), \end{aligned}$$

all terms having one sign; the constant is  $\zeta(6)/(3\pi^6) + \zeta(8)/(4\pi^8) + \dots = 0.000381\dots < 1/2600$ . Summing over  $n$ ,

$$\begin{aligned} \sum_{n \geq 0} z_n^2 &= 3t^2, \\ \sum_{n \geq 0} z_n^4 &= 9t^4 \frac{(1-r^2)^2}{1-r^4} = 9t^4 \frac{1-r^2}{1+r^2}, \\ \sum_{n \geq 0} |z_n|^6 &= 27|t|^6 \frac{(1-r^2)^3}{1-r^6} = 27|t|^6 \frac{(1-r^2)^2}{1+r^2+r^4}, \end{aligned}$$

so  $\log \Psi_r(t) = -\frac{1}{2}t^2 - \frac{1}{20} \frac{1-r^2}{1+r^2} t^4 + \varrho_r(t)$  with  $\varrho_r(t) = \sum_n R(z_n)$ , and  $|\varrho_r(t)| \leq \frac{1}{2600} \sum_n |z_n|^6$  is the first bound of (8.11); the second uses  $(1-r^2)^2 \leq 4(1-r)^2$ ,  $1+r^2+r^4 \geq 1$ , and  $108/2600 < 1/24$ . For the limit (8.12), fix  $t \neq 0$  (the case  $t = 0$  is trivial) and take  $T = |t|$ : the expansion applies for  $r > r_0(|t|)$ , and  $(1-r^2)/(1+r^2) = (1-r) + \mathcal{O}((1-r)^2)$ .  $\square$

## 8.6 The first Edgeworth correction, and the entropy deficit

Let  $H_k$  be the probabilists' Hermite polynomial. Fourier inversion of the cumulant expansion of theorem 8.4 predicts the first Edgeworth correction, which we write with explicit remainder terms,

$$f_{Z_r}(z) = \varphi(z) \left[ 1 + \frac{\lambda_4(r)}{24} H_4(z) \right] + \mathcal{R}_r(z), \quad (8.13)$$

and

$$\mathbb{P}(Z_r \leq z) = \Phi(z) - \frac{\lambda_4(r)}{24} H_3(z) \varphi(z) + \mathcal{R}_r^F(z), \quad \mathcal{R}_r^F(z) = \int_{-\infty}^z \mathcal{R}_r; \quad (8.14)$$

the two displays are consistent because  $\frac{d}{dz} [H_3(z) \varphi(z)] = -H_4(z) \varphi(z)$ . The fixed-frequency theorem above does not control these remainders uniformly in  $z$ : that requires a high-frequency estimate uniform in  $r$ , which the absorbed report did not supply and therefore conjectured. Part XI supplies it (lemma 5.3), and the conjecture is a theorem of this volume.

**Proposition 8.5** (Uniform first Edgeworth expansion). *There are  $r_0 < 1$  and, for each pair of integers  $j, K \geq 0$ , constants  $C_{j,K}$  and  $C_F$  such that for  $r_0 < r < 1$*

$$\sup_{z \in \mathbb{R}} (1+|z|)^K \left| \mathcal{R}_r^{(j)}(z) \right| \leq C_{j,K} (1-r)^2, \quad \sup_{z \in \mathbb{R}} \left| \mathcal{R}_r^F(z) \right| \leq C_F (1-r)^2. \quad (8.15)$$

The next term of the expansion is  $\varphi(z) [\lambda_6(r)H_6(z)/720 + \lambda_4(r)^2H_8(z)/1152]$ , after which the remainder is  $\mathcal{O}((1-r)^3)$  in the same seminorms.

*Proof.* This is [theorem 5.1](#) read through [table 1](#): Part XI’s  $Z_h$  is  $Z_r$  at  $r = e^{-h}$ , its  $\lambda_{2m}(h)$  is  $\lambda_{2m}(r)$  ([proposition 0.5.2](#)), and  $h = -\log r = (1-r)(1 + \mathcal{O}(1-r))$ , so powers of  $h$  and of  $1-r$  are interchangeable, up to constants, in every remainder. In the exact-cumulant Edgeworth polynomial (5.2) the multi-indices of weight at most 1 are the empty one and  $k_2 = 1$ , giving  $\mathcal{E}_{1,h} = 1 + \lambda_4 H_4/4!$ ; those of weight 2 are  $k_3 = 1$  and  $k_2 = 2$ , giving  $\lambda_6 H_6/6! + \lambda_4^2 H_8/(2! (4!)^2)$ , and  $2 \cdot 24^2 = 1152$ . The density bound is (5.4) with  $R = 1$  (respectively  $R = 2$ ); the distribution-function bound is its case  $j = 0$ ,  $K = 2$ , integrated:  $|\mathcal{R}_r^F(z)| \leq \int_{\mathbb{R}} |\mathcal{R}_r| \leq C_{0,2} h^2 \int_{\mathbb{R}} (1+|x|)^{-2} dx = 2C_{0,2} h^2$ .  $\square$

The absorbed report stated [proposition 8.5](#) as a conjecture; under the convention of [section 0.7](#) it is recorded here as a proposition proved in Part XI, and the same applies to the all-orders statement of [remark 10.1](#). The entropy deficit is a different matter: the first Edgeworth term is now proved, but its use inside a relative entropy needs global logarithmic control of  $f_{Z_r}/\varphi$ , which Part XI also lacks ([conjecture 11.5](#)).

**Conjecture 8.6** (Entropy deficit). *Let  $D(\text{Law}(Z_r) \| N(0, 1))$  be relative entropy. Then*

$$D(\text{Law}(Z_r) \| N(0, 1)) = \frac{\lambda_4(r)^2}{48} + o((1-r)^2) = \frac{3}{100}(1-r)^2 + o((1-r)^2). \quad (8.16)$$

Equivalently,

$$h(Z_r) = \frac{1}{2} \log(2\pi e) - \frac{3}{100}(1-r)^2 + o((1-r)^2).$$

The coefficient  $1/48$  follows formally by inserting  $f/\varphi = 1 + (\lambda_4/24)H_4 + \dots$  into relative entropy and using  $\mathbb{E}_\varphi H_4^2 = 4!$ . This is the leading term  $d_2 = 3/100$  of the full entropic expansion conjectured as [conjecture 11.5](#); a proof should combine [proposition 8.5](#) with tail control supplied by log-concavity ([theorem 4.2](#)) where  $f_{Z_r}/\varphi$  is small or large.

## 9 Numerical experiments

### 9.1 Reproducibility and deterministic error control

The script `assets/geometric-sinc-and-exponent-families/Fabius_Rvachev_Frontier_Report/numerical_experiments.py` uses NumPy, SciPy, and Matplotlib with seed 20260830. It truncates [equation \(2.1\)](#) only after the deterministic tail bound

$$\left| (1-q) \sum_{n \geq N} q^n U_n \right| \leq \frac{|1-q| |q|^N}{1-|q|} \quad (9.1)$$

falls below the requested tolerance. The coupling checks are not independent tests of the identities they accompany: reflected or reversed digits make the asserted equality exact by construction, so the reported errors measure floating-point roundoff and nothing else.

Table 22: Digit-reflection coupling for the negative/positive affine duality.

| $r$  | samples | terms | maximum coupling error |
|------|---------|-------|------------------------|
| 0.20 | 40000   | 21    | 7.772e-16              |
| 0.50 | 40000   | 46    | 1.776e-15              |
| 0.80 | 40000   | 137   | 6.661e-15              |

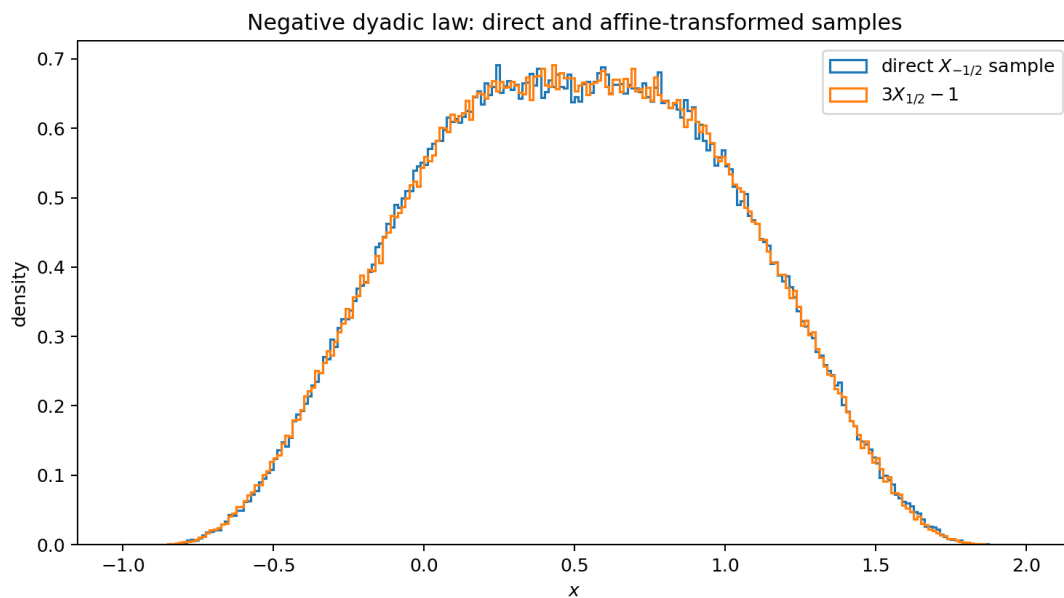


Table 23: Reversal coupling for finite reciprocal-base duality.

| $q$ | $N$ | samples | maximum coupling error |
|-----|-----|---------|------------------------|
| 2   | 14  | 35000   | 4.441e-16              |
| -2  | 14  | 35000   | 8.882e-16              |
| 4   | 14  | 35000   | 4.441e-16              |
| -4  | 14  | 35000   | 8.882e-16              |

Table 24: Standardized geometric-uniform laws approaching the Gaussian boundary.

| $r$  | samples | KS distance | empirical excess | exact excess |
|------|---------|-------------|------------------|--------------|
| 0.50 | 260000  | 0.0282      | -0.71908         | -0.72000     |
| 0.80 | 230000  | 0.0080      | -0.26110         | -0.26341     |
| 0.95 | 180000  | 0.0022      | -0.05054         | -0.06150     |
| 0.98 | 120000  | 0.0016      | -0.03496         | -0.02424     |

Figure 39: Independent simulations of  $X_{-1/2}$  and  $3X_{1/2} - 1$ . The visual agreement supplements the exact digit-reflection coupling in [table 22](#).

## 9.2 Negative dyadic collapse

## 9.3 The plateau phase

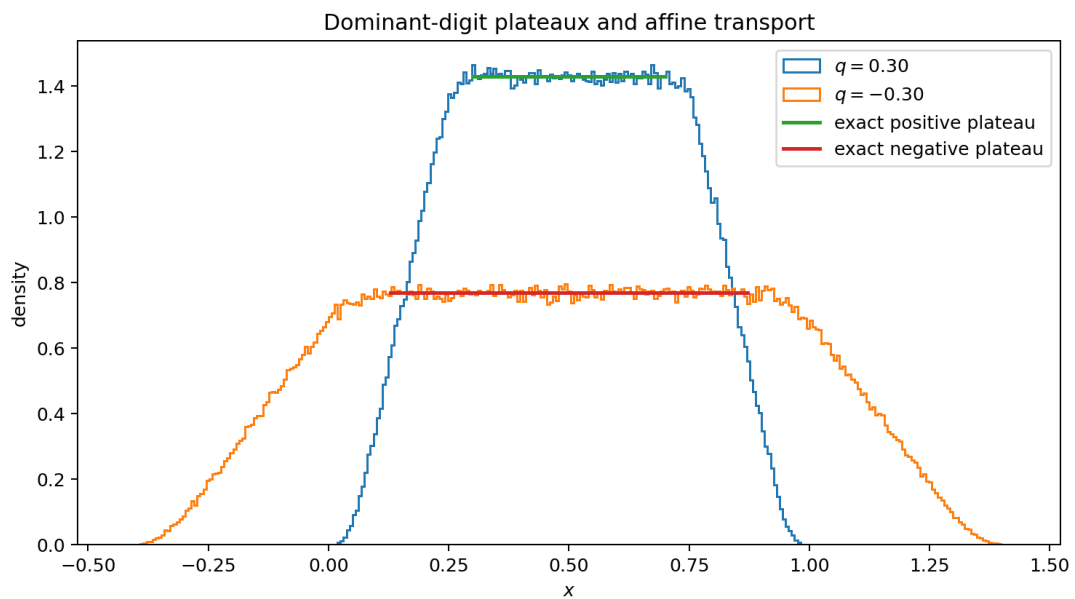


Figure 40: For  $r = 0.30$ , the exact positive plateau is  $[0.30, 0.70]$  at height  $1/0.70$ , while the negative plateau is  $[0.09/0.70, 1 - 0.09/0.70]$  at height  $1/1.30$ .

## 9.4 Approach to the Gaussian

The empirical excess kurtosis becomes noisy very near  $r = 1$ , because its true value is small and fourth moments converge more slowly than CDFs. The exact values in [table 24](#) are analytic, not fitted.

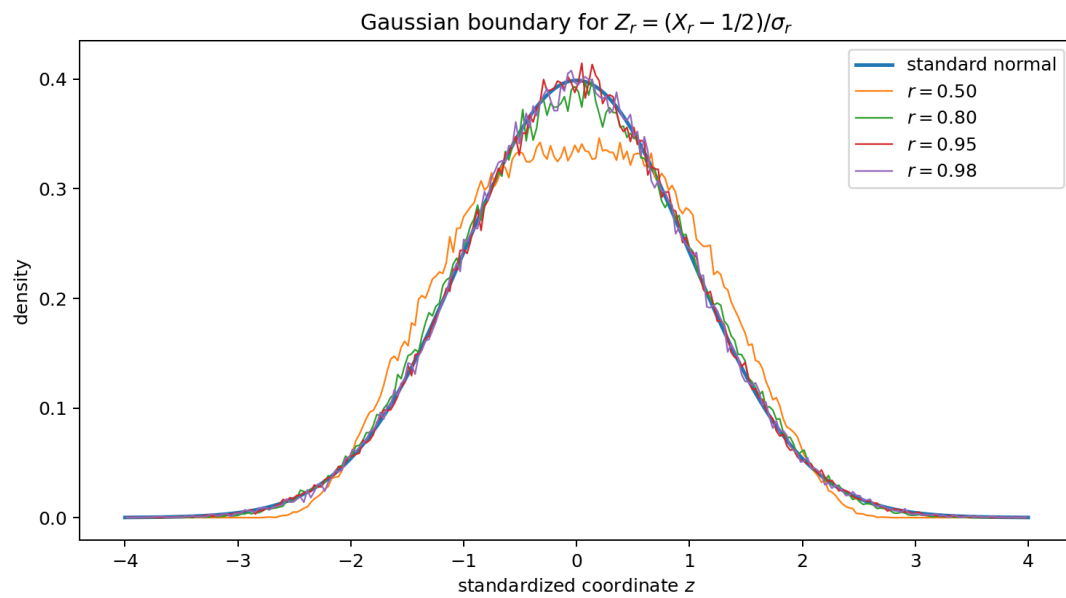


Figure 41: Standardized densities for increasing  $r$ , compared with the standard normal. The exact excess kurtosis is  $-\frac{6}{5}(1 - r^2)/(1 + r^2)$ , and the measured Kolmogorov distance decreases consistently with [theorem 8.1](#).

## 10 Further conjectures and frontier directions

### 10.1 All-orders Gaussian asymptotics

*Remark 10.1* (Complete Bernoulli–Bell–Hermite expansion: proved in Part XI). The absorbed report conjectured that for every integer  $M \geq 1$  the standardized density has an expansion through order  $(1-r)^M$  whose coefficients are finite Bell polynomials in the exact cumulants  $\lambda_4(r), \lambda_6(r), \dots, \lambda_{2M+2}(r)$ , expressed in the Hermite basis, uniform with polynomial weights and differentiable termwise a fixed number of times. This is exactly [theorem 5.1](#) together with [proposition 5.2](#), with  $h = -\log r$  as the expansion variable and the Bell-polynomial coefficient structure displayed in (5.2); following [section 0.7](#) the conjecture is demoted to this remark. The analytic estimate the report identified as crucial – uniform control of (8.2) outside the low-frequency window – is [lemma 5.3](#).

### 10.2 A generic- $q$ Lambert- $W$ endpoint theory

The all-orders endpoint programme is [conjecture 15.4](#), and its transform-level half is already proved: the negative Laplace transform of  $X_r$  is the geometric harmonic sum  $\mathbb{E}e^{-sX_r} = \prod_{n \geq 0} (1 - e^{-(1-r)r^n s}) / ((1-r)r^n s) -$  the product formula (0.3.4) at  $t = -s$ , since  $E(-a) = (1 - e^{-a})/a$ , the interchange of expectation and infinite product being justified there by locally uniform convergence – and [theorem 15.3](#) resolves its logarithm exactly into a quadratic polynomial in  $\log s$ , a real-analytic one-periodic fluctuation in  $\log_{1/r} s$  with explicit Gamma-zeta Fourier modes, and an exponentially small remainder. What remains open is the density-level saddle inversion, stated in the exact  $-W_{-1}$  coordinate as [conjecture 15.4](#); the negative cases transfer by [corollary 3.2](#), which carries the left endpoint of  $F_{-r}$  to that of  $F_r$  by  $x \mapsto (x - \ell_r)/c_r$ .

### 10.3 Legendre coefficients and endpoint saddles

Because each  $g_q$  is compactly supported and smooth, its affine Legendre coefficients decay faster than every power. The precise stretched-exponential envelope should be controlled by the same complex saddle geometry that governs endpoint flatness.

**Conjecture 10.2** (Legendre–Lambert asymptotic bridge). *For the even Legendre coefficients of  $g_r$ , the logarithm of the coefficient magnitude has a complete asymptotic expansion whose principal scale is obtained by balancing the large-degree Legendre boundary layer with the Lambert saddle of  $F_r$ . A smooth log-periodic factor survives in the coefficient asymptotics. Negative-base coefficients obey exactly the same law after the affine normalization in [equation \(4.10\)](#).*

This would connect three parts of the existing program that are currently largely parallel: endpoint Lambert asymptotics, Legendre expansions, and infinite sinc products.

### 10.4 Reciprocal-base algebra and $q$ -identities

The coefficient multiset proof of [theorem 7.1](#) suggests a broader principle: every symmetric polynomial or entire symmetric function of  $\{q^n/[N]_q : 0 \leq n < N\}$  is reciprocal-invariant. Power sums give cumulants; elementary symmetric functions give Gaussian-binomial expressions.

**Open problem 10.3** (A probability proof of a full Gaussian-binomial reciprocity package). [Appendix 11](#) and (13.1) already give the algebraic reciprocity, including  $[N]_q/[k]_q = q^{-k(N-k)} [N]_q/[k]_q$  and the row reversal of the  $q$ -Lagrange coefficients. What is asked here is a probabilistic realization: construct natural random variables or random polynomials whose elementary-symmetric statistics of the finite geometric weights recover  $[N]_q/[k]_q$ . Then seek mixed Bell/Gaussian identities obtained by converting between power sums and elementary symmetric functions.

## 10.5 A formalization roadmap

The strongest immediate formal targets are elementary enough to be realistic additions to the existing Lean development:

1. replace the negative-parameter density ambiguity by the sign-corrected theorem  $g_q = \text{sgn}(q)\tilde{f}_q$ ;
2. prove finite reciprocal-base duality by coefficient extensionality and list  $q = \pm 2, \pm 4$  wrappers;
3. formalize the exact plateau theorem from the CDF tails.

The remaining targets of the original list — the reflected-digit coupling and the affine pushforward  $\mu_{-r} = T_{r*}\mu_r$ , the CDF, quantile, support and smooth-density transports, the residue-class multiplication law with its characteristic-function corollary, and the Bernoulli cumulant and standardized-sign-collapse package — are tranches of Part VII's merged roadmap and are listed there once.

Log-concavity may require additional convex-measure infrastructure. Berry–Esseen and uniform Edgeworth theory are longer-term targets, but the exact cumulant and quartic formal-power-series identities can be separated from those analytic theorems.

## 11 Conclusions

The geometric-uniform family has more symmetry than is visible from the positive parameter alone. Centering turns every alternating geometric digit into a reflected uniform digit, and this collapses the entire negative half of the parameter interval to an affine copy of the positive half. That one observation resolves the negative density frontier, identifies the sign of the previously selected total function, and transports Fabius, inverse Fabius, Rvachev, Legendre, Lambert– $W$ , sinc-product, and Thue–Morse structures in one step.

Two further reorganizations complete the parameter picture. Residue classes produce a  $q$ -multiplication law, while finite digit reversal identifies  $q$  with  $q^{-1}$ . Thus negative parameters, bases outside the unit interval, and smaller positive bases are not separate theories but affine or reciprocal views of the same laws.

Finally, the parameter edge  $|q| \uparrow 1$  is a genuine Gaussian boundary. The rate is controlled by a closed Berry–Esseen ratio, and the departure from normality is encoded by Bernoulli cumulants and Bell partitions. The quartic coefficient  $-t^4/20$  is the first term of the all-orders Bernoulli–Bell–Hermite expansion proved in Part XI (theorem 5.1), which connects the probability family to the repository's existing Bell, Bernoulli, Lambert, Legendre, and formal saddle machinery.

## A Auxiliary algebra

### A.1 Support endpoints under negative affine transport

Since  $T_r(y) = 1/2 + c_r(y - 1/2)$ ,

$$T_r(0) = \frac{1 - c_r}{2} = -\frac{r}{1 - r}, \quad T_r(1) = \frac{1 + c_r}{2} = \frac{1}{1 - r}.$$

For the positive plateau endpoint  $y = r$ ,

$$T_r(r) = \frac{1}{2} + \frac{1 + r}{1 - r} \left( r - \frac{1}{2} \right) = \frac{r^2}{1 - r}.$$

These identities explain the apparently asymmetric support and the symmetric central plateau of the negative law.

## A.2 Finite negative/positive scale

Digit reflection in equation (7.1) gives

$$X_{-r,N} - \frac{1}{2} \stackrel{d}{=} \frac{[N]_r}{[N]_{-r}} \left( X_{r,N} - \frac{1}{2} \right),$$

where

$$\frac{[N]_r}{[N]_{-r}} = \frac{1+r}{1-r} \frac{1-r^N}{1-(-r)^N}.$$

For even  $N$ , this is exactly  $c_r$ ; for odd  $N$ , it converges exponentially to  $c_r$ .

## A.3 Low standardized cumulants

The first values from (0.4.6) are

$$\begin{aligned}\lambda_2(r) &= 1, \\ \lambda_4(r) &= -\frac{6}{5} \frac{1-r^2}{1+r^2}, \\ \lambda_6(r) &= \frac{48}{7} \frac{(1-r^2)^2}{1+r^2+r^4}, \\ \lambda_8(r) &= -\frac{432}{5} \frac{(1-r^2)^3}{1+r^2+r^4+r^6}.\end{aligned}$$

The last coefficient uses  $B_8 = -1/30$ . These feed directly into complete Bell polynomials and higher Edgeworth terms.

## B Numerical file manifest

This part's verification material is kept in the volume directory under `assets/geometric-sinc-and-exponent-families/Fabius_Rvachev_Frontier_Report/`:

- `assets/geometric-sinc-and-exponent-families/Fabius_Rvachev_Frontier_Report/numerical_experiments.py` — fully commented reproducible code;
- `assets/geometric-sinc-and-exponent-families/Fabius_Rvachev_Frontier_Report/numerical_results.tex` — the generated tables, read directly by this volume;
- `assets/geometric-sinc-and-exponent-families/Fabius_Rvachev_Frontier_Report/gaussian_convergence.csv`, `assets/geometric-sinc-and-exponent-families/Fabius_Rvachev_Frontier_Report/negative_coupling.csv`, and `assets/geometric-sinc-and-exponent-families/Fabius_Rvachev_Frontier_Report/reciprocal_coupling.csv` — the machine-readable results;
- `assets/geometric-sinc-and-exponent-families/Fabius_Rvachev_Frontier_Report/figures/` — PDF and PNG versions of all figures;
- `assets/geometric-sinc-and-exponent-families/Fabius_Rvachev_Frontier_Report/README.txt` — the absorbed package's own run notes.

The absorbed report's source and PDF were deleted when it was merged; git history is their archive. The volume itself is rebuilt from its own directory by running `pdflatex` on `geometric_q_fabius_frontiers.tex` three times, so that the table of contents and every cross-reference settle.

## Part IX

# Cyclotomic Radial Expansions, Natural Boundaries, and Inverse Branches

## Abstract

The dyadic Rvachev up-function has Fourier transform  $\prod_{j \geq 1} \text{sinc}_{\text{rad}}(z/2^j)$ , and the surrounding repository develops its Fabius, Thue–Morse,  $q$ -series, Bell–Bernoulli, inverse, sampling, and Lambert- $W$  structures in considerable depth. The current consolidated  $q$ -series frontier explicitly leaves two related tasks open: proving a generic natural boundary in the  $q$ -plane and constructing root-of-unity renormalizations. This report resolves a concrete scalar version of both tasks for the complex geometric-uniform transform

$$\Phi(q, z) = \prod_{j \geq 0} \text{sinc}_{\text{rad}}((1 - q)q^j z), \quad |q| < 1.$$

For a primitive  $m$ th root of unity  $\omega$  and  $d = \text{ord}_{\text{mult}}(\omega^2) = m / \gcd(m, 2)$ , we derive the radial expansion

$$\log \Phi(\omega(1 - \delta), z) = -\frac{\mathcal{A}_\omega(z)}{\delta} + \mathcal{B}_\omega(z) + O(\delta),$$

locally uniformly for  $|(1 - \omega)z| < \pi$ . The action is an explicit spectral dilogarithm,

$$\mathcal{A}_\omega(z) = \frac{1}{2d^2} \sum_{n \geq 1} \text{Li}_2 \left( \left( \frac{(1 - \omega)z}{\pi n} \right)^{2d} \right).$$

The action never vanishes for  $0 < |z| < \pi/2$  with  $z$  real or purely imaginary. Because roots of unity are dense, this proves that the unit circle is a natural boundary for  $q \mapsto \Phi(q, z)$  throughout that entire frequency interval, on both slices. The imaginary slice is the centered exponential germ  $C_q(t) = \Phi(q, -it/2)$  of Part VII, so the theorem settles [conjecture 17.1](#) for every real  $t$  with  $0 < |t| < \pi$  — for all such  $t$ , not merely generic ones — replacing that conjecture’s cumulant-cancellation strategy by an explicit action.

A double scaling  $q = \omega(1 - \varepsilon^d)$ ,  $z = \sqrt{\varepsilon} w$  then gives the locally uniform cyclotomic limit

$$\Phi(\omega(1 - \varepsilon^d), \sqrt{\varepsilon} w) \longrightarrow \exp(-A_\omega w^{2d}), \quad A_\omega = \frac{\zeta(2d)(1 - \omega)^{2d}}{2d^2 \pi^{2d}}.$$

We prove the first corrections and an all-orders Bell-polynomial expansion, identify the corresponding polyharmonic heat symbol, show that the reciprocal zero moments and Bernoulli cumulants collapse to a single order  $2d$ , derive the limiting Gould–Hopper Appell family, and construct formal inverse series both in the scaled frequency and in  $q$  itself. A universal analytic-germ theorem shows that the mechanism extends beyond sinc products to  $q$ -Pochhammer and other geometric products. High-precision experiments, source code, CSV data, and figures accompany the report.



# 1 Scope, status, and contribution boundary

## 1.1 The audited source landscape

The requested source tree is a living research corpus rather than a single paper. Its current top level contains the primary Fabijs–Rvachev exposition, a mathematical glossary, a non-elementarity study, a Lean walkthrough, imported papers, an archive, and a semi-formalized frontier system. During August 2026 many formerly separate frontier drafts were consolidated into thematic volumes. Consequently, a novelty audit based only on old filenames would be misleading: the correct comparison target is the current consolidated mathematics, supplemented by historical inventories for paths that were absorbed or archived.

The audit used three layers.

1. The current public GitHub tree, the current semi-formalized README, and the raw consolidated *Exponents and q-Series Frontiers* source were inspected at theorem and open-problem level.
2. A preserved recursive ledger of every `*.tex` path under `Analysis/FabijsFunction/docs` at commit `24da985eba70e46d862acdadd476793a428213e` was used to retain coverage of the frozen archive and earlier source-paper layout. The 57-path ledger is included in the companion archive as `corpus_inventory_2026-08-27.txt`.
3. Prior frontier reports in the user’s library were phrase- and formula-searched for the proposed concepts. In particular, Stein–Koopman calculus, transfer determinants, fractional transforms, dyadic-comb interpolation, inverse-Lambert expansions, Appell deconvolution, and spectral zero-divisor theory were treated as occupied territory and are not presented here as new.

The current consolidated  $q$ -series volume formulates a *generic natural-boundary conjecture* for its geometric characteristic function and separately asks for *root-of-unity renormalization*. The present report targets precisely that remaining scalar analytic gap. It does not claim to solve the full two-nome finite-law, matrix Mahler, or mixed Prouhet problem.

*Status note 1.1* (Meaning of “new”). Throughout this report, *new* means apparently new relative to the audited repository corpus. It is not a claim of worldwide priority. Classical ingredients such as Euler’s sine product, Bernoulli cumulants,  $q$ -Pochhammer notation, and radial root-of-unity methods are cited and clearly separated from the deductions made here.

## 1.2 Result ledger

The principal contributions are as follows.

- R1.** An exact logarithmic and  $q$ -Pochhammer representation of the complex geometric sinc product, organized so that cyclotomic resonance is visible coefficient by coefficient. The underlying product and the global spectral  $q$ -Pochhammer factorization now have exact Lean counterparts; the local master logarithm remains a manuscript theorem.
- R2.** A two-term radial expansion at every nontrivial root of unity, with an explicit action  $\mathcal{A}_\omega$  and constant term  $\mathcal{B}_\omega$ .
- R3.** Three equivalent descriptions of  $\mathcal{A}_\omega$ : a resonant zeta series, an infinite spectral dilogarithm, and a cyclic projection integral.
- R4.** A sign theorem for the action and, as a consequence, a proof that the unit circle is a natural boundary for every fixed real  $0 < |z| < \pi/2$ .
- R5.** A root-of-unity blow-up limit to  $\exp(-A_\omega w^{2d})$ , including subcritical corrections, the complete order- $\varepsilon^d$  layer, and an all-orders Bell-polynomial scheme.

- R6.** A universal analytic-germ theorem that recovers both the sinc result and a root-scaled  $q$ -Pochhammer limit.
- R7.** Spectral condensation: among all reciprocal zero power sums, exactly the  $2d$ th survives after cyclotomic scaling.
- R8.** Bernoulli–Bell condensation: among all scaled cumulants, exactly the  $2d$ th survives, forcing an explicit sparse moment sequence.
- R9.** A limiting Gould–Hopper/Appell family obtained from the inverse moment generating function.
- R10.** Polyharmonic heat-kernel and Stokes-sector interpretations, distinguishing the genuinely dissipative roots  $m \equiv 2 \pmod{4}$  from backward/rotated sectors.
- R11.** Formal inverse series in the scaled frequency and in  $q$ , with explicit first corrections; sectorial analytic realization and monodromy remain open.
- R12.** Reproducible high-precision tests whose fitted convergence exponents agree with the displayed asymptotic orders.

*Status note 1.2* (Current formal and analytic boundary). The repository already defines the complex geometric sinc product and its reciprocal-Gamma reflection factorization. The global spectral  $q$ -Pochhammer identity in [equation \(3.6\)](#) is now machine-checked in Lean; exact names and normalizations are recorded below. No later root-of-unity asymptotic, natural-boundary, condensation, Gould–Hopper, or inverse-branch claim in this report is thereby Lean-verified.

There is also an analytic distinction inside the manuscript. The fixed-index cyclotomic expansions do not by themselves justify summation over an unbounded index. The proof of [theorem 4.4](#) closes this gap with a uniform two-range estimate, splitting at  $k \asymp \delta^{-1/2}$  and using geometric decay beyond the cutoff. By contrast, the inverse formulas below record formal coefficients and sectorial targets, not established convergent analytic germs at the boundary.

## 2 The Fabius–Rvachev interface and its complex $q$ -extension

### 2.1 The dyadic probability law

Let  $V_0, V_1, \dots$  be independent random variables uniformly distributed on  $[-1, 1]$ . The centered dyadic random series

$$Y_{1/2} = \sum_{j \geq 0} 2^{-(j+1)} V_j \quad (2.1)$$

has support  $[-1, 1]$  and density Rvachev’s up-function  $\widehat{\text{up}}$ . Its characteristic function, with the convention  $\text{sinc}_{\text{rad}} z = \sin z/z$ , is

$$\widehat{\text{up}}_{\text{ang}}(-z) = \prod_{j \geq 0} \text{sinc}_{\text{rad}}\left(\frac{z}{2^{j+1}}\right). \quad (2.2)$$

The Fabius distribution on  $[0, 1]$  is obtained by the affine change  $X = (Y_{1/2} + 1)/2$ ; this is [proposition 0.2.5](#) in the canonical normalization. The repository develops this bridge, exact dyadic arithmetic, Thue–Morse derivative signs, moment and cumulant formulas, Legendre expansions, inverse asymptotics, and many related representations. We use those structures only as motivation and normalization.

### 2.2 The geometric family

*Status note 2.1* (Conventions of this part). This part is the one that works with *complex*  $q$ , and its natural frequency variable is therefore the one in which the sinc arguments carry no factor of  $\frac{1}{2}$ . Concretely it

uses seeds uniform on  $[-1, 1]$  and writes  $Y_q$  for what [definition 0.2.1](#) calls  $2X_q^\circ$ ; for real  $0 < q < 1$ ,

$$Y_q = 2X_q^\circ, \quad \mathbb{E}e^{izY_q} = \varphi_q(2z),$$

so every formula below converts to the canonical normalization by the single substitution  $z \mapsto t/2$ , as recorded in [table 1](#). In particular this part's cumulants are  $2^{2m}$  times [\(0.4.4\)](#) and its variance is  $(1 - q)/(3(1 + q))$ , four times [\(0.4.5\)](#). The letter  $Y_q$  denotes a different object in Parts VII and VIII — there the uncentered and the centered variable respectively — which is why [table 1](#) exists and why the notation catalogue's collision register carries a row for  $X, Y, Z$ .

For real  $0 < q < 1$ , define

$$Y_q = (1 - q) \sum_{j \geq 0} q^j V_j. \quad (2.3)$$

The coefficient sum is one, so  $Y_q$  is again supported on  $[-1, 1]$ . Independence gives

$$\mathbb{E}e^{izY_q} = \prod_{j \geq 0} \text{sinc}_{\text{rad}}((1 - q)q^j z). \quad (2.4)$$

For complex  $q$  there is generally no positive probability law, but the product itself remains a natural analytic object.

**Definition 2.2** (Complex  $q$ -Fabius–Rvachev sinc product). For  $q \in \mathbb{D} = \{q \in \mathbb{C} : |q| < 1\}$  and  $z \in \mathbb{C}$ , set

$$\Phi(q, z) = \prod_{j=0}^{\infty} \text{sinc}_{\text{rad}}((1 - q)q^j z). \quad (2.5)$$

Here  $\text{sinc}_{\text{rad}} 0 = 1$  and the empty power is read as  $q^0 = 1$  for every  $q$ , including  $q = 0$ . Thus at  $q = 0$  the factor  $j = 0$  contributes  $\text{sinc}_{\text{rad}} z$  and every factor with  $j \geq 1$  contributes  $\text{sinc}_{\text{rad}} 0 = 1$ ; this is the convention under which  $\Phi(0, z) = \text{sinc}_{\text{rad}} z$  below, and it matches the division-free reading of the zero set in [equation \(8.1\)](#).

**Proposition 2.3** (Holomorphy and normalization). *The product in [equation \(2.5\)](#) converges locally uniformly on  $\mathbb{D} \times \mathbb{C}$  and defines a holomorphic function there. Moreover,*

$$\Phi(q, 0) = 1, \quad \Phi(0, z) = \text{sinc}_{\text{rad}} z, \quad \Phi\left(\frac{1}{2}, z\right) = \widehat{\text{up}}_{\text{ang}}(-z).$$

*Proof.* On a compact set  $|q| \leq r < 1$ ,  $|z| \leq R$ , the  $j$ th argument is  $O(r^j)$  uniformly. Since  $\text{sinc}_{\text{rad}} w = 1 - w^2/6 + O(w^4)$ , the deviation of the  $j$ th factor from one is  $O(r^{2j})$ . The Weierstrass criterion gives locally uniform convergence of the product and of its holomorphic finite approximants. The three normalizations follow directly.  $\square$

**Remark 2.4** (Exact Lean normalization dictionary). Lean's existing product uses the frequency coordinate

$$\text{geometricSincProduct}(q, u) = \prod_{j \geq 0} \text{sinc}_{\text{rad}}(\pi q^j u).$$

Consequently the present convention is exactly

$$\Phi(q, z) = \text{geometricSincProduct}\left(q, \frac{(1 - q)z}{\pi}\right).$$

The definition and its module are

`Fabius.geometricSincProduct` in `FabiusFunction.GeometricReciprocalGamma`.

At the dyadic point, the theorem

`Fabius.centeredSincProduct_eq_rvachevFourierProduct`

records the further coordinate conversion

$$\Phi\left(\frac{1}{2}, z\right) = \text{centeredSincProduct}(z) = \text{rvachevFourierProduct}\left(\frac{z}{2\pi}\right).$$

Thus the manuscript's Fourier variable and Lean's canonical Rvachev product variable differ by the displayed factor  $2\pi$ .

### 3 Master logarithmic and $q$ -Pochhammer representations

#### 3.1 The exact logarithmic series

Euler's sine product and its logarithm give

$$\text{sinc}_{\text{rad}} w = \prod_{n \geq 1} \left(1 - \frac{w^2}{\pi^2 n^2}\right), \quad \log \text{sinc}_{\text{rad}} w = - \sum_{k \geq 1} c_k w^{2k}, \quad (3.1)$$

where

$$c_k = \frac{\zeta(2k)}{k\pi^{2k}} > 0. \quad (3.2)$$

The logarithm is the branch normalized by  $\log \text{sinc}_{\text{rad}} 0 = 0$  in  $|w| < \pi$ .

**Theorem 3.1** (Master logarithm). *On the domain*

$$\Omega = \{(q, z) \in \mathbb{D} \times \mathbb{C} : |(1 - q)z| < \pi\},$$

*one has the absolutely and locally uniformly convergent expansion*

$$\log \Phi(q, z) = - \sum_{k \geq 1} c_k \frac{(1 - q)^{2k}}{1 - q^{2k}} z^{2k}. \quad (3.3)$$

*Proof.* For  $(q, z) \in \Omega$  every argument  $(1 - q)q^j z$  lies in the zero-free disk  $|w| < \pi$ , so [equation \(3.1\)](#) may be inserted factor by factor. Fix a compact  $L \subset \Omega$  and choose  $r < 1$  and  $\rho < 1$  with  $|q| \leq r$  and  $|(1 - q)z| \leq \rho\pi$  on  $L$ . Since  $\zeta(2k) \leq \zeta(2)$ , so that  $c_k \leq \zeta(2)/(k\pi^{2k})$ , and since  $|q|^{2jk} \leq |q|^{2j}$  for  $k \geq 1$ ,

$$\sum_{j \geq 0} \sum_{k \geq 1} c_k |(1 - q)q^j z|^{2k} \leq \sum_{j \geq 0} r^{2j} \sum_{k \geq 1} \frac{\zeta(2)}{k} \rho^{2k} = \frac{\zeta(2)}{1 - r^2} \log \frac{1}{1 - \rho^2} < \infty \quad (3.4)$$

uniformly on  $L$ . The double family is therefore absolutely summable, so it may be summed in either order; summing first over  $j$  and using  $\sum_{j \geq 0} q^{2kj} = (1 - q^{2k})^{-1}$  gives [equation \(3.3\)](#). The same majorant, with the factor  $\sum_j r^{2j}$  deleted, dominates the right-hand side of [equation \(3.3\)](#) and shows that it too converges absolutely and uniformly on  $L$ . The logarithm on the left is the sum of the individual zero-normalized branches, which is the branch with  $\log \Phi(q, 0) = 0$ .  $\square$

The coefficient

$$R_k(q) := \frac{(1 - q)^{2k}}{1 - q^{2k}} \quad (3.5)$$

is simultaneously a geometric power sum, a cumulant multiplier, and the source of every root-of-unity singularity. The entire cyclotomic theory below is a careful organization of the resonances  $q^{2k} = 1$ .

### 3.2 A spectral $q$ -Pochhammer factorization

Let  $(a; Q)_\infty = \prod_{j \geq 0} (1 - aQ^j)$ . Reordering Euler's sine factors gives a second exact representation.

**Theorem 3.2** (Infinite spectral  $q$ -Pochhammer product). *For every  $q \in \mathbb{D}$  and  $z \in \mathbb{C}$ ,*

$$\Phi(q, z) = \prod_{n \geq 1} \left( \frac{(1-q)^2 z^2}{\pi^2 n^2}; q^2 \right)_\infty. \quad (3.6)$$

*The double product is locally normally convergent.*

*Proof.* Apply Euler's product to each sinc factor, obtaining the doubly indexed product

$$\Phi(q, z) = \prod_{j \geq 0} \prod_{n \geq 1} (1 + \alpha_{j,n}), \quad \alpha_{j,n} = -\frac{(1-q)^2 q^{2j} z^2}{\pi^2 n^2}.$$

Since  $\sum_{n \geq 1} n^{-2} = \zeta(2) = \pi^2/6$  and  $\sum_{j \geq 0} |q|^{2j} = (1 - |q|^2)^{-1}$ ,

$$\sum_{j \geq 0} \sum_{n \geq 1} |\alpha_{j,n}| = \frac{|(1-q)z|^2}{\pi^2} \cdot \frac{\zeta(2)}{1 - |q|^2} = \frac{|(1-q)z|^2}{6(1 - |q|^2)}, \quad (3.7)$$

which is finite for  $|q| < 1$  and bounded on compact subsets of  $\mathbb{D} \times \mathbb{C}$ . A product  $\prod (1 + \alpha_i)$  over a countable index set with  $\sum_i |\alpha_i| < \infty$  converges unconditionally, is unchanged by any bijective reindexing or by grouping the factors into blocks, and its value equals either iterated product; this is the standard criterion for normal convergence of infinite products, applied here to the index set  $\{(j, n)\}$ . Grouping by  $n$  and evaluating the inner product over  $j$  turns  $\prod_{j \geq 0} (1 - (1-q)^2 z^2 q^{2j} / (\pi^2 n^2))$  into the stated  $q^2$ -Pochhammer symbol. Because [equation \(3.7\)](#) is uniform on compact sets, the convergence is locally normal. No hypothesis on the vanishing of individual Euler factors is needed: absolute summability of the  $\alpha_{j,n}$  is what licenses the rearrangement, and a vanishing factor merely makes both sides zero. Hence the identity holds for every  $q \in \mathbb{D}$  and every  $z \in \mathbb{C}$ .  $\square$

*Remark 3.3* (Lean crosswalk: global factorization). The compiled theorem

`Fabius.geometricSincProduct_eq_tprod_complexQPochhammerInf`

in `FabiusFunction.RvachevPochhammerFactorization` states, for every  $q \in \mathbb{C}$  with  $|q| < 1$  and every  $u \in \mathbb{C}$ ,

$$\text{geometricSincProduct}(q, u) = \prod_{k \geq 0} \left( \frac{u^2}{(k+1)^2}; q^2 \right)_\infty.$$

Substituting  $u = (1-q)z/\pi$  and reindexing  $n = k+1$  gives [equation \(3.6\)](#) exactly. Its proof is global in  $u$ , includes  $q = 0$  without division by a geometric scale, and remains valid when individual Euler factors vanish. The older dyadic theorem `Fabius.rvachevFourierProduct_eq_tprod_complexQPochhammerInf` is now a direct specialization of this variable- $q$  result.

*Remark 3.4* (Why dilogarithms appear). Radial root-of-unity asymptotics of a single  $q$ -Pochhammer symbol are governed by a dilogarithmic action and a finite cyclic factor. Formula [equation \(3.6\)](#) superposes one such action for each spectral point  $\pi n$ . The result is the infinite spectral dilogarithm obtained below. Our direct proof from [equation \(3.3\)](#) avoids branch bookkeeping and reduces the needed uniformity to the explicit two-range estimate in [theorem 4.4](#).

## 4 Radial resonance at roots of unity

### 4.1 Cyclotomic notation

Fix a primitive  $m$ th root

$$\omega = \exp\left(\frac{2\pi i h}{m}\right), \quad (h, m) = 1, \quad m \geq 2, \quad (4.1)$$

and write

$$d = \text{ord}_{\text{mult}}(\omega^2) = \frac{m}{\gcd(m, 2)}, \quad a_\omega = 1 - \omega. \quad (4.2)$$

Thus  $\omega^{2k} = 1$  exactly when  $d \mid k$ . We approach the root radially from inside the disk by

$$q = \omega(1 - \delta), \quad \delta \downarrow 0. \quad (4.3)$$

**Lemma 4.1** (Resonant and nonresonant multipliers). *For fixed  $k \geq 1$ ,*

$$\frac{(1 - q)^{2k}}{1 - q^{2k}} = \frac{a_\omega^{2k}}{1 - \omega^{2k}} + O(\delta), \quad d \nmid k, \quad (4.4)$$

$$\frac{(1 - q)^{2k}}{1 - q^{2k}} = a_\omega^{2k} \left[ \frac{1}{2k\delta} + \frac{\omega}{1 - \omega} + \frac{2k - 1}{4k} + O(\delta) \right], \quad d \mid k. \quad (4.5)$$

*Both remainders are asserted here only for each fixed  $k$ ; no uniformity in an unbounded residue class is part of this lemma.*

*Proof.* The nonresonant denominator tends to the nonzero number  $1 - \omega^{2k}$ . In the resonant case,

$$\begin{aligned} (1 - q)^{2k} &= a_\omega^{2k} \left( 1 + 2k \frac{\omega}{a_\omega} \delta + O(k^2 \delta^2) \right), \\ 1 - q^{2k} &= 1 - (1 - \delta)^{2k} = 2k\delta - k(2k - 1)\delta^2 + O(k^3 \delta^3). \end{aligned}$$

Divide the two expansions. This proves the two fixed- $k$  statements. Passing from them to a summed expansion requires the separate uniform remainder estimate isolated in the appendix.  $\square$

### 4.2 The radial action and constant term

**Definition 4.2** (Cyclotomic action). For  $|a_\omega z| < \pi$ , define

$$\mathcal{A}_\omega(z) = \sum_{\ell \geq 1} \frac{c_\ell d}{2\ell d} a_\omega^{2\ell d} z^{2\ell d}. \quad (4.6)$$

Also define

$$\mathcal{B}_\omega(z) = - \sum_{\substack{k \geq 1 \\ d \nmid k}} c_k \frac{a_\omega^{2k}}{1 - \omega^{2k}} z^{2k} \quad (4.7)$$

$$- \sum_{\substack{k \geq 1 \\ d \mid k}} c_k a_\omega^{2k} \left( \frac{\omega}{1 - \omega} + \frac{2k - 1}{4k} \right) z^{2k}. \quad (4.8)$$

**Lemma 4.3** (Uniform multiplier estimates). *Write  $a = a_\omega$  and  $b = \omega/a$ , and for  $k \geq 1$  and  $q = \omega(1 - \delta)$  let*

$$S_k(\delta) = \frac{(1 - q)^{2k}}{1 - q^{2k}} = \frac{(a + \omega\delta)^{2k}}{1 - \omega^{2k}(1 - \delta)^{2k}} \quad (4.9)$$

*be the multiplier of  $z^{2k}$  in equation (3.3).*

(i) (Resonant factorization.) If  $d \mid k$ , then

$$S_k(\delta) = \frac{a^{2k}}{2k\delta} H_k(\delta), \quad H_k(\delta) = (1 + b\delta)^{2k} \frac{2k\delta}{1 - (1 - \delta)^{2k}}, \quad (4.10)$$

and  $H_k$  extends holomorphically to the closed disc  $|\delta| \leq 1/(4k)$ , where

$$|H_k(\delta)| \leq M_1 := 2e^{|b|/2}, \quad H_k(0) = 1, \quad [\delta^1] H_k = 2kb + \frac{2k-1}{2}. \quad (4.11)$$

(ii) (Nonresonant separation.) Suppose  $d \geq 2$  and  $d \nmid k$ . Then  $\omega^{2k}$  is a  $d$ th root of unity different from 1, and

$$\gamma := \min_{\substack{1 \leq r \leq d \\ 0 \leq t \leq 1}} |1 - \omega^{2r} t| \geq \sin \frac{\pi}{d} > 0. \quad (4.12)$$

In particular  $|1 - \omega^{2k}(1 - \delta)^{2k}| \geq \gamma$  for every real  $\delta \in [0, 1]$ .

(iii) (Nonresonant derivative.) Under the hypotheses of (ii), for  $0 \leq \delta \leq \delta_1 := |a|/2$ ,

$$|S'_k(\delta)| \leq 2k \Theta |a + \omega\delta|^{2k}, \quad \Theta := \frac{2}{\gamma|a|} + \frac{1}{\gamma^2}. \quad (4.13)$$

(iv) (Nonresonant analyticity.) Under the hypotheses of (ii), put

$$\gamma_0 = 2 \sin \frac{\pi}{d}, \quad \lambda_\omega = \min \left\{ \frac{1}{2}, \log \left( 1 + \frac{\gamma_0}{2} \right) \right\}. \quad (4.14)$$

Then  $S_k$  is holomorphic on  $|\delta| \leq \lambda_\omega/(2k)$  and

$$|S_k(\delta)| \leq M_2 |a|^{2k}, \quad M_2 := \frac{2e^{|b|/2}}{\gamma_0}, \quad (4.15)$$

on that disc.

(v) (Crude resonant lower bound.) If  $d \mid k$  and  $0 < \delta \leq 1$ , then

$$1 - (1 - \delta)^{2k} \geq 1 - e^{-2k\delta} \geq (1 - e^{-1}) \min(1, 2k\delta). \quad (4.16)$$

When  $d = 1$  every index is resonant, parts (ii)–(iv) are vacuous, and we then adopt the conventions  $\gamma = \gamma_0 = 1$  and  $\lambda_\omega = \frac{1}{2}$ , so that the constants  $\Theta$  and  $M_2$  named above are defined in every case.

*Proof.* Throughout,  $1 - q = 1 - \omega(1 - \delta) = a + \omega\delta$  and  $q^{2k} = \omega^{2k}(1 - \delta)^{2k}$ , which is [equation \(4.9\)](#).

(i) If  $d \mid k$  then  $\omega^{2k} = 1$ , and [equation \(4.10\)](#) is the identity  $(a + \omega\delta)^{2k} = a^{2k}(1 + b\delta)^{2k}$  together with the insertion of  $2k\delta$ . Write  $n = 2k$ . From the binomial theorem,

$$\frac{1 - (1 - \delta)^n}{n\delta} = \sum_{j \geq 0} p_j(n) \delta^j, \quad p_j(n) = \frac{(-1)^j}{n} \binom{n}{j+1} = \frac{(-1)^j (n-1) \cdots (n-j)}{(j+1)!}, \quad (4.17)$$

so  $p_0 = 1$ ,  $p_1(n) = -(n-1)/2$  and  $|p_j(n)| \leq n^j/(j+1)!$ . Hence for  $n|\delta| \leq \frac{1}{2}$ ,

$$\left| \frac{1 - (1 - \delta)^n}{n\delta} - 1 \right| \leq \sum_{j \geq 1} \frac{(n|\delta|)^j}{(j+1)!} = \frac{e^x - 1 - x}{x} \Big|_{x=n|\delta|} \leq \frac{e^{1/2} - \frac{3}{2}}{\frac{1}{2}} < 0.298 < \frac{1}{2},$$

the middle function being increasing in  $x > 0$ . Therefore the series [equation \(4.17\)](#) is a nonvanishing holomorphic function on  $|\delta| \leq 1/(2n) = 1/(4k)$ , its reciprocal  $G_n(\delta) = n\delta/(1 - (1 - \delta)^n)$  is holomorphic there with  $|G_n| \leq (1 - \frac{1}{2})^{-1} = 2$ , and  $[\delta^1] G_n = -p_1(n) = (n-1)/2$ . On the same disc



$|(1 + b\delta)^n| \leq e^{n|b||\delta|} \leq e^{|b|/2}$ , which gives the bound in [equation \(4.11\)](#); and  $[\delta^1] H_k = nb + (n - 1)/2$  with  $n = 2k$ , which is the last assertion.

(ii)  $\omega^2$  has multiplicative order  $d$ , so  $\omega^{2k}$  depends only on  $k \bmod d$  and equals 1 exactly when  $d \mid k$ ; for  $d \nmid k$  it is one of the numbers  $\omega^{2r}$ ,  $1 \leq r < d$ . Write  $\omega^{2r} = e^{i\phi}$ . The distance from 1 to the segment  $\{e^{i\phi}t : 0 \leq t \leq 1\}$  equals  $|\sin \phi|$  if  $\cos \phi \geq 0$  and equals 1 if  $\cos \phi < 0$ . In the second case the distance is  $1 \geq \sin(\pi/d)$ . In the first case  $\phi = 2\pi s/d$  with  $s \not\equiv 0 \pmod{d}$ , and writing  $2s \equiv u \pmod{2d}$  with  $0 \leq u < 2d$  we get  $|\sin \phi| = |\sin(\pi u/d)|$ ; here  $u = 0$  is excluded by  $s \not\equiv 0 \pmod{d}$ , and  $u = d$  forces  $\phi = \pi$ , hence  $\cos \phi < 0$ , which is the second case. For the remaining  $u$  we have  $|\sin(\pi u/d)| \geq \sin(\pi/d)$ . Since  $(1 - \delta)^{2k} \in [0, 1]$  for real  $\delta \in [0, 1]$ , the displayed lower bound on  $|1 - \omega^{2k}(1 - \delta)^{2k}|$  follows.

(iii) With  $D_k(\delta) = 1 - \omega^{2k}(1 - \delta)^{2k}$  one has  $D'_k(\delta) = 2k\omega^{2k}(1 - \delta)^{2k-1}$ , so

$$S'_k(\delta) = \frac{2k\omega(a + \omega\delta)^{2k-1}}{D_k(\delta)} - \frac{2k\omega^{2k}(1 - \delta)^{2k-1}(a + \omega\delta)^{2k}}{D_k(\delta)^2}.$$

For real  $\delta \in [0, 1]$  we have  $|(1 - \delta)^{2k-1}| \leq 1$  and, by (ii),  $|D_k| \geq \gamma$ . For  $0 \leq \delta \leq |a|/2$  also  $|a + \omega\delta| \geq |a| - \delta \geq |a|/2$ , so  $|a + \omega\delta|^{2k-1} \leq (2/|a|)|a + \omega\delta|^{2k}$ . Combining the three bounds gives [equation \(4.13\)](#).

(iv) The minimum of  $|1 - \eta|$  over  $d$ th roots of unity  $\eta \neq 1$  is  $|1 - e^{2\pi i/d}| = 2 \sin(\pi/d) = \gamma_0$ . For  $|\delta| \leq \lambda_\omega/(2k)$ ,

$$|(1 - \delta)^{2k} - 1| \leq (1 + |\delta|)^{2k} - 1 \leq e^{2k|\delta|} - 1 \leq e^{\lambda_\omega} - 1 \leq \frac{\gamma_0}{2},$$

so  $|1 - \omega^{2k}(1 - \delta)^{2k}| \geq \gamma_0 - \gamma_0/2 = \gamma_0/2 > 0$ ; in particular the denominator does not vanish and  $S_k$  is holomorphic there. The numerator obeys  $|(a + \omega\delta)^{2k}| \leq |a|^{2k} e^{2k|b||\delta|} \leq |a|^{2k} e^{|b|/2}$ , which gives [equation \(4.15\)](#).

(v)  $1 - \delta \leq e^{-\delta}$  gives the first inequality. For the second,  $x \mapsto 1 - e^{-x}$  is concave with value 0 at  $x = 0$ , hence lies above the chord  $x \mapsto (1 - e^{-1})x$  on  $[0, 1]$ , and it is increasing, hence at least  $1 - e^{-1}$  for  $x \geq 1$ .  $\square$

**Theorem 4.4** (Uniform two-term radial root-of-unity expansion). *Let  $K$  be a compact subset of  $\{z : |a_\omega z| < \pi\}$ . Uniformly for  $z \in K$ ,*

$$\log \Phi(\omega(1 - \delta), z) = -\frac{\mathcal{A}_\omega(z)}{\delta} + \mathcal{B}_\omega(z) + O_K(\delta). \quad (4.18)$$

*In particular, every nontrivial root of unity carries a full exponential-scale singularity unless  $\mathcal{A}_\omega(z) = 0$ .*

*Proof.* Put  $a = a_\omega$  and  $b = \omega/a$ , choose  $0 < \rho < \rho_1 < 1$  with  $|az| \leq \rho\pi$  on  $K$ , and shrink  $\delta_0 > 0$  so that  $\delta_0 \leq \min\{\frac{1}{64}, |a|/2\}$  and  $|(a + \omega\delta)z| \leq \rho_1\pi$  for  $z \in K$  and  $0 \leq \delta \leq \delta_0$ . Then  $(q, z) \in \Omega$  for  $q = \omega(1 - \delta)$  and  $z \in K$ , so [equation \(3.3\)](#) applies and

$$\log \Phi(\omega(1 - \delta), z) = -\sum_{k \geq 1} c_k S_k(\delta) z^{2k}$$

with  $S_k$  as in [equation \(4.9\)](#). We use the majorants  $c_k |az|^{2k} \leq \zeta(2)\rho^{2k}/k$  and  $c_k |(a + \omega\delta)z|^{2k} \leq \zeta(2)\rho_1^{2k}/k$ , both immediate from  $c_k \leq \zeta(2)/(k\pi^{2k})$ .

*Nonresonant indices.* Assume  $d \geq 2$ ; for  $d = 1$  there are none. For  $d \nmid k$  the function  $S_k$  is real-analytic on  $[0, \delta_0]$  and [lemma 4.3\(iii\)](#) applies, so the mean value theorem gives, for  $0 \leq \delta \leq \delta_0$ ,

$$c_k |z|^{2k} |S_k(\delta) - S_k(0)| \leq 2k\Theta \delta c_k \max_{0 \leq s \leq \delta} |(a + \omega s)z|^{2k} \leq C_K \delta \rho_1^{2k}, \quad C_K = 2\zeta(2)\Theta. \quad (4.19)$$

Summing over  $k$  bounds the total nonresonant deviation by  $C_K \rho_1^2 (1 - \rho_1^2)^{-1} \delta = O_K(\delta)$ , and the retained values  $S_k(0) = a^{2k}/(1 - \omega^{2k})$  contribute exactly [equation \(4.7\)](#).

*Resonant indices below the cutoff.* Let  $d \mid k$ , put  $n = 2k$ , and set  $N = \lfloor \delta^{-1/2} \rfloor$ . Since  $\delta \leq 1/64$  we have  $N \leq \delta^{-1/2} \leq 1/(8\delta)$ , so every  $k \leq N$  satisfies  $\delta \leq 1/(8k)$ , that is,  $\delta$  lies within half the radius of the disc of lemma 4.3(i). Cauchy's estimate on that disc, applied to  $H_k$  with  $|H_k| \leq M_1$  and radius  $1/(4k)$ , gives for  $|\delta| \leq 1/(8k)$

$$|H_k(\delta) - 1 - ([\delta^1] H_k)\delta| \leq M_1 \frac{(4k\delta)^2}{1 - 4k\delta} \leq 32M_1 k^2 \delta^2.$$

Dividing by  $2k\delta$  and using equations (4.10) and (4.11) therefore yields the uniform expansion

$$\frac{S_k(\delta)}{a^{2k}} = \frac{1}{2k\delta} + \frac{\omega}{a} + \frac{2k-1}{4k} + O(k\delta), \quad (4.20)$$

in which the implied constant is  $16M_1$ , independent of  $k \leq N$  and of  $\delta \leq \delta_0$ . Multiplying by  $c_k |az|^{2k}$  bounds the remainder of the  $k$ th term by  $16M_1 \zeta(2) \rho^{2k} \delta$ , so the sum of these remainders over the resonant indices  $k \leq N$  is at most  $16M_1 \zeta(2) \rho^2 (1 - \rho^2)^{-1} \delta = O_K(\delta)$ . The three retained terms of equation (4.20), summed over all resonant  $k$ , produce exactly  $-\mathcal{A}_\omega(z)/\delta$  and equation (4.8).

*Resonant indices beyond the cutoff.* By lemma 4.3(v),

$$c_k |z|^{2k} |S_k(\delta)| \leq \frac{\zeta(2) \rho_1^{2k}}{k(1 - e^{-1}) \min(1, 2k\delta)} \leq \frac{\zeta(2)}{1 - e^{-1}} \rho_1^{2k} \left( \frac{1}{k} + \frac{1}{2k^2 \delta} \right),$$

while the corresponding subtracted terms satisfy

$$c_k |az|^{2k} \left( \frac{1}{2k\delta} + |b| + \frac{2k-1}{4k} \right) \leq \zeta(2) \rho^{2k} \left( \frac{1}{2k^2 \delta} + \frac{|b|+1}{k} \right).$$

Summing both over  $k > N$  gives  $O_K(\delta^{-1} \rho_1^{2N})$ , and  $\rho_1^{2N} \leq \rho_1^{-2} (\rho_1^2)^{\delta^{-1/2}}$ , so by equation (A.2) this is  $o_K(\delta^M)$  for every  $M$ ; in particular it is  $O_K(\delta)$ .

Adding the three estimates proves equation (4.18), with the explicit constant  $2\zeta(2)\Theta\rho_1^2(1 - \rho_1^2)^{-1} + 16M_1\zeta(2)\rho^2(1 - \rho^2)^{-1} + 1$  for all sufficiently small  $\delta$ , the first summand being absent when  $d = 1$ .  $\square$

### 4.3 Dilogarithmic and cyclic descriptions

The action has two additional forms that expose its spectral and finite-cyclic content.

**Theorem 4.5** (Spectral dilogarithm). *For  $|a_\omega z| < \pi$ ,*

$$\boxed{\mathcal{A}_\omega(z) = \frac{1}{2d^2} \sum_{n \geq 1} \text{Li}_2 \left( \left( \frac{a_\omega z}{\pi n} \right)^{2d} \right)}. \quad (4.21)$$

*The series converges absolutely and locally uniformly.*

*Proof.* Since  $c_k = \zeta(2k)/(k\pi^{2k})$ , the  $\ell$ th term of equation (4.6) is

$$\frac{c_{\ell d}}{2\ell d} a_\omega^{2\ell d} z^{2\ell d} = \frac{\zeta(2\ell d)}{2(\ell d)^2} \left( \frac{a_\omega z}{\pi} \right)^{2\ell d} = \sum_{n \geq 1} \frac{1}{2(\ell d)^2} \left( \frac{a_\omega z}{\pi n} \right)^{2\ell d}.$$

Let  $K$  be a compact subset of  $\{z : |a_\omega z| < \pi\}$  and put  $\rho = \max_{z \in K} |a_\omega z|/\pi < 1$ . Because  $\zeta(2\ell d) \leq \zeta(2)$  for  $\ell d \geq 1$ , the resulting double family is absolutely summable, with

$$\sum_{n \geq 1} \sum_{\ell \geq 1} \frac{1}{2(\ell d)^2} \left| \frac{a_\omega z}{\pi n} \right|^{2\ell d} = \sum_{\ell \geq 1} \frac{\zeta(2\ell d)}{2(\ell d)^2} \rho^{2\ell d} \leq \frac{\zeta(2)}{2d^2} \sum_{\ell \geq 1} \frac{\rho^{2\ell d}}{\ell^2} \leq \frac{\zeta(2)^2}{2d^2} \quad (4.22)$$

uniformly for  $z \in K$ . A summable double family may be summed in either order, so

$$\mathcal{A}_\omega(z) = \sum_{n \geq 1} \sum_{\ell \geq 1} \frac{1}{2(\ell d)^2} \left( \frac{a_\omega z}{\pi n} \right)^{2\ell d} = \frac{1}{2d^2} \sum_{n \geq 1} \text{Li}_2 \left( \left( \frac{a_\omega z}{\pi n} \right)^{2d} \right),$$

the inner sum being the defining series  $\text{Li}_2(x) = \sum_{\ell \geq 1} x^\ell / \ell^2$  evaluated at  $x = (a_\omega z / (\pi n))^{2d}$ , which satisfies  $|x| \leq \rho^{2d} < 1$ . Since the bound [equation \(4.22\)](#) is uniform on  $K$  and  $K$  was an arbitrary compact subset, the convergence is absolute and locally uniform.  $\square$

**Theorem 4.6** (Cyclic projection formula). *For  $|a_\omega z| < \pi$ ,*

$$\mathcal{A}_\omega(z) = - \int_0^z \left[ \frac{1}{d} \sum_{r=0}^{d-1} \log \text{sinc}_{\text{rad}}(a_\omega \omega^r t) \right] \frac{dt}{t}. \quad (4.23)$$

Here every logarithm is the zero-normalized holomorphic branch from [equation \(3.1\)](#), and the integral is taken along the straight segment from 0 to  $z$  (equivalently, along any path in the disk). The integrand has a holomorphic removable extension at zero.

*Proof.* Fix  $z$  with  $|a_\omega z| < \pi$  and put  $\rho = |a_\omega z| / \pi < 1$ . For  $t$  on the segment  $[0, z]$  every argument  $a_\omega \omega^r t$  lies in  $\{|w| \leq \rho\pi\}$ , so [equation \(3.1\)](#) applies to each of the  $d$  logarithms and

$$\frac{1}{d} \sum_{r=0}^{d-1} \log \text{sinc}_{\text{rad}}(a_\omega \omega^r t) = - \sum_{k \geq 1} c_k a_\omega^{2k} t^{2k} \left( \frac{1}{d} \sum_{r=0}^{d-1} \omega^{2kr} \right) = - \sum_{\substack{k \geq 1 \\ d|k}} c_k a_\omega^{2k} t^{2k},$$

because  $\omega^2$  has order  $d$ , so the inner average is  $1_{d|k}$ . The surviving series has lowest term  $t^{2d}$ , so after division by  $t$  the integrand of [equation \(4.23\)](#) is  $O(t^{2d-1})$  and extends holomorphically through  $t = 0$ . Parametrizing the segment by  $t = sz$  with  $s \in [0, 1]$  turns the integrand into  $-\sum_{d|k} c_k a_\omega^{2k} z^{2k} s^{2k-1}$ , which is dominated by the summable constants  $\zeta(2)\rho^{2k}/k$  uniformly in  $s \in [0, 1]$  by [equation \(A.1\)](#). Uniform convergence on the compact parameter interval permits termwise integration, and  $\int_0^1 s^{2k-1} ds = 1/(2k)$  turns the  $k$ th term into  $c_k a_\omega^{2k} z^{2k} / (2k)$ . Summing over  $k = \ell d$  gives exactly [equation \(4.6\)](#). The integral is path independent inside the disc because the integrand is holomorphic there.  $\square$

*Remark 4.7.* Formula [equation \(4.23\)](#) is a root-of-unity filter in the scale index. The sinc function is even, so the relevant cycle is generated by  $\omega^2$ , not  $\omega$ ; this explains the compression from order  $m$  to  $d = m / \gcd(m, 2)$ .

## 5 A natural-boundary theorem

### 5.1 The sign of the action

The crucial point is that the action cannot vanish on a uniform real-frequency interval.

**Lemma 5.1** (Reality and sign of  $a_\omega^{2d}$ ). *The number  $a_\omega^{2d}$  is real and nonzero. For a primitive  $m$ th root,*

$$\text{sgn}(a_\omega^{2d}) = \begin{cases} -1, & m \text{ odd}, \\ (-1)^{d+1}, & m \text{ even}. \end{cases} \quad (5.1)$$

*Thus it is positive exactly when  $m \equiv 2 \pmod{4}$ .*

*Proof.* Write  $\omega = e^{i\theta}$ . Since  $1 - e^{i\theta} = -2ie^{i\theta/2} \sin(\theta/2)$ ,

$$a_\omega^{2d} = (-4)^d e^{id\theta} \sin^{2d}(\theta/2).$$

If  $m$  is odd then  $d = m$  and  $e^{id\theta} = 1$ , while  $d$  is odd. If  $m$  is even then  $d = m/2$ , the primitive numerator  $h$  is odd, and  $e^{id\theta} = e^{i\pi h} = -1$ . The signs follow.  $\square$

**Theorem 5.2** (Nonvanishing action on a universal interval). *Let  $\omega \neq 1$  be a primitive  $m$ th root of unity and let  $z$  be either real or purely imaginary, with*

$$0 < |z| < \frac{\pi}{2}. \quad (5.2)$$

*Then  $\mathcal{A}_\omega(z)$  is real, nonzero, and*

$$\operatorname{sgn} \mathcal{A}_\omega(z) = \begin{cases} \operatorname{sgn}(a_\omega^{2d}), & z \in \mathbb{R}, \\ (-1)^d \operatorname{sgn}(a_\omega^{2d}), & z \in i\mathbb{R}. \end{cases} \quad (5.3)$$

*Proof.* Because  $|a_\omega| \leq 2$ , condition [equation \(5.2\)](#) implies  $|a_\omega z/(\pi n)| < 1$  for every  $n \geq 1$ . Write  $z = \varsigma$  in the real case and  $z = i\varsigma$  with  $\varsigma$  real in the imaginary case; in both cases  $z^{2d} = \epsilon \varsigma^{2d}$  with  $\epsilon = 1$  and  $\epsilon = (-1)^d$  respectively, because  $i^{2d} = (-1)^d$ . Hence, by [equation \(5.1\)](#),

$$\left(\frac{a_\omega z}{\pi n}\right)^{2d} = a_\omega^{2d} \epsilon \left(\frac{\varsigma}{\pi n}\right)^{2d} = \sigma x_n, \quad \sigma = \epsilon \operatorname{sgn}(a_\omega^{2d}) \in \{-1, 1\}, \quad 0 < x_n < 1,$$

the sign  $\sigma$  being the same for every  $n$  and the bound  $x_n < 1$  coming from [equation \(5.2\)](#) exactly as in the real case, since  $|z| = |\varsigma|$ . For  $0 < x < 1$ ,  $\operatorname{Li}_2(x) > 0$  and  $\operatorname{Li}_2(-x) < 0$ , and the series [equation \(4.21\)](#) converges absolutely. Every summand therefore has sign  $\sigma$ , so the sum is real and nonzero with that sign, which is [equation \(5.3\)](#).  $\square$

*Remark 5.3* (Why the imaginary slice is worth the extra line). The real slice  $z \in \mathbb{R}$  is the characteristic-function slice; the imaginary slice  $z \in i\mathbb{R}$  is the *exponential* slice, on which  $\Phi$  becomes the centered moment generating germ of Part VII. The two are transverse, and the natural-boundary statement that Part VII conjectures lives on the second one. The only change in the proof is the factor  $i^{2d} = (-1)^d$ , which moves the common sign without ever making it depend on  $n$ ; that is what [corollary 5.5](#) needs.

## 5.2 Natural boundary

**Theorem 5.4** (Natural boundary on a universal frequency interval). *Fix  $z$  either real or purely imaginary with  $0 < |z| < \pi/2$ . Then the holomorphic function*

$$q \mapsto \Phi(q, z), \quad |q| < 1, \quad (5.4)$$

*admits no single-valued meromorphic continuation through any point of the unit circle. Explicitly: there exist no  $\zeta_0$  with  $|\zeta_0| = 1$ , no  $\varrho > 0$ , and no function  $F$  meromorphic on the disc  $\Delta(\zeta_0, \varrho) = \{q : |q - \zeta_0| < \varrho\}$  such that  $F = \Phi(\cdot, z)$  on the connected set  $\Delta(\zeta_0, \varrho) \cap \mathbb{D}$ . In particular the unit circle is a natural boundary, and no continuation crosses a nontrivial boundary arc.*

*Proof. Step 1:*  $\Phi(\cdot, z)$  is zero-free on  $\mathbb{D}$ . For  $|q| < 1$  we have  $|1 - q| < 2$ , hence  $|(1 - q)z| < 2 \cdot \pi/2 = \pi$ , so  $(q, z) \in \Omega$  and [equation \(3.3\)](#) represents  $\log \Phi(q, z)$  as a convergent series. Therefore  $\Phi(q, z) = e^{\log \Phi(q, z)} \neq 0$  for every  $q \in \mathbb{D}$ . The same inequality shows  $|a_\omega z| \leq 2|z| < \pi$  for every root of unity  $\omega$ , so  $z$  lies in the common domain of  $\mathcal{A}_\omega$  and  $\mathcal{B}_\omega$  and [equation \(4.18\)](#) may be applied at  $z$ .

*Step 2: no local continuation.* Suppose such a triple  $(\zeta_0, \varrho, F)$  existed. The lens  $\Delta(\zeta_0, \varrho) \cap \mathbb{D}$  is a nonempty connected open set on which  $F$  coincides with the zero-free function  $\Phi(\cdot, z)$ , so  $F$  does not vanish identically on the connected disc  $\Delta(\zeta_0, \varrho)$ . Roots of unity are dense on the unit circle, so  $\Delta(\zeta_0, \varrho)$  contains a root of unity  $\omega \neq 1$ ; fix one, primitive of some order  $m \geq 2$ . For all sufficiently small  $\delta > 0$  the radial point  $\omega(1 - \delta)$  lies in  $\Delta(\zeta_0, \varrho) \cap \mathbb{D}$ , so

$$F(\omega(1 - \delta)) = \Phi(\omega(1 - \delta), z). \quad (5.5)$$

*Step 3: the two growth rates are incompatible.* Because  $F$  is meromorphic near  $\omega$  and not identically zero, there are an integer  $\nu$  and a function  $G$  holomorphic near  $\omega$  with  $G(\omega) \neq 0$  and  $F(q) =$

$(q - \omega)^\nu G(q)$ ; this single formula covers a regular point ( $\nu = 0$ ), a zero ( $\nu > 0$ ) and a pole ( $\nu < 0$ ), and  $\nu$  is finite in each case. Since  $|\omega(1 - \delta) - \omega| = \delta$ ,

$$\log |F(\omega(1 - \delta))| = \nu \log \delta + O(1), \quad \delta \downarrow 0. \quad (5.6)$$

On the other hand  $A := \mathcal{A}_\omega(z)$  is real and nonzero by [theorem 5.2](#), and taking real parts in [equation \(4.18\)](#) with  $K = \{z\}$  gives

$$\log |\Phi(\omega(1 - \delta), z)| = -\frac{A}{\delta} + O(1). \quad (5.7)$$

By [equation \(5.5\)](#) the left-hand sides of [equations \(5.6\)](#) and [\(5.7\)](#) coincide. But the right-hand side of [equation \(5.6\)](#) is  $O(\log(1/\delta))$ , whereas that of [equation \(5.7\)](#) tends to  $-\infty$  at rate  $\delta^{-1}$  when  $A > 0$  and to  $+\infty$  at rate  $\delta^{-1}$  when  $A < 0$ . Dividing by  $\delta^{-1}$  and letting  $\delta \downarrow 0$  gives  $-A = 0$ , contradicting  $A \neq 0$ .

Hence no boundary point admits a single-valued meromorphic continuation. A continuation across a nontrivial arc would restrict to one at each of its points, so none exists; and since a meromorphic continuation is in particular weaker than a holomorphic one, the unit circle is a natural boundary.  $\square$

**Corollary 5.5** (The centered exponential germ of Part VII). *For every real  $t$  with  $0 < |t| < \pi$ , the unit circle is a natural boundary for*

$$q \mapsto C_q(t) = e^{-t/2} M_q(t), \quad |q| < 1,$$

*the centered exponential germ of [equation \(2.8\)](#), and no single-valued meromorphic continuation crosses a nontrivial boundary arc. In particular [conjecture 17.1](#) holds on that interval, with its hypothesis “generic fixed  $t \neq 0$ ” replaced by the explicit punctured interval  $0 < |t| < \pi$  and with no removable degeneracies to discard.*

*Proof.* By [equation \(2.5\)](#) and the dictionary [table 1](#),  $\Phi(q, z) = \varphi_q(2z) = \mathbb{E} e^{2izX_q^\circ}$ , while  $C_q(t) = \mathbb{E} e^{tX_q^\circ}$  by [proposition 0.3.2](#). Both expressions are entire in their argument, so setting  $z = -it/2$  gives the identity of holomorphic functions

$$C_q(t) = \Phi\left(q, -\frac{it}{2}\right), \quad t \in \mathbb{C}, |q| < 1. \quad (5.8)$$

For real  $t \neq 0$  the frequency  $-it/2$  is purely imaginary, and the constraint  $0 < |z| < \pi/2$  of [theorem 5.4](#) reads  $0 < |t| < \pi$ . Apply that theorem.  $\square$

*Remark 5.6* (Scope of the theorem). The theorem gives an explicit, root-independent interval  $0 < |z| < \pi/2$  on each of the two slices. It does not classify exceptional complex frequencies off them. There the action can cross dilogarithm branch cuts, and cancellation between spectral terms is a genuine possibility.

## 6 Cyclotomic blow-ups

The fixed-frequency expansion diverges like  $\delta^{-1}$ . A nontrivial local limit appears when the frequency shrinks at the exact resonant rate.

### 6.1 Leading blow-up theorem

Set

$$q_\varepsilon = \omega(1 - \varepsilon^d), \quad z = \sqrt{\varepsilon} w, \quad \varepsilon \downarrow 0, \quad (6.1)$$

and define

$$A_\omega = \frac{c_d a_\omega^{2d}}{2d} = \frac{\zeta(2d) a_\omega^{2d}}{2d^2 \pi^{2d}}. \quad (6.2)$$

**Lemma 6.1** (Uniform scaled-tail bound). *For every compact set  $W \subset \mathbb{C}$  there are  $C_W > 0$  and  $\varepsilon_0 > 0$  such that, for every integer  $M \geq d$ , every  $w \in W$ , and  $0 < \varepsilon \leq \varepsilon_0$ , the tail of the master logarithm satisfies*

$$\sum_{k \geq M} c_k \varepsilon^k |w|^{2k} \left| \frac{(a_\omega + \omega \varepsilon^d)^{2k}}{1 - \omega^{2k}(1 - \varepsilon^d)^{2k}} \right| \leq C_W \varepsilon^{M-d}. \quad (6.3)$$

*Proof.* Put  $s = \varepsilon^d$ . If  $d \mid k$ , then  $1 - (1 - s)^{2k} \geq s$ . If  $d \nmid k$ , the finitely many phases  $\omega^{2k} \neq 1$  keep  $|1 - \omega^{2k}(1 - s)^{2k}|$  uniformly away from zero. After reducing  $\varepsilon_0$  if necessary, both cases are therefore bounded below by  $c_\omega s$  for a fixed  $c_\omega > 0$ . If  $|w| \leq R$  on  $W$ , then  $c_k \leq \zeta(2)/(k\pi^{2k})$  turns the left side into at most

$$C \varepsilon^{-d} \sum_{k \geq M} \frac{1}{k} \left( \frac{|a_\omega + \omega \varepsilon^d|^2 R^2}{\pi^2} \varepsilon \right)^k.$$

The parenthesized base is at most  $B_W \varepsilon < 1/2$  for small  $\varepsilon$ . Summing the geometric tail proves equation (6.3).  $\square$

**Theorem 6.2** (Cyclotomic blow-up limit). *Locally uniformly for  $w \in \mathbb{C}$ ,*

$$\boxed{\Phi(q_\varepsilon, \sqrt{\varepsilon} w) \longrightarrow \exp(-A_\omega w^{2d})}. \quad (6.4)$$

*Proof.* In equation (3.3), the  $k$ th term acquires the factor  $\varepsilon^k$ . If  $d \nmid k$ , the multiplier equation (3.5) remains bounded and the term tends to zero. If  $k = \ell d$ , its multiplier is asymptotic to  $a_\omega^{2k}/(2k\varepsilon^d)$ , so the term has order  $\varepsilon^{(\ell-1)d}$ . Only  $k = d$  survives, giving  $-A_\omega w^{2d}$ . Taking  $M = d + 1$  in equation (6.3) makes every omitted term  $O_W(\varepsilon)$ , which proves local uniformity and permits exponentiation.  $\square$

## 6.2 Subcritical and order- $d$ corrections

For  $1 \leq k < d$  define

$$C_{k,\omega} = c_k \frac{a_\omega^{2k}}{1 - \omega^{2k}}, \quad (6.5)$$

and set

$$\eta_\omega = \frac{2d\omega}{1 - \omega} + \frac{2d - 1}{2}, \quad D_\omega = \frac{c_{2d} a_\omega^{4d}}{4d}. \quad (6.6)$$

**Theorem 6.3** (First complete correction layers). *Uniformly on compact  $w$ -sets,*

$$\log \Phi(q_\varepsilon, \sqrt{\varepsilon} w) = -A_\omega w^{2d} - \sum_{k=1}^{d-1} C_{k,\omega} \varepsilon^k w^{2k} \quad (6.7)$$

$$- \varepsilon^d (A_\omega \eta_\omega w^{2d} + D_\omega w^{4d}) + O(\varepsilon^{d+1}). \quad (6.8)$$

*Proof.* For  $k < d$  there is no resonance, so the constant term of the multiplier gives equation (6.5). At total order  $\varepsilon^d$ , two and only two resonant sources contribute. The next term in the  $k = d$  multiplier is the constant in equation (4.5), producing  $A_\omega \eta_\omega w^{2d}$ . The leading pole of the  $k = 2d$  multiplier produces  $D_\omega w^{4d}$ . Every omitted nonresonant term begins at  $k = d + 1$ , and every later resonant term begins at order at least  $2d$ . Taylor's theorem handles the finitely many displayed indices, while equation (6.3) with  $M = 2d + 1$  handles the infinite remainder, giving  $O(\varepsilon^{d+1})$ .  $\square$

## 6.3 All orders and Bell polynomials

Define the Laurent germ

$$\mathcal{R}_k(\varepsilon) = \frac{(a_\omega + \omega \varepsilon^d)^{2k}}{1 - \omega^{2k}(1 - \varepsilon^d)^{2k}}. \quad (6.9)$$

For  $d \nmid k$  it is analytic at zero; for  $d \mid k$  it has a simple pole in  $\varepsilon^d$ .



**Theorem 6.4** (All-orders logarithmic expansion). *There are even polynomials  $L_n(w)$ , with  $\deg L_n \leq 2(n + d)$ , such that for every compact  $W \subset \mathbb{C}$  and every integer  $N \geq 1$  there are  $C_{W,N} > 0$  and  $\varepsilon_{W,N} > 0$  with*

$$\left| \log \Phi(q_\varepsilon, \sqrt{\varepsilon} w) + A_\omega w^{2d} - \sum_{n=1}^N L_n(w) \varepsilon^n \right| \leq C_{W,N} \varepsilon^{N+1} \quad (6.10)$$

for all  $w \in W$  and  $0 < \varepsilon \leq \varepsilon_{W,N}$ . The constants depend on  $N$  and  $W$ , but not on  $w$  or  $\varepsilon$ . We abbreviate this locally uniform Poincaré expansion by

$$\log \Phi(q_\varepsilon, \sqrt{\varepsilon} w) \sim -A_\omega w^{2d} + \sum_{n \geq 1} L_n(w) \varepsilon^n, \quad (6.11)$$

with no claim that the series on the right converges. The coefficient is the finite sum

$$L_n(w) = - \sum_{k=1}^{n+d} c_k w^{2k} \left[ \varepsilon^{n-k} \right] \mathcal{R}_k(\varepsilon). \quad (6.12)$$

Consequently,

$$\Phi(q_\varepsilon, \sqrt{\varepsilon} w) \sim e^{-A_\omega w^{2d}} \left( 1 + \sum_{n \geq 1} P_n(w) \varepsilon^n \right), \quad (6.13)$$

with

$$P_n(w) = \frac{1}{n!} B_n(1! L_1(w), 2! L_2(w), \dots, n! L_n(w)), \quad (6.14)$$

where  $B_n$  is the complete exponential Bell polynomial.

*Proof.* Fix  $N \geq 1$  and a compact  $W$ . For small  $\varepsilon$  we have  $|(1 - q_\varepsilon)\sqrt{\varepsilon} w| < \pi$  on  $W$ , so [equation \(3.3\)](#) applies and  $\log \Phi(q_\varepsilon, \sqrt{\varepsilon} w) = - \sum_{k \geq 1} c_k w^{2k} \varepsilon^k \mathcal{R}_k(\varepsilon)$  with  $\mathcal{R}_k$  as in [equation \(6.9\)](#). The coefficient of  $\varepsilon^n$  in the  $k$ th term is the Laurent coefficient of  $\mathcal{R}_k$  with index  $n - k$ ; since the lowest exponent occurring in  $\mathcal{R}_k$  is  $-d$ , only  $k \leq n + d$  contributes. Hence [equation \(6.12\)](#) is a finite sum and  $\deg L_n \leq 2(n + d)$ , so each  $L_n$  is bounded on  $W$ .

For the remainder, split at  $k = N + d$ . Each of the finitely many germs  $f_k(\varepsilon) = \varepsilon^d \cdot \varepsilon^k \mathcal{R}_k(\varepsilon)$  with  $1 \leq k \leq N + d$  is holomorphic at  $\varepsilon = 0$ , because  $\varepsilon^k \mathcal{R}_k$  has a pole of order at most  $d$  there. Taylor's theorem applied to  $f_k$  on a disc  $|\varepsilon| \leq r_k$  gives  $\left| f_k(\varepsilon) - \sum_{j \leq N+d} [\varepsilon^j] f_k \varepsilon^j \right| \leq c_N^{(k)} \varepsilon^{N+d+1}$ ; dividing by  $\varepsilon^d$  shows that  $\varepsilon^k \mathcal{R}_k(\varepsilon)$  differs from its truncation through order  $\varepsilon^N$  by at most  $c_N^{(k)} \varepsilon^{N+1}$ . There are at most  $N + d$  such indices, and multiplying by the bounded factor  $c_k w^{2k}$  keeps the bound uniform on  $W$ . The indices  $k \geq N + d + 1$  are covered by [equation \(6.3\)](#) with  $M = N + d + 1$ , whose right-hand side is  $C_W \varepsilon^{N+1}$ . Adding the two estimates gives [equation \(6.10\)](#). Formula [equation \(6.14\)](#) is then the standard exponential formula for exponentiating a formal series, applied coefficientwise to [equation \(6.11\)](#).  $\square$

*Remark 6.5.* This is a second Bell structure in the theory. The repository's Bell polynomials encode moments from cumulants; here the same combinatorial transform encodes multiplicative cyclotomic corrections from logarithmic ones. The two uses meet in [appendix 9](#).

## 6.4 Four small root orders

**Example 6.6** (The root  $\omega = -1$ ). Here  $m = 2$ ,  $d = 1$ , and  $A_{-1} = 1/3$ . Thus

$$\log \Phi(-1 + \varepsilon, \sqrt{\varepsilon} w) = -\frac{w^2}{3} + \varepsilon \left( \frac{w^2}{6} - \frac{w^4}{45} \right) + O(\varepsilon^2). \quad (6.15)$$

The first cyclotomic limit is Gaussian.



**Example 6.7** (Primitive cube root). For  $\omega = e^{2\pi i/3}$ ,  $d = 3$  and  $A_\omega = -1/630$ . Moreover

$$C_{1,\omega} = -\frac{i}{2\sqrt{3}}, \quad C_{2,\omega} = -\frac{i}{20\sqrt{3}}.$$

Hence

$$\log \Phi(\omega(1 - \varepsilon^3), \sqrt{\varepsilon} w) = \frac{w^6}{630} + \frac{i\varepsilon w^2}{2\sqrt{3}} + \frac{i\varepsilon^2 w^4}{20\sqrt{3}} + O(\varepsilon^3). \quad (6.16)$$

The real-axis limit grows, so its transform interpretation belongs to rotated Stokes sectors.

**Example 6.8** (Fourth root). For  $\omega = i$ ,  $d = 2$ ,  $A_\omega = -1/180$ , and  $C_{1,\omega} = -i/6$ . Therefore

$$\log \Phi(i(1 - \varepsilon^2), \sqrt{\varepsilon} w) = \frac{w^4}{180} + \frac{i\varepsilon w^2}{6} \quad (6.17)$$

$$- \varepsilon^2 \left[ \left( \frac{1}{360} - \frac{i}{90} \right) w^4 + \frac{w^8}{18900} \right] + O(\varepsilon^3). \quad (6.18)$$

**Example 6.9** (Primitive sixth root). For  $\omega = e^{\pi i/3}$ ,  $d = 3$ ,  $A_\omega = 1/17010$ ,

$$C_{1,\omega} = -\frac{i}{6\sqrt{3}}, \quad C_{2,\omega} = \frac{i\sqrt{3}}{540}.$$

Thus

$$\log \Phi(\omega(1 - \varepsilon^3), \sqrt{\varepsilon} w) = -\frac{w^6}{17010} + \frac{i\varepsilon w^2}{6\sqrt{3}} - \frac{i\sqrt{3}\varepsilon^2 w^4}{540} + O(\varepsilon^3). \quad (6.19)$$

This is the first genuinely higher-order dissipative limit.

## 7 A universal analytic-germ principle

The sinc product is one instance of a more general mechanism.

**Theorem 7.1** (Universal resonant germ). *Let  $h$  be entire with  $h(0) = 1$ , and fix  $R_h > 0$  so small that  $h$  has no zero on the closed disc  $|w| \leq R_h$ . Then  $h$  has a holomorphic logarithm vanishing at the origin, and Cauchy's estimates give*

$$\log h(z) = \sum_{n \geq 1} b_n z^n, \quad |b_n| \leq M_h R_h^{-n}, \quad M_h = \max_{|w|=R_h} |\log h(w)|. \quad (7.1)$$

For  $|q| < 1$  define

$$\mathcal{P}_h(q, z) = \prod_{j \geq 0} h((1 - q)q^j z). \quad (7.2)$$

Fix a root of unity  $\omega \neq 1$ , of order  $m \geq 2$ , put

$$\gamma_h = \min \{ |1 - \zeta t| : \zeta^m = 1, \zeta \neq 1, 0 \leq t \leq 1 \} > 0, \quad (7.3)$$

and let

$$r = \min \{ n \geq 1 : b_n \neq 0 \text{ and } \omega^n = 1 \}. \quad (7.4)$$

Assume this set is nonempty. Along the radial approach  $q = \omega(1 - \delta)$  with  $\delta > 0$  and  $\delta \downarrow 0$ , interpret  $\delta^{1/r}$  as the positive real root. Then, locally uniformly for  $w$ ,

$$\boxed{\mathcal{P}_h(\omega(1 - \delta), \delta^{1/r} w) \longrightarrow \exp \left( \frac{b_r (1 - \omega)^r}{r} w^r \right)}. \quad (7.5)$$

*Proof.* That  $\gamma_h > 0$  is clear:  $|1 - \zeta t|$  is continuous and positive on the compact set  $(\mu_m \setminus \{1\}) \times [0, 1]$ , since  $\zeta t = 1$  with  $t \in [0, 1]$  forces  $t = 1$  and  $\zeta = 1$ .

Write  $a = 1 - \omega$  and  $q = \omega(1 - \delta)$ , so that  $1 - q = a + \omega\delta$  and  $|1 - q| \leq |a| + \delta \leq 3$  for  $0 < \delta \leq 1$ . Let  $W \subset \mathbb{C}$  be compact,  $R_W = \max_W |w|$ , and restrict to those  $\delta$  with

$$0 < \delta \leq \min \{1, \gamma_h\}, \quad 3R_W \delta^{1/r} \leq \frac{1}{2} R_h. \quad (7.6)$$

For  $z = \delta^{1/r} w$  with  $w \in W$  this gives  $|(1 - q)q^j z| \leq |(1 - q)z| \leq \frac{1}{2} R_h < R_h$  for every  $j \geq 0$ : all arguments of  $h$  lie in the disc on which  $\log h$  is the holomorphic branch of [equation \(7.1\)](#), which is the domain of the computation that follows.

Since  $|b_n| |(1 - q)z|^n |q|^{jn} \leq M_h 2^{-n} |q|^j$  for  $n \geq 1$ , the double family is absolutely summable and

$$\log \mathcal{P}_h(q, z) = \sum_{j \geq 0} \log h((1 - q)q^j z) = \sum_{j \geq 0} \sum_{n \geq 1} b_n (1 - q)^n q^{jn} z^n = \sum_{n \geq 1} b_n \frac{(1 - q)^n}{1 - q^n} z^n,$$

each partial product of [equation \(7.2\)](#) being the exponential of the corresponding partial sum. Put  $\theta = |(1 - q)z| / R_h \leq \frac{1}{2}$ , so that  $|b_n| |(1 - q)z|^n \leq M_h \theta^n$  and  $\theta \leq 3R_W \delta^{1/r} / R_h$ .

If  $\omega^n \neq 1$  then  $q^n = \omega^n (1 - \delta)^n$  with  $(1 - \delta)^n \in [0, 1]$ , so  $|1 - q^n| \geq \gamma_h$  by [equation \(7.3\)](#); if  $\omega^n = 1$  then  $1 - q^n = 1 - (1 - \delta)^n \geq \delta$ . Since  $\delta \leq \gamma_h$ , in both cases  $|1 - q^n|^{-1} \leq \delta^{-1}$ . Hence the tail  $n > r$  obeys

$$\sum_{n > r} \left| b_n \frac{(1 - q)^n}{1 - q^n} z^n \right| \leq \frac{M_h}{\delta} \sum_{n > r} \theta^n \leq \frac{2M_h}{\delta} \theta^{r+1} \leq \frac{2M_h}{\delta} \left( \frac{3R_W}{R_h} \right)^{r+1} \delta^{(r+1)/r} = O_W(\delta^{1/r}).$$

For  $1 \leq n < r$  with  $b_n \neq 0$  the index is nonresonant by the minimality in [equation \(7.4\)](#), so  $|1 - q^n| \geq \gamma_h$  and the term is at most  $M_h \gamma_h^{-1} \theta^n = O_W(\delta^{n/r}) = O_W(\delta^{1/r})$ . Finally  $z^r = \delta w^r$ , so the term  $n = r$  equals

$$b_r w^r (a + \omega\delta)^r \frac{\delta}{1 - (1 - \delta)^r} \rightarrow \frac{b_r a^r}{r} w^r$$

uniformly on  $W$ , because  $\delta / (1 - (1 - \delta)^r) \rightarrow 1/r$  and  $(a + \omega\delta)^r \rightarrow a^r$ . Adding the three estimates gives  $\log \mathcal{P}_h(\omega(1 - \delta), \delta^{1/r} w) \rightarrow b_r a^r w^r / r$  uniformly on  $W$ , and exponentiating, which is uniformly continuous on bounded sets, gives [equation \(7.5\)](#).  $\square$

**Corollary 7.2** (Sinc and  $q$ -Pochhammer specializations). *For  $h(z) = \text{sinc}_{\text{rad}} z$  one may take any  $R_h < \pi$ ; only even coefficients occur, with  $b_{2k} = -c_k$ , so the first resonance is  $r = \min \{2k : m \mid 2k\} = 2d$  and  $b_r(1 - \omega)^r / r = -c_d a_\omega^{2d} / (2d) = -A_\omega$ , recovering [equation \(6.4\)](#). For  $h(z) = 1 - z$  one may take any  $R_h < 1$ ; here  $b_n = -1/n$ , so for a primitive  $m$ th root  $\omega = m$  and*

$$\prod_{j \geq 0} \left( 1 - (1 - q)q^j \delta^{1/m} w \right) \rightarrow \exp \left( -\frac{(1 - \omega)^m}{m^2} w^m \right). \quad (7.7)$$

*Remark 7.3.* The theorem separates three ingredients: the local Taylor support of  $\log h$ , the cyclic order of  $\omega$ , and the geometric denominator  $1 - q^n$ . It therefore applies to many  $q$ -analog kernels, not merely to the Fabius family.

## 8 Spectral zero condensation

### 8.1 The zero cloud

For  $0 < |q| < 1$ , every sinc factor contributes the factor-labelled zeros

$$z_{\ell, j}(q) = \frac{\pi \ell}{(1 - q)q^j}, \quad \ell \in \mathbb{Z} \setminus \{0\}, \quad j \geq 0. \quad (8.1)$$

The division-free form

$$(1 - q)q^j z = \pi \ell$$

is valid also at  $q = 0$  and is the correct total statement there. Distinct pairs  $(\ell, j)$  may describe the same geometric zero; throughout this section reciprocal sums count the Euler factors with their resulting multiplicity. For positive  $\ell$ , define the reciprocal even power sum

$$S_k(q) = \sum_{j \geq 0} \sum_{\ell \geq 1} z_{\ell, j}(q)^{-2k}. \quad (8.2)$$

**Proposition 8.1** (Exact reciprocal zero moments). *For every  $k \geq 1$ ,*

$$S_k(q) = \frac{\zeta(2k)}{\pi^{2k}} \frac{(1 - q)^{2k}}{1 - q^{2k}}. \quad (8.3)$$

Moreover, the master logarithm is the canonical zero-moment expansion

$$\log \Phi(q, z) = - \sum_{k \geq 1} \frac{S_k(q)}{k} z^{2k}. \quad (8.4)$$

*Proof.* Insert [equation \(8.1\)](#), then separate the geometric and zeta sums. The exact identity

$$\frac{S_k(q)}{k} = c_k \frac{(1 - q)^{2k}}{1 - q^{2k}}$$

makes [equation \(8.4\)](#) identical to the master logarithm.  $\square$

**Theorem 8.2** (One surviving spectral moment). *Under the cyclotomic scaling [equation \(6.1\)](#),*

$$\varepsilon^k S_k(q_\varepsilon) \longrightarrow \begin{cases} \frac{\zeta(2d)}{\pi^{2d}} \frac{a_\omega^{2d}}{2d}, & k = d, \\ 0, & k \neq d. \end{cases} \quad (8.5)$$

*Proof.* If  $d \nmid k$ , the multiplier in [equation \(8.3\)](#) remains bounded. If  $k = \ell d$ , it grows like  $\varepsilon^{-d}$ , so multiplication by  $\varepsilon^k$  gives  $\varepsilon^{(\ell-1)d}$ . Only  $\ell = 1$  survives.  $\square$

**Remark 8.3** (Escaping zeros leave an exponential trace). The limiting entire function  $e^{-A_\omega w^{2d}}$  has no zeros. The zero cloud of the rescaled sinc product escapes to infinity, but its  $2d$ th reciprocal moment remains finite and becomes the exponential factor in the Hadamard limit. All other reciprocal moments disappear.

## 9 Bernoulli cumulants, Bell moments, and Appell collapse

### 9.1 Complex moment functional

For real  $0 < q < 1$ , the moment generating function of  $Y_q$  is

$$M(q, t) = \mathbb{E} e^{tY_q} = \Phi(q, it) = \prod_{j \geq 0} \frac{\sinh((1 - q)q^j t)}{(1 - q)q^j t}. \quad (9.1)$$

For complex  $q$ , [equation \(9.1\)](#) defines an entire generating function and hence a complex moment functional. Its cumulants are defined for every  $q \in \mathbb{D}$  by the coefficient prescription

$$\kappa_n(q) := n! [t^n] \log M(q, t) \in \mathbb{C}, \quad (9.2)$$

the logarithm being the branch with  $\log M(q, 0) = 0$ ; for real  $0 < q < 1$  this is the usual  $n$ th cumulant of  $Y_q$ .

**Proposition 9.1** (Bernoulli form of the cumulants). *For every  $q \in \mathbb{D}$  and every  $k \geq 1$ ,*

$$\kappa_{2k}(q) = \frac{2^{2k-1} B_{2k}}{k} \frac{(1-q)^{2k}}{1-q^{2k}}, \quad \kappa_{2k+1}(q) = 0. \quad (9.3)$$

*Proof.* Insert  $z = it$  into [equation \(3.3\)](#), which is legitimate for  $|(1-q)t| < \pi$ , and use  $(it)^{2k} = (-1)^k t^{2k}$ :

$$\log M(q, t) = \sum_{k \geq 1} (-1)^{k-1} c_k R_k(q) t^{2k}, \quad (9.4)$$

absolutely convergent there. Only even powers occur, so  $\kappa_{2k+1}(q) = 0$  and  $\kappa_{2k}(q) = (2k)! (-1)^{k-1} c_k R_k(q)$ . Euler's formula  $\zeta(2k) = (-1)^{k+1} B_{2k} (2\pi)^{2k} / (2(2k)!)$  in the Bernoulli convention [equation \(0.4.1\)](#) converts  $c_k = \zeta(2k)/(k\pi^{2k})$  into

$$(2k)! (-1)^{k-1} c_k = (-1)^{k-1} \frac{(2k)!}{k\pi^{2k}} \cdot \frac{(-1)^{k+1} B_{2k} 2^{2k} \pi^{2k}}{2(2k)!} = \frac{2^{2k} B_{2k}}{2k} = \frac{2^{2k-1} B_{2k}}{k},$$

which is [equation \(9.3\)](#). For real  $0 < q < 1$  this is exactly  $2^{2k}$  times [equation \(0.4.4\)](#), as [status note 2.1](#) requires, since  $Y_q = 2X_q^\circ$  and the  $n$ th cumulant is homogeneous of degree  $n$ . Both sides of [equation \(9.3\)](#) are rational in  $q$  off the set  $q^{2k} = 1$ , so the identity for real  $q$  already forces it for all  $q \in \mathbb{D}$ ; the direct derivation above is recorded because it also fixes the meaning of the left-hand side there.  $\square$

**Lemma 9.2** (Coefficientwise limits pass through products, exponentials and reciprocals). *Let  $f_\varepsilon, f, g_\varepsilon, g \in \mathbb{C}[[t]]$  and suppose that  $[t^n] f_\varepsilon \rightarrow [t^n] f$  and  $[t^n] g_\varepsilon \rightarrow [t^n] g$  as  $\varepsilon \downarrow 0$ , for every  $n \geq 0$ . Then, coefficientwise as  $\varepsilon \downarrow 0$ ,*

- (i)  $f_\varepsilon g_\varepsilon \rightarrow fg$ ;
- (ii)  $e^{f_\varepsilon} \rightarrow e^f$ , provided  $f_\varepsilon(0) = f(0) = 0$ ;
- (iii)  $1/f_\varepsilon \rightarrow 1/f$ , provided  $f_\varepsilon(0) = f(0) = 1$ .

*Proof.* In each case the  $n$ th coefficient of the output is one fixed polynomial, with coefficients not depending on  $\varepsilon$ , in the first  $n+1$  coefficients of the input. For (i) this is the Cauchy product  $[t^n](fg) = \sum_{i+j=n} f_i g_j$ . For (ii) it is the exponential formula  $[t^n] e^f = B_n(1!f_1, \dots, n!f_n)/n!$ , with  $B_n$  the complete exponential Bell polynomial, which is the identity already used in [equation \(6.14\)](#). For (iii) it is the recursion  $h_0 = 1$ ,  $h_n = -\sum_{j=1}^n f_j h_{n-j}$ , which exhibits  $h_n = [t^n](1/f)$  as a polynomial in  $f_1, \dots, f_n$ . Polynomials are continuous, so coefficientwise convergence is preserved.  $\square$

**Theorem 9.3** (Cyclotomic cumulant condensation). *Let  $Z_\varepsilon = \sqrt{\varepsilon} Y_{q_\varepsilon}$  at the level of formal moments. Then*

$$\kappa_n(Z_\varepsilon) \longrightarrow 0 \quad (n \neq 2d), \quad (9.5)$$

and

$$\boxed{\kappa_{2d}(Z_\varepsilon) \longrightarrow K_{2d,\omega} = \frac{2^{2d-2} B_{2d} a_\omega^{2d}}{d^2}.} \quad (9.6)$$

Equivalently, coefficientwise near  $t = 0$ ,

$$M(q_\varepsilon, \sqrt{\varepsilon} t) \longrightarrow \exp(B_\omega t^{2d}), \quad B_\omega = (-1)^{d+1} A_\omega = \frac{K_{2d,\omega}}{(2d)!}. \quad (9.7)$$

*Proof.* Scaling by  $\sqrt{\varepsilon}$  multiplies the  $n$ th cumulant by  $\varepsilon^{n/2}$ , so  $\kappa_n(Z_\varepsilon) = \varepsilon^{n/2} \kappa_n(q_\varepsilon)$ . All odd cumulants vanish by [proposition 9.1](#), and for  $n = 2k$

$$\kappa_{2k}(Z_\varepsilon) = \frac{2^{2k-1} B_{2k}}{k} \varepsilon^k \mathcal{R}_k(\varepsilon)$$

with  $\mathcal{R}_k$  as in [equation \(6.9\)](#). If  $d \nmid k$  then  $\mathcal{R}_k(\varepsilon) \rightarrow a_\omega^{2k}/(1 - \omega^{2k})$  stays bounded, so the factor  $\varepsilon^k$  forces the limit 0. If  $k = \ell d$  then  $1 - (1 - \varepsilon^d)^{2k} = 2k\varepsilon^d(1 + O(\varepsilon^d))$ , so

$$\varepsilon^k \mathcal{R}_k(\varepsilon) = \frac{a_\omega^{2k}}{2k} \varepsilon^{(\ell-1)d} (1 + O(\varepsilon^d)),$$

which tends to 0 for  $\ell \geq 2$  and to  $a_\omega^{2d}/(2d)$  for  $\ell = 1$ . This is [equation \(9.5\)](#), and at  $k = d$  it gives

$$\kappa_{2d}(Z_\varepsilon) \longrightarrow \frac{2^{2d-1} B_{2d}}{d} \cdot \frac{a_\omega^{2d}}{2d} = \frac{2^{2d-2} B_{2d} a_\omega^{2d}}{d^2} = K_{2d,\omega},$$

which is [equation \(9.6\)](#). Consequently  $\log M(q_\varepsilon, \sqrt{\varepsilon} t) = \sum_{k \geq 1} \kappa_{2k}(Z_\varepsilon) t^{2k}/(2k)!$  converges coefficientwise to  $K_{2d,\omega} t^{2d}/(2d)!$ , and [lemma 9.2\(ii\)](#) upgrades this to the coefficientwise limit [equation \(9.7\)](#) for  $M$  itself, with  $B_\omega = K_{2d,\omega}/(2d)!$ . Finally Euler's formula for  $\zeta(2d)$  gives

$$A_\omega = \frac{\zeta(2d) a_\omega^{2d}}{2d^2 \pi^{2d}} = \frac{(-1)^{d+1} B_{2d} 2^{2d} \pi^{2d} a_\omega^{2d}}{2(2d)! \cdot 2d^2 \pi^{2d}} = (-1)^{d+1} \frac{2^{2d-2} B_{2d} a_\omega^{2d}}{d^2 (2d)!} = (-1)^{d+1} B_\omega,$$

which is the remaining identity of [equation \(9.7\)](#). □

**Corollary 9.4** (Sparse Bell moment limit). *Let  $m_n(q)$  be the moments generated by  $M(q, t)$ . Then*

$$\varepsilon^{n/2} m_n(q_\varepsilon) \longrightarrow 0 \quad \text{unless } n = 2dr, \quad (9.8)$$

and for  $r \geq 0$ ,

$$\boxed{\varepsilon^{dr} m_{2dr}(q_\varepsilon) \longrightarrow \frac{(2dr)!}{r!} B_\omega^r.} \quad (9.9)$$

*Proof.* By definition  $m_n(q) = n! [t^n] M(q, t)$ , and replacing  $t$  by  $\sqrt{\varepsilon} t$  multiplies the  $n$ th coefficient by  $\varepsilon^{n/2}$ , so

$$\varepsilon^{n/2} m_n(q_\varepsilon) = n! [t^n] M(q_\varepsilon, \sqrt{\varepsilon} t).$$

By [equation \(9.7\)](#) the right-hand side converges to  $n! [t^n] e^{B_\omega t^{2d}}$ . Expanding  $e^{B_\omega t^{2d}} = \sum_{r \geq 0} B_\omega^r t^{2dr}/r!$  shows that this vanishes unless  $2d \mid n$  and equals  $(2dr)! B_\omega^r/r!$  when  $n = 2dr$ , which is [equations \(9.8\)](#) and [\(9.9\)](#). Equivalently, apply the complete Bell polynomial to a cumulant list with exactly one nonzero entry. □

## 9.2 Limiting Appell family

Let  $\mathcal{A}_n^{(q)}$  denote the inverse-moment Appell polynomials generated by

$$\frac{e^{xt}}{M(q, t)} = \sum_{n \geq 0} \mathcal{A}_n^{(q)}(x) \frac{t^n}{n!}. \quad (9.10)$$

Define the scaled family

$$\tilde{\mathcal{A}}_{n,\varepsilon}(x) = \varepsilon^{n/2} \mathcal{A}_n^{(q_\varepsilon)}(x/\sqrt{\varepsilon}). \quad (9.11)$$

**Theorem 9.5** (Gould–Hopper cyclotomic limit). *Coefficientwise as  $\varepsilon \downarrow 0$ ,*

$$\tilde{\mathcal{A}}_{n,\varepsilon}(x) \longrightarrow H_n^{(d, B_\omega)}(x), \quad (9.12)$$

where

$$\sum_{n \geq 0} H_n^{(d, B)}(x) \frac{t^n}{n!} = e^{xt - Bt^{2d}} \quad (9.13)$$

and explicitly

$$H_n^{(d,B)}(x) = n! \sum_{r=0}^{\lfloor n/(2d) \rfloor} \frac{(-B)^r x^{n-2dr}}{r!(n-2dr)!}. \quad (9.14)$$

These polynomials satisfy

$$H_n^{(d,B)} = e^{-BD^{2d}} x^n, \quad H_{n+1}^{(d,B)} = \left( x - 2dBD^{2d-1} \right) H_n^{(d,B)}, \quad D = \frac{d}{dx}. \quad (9.15)$$

*Proof.* Substituting  $(x, t) \mapsto (x/\sqrt{\varepsilon}, \sqrt{\varepsilon} t)$  in [equation \(9.10\)](#) leaves  $e^{xt}$  unchanged and multiplies the  $n$ th Taylor coefficient by  $\varepsilon^{n/2}$ , so by [equation \(9.11\)](#)

$$\sum_{n \geq 0} \tilde{\mathcal{A}}_{n,\varepsilon}(x) \frac{t^n}{n!} = \frac{e^{xt}}{M(q_\varepsilon, \sqrt{\varepsilon} t)}.$$

The series  $M(q_\varepsilon, \sqrt{\varepsilon} t)$  has constant term 1 and converges coefficientwise to  $e^{B_\omega t^{2d}}$  by [equation \(9.7\)](#), so [lemma 9.2\(iii\)](#) gives coefficientwise convergence of its reciprocal, and then part (i), applied with the  $\varepsilon$ -independent factor  $e^{xt}$ , gives

$$[t^n] \frac{e^{xt}}{M(q_\varepsilon, \sqrt{\varepsilon} t)} \longrightarrow [t^n] e^{xt - B_\omega t^{2d}}$$

for every  $n$ . Both sides are polynomials in  $x$  of degree at most  $n$ ; expanding  $e^{xt} = \sum_{j \geq 0} x^j t^j / j!$  exhibits the coefficient of  $x^j$  in  $[t^n]$  as  $[t^{n-j}]$  of the reciprocal divided by  $j!$ , so the convergence is also coefficient by coefficient in  $x$ , which is the assertion of [equation \(9.12\)](#). Expanding  $e^{xt - B t^{2d}} = e^{xt} \sum_{r \geq 0} (-B)^r t^{2dr} / r!$  and reading off  $[t^n]$  gives [equation \(9.14\)](#). For the operator identities write  $D = \frac{d}{dx}$  and note  $D^j e^{xt} = t^j e^{xt}$ . Hence

$$\sum_{n \geq 0} (e^{-BD^{2d}} x^n) \frac{t^n}{n!} = e^{-BD^{2d}} e^{xt} = e^{-Bt^{2d}} e^{xt},$$

which is [equation \(9.13\)](#), so  $H_n^{(d,B)} = e^{-BD^{2d}} x^n$ ; and differentiating [equation \(9.13\)](#) in  $t$  gives  $\sum_n H_{n+1}^{(d,B)} t^n / n! = (x - 2dBt^{2d-1}) e^{xt - Bt^{2d}} = (x - 2dBD^{2d-1}) e^{xt - Bt^{2d}}$ , whence  $H_{n+1}^{(d,B)} = (x - 2dBD^{2d-1}) H_n^{(d,B)}$ . Both identities are the content of [equation \(9.15\)](#).  $\square$

## 10 Polyharmonic heat symbols and Stokes sectors

### 10.1 Dissipative roots

By [equation \(5.1\)](#),  $A_\omega > 0$  exactly when the primitive root order satisfies  $m \equiv 2 \pmod{4}$ . In that case the cyclotomic limit decays on the real frequency axis.

**Definition 10.1** (Cyclotomic polyharmonic kernel). For  $A > 0$ ,  $d \geq 1$ , and  $\tau > 0$ , set

$$K_{d,A}(x, \tau) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{-\tau A \xi^{2d} - i x \xi} d\xi. \quad (10.1)$$

**Proposition 10.2** (Polyharmonic semigroup). The kernel is Schwartz in  $x$ , has total mass one, and solves

$$\partial_\tau K_{d,A} = (-1)^{d+1} A \partial_x^{2d} K_{d,A}, \quad K_{d,A}(\cdot, \tau) \rightarrow \delta_0 \quad (\tau \downarrow 0) \quad (10.2)$$

in the distributional sense. It obeys the scaling law

$$K_{d,A}(x, \tau) = (A\tau)^{-1/(2d)} K_{d,1} \left( x(A\tau)^{-1/(2d)}, 1 \right). \quad (10.3)$$

For  $d = 1$  it is Gaussian; for  $d \geq 2$  it necessarily changes sign.

*Proof. Smoothness and the equation.* Fix  $\tau_0 > 0$ . For  $\tau \geq \tau_0$  and all integers  $\alpha, \beta \geq 0$ ,

$$\left| \partial_\tau^\alpha \partial_x^\beta e^{-\tau A \xi^{2d} - i x \xi} \right| = A^\alpha |\xi|^{2d\alpha + \beta} e^{-\tau A \xi^{2d}} \leq A^\alpha |\xi|^{2d\alpha + \beta} e^{-\tau_0 A \xi^{2d}},$$

and the right-hand side is integrable in  $\xi$  and independent of  $(x, \tau)$ . Differentiation under the integral sign is therefore legitimate, and  $K_{d,A}$  is  $C^\infty$  on  $\mathbb{R} \times (0, \infty)$  with

$$\partial_\tau K_{d,A} = \frac{1}{2\pi} \int_{\mathbb{R}} (-A \xi^{2d}) e^{-\tau A \xi^{2d} - i x \xi} d\xi, \quad \partial_x^{2d} K_{d,A} = \frac{1}{2\pi} \int_{\mathbb{R}} (-i \xi)^{2d} e^{-\tau A \xi^{2d} - i x \xi} d\xi.$$

Since  $(-i)^{2d} = (-1)^d$ , multiplying the second identity by  $(-1)^{d+1} A$  reproduces the first, which is the equation in [equation \(10.2\)](#).

*Schwartz class and mass.* For fixed  $\tau > 0$  the function  $\xi \mapsto \xi^\beta e^{-\tau A \xi^{2d}}$  is Schwartz, so repeated integration by parts in  $\xi$  shows that  $x^\alpha \partial_x^\beta K_{d,A}(x, \tau)$  is bounded for all  $\alpha, \beta \geq 0$ ; hence  $K_{d,A}(\cdot, \tau) \in \mathcal{S}(\mathbb{R})$ . The symbol  $e^{-\tau A \xi^{2d}}$  is even, so [equation \(10.1\)](#) coincides with the inverse angular Fourier transform of [equation \(0.3.3\)](#), and Fourier inversion for Schwartz functions gives  $\mathcal{F}_{\text{ang}}[K_{d,A}(\cdot, \tau)](\xi) = e^{-\tau A \xi^{2d}}$ . Evaluating at  $\xi = 0$  gives total mass  $\int_{\mathbb{R}} K_{d,A}(x, \tau) dx = 1$ .

*Scaling and the initial value.* The substitution  $\xi = (A\tau)^{-1/(2d)} u$  in [equation \(10.1\)](#) gives [equation \(10.3\)](#) directly. For bounded continuous  $\varphi$  that law and  $K_{d,1}(\cdot, 1) \in L^1$  give

$$\int_{\mathbb{R}} K_{d,A}(x, \tau) \varphi(x) dx = \int_{\mathbb{R}} K_{d,1}(y, 1) \varphi(y(A\tau)^{1/(2d)}) dy \longrightarrow \varphi(0) \quad (\tau \downarrow 0)$$

by dominated convergence, which is the assertion  $K_{d,A}(\cdot, \tau) \rightarrow \delta_0$ . For  $d = 1$  the integral [equation \(10.1\)](#) is the classical Gaussian.

*Sign change for  $d \geq 2$ .* Suppose  $K_{d,A}(\cdot, 1) \geq 0$ . Being nonnegative with total mass one it is then the density of a probability measure  $\mu$ , whose characteristic function is  $\varphi(\xi) = e^{-A \xi^{2d}}$ . For  $\xi \neq 0$ ,

$$\int_{\mathbb{R}} \frac{2 - 2 \cos(\xi x)}{\xi^2} d\mu(x) = \frac{2 - \varphi(\xi) - \varphi(-\xi)}{\xi^2} = \frac{2 - 2e^{-A \xi^{2d}}}{\xi^2} \longrightarrow 0 \quad (\xi \rightarrow 0),$$

because  $2d \geq 4$ . The integrand is nonnegative and tends pointwise to  $x^2$ , so Fatou's lemma gives  $\int_{\mathbb{R}} x^2 d\mu(x) = 0$ , hence  $\mu = \delta_0$  and  $\varphi \equiv 1$  – contradicting  $\varphi(\xi) = e^{-A \xi^{2d}}$ . Therefore  $K_{d,A}(\cdot, 1)$  takes both signs, and by [equation \(10.3\)](#) so does  $K_{d,A}(\cdot, \tau)$  for every  $\tau > 0$ .  $\square$

**Corollary 10.3** (The sixth-order Rvachev cyclotomic heat limit). *At a primitive sixth root,  $d = 3$  and  $A_\omega = 1/17010$ . The rescaled complex  $q$ -Fabius transform converges to  $e^{-w^6/17010}$ , the time-one Fourier multiplier of*

$$\partial_\tau u = \frac{1}{17010} \partial_x^6 u; \tag{10.4}$$

*the multiplier at a general time  $\tau > 0$  is recovered from the same limit after the substitution  $w \mapsto \tau^{1/6} w$ . The associated fundamental solution has unit mass but is not of one sign: by the preceding proposition it changes sign, and the classical oscillatory asymptotics for polyharmonic kernels [43] show that it in fact oscillates.*

*Proof.* At  $\omega = e^{\pi i/3}$  we have  $m = 6$ ,  $\gcd(m, 2) = 2$  and  $d = 3$ , and  $\theta = \pi/3$ , so the computation in the proof of [equation \(5.1\)](#) gives  $a_\omega^6 = (-4)^3 e^{i\pi} \sin^6(\pi/6) = (-64)(-1) \cdot 2^{-6} = 1$ . Hence, by [equation \(6.2\)](#) and  $\zeta(6) = \pi^6/945$ ,

$$A_\omega = \frac{\zeta(6) a_\omega^6}{2 \cdot 3^2 \pi^6} = \frac{1}{945 \cdot 18} = \frac{1}{17010} > 0.$$

[equation \(6.4\)](#) therefore gives  $\Phi(q_\varepsilon, \sqrt{\varepsilon} w) \rightarrow e^{-w^6/17010}$  locally uniformly in  $w$ . By the proof of the preceding proposition,  $\mathcal{F}_{\text{ang}}[K_{3,A_\omega}(\cdot, \tau)](\xi) = e^{-\tau A_\omega \xi^6}$ , whose value at  $\tau = 1$  is exactly the limit just obtained; and by [equation \(10.2\)](#) with  $d = 3$  the kernel solves [equation \(10.4\)](#), since  $(-1)^{d+1} = 1$ . For a general  $\tau > 0$  apply the same limit at the frequency  $\tau^{1/6} w$ , which is legitimate because the convergence is locally uniform and  $\tau^{1/6} W$  is again compact. The mass and sign statements are those of the preceding proposition.  $\square$



## 10.2 Backward roots and contour rotation

If  $A_\omega < 0$  (all odd  $m$  and all  $m \equiv 0 \pmod{4}$ ), the real-axis symbol  $e^{-A_\omega w^{2d}}$  grows. It should not be interpreted as a tempered Fourier multiplier on  $\mathbb{R}$ . It is nevertheless an entire function with canonical decaying sectors

$$\left\{ w : \Re(A_\omega w^{2d}) > 0 \right\}. \quad (10.5)$$

More precisely, let  $\Gamma$  be a piecewise smooth contour with two unbounded ends and suppose that, outside a compact set,  $\Re(A_\omega w^{2d}) \geq c|w|^{2d}$  for some  $c > 0$ . Then

$$K_\Gamma(x, \tau) = \frac{1}{2\pi} \int_\Gamma e^{-\tau A_\omega w^{2d} - i x w} dw$$

converges absolutely and locally uniformly in  $x \in \mathbb{C}$  for  $\tau > 0$ . The degree- $2d$  decay dominates the linear  $xw$  term, so differentiation under the integral is justified and  $K_\Gamma$  solves the same polyharmonic equation as in [equation \(10.2\)](#). The sector boundaries  $\Re(A_\omega w^{2d}) = 0$  are Stokes lines. This produces a clean arithmetic/Stokes dichotomy: the residue class of  $m$  modulo four determines whether the canonical radial blow-up is forward dissipative or backward/rotated.

**Open problem 10.4** (Uniform Stokes connection). *Construct a single Riemann–Hilbert or steepest-descent description that connects the real dissipative kernels to the rotated kernels across the sign change of  $A_\omega$ , and determine the Stokes multipliers induced by the finite cyclic factor hidden in  $\mathcal{B}_\omega$ .*

## 11 Inverse cyclotomic branches

Throughout this section,  $O(\varepsilon^N)$  and  $O(z^N)$  denote formal order: the omitted series belongs to the corresponding principal ideal in the indicated formal power-series ring. No analytic remainder estimate is implicit.

### 11.1 Inverse frequency at fixed radial scaling

Fix a nonzero target  $y \in \mathbb{C}$  and a logarithm branch  $\ell = \text{Log } y$  with  $\ell \neq 0$ :

$$\ell = \text{Log } y. \quad (11.1)$$

Choose one of the  $2d$  roots  $w_0 \neq 0$  satisfying

$$-A_\omega w_0^{2d} = \ell. \quad (11.2)$$

**Proposition 11.1** (Formal inverse frequency series). *There is a unique formal series  $w(\varepsilon) \in w_0 + \varepsilon\mathbb{C}[[\varepsilon]]$  which solves, coefficient by coefficient in the formal expansion [equation \(6.11\)](#),*

$$\Phi(q_\varepsilon, \sqrt{\varepsilon} w(\varepsilon)) = y, \quad w(0) = w_0. \quad (11.3)$$

If  $d \geq 2$ , then

$$w(\varepsilon) = w_0 - \frac{C_{1,\omega}}{2dA_\omega} w_0^{3-2d} \varepsilon + O(\varepsilon^2). \quad (11.4)$$

For  $d = 1$ ,

$$w(\varepsilon) = w_0 - \frac{\eta_\omega}{2} w_0 \varepsilon - \frac{D_\omega}{2A_\omega} w_0^3 \varepsilon + O(\varepsilon^2). \quad (11.5)$$

All higher coefficients are obtained recursively from [equation \(6.12\)](#) and partial Bell polynomials.

*Proof.* Use the prescribed logarithm value  $\ell$  and substitute a formal series for  $w$  in [equation \(6.11\)](#). The derivative of the constant-term equation with respect to  $w$  is  $-2dA_\omega w_0^{2d-1} \neq 0$ , hence it is a unit and determines each successive coefficient triangularly (equivalently, apply the formal implicit-function theorem). At order  $\varepsilon$ , compare the derivative contribution with the  $-C_{1,\omega}w^2$  term when  $d \geq 2$ . For  $d = 1$ , use the complete order-one expression in [equation \(6.8\)](#).  $\square$

*Remark 11.2* (Analytic realization remains open). The proposition is formal: it does not assert a convergent germ in a full complex neighborhood of  $\varepsilon = 0$ . The denominators  $1 - (1 - \varepsilon^d)^{2k}$  have complex singularities accumulating at the origin as  $k \rightarrow \infty$ . Thus the direct analytic implicit-function argument is not available without proving cancellations among these singularities. Any analytic realization must supply such control, or else be formulated sectorially or asymptotically with a uniform remainder estimate.

## 11.2 Inverse $q$ at small frequency

The more unusual inverse asks for  $q$  as a function of  $z$  while the transform value is held fixed.

**Lemma 11.3** (The root-centered substitution is well defined). *Let  $\rho(s) \in \mathbb{C}[[s]]$  with  $\rho(0) \neq 0$  and set  $\delta = z^{2d}\rho(z^2)$  and  $q = \omega(1 - \delta)$ . Then for every  $k \geq 1$  the  $k$ th term  $-c_k z^{2k} \mathcal{R}_k$  of [equation \(3.3\)](#), with  $\mathcal{R}_k$  as in [equation \(6.9\)](#) and  $\varepsilon^d$  replaced by  $\delta$ , is a well-defined element of  $\mathbb{C}[[z^2]]$  whose lowest occurring power of  $z$  is  $z^{2k-2d}$  when  $d \mid k$  and  $z^{2k}$  otherwise. Consequently  $[z^N] \log \Phi(q(z), z)$  receives contributions only from the indices  $k \leq N/2 + d$ , the substitution is a well-defined map  $\mathbb{C}[[s]] \rightarrow \mathbb{C}[[z^2]]$ , and its constant term equals  $-A_\omega/\rho(0)$ .*

*Proof.* The numerator  $(a_\omega + \omega\delta)^{2k}$  is a power series in  $\delta$  with constant term  $a_\omega^{2k} \neq 0$ , and  $\delta \in z^{2d}\mathbb{C}[[z^2]]$ . If  $d \nmid k$  the denominator  $1 - \omega^{2k}(1 - \delta)^{2k}$  has constant term  $1 - \omega^{2k} \neq 0$ , hence is invertible in  $\mathbb{C}[[\delta]]$ , and  $-c_k z^{2k} \mathcal{R}_k \in z^{2k}\mathbb{C}[[z^2]]$ . If  $d \mid k$  the denominator is  $1 - (1 - \delta)^{2k} = 2k\delta u_k(\delta)$  with  $u_k \in \mathbb{C}[[\delta]]$  and  $u_k(0) = 1$ , so  $u_k$  is invertible and

$$-c_k z^{2k} \mathcal{R}_k = -\frac{c_k z^{2k-2d}}{2k \rho(z^2)} (a_\omega + \omega\delta)^{2k} u_k(\delta)^{-1} \in z^{2k-2d} \mathbb{C}[[z^2]],$$

$\rho$  being invertible because  $\rho(0) \neq 0$ . Since  $d \mid k$  forces  $k \geq d$ , the exponent  $2k - 2d$  is nonnegative in every case, so the sum over  $k$  lies in  $\mathbb{C}[[z^2]]$  and a term contributes to  $[z^N]$  only when  $2k - 2d \leq N$ , that is  $k \leq N/2 + d$ . The unique index with lowest power  $z^0$  is  $k = d$ , and its contribution there is  $-c_d a_\omega^{2d}/(2d\rho(0)) = -A_\omega/\rho(0)$  by [equation \(6.2\)](#).  $\square$

**Proposition 11.4** (Formal inverse  $q$ -series at a root of unity). *Assume  $d \geq 2$ , fix  $y$  and  $\ell$  as above, and choose the root-centered coordinate*

$$q(z) = \omega(1 - \delta(z)). \quad (11.6)$$

*There is a unique formal series*

$$\rho(s) \in -\frac{A_\omega}{\ell} + s\mathbb{C}[[s]], \quad \delta(z) = z^{2d}\rho(z^2),$$

*such that, after substitution into the master logarithm on the prescribed logarithm sheet, the identity*

$$\Phi(q(z), z) = y \quad (11.7)$$

*holds coefficient by coefficient. Its first terms are*

$$\delta(z) = -\frac{A_\omega}{\ell} z^{2d} \left( 1 - \frac{C_{1,\omega}}{\ell} z^2 + O(z^4) \right), \quad (11.8)$$

*and therefore*

$$q(z) = \omega \left[ 1 + \frac{A_\omega}{\ell} z^{2d} - \frac{A_\omega C_{1,\omega}}{\ell^2} z^{2d+2} + O(z^{2d+4}) \right]. \quad (11.9)$$

*Proof.* By [lemma 11.3](#) the left-hand side of [equation \(11.7\)](#), read through the master logarithm on the prescribed logarithm sheet, is a well-defined element  $\Lambda(\rho) \in \mathbb{C}[[z^2]]$  whose coefficient of  $z^N$  depends polynomially on  $\rho_0^{-1}$  and on finitely many  $\rho_i$ , and whose constant term is  $-A_\omega/\rho_0$ . The scalar equation  $-A_\omega/\rho_0 = \ell$  has the unique solution  $\rho_0 = -A_\omega/\ell \neq 0$ , at which the derivative of  $\rho_0 \mapsto -A_\omega/\rho_0$  equals  $A_\omega/\rho_0^2 \neq 0$ ; so the formal implicit-function theorem applies and determines  $\rho_1, \rho_2, \dots$  triangularly from the coefficients of  $z^2, z^4, \dots$ . The explicit low-order coefficients are computed in [appendix B](#), which is the official derivation of [equation \(11.8\)](#) and hence of [equation \(11.9\)](#): only the master logarithm is used there, never the two-term radial expansion, whose  $O(\delta)$  remainder is not an all-orders series. No analytic remainder is claimed at any step.  $\square$

*Remark 11.5* (No convergent boundary germ is claimed). The formal series is meaningful because every requested coefficient involves only finitely many orders of the master expansion. It does not by itself define an analytic germ through  $z = 0$ : the root-of-unity singular loci of the individual denominators accumulate there. Establishing sectorial sums, their domains, and their Stokes transitions is a separate problem.

*Remark 11.6* (Formal branch geometry). For the inverse-frequency problem, each fixed root  $\omega$  and logarithm value  $\ell + 2\pi i k$  has  $2d$  choices of  $w_0$ , and each choice produces one formal series. For the inverse- $q$  problem, fixing  $\omega$  and a nonzero logarithm value determines a unique formal series  $\rho$ . Turning either algebraic indexing into analytic continuation or monodromy requires sectorial realizations and a careful separation of logarithm-sheet changes from critical-value branching; neither is asserted here.

*Status note 11.7* (What an analytic inverse branch would require). The two propositions of this section are **proved here** as statements about formal series and **conjectural/open** as statements about functions; nothing in this part asserts an analytic inverse branch. The missing ingredient can be named precisely. For the inverse frequency  $w(\varepsilon) = \sum_{n \geq 0} w_n \varepsilon^n$  of [equation \(11.3\)](#) one would need constants  $C, A > 0$ , an exponent  $s \geq 0$ , a sector

$$\Sigma = \{\varepsilon \in \mathbb{C} : 0 < |\varepsilon| < \varepsilon_1, |\arg \varepsilon| < \vartheta\},$$

and a holomorphic  $\mathbf{w} : \Sigma \rightarrow \mathbb{C}$  with  $\Phi(q_\varepsilon, \sqrt{\varepsilon} \mathbf{w}(\varepsilon)) = y$  on  $\Sigma$  and

$$\left| \mathbf{w}(\varepsilon) - \sum_{n < N} w_n \varepsilon^n \right| \leq C A^N (N!)^s |\varepsilon|^N \quad (\varepsilon \in \Sigma, N \geq 1); \quad (11.10)$$

a Gevrey estimate of this shape, with  $\vartheta > \pi s/2$ , would identify  $\mathbf{w}$  as the Borel–Laplace sum of the formal series and would settle the analytic realization. The obstruction to producing [equation \(11.10\)](#) by the present method is explicit rather than technical: the denominators  $1 - (1 - \varepsilon^d)^{2k}$  of [equation \(6.9\)](#) vanish where  $\varepsilon^d = 1 - \zeta$  with  $\zeta^{2k} = 1$ , and the nearest such point satisfies  $|1 - \zeta| = 2 \sin(\pi/(2k)) \asymp 1/k$ , so the family  $\{\mathcal{R}_k\}_{k \geq 1}$  has no common disc of analyticity about  $\varepsilon = 0$  and no  $k$ -independent domain on which an implicit-function argument could be run. [theorem A.1](#) does not repair this: it is an asymptotic expansion at fixed frequency whose constants  $C_{K,R}$  are allowed to depend arbitrarily on the order  $R$ , and [equation \(11.10\)](#) is exactly a statement about how those constants grow. Proving a Gevrey version of [theorem A.1](#), with  $C_{K,R} \leq C A^R (R!)^s$ , is therefore the missing estimate; it is the analytic content of the open problem *Cyclotomic inverse monodromy* stated below.

## 12 Numerical experiments

### 12.1 Evaluation strategy

Direct products are poorly conditioned near roots of unity. The companion script evaluates the absolutely convergent master series [equation \(3.3\)](#) with 80 decimal digits, and computes actions and constants from their own convergent series. No numerical quadrature or random sampling is used. The four main diagnostics are:

1. the residual in the two-term fixed-frequency radial expansion;
2. the residual after removing every displayed blow-up correction through order  $\varepsilon^d$ ;
3. the error in the surviving Bernoulli cumulant and Bell moments;
4. the errors in the inverse frequency and inverse  $q$  branches.

## 12.2 Radial expansion

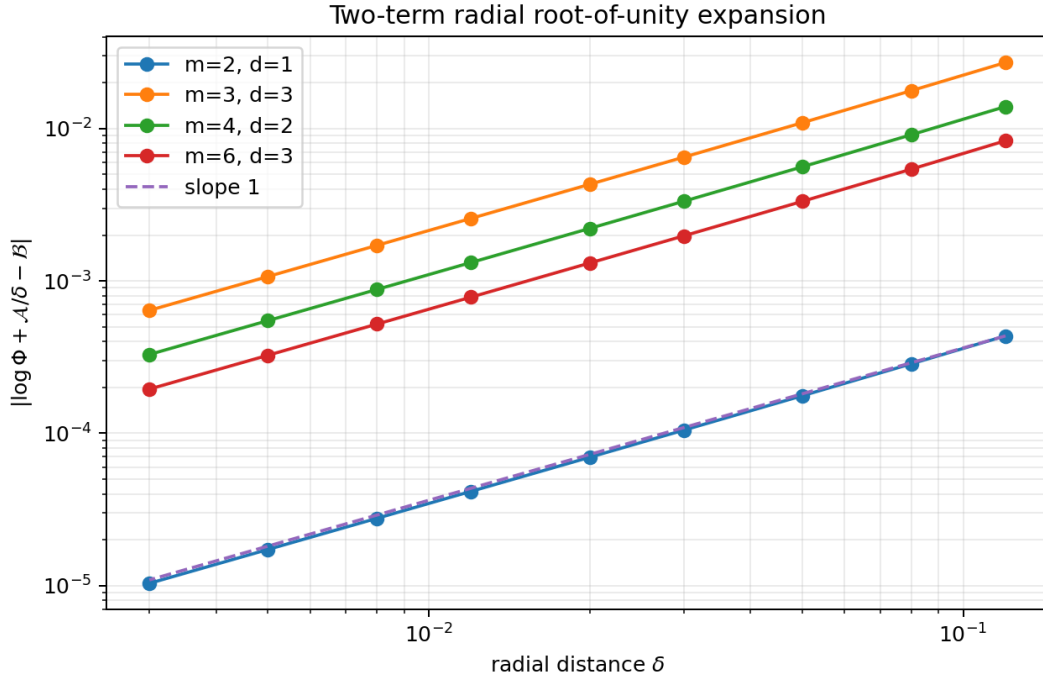


Figure 42: For  $z = 0.8$ , the residual  $|\log \Phi + \mathcal{A}/\delta - \mathcal{B}|$  is asymptotically linear in  $\delta$  at four root orders. The fitted slopes over the five smallest radial distances are all between 1.0043 and 1.0045.

The calculation tests both the divergent action and the finite constant term. It is notably sensitive to the  $(2k - 1)/(4k)$  term in [equation \(4.5\)](#); omitting that term changes the residual from  $O(\delta)$  to  $O(1)$ .

### 12.3 Blow-up and inverse errors

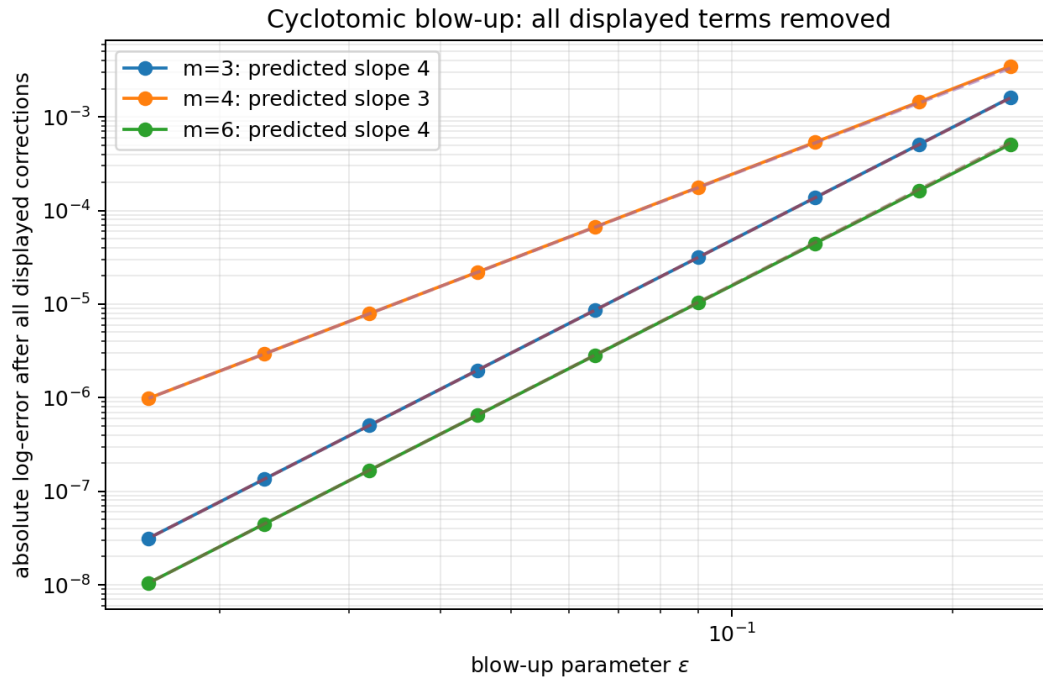


Figure 43: Absolute logarithmic error after removing the limit, all subcritical terms, and the complete order- $\varepsilon^d$  layer. The predicted next exponents are  $d + 1$ : four for  $m = 3, 6$  and three for  $m = 4$ .

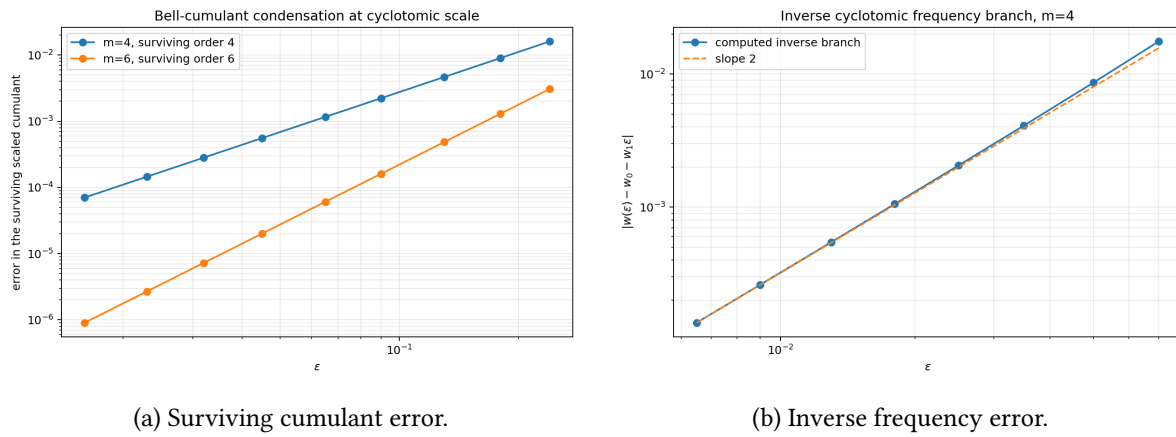


Figure 44: The Bernoulli cumulant converges to the unique order- $2d$  limit; the first-order inverse frequency formula has the predicted quadratic remainder.

## 12.4 Fitted exponents

Table 25: Log–log slopes fitted to the five smallest parameters. The final column gives the absolute error at the smallest tested parameter.

| diagnostic                   | predicted slope | fitted slope | last error    |
|------------------------------|-----------------|--------------|---------------|
| fixed radial, $m = 2$        | 1               | 1.0043       | $1.034e - 05$ |
| fixed radial, $m = 3$        | 1               | 1.0044       | $6.407e - 04$ |
| fixed radial, $m = 4$        | 1               | 1.0044       | $3.289e - 04$ |
| fixed radial, $m = 6$        | 1               | 1.0045       | $1.949e - 04$ |
| blow-up, $m = 3$             | 4               | 4.0001       | $3.146e - 08$ |
| blow-up, $m = 4$             | 3               | 3.0041       | $9.846e - 07$ |
| blow-up, $m = 6$             | 4               | 3.9934       | $1.045e - 08$ |
| inverse frequency, $m = 4$   | 2               | 2.0210       | $1.350e - 04$ |
| inverse $q$ -branch, $m = 4$ | 8               | 7.9960       | $3.534e - 08$ |

The inverse  $q$  experiment at  $\omega = i$  fixes  $y = 1.1$  and solves for  $q(z) = i(1 - \delta(z))$ . The leading approximation has error  $O(z^6)$ ; after the  $C_{1,\omega}$  correction, the fitted exponent is 7.9960, matching the predicted  $O(z^8)$  remainder. At  $z = 0.145$ , the computed local  $q$ -coordinate was

$$\delta = 2.57314407304007927 \times 10^{-5} + 9.47348632511157983 \times 10^{-7}i,$$

while the two-term absolute error was  $3.53 \times 10^{-8}$ .

*Status note 12.1* (Role of computation). The numerical experiments do not substitute for proofs. They check constants, signs, and predicted asymptotic orders, and they are especially useful for detecting missing resonant correction terms. In particular, they do not prove the uniform radial estimate established analytically above, convergence of either formal inverse series, or any Lean theorem. Only the global spectral  $q$ -Pochhammer factorization has the exact Lean crosswalk recorded above; the remaining cyclotomic assertions retain manuscript or frontier status according to the boundary note at the start of the report.

## 13 Connections with Thue–Morse and cyclic digital structure

The original dyadic theory combines two kinds of scale symmetry. The sinc product uses geometric scales  $2^{-j}$ , while the Thue–Morse polynomial  $\prod_{j < N} (1 - z^{2^j})$  encodes binary digit parity. The cyclotomic analysis introduces a third viewpoint: a periodic character in the *scale index*  $j$ .

At  $q = \omega(1 - \delta)$ , the leading phases  $q^j \approx \omega^j$  repeat with period  $m$ . Because sinc is even, the logarithm sees  $\omega^{2j}$  and the effective period is  $d$ . The cyclic projector in [equation \(4.23\)](#) is precisely the finite Fourier projection onto the resonant scale characters. At  $\omega = -1$ , the period-two character collapses to  $d = 1$  and produces the Gaussian limit. Higher roots produce higher-order heat symbols.

This does not identify the cyclotomic scale character with the Thue–Morse sequence itself: Thue–Morse parity is generated by binary digits of the *summation index*, whereas  $\omega^j$  is periodic in the *scale position*. A promising bridge is to combine the two characters in a two-nome product, exactly the larger problem left open in the repository.

**Open problem 13.1** (Thue–Morse/cyclotomic coupling). *For finite products carrying both a Thue–Morse sign and a scale phase  $\omega^j$ , determine the resonant cumulant order, the analogue of  $\mathcal{A}_\omega$ , and whether Prouhet cancellation can suppress the leading spectral dilogarithm. Classify the resulting limit symbols by the joint character group.*

## 14 Further conjectures and research directions

### 14.1 Maximal natural-boundary domain

**Open problem 14.1** (Generic natural boundary and action-zero control). *Determine a precise exceptional set of frequencies for which enough root-of-unity actions  $\mathcal{A}_\omega(z)$  vanish to obstruct the natural-boundary argument. Prove at least a residual or full-measure generic statement, and identify additional uniform arithmetic hypotheses that would give an explicit complement. A locally finite exceptional set is not assumed: a countable union of discrete action-zero sets need not be locally finite.*

The theorem proved here covers  $0 < |z| < \pi/2$  uniformly. Outside that interval, the dilogarithm arguments may leave the unit disk or encounter branch cuts, and the sign argument is unavailable. On its defining disk each action is already a nonzero analytic germ, since  $\mathcal{A}_\omega(z) = A_\omega z^{2d} + O(z^{4d})$  with  $A_\omega \neq 0$ , so its zeros there are discrete. The genuine target is simultaneous control of these zero sets across a dense family of roots, on compatible larger frequency domains; a Baire-category argument might then yield generic natural boundary, while arithmetic control of intersections could sharpen “generic” to an explicit complement.

### 14.2 Cyclic quantum dilogarithm in the finite term

The leading action is dilogarithmic, while standard root-of-unity asymptotics of  $q$ -Pochhammer symbols also contain a finite cyclic quantum dilogarithm. Formula [equations \(4.7\) and \(4.8\)](#) should admit a factorized form obtained by multiplying the cyclic factors over  $n$  in [equation \(3.6\)](#).

**Open problem 14.2** (Spectral cyclic dilogarithm). *Find a convergent regularized product  $\mathcal{D}_\omega(z)$  such that*

$$e^{\mathcal{B}_\omega(z)} = \mathcal{E}_\omega(z) \mathcal{D}_\omega(z),$$

where  $\mathcal{E}_\omega$  is an elementary square-root factor and  $\mathcal{D}_\omega$  is an infinite spectral product of finite cyclic quantum dilogarithms. Determine its functional equations and zero divisor.

### 14.3 Irrational boundary phases and small divisors

At an irrational point  $q = e^{2\pi i\alpha}$  there is no exact resonance, but denominators  $1 - q^{2k}$  can be arbitrarily small. Diophantine properties of  $\alpha$  should govern the boundary growth.

**Conjecture 14.3** (Diophantine phase transition). *For badly approximable  $\alpha$ , suitably Abel-regularized logarithms of  $\Phi(re^{2\pi i\alpha}, z)$  admit polynomial bounds as  $r \uparrow 1$  on a small frequency disk. For Liouville  $\alpha$ , there exist frequency values with superpolynomial subsequential growth generated by near-resonant cumulants.*

This would interpolate between the exact root-of-unity action and generic natural-boundary behavior through small-divisor analysis.

### 14.4 Arithmetic of the cyclotomic constants

The constants

$$A_\omega = \frac{\zeta(2d)}{2d^2\pi^{2d}}(1 - \omega)^{2d}$$

are algebraic because  $\zeta(2d)/\pi^{2d}$  is rational. Their signs are elementary, but their Galois traces, norms, and divisibility encode nontrivial cyclotomic arithmetic.

**Open problem 14.4** (Galois packets). *For fixed  $m$ , determine the trace and norm of  $A_\omega$  over all primitive  $m$ th roots, and similarly for the correction coefficients  $C_{k,\omega}$ . Relate their prime valuations to cyclotomic units and Bernoulli numerators.*



### 14.5 Uniform double scaling in root order

The present analysis fixes  $m$ . When  $m \rightarrow \infty$ , the resonance order  $d$  grows,  $A_\omega$  becomes very small, and the limit exponent has increasing degree. There should be a second asymptotic regime balancing  $d$ ,  $\varepsilon$ , and  $w$ .

**Open problem 14.5** (Large-order cyclotomic saddle). *Determine a scaling in which  $A_\omega w^{2d}$  remains nontrivial as  $d \rightarrow \infty$ , and identify whether the limiting profile is a hard cutoff, an extreme-value law, or a new Lambert- $W$  saddle. Track the transition between polyharmonic kernels and the endpoint asymptotics already developed in the repository.*

### 14.6 Inverse monodromy and branch braiding

The formal inverse-frequency series have  $2d$  algebraic choices on each logarithm sheet. If sectorial analytic realizations exist, continuation may braid them around critical values of the frequency map, while winding the target around zero changes the logarithm sheet.

**Open problem 14.6** (Cyclotomic inverse monodromy). *First construct sectorial analytic realizations of the formal inverse branches. Then determine their continuation around simple critical values, where inverse branches may transpose, and separately track the logarithm-sheet shift  $\ell \mapsto \ell + 2\pi i$  caused by winding around  $y = 0$ . The rotation  $w_0 \mapsto e^{\pi i/d} w_0$  permutes the  $2d$  roots on a fixed nonzero logarithm sheet; it should not be identified with the sheet shift without an additional continuation argument. Classify any resulting braid action and its compatibility with the escaping zero cloud.*

### 14.7 Two-nome extension

The repository's open problem asks for a two-nome phase diagram and canonical root-of-unity limits. The scalar theorem suggests the correct first invariant: identify the lowest Taylor monomial whose bidegree resonates with the pair of roots. The universal germ calculation suggests studying

$$\prod_{j \geq 0} h((1 - \rho)\rho^j z, (1 - \sigma)\sigma^j w),$$

but the resonance set is now a lattice rather than a single arithmetic progression. Its Newton polygon should determine the anisotropic scaling and the limiting exponential polynomial.

**Conjecture 14.7** (Newton-polygon resonance principle). *For an analytic two-variable germ, the canonical two-nome root-of-unity blow-up is determined by the lower convex hull of resonant Taylor exponents. A single vertex gives a monomial exponential; an edge gives a genuinely multivariate exponential polynomial and a coupled polyharmonic PDE.*

### 14.8 Formalization roadmap

The new theory has a favorable dependency graph. The product layer is already represented by `Fabius.geometricSincProduct_eq_tprod_complexQPochhammerInf`, while the general root-of-unity filter is available as `Fabius.sum_pow_eq_ite` and `Fabius.sum_pow_mul_inv_pow_eq_ite`. Thus the remaining roadmap begins after these reusable foundations.

1. **Finite algebra.** Formalize  $d = \text{ord}_{\text{mult}}(\omega^2)$ , the resonant/nonresonant split, and the exact first two coefficients of [equation \(4.5\)](#).
2. **Formal power series.** Prove the coefficient identity [equation \(6.12\)](#) in a Laurent-series ring; Bell exponentiation is already well suited to existing combinatorial infrastructure.
3. **Finite cyclic projection.** Specialize the existing root-of-unity filter to  $d^{-1} \sum_{r < d} \omega^{2kr} = \mathbf{1}_{d|k}$  and derive a finite-truncation version of [equation \(4.23\)](#).

4. **Analytic tails.** Formalize the square-root cutoff proof of [theorem 4.4](#), including the finite phase separation and exponential tail bound.
5. **Sign and natural boundary.** Formalize the sign of  $a_\omega^{2d}$  and  $\text{Li}_2(\pm x)$  on  $0 < x < 1$ , then use the finite-order zero/pole contradiction.
6. **Moments.** Reuse the repository’s Bernoulli cumulant and Bell moment APIs to prove the one-cumulant limit.
7. **Inverse branches.** Begin with formal coefficient recursion; defer the analytic implicit-function theorem until the local complex-analysis library is convenient.

A compact first formal milestone would prove the blow-up limit and sparse moment formula without dilogarithms. The natural-boundary theorem can then be added after both the analytic tail estimate and the sign representation are available in Lean.

## 15 Conclusion

The complex  $q$ -plane reveals a structure invisible on the positive real interval. At every root of unity, the geometric sinc product organizes itself by the order of  $\omega^2$ . Fixed frequencies see a spectral dilogarithmic action of size  $1/\delta$ ; shrinking frequencies at the exact cyclotomic rate isolate one resonant coefficient and produce a pure exponential  $e^{-A_\omega w^{2d}}$ . The same resonance appears simultaneously in the zero divisor, Bernoulli cumulants, Bell moments, and inverse Appell polynomials.

This organizes several themes of the Fabius–Rvachev corpus while explicitly reusing its existing product, root-of-unity-filter, Bell, Appell, and dyadic infrastructure. The sinc product supplies the spectral side;  $q$ -Pochhammer factorization explains the dilogarithmic action; Bell and Bernoulli polynomials control formal corrections and moments; and root order predicts a polyharmonic limit symbol. The two-range estimate below completes the radial theorem and hence the natural-boundary layer. Constructing sectorial inverse branches remains a separate analytic problem.

## A Uniform estimates behind the radial theorem

Let  $K \subseteq \{z : |a_\omega z| < \pi\}$  and choose  $\rho < 1$  such that  $|a_\omega z| \leq \rho\pi$  on  $K$ . The coefficient bound

$$c_k |a_\omega z|^{2k} \leq \frac{\zeta(2)}{k} \rho^{2k} \quad (\text{A.1})$$

provides the basic summable majorant. When  $d > 1$ , the nonresonant phases  $\omega^{2k}$  range over a finite set not containing one, and [lemma 4.3\(ii\)](#) bounds the denominator of  $S_k(\delta)$  below by the constant  $\gamma \geq \sin(\pi/d)$ , uniformly in  $k$  and in  $\delta \in [0, 1]$ . Part (iii) of the same lemma bounds  $|S'_k(\delta)|$  by  $2k\Theta |a_\omega + \omega\delta|^{2k}$ , the single factor  $k$  cancelling the  $1/k$  in [equation \(A.1\)](#). This is the detailed justification of [equation \(4.19\)](#), with the explicit constant  $C_K = 2\zeta(2)\Theta$  and  $\Theta = 2/(\gamma |a_\omega|) + \gamma^{-2}$ ; when  $d = 1$  the nonresonant part is empty.

For resonant  $k$ , the cutoff  $N = \lfloor \delta^{-1/2} \rfloor$  is deliberately smaller than the tempting scale  $\delta^{-1}$ . On  $k \leq N$  the true expansion parameter  $k\delta$  tends uniformly to zero, so Taylor’s theorem with remainder gives [equation \(4.20\)](#). On  $k > N$ , geometric decay wins against every power of  $\delta$ : for each  $M > 0$  and  $0 < r < 1$ ,

$$\delta^{-M} r^{\delta^{-1/2}} \longrightarrow 0. \quad (\text{A.2})$$

This proves, rather than merely predicts, the uniform  $O_K(\delta)$  remainder in [theorem 4.4](#).

The same cutoff method can be repeated to any prescribed finite order, once the order-dependent Taylor constants are kept explicit. The uniform bounds of [lemma 4.3](#) are exactly what is needed,

because they control every Taylor coefficient of the multipliers, not merely the first two. Define, for  $r \geq 0$ ,

$$B_{r,\omega}(z) = - \sum_{\substack{k \geq 1 \\ d \nmid k}} c_k [\delta^r] S_k z^{2k} - \sum_{\substack{k \geq 1 \\ d \mid k}} \frac{c_k a_\omega^{2k}}{2k} [\delta^{r+1}] H_k z^{2k}, \quad (\text{A.3})$$

with  $S_k$  and  $H_k$  as in equations (4.9) and (4.10).

**Theorem A.1** (All-orders fixed-frequency radial expansion). *Every series equation (A.3) converges absolutely and locally uniformly on  $\{z : |a_\omega z| < \pi\}$ , and  $B_{0,\omega} = \mathcal{B}_\omega$ . Moreover, for every compact  $K \Subset \{z : |a_\omega z| < \pi\}$  and every integer  $R \geq 0$  there are  $C_{K,R} > 0$  and  $\delta_{K,R} > 0$  such that*

$$\left| \log \Phi(\omega(1-\delta), z) + \frac{\mathcal{A}_\omega(z)}{\delta} - \sum_{r=0}^R B_{r,\omega}(z) \delta^r \right| \leq C_{K,R} \delta^{R+1} \quad (\text{A.4})$$

for all  $z \in K$  and  $0 < \delta \leq \delta_{K,R}$ . We abbreviate this by the locally uniform Poincaré expansion

$$\log \Phi(\omega(1-\delta), z) \sim -\frac{\mathcal{A}_\omega(z)}{\delta} + \sum_{r \geq 0} B_{r,\omega}(z) \delta^r. \quad (\text{A.5})$$

The constant  $C_{K,R}$  is allowed to depend on  $R$ ; no claim is made that the series in equation (A.5) converges, nor that its constants grow at any prescribed rate in  $R$ .

*Proof.* Fix  $0 < \rho < \rho_1 < 1$  with  $|a_\omega z| \leq \rho\pi$  on  $K$  and  $|(a_\omega + \omega\delta)z| \leq \rho_1\pi$  for  $z \in K$  and  $0 \leq \delta \leq \delta_0$ , and abbreviate  $a = a_\omega$ ,  $b = \omega/a$ .

*Step 1: coefficient bounds.* By lemma 4.3(i),  $H_k$  is holomorphic with  $|H_k| \leq M_1$  on the closed disc  $|\delta| \leq 1/(4k)$ , so Cauchy's estimates give

$$|[\delta^r] H_k| \leq M_1 (4k)^r, \quad \left| H_k(\delta) - \sum_{r=0}^{R+1} [\delta^r] H_k \delta^r \right| \leq 2M_1 (4k\delta)^{R+2} \quad (\text{A.6})$$

for  $0 \leq \delta \leq 1/(8k)$ , the second bound being the geometric estimate of the Taylor tail on half the radius. By lemma 4.3(iv), for  $d \nmid k$  the function  $S_k$  is holomorphic with  $|S_k| \leq M_2 |a|^{2k}$  on  $|\delta| \leq \lambda_\omega/(2k)$ , so likewise

$$|[\delta^r] S_k| \leq M_2 |a|^{2k} \left( \frac{2k}{\lambda_\omega} \right)^r, \quad \left| S_k(\delta) - \sum_{r=0}^R [\delta^r] S_k \delta^r \right| \leq 2M_2 |a|^{2k} \left( \frac{2k\delta}{\lambda_\omega} \right)^{R+1} \quad (\text{A.7})$$

for  $0 \leq \delta \leq \lambda_\omega/(4k)$ . Combining the left-hand bounds with equation (A.1) dominates the two families in equation (A.3) by  $M_2 \zeta(2) (2/\lambda_\omega)^r k^{r-1} \rho^{2k}$  and by  $M_1 4^{r+1} \zeta(2) k^{r-1} \rho^{2k}/2$  respectively, both summable in  $k$  because  $\rho < 1$ , and uniformly so on  $K$ . Hence each  $B_{r,\omega}$  converges absolutely and locally uniformly. For  $r = 0$  the coefficients are  $[\delta^0] S_k = a^{2k}/(1 - \omega^{2k})$  and, by equation (4.11),  $[\delta^1] H_k = 2kb + (2k - 1)/2$ , so equation (A.3) reduces termwise to equations (4.7) and (4.8); that is  $B_{0,\omega} = \mathcal{B}_\omega$ .

Note also that, for  $d \mid k$ , expanding  $H_k$  through order  $\delta^{R+1}$  and dividing by  $2k\delta$  expands  $S_k$  through order  $\delta^R$  starting at  $\delta^{-1}$ , with coefficient  $a^{2k}/(2k)$  there; multiplied by  $-c_k z^{2k}$  and summed over the resonant  $k$ , that coefficient reproduces exactly  $-\mathcal{A}_\omega(z)$  by equation (4.6). Thus the three displayed objects in equation (A.4) are the true sum, its  $\delta^{-1}$  coefficient, and its coefficients of  $\delta^0, \dots, \delta^R$ .

*Step 2: the remainder below the cutoff.* Split every sum over  $k$  at  $N = \lfloor \delta^{-1/2} \rfloor$ , and shrink  $\delta_{K,R} \leq \delta_0$  so that  $\delta_{K,R} \leq \min \{1/64, \lambda_\omega^2/16\}$ , which is positive because  $\lambda_\omega > 0$  by equation (4.14) and the convention recorded in lemma 4.3; then  $N \leq \delta^{-1/2} \leq \min \{1/(8\delta), \lambda_\omega/(4\delta)\}$ , so equations (A.6)

and (A.7) apply for every  $k \leq N$ . Using  $S_k = a^{2k} H_k / (2k\delta)$  in the resonant case and equation (A.1) throughout, the resonant part of the left-hand side of equation (A.4) restricted to  $k \leq N$  is at most

$$\sum_{d|k, k \leq N} \frac{c_k |az|^{2k}}{2k\delta} \cdot 2M_1(4k\delta)^{R+2} \leq M_1 \zeta(2) 4^{R+2} \delta^{R+1} \sum_{k \geq 1} k^R \rho^{2k},$$

and the nonresonant part at most

$$\sum_{d \nmid k, k \leq N} c_k |z|^{2k} \cdot 2M_2 |a|^{2k} \left( \frac{2k\delta}{\lambda_\omega} \right)^{R+1} \leq 2M_2 \zeta(2) \left( \frac{2}{\lambda_\omega} \right)^{R+1} \delta^{R+1} \sum_{k \geq 1} k^R \rho^{2k}.$$

Both sums over  $k$  are finite because  $\rho < 1$ , so this contribution is  $O_{K,R}(\delta^{R+1})$ .

*Step 3: the remainder beyond the cutoff.* For  $k > N$  the true terms are bounded exactly as in the proof of theorem 4.4: by lemma 4.3(ii) and (v),

$$c_k |z|^{2k} |S_k(\delta)| \leq C \rho_1^{2k} \left( \frac{1}{k} + \frac{1}{k^2 \delta} \right),$$

with  $C$  depending only on  $\gamma$  and  $\omega$ . The subtracted terms with index  $k > N$  — the  $k$ th term of  $\mathcal{A}_\omega(z)/\delta$  together with the  $k$ th terms of  $B_{0,\omega}, \dots, B_{R,\omega}$  — are bounded, after summation over  $0 \leq r \leq R$  and using the left-hand bounds of equations (A.6) and (A.7) together with  $\delta \leq 1$ , by  $C_R \rho^{2k} k^R (1 + (k\delta)^{-1})$ . Summing both over  $k > N$  gives  $O_R(\delta^{-1} N^R \rho_1^{2N})$ , and  $\rho_1^{2N} \leq \rho_1^{-2} (\rho_1^2)^{\delta^{-1/2}}$ , so by equation (A.2) this is  $o(\delta^M)$  for every  $M > 0$ ; in particular it is  $O(\delta^{R+1})$ .

Adding the estimates of Steps 2 and 3 proves equation (A.4), and equation (A.5) is its abbreviation.  $\square$

The bound obtained above for  $C_{K,R}$  grows like  $4^R R!$ , the factorial entering through  $\sum_k k^R \rho^{2k}$ . Whether it can be held to a Gevrey rate is a separate question, and an open one; it is what an analytic — as opposed to asymptotic — theory of the inverse branches would need, as recorded in status note 11.7.

## B A direct coefficient derivation for the inverse $q$ -branch

Set  $\delta = \rho z^{2d}$ . The  $k = d$  term in the logarithm is

$$-c_d z^{2d} \frac{(a_\omega + \omega \rho z^{2d})^{2d}}{1 - (1 - \rho z^{2d})^{2d}} = -\frac{A_\omega}{\rho} + O(z^{2d}). \quad (\text{B.1})$$

For  $1 \leq k < d$ , the nonresonant terms are

$$-C_{k,\omega} z^{2k} + O(z^{2k+2d}). \quad (\text{B.2})$$

Write

$$\rho(z) = \rho_0 + \rho_1 z^2 + O(z^4), \quad \rho_0 = -\frac{A_\omega}{\ell}. \quad (\text{B.3})$$

Then

$$-\frac{A_\omega}{\rho(z)} = \ell + \ell \left( -\frac{\rho_1}{\rho_0} \right) z^2 + O(z^4).$$

Matching the  $-C_{1,\omega} z^2$  term gives  $\rho_1/\rho_0 = -C_{1,\omega}/\ell$ , which is equation (11.8). Higher coefficients follow triangularly because the derivative in  $\rho$  never vanishes at  $\rho_0$ .

## C Exact small-order constants

Table 26: Leading constants for selected primitive roots  $\omega = e^{2\pi i/m}$ .

| $m$ | $d$ | $a_\omega^{2d}$ | $A_\omega$ | real-axis type          |
|-----|-----|-----------------|------------|-------------------------|
| 2   | 1   | 4               | 1/3        | dissipative Gaussian    |
| 3   | 3   | -27             | -1/630     | backward sixth order    |
| 4   | 2   | -4              | -1/180     | backward fourth order   |
| 6   | 3   | 1               | 1/17010    | dissipative sixth order |

For general primitive order  $m$ , the sign rule is

$$A_\omega > 0 \iff m \equiv 2 \pmod{4}, \quad A_\omega < 0 \iff m \text{ is odd or } 4 \mid m.$$

The magnitude can be written

$$|A_\omega| = \frac{\zeta(2d)}{2d^2\pi^{2d}} \left( 2 \sin \frac{\pi h}{m} \right)^{2d}. \quad (\text{C.1})$$

## D Reproducibility manifest

This part’s verification material is kept in the volume directory under `assets/geometric-sinc-and-exponent-families/Cyclotomic_q_Fabius_Rvachev_Frontier/`. The absorbed report’s own source and PDF were deleted when it was merged; git history is their archive. The directory holds:

- `assets/geometric-sinc-and-exponent-families/Cyclotomic_q_Fabius_Rvachev_Frontier/numerical_experiments.py` – commented high-precision experiments;
- `assets/geometric-sinc-and-exponent-families/Cyclotomic_q_Fabius_Rvachev_Frontier/data/` – the raw deterministic numerical tables, the generated `numerical_slopes_table.tex` read by this volume, and the diagnostic `experiment_summary.txt`;
- `assets/geometric-sinc-and-exponent-families/Cyclotomic_q_Fabius_Rvachev_Frontier/figures/` – vector and raster figures;
- `assets/geometric-sinc-and-exponent-families/Cyclotomic_q_Fabius_Rvachev_Frontier/corpus_audit_note.txt` – audit method and novelty boundary;
- `assets/geometric-sinc-and-exponent-families/Cyclotomic_q_Fabius_Rvachev_Frontier/corpus_inventory_2026-08-27.txt` – the preserved 57-path recursive ledger;
- `assets/geometric-sinc-and-exponent-families/Cyclotomic_q_Fabius_Rvachev_Frontier/requirements.txt` – Python dependencies;
- `assets/geometric-sinc-and-exponent-families/Cyclotomic_q_Fabius_Rvachev_Frontier/README.md`, `assets/geometric-sinc-and-exponent-families/Cyclotomic_q_Fabius_Rvachev_Frontier/README.txt`, and `assets/geometric-sinc-and-exponent-families/Cyclotomic_q_Fabius_Rvachev_Frontier/PDF_VALIDATION.txt` – the absorbed package’s own notes.

The script uses the exact convergent logarithmic series rather than a truncated product. To reproduce the tables and figures, run

```
python numerical_experiments.py --dps 80
```

from that directory; this volume reads the regenerated figures and data table in place. The volume itself is rebuilt from its own directory by running `pdflatex` on `geometric_q_fabius_frontiers.tex` three times, so that the table of contents and every cross-reference settle.

## Part X

# Parameter Flow, Convex Order, and Large Deviations

## Abstract

Let  $U_0, U_1, \dots$  be independent  $\text{Unif}[-1, 1]$  variables and

$$X_q = (1 - q) \sum_{j \geq 0} q^j U_j, \quad 0 < q < 1.$$

At  $q = 1/2$  the density of  $X_q$  is the standard centered Rvachev up-function, and hence lies on the usual Fabius–Rvachev bridge. The current ProveIt corpus already contains an extensive theory of this family: its infinite-sinc transform, Bernoulli cumulants,  $q$ -Pochhammer factorizations, parameter limits, fixed-base endpoint asymptotics, finite prefix transport, and a first Gaussian correction. After a source-level crosswalk against that corpus, this report develops a different parameter layer.

First, for  $0 < r < s < 1$  we construct an explicit probability kernel  $p_{r,s}$  whose generating function is

$$P_{r,s}(z) = \frac{1 - s}{1 - r} \frac{1 - rz}{1 - sz}.$$

It transports the geometric coefficient profile at  $r$  to that at  $s$  and forms a convolution cocycle. A random tail shift with law  $p_{r,s}$  gives an explicit martingale coupling, proving the universal convex-order flow  $X_s \preceq_{\text{cx}} X_r$ . The kernels are the increments of an inhomogeneous compound-Poisson subordinator with geometric jump law.

Second, writing  $q = e^{-\delta}$ , we prove a full large-deviation principle at speed  $1/\delta$ . Its limiting scaled cumulant transform is

$$\Lambda(\theta) = \int_0^\infty \log\left(\frac{\sinh(\theta e^{-t})}{\theta e^{-t}}\right) dt = \int_0^\theta \frac{\log(\sinh u/u)}{u} du,$$

and the rate is  $I = \Lambda^*$ . The finite- $q$  transform is an exact midpoint Riemann sum. Midpoint Euler–Maclaurin therefore yields an all-orders expansion containing only even powers of  $\delta$  and Bernoulli polynomials at  $1/2$ ; in particular

$$\Lambda_\delta(\theta) = \Lambda(\theta) - \frac{\delta^2 \theta^2}{24} \Lambda''(\theta) + O(\delta^4).$$

The rate has an explicit rational Taylor series at the origin and a new Lambert- $W_{-1}$  boundary law as  $x \uparrow 1$ .

Third, we prove a uniform all-orders Edgeworth expansion for the *continuous* limit  $q \uparrow 1$ , valid after any fixed number of density derivatives. This is distinct from the discrete  $\text{Fup}_n$  Edgeworth theorem already present in the repository. Exact Bernoulli cumulants produce the Hermite coefficients, and a geometric infinite-sinc tail lemma supplies the global Fourier control. Numerical experiments reproduce the predicted orders 1, 2, 3, 4 and the midpoint orders 2, 4. The report closes with a moderate-deviation bridge, a sharp saddlepoint conjecture, the exact sign pattern of the rate series — the absorbed report’s positivity conjecture is refuted at degree 24 —, and a formalization roadmap. All theorem proofs here are paper-level arguments; no claim of Lean verification is made.



# 1 Scope, corpus audit, and nonduplication boundary

## 1.1 What was compared

The source tree under

<https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction/docs>

contains the expository Fabius and Rvachev documents, the Lean walkthrough, archival material, the canonical semi-formalized frontier volume, and a large set of current standalone drafts. The working comparison used the live source tree, the canonical frontier volume and its group READMEs, and the global draft manifest dated 28–30 August 2026 [82, 72]. The purpose of the comparison was not merely to avoid repeated wording: it was to remove mathematical layers whose main claims were already present under different titles.

The following table records the decisive exclusions and the boundary of the present contribution. “New here” always means *not found in the live repository crosswalk used for this report*; it is not an unqualified claim of priority over all mathematical literature.

| Corpus layer                                                    | Already present                                                                                                                                                                               | Layer developed here                                                                                                               |
|-----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Exponent- and $q$ -series frontiers, Parts VI–VII               | The geometric family, infinite sinc products, exact Bernoulli cumulants, $q$ -Pochhammer structure, quantitative $q \uparrow 1$ Gaussian convergence, and the first characteristic correction | A complete continuous-parameter Edgeworth hierarchy, uniform on $\mathbb{R}$ and after arbitrary fixed derivatives                 |
| Finite-sinc transport, Part V                                   | Convex order, peakedness, stop-loss identities, and exact metrics along the <i>prefix index</i> $N$                                                                                           | Convex order along the <i>geometric parameter</i> $q$ , with an explicit tail-shift martingale and a compound-Poisson cocycle      |
| Atomic-function/Fup hierarchy                                   | A proved all-orders Edgeworth theorem for the discrete family $\text{Fup}_n$                                                                                                                  | A separate theorem for $X_q$ as the continuous parameter $q \uparrow 1$ ; the triangular arrays and cumulant weights differ        |
| Fixed-base endpoint/Lambert theory                              | Spatial endpoint asymptotics at fixed $q$ or fixed base, including Lambert- $W_{-1}$ inversion and periodic transform terms                                                                   | A parameter-boundary large-deviation rate as $q \uparrow 1$ , whose own $x \uparrow 1$ saddle has a distinct Lambert- $W_{-1}$ law |
| Shape, native orthogonal-polynomial, and noncommutative reports | Strict log-concavity, native Jacobi/Hankel theory, free and Boolean cumulants, and non-free-infinite-divisibility certificates                                                                | Explicitly omitted after the final cross-check                                                                                     |

## 1.2 Main claims of the report

The central claims are the following.

- R1.** An exact coefficient kernel  $p_{r,s}$ , its cocycle, its compound-Poisson generator, and an explicit martingale proof of  $X_s \preceq_{\text{cx}} X_r$ .
- R2.** An all-orders midpoint Euler–Maclaurin expansion for the scaled  $q \uparrow 1$  cumulant transform.

- R3.** A Gärtner–Ellis large-deviation principle with an explicit integral transform, rational central rate series, and Lambert- $W_{-1}$  edge inversion.
- R4.** A differentiated, uniform, all-orders Edgeworth expansion for the continuous geometric family.
- R5.** Matching between central and large deviations, together with saddlepoint and transseries conjectures and the refutation of the rate-series positivity conjecture.

The constructions extend verbatim to many bounded seed distributions. The uniform seed is singled out because it is exactly the Fabius–Rvachev case and because its cumulant transform is controlled by Bernoulli numbers.

## 2 The geometric-uniform law and the Fabius–Rvachev point

### 2.1 Random series, transforms, and fixed point

Let  $U_0, U_1, \dots$  be independent variables with density  $\frac{1}{2}\mathbf{1}_{[-1,1]}$ . For  $0 < q < 1$  define

$$X_q = (1 - q) \sum_{j=0}^{\infty} q^j U_j. \quad (2.1)$$

Since  $X_q = 2X_q^\circ$  by [table 1](#), [proposition 0.2.2](#) gives that the series converges absolutely for every realization (its terms are dominated by  $(1 - q)q^j$ ), that  $\mathbb{E}X_q = 0$ , and that  $|X_q| \leq 1$  with support exactly  $[-1, 1]$ . Its distribution solves

$$X_q \stackrel{d}{=} (1 - q)U + qX'_q, \quad (2.2)$$

where  $X'_q$  is an independent copy, and by [proposition 0.2.3](#), transported by [lemma 0.5.1](#) with  $\alpha = 2$ , it is the unique solution of (2.2) with finite first moment – in particular the unique compactly supported one. We use

$$\text{sinc}_{\text{rad}} z = \frac{\sin z}{z}, \quad \text{sinhc} z = \frac{\sinh z}{z},$$

with their continuous values at zero. The characteristic and moment generating functions are

$$\phi_q(t) = \mathbb{E}e^{itX_q} = \prod_{j=0}^{\infty} \text{sinc}_{\text{rad}}((1 - q)q^j t), \quad (2.3)$$

$$M_q(t) = \mathbb{E}e^{tX_q} = \prod_{j=0}^{\infty} \text{sinhc}((1 - q)q^j t). \quad (2.4)$$

At  $q = 1/2$  the first product is  $\prod_{k \geq 1} \text{sinc}_{\text{rad}}(t/2^k)$ , the standard Fourier transform of the centered Rvachev up-function. Thus  $X_{1/2}$  is exactly the centered up-law; the usual piecewise shift/reflection relation then connects its density to the Fabius function [\[36, 61, 32\]](#).

### 2.2 Bernoulli cumulants

Let  $B_n$  denote the Bernoulli numbers with  $B_2 = 1/6$ . Since

$$\log \text{sinhc} z = \sum_{m=1}^{\infty} \frac{2^{2m-1} B_{2m}}{m(2m)!} z^{2m}, \quad |z| < \pi, \quad (2.5)$$

the even cumulants of  $U$  and  $X_q$  are

$$\kappa_{2m}(U) = \frac{2^{2m-1} B_{2m}}{m}, \quad (2.6)$$

$$\kappa_{2m}(X_q) = \frac{2^{2m-1} B_{2m}}{m} \frac{(1 - q)^{2m}}{1 - q^{2m}}, \quad \kappa_{2m+1}(X_q) = 0. \quad (2.7)$$

In particular

$$\sigma_q^2 := \text{Var}(X_q) = \frac{1-q}{3(1+q)}. \quad (2.8)$$

The moments are complete Bell polynomials in the cumulants [41], so every even moment is a rational function of  $q$  with Bernoulli input. This exact cumulant layer is part of the established corpus; our use of it below is to build the new parameter asymptotics.

### 2.3 The natural small parameter

Set

$$q = e^{-\delta}, \quad \delta = -\log q. \quad (2.9)$$

Then

$$1-q = \delta s_\delta e^{-\delta/2}, \quad s_\delta := \frac{2 \sinh(\delta/2)}{\delta} = 1 + \frac{\delta^2}{24} + \frac{\delta^4}{1920} + O(\delta^6). \quad (2.10)$$

This exact half-step factorization is the source of the midpoint Euler–Maclaurin cancellation in [appendix 4](#). Notice also

$$\sigma_q^2 = \frac{1}{3} \tanh \frac{\delta}{2} = \frac{\delta}{6} - \frac{\delta^3}{72} + \frac{\delta^5}{720} + O(\delta^7). \quad (2.11)$$

Thus the unstandardized law collapses to zero at scale  $\sqrt{\delta}$ , while its standardized form tends to a Gaussian.

## 3 An exact parameter-flow kernel and martingale transport

The finite-prefix transport theory already in the corpus orders successive truncations. Here the number of factors is infinite and the parameter itself moves. The key observation is that the geometric coefficient profile admits an exact stochastic factorization.

### 3.1 Coefficient cocycle

Write

$$a_j(q) = (1-q)q^j, \quad A_q(z) = \sum_{j \geq 0} a_j(q)z^j = \frac{1-q}{1-qz}. \quad (3.1)$$

**Theorem 3.1** (Parameter-flow kernel). *For  $0 < r < s < 1$ , define a probability mass function on  $\mathbb{N}_0$  by*

$$p_{r,s}(0) = \frac{1-s}{1-r}, \quad p_{r,s}(k) = \frac{(s-r)(1-s)}{1-r} s^{k-1}, \quad k \geq 1. \quad (3.2)$$

*Its probability generating function is*

$$P_{r,s}(z) = \sum_{k \geq 0} p_{r,s}(k)z^k = \frac{1-s}{1-r} \frac{1-rz}{1-sz}. \quad (3.3)$$

*The geometric coefficient profiles satisfy*

$$a(s) = p_{r,s} * a(r), \quad A_s(z) = P_{r,s}(z)A_r(z), \quad (3.4)$$

*and the kernels form the convolution cocycle*

$$p_{r,t} = p_{s,t} * p_{r,s}, \quad 0 < r < s < t < 1. \quad (3.5)$$

*Proof.* The coefficients in (3.2) are nonnegative and

$$p_{r,s}(0) + \sum_{k \geq 1} p_{r,s}(k) = \frac{1-s}{1-r} + \frac{s-r}{1-r} = 1.$$

Dividing the two rational functions in (3.1) gives

$$\frac{A_s(z)}{A_r(z)} = \frac{1-s}{1-r} \frac{1-rz}{1-sz},$$

whose geometric-series expansion is exactly (3.2). This proves (3.4). The cocycle follows immediately from  $P_{r,t} = A_t/A_r = (A_t/A_s)(A_s/A_r)$ .  $\square$

### 3.2 Compound-Poisson interpretation

The cocycle is not merely algebraic. Differentiating its logarithm in the upper parameter gives

$$\begin{aligned} \frac{\partial}{\partial s} \log P_{r,s}(z) &= \frac{z}{1-sz} - \frac{1}{1-s} \\ &= \sum_{k \geq 1} s^{k-1} (z^k - 1). \end{aligned} \quad (3.6)$$

Consequently

$$P_{r,s}(z) = \exp \left\{ \int_r^s \sum_{k \geq 1} u^{k-1} (z^k - 1) du \right\}. \quad (3.7)$$

**Proposition 3.2** (Inhomogeneous geometric-jump subordinator). *There exists an increasing process  $(N_q)_{0 < q < 1}$  with independent increments such that  $N_s - N_r$  has law  $p_{r,s}$ . At parameter value  $u$ , jumps occur with total intensity  $(1-u)^{-1} du$ , and conditional on a jump the size has geometric law*

$$\mathbb{P}\{J = k \mid u\} = (1-u)u^{k-1}, \quad k \geq 1. \quad (3.8)$$

*Proof.* The process is constructed, not merely read off (3.7). Let  $\mathcal{N}$  be a Poisson random measure on  $(0, 1) \times \{1, 2, \dots\}$  with intensity  $\nu(du, \{k\}) = u^{k-1} du$ . For  $0 < r < s < 1$  the intensity of the strip  $(r, s] \times \{1, 2, \dots\}$  is

$$\sum_{k \geq 1} \int_r^s u^{k-1} du = \int_r^s \frac{du}{1-u} = \log \frac{1-r}{1-s} < \infty,$$

so  $\mathcal{N}$  has finitely many points in every such strip almost surely (the mass diverges only as  $s \uparrow 1$ , which is why the process is indexed by  $q < 1$ ). Put

$$N_q = \int_{(0,q] \times \{1,2,\dots\}} k \mathcal{N}(du, dk), \quad 0 < q < 1,$$

a finite sum of the jump sizes in  $(0, q]$ . It is nondecreasing in  $q$ , and since  $\mathcal{N}$  restricted to disjoint strips is independent, the increments over disjoint parameter intervals are independent. The exponential formula for Poisson random measures gives, for  $0 \leq z \leq 1$ ,

$$\mathbb{E} z^{N_s - N_r} = \exp \left\{ \int_r^s \sum_{k \geq 1} u^{k-1} (z^k - 1) du \right\} = P_{r,s}(z)$$

by (3.7), so  $N_s - N_r$  has law  $p_{r,s}$ ; the cocycle (3.5) is then the additivity of increments. At parameter value  $u$  the total jump rate is  $\sum_{k \geq 1} u^{k-1} = (1-u)^{-1}$ , and normalizing the intensity by it gives (3.8).  $\square$

The expected number of jumps between  $r$  and  $s$  is  $\log((1-r)/(1-s))$ . Thus the apparently static operation of making the coefficient sequence more diffuse is governed by a simple pure-jump flow.

### 3.3 Explicit martingale coupling

The kernel acts by random shifts of the independent environment. This gives a direct convex-order proof, stronger than a comparison of finitely many moments.

**Theorem 3.3** (Tail-shift martingale transport). *Let  $V_0, V_1, \dots$  be iid integrable real random variables and define*

$$X_q^{(V)} = (1 - q) \sum_{j \geq 0} q^j V_j. \quad (3.9)$$

For  $0 < r < s < 1$ , let  $N$  be independent of the sequence and have law  $p_{r,s}$ . Put

$$Y_{r,s} = \sum_{j \geq 0} a_j(r) V_{j+N}. \quad (3.10)$$

Then

$$Y_{r,s} \stackrel{d}{=} X_r^{(V)}, \quad \mathbb{E}[Y_{r,s} \mid V_0, V_1, \dots] = X_s^{(V)}. \quad (3.11)$$

Consequently

$$X_s^{(V)} \preceq_{\text{cx}} X_r^{(V)}. \quad (3.12)$$

*Proof.* First, the series defining  $X_q^{(V)}$  and  $Y_{r,s}$  converge absolutely almost surely: since  $\sum_j a_j(q) = 1$ ,  $\mathbb{E} \sum_{j \geq 0} a_j(q) |V_j| = \mathbb{E} |V| < \infty$ , so  $\sum_j a_j(q) |V_j| < \infty$  a.s., and the same holds for every shifted tail. For every fixed  $n$ , the tail  $(V_n, V_{n+1}, \dots)$  has the same joint law as the original iid sequence, hence the conditional-on- $N = n$  variable in (3.10) has law  $X_r^{(V)}$ . Thus  $Y_{r,s} \stackrel{d}{=} X_r^{(V)}$ . Conditioning instead on the entire environment, the interchange of the two sums below is Fubini's theorem for the signed double series, justified by the integrable majorant

$$\mathbb{E} \sum_{n \geq 0} p_{r,s}(n) \sum_{j \geq 0} a_j(r) |V_{j+n}| = \sum_{n \geq 0} p_{r,s}(n) \sum_{j \geq 0} a_j(r) \mathbb{E} |V| = \mathbb{E} |V| < \infty;$$

it gives

$$\begin{aligned} \mathbb{E}[Y_{r,s} \mid (V_j)_{j \geq 0}] &= \sum_{n \geq 0} p_{r,s}(n) \sum_{j \geq 0} a_j(r) V_{j+n} \\ &= \sum_{k \geq 0} \left( \sum_{n=0}^k p_{r,s}(n) a_{k-n}(r) \right) V_k \\ &= \sum_{k \geq 0} a_k(s) V_k = X_s^{(V)}, \end{aligned}$$

where the penultimate equality is (3.4). Therefore  $X_s^{(V)}$  is the conditional expectation of a variable having law  $X_r^{(V)}$ . Jensen's inequality yields  $\mathbb{E}\Phi(X_s^{(V)}) \leq \mathbb{E}\Phi(X_r^{(V)})$  for every convex  $\Phi$  for which the expectations exist, which is exactly convex order.  $\square$

**Corollary 3.4** (Stop-loss and moment inequalities). *For  $0 < r < s < 1$  and every  $t \in \mathbb{R}$ ,*

$$\mathbb{E}(X_s - t)_+ \leq \mathbb{E}(X_r - t)_+. \quad (3.13)$$

*If the seed is centered and has finite variance, then*

$$\text{Var}(X_q^{(V)}) = \text{Var}(V) \frac{1-q}{1+q}, \quad (3.14)$$

*and this variance is strictly decreasing unless  $V$  is degenerate. All even convex moments that exist are nonincreasing in  $q$ .*

*Remark 3.5* (Why the proof is genuinely infinite-dimensional). The right shift of an infinite coefficient sequence is not a finite permutation. Nevertheless every shifted iid tail produces exactly the same weighted-sum law. Averaging over the shift kernel therefore supplies the martingale directly and bypasses finite-dimensional approximation. The finite-prefix convex-order chain in the repository and the parameter flow (3.12) are complementary rather than duplicate statements.

## 4 The scaled cumulant transform as a midpoint $q$ -integral

### 4.1 Exact half-step representation

For the uniform seed set

$$g(z) = \log \sinh z = \log \frac{\sinh z}{z}. \quad (4.1)$$

The scaled cumulant generating function at large-deviation speed  $1/\delta$  is

$$\Lambda_\delta(\theta) := \delta \log \mathbb{E} \exp \left( \frac{\theta X_{e^{-\delta}}}{\delta} \right). \quad (4.2)$$

Using (2.10), the  $j$ th argument of  $g$  is

$$\frac{\theta(1 - e^{-\delta})e^{-j\delta}}{\delta} = \theta s_\delta e^{-(j+1/2)\delta}.$$

Hence the exact identity

$$\Lambda_\delta(\theta) = \delta \sum_{j=0}^{\infty} g \left( \theta s_\delta e^{-(j+1/2)\delta} \right). \quad (4.3)$$

This is a midpoint Riemann sum on the logarithmic scale. It is the reason the first correction is  $O(\delta^2)$  rather than  $O(\delta)$ .

### 4.2 Limit transform

**Theorem 4.1** (Limiting scaled cumulant transform). *Uniformly for  $\theta$  in compact subsets of  $\mathbb{R}$ ,*

$$\Lambda_\delta(\theta) \longrightarrow \Lambda(\theta) := \int_0^\infty g(\theta e^{-t}) dt = \int_0^\theta \frac{g(u)}{u} du. \quad (4.4)$$

More precisely, with  $s_\delta$  as in (2.10), for every  $\theta \in \mathbb{R}$  and every  $\delta > 0$ ,

$$|\Lambda_\delta(\theta) - \Lambda(\theta)| \leq \frac{s_\delta \delta^2 |\theta|}{6}, \quad (4.5)$$

so the convergence is of order  $\delta^2$ , locally uniformly. The function  $\Lambda$  is even, real analytic, and strictly convex, with

$$\Lambda'(\theta) = \frac{1}{\theta} \log \frac{\sinh \theta}{\theta}, \quad (4.6)$$

$$\Lambda''(\theta) = \frac{\theta \coth \theta - \log(\sinh \theta / \theta) - 1}{\theta^2} > 0, \quad (4.7)$$

$$\Lambda'(\mathbb{R}) = (-1, 1). \quad (4.8)$$

At the origin,

$$\Lambda(\theta) = \sum_{m \geq 1} \frac{2^{2m-2} B_{2m}}{m^2 (2m)!} \theta^{2m} = \frac{\theta^2}{12} - \frac{\theta^4}{720} + \frac{\theta^6}{17010} - \cdots. \quad (4.9)$$

*Proof. Elementary facts about  $g$ .* On  $(0, \infty)$ ,  $g'(u) = \coth u - u^{-1}$  and  $g''(u) = u^{-2} - \sinh^{-2} u$ ;  $g''$  decreases from  $g''(0) = \frac{1}{3}$  to 0, so  $0 < g'' \leq \frac{1}{3}$  and, integrating from 0,  $0 < g'(u) \leq u/3$ . With the Euler operator  $D = u \, d/du$  one has  $Dg(u) = u \coth u - 1$  and  $D^2g(u) = u(Dg)'(u) = u \coth u - u^2 \sinh^{-2} u$ , both nonnegative, with  $Dg(u) \leq u$  because  $\coth u \leq 1 + u^{-1}$ . All of  $g, Dg, D^2g$  are  $O(u^2)$  as  $u \rightarrow 0$ , by (2.5).

*The two integrals in (4.4).* For  $\theta \neq 0$  the substitution  $u = |\theta| e^{-t}$  turns  $\int_0^\infty g(\theta e^{-t}) dt$  into  $\int_0^{|\theta|} g(u) u^{-1} du$ , which is absolutely convergent because  $g(u)/u = O(u)$  at 0; both sides are even in  $\theta$ .

*The rate (4.5).* By evenness take  $\theta > 0$  and put  $w = \theta s_\delta$  and  $f(t) = g(we^{-t})$ , so that (4.3) reads  $\Lambda_\delta(\theta) = \delta \sum_{j \geq 0} f((j + \frac{1}{2})\delta)$  and  $\int_0^\infty f = \Lambda(w)$ . The chain rule gives  $f''(t) = D^2g(we^{-t}) \geq 0$ , so  $f \in C^2[0, \infty)$  with

$$\int_0^\infty |f''(t)| dt = \int_0^w D^2g(u) \frac{du}{u} = \int_0^w (Dg)'(u) du = Dg(w) = w \coth w - 1 \leq w.$$

On each cell  $[j\delta, (j+1)\delta]$  with midpoint  $m_j$ , the midpoint rule has the Peano representation  $\int_{j\delta}^{(j+1)\delta} f - \delta f(m_j) = \int_{j\delta}^{(j+1)\delta} f''(t) k_j(t) dt$  with the kernel  $k_j(t) = \frac{1}{2}(\frac{\delta}{2} - |t - m_j|)^2$ ,  $0 \leq k_j \leq \delta^2/8$ . Summing over  $j$ ,

$$|\Lambda_\delta(\theta) - \Lambda(\theta s_\delta)| \leq \frac{\delta^2}{8} \int_0^\infty |f''| \leq \frac{\delta^2}{8} \theta s_\delta.$$

Next,  $\Lambda'(u) = g(u)/u$  (fundamental theorem of calculus applied to the second integral in (4.4)), and  $0 \leq g(u)/u < 1$  for  $u > 0$  because  $g(u) = \int_0^u g' < u$ ; hence  $|\Lambda(\theta s_\delta) - \Lambda(\theta)| \leq \theta(s_\delta - 1)$ . Finally  $s_\delta - 1 = \sum_{k \geq 1} (\delta/2)^{2k} / (2k+1)!$  and  $\frac{\delta^2}{24} s_\delta = \sum_{k \geq 1} (\delta/2)^{2k} / (6(2k-1)!)$  with  $(2k+1)! \geq 6(2k-1)!$ , so  $s_\delta - 1 \leq \delta^2 s_\delta / 24$ . Adding the two estimates gives  $\delta^2 \theta s_\delta (\frac{1}{8} + \frac{1}{24}) = \delta^2 \theta s_\delta / 6$ , which is (4.5); the convergence in (4.4) follows.

*Regularity, derivatives, and convexity.* From  $\Lambda'(\theta) = g(\theta)/\theta$  one reads off (4.6), and differentiating once more gives (4.7), whose numerator is  $Dg(\theta) - g(\theta) = \int_0^\theta (g'(\theta) - g'(u)) du > 0$  for  $\theta > 0$  because  $g'$  is strictly increasing; at the origin  $\Lambda''(0) = g''(0)/2 = \frac{1}{6}$  (the series (4.9)). Hence  $\Lambda'' > 0$  everywhere, and  $\Lambda$  is real analytic on  $\mathbb{R}$  because  $g(u)/u$  is (the entire function  $\sinh u/u$  is strictly positive on the real axis). The integral representation  $\Lambda''(\theta) = \int_0^\infty e^{-2t} g''(\theta e^{-t}) dt$  is also legitimate: the  $\theta$ -derivatives of the integrand are  $e^{-t} g'(\theta e^{-t})$  and  $e^{-2t} g''(\theta e^{-t})$ , dominated for  $|\theta| \leq R$  by  $Re^{-2t}/3$  and  $e^{-2t}/3$  respectively, so differentiation under the integral sign is covered by dominated convergence. Finally  $g(\theta) = \theta - \log(2\theta) + O(e^{-2\theta})$  as  $\theta \rightarrow +\infty$ , so  $\Lambda'(\theta) = g(\theta)/\theta \rightarrow 1$  while staying below 1; oddness gives the lower endpoint, and strict monotonicity of  $\Lambda'$  gives (4.8). Integrating (2.5) termwise on  $|\theta| < \pi$  gives (4.9).  $\square$

### 4.3 All-orders midpoint Euler–Maclaurin expansion

Let  $B_n(x)$  be the Bernoulli polynomials and let  $D = \theta \frac{d}{d\theta}$  be the Euler operator.

**Theorem 4.2** (All-order midpoint expansion). *For every integer  $N \geq 1$  and every compact  $K \subset \mathbb{R}$ ,*

$$\Lambda_\delta(\theta) = \Lambda(\theta s_\delta) + \sum_{m=1}^{N-1} \frac{B_{2m}(1/2)}{(2m)!} \delta^{2m} D^{2m-1} g(\theta s_\delta) + O_{K,N}(\delta^{2N}) \quad (4.10)$$

*uniformly for  $\theta \in K$ . In particular the expansion contains only even powers of  $\delta$ . The first two nontrivial terms are*

$$\Lambda_\delta(\theta) = \Lambda(\theta) - \frac{\delta^2 \theta^2}{24} \Lambda''(\theta) + \delta^4 C_4(\theta) + O_K(\delta^6), \quad (4.11)$$

$$C_4(\theta) = \frac{7D^3g - 10D^2g + 5Dg - 2g}{5760}(\theta). \quad (4.12)$$



*Proof. Derivatives of the summand.* Put  $w = \theta s_\delta$  and  $f(t) = f_{\delta, \theta}(t) = g(we^{-t})$ . Under  $u = we^{-t}$  one has  $d/dt = -u d/du = -D$ , so by induction

$$f^{(n)}(t) = (-1)^n (D^n g)(we^{-t}), \quad f^{(n)}(0) = (-1)^n D^n g(w), \quad (4.13)$$

which is the identity  $f^{(2m-1)}(0) = -D^{2m-1}g(\theta s_\delta)$  used below. On  $|u| < \pi$  the series (2.5) gives  $D^n g(u) = \sum_{m \geq 1} (2m)^n \frac{2^{2m-1} B_{2m}}{m(2m)!} u^{2m} = O(u^2)$ , and  $D^n g$  is continuous on  $\mathbb{R}$ ; hence every  $f^{(n)}(t)$  is  $O(e^{-2t})$  as  $t \rightarrow \infty$ , so all derivatives of  $f$  vanish at infinity and are integrable on  $[0, \infty)$ , and the quantity

$$V_n(w) := \int_0^\infty |f^{(n)}(t)| dt = \int_0^{|w|} |D^n g(u)| \frac{du}{u} \quad (4.14)$$

is finite, continuous, and nondecreasing in  $|w|$ .

*Midpoint Euler–Maclaurin with kernel remainder.* For  $F \in C^{2N}[0, 1]$  and  $\omega \in [0, 1]$  the Bernoulli-polynomial expansion [19]

$$F(\omega) = \int_0^1 F + \sum_{k=1}^{2N} \frac{B_k(\omega)}{k!} [F^{(k-1)}(1) - F^{(k-1)}(0)] - \frac{1}{(2N)!} \int_0^1 \tilde{B}_{2N}(\omega - x) F^{(2N)}(x) dx,$$

with  $\tilde{B}_n$  the 1-periodic Bernoulli function, applied on the cell  $[j\delta, (j+1)\delta]$  to  $F(x) = f((j+x)\delta)$  at  $\omega = \frac{1}{2}$  gives, after multiplication by  $\delta$ ,

$$\delta f((j+\tfrac{1}{2})\delta) = \int_{j\delta}^{(j+1)\delta} f + \sum_{k=1}^{2N} \frac{B_k(\frac{1}{2})}{k!} \delta^k [f^{(k-1)}]_{j\delta}^{(j+1)\delta} - \frac{\delta^{2N}}{(2N)!} \int_{j\delta}^{(j+1)\delta} \tilde{B}_{2N}\left(\frac{t}{\delta} - \frac{1}{2}\right) f^{(2N)}(t) dt,$$

where  $\tilde{B}_{2N}(\frac{1}{2} - t/\delta + j) = \tilde{B}_{2N}(t/\delta - \frac{1}{2})$  by periodicity and evenness. Summing over  $j \geq 0$ , the bracketed terms telescope to  $-f^{(k-1)}(0)$  because  $f^{(k-1)}(t) \rightarrow 0$ , and the odd Bernoulli values  $B_{2m+1}(\frac{1}{2})$  vanish. Hence the exact identity

$$\delta \sum_{j \geq 0} f((j+\tfrac{1}{2})\delta) = \int_0^\infty f - \sum_{m=1}^N \frac{B_{2m}(\frac{1}{2})}{(2m)!} \delta^{2m} f^{(2m-1)}(0) - R_N, \quad (4.15)$$

where

$$R_N = \frac{\delta^{2N}}{(2N)!} \int_0^\infty \tilde{B}_{2N}\left(\frac{t}{\delta} - \frac{1}{2}\right) f^{(2N)}(t) dt.$$

Since  $\sup |\tilde{B}_{2N}| = |B_{2N}|$ , the remainder obeys

$$|R_N| \leq \frac{|B_{2N}|}{(2N)!} \delta^{2N} V_{2N}(\theta s_\delta), \quad (4.16)$$

and for  $\theta \in K$  and  $0 < \delta \leq 1$  the right side is at most  $\frac{|B_{2N}|}{(2N)!} \delta^{2N} V_{2N}(s_1 \max_K |\theta|)$  with  $s_1 = 2 \sinh \frac{1}{2}$ , a bound of the form  $O_{K,N}(\delta^{2N})$ . The integral in (4.15) is  $\Lambda(\theta s_\delta)$ , and (4.13) converts  $-f^{(2m-1)}(0)$  into  $+D^{2m-1}g(\theta s_\delta)$ ; the term  $m = N$  is itself  $O_{K,N}(\delta^{2N})$  by continuity of  $D^{2N-1}g$ , and absorbing it gives (4.10). The even-power structure is the vanishing of  $B_{2m+1}(\frac{1}{2})$ ; explicitly  $B_{2m}(\frac{1}{2}) = (2^{1-2m} - 1)B_{2m}$ , so  $B_2(\frac{1}{2}) = -\frac{1}{12}$  and  $B_4(\frac{1}{2}) = \frac{7}{240}$ .

*The first two corrections.* Write  $\theta s_\delta = \theta e^\lambda$  with

$$\lambda = \log s_\delta = \frac{\delta^2}{24} - \frac{\delta^4}{2880} + O(\delta^6),$$

from  $s_\delta = 1 + \delta^2/24 + \delta^4/1920 + O(\delta^6)$  in (2.10). Because  $D\Lambda = \theta\Lambda' = g$  and  $D^2\Lambda = Dg$ , Taylor expansion along the Euler flow gives  $\Lambda(\theta e^\lambda) = \Lambda + \lambda g + \frac{1}{2}\lambda^2 Dg + O(\delta^6)$ , and likewise  $Dg(\theta e^\lambda) = Dg + \lambda D^2g + O(\delta^4)$ , all evaluated at  $\theta$ . Inserting these into (4.10) with  $N = 3$ :

$$\begin{aligned}\Lambda(\theta s_\delta) &= \Lambda + \frac{\delta^2}{24}g + \delta^4\left(-\frac{g}{2880} + \frac{Dg}{1152}\right) + O(\delta^6), \\ \frac{B_2(\frac{1}{2})}{2!}\delta^2 Dg(\theta s_\delta) &= -\frac{\delta^2}{24}Dg - \frac{\delta^4}{576}D^2g + O(\delta^6), \\ \frac{B_4(\frac{1}{2})}{4!}\delta^4 D^3g(\theta s_\delta) &= \frac{7\delta^4}{5760}D^3g + O(\delta^6).\end{aligned}$$

The  $\delta^2$  terms combine to  $\frac{\delta^2}{24}(g - Dg)$ , and  $Dg - g = \theta g'(\theta) - g(\theta) = \theta^2 \Lambda''(\theta)$  by (4.7), which is the first correction in (4.11). The  $\delta^4$  terms combine to  $\frac{-2g+5Dg-10D^2g+7D^3g}{5760}$ , which is (4.12).  $\square$

*Remark 4.3* (A  $q$ -calculus reading). Equation (4.3) is a logarithmic Jackson-type sum whose mesh is  $\delta = -\log q$ , and it is the exponential factor  $e^{-\delta/2}$  of the exact factorization (2.10) that centers the sampling at half steps; the residual amplitude  $s_\delta$  moves no node and only rescales the argument,  $\theta \mapsto \theta s_\delta$ . The resulting coefficients are Bernoulli polynomials at  $1/2$ , not merely Bernoulli numbers at zero. This provides a direct bridge between the repository's  $q$ -product layer and its Bernoulli/Euler–Maclaurin layer [22].

## 5 A large-deviation principle as $q \uparrow 1$

### 5.1 The rate function

**Theorem 5.1** (Large-deviation principle). *The family  $X_{e^{-\delta}}$  satisfies a large-deviation principle on  $\mathbb{R}$  with speed  $1/\delta$  and good rate function*

$$I(x) = \sup_{\theta \in \mathbb{R}} \{\theta x - \Lambda(\theta)\}. \quad (5.1)$$

*The function  $I$  is even, strictly convex on  $(-1, 1)$ , satisfies  $I(0) = 0$ , and is infinite for  $|x| \geq 1$ . For  $|x| < 1$ , let  $\theta_x$  be the unique solution of*

$$x = \Lambda'(\theta_x) = \frac{1}{\theta_x} \log \frac{\sinh \theta_x}{\theta_x}. \quad (5.2)$$

*Then*

$$I(x) = \theta_x x - \Lambda(\theta_x), \quad I'(x) = \theta_x, \quad I''(x) = \frac{1}{\Lambda''(\theta_x)}. \quad (5.3)$$

*Proof.* The hypotheses of the Gärtner–Ellis theorem [56, 2] are checked one by one. The scaled logarithmic moment generating functions of the family at speed  $1/\delta$  are exactly  $\Lambda_\delta$  in (4.2), and (4.5) gives  $\Lambda_\delta(\theta) \rightarrow \Lambda(\theta)$  for every  $\theta \in \mathbb{R}$ , not only on compacts. The limit  $\Lambda$  is finite on all of  $\mathbb{R}$ , so its effective domain is  $\mathbb{R}$ , whose boundary is empty;  $\Lambda$  is differentiable there by theorem 4.1. Hence  $\Lambda$  is lower semicontinuous and essentially smooth, the steepness condition being vacuous, and 0 is an interior point of the domain. Since every  $X_{e^{-\delta}}$  is supported in the fixed compact interval  $[-1, 1]$ , the family is exponentially tight. The theorem therefore gives the full large-deviation principle, upper and lower bounds included, with the good rate function (5.1);  $I$  is even because  $\Lambda$  is, and  $I(0) = 0$  because  $\Lambda \geq 0 = \Lambda(0)$  and  $\Lambda'(0) = 0$ .

For the parametric form,  $\Lambda'$  is real analytic with  $\Lambda'' > 0$  and range  $(-1, 1)$  by (4.7)–(4.8), so for  $|x| < 1$  the saddle equation (5.2) has exactly one solution  $\theta_x$ , and by the implicit function theorem  $x \mapsto \theta_x$  is real analytic on  $(-1, 1)$  with  $\theta'_x = 1/\Lambda''(\theta_x)$ . The concave function  $\theta \mapsto \theta x - \Lambda(\theta)$  has its maximum at its unique critical point  $\theta_x$ , giving  $I(x) = \theta_x x - \Lambda(\theta_x)$ ; differentiating,  $I'(x) =$

$\theta_x + \theta'_x(x - \Lambda'(\theta_x)) = \theta_x$  and  $I''(x) = \theta'_x = 1/\Lambda''(\theta_x) > 0$ , which is (5.3) and the strict convexity of  $I$  on  $(-1, 1)$ . Outside  $[-1, 1]$  the rate is infinite by support. At  $x = 1$ ,  $\theta - \Lambda(\theta) \rightarrow \infty$  as  $\theta \rightarrow \infty$  by the expansion (6.6) proved in theorem 6.2; hence  $I(1) = \infty$ , and similarly at  $-1$ .  $\square$

Part XI proves the same principle on its own compact scale, and the two statements are exact transcriptions of one another rather than two theorems. By table 1,  $X_q = 2X_q^\circ$ , so the standardized variable  $X_q/\sigma_q$ , with  $\sigma_q$  as in (2.8), is  $X_q^*$ , which is Part XI's  $Z_h$  at  $h = \delta$ ; by (2.11),  $\sqrt{\delta}/\sigma_q = \sqrt{6}(1 + O(\delta^2))$ , so Part XI's compact variable  $Y_h = \sqrt{\delta} Z_h$  of (7.4) is  $\sqrt{6} X_{e^{-\delta}}(1 + O(\delta^2))$ . Accordingly theorem 7.2 is the present theorem after  $y = \sqrt{6}x$ : the two speeds  $1/h$  and  $1/\delta$  coincide, the transforms are related by  $\Lambda_{\text{XI}}(s) = \Lambda(\sqrt{6}s)$ , because (7.8) reads  $\Lambda_{\text{XI}}(s) = J(\sqrt{6}s)$  and  $J = \Lambda$  by proposition 6.1, and the rate functions by  $I_{\text{XI}}(y) = I(y/\sqrt{6})$ ; in particular (4.8) becomes the range  $(-\sqrt{6}, \sqrt{6})$  of (7.9), and (5.6) becomes (7.15). Under the same substitution the parametric identities (5.3) are theorem 7.3: the saddle  $\theta_x$  of (5.2) is Part XI's  $a$ , the level is  $y(a) = \sqrt{6} \ell(a)/a$  of (7.12), and  $I(x) = \theta_x x - \Lambda(\theta_x)$  is (7.13). Both coordinates are carried by the volume on purpose: the present one has support  $[-1, 1]$  and is the one used in appendix 6 and appendix 8, while Part XI's has support  $[-\sqrt{6}, \sqrt{6}]$  and is the one in which its Edgeworth matching is stated.

Thus for regular sets  $A$  the logarithmic probability scale is

$$\mathbb{P}\{X_q \in A\} = \exp \left[ -\frac{\inf_{x \in A} I(x) + o(1)}{-\log q} \right]. \quad (5.4)$$

This is a parameter-concentration law, not the fixed- $q$  spatial endpoint asymptotic already developed in the corpus.

## 5.2 Central rate series

The inverse saddle map in (5.2) is analytic at zero. Series reversion gives

$$\theta_x = 6x + \frac{36}{5}x^3 + \frac{1656}{175}x^5 + \frac{9504}{875}x^7 + \frac{709344}{67375}x^9 + \frac{18086976}{1990625}x^{11} + O(x^{13}), \quad (5.5)$$

$$I(x) = 3x^2 + \frac{9}{5}x^4 + \frac{276}{175}x^6 + \frac{1188}{875}x^8 + \frac{354672}{336875}x^{10} + \frac{1507248}{1990625}x^{12} + O(x^{14}). \quad (5.6)$$

All displayed coefficients are positive, and the companion symbolic script checks positivity through the  $x^{16}$  coefficient. The pattern does not persist.

*Remark 5.2* (Sign pattern and radius of the rate series). Write  $I(x) = \sum_{n \geq 1} c_n x^{2n}$ . The absorbed report conjectured that every  $c_n$  is strictly positive. That conjecture is *false*: exact rational reversion of (4.9), carried out for this volume to order  $x^{28}$  and reproducing the eight coefficients of the appendix, gives  $c_9, c_{10}, c_{11} > 0$  but

$$\begin{aligned} c_{12} &= -\frac{3032147123502547968}{3562005918173828125} = -0.85124707\dots, \\ c_{13} &= -\frac{58363635922427846541312}{89139198102300048828125} = -0.65474715\dots, \end{aligned}$$

while  $c_{14} > 0$  again; the values are listed in the appendix. The sign of  $c_{12}$  was confirmed independently of the series arithmetic by evaluating  $I$  from its exact saddle at sixty digits: at  $x = 0.3$  the residual  $I(x) - \sum_{n \leq 11} c_n x^{2n} = -2.514 \times 10^{-13}$  agrees with  $c_{12}x^{24} + c_{13}x^{26} + c_{14}x^{28} = -2.526 \times 10^{-13}$  to the size of the first omitted term. Convexity of  $I$  is of course unaffected.

The radius of convergence  $R_I$  of the series is positive, because  $\theta_x$  is the analytic inverse of  $\Lambda'$  at a point where  $\Lambda'' \neq 0$ , and  $R_I \leq 1$ , because  $I$  is real analytic on  $(-1, 1)$  and  $I(x) \rightarrow \infty$  as  $x \uparrow 1$  (theorem 6.2), whereas a radius larger than 1 would make the series bounded near  $x = 1$ . The fourteen computed values of  $c_n^{1/(2n)}$  lie between 0.92 and 1.74 and cluster near 1, consistent with  $R_I = 1$ ; the exact value is not determined here, and because the signs alternate irregularly Pringsheim's theorem does not apply.

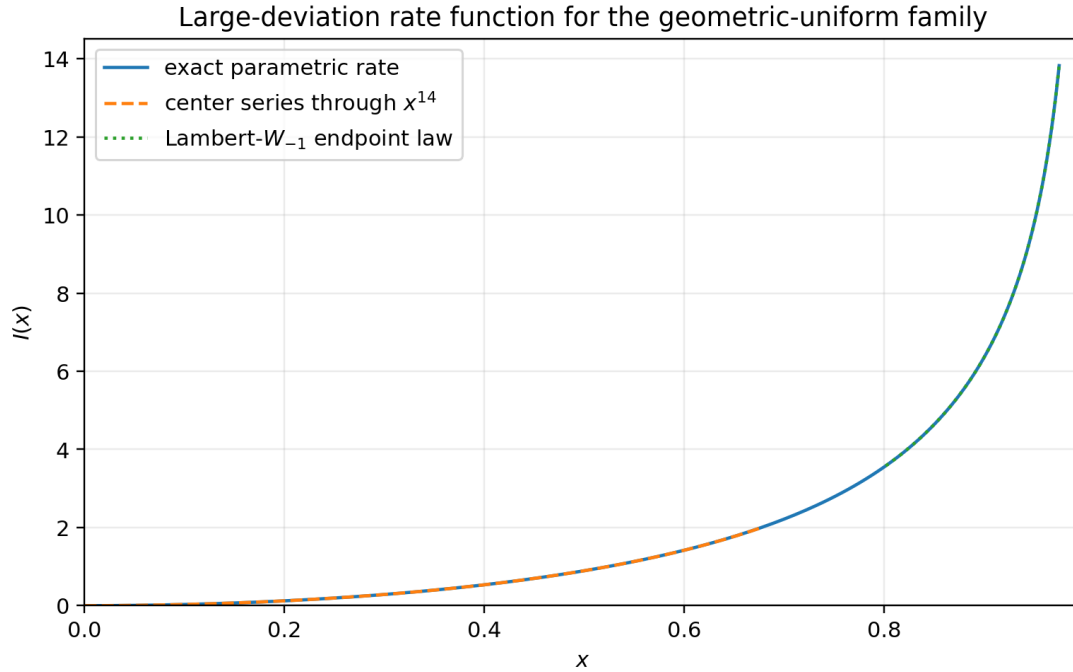


Figure 45: The exact parametric rate  $I(x)$ , its rational center expansion, and the Lambert- $W_{-1}$  boundary approximation. The latter is already visually indistinguishable from the exact curve for  $x \gtrsim 0.8$ .

## 6 Lambert- $W_{-1}$ behavior at the rate boundary

The fixed-base Fabius endpoint theory uses Lambert  $W$  to invert flat spatial tails. The rate function develops a second, conceptually different Lambert mechanism: the saddle parameter diverges as the deviation level approaches the fixed support boundary.

### 6.1 Asymptotics of the limiting transform

For  $u > 0$ ,

$$g(u) = u - \log(2u) + \log(1 - e^{-2u}). \quad (6.1)$$

Define the convergent constant

$$C_\Lambda = \int_0^1 \frac{\log(\sinh u/u)}{u} du - 1 + \int_1^\infty \frac{\log(1 - e^{-2u})}{u} du. \quad (6.2)$$

Numerically,

$$C_\Lambda = -0.96892042396303131827115715418362428628 \dots \quad (6.3)$$

The constant (6.2) is not a new transcendental. Throughout, Bernoulli numbers are those of (0.4.1), so that (2.5) holds, and  $\gamma_1$  denotes the *first Stieltjes constant* in the convention fixed by Part XI in (8.1), namely the one in which  $\zeta(1+s) = s^{-1} + \gamma - \gamma_1 s + O(s^2)$ ; in that convention  $\gamma_1 = -0.0728158454836767248605 \dots$

**Proposition 6.1** (Closed form of the endpoint constant). *The limiting transform (4.4) is Part XI's transform  $J$  of (7.5): one has  $\Lambda = J$  on all of  $\mathbb{R}$ . The constant (6.2) has the closed form*

$$C_\Lambda = \mathfrak{C} - \frac{(\log 2)^2}{2} = \gamma_1 + \frac{\gamma^2}{2} - \frac{\pi^2}{12} - \frac{(\log 2)^2}{2}, \quad (6.4)$$

where  $\gamma$  is Euler's constant,  $\gamma_1$  is the *first Stieltjes constant* of (8.1), and  $\mathfrak{C}$  is the renormalized Mellin constant (8.2) of Part XI.

*Proof.* The integrand of (4.4) is  $g(u)/u$  with  $g = \log \sinh c$  from (4.1), and Part XI's  $\ell$  in (7.5) is the same function of the same variable. Hence  $\Lambda(\theta) = \int_0^\theta g(u)u^{-1} du = \int_0^\theta \ell(v)v^{-1} dv = J(\theta)$ , both sides being even.

For the constant, substitute  $x = 2u$ , under which  $du/u = dx/x$ . By (6.1),

$$\int_0^1 \frac{g(u)}{u} du = 1 + \int_0^1 \frac{\log(1 - e^{-2u}) - \log(2u)}{u} du = 1 + \int_0^2 \frac{\log(1 - e^{-x}) - \log x}{x} dx,$$

while the second integral of (6.2) becomes  $\int_2^\infty \log(1 - e^{-x})x^{-1} dx$ . The two occurrences of 1 in (6.2) cancel, so

$$C_\Lambda = \int_0^2 \frac{\log(1 - e^{-x}) - \log x}{x} dx + \int_2^\infty \frac{\log(1 - e^{-x})}{x} dx. \quad (6.5)$$

This is (8.8) with the splitting point 2 in place of 1, and the two differ by an elementary integral:

$$\mathfrak{C} - C_\Lambda = \int_1^2 \frac{\log(1 - e^{-x})}{x} dx - \int_1^2 \frac{\log(1 - e^{-x}) - \log x}{x} dx = \int_1^2 \frac{\log x}{x} dx = \frac{(\log 2)^2}{2}.$$

lemma 8.1 identifies  $\mathfrak{C}$  with the finite part of  $-\Gamma(s)\zeta(s+1)$  at  $s = 0$ , computed there from (8.6) and (8.1) as (8.7), giving  $\mathfrak{C} = \gamma_1 + \frac{1}{2}\gamma^2 - \frac{1}{12}\pi^2$ . This proves (6.4).

The same statement read through the two logarithmic variables explains why a  $(\log 2)^2$  appears at all. lemma 8.1 expands  $J$  in  $L = \log(2a)$ , giving  $J(\theta) = \theta - \frac{1}{2}L^2 + \mathfrak{C} + O(e^{-2\theta}/\theta)$ , whereas (6.6) expands the same function in  $\log \theta$ . Since  $\frac{1}{2}L^2 = \frac{1}{2}(\log \theta)^2 + (\log 2)\log \theta + \frac{1}{2}(\log 2)^2$ , the elementary terms agree and the constants must differ by exactly  $\frac{1}{2}(\log 2)^2$ , the additive constant of such an expansion being unique because  $\theta$ ,  $(\log \theta)^2$ ,  $\log \theta$  and 1 are asymptotically independent and the remainder vanishes.  $\square$

Numerically (6.4) reproduces (6.3): by (8.2),  $\mathfrak{C} = -0.72869391700393060593\dots$ , and subtracting  $\frac{1}{2}(\log 2)^2 = 0.24022650695910071233\dots$  gives  $-0.96892042396303131827\dots$ . The closed form and the two absolutely convergent integrals of (6.2) were evaluated independently to sixty decimal places and agree to that precision.

**Theorem 6.2** (Transform and saddle endpoint). *As  $\theta \rightarrow +\infty$ ,*

$$\Lambda(\theta) = \theta - \frac{1}{2}(\log \theta)^2 - (\log 2)\log \theta + C_\Lambda + O\left(\frac{e^{-2\theta}}{\theta}\right), \quad (6.6)$$

$$1 - \Lambda'(\theta) = \frac{\log(2\theta)}{\theta} + O\left(\frac{e^{-2\theta}}{\theta}\right). \quad (6.7)$$

Let  $\eta = 1 - x \downarrow 0$  and

$$\theta_0(\eta) = -\frac{1}{\eta}W_{-1}(-\eta/2). \quad (6.8)$$

Then, with  $L_\eta = \log \theta_0(\eta)$ , the exact saddle satisfies, for  $0 < \eta \leq \frac{1}{2}$ ,

$$\theta_{1-\eta} = \theta_0(\eta)(1 + \varepsilon_\eta), \quad 0 < \varepsilon_\eta \leq \frac{2e^{-2\theta_0(\eta)}}{(1 - e^{-2\theta_0(\eta)})L_\eta}, \quad (6.9)$$

in particular  $\theta_{1-\eta} = \theta_0(\eta)(1 + O(e^{-2\theta_0(\eta)}/L_\eta))$ , and

$$I(1 - \eta) = \frac{1}{2}L_\eta^2 + (\log 2 - 1)L_\eta - \log 2 - C_\Lambda + O(e^{-2\theta_0(\eta)}), \quad (6.10)$$

the remainder being at most  $4e^{-2\theta_0(\eta)}$  in absolute value for  $0 < \eta \leq \frac{1}{2}$ .

*Proof.* Integrate (6.1) from 1 to  $\theta$ . The elementary terms give  $\theta - \frac{1}{2}(\log \theta)^2 - (\log 2) \log \theta$ , while the two convergent remainders combine exactly into (6.2), leaving the tail

$$T(\theta) := \int_{\theta}^{\infty} \frac{-\log(1 - e^{-2u})}{u} du, \quad 0 < T(\theta) \leq \frac{e^{-2\theta}}{2\theta(1 - e^{-2\theta})},$$

so that  $\Lambda(\theta) = \theta - \frac{1}{2}(\log \theta)^2 - (\log 2) \log \theta + C_{\Lambda} + T(\theta)$  exactly; this is (6.6), and dividing (6.1) by  $\theta$  gives (6.7).

For  $\theta > 0$  the saddle equation  $1 - \eta = \Lambda'(\theta) = g(\theta)/\theta$  is, by (6.1), equivalent to

$$\eta\theta = \log(2\theta) - \log(1 - e^{-2\theta}). \quad (6.11)$$

Dropping the exponentially small last term yields  $\eta\theta = \log(2\theta)$ , equivalent to  $(-\eta\theta)e^{-\eta\theta} = -\eta/2$ . The diverging positive solution is (6.8) on branch  $W_{-1}$  [52]; it is a decreasing function of  $\eta$  with  $\theta_0(\frac{1}{2}) = 4.3065\dots$ , so  $\theta_0 \geq 4$  for  $0 < \eta \leq \frac{1}{2}$ .

*The displacement.* Write  $\theta = \theta_0(1 + \varepsilon)$  and use  $\eta\theta_0 = \log(2\theta_0)$ : equation (6.11) becomes

$$h(\varepsilon) := \varepsilon \log(2\theta_0) - \log(1 + \varepsilon) = r(\varepsilon) := -\log(1 - e^{-2\theta_0(1+\varepsilon)}).$$

For  $\varepsilon \geq 0$  the left side is increasing (its derivative is  $\log(2\theta_0) - (1 + \varepsilon)^{-1} > 0$  because  $\log(2\theta_0) > 1$ ) with  $h(0) = 0$ , while  $r$  is positive and decreasing; hence  $h - r$  increases strictly from  $-r(0) < 0$  to  $+\infty$  on  $[0, \infty)$  and has exactly one zero  $\varepsilon_{\eta} > 0$  there. Since (6.11) has the unique solution  $\theta_{1-\eta}$ , this zero is it, which gives the positivity in (6.9). For the upper bound,  $\log(1 + \varepsilon) \leq \varepsilon$  gives  $\varepsilon_{\eta}(\log(2\theta_0) - 1) \leq h(\varepsilon_{\eta}) = r(\varepsilon_{\eta}) \leq r(0) \leq e^{-2\theta_0}/(1 - e^{-2\theta_0})$ , and  $\log(2\theta_0) - 1 = L_{\eta} + \log 2 - 1 \geq L_{\eta}/2$  as soon as  $L_{\eta} \geq 2(1 - \log 2)$ , i.e.  $\theta_0 \geq 1.85$ , which holds for  $\eta \leq \frac{1}{2}$ .

*The edge law.* Put  $\theta_x = \theta_{1-\eta}$  and  $x = 1 - \eta$ . By (5.3) and the exact form of (6.6),

$$I(1 - \eta) = \theta_{xx} - \Lambda(\theta_x) = -\eta\theta_x + \frac{1}{2}(\log \theta_x)^2 + (\log 2) \log \theta_x - C_{\Lambda} - T(\theta_x),$$

and substituting (6.11) for  $\eta\theta_x$ ,

$$I(1 - \eta) = \frac{1}{2}(\log \theta_x)^2 + (\log 2 - 1) \log \theta_x - \log 2 - C_{\Lambda} + \log(1 - e^{-2\theta_x}) - T(\theta_x),$$

an identity with no error term. The last two terms are together at most  $(1 + \frac{1}{2\theta_0})e^{-2\theta_0}/(1 - e^{-2\theta_0})$  in absolute value, since  $\theta_x \geq \theta_0$ . Finally  $\log \theta_x = L_{\eta} + \log(1 + \varepsilon_{\eta})$  with  $0 < \log(1 + \varepsilon_{\eta}) \leq \varepsilon_{\eta}$ , so replacing  $\theta_x$  by  $\theta_0$  in the two elementary terms changes them by  $(L_{\eta} + \log 2 - 1 + \frac{1}{2} \log(1 + \varepsilon_{\eta})) \log(1 + \varepsilon_{\eta}) \leq L_{\eta} \varepsilon_{\eta} \leq 2e^{-2\theta_0}/(1 - e^{-2\theta_0})$  by (6.9). Adding the two bounds gives a remainder of at most  $(3 + \frac{1}{2\theta_0})e^{-2\theta_0}/(1 - e^{-2\theta_0}) \leq 4e^{-2\theta_0}$  for  $\theta_0 \geq 4$ , which is (6.10).  $\square$

*Remark 6.3* (Double logarithmic geometry). The standard branch expansion gives

$$-W_{-1}(-\eta/2) = \log(2/\eta) + \log \log(2/\eta) + o(1),$$

and consequently

$$L_{\eta} = \log(1/\eta) + \log \log(1/\eta) + o(\log \log(1/\eta)).$$

Thus the rate diverges primarily like  $\frac{1}{2} \log^2(1/\eta)$ , with a forced  $\log(1/\eta) \log \log(1/\eta)$  correction. This is the parameter-LDP counterpart of the log-squared endpoint structures elsewhere in the Fabius–Rvachev theory.

The edge law (6.10) is theorem 8.2 written in the coordinate of this part. The edge parameters agree, because  $y = \sqrt{6}x$  turns Part XI's  $1 - y/\sqrt{6}$  into  $\eta = 1 - x$ , and Part XI's Lambert variable (8.11) is, by (6.8) and  $\eta\theta_0 = \log(2\theta_0)$ ,

$$w(\eta) = -W_{-1}(-\eta/2) = \eta\theta_0(\eta) = \log(2\theta_0(\eta)) = L_{\eta} + \log 2.$$

Substituting this together with (6.4) into (8.12) returns (6.10) identically:

$$\frac{1}{2}w(\eta)^2 - w(\eta) - \mathfrak{C} = \frac{1}{2}L_\eta^2 + (\log 2 - 1)L_\eta - \log 2 + \frac{1}{2}(\log 2)^2 - \mathfrak{C} = \frac{1}{2}L_\eta^2 + (\log 2 - 1)L_\eta - \log 2 - C_\Lambda.$$

Part XI's (8.13) states the same exponentially small error at the exact saddle  $a = \theta_{1-\eta}$ ; (6.10) carries it over to the Lambert approximation  $\theta_0(\eta)$  through the explicit displacement bound (6.9), at the price of the constant 4.

## 7 Uniform all-orders Edgeworth expansion for $q \uparrow 1$

### 7.1 Standardized infinite-sinc product

Set

$$Z_q = \frac{X_q}{\sigma_q}, \quad b_q = \frac{1-q}{\sigma_q} = \sqrt{3(1-q^2)}. \quad (7.1)$$

Then

$$\chi_q(t) := \mathbb{E} e^{itZ_q} = \prod_{j \geq 0} \text{sinc}_{\text{rad}}(b_q q^j t). \quad (7.2)$$

This standardized layer is shared with Part XI, and the two parts use the same variable, not merely proportional ones. Indeed  $X_q = 2X_q^\circ$  by table 1, so  $\sigma_q$  of (2.8) is twice the canonical standard deviation and  $Z_q = X_q^*$ ; with  $r = q = e^{-\delta}$  and  $h = \delta$  this is exactly Part XI's  $Z_h$  of (1.3). Consequently  $b_q = a_h = \sqrt{3(1-q^2)}$  and  $\chi_q = \Psi_h$  of (3.3), and everything proved in appendix 5 applies to  $\chi_q$  verbatim after the substitution  $h = \delta$ .

Let  $\lambda_n(q)$  be the standardized cumulants of  $Z_q$ . From (2.7), for  $m \geq 2$ ,

$$\lambda_{2m}(q) = \frac{3^m 2^{2m-1} B_{2m}}{m} \frac{(1-q^2)^m}{1-q^{2m}} \quad (7.3)$$

$$= \frac{3^m 2^{3m-2} B_{2m}}{m} \frac{\sinh^m \delta}{\sinh(m\delta)}, \quad q = e^{-\delta}. \quad (7.4)$$

Therefore  $\lambda_{2m} = O(\delta^{m-1})$ . Both forms are proposition 4.1, where they are proved; they are repeated here because the hyperbolic parity of (7.4) is read off below. The first values are

$$\lambda_4 = -\frac{6}{5} \tanh \delta = -\frac{6}{5} \delta + \frac{2}{5} \delta^3 + O(\delta^5), \quad (7.5)$$

$$\lambda_6 = \frac{64}{7} \delta^2 - \frac{64}{7} \delta^4 + O(\delta^6), \quad (7.6)$$

$$\lambda_8 = -\frac{864}{5} \delta^3 + \frac{1728}{5} \delta^5 + O(\delta^7). \quad (7.7)$$

### 7.2 Weighted Hermite polynomial

Let  $H_n$  denote the probabilists' Hermite polynomials,  $\phi(x) = (2\pi)^{-1/2} e^{-x^2/2}$ , and let a multi-index  $\mathbf{k} = (k_2, k_3, \dots)$  have Edgeworth weight and Hermite degree

$$w(\mathbf{k}) = \sum_{m \geq 2} (m-1)k_m, \quad |\mathbf{k}|_* = \sum_{m \geq 2} m k_m. \quad (7.8)$$

For  $R \geq 0$  define

$$\mathcal{E}_{R,q}(x) = \sum_{w(\mathbf{k}) \leq R} H_{2|\mathbf{k}|_*}(x) \prod_{m \geq 2} \frac{1}{k_m!} \left( \frac{\lambda_{2m}(q)}{(2m)!} \right)^{k_m}. \quad (7.9)$$

Only finitely many indices contribute. Keeping exact finite- $q$  cumulants is useful: it automatically resums many higher powers of  $\delta$ . The weight  $w$  and the degree  $|\mathbf{k}|_*$  are Part XI's  $w$  and  $\nu$  of (5.1), so (7.9) is (5.2) at  $h = \delta$ :  $\mathcal{E}_{R,q} = \mathcal{E}_{R,h}$ .



### 7.3 Global Fourier bounds

*Remark 7.1* (Geometric infinite-sinc tails). The global frequency control needed below is proved once, in Part XI, for the very same product: by (7.2) and table 1 one has  $\chi_q = \Psi_h$  at  $h = \delta$ , so lemma 5.3 applies verbatim. In the letters of this part it gives, for each  $q_0 \in (0, 1)$  and each integer  $L \geq 0$ , positive constants  $c, C_L$  such that for  $q_0 \leq q < 1$

$$|\chi_q(t)| \leq \begin{cases} e^{-ct^2}, & |t| \leq b_q^{-1}, \\ C_L \exp\left(-\frac{c}{1-q}\right) (1 + b_q|t|)^{-L}, & |t| \geq b_q^{-1}. \end{cases} \quad (7.10)$$

The first line is (5.8) together with the exact identity (5.10), which is where the normalization  $\sum_{j \geq 0} (b_q q^j)^2 = 3$  enters; the exponential factor on the second line is (5.11) combined with  $1 - q^2 \asymp \delta$ ; and the polynomial factor is the elementary bound  $|\operatorname{sinc}_{\text{rad}} u| \leq |u|^{-1}$  applied to the first  $L$  factors of the product. Lemma 5.3 is strictly stronger than (7.10) in two respects that the sup-norm statement below does not display: it bounds every fixed frequency derivative  $\partial_t^p \Psi_h$ , which is what a Schwartz-seminorm conclusion needs, and on the outer zone the proof of theorem 5.1 sharpens the tail to  $e^{-c/\delta} (1 + |t|)^{-N}$  for every prescribed  $N$ .

### 7.4 The differentiated theorem

*Remark 7.2* (Continuous-parameter Edgeworth expansion). The differentiated Edgeworth expansion this part uses is the sup-norm specialization of the theorem proved in Part XI, and the volume states it only once. Since  $Z_q$  is Part XI's  $Z_h$  at  $h = \delta$  and  $\mathcal{E}_{R,q} = \mathcal{E}_{R,h}$ , theorem 5.1 with  $j = r$  and  $K = 0$  reads: writing  $f_q$  for the density of  $Z_q$ , for every pair of nonnegative integers  $R$  and  $r$  there exist  $q_{R,r} < 1$  and  $C_{R,r} < \infty$  such that

$$\sup_{x \in \mathbb{R}} \left| f_q^{(r)}(x) - \frac{d^r}{dx^r} [\phi(x) \mathcal{E}_{R,q}(x)] \right| \leq C_{R,r} \delta^{R+1}, \quad q_{R,r} < q < 1, \quad (7.11)$$

where  $\delta = -\log q$ . Part XI's (5.4) is strictly stronger, since it carries an arbitrary fixed polynomial weight  $(1 + |x|)^K$  alongside the  $r$  spatial derivatives; (7.11) is its  $K = 0$  corollary.

One ingredient deserves to be named, because the report absorbed into this part used it as an assumption rather than proving it. Its low-frequency step was the expansion

$$\log \chi_q(t) = -\frac{t^2}{2} + \sum_{m=2}^{R+1} \frac{\lambda_{2m}(q)}{(2m)!} (it)^{2m} + \rho_{R,q}(t), \quad |\rho_{R,q}(t)| \leq C_R \delta^{R+1} (1 + |t|^{2R+4}),$$

with the frequency window on which it holds left to be arranged afterwards. Nothing in this volume rests on that as an assumption: the estimate is lemma 5.4, proved there on the explicit window  $b_q|t| \leq \varepsilon_0$  for any fixed  $\varepsilon_0 < \pi$ , and for every fixed frequency derivative, as (5.14) and (5.15) at  $h = \delta$ . (Part XI writes  $\delta$  for that frequency-split constant; it is unrelated to the  $\delta = -\log q$  of this part, which is why it is called  $\varepsilon_0$  here.) The global envelope that accompanies it is lemma 5.3, quoted above as (7.10).

### 7.5 First explicit corrections

Expanding the exact-cumulant form in the natural variable  $\delta$  gives

$$f_q(x) = \phi(x) [1 + \delta Q_1(x) + \delta^2 Q_2(x) + \delta^3 Q_3(x) + O_{Cr}(\delta^4)], \quad (7.12)$$

where

$$Q_1 = -\frac{1}{20} H_4, \quad (7.13)$$

$$Q_2 = \frac{4}{315} H_6 + \frac{1}{800} H_8, \quad (7.14)$$

$$Q_3 = \frac{1}{60} H_4 - \frac{3}{700} H_8 - \frac{1}{1575} H_{10} - \frac{1}{48000} H_{12}. \quad (7.15)$$

The absence of an  $H_4$  term in  $Q_2$  and of an  $H_6$  term in  $Q_3$  reflects the hyperbolic parity visible in (7.4). If one instead uses  $1 - q$  as the small parameter, these cancellations are obscured by the change of variables. The same three orders appear in Part XI as (5.27), written in  $h = \delta$  and with the remainder measured in every fixed Schwartz seminorm; the parity reading above is what this part adds to them.

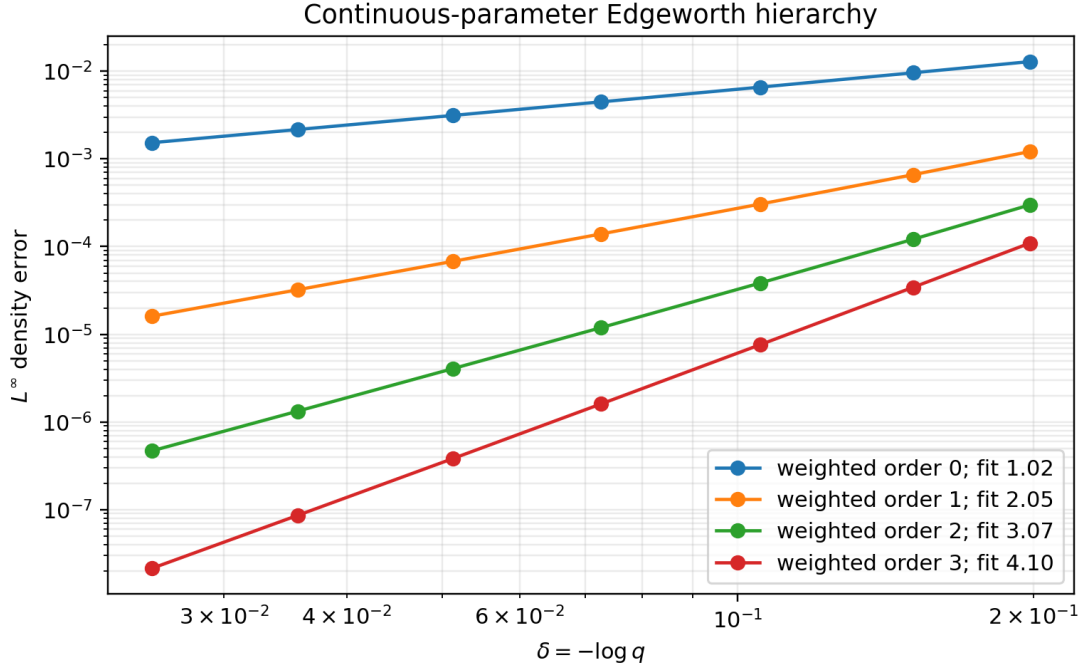


Figure 46: Sup-norm density errors from direct Fourier inversion. Fitted slopes 1.02, 2.05, 3.07, and 4.10 agree with (7.11) and theorem 5.1.

## 8 Central, moderate, and large deviations in one scale diagram

The Edgeworth and large-deviation results are not separate asymptotic stories. The center expansion of  $I$  contains the same cumulant information as the Hermite correction.

Substitute  $x = \sigma_q z$  into (5.6) and use (2.11). The rate  $I$  is real analytic on  $(-1, 1)$ , because  $\theta_x$  is the analytic inverse of  $\Lambda'$  (proof of theorem 5.1), so the series (5.6) has a positive radius of convergence  $R_I$  (remark 5.2). Fix  $0 < \rho < \rho' < R_I$  and put  $M = \max_{|x|=\rho'} |I(x)|$ ; the Cauchy estimates  $|c_n| \leq M \rho'^{-2n}$  give

$$\left| I(x) - \sum_{n \leq 3} c_n x^{2n} \right| \leq \frac{M}{\rho'^8 (1 - \rho^2/\rho'^2)} x^8, \quad |x| \leq \rho.$$

Then, uniformly on the range  $\sigma_q |z| \leq \rho$ , that is for  $|z| \leq \rho \sigma_q^{-1}$ , which by (2.11) is a window of width of order  $\delta^{-1/2}$ , the three retained terms expand as  $c_1 \sigma_q^2 / \delta = \frac{1}{2} - \frac{\delta^2}{24} + O(\delta^4)$ ,  $c_2 \sigma_q^4 / \delta = \frac{\delta}{20} + O(\delta^3)$  and  $c_3 \sigma_q^6 / \delta = \frac{23\delta^2}{3150} + O(\delta^4)$ , while the remainder is  $O(\sigma_q^8 z^8 / \delta) = O(\delta^3 z^8)$ ; collecting terms,

$$\frac{I(\sigma_q z)}{\delta} = \frac{z^2}{2} + \frac{\delta z^4}{20} + \delta^2 \left( \frac{23z^6}{3150} - \frac{z^2}{24} \right) + O(\delta^3 (1 + |z|^8)). \quad (8.1)$$

The remainder carries the growing factor  $1 + |z|^8$  because (8.1) is used at  $|z| \rightarrow \infty$ , in (8.2) and in principle 8.1(ii); it is exactly the size of the next term  $c_4 \sigma_q^8 z^8 / \delta$  of the rate series, with  $c_4 = \frac{1188}{875}$ , and each displayed term dominates the next precisely while  $\delta z^2 \rightarrow 0$ , that is on  $|z| = o(\delta^{-1/2})$ . For bounded  $z$  the statement degenerates to the uninformative  $O(\delta^3)$ . The first exponential correction

$e^{-\delta z^4/20} = 1 - \delta z^4/20 + \dots$  matches the leading Hermite term  $-\delta H_4/20 = -\delta(z^4 - 6z^2 + 3)/20$  of (7.13) in its two top degrees: the  $z^4$  terms agree outright, and the  $z^2$  term  $+3\delta z^2/10$  is produced by the saddlepoint prefactor, since  $\sigma_q(2\pi\delta\Lambda''(\theta_{\sigma_q z}))^{-1/2} = (2\pi)^{-1/2}(1 + \frac{3}{10}\delta z^2 + O(\delta^2))$  by  $\Lambda''(\theta) = \frac{1}{6} - \frac{\theta^2}{60} + O(\theta^4)$  and  $\theta_x = 6x + O(x^3)$ . The constant term  $-3\delta/20$  is not reached by these two ingredients: it is the value  $c_1(0) = -\frac{3}{20}$  that the conjectural first saddlepoint coefficient of (8.3) must take, in agreement with the generic formula  $c_1(0) = \lambda_4/8$  for the standardized fourth cumulant  $\lambda_4 = \Lambda''''(0)/\Lambda''(0)^2 = -\frac{6}{5}$  of the transform. The Gaussian moderate-deviation window suggested by the quartic term is

$$|z| = o(\delta^{-1/4}). \quad (8.2)$$

Beyond that scale, the rate function rather than a finite Edgeworth polynomial becomes the stable description.

**Principle 8.1** (Three nested regimes). As  $q = e^{-\delta} \uparrow 1$ :

- (i) fixed standardized  $z$  is governed by the differentiated Edgeworth series;
- (ii)  $1 \ll |z| \ll \delta^{-1/2}$  is governed by the Taylor expansion of  $I(\sigma_q z)/\delta$  and its moderate-deviation resummations; on this window the corresponding large-deviation principle is not heuristic but proved, as [theorem 7.1](#), whose scale  $b_h$  with  $b_h \rightarrow \infty$  and  $hb_h^2 \rightarrow 0$  is precisely  $1 \ll b_h \ll \delta^{-1/2}$  once its parameter is read as  $h = \delta$  and its variable as  $Z_h = Z_q$ ;
- (iii) fixed  $x \in (-1, 1) \setminus \{0\}$  is governed by  $e^{-I(x)/\delta}$ , with Lambert- $W_{-1}$  behavior as  $x$  approaches the support edge.

**Conjecture 8.2** (Uniform sharp saddlepoint expansion). *For every compact  $K \Subset (-1, 1)$ , the density of  $X_{e^{-\delta}}$  has an all-orders expansion*

$$f_{X_q}(x) \sim \frac{e^{-I(x)/\delta}}{\sqrt{2\pi\delta\Lambda''(\theta_x)}} \left( 1 + \sum_{n \geq 1} c_n(x)\delta^n \right), \quad x \in K, \quad (8.3)$$

*uniformly with all fixed  $x$ -derivatives. The coefficients should be rational differential polynomials in the tilted cumulants  $g^{(k)}(\theta_x e^{-t})$  integrated over  $t$ .*

[Conjecture 11.1](#) is this same conjecture in Part XI's compact coordinate. Under  $y = \sqrt{6}x$  of [table 1](#) the densities differ by the Jacobian  $\sqrt{6}$  and the curvatures by  $\Lambda''_{\text{XI}}(s(y)) = 6\Lambda''(\theta_x)$ , so the two prefactors agree; both are open. This part keeps (8.3) because [appendix 10](#) refines it in the coordinate with support  $[-1, 1]$ , and the matching statement that belongs with it is [conjecture 11.2](#), of which [principle 8.1](#) is the scale-diagram form.

A proof should combine exponential tilting with a local Edgeworth theorem for the resulting infinitesimal triangular array. The midpoint expansion (4.11) implies that no order-one exponential amplitude is hidden in the finite- $q$  cumulant transform.

## 9 Numerical experiments and reproducibility

This part's verification material is kept in the volume directory under `assets/q-fabius-parameter-deformations/fabius_q_frontiers_report/`, which holds two documented programs; paths written `data/...` and `figures/...` below are relative to it.

- `assets/q-fabius-parameter-deformations/fabius_q_frontiers_report/experiments.py` performs direct infinite-product Fourier inversion, midpoint cumulant-transform comparisons, rate-function and Lambert computations, and a Monte-Carlo stop-loss diagnostic.
- `assets/q-fabius-parameter-deformations/fabius_q_frontiers_report/symbolic_checks.py` uses exact SymPy rational arithmetic to verify cumulants, Hermite coefficients, the rate series, the kernel cocycle, and the first Euler–Maclaurin operators.

The main numerical outcomes are summarized in [table 28](#).

Table 28: Observed convergence orders and consistency checks.

| Quantity                                          | Numerical result           |
|---------------------------------------------------|----------------------------|
| Uncorrected scaled-CGF order                      | 1.99999                    |
| After $-\delta^2\theta^2\Lambda''/24$             | 4.00134                    |
| Gaussian density error                            | 1.01783                    |
| Edgeworth weighted order 1                        | 2.05126                    |
| Edgeworth weighted order 2                        | 3.07181                    |
| Edgeworth weighted order 3                        | 4.09974                    |
| Endpoint constant $C_\Lambda$                     | $-0.968920423963031$       |
| Minimum sampled stop-loss gap                     | 0 (to displayed precision) |
| Maximum sampled stop-loss gap, $r = .35, s = .72$ | 0.0757670                  |

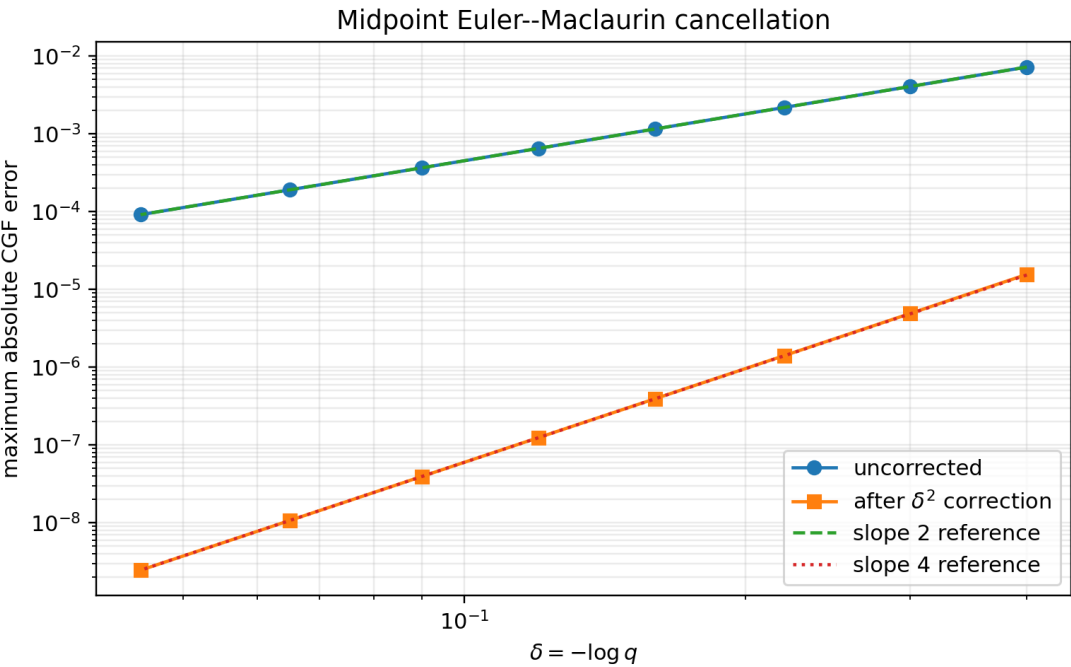


Figure 47: The raw midpoint-sum error is quadratic in  $\delta$ ; subtracting the proved first correction exposes the quartic remainder.

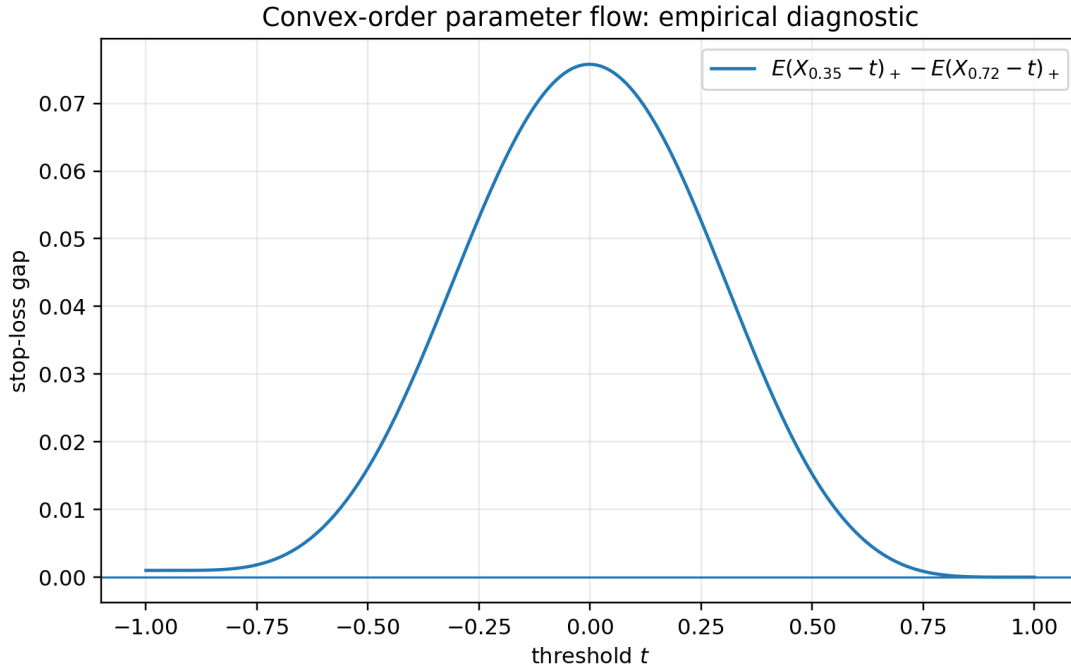


Figure 48: Empirical stop-loss gap for two parameter values. The plot is only a diagnostic; nonnegativity is proved exactly by the martingale coupling in [theorem 3.3](#).

## 9.1 Numerical methodology

The standardized density is computed from

$$f_q(x) = \frac{1}{\pi} \int_0^\infty \chi_q(t) \cos(tx) dt.$$

The product is truncated only after the first omitted argument is below  $10^{-10}$ , and the quadratic logarithmic tail is inserted analytically. A cosine quadrature on  $[0, 30]$  then supplies the density. The values of  $|\chi_q(30)|$  stored in `data/edgeworth_errors.csv` are far below the reported approximation errors.

The finite- $q$  scaled cumulant transform is evaluated in its exact midpoint form (4.3). The endpoint constant is evaluated using the two absolutely convergent integrals in (6.2); the quadrature is reported to the fifteen decimals its double-precision arithmetic supports, and agrees there with the closed form (6.4) and with the thirty-eight digits of (6.3). The stop-loss experiment uses independent samples and a sorted-sample suffix-sum algorithm; it is not used in any proof.

## 10 Extensions, conjectures, and research program

### 10.1 General bounded seeds

The two structural results are not special to the uniform law.

**Proposition 10.1** (Seed-universal transform limit). *Let  $V$  be bounded and nondegenerate and let  $K_V(\theta) = \log \mathbb{E} e^{\theta V}$ . For  $X_q^{(V)}$  from (3.9), locally uniformly in  $\theta$  and with an error  $O_K(\delta^2)$ ,*

$$\delta \log \mathbb{E} \exp \left( \frac{\theta X_q^{(V)}}{\delta} \right) \longrightarrow \Lambda_V(\theta) := \int_0^\infty K_V(\theta e^{-t}) dt = \int_0^\theta \frac{K_V(u)}{u} du. \quad (10.1)$$

The midpoint Euler–Maclaurin expansion (4.10) holds with  $g$  replaced by  $K_V$  and with the same kernel remainder (4.16). The function  $\Lambda_V$  is real analytic and strictly convex on  $\mathbb{R}$  with  $\Lambda'_V(\mathbb{R}) = (\text{ess inf } V, \text{ess sup } V)$ , and the family  $X_{e^{-\delta}}^{(V)}$  satisfies a full large-deviation principle at speed  $1/\delta$  with good rate function  $\Lambda_V^*$ , finite on that open interval and infinite outside its closure. Independently, the convex-order flow (3.12) holds for every integrable seed.

*Proof.* The coefficient identity and martingale proof in appendix 3 do not use the distribution of the seed. For the transform, since  $V$  is bounded the series  $X_q^{(V)}$  converges absolutely, independence and dominated convergence give  $\mathbb{E}e^{\theta X_q^{(V)}/\delta} = \prod_j \mathbb{E}e^{\theta a_j(q)V_j/\delta}$ , and the exact half-step calculation of appendix 4 turns the logarithm into

$$\delta \sum_{j \geq 0} K_V(\theta s_\delta e^{-(j+1/2)\delta}),$$

the series converging because  $K_V(u) = O(u)$  at 0. The function  $K_V$  is real analytic on  $\mathbb{R}$ , the moment generating function of a bounded variable being entire and positive on the real axis, with  $K_V(u) = \sum_{m \geq 1} \kappa_m(V)u^m/m!$  near 0; hence  $D^n K_V(u) = \sum_m m^n \kappa_m(V)u^m/m! = O(u)$  at 0, so for  $f(t) = K_V(we^{-t})$  every derivative  $f^{(n)}(t) = (-1)^n D^n K_V(we^{-t})$  is  $O(e^{-t})$ , and the quantities  $\int_0^{|w|} |D^n K_V(u)| u^{-1} du$  are finite and locally bounded in  $w$ . These are exactly the properties of  $g$  used in the proofs of theorems 4.1 and 4.2 (there  $D^n g = O(u^2)$ , and  $O(u)$  suffices), so the Peano bound gives the  $O_K(\delta^2)$  convergence in (10.1), and the kernel identity (4.15) gives the all-orders expansion with the remainder (4.16).

For the regularity of  $\Lambda_V$ :  $K_V(u)/u$  is real analytic on  $\mathbb{R}$  (it equals  $\mathbb{E}V$  at 0), so  $\Lambda_V$  is real analytic, and  $\Lambda'_V(\theta) = K_V(\theta)/\theta$  is the difference quotient of the strictly convex function  $K_V$  from the base point 0, hence strictly increasing; thus  $\Lambda''_V > 0$ . As  $\theta \rightarrow \pm\infty$ ,  $K_V(\theta)/\theta \rightarrow \text{ess sup } V$  and  $\text{ess inf } V$  respectively, which gives the range of  $\Lambda'_V$ . For the large-deviation principle, the Gärtner–Ellis hypotheses hold as in the proof of theorem 5.1: the scaled transforms converge for every  $\theta$  to  $\Lambda_V$ , which is finite and differentiable on all of  $\mathbb{R}$  and therefore essentially smooth with vacuous steepness, and the family is exponentially tight because  $|X_q^{(V)}| \leq \|V\|_\infty$ . The rate  $\Lambda_V^*$  is finite exactly on the range of  $\Lambda'_V$  and infinite outside the closed interval  $[\text{ess inf } V, \text{ess sup } V]$ , which carries the whole family.  $\square$

This suggests a broad “geometric averaging universality class” indexed by the seed cumulant transform. The uniform seed is distinguished by the Bernoulli and infinite-sinc structures.

## 10.2 Sharp and beyond-all-orders questions

**Conjecture 10.2** (Interior all-orders saddlepoint theorem). *The expansion (8.3) is valid and Borel summable for each fixed compact subinterval of  $(-1, 1)$ . Its Borel singularities are governed by the complex zeros of  $\sinh z/z$ , hence by the same arithmetic zero lattice that controls the sinc-product Fourier theory.*

**Conjecture 10.3** (Endpoint transseries). *Let  $\eta = 1 - x$  and use the exact Lambert variable  $\theta_0 = -W_{-1}(-\eta/2)/\eta$ . The rate and saddle admit full transseries in inverse powers of  $\theta_0$  and exponentials  $e^{-2k\theta_0}$ , with no additional logarithmic scale after  $\theta_0$  is adopted. Matching this transseries to the finite- $q$  saddlepoint expansion should produce a uniform double-scaling theory when  $\eta$  and  $\delta$  tend to zero together.*

**Open problem 10.4** (Critical boundary layer). *Determine the correct uniform asymptotics when*

$$1 - x \asymp \delta \log(1/\delta).$$

*This is the scale on which the Lambert saddle  $\theta_x \asymp 1/\delta$  meets the largest individual Fourier/MGF argument. A change of local model, and possibly a Stokes transition, should occur.*

### 10.3 Parameter-flow geometry

The exact shift subordinator invites several extensions.

**Open problem 10.5** (Metric contraction along  $q$ ). *For which probability metrics  $d$  does the map  $q \mapsto \text{Law}(X_q)$  admit a closed differential law or sharp contraction coefficient? Convex order settles all stop-loss functionals, while the corpus settles several metrics along finite truncations; the parameter analogue for Wasserstein, Zolotarev, entropy, and Fisher metrics remains open.*

**Conjecture 10.6** (Strict stop-loss interior). *For the uniform seed and  $0 < r < s < 1$ ,*

$$\mathbb{E}(X_s - t)_+ < \mathbb{E}(X_r - t)_+$$

*for every  $t \in (-1, 1)$ . More generally, the difference should be strictly unimodal in  $t$  and inherit a total-positivity property from the random-shift kernel.*

**Open problem 10.7** (Generator on laws). *Translate the coefficient jump generator*

$$\mathcal{L}_q F(n) = \sum_{k \geq 1} q^{k-1} [F(n+k) - F(n)]$$

*into a closed evolution equation for the density or its transform. The transform equation is explicit, but a useful physical-space equation may connect the parameter flow to the Stein–Koopman and diffusion layers already present in the corpus.*

### 10.4 Rate-series arithmetic

The rational coefficients in (5.6) arise from compositional inversion of a Bernoulli series. Their signs, denominators, and  $p$ -adic valuations appear to be a new arithmetic sequence.

**Open problem 10.8** (Lagrange–Bernoulli formula). *Find a closed partition formula for the coefficient  $c_n$  in  $I(x) = \sum_{n \geq 1} c_n x^{2^n}$ , and determine the exact denominator and 2-adic valuation. A useful formula should explain the sign pattern of [remark 5.2](#) — eleven positive coefficients, then two negative ones — and identify the singularities of  $I$  on its circle of convergence, whose radius  $R_I \leq 1$  is not yet known exactly.*

### 10.5 Formalization roadmap

The parameter-flow theorem is especially suitable for formalization:

1. define the geometric coefficient sequence in  $\ell^1$  and prove the rational generating-function identity;
2. define  $p_{r,s}$  and prove positivity, normalization, and cocycle;
3. construct the random shift on an iid sequence and justify exchange of the two absolutely convergent sums;
4. apply conditional Jensen to obtain convex order;
5. formalize the midpoint identity as an algebraic rewrite before introducing analytic Euler–Maclaurin estimates.

The LDP and all-order Edgeworth proofs require substantially more analytic infrastructure. A practical formal route would first isolate the transform limit, strict convexity, and the first  $O(\delta^2)$  correction; the full uniform derivative theorem can then be modularized around [lemma 5.3](#) and [lemma 5.4](#).



## 11 Conclusion

The geometric-uniform family is more than a deformation that happens to pass through the Fabius–Rvachev point. Its coefficient profile carries an exact probabilistic flow: increasing  $q$  is random tail shifting, and random tail shifting is martingale averaging. On the analytic side, the same parameter turns the logarithm of the  $q$ -product into a midpoint integral. That single observation explains the even Euler–Maclaurin hierarchy, produces a new large-deviation transform, and organizes the continuous Edgeworth expansion.

The resulting picture has three compatible scales. At the center, Bernoulli cumulants become Hermite corrections. At fixed deviations, their integrated limit becomes a Legendre rate. At the support edge, the saddle is inverted by  $W_{-1}$ . These layers are distinct from, but structurally aligned with, the finite-prefix transport, fixed-base endpoint, and discrete  $\text{Fup}_n$  results already in the repository.

## A A compact proof ledger

For ease of audit, the logical dependencies of the main results are:

| Result                                 | Dependencies                                                                                                                                                                                        |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">theorem 3.1</a>            | Rational generating functions only                                                                                                                                                                  |
| <a href="#">theorem 3.3</a>            | <a href="#">theorem 3.1</a> , iid tail invariance, conditional Jensen                                                                                                                               |
| <a href="#">theorem 4.2</a>            | Exact half-step factorization, midpoint Euler–Maclaurin, exponential derivative decay                                                                                                               |
| <a href="#">theorem 5.1</a>            | <a href="#">theorem 4.1</a> , compact support, Gärtner–Ellis; and <a href="#">theorem 6.2</a> for $I(\pm 1) = \infty$                                                                               |
| <a href="#">theorem 6.2</a>            | Elementary decomposition of $\log \sinh c$ , $W_{-1}$ inversion                                                                                                                                     |
| <a href="#">proposition 6.1 (7.11)</a> | (6.1), the substitution $x = 2u$ , and <a href="#">lemma 8.1</a> for the Mellin finite part<br><a href="#">theorem 5.1</a> , which rests on <a href="#">lemma 5.3</a> and <a href="#">lemma 5.4</a> |

## B Explicit rate coefficients through order twenty-eight

The symbolic companion gives, through order sixteen,

$$I(x) = 3x^2 + \frac{9}{5}x^4 + \frac{276}{175}x^6 + \frac{1188}{875}x^8 + \frac{354672}{336875}x^{10} \\ + \frac{1507248}{1990625}x^{12} + \frac{496585728}{766390625}x^{14} + \frac{53214713856}{65143203125}x^{16} + O(x^{18}),$$

and the exact reversion carried out for this volume continues the series with

$$\begin{aligned} c_9 &= \frac{514221151037184}{476522530859375}, & c_{10} &= \frac{6034173698304}{6188604296875}, \\ c_{11} &= \frac{9630466930063899648}{54854891139876953125}, & c_{12} &= -\frac{3032147123502547968}{3562005918173828125}, \\ c_{13} &= -\frac{58363635922427846541312}{89139198102300048828125}, & c_{14} &= \frac{802863515660126853906432}{409394959610111083984375}, \end{aligned}$$

so that  $c_{12}$  and  $c_{13}$  are the first negative coefficients ([remark 5.2](#)). The reversion inverts  $x = \Lambda'(\theta)$  from the Bernoulli series (4.9) truncated at  $\theta^{28}$ , which determines  $c_1, \dots, c_{14}$  exactly. Further, and

$$\theta_x = 6x + \frac{36}{5}x^3 + \frac{1656}{175}x^5 + \frac{9504}{875}x^7 + \frac{709344}{67375}x^9 \\ + \frac{18086976}{1990625}x^{11} + \frac{993171456}{109484375}x^{13} + \frac{851435421696}{65143203125}x^{15} + O(x^{17}).$$

## C Files in the reproducibility package

This part’s verification material is kept in the volume directory under `assets/q-fabius-parameter-deformations/fabius_q_frontiers_report/`; the absorbed report’s own source and PDF were deleted when it was merged, and git history is their archive. The directory holds:

- `assets/q-fabius-parameter-deformations/fabius_q_frontiers_report/experiments.py`, `assets/q-fabius-parameter-deformations/fabius_q_frontiers_report/symbolic_checks.py`, and `assets/q-fabius-parameter-deformations/fabius_q_frontiers_report/requirements.txt`;
- CSV tables under `assets/q-fabius-parameter-deformations/fabius_q_frontiers_report/data/` for the cumulant-transform, rate, Edgeworth, and stop-loss experiments;
- `assets/q-fabius-parameter-deformations/fabius_q_frontiers_report/data/numerical_summary.txt` and `assets/q-fabius-parameter-deformations/fabius_q_frontiers_report/data/symbolic_checks.txt`;
- the four figures, in PDF and PNG, under `assets/q-fabius-parameter-deformations/fabius_q_frontiers_report/figures/`;
- `assets/q-fabius-parameter-deformations/fabius_q_frontiers_report/README.md`, the absorbed package’s own run notes.

The volume itself is rebuilt from its own directory by running `pdflatex` on `geometric_q_fabius_frontiers.tex` three times, so that the table of contents and every cross-reference settle.

## Part XI

# Continuous-Parameter Edgeworth Theory and Quadratic $q$ -Gevrey Regularity

## Abstract

The Fabius function, Rvachev’s compactly supported function up, and their geometric  $q$ -deformations sit at an unusually dense intersection of probability, infinite sinc products, automatic sequences,  $q$ -series, exact dyadic arithmetic, inverse-function asymptotics, and ultradifferentiable analysis. This report audits the current LaTeX corpus under `Analysis/FabiusFunction/docs` in the PROVEIt repository and develops results selected specifically to avoid duplicating the canonical exposition and its frontier drafts.

The first result closes an explicitly recorded gap in the repository: for the standardized geometric-uniform law with  $r = e^{-h} \uparrow 1$ , we prove an all-orders Edgeworth expansion in every fixed Schwartz seminorm. Its coefficients are finite Bernoulli–Bell–Hermite polynomials generated by exact standardized cumulants. We derive the first three corrections, uniform CDF expansions, and an all-orders central Cornish–Fisher theory for the inverse distribution. The second group of results identifies two new probability scales. We prove Gaussian moderate deviations and a compact-scale large-deviation principle with speed  $h^{-1}$ . The limiting cumulant generating function is

$$\Lambda(s) = \int_0^1 \frac{\log(\sinh(\sqrt{6} su)/(\sqrt{6} su))}{u} du.$$

Its Legendre transform admits an exact saddle parametrization. At the limiting support edge the saddle inversion is governed by  $W_{-1}$ , and the rate contains the renormalized constant

$$\gamma_1 + \frac{\gamma^2}{2} - \frac{\pi^2}{12},$$

where  $\gamma_1$  is the first Stieltjes constant. This furnishes a direct bridge between the Gaussian  $q \uparrow 1$  regime and the Lambert– $W$  endpoint machinery emphasized elsewhere in the corpus.

Finally, using the repository’s exact Thue–Morse derivative norms, we identify the sharp global Denjoy–Carleman class of up and of the separated-base atomic functions  $h_a$ ,  $a \geq 2$ . These functions are not Gevrey of any finite order; their optimal derivative scale is quadratic-exponential,  $M_n \asymp a^{n^2/2}$ , and the corresponding class is nonquasianalytic. Reproducible Python computations verify the predicted successive error orders and the Lambert– $W$  edge law. The final sections formulate a matched central/moderate/large/endpoint asymptotic program, local symbolic Carleman spectra, and several arithmetic and transseries conjectures.

---

**Novelty convention.** “New” in this report means absent from the inspected PROVEIt snapshot and from the source papers used as its baseline. No claim of global literature priority is made without a separate exhaustive bibliographic search. All theorem statements below are proved in the report; conjectures and heuristic extensions are labeled explicitly.

---

# 1 Executive summary and principal contributions

## 1.1 The geometric-uniform family

Let  $(U_n)_{n \geq 0}$  be independent  $\text{Unif}[0, 1]$  random variables and let

$$X_r = (1 - r) \sum_{n=0}^{\infty} r^n U_n, \quad 0 < r < 1. \quad (1.1)$$

The weights sum to one, so  $X_r \in [0, 1]$ . At  $r = 1/2$ , its distribution function is the Fabius function  $F$ , and its density is the affine rescaling [36, 33, 69]

$$F'(x) = 2 \text{up}(2x - 1), \quad 0 < x < 1. \quad (1.2)$$

Thus (1.1) is simultaneously a probability deformation of  $F$ , a  $q$ -analog with  $q = r$ , and an infinite-convolution deformation of Rvachev's up.

Put

$$r = e^{-h}, \quad \sigma_r^2 = \frac{1 - r}{12(1 + r)}, \quad Z_h = \frac{X_r - 1/2}{\sigma_r}. \quad (1.3)$$

The report establishes the following package.

---

### Proved frontier results.

- (I) **All-orders continuous-parameter Edgeworth theorem.** The density of  $Z_h$  has an expansion to arbitrary order in  $h$ , uniformly with polynomial weights and after any fixed number of derivatives; see [Theorem 5.1](#).
  - (II) **Bernoulli–Bell coefficient calculus.** Every standardized cumulant has an explicit all-orders  $h$ -expansion whose coefficients are complete Bell polynomials in Bernoulli numbers; see [Theorem 4.2](#).
  - (III) **Central inverse theory.** The CDF and quantile admit all-orders uniform central expansions. The first two Cornish–Fisher terms are explicit; see [Theorem 6.2](#).
  - (IV) **Moderate deviations.** Every scale  $b_h \rightarrow \infty$  with  $hb_h^2 \rightarrow 0$  has the Gaussian rate  $x^2/2$ ; see [Theorem 7.1](#).
  - (V) **Compact-scale large deviations.**  $Y_h = \sqrt{h} Z_h$  satisfies an LDP with speed  $h^{-1}$ , compact limiting support  $[-\sqrt{6}, \sqrt{6}]$ , and an explicit integral cumulant generator; see [Theorem 7.2](#).
  - (VI) **Lambert– $W$  support-edge law.** The large-deviation rate has an exact saddle parametrization and a  $W_{-1}$  edge expansion containing a Stieltjes constant; see [Theorems 7.3 and 8.2](#).
  - (VII) **Sharp quadratic  $q$ -Gevrey classification.** The exact Thue–Morse derivative norms place up and  $h_a$ ,  $a \geq 2$ , in optimal nonquasianalytic Denjoy–Carleman classes and in no finite Gevrey class; see [Theorems 9.1 and 9.3](#).
- 

The central Edgeworth theorem resolves a program that the repository itself marks as open: its discrete finite-level analog was completed, while the continuous parameter limit  $a \downarrow 1$ , equivalently  $r \uparrow 1$ , remained open. A companion frontier draft records the first fixed-frequency quartic correction and conjectures precisely the weighted, differentiated, all-orders statement proved here [68, 66].

## 1.2 Why the new scales are natural

The standard CLT scale keeps  $Z_h = O(1)$ . The standardized support, however, is

$$|Z_h| \leq A_h, \quad A_h = \frac{1}{2\sigma_r} = \sqrt{3 \frac{1+r}{1-r}} = \sqrt{3 \coth(h/2)} \sim \sqrt{\frac{6}{h}}. \quad (1.4)$$

Hence  $Y_h = \sqrt{h}Z_h$  has a nondegenerate compact limiting support. Three nested regimes appear:

$$Z_h = O(1) \longrightarrow Z_h = b_h x, \ 1 \ll b_h \ll h^{-1/2} \longrightarrow Y_h = \sqrt{h}Z_h = O(1).$$

They are governed respectively by Edgeworth expansions, Gaussian moderate deviations, and a nonquadratic compact-support rate function. Letting the compact coordinate approach  $\sqrt{6}$  then creates the Lambert– $W_{-1}$  phase. This four-layer structure is the conceptual backbone of the report.

## 2 Repository audit and nonduplication protocol

### 2.1 Corpus map

The inspected repository snapshot was the public main branch on 30 August 2026. The live documentation tree contains five canonical LaTeX works, a large thematically reorganized draft corpus, imported source papers, and a frozen historical archive. The audit used the following hierarchy.

| Corpus stratum                          | Role in the nonduplication audit                                                                                                                                                                                                                               |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Fabius_Function_and_Rvachev_Up          | Primary synthesis: construction, probability, Fourier products, moments, exact dyadic values, $q$ -specialization, inverse transport, Lambert– $W$ endpoint asymptotics, and many auxiliary transforms.                                                        |
| semi-formalized-research-frontiers      | Canonical frontier volume: probabilistic engines, repeated integration, exact dyadic web, Thue–Morse prefix sums, endpoint and inverse asymptotics.                                                                                                            |
| Non_Elementarity_of_the_Fabius_Function | Nowhere-local-elementarity and formalization-backed analytic-locus arguments.                                                                                                                                                                                  |
| fabius_lean_walkthrough                 | Lean architecture, definitions, theorem dependencies, and proof-engineering constraints.                                                                                                                                                                       |
| FabiusFunction_Mathematical_Glossary    | Notation and cross-document terminology.                                                                                                                                                                                                                       |
| Draft manifest and raw candidate drafts | Fourier decay, $q$ -Pochhammer and $q$ -binomial inverses, Legendre/Jacobi/Padé systems, total positivity, interpolation on geometric combs, atomic-function general bases, Gaussian boundary limits, reciprocal transforms, and many archived research waves. |
| Imported papers                         | Arias de Reyna on the compactly supported function and Fabius arithmetic; Allouche–Shallit on Thue–Morse; lacunary-product papers; a Thue–Morse Dirichlet-series paper; and related source material.                                                           |
| Historical archive                      | Frozen first-introduction versions grouped by subject. Byte-identical copies already represented in the live tree are deliberately pruned by the repository’s own archive policy.                                                                              |

The imported baseline includes the classical compact-support and arithmetic papers, the standard Thue–Morse survey, lacunary-product work, and a recent Thue–Morse Dirichlet-series paper [33, 31, 39, 11, 10, 42].

The archive and draft manifest are important methodologically: they prevent a recent theorem hidden in a prior wave from being misidentified as new. Historical copies were treated as lineage evidence, not as independent mathematical content.

## 2.2 Results deliberately excluded as already present

The following substantial themes were excluded from the novelty claims because the corpus already develops them:

- construction, uniqueness, refinement equations, support, symmetry, monotonicity, flatness, and nonanalyticity of  $F$  and up;
- probability representations by weighted uniform sums, finite spline approximants, weak convergence, and general absolutely summable weights;
- the geometric  $q$ -family at fixed  $0 < q < 1$ , including negative-parameter dualities, support geometry, plateaux, reciprocal bases, and the first Gaussian correction;
- infinite sinc products, log-Gaussian Fourier envelopes, Poisson summation, moments, Legendre expansions, and exact derivative hierarchies;
- exact dyadic rational formulas, 2-adic valuations,  $q$ -binomial identities, and geometric interpolation;
- fixed-parameter endpoint asymptotics with periodic fluctuations and Lambert– $W$  inversion;
- discrete finite-level Edgeworth expansions for the spline approximants;
- exact derivative norms and equimeasurability for separated-base atomic functions  $h_a$ ,  $a \geq 2$ .

The new work starts where those results stop. In particular, the corpus states that the continuous-parameter Edgeworth program remains open, and another draft conjectures a complete Bernoulli–Bell–Hermite expansion with polynomial weights and differentiated uniformity. Searches of the canonical and frontier sources found no compact-scale large-deviation principle, no Gärtner–Ellis rate integral, no Cornish–Fisher expansion, and no completed Denjoy–Carleman classification. The exact derivative norms used in [Appendix 9](#) are therefore inputs from the repository; the functional-class consequences are new deductions.

## 2.3 Audit limitations and reproducibility

The tree includes reorganized drafts and frozen versions whose text is intentionally duplicated. The audit therefore used the canonical documents as the semantic baseline, the manifest and archive inventory to identify equivalence classes, and raw-source inspection whenever a proposed result was close to an existing topic. This is a content audit, not a claim that every historical byte was independently reread after duplication was established.

All computations in [Appendix 10](#) are reproduced by

```
assets/q-fabius-parameter-deformations/Continuous_Parameter_Edgeworth_and_q
    _Gevrey_Frontier/numerics/fabius_q_edgeworth.py.
```

That directory, under the volume, holds the script together with the generated CSV and LaTeX tables and the vector/PNG figures.

## 3 Preliminaries: Fabius, Rvachev, and the geometric $q$ -family

### 3.1 Random-series identities

Write  $V_n = U_n - 1/2$ , so that  $V_n \sim \text{Unif}[-1/2, 1/2]$ . Then

$$X_r - \frac{1}{2} = (1-r) \sum_{n \geq 0} r^n V_n. \quad (3.1)$$

The variance follows immediately:

$$\sigma_r^2 = \frac{(1-r)^2}{12} \sum_{n \geq 0} r^{2n} = \frac{1-r}{12(1+r)}. \quad (3.2)$$

The moment generating function of  $V_n$  is  $\sinh(t/2)/(t/2)$ , while its characteristic function is  $\text{sinc}_{\text{rad}}(t/2) = \sin(t/2)/(t/2)$ . Hence the standardized characteristic function is

$$\Psi_h(t) = \mathbb{E} e^{itZ_h} = \prod_{n=0}^{\infty} \text{sinc}_{\text{rad}}(a_h r^n t), \quad a_h = \sqrt{3(1-r^2)}, \quad r = e^{-h}. \quad (3.3)$$

The normalization is exact because

$$\sum_{n \geq 0} (a_h r^n)^2 = 3. \quad (3.4)$$

### 3.2 Smoothness and Fourier conventions

We use

$$\widehat{f}_{\text{ang}}(-t) = \int_{\mathbb{R}} e^{itx} f(x) \, dx, \quad f(x) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{-itx} \widehat{f}_{\text{ang}}(-t) \, dt. \quad (3.5)$$

For every fixed  $h > 0$ , the product (3.3) decays faster than every negative power, and therefore  $Z_h$  has a compactly supported  $C^\infty$  density  $f_h$  [8, 33]. A quantitative version is proved in Lemma 5.3.

The probabilists' Hermite polynomials are

$$H_n(x) = (-1)^n e^{x^2/2} \frac{d^n}{dx^n} e^{-x^2/2}, \quad \varphi(x) = \frac{e^{-x^2/2}}{\sqrt{2\pi}}. \quad (3.6)$$

Thus  $\varphi^{(n)} = (-1)^n H_n \varphi$ , and Fourier inversion in the positive-exponent convention (3.5) gives

$$\frac{1}{2\pi} \int_{\mathbb{R}} e^{-itx} (it)^n e^{-t^2/2} \, dt = H_n(x) \varphi(x). \quad (3.7)$$

### 3.3 Cumulants of a centered uniform variable

With Bernoulli numbers defined by

$$\frac{z}{e^z - 1} = \sum_{n=0}^{\infty} B_n \frac{z^n}{n!}, \quad (3.8)$$

we have

$$\log \frac{\sinh(z/2)}{z/2} = \sum_{m=1}^{\infty} \frac{B_{2m}}{2m} \frac{z^{2m}}{(2m)!}, \quad |z| < 2\pi. \quad (3.9)$$

Accordingly, the  $2m$ -th cumulant of  $V_n$  is  $B_{2m}/(2m)$ , and all odd cumulants vanish.



## 4 Exact cumulants and the Bernoulli–Bell bridge

**Proposition 4.1** (Exact standardized cumulants). *For every  $m \geq 2$ , the standardized cumulant of order  $2m$  of  $Z_h$  is*

$$\lambda_{2m}(r) = \frac{B_{2m}}{2m} 12^m \frac{(1-r^2)^m}{1-r^{2m}}. \quad (4.1)$$

Equivalently, for  $r = e^{-h}$ ,

$$\lambda_{2m}(h) = \frac{B_{2m}}{2m} 12^m 2^{m-1} \frac{\sinh^m h}{\sinh(mh)}. \quad (4.2)$$

In particular,  $\lambda_{2m}(h) = \mathcal{O}(h^{m-1})$  as  $h \downarrow 0$ .

*Proof.* Cumulants add under independent summation and scale homogeneously. From (3.1) and (3.9),

$$\kappa_{2m}(X_r - 1/2) = \frac{B_{2m}}{2m} (1-r)^{2m} \sum_{n \geq 0} r^{2mn} = \frac{B_{2m}}{2m} \frac{(1-r)^{2m}}{1-r^{2m}}.$$

Dividing by  $\sigma_r^{2m}$ , with (3.2), proves (4.1). Since

$$1 - e^{-2h} = 2e^{-h} \sinh h, \quad 1 - e^{-2mh} = 2e^{-mh} \sinh(mh),$$

the exponential factors cancel and give (4.2). Finally  $\sinh^m h / \sinh(mh) \sim h^{m-1}/m$ .  $\square$

The hyperbolic representation makes a second layer of structure visible. Set

$$c_k = \frac{2^{2k-1} B_{2k}}{k(2k)!}, \quad d_{m,k} = c_k (m - m^{2k}). \quad (4.3)$$

The classical identity

$$\log \frac{\sinh z}{z} = \sum_{k=1}^{\infty} c_k z^{2k}, \quad |z| < \pi, \quad (4.4)$$

turns every cumulant into an exponential Bell series.

**Theorem 4.2** (Bernoulli–Bell expansion of every cumulant). *Let  $Y_\ell$  denote the complete exponential Bell polynomial, characterized by*

$$\exp \left( \sum_{k \geq 1} x_k \frac{z^k}{k!} \right) = \sum_{\ell \geq 0} Y_\ell(x_1, \dots, x_\ell) \frac{z^\ell}{\ell!}. \quad (4.5)$$

For  $m \geq 2$ , define

$$A_m = \frac{B_{2m}}{2m} 12^m \frac{2^{m-1}}{m}. \quad (4.6)$$

Then, for every  $h$  with  $0 < h < \pi/m$ ,

$$\lambda_{2m}(h) = A_m h^{m-1} \exp \left( \sum_{k \geq 1} d_{m,k} h^{2k} \right) \quad (4.7)$$

$$= A_m h^{m-1} \sum_{\ell=0}^{\infty} \frac{Y_\ell(1!d_{m,1}, 2!d_{m,2}, \dots, \ell!d_{m,\ell})}{\ell!} h^{2\ell}. \quad (4.8)$$

Thus the complete continuous-parameter cumulant calculus is generated by Bernoulli numbers followed by one Bell-polynomial exponentiation.

*Proof.* Rewrite (4.2) as

$$\lambda_{2m}(h) = A_m h^{m-1} \left( \frac{\sinh h}{h} \right)^m \frac{mh}{\sinh(mh)}.$$

Taking logarithms of the factor after  $A_m h^{m-1}$  and applying (4.4) gives

$$m \log \frac{\sinh h}{h} - \log \frac{\sinh(mh)}{mh} = \sum_{k \geq 1} c_k (m - m^{2k}) h^{2k},$$

which is (4.7). Apply (4.5) with  $z = h^2$  and  $x_k = k! d_{m,k}$ .

The stated range is exact and cannot be replaced by a single interval independent of  $m$ . By Lemma 0.4.1 the series (4.4) has radius exactly  $\pi$ ; it is applied here at the two arguments  $h$  and  $mh$ , and the second is the binding one, so (4.7) and (4.8) converge precisely for  $0 < h < \pi/m$ . The threshold degenerates as  $m \rightarrow \infty$ , which is harmless downstream: Theorem 5.1 and Proposition 5.2 use (4.8) only for  $2 \leq m \leq R+1$  with  $R$  fixed in advance, hence only for  $0 < h < \pi/(R+1)$ .  $\square$

The first cumulants, useful later, are

$$\lambda_4(h) = -\frac{6}{5} \tanh h = -\frac{6}{5}h + \frac{2}{5}h^3 - \frac{4}{25}h^5 + \mathcal{O}(h^7), \quad (4.9)$$

$$\lambda_6(h) = \frac{192}{7} \frac{\sinh^3 h}{\sinh 3h} = \frac{64}{7}h^2 - \frac{64}{7}h^4 + \mathcal{O}(h^6), \quad (4.10)$$

$$\lambda_8(h) = -\frac{3456}{5} \frac{\sinh^4 h}{\sinh 4h} = -\frac{864}{5}h^3 + \frac{1728}{5}h^5 + \mathcal{O}(h^7). \quad (4.11)$$

## 5 All-orders Edgeworth expansion in Schwartz seminorms

### 5.1 The exact-cumulant Edgeworth polynomial

For a multi-index  $\mathbf{k} = (k_2, \dots, k_{R+1}) \in \mathbb{N}_0^R$ , define its Edgeworth weight and Hermite degree by

$$w(\mathbf{k}) = \sum_{m=2}^{R+1} (m-1)k_m, \quad \nu(\mathbf{k}) = \sum_{m=2}^{R+1} mk_m. \quad (5.1)$$

Define

$$\mathcal{E}_{R,h}(x) = \sum_{\substack{\mathbf{k} \in \mathbb{N}_0^R \\ w(\mathbf{k}) \leq R}} H_{2\nu(\mathbf{k})}(x) \prod_{m=2}^{R+1} \frac{1}{k_m!} \left( \frac{\lambda_{2m}(h)}{(2m)!} \right)^{k_m}. \quad (5.2)$$

The zero multi-index contributes 1. The polynomial uses the *exact* cumulants, not their truncated power series. This makes the theorem compact and automatically incorporates many higher-order parameter terms.

For integers  $j, K \geq 0$ , write

$$\|g\|_{j,K} = \sup_{x \in \mathbb{R}} (1 + |x|)^K \left| g^{(j)}(x) \right|. \quad (5.3)$$

**Theorem 5.1** (All-orders continuous-parameter Edgeworth expansion). *Fix integers  $R, j, K \geq 0$ . There exist  $h_0 > 0$  and  $C_{R,j,K} < \infty$  such that, for  $0 < h \leq h_0$ ,*

$$\|f_h - \varphi \mathcal{E}_{R,h}\|_{j,K} \leq C_{R,j,K} h^{R+1}. \quad (5.4)$$

Consequently  $f_h$  has a complete asymptotic expansion in the Schwartz topology, and every coefficient is an explicit Bernoulli–Bell–Hermite polynomial.

Relative to the classical Edgeworth framework [53, 94], this theorem strengthens the usual local statement in three ways: it is uniform on all of  $\mathbb{R}$ , survives any fixed number of spatial derivatives, and carries arbitrary fixed polynomial weights.

The passage from the exact-cumulant polynomial  $\mathcal{E}_{R,h}$  to a genuine power series in  $h$  — which is what the last sentence of the theorem asserts, and what the quantile expansion below consumes — requires re-expanding each  $\lambda_{2m}(h)$  and re-truncating. That change of remainder is estimated once, here.

**Proposition 5.2** (Re-expansion in powers of  $h$ ). *Let  $P_N$  be the polynomial defined by (B.1). For every  $R \geq 0$  the difference  $\mathcal{E}_{R,h} - \sum_{N=0}^R h^N P_N$  is a polynomial in  $x$  of degree at most  $D_R = 4R$ , all of whose coefficients are  $\mathcal{O}_R(h^{R+1})$  as  $h \downarrow 0$ . Consequently, for all  $j, K \geq 0$ ,*

$$\left\| f_h - \varphi \sum_{N=0}^R h^N P_N \right\|_{j,K} \leq C_{R,j,K} h^{R+1}, \quad (5.5)$$

so  $f_h$  does have a complete asymptotic expansion in powers of  $h$  in every Schwartz seminorm, with coefficients  $\varphi P_N$ .

*Proof.* The term of (5.2) indexed by  $\mathbf{k}$  is

$$H_{2\nu(\mathbf{k})}(x) \prod_{m=2}^{R+1} \frac{1}{k_m!} \left( \frac{\lambda_{2m}(h)}{(2m)!} \right)^{k_m}.$$

Its  $x$ -independent factor is real analytic in  $h$  on  $0 < h < \pi/(R+1)$  by Theorem 4.2, and vanishes to order  $w(\mathbf{k})$  at  $h = 0$  because  $\lambda_{2m}(h) = A_m h^{m-1}(1 + \mathcal{O}(h^2))$ . Only the finitely many  $\mathbf{k}$  with  $w(\mathbf{k}) \leq R$  occur, so Taylor's theorem with remainder at order  $R$ , applied to each such coefficient function of  $h$ , replaces every coefficient by a polynomial in  $h$  of degree  $R$  up to  $\mathcal{O}_R(h^{R+1})$ . The Hermite degrees are bounded by  $2\nu(\mathbf{k}) = 2(w(\mathbf{k}) + \sum_m k_m) \leq 4w(\mathbf{k}) \leq 4R$ , because  $m-1 \geq 1$  gives  $\sum_m k_m \leq w(\mathbf{k})$ ; this is  $D_R$ . That the resulting coefficient of  $h^N$  equals  $P_N$  for every  $N \leq R$ , independently of  $R$ , is (B.1): multi-indices with  $w(\mathbf{k}) \geq R+1$  cannot contribute to  $h^N$  with  $N \leq R$ . Finally  $\|\varphi x^d\|_{j,K} < \infty$  for every  $d \leq 4R$ , so multiplying the coefficient bound by  $\varphi$  and adding (5.4) gives (5.5).  $\square$

## 5.2 Global sinc-product bounds

The proof starts with a uniform frequency envelope.

**Lemma 5.3** (Global sinc envelope). *The constant  $c = \frac{1}{6}$  is admissible throughout. First,*

$$|\text{sinc}_{\text{rad}} u| \leq \exp(-c \min(u^2, 1)), \quad u \in \mathbb{R}. \quad (5.6)$$

Let

$$S_h(t) = \sum_{n \geq 0} \min((a_h r^n t)^2, 1). \quad (5.7)$$

Then

$$|\Psi_h(t)| \leq e^{-c S_h(t)}. \quad (5.8)$$

Moreover, with the same  $c$  and with  $A_h$  the support constant of (1.4), for every integer  $q \geq 0$ ,

$$|\partial_t^q \Psi_h(t)| \leq (e^c A_h)^q e^{-c S_h(t)}, \quad A_h = \sum_{n \geq 0} a_h r^n = \sqrt{3 \coth(h/2)}, \quad A_h \leq \sqrt{\frac{13}{2h}} \quad (0 < h \leq 1). \quad (5.9)$$

Since  $A_h \leq \sqrt{13/(2h)}$ , this implies the weaker form  $|\partial_t^q \Psi_h| \leq C_q h^{-q/2} (1 + |t|)^q e^{-cS_h}$  in which the estimate is quoted elsewhere; the sharper form carries no factor  $(1 + |t|)^q$ . Finally, for any fixed  $0 < \delta < 1$  and with  $c_\delta = \frac{1}{2} \min(\delta^2, e^{-2})$ ,

$$S_h(t) = 3t^2, \quad |t| \leq a_h^{-1}, \quad (5.10)$$

$$S_h(t) \geq c_\delta h^{-1}, \quad |t| \geq \delta a_h^{-1}, \quad (5.11)$$

$$S_h(t) \geq \frac{\log(a_h |t|)}{h} + \frac{e^{-2}}{2h}, \quad a_h |t| \geq 1, \quad 0 < h \leq 1. \quad (5.12)$$

*Proof. The scalar envelope.* From the Euler product  $\text{sinc}_{\text{rad}} u = \prod_{k \geq 1} (1 - u^2/(k\pi)^2)$  and  $\log(1 - x) \leq -x$ ,

$$\log |\text{sinc}_{\text{rad}} u| \leq - \sum_{k \geq 1} \frac{u^2}{k^2 \pi^2} = -\frac{u^2}{6}, \quad |u| < \pi,$$

which gives (5.6) with  $c = \frac{1}{6}$  on  $|u| \leq 1$ . For  $|u| \geq 1$  the function  $\text{sinc}_{\text{rad}}$  decreases on  $[1, \pi]$  and satisfies  $|\text{sinc}_{\text{rad}} u| \leq |u|^{-1} \leq \pi^{-1}$  beyond, so

$$\sup_{|u| \geq 1} |\text{sinc}_{\text{rad}} u| = \sin 1 = 0.8414709 \dots < 0.8464817 \dots = e^{-1/6},$$

which is (5.6) on  $|u| \geq 1$ . Multiplying the factors of (3.3) proves (5.8).

*Derivatives.* Write  $\alpha_n = a_h r^n$  and  $\Psi_{h,N}(t) = \prod_{n < N} \text{sinc}_{\text{rad}}(\alpha_n t)$ . By [proposition 0.3.1](#) the partial products converge to  $\Psi_h$  locally uniformly on  $\mathbb{C}$ , and every  $\Psi_{h,N}$  is entire; by Weierstrass' theorem  $\partial_t^q \Psi_{h,N} \rightarrow \partial_t^q \Psi_h$  locally uniformly for every  $q$ . It therefore suffices to bound the partial products uniformly in  $N$ , and no interchange beyond this needs to be assumed. Since  $\text{sinc}_{\text{rad}} u = \int_0^1 \cos(us) \, ds$ ,

$$\left\| \text{sinc}_{\text{rad}}^{(m)} \right\|_\infty \leq \int_0^1 s^m \, ds = \frac{1}{m+1} \leq 1. \quad (5.13)$$

Applying the multinomial form of Leibniz's rule to a finite product – which, unlike the informal count “one factor per differentiation”, allows several derivatives to land on the same factor –

$$|\partial_t^q \Psi_{h,N}(t)| \leq \sum_{\substack{(m_n)_{n < N} \in \mathbb{N}_0^N \\ \sum_n m_n = q}} \frac{q!}{\prod_n m_n!} \prod_{n: m_n \geq 1} \alpha_n^{m_n} \prod_{n: m_n = 0} |\text{sinc}_{\text{rad}}(\alpha_n t)|,$$

where (5.13) was used on each differentiated factor. At most  $q$  indices carry  $m_n \geq 1$ , and each index contributes at most 1 to  $S_h$ , so by (5.6) the undifferentiated block is at most  $e^{cq} e^{-cS_h(t)}$ . What remains is the multinomial expansion of  $(\sum_{n < N} \alpha_n)^q$ . Letting  $N \rightarrow \infty$  and using

$$\sum_{n \geq 0} a_h r^n = \frac{a_h}{1-r} = \sqrt{\frac{3(1-r^2)}{(1-r)^2}} = \sqrt{\frac{3(1+r)}{1-r}} = \sqrt{3 \coth(h/2)} = A_h$$

gives (5.9); and since  $x \coth x$  increases,  $hA_h^2 = 3h \coth(h/2) \leq 3 \coth(\frac{1}{2}) = 6.4918 \dots < \frac{13}{2}$  for  $0 < h \leq 1$ .

*Lower bounds for  $S_h$ .* If  $a_h |t| \leq 1$ , every term in (5.7) is quadratic, so (3.4) gives (5.10). If  $\delta a_h^{-1} \leq |t| \leq a_h^{-1}$ , then by (5.10),  $a_h^2 = 3(1-r^2)$  and  $1 - e^{-2h} \leq 2h$ ,

$$S_h(t) = 3t^2 \geq \frac{3\delta^2}{a_h^2} = \frac{\delta^2}{1 - e^{-2h}} \geq \frac{\delta^2}{2h} \geq \frac{c_\delta}{h}.$$

If  $a_h |t| > 1$ , let  $n_1 \geq 1$  be least with  $a_h r^{n_1} |t| < 1$ ; the indices  $n < n_1$  contribute exactly 1 each, so  $S_h(t) \geq n_1 > h^{-1} \log(a_h |t|)$ , while  $a_h r^{n_1} |t| \geq r a_h r^{n_1-1} |t| \geq r$  makes the quadratic tail

$$\sum_{n \geq n_1} (a_h r^n t)^2 = \frac{(a_h r^{n_1} t)^2}{1-r^2} \geq \frac{r^2}{1-r^2} = \frac{1}{e^{2h} - 1} \geq \frac{e^{-2h}}{2h} \quad (\text{by } e^x - 1 \leq x e^x).$$

Adding the two blocks gives (5.12) for  $0 < h \leq 1$ . It also gives (5.11) on this range for  $0 < h \leq 1$ , because  $e^{-2h}/(2h) \geq e^{-2}/(2h) \geq c_\delta/h$ ; and for  $h > 1$  one has  $S_h(t) \geq 1 \geq h^{-1} \geq c_\delta h^{-1}$  since  $c_\delta < 1$ . Thus (5.11) holds for every  $h > 0$ .  $\square$

### 5.3 Low-frequency cumulant control

**Lemma 5.4** (Uniform cumulant remainder). *Fix  $R, q \geq 0$  and  $0 < \delta < \pi$ . Uniformly for  $a_h |t| \leq \delta$ ,*

$$\log \Psi_h(t) = -\frac{t^2}{2} + \sum_{m=2}^{R+1} \frac{\lambda_{2m}(h)}{(2m)!} (it)^{2m} + \rho_{R,h}(t), \quad (5.14)$$

where

$$|\partial_t^q \rho_{R,h}(t)| \leq C_{R,q,\delta} h^{R+1} (1 + |t|)^{2R+4}. \quad (5.15)$$

The identity and the bound hold verbatim for complex  $t$  with  $a_h |t| \leq \delta$ , the logarithm being the principal branch; only holomorphy of  $\log \operatorname{sinc}_{\text{rad}}$  on  $|z| < \pi$  is used.

*Proof.* Since  $\operatorname{sinc}_{\text{rad}} z = \sinh c(\operatorname{iz})$ , (4.4) gives  $\log \operatorname{sinc}_{\text{rad}} z = \sum_{k \geq 1} (-1)^k c_k z^{2k}$  on  $|z| < \pi$ , where the logarithm is holomorphic because  $\operatorname{sinc}_{\text{rad}}$  has no zero there. Put

$$g_R(z) = \log \operatorname{sinc}_{\text{rad}} z - \sum_{k=1}^{R+1} (-1)^k c_k z^{2k},$$

holomorphic on  $|z| < \pi$  and vanishing to order  $2R+4$  at the origin, so  $g_R(z) = z^{2R+4} \tilde{g}_R(z)$  with  $\tilde{g}_R$  holomorphic there. Fix  $\delta < \delta' < \pi$  and set  $M = \sup_{|z| \leq \delta'} |\tilde{g}_R(z)| < \infty$ . Cauchy's estimate on circles of radius  $\delta' - \delta$  gives  $|\tilde{g}_R^{(i)}(z)| \leq i! M (\delta' - \delta)^{-i}$  for  $|z| \leq \delta$ ; expanding  $\partial_z^q (z^{2R+4} \tilde{g}_R)$  by Leibniz's rule and using  $|z| \leq \delta$  to absorb the lower powers therefore yields the derivative-uniform remainder estimate

$$|g_R^{(q)}(z)| \leq C_{R,q,\delta} |z|^{\max(0, 2R+4-q)}, \quad |z| \leq \delta, \quad q \geq 0. \quad (5.16)$$

This is the statement that “differentiation lowers powers of  $t$ ”, now proved rather than asserted.

For  $a_h |t| \leq \delta$  all factors of (3.3) are zero free, so  $\log \Psi_h(t) = \sum_{n \geq 0} \log \operatorname{sinc}_{\text{rad}}(a_h r^n t)$ , the series converging locally uniformly (its  $n$ th term is  $\mathcal{O}((a_h r^n)^2 t^2)$ ) and hence differentiable term by term. Splitting each summand into its polynomial part and  $g_R$  gives (5.14) with  $\rho_{R,h}(t) = \sum_{n \geq 0} g_R(a_h r^n t)$ , because

$$(-1)^k c_k \frac{a_h^{2k}}{1 - r^{2k}} t^{2k} = \frac{B_{2k}}{2k} \frac{12^k}{(2k)!} \frac{(1 - r^2)^k}{1 - r^{2k}} (it)^{2k} = \frac{\lambda_{2k}(h)}{(2k)!} (it)^{2k}$$

by (4.3),  $a_h^2 = 3(1 - r^2)$  and Proposition 4.1; the term  $k = 1$  is  $-t^2/2$  since  $\lambda_2 = 1$ .

For the bound,  $\partial_t^q \rho_{R,h}(t) = \sum_n (a_h r^n)^q g_R^{(q)}(a_h r^n t)$ . Because  $a_h^2 = 3(1 - e^{-2h}) \leq 6h$  and  $1 - e^{-x} \geq x/2$  for  $0 \leq x \leq 1$ ,

$$\sum_{n \geq 0} (a_h r^n)^{2k} = \frac{a_h^{2k}}{1 - r^{2k}} \leq \frac{(6h)^k}{kh} = \frac{6^k}{k} h^{k-1} \quad (2kh \leq 1). \quad (5.17)$$

If  $q \leq 2R+4$ , (5.16) and (5.17) at  $k = R+2$  give

$$|\partial_t^q \rho_{R,h}(t)| \leq C_{R,q,\delta} |t|^{2R+4-q} \sum_{n \geq 0} (a_h r^n)^{2R+4} \leq C_{R,q,\delta} \frac{6^{R+2}}{R+2} h^{R+1} |t|^{2R+4-q};$$

if  $q > 2R+4$ , the same two displays give  $|\partial_t^q \rho_{R,h}| \leq C \sum_n (a_h r^n)^q \leq C' h^{q/2-1} \leq C' h^{R+1}$  for  $h \leq 1$ , since  $q/2 - 1 \geq R + \frac{3}{2}$ . Both cases are (5.15). Every step uses only  $|z| \leq \delta$  and holomorphy, so the argument is unchanged for complex  $t$  with  $a_h |t| \leq \delta$ .  $\square$

## 5.4 Proof of the Edgeworth theorem

*Proof of Theorem 5.1.* Let

$$A_{R,h}(t) = e^{-t^2/2} \sum_{\substack{\mathbf{k} \in \mathbb{N}_0^R \\ w(\mathbf{k}) \leq R}} \prod_{m=2}^{R+1} \frac{1}{k_m!} \left( \frac{\lambda_{2m}(h)(it)^{2m}}{(2m)!} \right)^{k_m}. \quad (5.18)$$

By (3.7), its inverse Fourier transform is  $\varphi \mathcal{E}_{R,h}$ .

Fix  $\varepsilon = \frac{1}{8(R+2)}$ , so that  $\varepsilon < \frac{1}{4}$  and  $\varepsilon(2R+4) = \frac{1}{4} < R+1$ , and split the frequency axis into

$$\mathcal{Z}_0 = \{|t| \leq h^{-\varepsilon}\}, \quad \mathcal{Z}_1 = \{h^{-\varepsilon} < |t| \leq \delta/a_h\}, \quad \mathcal{Z}_2 = \{|t| > \delta/a_h\},$$

where  $0 < \delta < 1$  is fixed.

On  $\mathcal{Z}_0$  we work on the closed complex discs  $D_t = \{z : |z - t| \leq 1\}$ ,  $t \in \mathcal{Z}_0$ . Since  $a_h^2 = 3(1 - e^{-2h}) \leq 6h$ , we have  $a_h(h^{-\varepsilon} + 1) \leq \sqrt{6h}(h^{-\varepsilon} + 1) \rightarrow 0$  because  $\varepsilon < \frac{1}{2}$ , so for small  $h$  every such disc lies in the window  $a_h|z| \leq \delta$  on which Lemma 5.4 is valid. There write  $\Psi_h = e^{-z^2/2}e^u$  with  $u = P_h + \rho_{R,h}$  and  $P_h(z) = \sum_{m=2}^{R+1} \lambda_{2m}(h)(iz)^{2m}/(2m)!$ , and  $A_{R,h} = e^{-z^2/2}\Sigma_{R,h}$  with  $\Sigma_{R,h}$  the sum in (5.18); then

$$e^{z^2/2}(\Psi_h - A_{R,h})(z) = (e^{P_h} - \Sigma_{R,h}) + e^{P_h}(e^{\rho_{R,h}} - 1). \quad (5.19)$$

The omitted monomials form an infinite family of unbounded degree in  $z$ , so they must be summed, not merely counted. Put  $\theta = h(1 + |z|^2)^2$  and

$$C_R^* = \max_{2 \leq m \leq R+1} \sup_{0 < h \leq 1} \frac{|\lambda_{2m}(h)|}{(2m)! h^{m-1}},$$

which is finite because by Proposition 4.1 the function  $\sinh^m h / \sinh(mh)$  is continuous on  $(0, 1]$  with limit  $h^{m-1}/m$  at the origin. Since  $|z|^{2m} \leq (1 + |z|^2)^{2m-2}$  for  $m \geq 2$ ,

$$\beta_m := \frac{|\lambda_{2m}(h)| |z|^{2m}}{(2m)!} \leq C_R^* \theta^{m-1}, \quad 2 \leq m \leq R+1. \quad (5.20)$$

In the expansion  $e^{P_h} = \sum_{\mathbf{k} \in \mathbb{N}_0^R} \prod_m k_m!^{-1} (\lambda_{2m}(iz)^{2m}/(2m)!)^{k_m}$  the multi-index  $\mathbf{k}$  contributes at most  $\prod_m \beta_m^{k_m}/k_m! \leq (\max(1, C_R^*)\theta)^{w(\mathbf{k})}$ , because  $m-1 \geq 1$  forces  $\sum_m k_m \leq w(\mathbf{k})$ ; and at most  $(W+1)^R$  multi-indices have  $w(\mathbf{k}) = W$ . Hence, once  $\max(1, C_R^*)\theta \leq \frac{1}{2}$ ,

$$|e^{P_h} - \Sigma_{R,h}| \leq \sum_{W \geq R+1} (W+1)^R (\max(1, C_R^*)\theta)^W \leq C_R \theta^{R+1} = C_R h^{R+1} (1 + |z|^2)^{2R+2}. \quad (5.21)$$

On  $\mathcal{Z}_0$  and its discs,  $1 + |z|^2 \leq 4h^{-2\varepsilon}$  for small  $h$ , so  $\theta \leq 16h^{1-4\varepsilon} \rightarrow 0$  and the hypothesis of (5.21) holds; the same estimate gives  $|P_h| \leq \sum_m \beta_m \leq 2C_R^* \theta \leq 1$ , while (5.15) with  $q = 0$  gives  $|\rho_{R,h}| \leq C_R h^{R+1-\varepsilon(2R+4)} = C_R h^{R+3/4} \leq 1$ . Therefore the second term of (5.19) is at most  $e^2 |\rho_{R,h}| \leq C_R h^{R+1} (1 + |z|)^{2R+4}$ . Finally  $|e^{-z^2/2}| = e^{-\Re(z^2)/2} \leq e^{1/2} e^{|t|} e^{-t^2/2} \leq e^{3/2} e^{-t^2/4}$  for  $|z - t| \leq 1$ , and  $(1 + |z|) \leq 2 + |t|$ , so

$$\sup_{|z-t| \leq 1} |(\Psi_h - A_{R,h})(z)| \leq C_R h^{R+1} (1 + |t|)^{4R+4} e^{-t^2/4}, \quad t \in \mathcal{Z}_0. \quad (5.22)$$

Both  $\Psi_h$  and  $A_{R,h}$  are entire, so Cauchy's estimate  $|\partial_t^q g(t)| \leq q! \sup_{|z-t|=1} |g(z)|$  upgrades (5.22) to every frequency derivative at once, with no Faà di Bruno bookkeeping:

$$|\partial_t^q (\Psi_h - A_{R,h})(t)| \leq C h^{R+1} (1 + |t|)^M e^{-t^2/4} \quad (t \in \mathcal{Z}_0), \quad (5.23)$$

with  $M = 4R + 4$  and  $C = C(R, q)$ . The right side is integrable with norm  $\mathcal{O}(h^{R+1})$ .

On  $\mathcal{Z}_1$  one has  $a_h |t| \leq \delta < 1$ , so (5.10) gives  $S_h(t) = 3t^2$  and (5.9) gives  $|\partial_t^q \Psi_h(t)| \leq (e^c A_h)^q e^{-t^2/2}$  with  $c = \frac{1}{6}$ . The approximant  $A_{R,h}$  is  $e^{-t^2/2}$  times a polynomial of degree at most  $4R$  whose coefficients are bounded uniformly for  $0 < h \leq 1$ . Hence for all  $j, q \geq 0$

$$\int_{\mathcal{Z}_1} (1 + |t|)^j |\partial_t^q (\Psi_h - A_{R,h})| dt \leq C_{j,q,R} h^{-q/2} \int_{|t| > h^{-\varepsilon}} (1 + |t|)^{j+4R} e^{-t^2/2} dt = \mathcal{O}(e^{-h^{-2\varepsilon}/4}),$$

which is  $\mathcal{O}(h^N)$  for every  $N$ .

On  $\mathcal{Z}_2$  the estimate must be made, not merely indicated; split it at  $a_h |t| = 1$ . For  $\delta a_h^{-1} < |t| \leq a_h^{-1}$ , (5.11) gives  $S_h \geq c_\delta h^{-1}$ , the zone has length at most  $2a_h^{-1}$ , and  $(1 + |t|)^j \leq (1 + a_h^{-1})^j$ ; since  $a_h^{-1} \leq (3h)^{-1/2}$  for  $h \leq 1$ , the corresponding integral of  $(1 + |t|)^j |\partial_t^q \Psi_h|$  is at most  $C_{j,q} h^{-(j+q+1)/2} e^{-cc_\delta/h}$ .

For  $|t| > a_h^{-1}$  put  $T = a_h |t| \geq 1$ . Then (5.12) and (5.9) give the explicit outer decay

$$|\partial_t^q \Psi_h(t)| \leq (e^c A_h)^q e^{-ce^{-2}/(2h)} T^{-c/h}, \quad c = \frac{1}{6}, \quad (5.24)$$

a power of  $T$  whose exponent  $c/h$  diverges: this is the precise form of “the number of sinc factors with argument at least one grows like  $h^{-1} \log(a_h |t|)$ ”. Using  $1 + |t| \leq 2T/a_h$  there (legitimate since  $|t| \geq a_h^{-1} \geq 1$  for  $h \leq \frac{1}{6}$ ) and substituting  $t = T/a_h$ ,

$$\int_{|t| > a_h^{-1}} (1 + |t|)^j |\partial_t^q \Psi_h| dt \leq \frac{2^{j+1} (e^c A_h)^q}{a_h^{j+1}} e^{-ce^{-2}/(2h)} \int_1^\infty T^{j-c/h} dT \leq \frac{2^{j+2} (e^c A_h)^q}{a_h^{j+1}} e^{-ce^{-2}/(2h)}$$

as soon as  $c/h \geq j + 2$ , that is  $h \leq \frac{1}{6(j+2)}$ . The prefactor is  $\mathcal{O}(h^{-(j+q+1)/2})$  and is absorbed by  $e^{-e^{-2}/(12h)}$ . The approximant  $A_{R,h}$  carries the factor  $e^{-t^2/2}$  with  $|t| > a_h^{-1} \geq (6h)^{-1/2}$  and is smaller still. The quantifiers now match the statement of the theorem:  $R, j$  and  $K$  are fixed first, only derivative orders  $q \leq K$  are needed, and  $h_0$  is then constrained by  $h_0 \leq \min(\frac{1}{6}, \frac{1}{6(j+2)})$  together with the finitely many smallness conditions used on  $\mathcal{Z}_0$ ; no bound is required for an unquantified  $N$ .

Collecting the three zones and applying Leibniz’s rule – which writes  $\partial_t^q [(it)^j g]$  as a combination of  $t^{j-i} \partial_t^{q-i} g$  with  $0 \leq i \leq \min(j, q)$  – we obtain, for every fixed  $q$  and  $j$ ,

$$\|\partial_t^q [(it)^j (\Psi_h - A_{R,h})]\|_{L^1(\mathbb{R})} \leq C_{R,j,q} h^{R+1}. \quad (5.25)$$

It remains to convert (5.25) into the seminorm bound; since the Schwartz-topology claim rests entirely on this step, it is written out. Let  $g = f_h - \varphi \mathcal{E}_{R,h}$ , so that  $\widehat{g}_{\text{ang}}(-t) = \Psi_h - A_{R,h} =: G(t)$  by (3.7), and  $G$  together with all its derivatives is integrable by the three zone estimates. By (3.5),  $g^{(j)}(x) = \frac{1}{2\pi} \int (-it)^j e^{-itx} G(t) dt$ . From  $x e^{-itx} = i \partial_t e^{-itx}$ ,  $K$  integrations by parts – with no boundary contribution, because  $\partial_t^i [(it)^j G(t)] \rightarrow 0$  as  $|t| \rightarrow \infty$  for  $i \leq K$  – give

$$x^K g^{(j)}(x) = \frac{(-i)^K}{2\pi} \int_{\mathbb{R}} e^{-itx} \partial_t^K [(-it)^j G(t)] dt, \quad \sup_x |x^K g^{(j)}(x)| \leq \frac{1}{2\pi} \|\partial_t^K [(it)^j G]\|_{L^1}. \quad (5.26)$$

The case  $K = 0$  is plain Fourier inversion. Since  $(1 + |x|)^K \leq 2^K (1 + |x|^K)$ ,

$$\|g\|_{j,K} \leq \frac{2^K}{2\pi} (\|(it)^j G\|_{L^1} + \|\partial_t^K [(it)^j G]\|_{L^1}) \leq C_{R,j,K} h^{R+1}$$

by (5.25) with  $q = 0$  and  $q = K$ . This is (5.4).  $\square$



## 5.5 First explicit orders

Expanding the exact-cumulant polynomial in powers of  $h$  gives

$$\begin{aligned} f_h(x) = \varphi(x) & \left[ 1 - \frac{h}{20} H_4(x) \right. \\ & + h^2 \left( \frac{4}{315} H_6(x) + \frac{1}{800} H_8(x) \right) \\ & \left. + h^3 \left( \frac{1}{60} H_4(x) - \frac{3}{700} H_8(x) - \frac{1}{1575} H_{10}(x) - \frac{1}{48000} H_{12}(x) \right) \right] + \mathcal{O}_S(h^4). \end{aligned} \quad (5.27)$$

Here  $\mathcal{O}_S$  means an error of that order in every fixed Schwartz seminorm. The second-order terms are the familiar Bell combination

$$\frac{\lambda_6}{6!} H_6 + \frac{1}{2} \left( \frac{\lambda_4}{4!} \right)^2 H_8,$$

but [Theorem 5.1](#) establishes the complete continuous-parameter hierarchy and its global analytic control.

## 6 CDF and inverse-distribution asymptotics

### 6.1 Uniform CDF expansion

Because the error in [Theorem 5.1](#) has polynomially weighted uniform control, it is integrable with the same order. Since

$$\int_{-\infty}^x H_n(u) \varphi(u) \, du = -H_{n-1}(x) \varphi(x), \quad n \geq 1, \quad (6.1)$$

we obtain the following.

**Corollary 6.1** (All-orders CDF expansion). *Let  $F_h(x) = \mathbb{P}(Z_h \leq x)$ . For every  $R \geq 0$ , uniformly in  $x \in \mathbb{R}$ ,*

$$F_h(x) = \Phi(x) - \varphi(x) \sum_{\substack{\mathbf{k} \in \mathbb{N}_0^R \\ 1 \leq w(\mathbf{k}) \leq R}} H_{2\nu(\mathbf{k})-1}(x) \prod_{m=2}^{R+1} \frac{1}{k_m!} \left( \frac{\lambda_{2m}(h)}{(2m)!} \right)^{k_m} + \mathcal{O}(h^{R+1}). \quad (6.2)$$

Moreover, for every fixed  $K \geq 0$  the error term  $\mathcal{R}_{R,h}(x)$  in (6.2) obeys the weighted bound

$$\sup_{x \in \mathbb{R}} (1 + |x|)^K |\mathcal{R}_{R,h}(x)| \leq C_{R,K} h^{R+1}. \quad (6.3)$$

*Proof.* Let  $D_{R,h}$  be the right-hand side of (6.2) without the error term, and  $\mathcal{R}_{R,h} = F_h - D_{R,h}$ . Since the zero multi-index contributes  $H_0 = 1$  to (5.2), (6.1) gives  $D_{R,h}(x) = \int_{-\infty}^x \varphi \mathcal{E}_{R,h}$ , whence  $\mathcal{R}_{R,h}(x) = \int_{-\infty}^x (f_h - \varphi \mathcal{E}_{R,h})$ .

The weighted claim does *not* follow from (5.4) alone: integrating a density error of size  $h^{R+1}(1 + |x|)^{-K-1}$  from  $-\infty$  leaves, for a generic integrand, a constant that no polynomial weight can beat. What makes it true here is the exact cancellation of total mass,

$$\int_{\mathbb{R}} (f_h - \varphi \mathcal{E}_{R,h}) \, dx = 0, \quad (6.4)$$

which holds because  $\int f_h = 1$ ,  $\int \varphi = 1$ , and  $\int H_n \varphi = 0$  for every  $n \geq 1$ , so that only the zero multi-index of (5.2) survives integration.

Granted (6.4), integrate from the near side of the axis. For  $x \geq 0$ ,  $\mathcal{R}_{R,h}(x) = -\int_x^\infty (f_h - \varphi \mathcal{E}_{R,h})$ , so by Theorem 5.1 with  $j = 0$  and weight  $K + 1$ ,

$$|\mathcal{R}_{R,h}(x)| \leq \|f_h - \varphi \mathcal{E}_{R,h}\|_{0,K+1} \int_x^\infty \frac{du}{(1+u)^{K+1}} = \frac{\|f_h - \varphi \mathcal{E}_{R,h}\|_{0,K+1}}{K(1+x)^K}$$

for  $K \geq 1$ , and symmetrically for  $x \leq 0$  from  $\mathcal{R}_{R,h}(x) = \int_{-\infty}^x (f_h - \varphi \mathcal{E}_{R,h})$ . For  $K = 0$  the same two formulas give the unweighted statement (6.2). This proves (6.3).  $\square$

Through second order,

$$F_h(x) = \Phi(x) + \varphi(x) \left[ \frac{h}{20} H_3(x) - h^2 \left( \frac{4}{315} H_5(x) + \frac{1}{800} H_7(x) \right) \right] + \mathcal{O}_{\text{weighted}}(h^3). \quad (6.5)$$

## 6.2 Central Cornish–Fisher inversion

Let  $Q_h(u) = F_h^{-1}(u)$ ,  $0 < u < 1$ , and  $z = \Phi^{-1}(u)$ . The inverse Fabius function is the special member  $r = 1/2$  after the appropriate affine normalization; the theorem below is an inverse-distribution asymptotic at the opposite, Gaussian boundary  $r \uparrow 1$ .

**Theorem 6.2** (All-orders central quantile expansion). *Fix  $0 < \eta < 1/2$  and  $R \geq 1$ . Uniformly for  $u \in [\eta, 1 - \eta]$ ,*

$$Q_h(u) = z + \sum_{n=1}^R h^n q_n(z) + \mathcal{O}_{R,\eta}(h^{R+1}), \quad z = \Phi^{-1}(u), \quad (6.6)$$

where each  $q_n$  is an explicitly computable polynomial. The first two are

$$q_1(z) = -\frac{1}{20} H_3(z), \quad (6.7)$$

$$q_2(z) = \frac{z H_3(z)^2}{800} - \frac{H_4(z) H_3(z)}{400} + \frac{4}{315} H_5(z) + \frac{1}{800} H_7(z). \quad (6.8)$$

All higher  $q_n$  follow recursively by coefficient extraction from (6.11) below.

*Proof.* A window with a uniform lower bound on the density. Let  $z_\eta = \Phi^{-1}(1 - \eta)$  and  $I_\eta = [-z_\eta - 1, z_\eta + 1]$ ; for  $u \in [\eta, 1 - \eta]$  the normal quantile  $z = \Phi^{-1}(u)$  lies in  $[-z_\eta, z_\eta]$ . Put  $2c_\eta = \varphi(z_\eta + 1) > 0$ . By Theorem 5.1 with  $R = j = K = 0$  (where  $\mathcal{E}_{0,h} \equiv 1$ ) we have  $\|f_h - \varphi\|_\infty \leq C_0 h$ , so  $f_h \geq c_\eta$  on  $I_\eta$  once  $h \leq h_1(\eta)$ . Hence  $F_h$  is strictly increasing on  $I_\eta$  with  $F'_h \geq c_\eta$ , and  $Q_h$  is well defined and smooth on  $F_h(I_\eta) \supseteq [\eta, 1 - \eta]$ . This much is the implicit function theorem; by itself it produces no expansion, and in particular no uniform remainder. The next two steps supply both.

*An approximate inverse.* By Corollary 6.1 together with the re-expansion of Proposition 5.2, applied to the  $h$ -dependent coefficients in (6.2), there are polynomials  $A_1, A_2, \dots$  with

$$\sup_{x \in \mathbb{R}} \left| F_h(x) - \Phi(x) - \varphi(x) \sum_{n=1}^N h^n A_n(x) \right| \leq C_N h^{N+1} \quad (6.9)$$

for every  $N$ . Let  $q_1, \dots, q_N$  be given by (6.11) below and set  $\tilde{Q}_N(h, z) = z + \sum_{n=1}^N h^n q_n(z)$ . The functions  $\Phi, \varphi, A_n, q_n$  are smooth, so on the compact set  $I_\eta$  they and their derivatives through order  $N + 1$  are bounded by some  $B = B(N, \eta)$ . Since  $\tilde{Q}_N(h, z) - z = \mathcal{O}(h)$  uniformly for  $z \in [-z_\eta, z_\eta]$ , the increment stays in  $I_\eta$  for small  $h$  and Taylor expansion of  $\Phi$  and of  $\varphi A_n$  about  $z$  in that increment, truncated after order  $N$ , is legitimate with remainder  $\mathcal{O}_{N,\eta}(h^{N+1})$ . Recursion (6.11) says precisely that the coefficients of  $h^1, \dots, h^N$  in the resulting expansion of  $F_h(\tilde{Q}_N(h, z)) - \Phi(z)$  all vanish. Therefore

$$\sup_{u \in [\eta, 1-\eta]} \left| F_h(\tilde{Q}_N(h, z)) - u \right| \leq C_{N,\eta} h^{N+1}. \quad (6.10)$$

*Transfer to the exact quantile.* For small  $h$  both  $\tilde{Q}_N(h, z)$  and  $Q_h(u)$  lie in  $I_\eta$ , where  $F'_h \geq c_\eta$ . The mean value theorem and  $F_h(Q_h(u)) = u$  give

$$\left| Q_h(u) - \tilde{Q}_N(h, z) \right| \leq \frac{1}{c_\eta} \left| F_h(\tilde{Q}_N(h, z)) - F_h(Q_h(u)) \right| \leq \frac{C_{N,\eta}}{c_\eta} h^{N+1},$$

uniformly in  $u \in [\eta, 1 - \eta]$ ; the remainder order survives because the lower bound  $F'_h \geq c_\eta$  is uniform on the window. Taking  $N = R$  proves (6.6).

For the explicit recursion, write

$$F_h(x) = \Phi(x) + \varphi(x) \sum_{n \geq 1} h^n A_n(x), \quad q_{<N}(h, z) = \sum_{n=1}^{N-1} h^n q_n(z).$$

Then

$$q_N(z) = -[h^N] \left\{ \frac{\Phi(z + q_{<N}) - \Phi(z)}{\varphi(z)} + \frac{\varphi(z + q_{<N})}{\varphi(z)} \sum_{n=1}^N h^n A_n(z + q_{<N}) \right\}, \quad (6.11)$$

where  $[h^N]$  extracts the coefficient. Equivalently, insert  $q_{<N} + h^N q_N$  into the identity  $F_h(Q_h(u)) = u$  and isolate the unique linear contribution  $h^N q_N$ . This is a Faà di Bruno/Bell recursion and proves all-orders computability.

From (6.5),

$$A_1 = \frac{1}{20} H_3, \quad A_2 = -\frac{4}{315} H_5 - \frac{1}{800} H_7.$$

At first order,  $q_1 = -A_1$ , giving (6.7). At second order,

$$q_2 = - \left[ \frac{1}{2} \frac{\varphi'}{\varphi} q_1^2 + \frac{(\varphi A_1)'}{\varphi} q_1 + A_2 \right].$$

Use  $\varphi'/\varphi = -z$ ,  $(\varphi H_3)' = -\varphi H_4$ , and (6.7); simplification gives (6.8).  $\square$

*Remark 6.3* (Connection to Lagrange inversion). The recursion (6.11) is a moving-base analogue of Lagrange inversion: the base map is  $\Phi$ , and the perturbation is organized by Hermite polynomials. It provides a concrete meeting point between the repository's inverse-function program, Bell polynomials, and its geometric-comb interpolation program.

## 7 Moderate deviations and the compact-scale large-deviation principle

### 7.1 Gaussian moderate deviations

**Theorem 7.1** (Moderate-deviation principle). *Let  $b_h \rightarrow \infty$  and  $hb_h^2 \rightarrow 0$  as  $h \downarrow 0$ . Then  $Z_h/b_h$  satisfies a large-deviation principle with speed  $b_h^2$  and good rate function*

$$I_{\text{mod}}(x) = \frac{x^2}{2}. \quad (7.1)$$

*Equivalently, for each Borel set  $A$ , the usual LDP upper and lower bounds hold with speed  $b_h^2$  and rate  $x^2/2$ .*

*Proof.* Avoid the cumulant series, whose radius of convergence would itself have to be controlled, and use the exact product form instead. By (3.1) and proposition 0.5.2 (ii), the coefficient of  $V_n$  in  $Z_h$  is  $2a_h r^n$ , so with  $\ell(a) = \log(\sinh a/a)$  as in (7.5) below,

$$\frac{1}{b_h^2} \log \mathbb{E} e^{tb_h Z_h} = \frac{1}{b_h^2} \sum_{n \geq 0} \ell(a_h r^n t b_h).$$

The Euler product  $\sinh z/z = \prod_{k \geq 1} (1 + z^2/(k\pi)^2)$  together with  $0 \leq x - \log(1+x) \leq x^2/2$  for  $x \geq 0$  gives a two-sided estimate valid for *all* real  $z$ , with no smallness restriction:

$$0 \leq \frac{z^2}{6} - \ell(z) \leq \frac{z^4}{2} \sum_{k \geq 1} \frac{1}{(k\pi)^4} = \frac{z^4}{2} \cdot \frac{\zeta(4)}{\pi^4} = \frac{z^4}{180}, \quad z \in \mathbb{R}. \quad (7.2)$$

Summing (7.2) over  $n$ , using  $\sum_n (a_h r^n)^2 = 3$  from (3.4) and

$$\sum_{n \geq 0} (a_h r^n)^4 = \frac{a_h^4}{1 - r^4} = \frac{9(1 - r^2)}{1 + r^2} \leq 18h,$$

we obtain the explicit, locally uniform estimate

$$\left| \frac{1}{b_h^2} \log \mathbb{E} e^{tb_h Z_h} - \frac{t^2}{2} \right| \leq \frac{t^4}{10} h b_h^2 \quad (t \in \mathbb{R}), \quad (7.3)$$

whose right side tends to 0 uniformly on compact  $t$ -sets exactly under the hypothesis  $h b_h^2 \rightarrow 0$ . The limit  $t \mapsto t^2/2$  is finite and differentiable on all of  $\mathbb{R}$ , hence lower semicontinuous and essentially smooth (its essential domain has empty boundary), so the Gärtner–Ellis theorem [2] applies in its full form and gives the LDP at speed  $b_h^2$  with the good rate function  $\sup_t \{tx - t^2/2\} = x^2/2$ .  $\square$

## 7.2 The compact scale

Define

$$Y_h = \sqrt{h} Z_h. \quad (7.4)$$

By (1.4), its support converges to  $[-\sqrt{6}, \sqrt{6}]$ . Put

$$\ell(a) = \log \frac{\sinh a}{a}, \quad \ell(0) = 0, \quad J(a) = \int_0^a \frac{\ell(v)}{v} dv. \quad (7.5)$$

The function  $J$  is even, because  $\ell(v)/v$  is odd.

**Theorem 7.2** (Compact-scale LDP). *The family  $(Y_h)_{h \downarrow 0}$  satisfies a large-deviation principle with speed  $h^{-1}$  and good convex rate function*

$$I(y) = \sup_{s \in \mathbb{R}} \{sy - \Lambda(s)\}, \quad (7.6)$$

where

$$\Lambda(s) = \int_0^1 \frac{\ell(\sqrt{6} su)}{u} du \quad (7.7)$$

$$= J(\sqrt{6} s). \quad (7.8)$$

The function  $\Lambda$  is even, real analytic, and strictly convex. Its derivative maps  $\mathbb{R}$  bijectively onto  $(-\sqrt{6}, \sqrt{6})$ . Consequently

$$I(y) < \infty \iff |y| < \sqrt{6}, \quad I(y) = \infty \text{ for } |y| \geq \sqrt{6}. \quad (7.9)$$

*Proof.* The limiting generator, by a two-sided Riemann bracket. The coefficient of  $V_n$  in  $Z_h$  is  $2a_h r^n$ , so  $\log \mathbb{E} e^{\theta Z_h} = \sum_{n \geq 0} \ell(a_h r^n \theta)$ ; at  $\theta = s/\sqrt{h}$ ,

$$\Lambda_h(s) := h \log \mathbb{E} \exp\left(\frac{s Y_h}{h}\right) = h \sum_{n \geq 0} \ell(\beta_h s e^{-nh}), \quad \beta_h = \frac{a_h}{\sqrt{h}} = \sqrt{\frac{3(1 - e^{-2h})}{h}} \rightarrow \sqrt{6}. \quad (7.10)$$

(The scale is written  $\beta_h$ , not  $b_h$ , to keep it distinct from the moderate-deviation scale of [Theorem 7.1](#).) Fix  $s > 0$ . Because  $\ell$  is even, nonnegative, and increasing on  $[0, \infty)$ , the function  $\varphi_h(x) = \ell(\beta_h s e^{-x})$  is nonnegative and *decreasing* on  $[0, \infty)$ ; for such a function the Riemann sum brackets the integral,

$$\int_0^\infty \varphi_h \leq h \sum_{n \geq 0} \varphi_h(nh) \leq \int_0^\infty \varphi_h + h \varphi_h(0),$$

so no domination argument and no fixed majorant are needed even though  $\beta_h$  moves with  $h$ . Substituting  $u = e^{-x}$  turns  $\int_0^\infty \varphi_h$  into  $\int_0^1 \ell(\beta_h s u) du/u = J(\beta_h s)$ , giving

$$J(\beta_h s) \leq \Lambda_h(s) \leq J(\beta_h s) + h \ell(\beta_h s). \quad (7.11)$$

Both  $J$  and  $\ell$  are continuous,  $\beta_h \rightarrow \sqrt{6}$ , and  $h \ell(\beta_h s) \rightarrow 0$  for each fixed  $s$ ; hence  $\Lambda_h(s) \rightarrow J(\sqrt{6} s) = \Lambda(s)$ , which is (7.7) after  $u = e^{-x}$  and (7.8) after  $v = \sqrt{6} s u$ . The case  $s < 0$  follows from evenness of  $\ell$ , and  $s = 0$  is trivial. This verifies the hypothesis of the Gärtner–Ellis theorem for every  $s \in \mathbb{R}$ .

*Regularity of  $\Lambda$ , without differentiating under an integral.* The function  $v \mapsto \ell(v)/v$ , set to 0 at  $v = 0$ , is real analytic on  $\mathbb{R}$ :  $\sinh v/v$  is entire and strictly positive on  $\mathbb{R}$ , so  $\ell$  is real analytic and vanishes to second order at the origin. Hence its primitive  $J$ , and therefore  $\Lambda = J(\sqrt{6} \cdot)$ , are real analytic on all of  $\mathbb{R}$ ; evenness of  $\Lambda$  follows from that of  $J$ . For strict convexity,  $\Lambda''(s) = 6J''(\sqrt{6} s)$  and  $J'(a) = \ell(a)/a = (\ell(a) - \ell(0))/(a - 0)$  is the difference quotient of  $\ell$  based at the origin. Now  $\ell$  is the logarithmic moment generating function of  $2V$  with  $V$  uniform on  $[-\frac{1}{2}, \frac{1}{2}]$ , a nondegenerate variable, so  $\ell$  is strictly convex; a difference quotient of a strictly convex function from a fixed base point is strictly increasing, so  $J'' > 0$  and  $\Lambda'' > 0$ .

*Derivative range, essential smoothness, and the support of  $I$ .* From  $\ell(a) = a - \log(2a) + \log(1 - e^{-2a})$  one gets  $\ell(a)/a \rightarrow 1$  as  $a \rightarrow +\infty$ , so  $\Lambda'(s) = \sqrt{6} \ell(\sqrt{6} s)/(\sqrt{6} s)$  increases strictly from  $-\sqrt{6}$  to  $\sqrt{6}$  and  $\Lambda'(\mathbb{R}) = (-\sqrt{6}, \sqrt{6})$  exactly. Since  $\Lambda$  is finite and differentiable on all of  $\mathbb{R}$ , its essential domain is  $\mathbb{R}$  and has empty boundary;  $\Lambda$  is therefore lower semicontinuous and essentially smooth, and the steepness condition is vacuous. Consequently the Gärtner–Ellis theorem [2] yields the *full* large-deviation principle, lower bound included, with the good convex rate function (7.6);  $I$  is good because  $\Lambda$  is finite everywhere, and in any case  $|Y_h| \leq \sqrt{h} A_h = \sqrt{3h \coth(h/2)} \leq \sqrt{13/2}$  for  $0 < h \leq 1$ , so the family is exponentially tight. Finally  $I$  is finite exactly on the derivative range: for  $|y| < \sqrt{6}$  the saddle equation  $\Lambda'(s) = y$  is solvable and  $I(y)$  is the finite value there, while at  $y = \sqrt{6}$ , writing  $a = \sqrt{6} s$ ,

$$\sqrt{6} s - \Lambda(s) = a - J(a) = \int_0^a \left(1 - \frac{\ell(v)}{v}\right) dv \geq \int_1^a \frac{\log(2v)}{v} dv \rightarrow \infty$$

because  $1 - \ell(v)/v = (\log(2v) - \log(1 - e^{-2v}))/v \geq \log(2v)/v$  for  $v \geq 1$ ; a fortiori  $I(y) = \infty$  for  $y > \sqrt{6}$ . This is (7.9).  $\square$

### 7.3 Exact saddle parametrization

**Theorem 7.3** (Parametric rate formula). *For  $a > 0$ , define*

$$y(a) = \sqrt{6} \frac{\ell(a)}{a}. \quad (7.12)$$

*Then  $y(a)$  is strictly increasing from 0 to  $\sqrt{6}$ , and*

$$\boxed{I(y(a)) = \ell(a) - J(a)}. \quad (7.13)$$

*The negative half of the rate function follows by symmetry.*

*Proof.* Set  $a = \sqrt{6}s$ . Since  $J'(a) = \ell(a)/a$ , the saddle equation  $y = \Lambda'(s)$  becomes (7.12). Strict convexity of  $\Lambda$  makes  $y(a)$  strictly increasing. At the saddle,

$$sy - \Lambda(s) = \frac{a}{\sqrt{6}} \left( \sqrt{6} \frac{\ell(a)}{a} \right) - J(a) = \ell(a) - J(a).$$

□

The Taylor expansion

$$\Lambda(s) = \frac{s^2}{2} - \frac{s^4}{20} + \frac{4}{315}s^6 - \frac{3}{700}s^8 + \frac{16}{9625}s^{10} + \mathcal{O}(s^{12}) \quad (7.14)$$

reuses exactly the Bernoulli coefficients that drive the Edgeworth expansion. Series reversion gives

$$I(y) = \frac{y^2}{2} + \frac{y^4}{20} + \frac{23}{3150}y^6 + \frac{11}{10500}y^8 + \frac{821}{6063750}y^{10} + \mathcal{O}(y^{12}). \quad (7.15)$$

The positive quartic term shows how compact support makes deviations more expensive than Gaussian ones as soon as one leaves the infinitesimal center.

## 8 Lambert– $W$ inversion at the limiting support edge

The parametric formula in Theorem 7.3 permits a precise edge analysis. It is here that the Gaussian boundary, the lower real branch of Lambert  $W$  [52], and the zeta/Stieltjes constants meet.

### 8.1 A renormalized Mellin constant

Use the standard Stieltjes convention

$$\zeta(1+s) = \frac{1}{s} + \gamma - \gamma_1 s + \mathcal{O}(s^2). \quad (8.1)$$

Define

$$\mathfrak{C} = \gamma_1 + \frac{\gamma^2}{2} - \frac{\pi^2}{12} = -0.7286939170039306059 \dots \quad (8.2)$$

**Lemma 8.1** (Large-saddle expansion of  $J$ ). *As  $a \rightarrow +\infty$ , with  $L = \log(2a)$ ,*

$$\ell(a) = a - L + \mathcal{O}(e^{-2a}), \quad (8.3)$$

$$J(a) = a - \frac{1}{2}L^2 + \mathfrak{C} + \mathcal{O}(e^{-2a}/a). \quad (8.4)$$

*Proof.* The first identity follows from

$$\ell(a) = a - \log(2a) + \log(1 - e^{-2a}).$$

For the second, put  $x = 2v$  in (7.5):

$$J(a) = a + \int_0^{2a} \frac{\log(1 - e^{-x}) - \log x}{x} dx. \quad (8.5)$$

The Mellin transform

$$\int_0^\infty x^{s-1} \log(1 - e^{-x}) dx = -\Gamma(s)\zeta(s+1), \quad \Re s > 0, \quad (8.6)$$

continues meromorphically to  $s = 0$ . Using

$$\Gamma(s) = \frac{1}{s} - \gamma + \left( \frac{\gamma^2}{2} + \frac{\pi^2}{12} \right) s + \mathcal{O}(s^2)$$

and (8.1),

$$-\Gamma(s)\zeta(s+1) = -\frac{1}{s^2} + \mathfrak{C} + \mathcal{O}(s). \quad (8.7)$$

More explicitly, the finite part in (8.7) is the absolutely convergent quantity

$$\mathfrak{C} = \int_0^1 \frac{\log(1 - e^{-x}) - \log x}{x} dx + \int_1^\infty \frac{\log(1 - e^{-x})}{x} dx. \quad (8.8)$$

Indeed, for  $A > 1$ ,

$$\int_0^A \frac{\log(1 - e^{-x}) - \log x}{x} dx + \frac{1}{2} \log^2 A = \int_0^1 \frac{\log(1 - e^{-x}) - \log x}{x} dx + \int_1^A \frac{\log(1 - e^{-x})}{x} dx.$$

Taking  $A = 2a$  and subtracting (8.8) leaves

$$-\int_{2a}^\infty \frac{\log(1 - e^{-x})}{x} dx = \mathcal{O}(e^{-2a}/a),$$

which proves (8.4). □

## 8.2 The lower Lambert branch

Let

$$\delta = 1 - \frac{y}{\sqrt{6}}, \quad y \uparrow \sqrt{6}. \quad (8.9)$$

From (7.12),

$$a\delta = \log(2a) - \log(1 - e^{-2a}). \quad (8.10)$$

The exponentially small second term is the sole difference from the exactly Lambert-invertible equation  $a\delta = \log(2a)$ .

**Theorem 8.2** (Lambert- $W_{-1}$  support-edge rate). *Let*

$$w(\delta) = -W_{-1}(-\delta/2), \quad \delta \downarrow 0. \quad (8.11)$$

*Then*

$$\boxed{I(\sqrt{6}(1 - \delta)) = \frac{1}{2}w(\delta)^2 - w(\delta) - \mathfrak{C} + o(1).} \quad (8.12)$$

*More precisely, if  $a$  is the exact saddle determined by (8.10), then*

$$I(y(a)) = \frac{1}{2} \log^2(2a) - \log(2a) - \mathfrak{C} + \mathcal{O}(e^{-2a}). \quad (8.13)$$

*Proof.* Subtract (8.4) from (8.3) and use (7.13); this proves (8.13). Ignoring the exponentially small term in (8.10), write  $L = \log(2a)$ . The equation  $a\delta = L$  and  $2a = e^L$  give

$$Le^{-L} = \frac{\delta}{2}, \quad L = -W_{-1}(-\delta/2) = w(\delta).$$

For the exact equation, put  $L = \log(2a)$  and  $F(L) = Le^{-L}$ . Multiplying (8.10) by  $e^{-L}$  gives

$$\frac{\delta}{2} = F(L) - e^{-L} \log(1 - e^{-e^L}).$$



If  $w = w(\delta)$ , then  $F(w) = \delta/2$ . Since  $F'(u) = (1-u)e^{-u}$ , the mean-value theorem and  $-\log(1-z) \asymp z$  yield

$$L - w = \mathcal{O}\left(\frac{e^{-e^w}}{w}\right). \quad (8.14)$$

In particular  $w(L - w) = o(1)$ , so substituting  $L = w + \mathcal{O}(e^{-e^w}/w)$  into (8.13) proves (8.12).  $\square$

*Remark 8.3* (A new three-way connection). The coefficient  $\mathfrak{C}$  comes from the finite part of  $-\Gamma(s)\zeta(s+1)$  at  $s = 0$ , the saddle coordinate comes from an infinite geometric uniform convolution, and the edge inversion is controlled by  $W_{-1}$ . This is distinct from the fixed- $r$  periodic Lambert- $W$  endpoint expansion in the repository: here  $r \uparrow 1$  first, producing a smooth nonperiodic limiting rate, and only then does the compact coordinate approach its support edge.

## 9 Quadratic $q$ -Gevrey and Denjoy–Carleman regularity

### 9.1 A convention for global ultradifferentiable classes

Let  $M = (M_n)_{n \geq 0}$  be positive with  $M_0 = 1$ . Define the global Roumieu class

$$\mathcal{B}^{\{M\}}(\mathbb{R}) = \left\{ f \in C^\infty(\mathbb{R}) : \exists C, A > 0 \forall n \geq 0, \|f^{(n)}\|_\infty \leq CA^n M_n \right\}. \quad (9.1)$$

For log-convex  $M$ , the Denjoy–Carleman criterion in this convention [27] says that the class is quasianalytic exactly when

$$\sum_{n \geq 1} \frac{M_{n-1}}{M_n} = \infty. \quad (9.2)$$

A Gevrey class of finite order  $s$  corresponds, up to exponential factors, to  $M_n = (n!)^s$ .

### 9.2 The Rvachev function

The exact norm identity is not imported here as an axiom: in this volume it is [Theorem 11.3](#) specialized to  $a = 2$ , where  $h_2 = \text{up}$ . Combining (11.8) and (11.9) at  $a = 2$ , and using  $\|h_2\|_\infty = 1$ , gives

$$\|\text{up}^{(n)}\|_\infty = \|\text{up}^{(n)}\|_1 = 2^{n(n+1)/2}. \quad (9.3)$$

Its finite signed derivative decomposition is (11.5), indexed by the signed Thue–Morse sequence [39]. Thus automatic-sequence cancellation and support tiling produce an exact global smoothness scale, and the two theorems of this section are **proved here** within the volume rather than **existing repository input**.

**Theorem 9.1** (Sharp Denjoy–Carleman class of  $\text{up}$ ). *Let*

$$M_n^{\text{up}} = 2^{n(n+1)/2}. \quad (9.4)$$

*Then:*

- (i)  $\text{up} \in \mathcal{B}^{\{M^{\text{up}}\}}(\mathbb{R})$ , with the sharp constants  $C = A = 1$ .
- (ii) The class is optimal modulo the natural exponential equivalence of weight sequences: if  $\text{up} \in \mathcal{B}^{\{N\}}$ , then  $N_n \geq C^{-1}A^{-n}M_n^{\text{up}}$  for suitable  $C, A > 0$ .
- (iii)  $\text{up}$  belongs to no Gevrey class of finite order.
- (iv)  $\mathcal{B}^{\{M^{\text{up}}\}}$  is nonquasianalytic, since

$$\sum_{n \geq 1} \frac{M_{n-1}^{\text{up}}}{M_n^{\text{up}}} = \sum_{n \geq 1} 2^{-n} = 1. \quad (9.5)$$

*Proof.* Parts (i) and (ii) are immediate from the equality (9.3). If up were Gevrey of finite order  $s$ , then for some  $C, A$ ,

$$2^{n(n+1)/2} \leq CA^n(n!)^s.$$

The logarithm of the left side is  $(\log 2)n^2/2 + \mathcal{O}(n)$ , whereas the right side has logarithm  $\mathcal{O}(n \log n)$ , a contradiction. Finally

$$M_n^{\text{up}}/M_{n-1}^{\text{up}} = 2^n,$$

which is increasing in  $n$ ; together with  $M_0^{\text{up}} = 1$  this is the log-convexity hypothesis under which (9.2) was stated, and which must be verified before the criterion is used. The sum (9.5) is then finite, so the Denjoy–Carleman criterion proves nonquasianalyticity.  $\square$

**Definition 9.2** (Quadratic  $q$ -Gevrey scale). In this report, a derivative weight with

$$\log M_n = \frac{\log(1/q)}{2}n^2 + \mathcal{O}(n), \quad 0 < q < 1,$$

is called *quadratic  $q$ -Gevrey*. This terminology is descriptive and is not intended to replace other established  $q$ -Gevrey conventions.

For up,  $q = 1/2$ . The name records the same quadratic exponent that appears in  $q$ -Pochhammer products, Stieltjes–Wigert moments, and log-Gaussian Fourier envelopes.

### 9.3 Separated-base atomic functions

For the general atomic functions  $h_a$ ,  $a \geq 2$ , Theorem 11.3 proves

$$\|h_a^{(n)}\|_{\infty} = \frac{a^{n(n+3)/2+1}}{2^{n+1}}, \quad \|h_a^{(n)}\|_1 = a^{n(n+1)/2}. \quad (9.6)$$

**Theorem 9.3** (Sharp class for  $h_a$ ,  $a \geq 2$ ). Set

$$M_n^{(a)} = \frac{a^{n(n+3)/2}}{2^n}. \quad (9.7)$$

Then  $h_a \in \mathcal{B}^{\{M^{(a)}\}}$  sharply modulo exponential equivalence, belongs to no finite Gevrey class, and its class is nonquasianalytic because

$$\sum_{n \geq 1} \frac{M_{n-1}^{(a)}}{M_n^{(a)}} = \sum_{n \geq 1} \frac{2}{a^{n+1}} < \infty. \quad (9.8)$$

Its quadratic  $q$ -Gevrey parameter is  $q = a^{-1}$ .

*Proof.* The first identity in (9.6) says  $\|h_a^{(n)}\|_{\infty} = (a/2)M_n^{(a)}$ , so  $h_a \in \mathcal{B}^{\{M^{(a)}\}}$  with  $C = a/2$ ,  $A = 1$ , and conversely  $h_a \in \mathcal{B}^{\{N\}}$  forces  $N_n \geq (2/a)C^{-1}A^{-n}M_n^{(a)}$ ; this is membership together with optimality. (The second identity in (9.6), the  $L^1$  norm  $a^{n(n+1)/2}$ , is not  $(a/2)M_n^{(a)}$  – at  $a = 3$ ,  $n = 1$  the two are 3 and  $27/4$  – and it enters only through the Fourier envelope (9.11).) Quadratic growth again excludes every finite Gevrey order. Finally  $M_0^{(a)} = 1$  and  $M_n^{(a)}/M_{n-1}^{(a)} = a^{n+1}/2$  is increasing, so  $M^{(a)}$  is log-convex and (9.2) may be applied; summing the reciprocals gives (9.8).  $\square$

## 9.4 Associated functions and Fourier decay

For a weight  $M$ , define its associated function

$$\omega_M(t) = \sup_{n \geq 0} \log \frac{t^n}{M_n}. \quad (9.9)$$

For  $M_n^{\text{up}} = 2^{n(n+1)/2}$ , elementary optimization gives

$$\omega_{M^{\text{up}}}(t) = \frac{(\log t)^2}{2 \log 2} - \frac{1}{2} \log t + \mathcal{O}(1), \quad t \rightarrow \infty. \quad (9.10)$$

Integration by parts and (9.3) therefore imply a log-Gaussian Fourier envelope

$$|\widehat{\text{up}}_{\text{ang}}(-\xi)| \leq C \exp \left[ -\frac{(\log(C' |\xi|))^2}{2 \log 2} + \mathcal{O}(\log |\xi|) \right]. \quad (9.11)$$

For  $h_a$ , the coefficient  $(2 \log a)^{-1}$  replaces  $(2 \log 2)^{-1}$ . The repository develops the Fourier envelope by product methods; the new point here is that the same quadratic law is precisely the associated function of the optimal Denjoy–Carleman class. The derivative and Fourier pictures are Legendre dual.

## 10 Numerical experiments

### 10.1 Method

The exact standardized characteristic function (3.3) was evaluated on a centered FFT frequency grid. The infinite product was truncated at the first index  $N$  whose sinc argument fell below  $10^{-14}$ ; by (5.6) the discarded tail multiplies the product by a factor in  $[\exp(-10^{-28}/(6(1-r^2))), 1]$ , which for every tabulated  $h \geq 0.05$  is within  $10^{-27}$  of 1 and so far below the smallest reported error,  $1.83 \times 10^{-7}$ . Fourier inversion used  $2^{15}$  grid points with physical spacing 0.002, i.e. a window  $|x| \leq 32.768$ ; since  $Z_h$  is supported in  $|x| \leq A_h = \sqrt{3 \coth(h/2)}$ , which is 10.96 at  $h = 0.05$  and smaller for larger  $h$ , the window contains the whole support and there is no wrap-around. The numerical density mass was 0.99999999999926 for every reported  $h$ . No Monte Carlo sampling was used.

For  $E_R(h)$ , define the maximum density error on  $|x| \leq 8$  after retaining terms through  $h^R$ . The results are:

Table 30: Successive uniform density errors, with predicted orders  $h, h^2, h^3, h^4$ .

| $h$  | Gaussian error          | first order             | second order            | third order             |
|------|-------------------------|-------------------------|-------------------------|-------------------------|
| 0.30 | $2.0332 \times 10^{-2}$ | $2.8882 \times 10^{-3}$ | $7.7617 \times 10^{-4}$ | $4.1090 \times 10^{-4}$ |
| 0.20 | $1.2983 \times 10^{-2}$ | $1.1427 \times 10^{-3}$ | $1.9456 \times 10^{-4}$ | $6.9283 \times 10^{-5}$ |
| 0.15 | $9.5350 \times 10^{-3}$ | $6.1257 \times 10^{-4}$ | $7.3140 \times 10^{-5}$ | $1.8947 \times 10^{-5}$ |
| 0.10 | $6.2279 \times 10^{-3}$ | $2.6129 \times 10^{-4}$ | $1.9479 \times 10^{-5}$ | $3.2761 \times 10^{-6}$ |
| 0.07 | $4.3071 \times 10^{-3}$ | $1.2518 \times 10^{-4}$ | $6.3096 \times 10^{-6}$ | $7.3484 \times 10^{-7}$ |
| 0.05 | $3.0520 \times 10^{-3}$ | $6.2957 \times 10^{-5}$ | $2.2183 \times 10^{-6}$ | $1.8343 \times 10^{-7}$ |

The normalized errors  $E_R(h)/h^{R+1}$  remain bounded and settle toward finite values as  $h \downarrow 0$ , exactly as Theorem 5.1 predicts.

### 10.2 Central quantiles

At the seven levels  $u \in \{0.05, 0.10, 0.25, 0.50, 0.75, 0.90, 0.95\}$ , the maximum errors are: The second-order error behaves consistently with  $\mathcal{O}(h^3)$ , while the first-order error behaves as  $\mathcal{O}(h^2)$ .

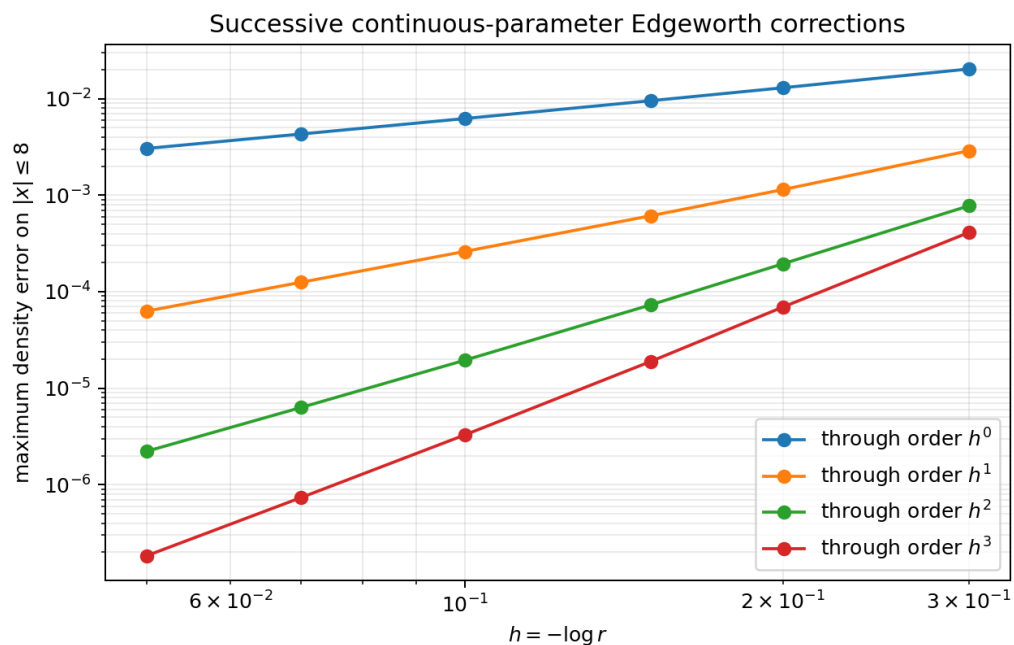


Figure 49: Log-log error curves for the Gaussian approximation and the first three continuous-parameter Edgeworth corrections.

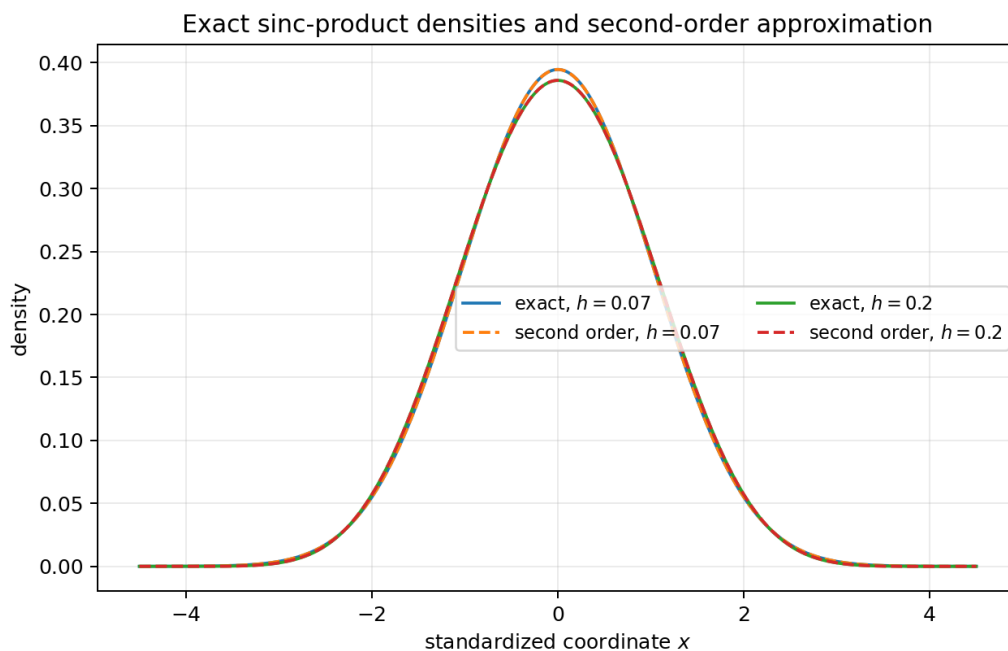


Figure 50: Exact sinc-product densities and the second-order approximation for two values of  $h$ . The curves are visually indistinguishable near the center at  $h = 0.07$ .

Table 31: Central quantile errors for the normal, first-order, and second-order Cornish–Fisher approximations.

| $h$  | Gaussian quantile       | first order             | second order            |
|------|-------------------------|-------------------------|-------------------------|
| 0.30 | $3.0365 \times 10^{-2}$ | $4.6160 \times 10^{-3}$ | $2.2050 \times 10^{-3}$ |
| 0.20 | $1.9087 \times 10^{-2}$ | $1.9206 \times 10^{-3}$ | $4.8685 \times 10^{-4}$ |
| 0.15 | $1.3929 \times 10^{-2}$ | $1.0540 \times 10^{-3}$ | $1.8484 \times 10^{-4}$ |
| 0.10 | $9.0414 \times 10^{-3}$ | $4.5827 \times 10^{-4}$ | $4.9114 \times 10^{-5}$ |
| 0.07 | $6.2435 \times 10^{-3}$ | $2.2203 \times 10^{-4}$ | $1.5399 \times 10^{-5}$ |
| 0.05 | $4.4288 \times 10^{-3}$ | $1.1264 \times 10^{-4}$ | $5.2626 \times 10^{-6}$ |

### 10.3 Lambert– $W$ edge check

The exact parametric rate (7.13) was compared with the edge form (8.12). Here  $a$  is the saddle and  $\delta = 1 - y(a)/\sqrt{6}$ . The difference is already below  $10^{-10}$  at  $a = 10$ , despite  $\delta \approx 0.30$ . This unusually

Table 32: Rapid onset of the Lambert– $W$  support-edge asymptotic.

| $a$ | $\delta = 1 - y/\sqrt{6}$ | exact $I(y)$ | exact minus Lambert– $W$ form |
|-----|---------------------------|--------------|-------------------------------|
| 2   | 0.7023899                 | 0.2810218    | $-4.246 \times 10^{-3}$       |
| 3   | 0.5980804                 | 0.5392933    | $-3.642 \times 10^{-4}$       |
| 5   | 0.4605261                 | 1.0770083    | $-4.158 \times 10^{-6}$       |
| 8   | 0.3465736                 | 1.7997292    | $-6.640 \times 10^{-9}$       |
| 10  | 0.2995732                 | 2.2201676    | $-9.836 \times 10^{-11}$      |
| 20  | 0.1844440                 | 3.8437303    | $-1.037 \times 10^{-19}$      |

rapid convergence reflects the exponentially small correction in (8.10).

## 11 Conjectures and frontier directions

The theorems above open several directions that are not merely requests for more coefficients. They concern matching between asymptotic regimes, local symbolic regularity, and arithmetic structure.

### 11.1 A matched four-regime asymptotic theory

**Conjecture 11.1** (Uniform saddlepoint local large deviations). *For every compact  $K \subset (-\sqrt{6}, \sqrt{6})$ , the density of  $Y_h$  has the uniform saddlepoint expansion*

$$f_{Y_h}(y) \sim \frac{\exp[-I(y)/h]}{\sqrt{2\pi h \Lambda''(s(y))}} \sum_{n=0}^{\infty} h^n A_n(y), \quad y \in K, \quad (11.1)$$

where  $s(y)$  is the unique solution of  $\Lambda'(s) = y$ , and each  $A_n$  is an explicit differential polynomial in  $\Lambda$ .

Conjecture 8.2 is the same conjecture in Part X’s coordinate, obtained by the substitution  $y = \sqrt{6}x$  of Table 1; the density picks up the Jacobian  $\sqrt{6}$  and  $\Lambda''(s(y)) = 6\Lambda''(\theta_x)$ , so the two prefactors agree.

**Conjecture 11.2** (Edgeworth–large-deviation matching). *When  $y = \sqrt{h}x \rightarrow 0$ , the saddlepoint expansion (11.1), re-expanded in  $h$ , matches the Bernoulli–Bell–Hermite Edgeworth series term by term. In the overlap  $1 \ll x \ll h^{-1/2}$ , it matches the Gaussian moderate-deviation expansion. This matching is Borel summable in a sector of the complex  $x$ -plane avoiding the images of the nearest sinc zeros.*

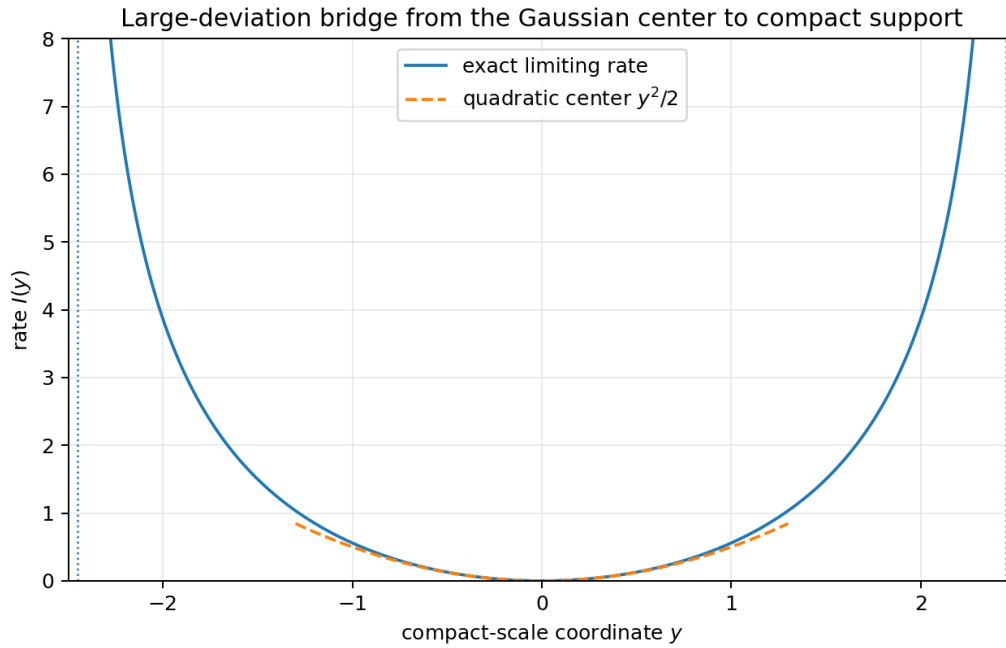


Figure 51: The exact limiting compact-scale rate. It is quadratic near the origin and diverges at the limiting support endpoints  $\pm\sqrt{6}$ .

**Conjecture 11.3** (Double matching at the compact edge). *There is a two-parameter uniform asymptotic that matches the compact-scale Lambert– $W$  law (8.12) to the fixed- $r$  endpoint asymptotic with its periodic logarithmic profile. The crossover occurs when  $\delta \downarrow 0$  and  $h \downarrow 0$  jointly so that the exact saddle reaches the first geometric lattice scale before the continuum Riemann-sum approximation loses uniformity.*

The last conjecture should reveal how the periodic fluctuations at fixed  $r$  disappear in the continuum  $r \uparrow 1$  limit. Poisson summation in the logarithmic index is a promising mechanism: nonzero Fourier modes should be exponentially small in  $1/h$  in the interior but become visible near the edge.

## 11.2 Beyond all orders and sinc-zero geometry

**Conjecture 11.4** (Resurgent frequency corrections). *After optimal truncation of the continuous Edgeworth series, the remainder is controlled by complex saddles associated with the nearest zeros of  $\text{sinc}_{\text{rad}}(a_h r^n t)$ . Their actions form a geometric lattice, and the Stokes multipliers admit a  $q$ -product description. The first exponentially small scale is  $\exp(-C/h)$ , consistent with the outer-frequency estimate in Lemma 5.3.*

A practical route is to apply logarithmic Poisson summation to

$$\sum_{n \geq 0} \log \text{sinc}_{\text{rad}}(a_h e^{-nh} t)$$

and isolate its nonzero dual modes. This should be the Gaussian-boundary counterpart of the periodic endpoint profiles already present at fixed base.

## 11.3 Entropy and information geometry

The repository conjectures the leading entropy deficit. Theorem 5.1 supplies the missing local expansion, but a complete proof still requires global logarithmic control where  $f_h/\varphi$  is small or large.

**Conjecture 11.5** (Full entropic expansion). *The relative entropy has a complete expansion*

$$D(\text{Law}(Z_h) \| N(0, 1)) \sim \sum_{n \geq 2} d_n h^n, \quad d_2 = \frac{3}{100}, \quad (11.2)$$

where every  $d_n$  is a finite contraction of Bernoulli-generated Edgeworth coefficients using Hermite orthogonality. The same holds for Fisher information and for Rényi divergences in a nontrivial interval of orders.

A proof should combine the weighted Edgeworth theorem with a saddlepoint majorant on the scale  $x = O(h^{-1/2})$ . The LDP from [Theorem 7.2](#) provides exactly the missing tail geometry.

#### 11.4 Local symbolic Carleman spectra

The global weight in [Theorem 9.3](#) is sharp, but local derivative growth can be much smaller at points whose symbolic orbit repeatedly enters regions where the base profile is tiny.

**Open problem 11.6** (Multifractal local regularity). *For  $a \geq 2$  and  $x$  in the symbolic attractor, determine the spectrum of*

$$\underline{\alpha}(x) = \liminf_{n \rightarrow \infty} \frac{\log |h_a^{(n)}(x)|}{n^2}, \quad \bar{\alpha}(x) = \limsup_{n \rightarrow \infty} \frac{\log |h_a^{(n)}(x)|}{n^2}. \quad (11.3)$$

Relate the Hausdorff dimensions of level sets to recurrence rates of the digit shift and to thermodynamic pressure.

**Conjecture 11.7** (Typical quadratic exponent). *For the natural Bernoulli measure on the symbolic attractor of  $h_a$ ,*

$$\underline{\alpha}(x) = \bar{\alpha}(x) = \frac{1}{2} \log a$$

for almost every  $x$ , while exceptional recurrent orbits realize a nontrivial interval of lower subleading exponents.

#### 11.5 The overlapping phase $1 < a < 2$

The exact norm formula uses disjoint cylinder supports and therefore stops at  $a = 2$ . Fourier methods still give the matching upper bound, and in fact with a linear rather than an  $n \log n$  correction. Write  $T = 2\pi |\xi|$  and  $L = \log_a T$ , and use  $|\text{sinc}_{\text{rad}} u| \leq \min(1, |u|^{-1})$  in Part VI's product  $\widehat{h_{a/2\pi}}(\xi) = \prod_{j \geq 1} \text{sinc}_{\text{rad}}(T a^{-j})$ , keeping only the  $N = \lfloor L \rfloor$  factors whose argument is at least 1: for  $T \geq a$ , with  $\phi = L - N \in [0, 1)$ ,

$$\left| \widehat{h_{a/2\pi}}(\xi) \right| \leq \frac{a^{N(N+1)/2}}{T^N} = \exp\left(-\frac{\log a}{2}(L^2 - L - \phi^2 + \phi)\right) \leq a^{1/8} T^{1/2} \exp\left(-\frac{\log^2 T}{2 \log a}\right),$$

since  $\phi^2 - \phi \geq -\frac{1}{4}$ . Hence, substituting  $T = e^v$ ,

$$\left\| h_a^{(n)} \right\|_{\infty} \leq \int_{\mathbb{R}} (2\pi |\xi|)^n \left| \widehat{h_{a/2\pi}}(\xi) \right| d\xi \leq \frac{a^{n+1}}{\pi(n+1)} + \frac{a^{1/8}}{\pi} \int_{-\infty}^{\infty} e^{(n+3/2)v - v^2/(2 \log a)} dv,$$

and completing the square in the Gaussian integral gives

$$\log \left\| h_a^{(n)} \right\|_{\infty} \leq \frac{\log a}{2} n^2 + \mathcal{O}_a(n), \quad 1 < a < 2. \quad (11.4)$$

This much is **proved here**. It is one half of (11.5); the matching lower bound, which the disjointness argument supplies only for  $a \geq 2$ , is what the following conjecture asserts.



**Conjecture 11.8** (Sharp quadratic exponent under overlap). *For every  $1 < a < 2$ ,*

$$\lim_{n \rightarrow \infty} \frac{1}{n^2} \log \left\| h_a^{(n)} \right\|_{\infty} = \frac{1}{2} \log a. \quad (11.5)$$

*The subleading term changes at parameters with exceptional Bernoulli-convolution overlap, and may encode arithmetic properties of  $a$ .*

This conjecture would extend the quadratic  $q$ -Gevrey classification through the entire smooth geometric family, while leaving room for arithmetic phase transitions in the  $n \log n$  or linear term.

## 11.6 Arithmetic and Thue–Morse directions

**Conjecture 11.9** (Cumulant-coefficient denominators). *After the natural normalization by  $(2m)!$ , the coefficients in the  $h$ -expansion (4.8) have denominators controlled by von Staudt–Clausen primes from the Bernoulli numbers and by the divisors of  $m^{2^k} - m$ . There is a finite-prime valuation formula compatible with the dyadic arithmetic of  $F(2^{-n})$ .*

**Open problem 11.10** (Automatic-sequence Edgeworth analogs). *Replace the uniform digits in (1.1) by signed or block-dependent digits generated by Thue–Morse and other automatic sequences. Determine when a Gaussian boundary survives, how automatic cancellations alter the first nonzero cumulant, and whether the resulting Edgeworth polynomials inherit finite-kernel recurrences.*

**Open problem 11.11** (Legendre–Hermite contraction). *The repository develops compact-support Legendre and Jacobi expansions, whereas the Gaussian boundary naturally selects the Hermite basis. Determine a joint limit in the polynomial degree and in  $h \downarrow 0$  under which the appropriately rescaled Legendre/Jacobi coefficients contract to the Bernoulli–Bell–Hermite Edgeworth coefficients. A successful contraction formula would connect the global compact-support spectral theory to the local Gaussian asymptotic theory without passing through pointwise Fourier inversion.*

## 11.7 Formalization targets

The analytic arguments naturally split into reusable lemmas suitable for Lean or another proof assistant:

1. exact cumulant algebra and hyperbolic simplification;
2. the global sinc envelope and geometric power-sum bounds;
3. weighted Fourier inversion from  $W^{K,1}$  estimates;
4. the Riemann-sum limit for  $\Lambda_h$ ;
5. strict convexity and Legendre duality;
6. the Mellin finite-part computation of  $\mathfrak{C}$ ;
7. the Carleman ratio test for the exact norm sequence.

The most delicate formalization component is the all-orders bookkeeping in [Theorem 5.1](#); encoding the weight  $w(\mathbf{k})$  as a graded finite-support map should make the Bell combinatorics manageable.

## 12 Promising research program

A productive sequence of projects, ordered by mathematical dependency rather than by perceived difficulty, is:

- P1. Local saddlepoint theorem.** Prove [Conjecture 11.1](#) by complex Fourier inversion around the unique imaginary saddle. Establish uniformity on compact subsets of the open limiting support.

- P2. Central matching.** Expand the saddlepoint coefficients at  $y = 0$  and identify them combinatorially with the Bernoulli–Bell–Hermite coefficients of [Theorem 5.1](#).
- P3. Entropic CLT with coefficients.** Combine weighted central control and the compact-scale LDP to prove [Conjecture 11.5](#), first through  $h^3$ , then to all orders.
- P4. Logarithmic Poisson modes.** Derive the exponentially small correction between the geometric Riemann sum and its continuum integral. Identify the first dual mode and its Stokes geometry.
- P5. Edge crossover.** Construct the double-scaling asymptotic of [Conjecture 11.3](#), retaining the fixed-base periodic phase until it is exponentially suppressed.
- P6. Overlapping quadratic regularity.** Prove (11.5); investigate whether Pisot, Salem, or exact-overlap parameters alter subleading derivative growth.
- P7. Local Carleman thermodynamics.** Translate the exact symbolic derivative formula into a subadditive pressure problem and compute the multifractal spectrum in [Open problem 11.6](#).
- P8. Arithmetic coefficient package.** Implement exact Bernoulli–Bell coefficient generation, valuation analysis, and conjecture discovery for (4.8) and the quantile polynomials.
- P9. Legendre–Hermite contraction.** Derive the double-scaling limit in the preceding problem and use it to transfer spectral estimates between compact-support and Gaussian-boundary regimes.

The first five projects form a single asymptotic arc from the Gaussian center to the fixed-parameter endpoint. Projects six through eight exploit the independent but structurally parallel quadratic growth appearing in derivatives,  $q$ -products, and arithmetic denominators; the ninth connects the repository’s orthogonal-polynomial and Gaussian asymptotic languages.

## 13 Conclusion

The geometric-uniform deformation turns the Fabius/Rvachev pair into a parameterized laboratory in which several asymptotic languages coexist. The exact cumulants make Bernoulli numbers primary; exponential composition turns them into Bell polynomials; Fourier inversion turns Bell monomials into Hermite corrections; inversion of the CDF produces Cornish–Fisher polynomials; rescaling to the compact support produces a Gärtner–Ellis rate; and approaching that support edge produces the lower Lambert branch and a Stieltjes constant. Independently, the Thue–Morse derivative hierarchy produces an exact quadratic-exponential Carleman weight, whose Legendre dual is the log-Gaussian Fourier envelope.

The central result is therefore not only a missing Edgeworth estimate. It is a bridge between the repository’s probability,  $q$ -series, inverse-function, Lambert– $W$ , Bernoulli/Bell, Fourier-product, and automatic-sequence components. The large-deviation and quadratic  $q$ -Gevrey theorems show that the same geometric scaling controls both the parameter boundary  $r \uparrow 1$  and the derivative order  $n \rightarrow \infty$ , but through different Legendre transforms. This duality appears to be a fertile organizing principle for the next phase of the Fabius–Rvachev program.

## A Explicit Hermite polynomials and third-order formulas

The probabilists’ Hermite polynomials used above begin

$$\begin{aligned}
 H_3(x) &= x^3 - 3x, \\
 H_4(x) &= x^4 - 6x^2 + 3, \\
 H_5(x) &= x^5 - 10x^3 + 15x, \\
 H_6(x) &= x^6 - 15x^4 + 45x^2 - 15,
 \end{aligned}$$

$$\begin{aligned}
H_7(x) &= x^7 - 21x^5 + 105x^3 - 105x, \\
H_8(x) &= x^8 - 28x^6 + 210x^4 - 420x^2 + 105.
\end{aligned}$$

The third-order CDF expansion obtained by integrating (5.27) is

$$\begin{aligned}
F_h(x) &= \Phi(x) + \varphi(x) \left[ \frac{h}{20} H_3(x) - h^2 \left( \frac{4}{315} H_5(x) + \frac{1}{800} H_7(x) \right) \right. \\
&\quad \left. + h^3 \left( -\frac{1}{60} H_3(x) + \frac{3}{700} H_7(x) + \frac{1}{1575} H_9(x) + \frac{1}{48000} H_{11}(x) \right) \right] \\
&\quad + \mathcal{O}_{\text{weighted}}(h^4).
\end{aligned} \tag{A.1}$$

## B A coefficient-extraction form of the full Edgeworth series

Treat  $h$  as a formal variable and write the cumulant expansions from Theorem 4.2 as

$$\lambda_{2m}(h) = \sum_{n \geq m-1} \lambda_{2m,n} h^n.$$

Then the coefficient of  $h^N$  in the density expansion is

$$P_N(x) = [h^N] \exp_{\mathcal{H}} \left( \sum_{m \geq 2} \frac{\lambda_{2m}(h)}{(2m)!} H_{2m} \right), \tag{B.1}$$

where  $\exp_{\mathcal{H}}$  means that multiplication is performed before replacing  $(it)^{2k}$  by  $H_{2k}(x)$ . More explicitly,

$$\begin{aligned}
P_N(x) &= \sum_{\substack{k_2, k_3, \dots \geq 0 \\ n_{m,j} \geq 0 \\ \sum_{m,j} j n_{m,j} = N \\ \sum_j n_{m,j} = k_m}} H_{2 \sum_m m k_m}(x) \prod_{m \geq 2} \frac{1}{k_m! (2m)!^{k_m}} \frac{k_m!}{\prod_j n_{m,j}!} \prod_j \lambda_{2m,j}^{n_{m,j}}.
\end{aligned} \tag{B.2}$$

Only finitely many indices occur for each  $N$ , because  $\lambda_{2m,j} = 0$  for  $j < m - 1$ . Formula (B.2) is suitable for exact symbolic generation.

## C Large-deviation series derivation

From (4.4),

$$\begin{aligned}
\Lambda(s) &= \int_0^1 \sum_{k \geq 1} c_k (\sqrt{6} su)^{2k} \frac{du}{u} \\
&= \sum_{k \geq 1} \frac{c_k 6^k}{2k} s^{2k}.
\end{aligned}$$

This yields (7.14). To compute  $I$ , solve  $y = \Lambda'(s)$  recursively:

$$s(y) = y + \frac{1}{5}y^3 + \frac{23}{525}y^5 + \frac{22}{2625}y^7 + \frac{821}{606375}y^9 + \mathcal{O}(y^{11}). \tag{C.1}$$

Substitution into  $I(y) = ys(y) - \Lambda(s(y))$  gives (7.15).

## D Reproducibility and files

This part’s verification material is kept in the volume directory under `assets/q-fabius-parameter-deformations/Continuous_Parameter_Edgeworth_and_q_Gevrey_Frontier/`. The absorbed report’s own source and PDF were deleted when it was merged; git history is their archive. The directory holds:

- `assets/q-fabius-parameter-deformations/Continuous_Parameter_Edgeworth_and_q_Gevrey_Frontier/numerics/fabius_q_edgeworth.py` and `assets/q-fabius-parameter-deformations/Continuous_Parameter_Edgeworth_and_q_Gevrey_Frontier/numerics/README.md`;
- `assets/q-fabius-parameter-deformations/Continuous_Parameter_Edgeworth_and_q_Gevrey_Frontier/data/edgeworth_errors.csv` and `assets/q-fabius-parameter-deformations/Continuous_Parameter_Edgeworth_and_q_Gevrey_Frontier/data/edgeworth_table.tex`;
- `assets/q-fabius-parameter-deformations/Continuous_Parameter_Edgeworth_and_q_Gevrey_Frontier/data/quantile_errors.csv` and `assets/q-fabius-parameter-deformations/Continuous_Parameter_Edgeworth_and_q_Gevrey_Frontier/data/quantile_table.tex`;
- `assets/q-fabius-parameter-deformations/Continuous_Parameter_Edgeworth_and_q_Gevrey_Frontier/data/large_deviation_edge.csv` and `assets/q-fabius-parameter-deformations/Continuous_Parameter_Edgeworth_and_q_Gevrey_Frontier/data/ldp_edge_table.tex`;
- `assets/q-fabius-parameter-deformations/Continuous_Parameter_Edgeworth_and_q_Gevrey_Frontier/figures/edgeworth_error_scaling.pdf`, `assets/q-fabius-parameter-deformations/Continuous_Parameter_Edgeworth_and_q_Gevrey_Frontier/figures/density_edgeworth_comparison.pdf`, and `assets/q-fabius-parameter-deformations/Continuous_Parameter_Edgeworth_and_q_Gevrey_Frontier/figures/large_deviation_rate.pdf`, each with a PNG raster beside it;
- `assets/q-fabius-parameter-deformations/Continuous_Parameter_Edgeworth_and_q_Gevrey_Frontier/README.md`, `assets/q-fabius-parameter-deformations/Continuous_Parameter_Edgeworth_and_q_Gevrey_Frontier/requirements.txt`, and `assets/q-fabius-parameter-deformations/Continuous_Parameter_Edgeworth_and_q_Gevrey_Frontier/MANIFEST.txt`.

The three generated LaTeX tables and the three figures are read from those locations by this volume. To install the numerical dependencies and regenerate every CSV table, LaTeX table, and figure, run from that directory:

```
python -m pip install -r requirements.txt
python numerics/fabius_q_edgeworth.py --output-dir .
```

The volume itself is rebuilt from its own directory by running `pdflatex` on `geometric_q_fabius_frontiers.tex` three times, so that the table of contents and every cross-reference settle.

## E Repository source inventory used in the audit

The main live source files and inventory documents consulted were (each canonical file is in its same-named directory):

- `FabiusFunction_Mathematical_Glossary.tex`;

- `Fabius_Function_and_Rvachev_Up.tex`;
- `Non_Elementarity_of_the_Fabius_Function.tex`;
- `fabius_lean_walkthrough.tex`;
- `semi-formalized-research-frontiers.tex`;
- the draft `MANIFEST.md`, with raw inspection of the Gaussian-boundary report, the consolidated exponents-and- $q$ -series volume, the general-base atomic-function chapters, inverse- $q$  reports, Fourier-decay audits, and candidate overlap documents;
- imported LaTeX sources for Arias de Reyna’s compact-support and arithmetic papers, Allouche–Shallit’s Thue–Morse survey, the lacunary-product papers, and the Thue–Morse Dirichlet-series paper;
- the historical archive README and category inventories.

The public path is

<https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction/docs>.

## Part XII

# Flat Parameter Response and Smooth $q$ -Dynamics at the Fabius Point

## Abstract

This report develops three new, mutually reinforcing directions in the Fabius–Rvachev system. The first concerns the normalized geometric-uniform family

$$X_q = (1 - q) \sum_{j \geq 0} q^j U_j, \quad 0 < q < 1,$$

whose half-base member is the Fabius distribution. We prove that its density is jointly  $C^\infty$  in the parameter and space variables, but that the exact central plateau for  $q \leq 1/2$  produces a continuum of flat nonanalytic fronts. Every mixed derivative of the plateau defect vanishes on those fronts, while the defect is strictly positive immediately on the other side. In particular,

$$\partial_q^r f_q\left(\frac{1}{2}\right)\Big|_{q=1/2} = 2^{r+1}r! \quad (r \geq 0),$$

yet the Taylor series does not represent the right-hand germ. This gives a parameter-space counterpart of the classical nowhere-analytic smoothness of the Fabius function.

The second direction is a linear-response or susceptibility theory at  $q = 1/2$ . A pathwise parameter derivative yields a canonical signed tangent measure. We derive its contractive resolvent expansion, exact CDF and density refinement equations, conditional-transport form, infinite-sinc Fourier transform, Bernoulli cumulants, Bell-polynomial moment jets, and a rational shifted-Legendre spectrum. The first several exact Legendre coefficients are computed; a tempting alternating-sign pattern is shown to fail at degree 16.

The third direction treats composition dynamics. We prove a  $C^\infty$  flat Koenigs linearization theorem for the inverse Fabius map at the midpoint, statement (K), which the absorbed report had left as a hypothesis. Together with the established strict convexity of  $F$ , exact midpoint-endpoint transfer, and complete affine midpoint jet, it gives a unique global  $C^\infty$  diffeomorphism  $\Theta : (0, 1) \rightarrow \mathbb{R}$  satisfying

$$\Theta(F(x)) = 2\Theta(x), \quad \Theta(F^{-1}(x)) = \tfrac{1}{2}\Theta(x), \quad \Theta'(1/2) = 1.$$

The coordinate differs from  $x - 1/2$  by a nonzero flat function. Consequently inverse iterates have the beyond-all-orders expansion

$$F^{-n}(x) = \frac{1}{2} + 2^{-n}\Theta(x) + \mathcal{O}_{x,N}(2^{-Nn}) \quad \text{for every } N,$$

with no intermediate algebraic correction terms. Finally, the known quadratic endpoint law is converted into a logarithmic Böttcher-type height, giving an exact doubling cocycle and a double-exponential asymptotic for forward iterates, and its quotient by  $-\Theta$  is an exact one-periodic dynamical factor. The report ends with explicit conjectures, numerical checks, and a Lean formalization roadmap.

# 1 Scope, corpus audit, and novelty boundary

## 1.1 What was surveyed

The anti-duplication audit used the live documentation tree under `Analysis/FabiusFunction/docs`, with the following hierarchy.

1. The primary exposition `Fabius_Function_and_Rvachev_Up/Fabius_Function_and_Rvachev_Up.tex`, its mathematical glossary, and the separate non-elementarity article were treated as the formalization-backed baseline.
2. The canonical semi-formalized-research-frontiers/semi-formalized-research-frontiers.tex was treated as the principal index of deductions, experiments, and open obligations not yet promoted to the primary exposition.
3. The current draft manifest and the thematic consolidation directories were used as the index of later work on Thue–Morse formulae, exponent sequences and  $q$ -series, finite sinc products, atomic functions beyond the dyadic case, signed/reciprocal  $q$ -Fabius orbits, Lambert  $W$ , spectra and arithmetic, transforms, inverse and sampling problems, Legendre/Lagrange/Rvachev representations, and Fourier decay.
4. The frozen archive was inspected as provenance, not counted as a separate source of new claims. Its own README states that live byte-identical versions were pruned and that the remaining entries are historical snapshots. Re-reading every frozen duplicate as an independent article would therefore bias the duplication audit.

The live frontier README records that eleven former notebooks were consolidated into six canonical thematic syntheses plus a gap register, preserving distinct arguments, numerical evidence, warnings, and proof obligations while compressing redundant repetitions [85]. The current manifest adds the larger thematic draft volumes and recent representation reports [71]. Accordingly, the phrase *new result* below means “not found in the active primary exposition, canonical frontier volume, or current manifest under the audited concepts and equivalent terminology.” It is not a claim of absolute priority over all mathematical literature.

---

**Repository-novel axes identified by the audit.** Exact searches in the active primary and canonical frontier sources found no Fabius-specific Koenigs/Schröder conjugacy, no Böttcher treatment of composition orbits, and no systematic parameter derivative/linear-response theory for the geometric-uniform family. The corpus does contain the ingredients used here: the geometric law and its smooth density, the exact plateau phase, the midpoint transfer, inverse midpoint defect identities, strict shape theorems, Bernoulli–Bell machinery, Legendre expansions, and sharp endpoint asymptotics.

---

## 1.2 Existing inputs deliberately not reproved from first principles

We use the following results from the repository as named inputs.

- The bounded Fabius function  $F$  is  $C^\infty$ , strictly increasing on  $[0, 1]$ , constant outside that interval, symmetric by  $F(1 - x) = 1 - F(x)$ , and satisfies

$$F'(x) = 2F(2x) \quad (0 \leq x \leq 1/2).$$

It is strictly convex on  $[0, 1/2]$ , strictly concave on  $[1/2, 1]$ , and all positive-order endpoint derivatives vanish [90].



- The exact midpoint transfer is

$$F\left(\frac{1}{2} + h\right) = \frac{1}{2} + 2h - F(h), \quad F\left(\frac{1}{2} - h\right) = \frac{1}{2} - 2h + F(h), \quad 0 \leq h \leq \frac{1}{2}, \quad (1.1)$$

and  $F^{(k)}(1/2) = 0$  for every  $k \geq 2$  [90].

- For  $0 < q < 1$ , the normalized geometric-uniform law has a compactly supported  $C^\infty$  density  $f_q$ , a continuous CDF  $F_q$ , reflection symmetry, exact support  $[0, 1]$ , and the affine refinement equation stated in [appendix 2](#) [90].
- At the dyadic endpoint,

$$\log F(2^{-t}) \sim -\frac{\log 2}{2} t^2 \quad (t \rightarrow \infty), \quad (1.2)$$

while the sharp expansion contains a nonconstant periodic correction in an exact lower-Lambert phase [90, 85].

Everything labeled **proved here** is derived in this report from these inputs and standard real/Fourier analysis. Conditional statements and conjectures are explicitly separated. No new theorem in this report has an exact current Lean counterpart; the formal files listed below certify inputs only.

### 1.3 Exact formalization boundary

All paths in this paragraph are relative to `Analysis/FabiusFunction/Lean/FabiusFunction/`. The fixed-parameter probability law, reflection, support, and affine self-similarity are in `GeometricUniformLaw.lean`; the fixed- $q$  CDF and density and their spatial smoothness are in `GeometricUniformCDF.lean`; the half-base identifications are in `ProbabilityRepresentation.lean`; and the fixed-parameter prefix sinc convergence is in `GeometricSincFactorization.lean`. The affine operator and uniqueness theorem are in `AffineIndependentCopy.lean`, while the shape and inverse inputs are in `Convexity.lean` and `FabiusInverse.lean`. Exact midpoint transfer and inverse midpoint defect are in `MidpointEndpointTransfer.lean` and `InverseMidpointDefect.lean`; endpoint flatness and the leading quadratic logarithmic law are in `FabiusFlatness.lean` and `FabiusLogSquaredAsymptotic.lean`. Finally, `CenteredMomentCumulants.lean`, `ExponentialBell.lean`, and `GeometricUniformDictionary.lean` provide fixed-base algebraic infrastructure.

Not presently formalized are joint parameter-space smoothness, the flat parameter fronts, the tangent measure and susceptibility/refinement theory, the response sinc/cumulant/moment/ Legendre calculations, the higher-response source regularity, the flat Koenigs linearization theorem, the global coordinate, the endpoint height, or the dynamical periodic factor. In particular, fixed- $q$  spatial smoothness in `GeometricUniformCDF.lean` must not be cited as joint  $q, x$  smoothness.

### 1.4 Relation to the major existing themes

The new constructions are not isolated from the established corpus. They create a set of bridges:

| Existing theme                | New object                      | Bridge                                                           |
|-------------------------------|---------------------------------|------------------------------------------------------------------|
| Geometric $q$ -law            | Parameter tangent measure $\nu$ | Differentiation of the affine fixed point; contractive resolvent |
| Exact plateau phase           | Flat nonanalytic fronts         | Endpoint flatness transported into parameter space               |
| Infinite sinc product         | Tangent Fourier product         | One marked factor plus a logarithmic derivative sum              |
| Bernoulli cumulants           | Moment $q$ -jets                | $q$ -integers and complete Bell polynomials                      |
| Legendre expansion            | Tangent Legendre spectrum       | Exact rational coefficients from moment derivatives              |
| Midpoint–endpoint transfer    | Global Schröder coordinate      | Flat local linearization theorem extended by inverse iteration   |
| Lambert/periodic endpoint law | Dynamical height $B$            | Doubling cocycle; periodic quotient by the global coordinate     |
| Thue–Morse/dyadic structure   | Composition clocks              | Multiplication by 2 in the global coordinate                     |

## 2 Conventions of this part

This part works in the canonical normalization of Part 0, restricted to  $0 < q < 1$ . In the notation of [definition 0.2.1](#) it writes

$$c_j(q) = (1 - q)q^j, \quad X_q = \sum_{j \geq 0} c_j(q)U_j = \mathsf{X}_q, \quad Z_q = X_q - \frac{1}{2} = \mathsf{X}_q^\circ, \quad (2.1)$$

with  $(U_j)_{j \geq 0}$  independent and uniform on  $[0, 1]$ , and

$$F_q(x) = \mathbb{P}(X_q \leq x), \quad f_q(x) = F'_q(x),$$

so  $F_q = G_q$  and  $f_q = g_q$  in Part 0's letters. Everything this part needs from the shared layer is proved there:  $X_q \in [0, 1]$  with  $\mathbb{E}X_q = \frac{1}{2}$  ([proposition 0.2.2](#)), the identification  $F_{1/2} = F_{\text{bd}}$  with  $f_{1/2} = 2 \text{ up}(2 \cdot -1)$  ([proposition 0.2.5](#)), the independent-copy decomposition

$$X_q \stackrel{d}{=} (1 - q)U_0 + qX'_q \quad (2.2)$$

together with its uniqueness ([proposition 0.2.3](#)), and the centered characteristic function

$$\Phi_q(t) = \mathbb{E}e^{itZ_q} = \prod_{j \geq 0} \text{sinc}_{\text{rad}}\left(\frac{c_j(q)t}{2}\right) = \varphi_q(t) \quad (2.3)$$

with its locally uniform convergence in  $(q, t)$  ([proposition 0.3.1](#)). Cumulants and variance are [theorem 0.4.2](#) and [\(0.4.5\)](#). The one consequence of [\(2.2\)](#) that this part uses and Part 0 does not record is the pair of exact distribution and density relations obtained by conditioning on  $U_0$ ,

$$F_q(x) = \int_0^1 F_q\left(\frac{x - (1 - q)u}{q}\right) \mathrm{d}u, \quad (2.4)$$

$$f_q(x) = \frac{F_q(x/q) - F_q((x - (1 - q))/q)}{1 - q}, \quad (2.5)$$

the second being the  $u$ -derivative of the first after the substitution  $u \mapsto (x - qy)/(1 - q)$ ; both are used throughout this part's response calculus.

### 3 Joint smoothness and flat nonanalytic parameter fronts

#### 3.1 Uniform Schwartz bounds in the parameter

The repository proves smoothness in  $x$  for each fixed  $q$ . The first new step is to make the parameter dependence simultaneous.

**Theorem 3.1** (Joint smoothness of the geometric-uniform density). *The map*

$$(q, x) \mapsto f_q(x)$$

is  $C^\infty$  on  $(0, 1) \times \mathbb{R}$ . More quantitatively, for every compact  $K \Subset (0, 1)$  and every  $a, b, N \in \mathbb{N}_0$ ,

$$\sup_{q \in K} \sup_{t \in \mathbb{R}} (1 + |t|)^N \left| \partial_q^a \partial_t^b \Phi_q(t) \right| < \infty. \quad (3.1)$$

Consequently  $q \mapsto f_q$  is  $C^\infty$  as a map from  $(0, 1)$  into every  $C^m(\mathbb{R})$ , and all parameter derivatives are supported in  $[0, 1]$ .

*Proof. Weights.* Fix  $K = [\alpha, \beta] \Subset (0, 1)$ . Differentiating  $c_j(q) = q^j - q^{j+1}$  termwise, for  $0 \leq l \leq r$  and  $j \geq l$ ,  $\sup_K |\partial_q^l c_j| \leq 2(j+1)^l \beta^{j-l}$ . Hence

$$\sum_{j \geq 0} \sup_{q \in K} |\partial_q^r c_j(q)| < \infty \quad (r \geq 0), \quad (3.2)$$

and in particular  $q \mapsto X_q$  is  $C^\infty$  in  $L^\infty$  under the common-digit coupling. Put  $\varepsilon_j^{(r)} = \max(\sup_K c_j, \max_{1 \leq l \leq r} \sup_K |\partial_q^l c_j|)$ , a summable sequence for each  $r$ .

*One factor.* Since  $\text{sinc}_{\text{rad}} z = \int_0^1 \cos(zs) \, ds$ , differentiation under the integral gives, for real  $z$  and every  $m \geq 0$ ,

$$|\text{sinc}_{\text{rad}}^{(m)}(z)| \leq \int_0^1 s^m \, ds = \frac{1}{m+1}. \quad (3.3)$$

Let  $F_j(q, t) = \text{sinc}_{\text{rad}}(c_j(q)t/2)$ . Then  $\partial_t^b F_j = (c_j/2)^b \text{sinc}_{\text{rad}}^{(b)}(c_j t/2)$ , and applying  $\partial_q^k$  by the Leibniz and Faà di Bruno formulas produces a finite sum of terms, each a product of factors  $\partial_q^l c_j$  ( $l \leq k$ ) with a factor  $(t/2)^i \text{sinc}_{\text{rad}}^{(b+i)}(c_j t/2)$ ,  $i \leq k$ . Every term with  $(k, b) \neq (0, 0)$  contains at least one factor  $c_j$  or  $\partial_q^l c_j$  with  $l \geq 1$ , and the remaining weight factors are bounded uniformly in  $j$  and  $q \in K$ . With (3.3) this gives

$$|\partial_q^k \partial_t^b F_j(q, t)| \leq C_{k,b} (1 + |t|)^k \varepsilon_j^{(k)} \quad ((k, b) \neq (0, 0)), \quad |F_j(q, t)| \leq \min\left(1, \frac{2}{c_j(q)|t|}\right). \quad (3.4)$$

*The tail.* For  $M \leq M'$  let  $P_{M,M'} = \prod_{M \leq j < M'} F_j$ . By the Leibniz rule,  $\partial_q^a \partial_t^b P_{M,M'}$  is a sum, over the ways of distributing the  $a$  parameter and  $b$  frequency derivatives among the factors, of multinomial coefficients times products in which at most  $a + b$  factors are marked by a derivative. Bounding every unmarked factor by 1 and every marked one by (3.4), and grouping the terms by the set  $S$  of marked indices ( $|S| \leq a + b$ ), one obtains

$$|\partial_q^a \partial_t^b P_{M,M'}(q, t)| \leq C_{a,b} (1 + |t|)^a \sum_{s \leq a+b} \frac{1}{s!} \left( \sum_{j \geq M} \varepsilon_j^{(a)} \right)^s \leq C'_{a,b} (1 + |t|)^a, \quad (3.5)$$

uniformly in  $M \leq M'$  and  $q \in K$ . As  $M' \rightarrow \infty$ ,  $P_{M,M'}$  converges locally uniformly in  $(q, t)$  to  $P_M = \prod_{j \geq M} F_j$  (proposition 0.3.1); since all mixed derivatives of order  $\leq m+1$  are bounded uniformly in  $M'$  on compact sets, the derivatives of order  $\leq m$  are equicontinuous, and the Arzelà–Ascoli theorem together with the uniqueness of the limit shows that the convergence holds in  $C^m$  on compact sets for every  $m$ . Hence  $P_M$  is  $C^\infty$  and satisfies (3.5).

*The prefix.* For  $\Phi_{q,M} = \prod_{j < M} F_j$  and  $|t| \geq 1$ , a term of the Leibniz expansion of  $\partial_q^{a'} \partial_t^{b'} \Phi_{q,M}$  has at most  $a' + b'$  marked factors, each  $\leq C(1 + |t|)^{a'}$ , and at least  $M - a' - b'$  unmarked ones, each  $\leq 2/(c_j |t|) \leq 2/((1 - \beta)\alpha^M |t|)$  because  $c_j(q) \geq (1 - \beta)\alpha^j \geq (1 - \beta)\alpha^M$  for  $j < M$ . Thus  $|\partial_q^{a'} \partial_t^{b'} \Phi_{q,M}| \leq C_{M,a',b'}(1 + |t|)^{a' - (M - a' - b')}$  on  $K \times \mathbb{R}$ , the case  $|t| \leq 1$  being trivial.

*The product.* Writing  $\Phi_q = \Phi_{q,M} P_M$  and applying the Leibniz rule once more, each term is bounded by  $C(1 + |t|)^{2a' + b' - M}(1 + |t|)^{a - a'} \leq C(1 + |t|)^{2a + b - M}$ . Choosing  $M \geq N + 2a + b$  proves (3.1).

*Inversion.* The characteristic function of  $X_q = Z_q + \frac{1}{2}$  is  $e^{it/2} \Phi_q(t)$ , so that in the angular convention (0.3.3) of Part 0 one has  $\widehat{f}_{q,\text{ang}}(t) = e^{-it/2} \Phi_q(t)$ , and since  $\Phi_q \in L^1$  by (3.1) with  $N = 2$ ,

$$f_q(x) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{-it(x-1/2)} \Phi_q(t) dt.$$

Applying  $\partial_q^a \partial_x^b$  to the integrand gives  $(2\pi)^{-1} (-it)^b e^{-it(x-1/2)} \partial_q^a \Phi_q(t)$ , whose modulus is bounded, uniformly for  $(q, x) \in K \times \mathbb{R}$ , by  $(2\pi)^{-1} |t|^b \sup_K |\partial_q^a \Phi_q(t)| \leq C_{a,b}(1 + |t|)^{-2}$  by (3.1) with  $N = b + 2$ . Differentiation under the integral sign is therefore justified by dominated convergence,  $f$  is jointly  $C^\infty$ , and

$$\sup_{q \in K} \sup_{x \in \mathbb{R}} |\partial_q^a \partial_x^b f_q(x)| \leq \frac{1}{2\pi} \int_{\mathbb{R}} (1 + |t|)^b \sup_{q \in K} |\partial_q^a \Phi_q(t)| dt < \infty. \quad (3.6)$$

Since  $F_q$  is identically zero to the left of zero and identically one to the right of one for all  $q \in (0, 1)$ , every parameter derivative of the density vanishes off  $[0, 1]$ .  $\square$

*Remark 3.2.* The theorem is a smooth, not analytic, parameter statement. The next results show that this distinction is essential, not technical.

### 3.2 The exact plateau and its moving boundary

For  $q \leq 1/2$ , the head interval  $(1 - q)U_0$  is at least as long as the support  $[0, q]$  of the tail  $qX'_q$ . The convolution therefore has a central plateau. This plateau is already present in the repository's general-base analysis. Its parameter-space consequence is new.

**Proposition 3.3** (Exact plateau and edge defects). *For  $0 < q \leq 1/2$ ,*

$$f_q(x) = \frac{1}{1 - q} \quad (q \leq x \leq 1 - q). \quad (3.7)$$

*Fix  $x \in (0, 1/2)$ , and define  $\Delta(q, x) = (1 - q)^{-1} - f_q(x)$ . Then, in the full neighborhood  $0 < q < 1 - x$  of the left front  $q = x$ ,*

$$\Delta(q, x) = \begin{cases} 0, & 0 < q \leq x, \\ \frac{F_q((q - x)/q)}{1 - q}, & x < q < 1 - x. \end{cases} \quad (3.8)$$

*At the meeting point  $x = 1/2$ ,*

$$\Delta\left(q, \frac{1}{2}\right) = \begin{cases} 0, & 0 < q \leq 1/2, \\ \frac{2F_q((2q - 1)/(2q))}{1 - q}, & 1/2 < q < 1. \end{cases} \quad (3.9)$$

*The right front is obtained by  $x \mapsto 1 - x$ .*

*Proof.* If  $q \leq x \leq 1 - q$ , then  $x/q \geq 1$  and  $(x - (1 - q))/q \leq 0$ . Formula (2.5) gives (3.7).

Now let  $x < 1/2$ . For  $x < q < 1 - x$ , the second CDF argument in (2.5) is negative, while  $x/q \in (0, 1)$ . Thus

$$f_q(x) = \frac{F_q(x/q)}{1 - q} = \frac{1 - F_q(1 - x/q)}{1 - q},$$

which is (3.8). At  $x = 1/2$ , reflection makes the two missing edge masses equal and gives (3.9).  $\square$

**Theorem 3.4** (Flat nonanalytic fronts). *The defect  $\Delta(q, x) = (1 - q)^{-1} - f_q(x)$  is flat on each plateau front:*

$$\partial_q^r \partial_x^s \Delta(q, x)|_{q=x} = 0 \quad (0 < x < 1/2), \quad (3.10)$$

$$\partial_q^r \partial_x^s \Delta(q, x)|_{q=1-x} = 0 \quad (1/2 < x < 1), \quad (3.11)$$

for all  $r, s \geq 0$ , and the same statement holds at  $(q, x) = (1/2, 1/2)$ . Nevertheless  $\Delta$  is strictly positive immediately outside the plateau. Therefore:

1.  $(q, x) \mapsto f_q(x)$  is not real analytic at any point of the two fronts;
2. for every fixed  $x \in (0, 1)$ , the section  $q \mapsto f_q(x)$  is not real analytic at  $q_c = \min(x, 1 - x)$ ;
3. the  $C^\infty(\mathbb{R})$ -valued map  $q \mapsto f_q$  is not real analytic at any  $q \in (0, 1/2]$ .

*Proof.* By [theorem 3.1](#),  $\Delta$  is jointly  $C^\infty$ . On one open side of each front it is identically zero by (3.7). Every mixed derivative is therefore zero on that side and, by continuity, on the boundary. This proves flatness.

For  $x < 1/2$  and  $q > x$  sufficiently close to  $x$ , formula (3.8) has a positive argument. The support of  $X_q$  is the entire interval  $[0, 1]$ , and a direct finite-digit event also shows  $F_q(y) > 0$  for every  $y > 0$ ; hence the defect is strictly positive. The center formula is similarly positive for  $q > 1/2$ .

A real-analytic function whose full Taylor series is zero at a point vanishes in a neighborhood. Thus the positive defect contradicts analyticity in one or two variables. For the function-space statement, a map  $q \mapsto f_q \in C^\infty(\mathbb{R})$  is called real analytic at  $q_0$  if there are  $\varphi_n \in C^\infty(\mathbb{R})$  and  $r > 0$  such that  $f_q = \sum_{n \geq 0} \varphi_n (q - q_0)^n$  for  $|q - q_0| < r$ , the series converging in every seminorm  $\sup_{[-A, A]} |\partial_x^m \cdot|$ ; this is the standard notion for a map into a Fréchet space. Composing such a map with a continuous linear functional yields an ordinary real-analytic scalar function, because the functional passes termwise through the convergent series. Apply this to the evaluation functional at  $x = q_0$  when  $q_0 < 1/2$ , or at  $x = 1/2$  when  $q_0 = 1/2$ : the resulting scalar function  $q \mapsto f_q(x)$  would be real analytic at  $q_0$ , contradicting (2).  $\square$

**Corollary 3.5** (Exact front jets). *For every  $r \geq 0$ ,*

$$\partial_q^r f_q(x)|_{q=x} = \frac{r!}{(1-x)^{r+1}}, \quad 0 < x < 1/2, \quad (3.12)$$

$$\partial_q^r f_q(x)|_{q=1-x} = \frac{r!}{x^{r+1}}, \quad 1/2 < x < 1, \quad (3.13)$$

$$\partial_q^r f_q\left(\frac{1}{2}\right)\bigg|_{q=1/2} = 2^{r+1} r!. \quad (3.14)$$

*Every mixed jet containing at least one  $x$ -derivative vanishes on the front.*

*Proof.* Subtract the flat defect from  $(1 - q)^{-1}$  and differentiate. The last statement follows because the plateau branch has no  $x$ -dependence.  $\square$

**Remark 3.6** (A smooth phase transition). At  $(1/2, 1/2)$ , the Taylor series in  $q - 1/2$  is exactly the convergent Taylor series of  $(1 - q)^{-1}$ . It therefore predicts the plateau continuation on both sides, while the true right-hand function is smaller by the positive, flat term in (3.9). This is a particularly transparent example of a convergent Taylor series that fails to represent its  $C^\infty$  function.

## 4 The first $q$ -response at the Fabius point

#### 4.1 Pathwise differentiation and the tangent measure

Differentiate the weights in the common-digit coupling. At  $q = 1/2$ ,

$$d_j := c'_j(1/2) = \begin{cases} -1, & j = 0, \\ 0, & j = 1, \\ (j-1)2^{-j}, & j \geq 2. \end{cases} \quad \sum_{j \geq 0} d_j = 0. \quad (4.1)$$

Define

$$X = X_{1/2}, \quad Z = X - \frac{1}{2}, \quad V = \sum_{j \geq 0} d_j U_j = \sum_{j \geq 0} d_j \left( U_j - \frac{1}{2} \right). \quad (4.2)$$

**Theorem 4.1** (Pathwise linear response). *For every  $\varphi \in C^1(\mathbb{R})$  whose derivative is bounded,*

$$\left. \frac{d}{dq} \mathbb{E} \varphi(X_q) \right|_{q=1/2} = \mathbb{E}[\varphi'(X)V]. \quad (4.3)$$

*There is a unique compactly supported smooth signed density*

$$\rho(x) = \partial_q f_q(x)|_{q=1/2} \quad (4.4)$$

*representing this functional, and its cumulative response*

$$G(x) = \partial_q F_q(x)|_{q=1/2} = \int_{-\infty}^x \rho(t) dt \quad (4.5)$$

*satisfies*

$$G(1-x) = -G(x), \quad \rho(1-x) = \rho(x), \quad (4.6)$$

$$\int \rho = 0, \quad \int \left( x - \frac{1}{2} \right) \rho(x) dx = 0, \quad \rho\left(\frac{1}{2}\right) = 4. \quad (4.7)$$

*Proof.* Let  $K$  be a compact neighborhood of  $1/2$  in  $(0, 1)$ . By (3.2),  $q \mapsto X_q(\omega) = \sum_j c_j(q)U_j(\omega)$  is  $C^1$  on  $K$  for every  $\omega$ , with derivative  $\sum_j c'_j(q)U_j$ , and the mean value theorem gives the  $\omega$ -independent bound

$$\left| \frac{\varphi(X_{q+h}) - \varphi(X_q)}{h} \right| \leq \|\varphi'\|_\infty \sup_{q' \in K} \left| \sum_j c'_j(q')U_j \right| \leq \|\varphi'\|_\infty \sum_j \sup_K |c'_j| < \infty$$

for  $q, q+h \in K$ . Dominated convergence therefore permits the limit  $h \rightarrow 0$  inside the expectation, and the chain rule gives (4.3) with  $V = \sum_j d_j U_j$ . On the other hand  $\mathbb{E} \varphi(X_q) = \int_0^1 \varphi f_q$ , and (3.6) with  $(a, b) = (1, 0)$  lets one differentiate under the integral: the functional is represented by  $\rho = \partial_q f_q|_{q=1/2}$ , which is smooth and supported in  $[0, 1]$  by theorem 3.1. It is the only continuous compactly supported representative, since a continuous compactly supported function integrating to zero against every  $\varphi \in C_c^\infty$  vanishes. Likewise  $F_q(x) = \int_{-\infty}^x f_q$  may be differentiated in  $q$  under the integral, the integrand  $\partial_q f_q$  being jointly continuous and supported in  $[0, 1]$ , which gives (4.5).

Replacing every digit by  $1 - U_j$  sends  $X$  to  $1 - X$  and, because  $\sum d_j = 0$ , sends  $V$  to  $-V$ . This gives (4.6). Total mass and mean are independent of  $q$ , proving the two integral identities. The center value is the case  $r = 1$  of (3.14).  $\square$

#### 4.2 A contractive susceptibility resolvent

The response is not only a derivative of an infinite product; it is the solution of a contractive fixed-point equation.

Let  $\mathcal{M}_0$  be the space of finite signed measures of total mass zero and finite first moment, equipped with the Kantorovich–Rubinstein norm

$$\|\eta\|_{\text{KR}} = \sup \left\{ \left| \int \phi \, d\eta \right| : \phi(0) = 0, \text{Lip}(\phi) \leq 1 \right\}. \quad (4.8)$$

For  $0 < q < 1$ , define

$$\langle \phi, R_q \eta \rangle = \int_0^1 \int \phi((1-q)u + qx) \, d\eta(x) \, du. \quad (4.9)$$

**Theorem 4.2** (Susceptibility series). *The operator  $R_q$  preserves  $\mathcal{M}_0$  and is a contraction there:*

$$\|R_q \eta\|_{\text{KR}} \leq q \|\eta\|_{\text{KR}}. \quad (4.10)$$

Let  $\mu_q$  be the law of  $X_q$ . The derivative  $\nu_q = \partial_q \mu_q$  exists in total variation (and hence in the KR norm), has smooth compactly supported density  $\partial_q f_q$ , and has total mass zero. Define the finite zero-mass measure

$$\sigma_q := \nu_q - R_q \nu_q. \quad (4.11)$$

For every  $\phi \in C_c^1(\mathbb{R})$ , this measure has the source formula

$$\langle \phi, \sigma_q \rangle = \int_0^1 \int \phi'((1-q)u + qx)(x-u) \, d\mu_q(x) \, du. \quad (4.12)$$

It has a smooth compactly supported density. Moreover,  $\nu_q$  is the unique zero-mass solution of

$$\nu_q = R_q \nu_q + \sigma_q, \quad (4.13)$$

and

$$\nu_q = (I - R_q)^{-1} \sigma_q = \sum_{n=0}^{\infty} R_q^n \sigma_q \quad (4.14)$$

with convergence in  $\|\cdot\|_{\text{KR}}$ .

*Proof.* Write

$$T_q \phi(x) = \int_0^1 \phi((1-q)u + qx) \, du.$$

If  $\text{Lip}(\phi) \leq 1$ , then  $T_q \phi$  has Lipschitz constant at most  $q$ . It need not vanish at the KR basepoint. Since  $\eta(\mathbb{R}) = 0$ , however,

$$\int T_q \phi \, d\eta = \int (T_q \phi - T_q \phi(0)) \, d\eta,$$

and the parenthesized function does vanish at zero. Thus  $\|R_q \eta\|_{\text{KR}} \leq q \|\eta\|_{\text{KR}}$ , proving (4.10); preservation of total mass is immediate.

*Differentiability in total variation.* Fix a compact  $K \Subset (0, 1)$  containing  $q$  in its interior. Taylor's formula with integral remainder in the parameter gives, for  $q + h \in K$  and every  $x$ ,

$$f_{q+h}(x) - f_q(x) - h \partial_q f_q(x) = \int_0^h (h-s) \partial_q^2 f_{q+s}(x) \, ds,$$

and all functions involved vanish off  $[0, 1]$ . Integrating in  $x$  over  $[0, 1]$  and using (3.6) with  $(a, b) = (2, 0)$ ,

$$\|f_{q+h} - f_q - h \partial_q f_q\|_{L^1(\mathbb{R})} \leq \frac{h^2}{2} \sup_{q' \in K} \sup_x |\partial_q^2 f_{q'}(x)|,$$

which is differentiability of  $q \mapsto \mu_q = f_q \, dx$  in total variation with derivative  $\nu_q = \partial_q f_q \, dx$ ; its total mass is  $\partial_q \int f_q = 0$ . Convergence in total variation implies convergence in the KR norm for



zero-mass measures carried by  $[0, 1]$ : a test function with  $\phi(0) = 0$  and  $\text{Lip}(\phi) \leq 1$  satisfies  $|\phi| \leq 1$  on  $[0, 1]$ , so  $\|\eta\|_{\text{KR}} \leq \|\eta\|_{\text{TV}}$  there. The measure  $R_q \nu_q$  has the smooth compactly supported density  $y \mapsto q^{-1} \int_0^1 \partial_q f_q((y - (1 - q)u)/q) du$ , so  $\sigma_q$  in (4.11) is well defined, has zero mass, and has a smooth density supported in  $[0, 1]$ .

*The source formula.* For  $\phi \in C_c^1(\mathbb{R})$  the fixed point (2.2) reads  $\int \phi d\mu_q = \int_0^1 \int_0^1 \phi((1 - q)u + qx) f_q(x) dx du$ . The integrand is jointly  $C^1$  in  $(q, u, x)$  on the compact set  $K \times [0, 1]^2$ , so the right side may be differentiated in  $q$  under the integral. The derivative falling on  $f_q$  gives  $\langle \phi, R_q \nu_q \rangle$ ; the derivative falling on the affine argument gives  $\int_0^1 \int \phi'((1 - q)u + qx)(x - u) d\mu_q(x) du$ . Since the left side differentiates to  $\langle \phi, \nu_q \rangle$ , this is (4.13) with the source (4.12).

*Uniqueness and the series.* If  $\eta \in \mathcal{M}_0$  solves  $\eta = R_q \eta + \sigma_q$ , then  $\eta - \nu_q = R_q(\eta - \nu_q)$ , so by (4.10)  $\|\eta - \nu_q\|_{\text{KR}} \leq q \|\eta - \nu_q\|_{\text{KR}}$ , forcing  $\|\eta - \nu_q\|_{\text{KR}} = 0$  and hence  $\eta = \nu_q$ , because a finite signed measure with finite first moment that integrates every Lipschitz function to zero is zero. For the series, note that  $(\mathcal{M}_0, \|\cdot\|_{\text{KR}})$  is not complete, so a Neumann-series argument is not available; instead iterate (4.13) to the exact identity

$$\nu_q - \sum_{n=0}^{N-1} R_q^n \sigma_q = R_q^N \nu_q, \quad \left\| \nu_q - \sum_{n=0}^{N-1} R_q^n \sigma_q \right\|_{\text{KR}} \leq q^N \|\nu_q\|_{\text{KR}} \longrightarrow 0,$$

which is (4.14), the resolvent notation standing for the unique solution just identified.  $\square$

**Proposition 4.3** (Higher response hierarchy). *For  $s \geq 0$  the measure  $\nu_q^{(s)} = \partial_q^s \mu_q$  exists in total variation and has the smooth density  $\partial_q^s f_q$  supported in  $[0, 1]$ ; it has zero mass for  $s \geq 1$ . For  $j \geq 0$  and a finite signed measure  $\eta$  carried by  $[0, 1]$ , define  $R_q^{(j)} \eta$  by*

$$\begin{aligned} \langle \phi, R_q^{(j)} \eta \rangle &= \int_0^1 \int \phi^{(j)}((1 - q)u + qx)(x - u)^j d\eta(x) du \\ &= \int_0^1 \int \partial_q^j [\phi((1 - q)u + qx)] d\eta(x) du, \quad \phi \in C_c^\infty(\mathbb{R}), \end{aligned} \quad (4.15)$$

so that  $R_q^{(0)} = R_q$ . If  $\eta = h dx$  with  $h \in C^\infty$  supported in  $[0, 1]$ , then  $R_q^{(j)} \eta$  is the finite signed measure with smooth density  $(-1)^j H_{q,j}^{(j)}$  supported in  $[0, 1]$ , where

$$H_{q,j}(y) = q^{-j-1} \int_0^1 (y - u)^j h\left(\frac{y - (1 - q)u}{q}\right) du, \quad (4.16)$$

and it has zero mass when  $j \geq 1$ . For every  $s \geq 1$ ,

$$(I - R_q) \nu_q^{(s)} = \sum_{j=1}^s \binom{s}{j} R_q^{(j)} \nu_q^{(s-j)} := S_{q,s}, \quad (4.17)$$

where  $S_{q,s}$  is a finite zero-mass measure with smooth density supported in  $[0, 1]$ , and consequently

$$\nu_q^{(s)} = (I - R_q)^{-1} S_{q,s} = \sum_{n=0}^{\infty} R_q^n S_{q,s} \quad (4.18)$$

with convergence in the KR norm,  $\nu_q^{(s)}$  being the unique zero-mass solution of  $\nu = R_q \nu + S_{q,s}$ .

*Proof.* The first assertion is the Taylor argument of the proof of theorem 4.2 applied to  $\partial_q^{s-1} f_q$ , using (3.6) with  $(a, b) = (s + 1, 0)$ ; zero mass follows from  $\int f_q = 1$ . In (4.15) the two expressions agree because the argument  $(1 - q)u + qx$  is affine in  $q$  with  $q$ -derivative  $x - u$ , and the right side is a

continuous linear functional on  $C_c^\infty$  since  $\eta$  is finite and carried by  $[0, 1]$ . For  $\eta = h \, dx$ , substitute  $y = (1 - q)u + qx$ , so that  $x = (y - (1 - q)u)/q$ ,  $dx = dy/q$  and  $x - u = (y - u)/q$ :

$$\langle \phi, R_q^{(j)} \eta \rangle = \int_{\mathbb{R}} \phi^{(j)}(y) H_{q,j}(y) \, dy = (-1)^j \int_{\mathbb{R}} \phi(y) H_{q,j}^{(j)}(y) \, dy,$$

after  $j$  integrations by parts,  $H_{q,j}$  being smooth and supported in  $[0, 1]$  because  $y = (1 - q)u + qx \in [0, 1]$  whenever  $u, x \in [0, 1]$ . This is (4.16); for  $j \geq 1$  the density is a derivative of a compactly supported function, so its integral vanishes.

For (4.17), write the fixed point as  $\int \phi \, d\mu_q = \int_0^1 \int_0^1 \phi((1 - q)u + qx) f_q(x) \, dx \, du$ . The integrand is jointly  $C^\infty$  on  $K \times [0, 1]^2$  for every compact  $K \Subset (0, 1)$ , so the right side may be differentiated  $s$  times in  $q$  under the integral, and the Leibniz rule distributes the derivatives between the argument and  $f_q$ :

$$\langle \phi, \nu_q^{(s)} \rangle = \sum_{j=0}^s \binom{s}{j} \int_0^1 \int_0^1 \partial_q^j [\phi((1 - q)u + qx)] \partial_q^{s-j} f_q(x) \, dx \, du = \sum_{j=0}^s \binom{s}{j} \langle \phi, R_q^{(j)} \nu_q^{(s-j)} \rangle.$$

Moving the  $j = 0$  term to the left gives (4.17), and by the density formula each summand of  $S_{q,s}$  is a finite zero-mass measure with smooth density supported in  $[0, 1]$ . The uniqueness and telescoping arguments of theorem 4.2, which used only the contraction (4.10) and the fact that  $\nu_q^{(s)} \in \mathcal{M}_0$ , then give (4.18).  $\square$

### 4.3 Exact refinement equations for the tangent CDF and density

Put  $f = F'$ . Differentiating (2.4) at  $q = 1/2$ , then changing variables from  $u$  to  $v = 2x - u$ , gives the following.

**Theorem 4.4** (Tangent refinement system). *Here  $F \equiv 0$  on  $(-\infty, 0]$  and  $F \equiv 1$  on  $[1, \infty)$ , while  $f, G$  and  $\rho$  vanish off  $[0, 1]$ ; these are the values the four functions take on  $\mathbb{R}$ , because  $F_q$  is constant off  $[0, 1]$  for every  $q$ . Then*

$$G(x) = \int_{2x-1}^{2x} G(v) \, dv + 4 \int_{2x-1}^{2x} (x - v) f(v) \, dv. \quad (4.19)$$

The source can be written without  $f$  as

$$4F(x) - 4xF(2x) - 4(1 - x)F(2x - 1). \quad (4.20)$$

Differentiation in  $x$  yields

$$\rho(x) = 2(G(2x) - G(2x - 1)) + 2f(x) - 8xf(2x) - 8(1 - x)f(2x - 1). \quad (4.21)$$

*Proof.* Put  $A(q, u, x) = (x - (1 - q)u)/q$ , so that  $\partial_q A = (uq - (x - (1 - q)u))/q^2 = (u - x)/q^2$  and  $A(\frac{1}{2}, u, x) = 2x - u$ ,  $\partial_q A|_{q=1/2} = 4(u - x)$ . The map  $(q, y) \mapsto F_q(y)$  is jointly  $C^\infty$ , since  $F_q(y) = \int_{-\infty}^y f_q$  and  $\partial_q f_q$  is jointly continuous with support in  $[0, 1]$  (theorem 3.1); hence  $(q, u) \mapsto F_q(A(q, u, x))$  is jointly  $C^1$  on a compact neighborhood of  $\{\frac{1}{2}\} \times [0, 1]$ , and (2.4) may be differentiated in  $q$  under the integral over  $[0, 1]$ . With  $G_q = \partial_q F_q$  and  $f_q = \partial_y F_q$  this gives, at  $q = \frac{1}{2}$ ,

$$G(x) = \int_0^1 G(2x - u) \, du + 4 \int_0^1 (u - x) f(2x - u) \, du,$$

and the substitution  $v = 2x - u$ , under which  $u - x = x - v$ , is (4.19). Integrating the source by parts,  $4 \int_{2x-1}^{2x} (x - v) f(v) \, dv = 4[(x - v)F(v)]_{2x-1}^{2x} + 4 \int_{2x-1}^{2x} F(v) \, dv$ , and the half-base CDF identity

$F(x) = \int_{2x-1}^{2x} F(v) dv$  (the case  $q = \frac{1}{2}$  of (2.4)) turns this into (4.20). Finally,  $G$  and  $v \mapsto (x - v)f(v)$  are continuous on  $\mathbb{R}$ , so the Leibniz rule for the variable limits applies to (4.19) and gives

$$G'(x) = 2G(2x) - 2G(2x - 1) + 4 \int_{2x-1}^{2x} f(v) dv + 4[2(x - 2x)f(2x) - 2(x - 2x + 1)f(2x - 1)],$$

where  $4 \int_{2x-1}^{2x} f = 4(F(2x) - F(2x - 1)) = 2f(x)$  by the case  $q = \frac{1}{2}$  of (2.5); this is (4.21).  $\square$

*Remark 4.5.* At  $x = 1/2$ , the boundary values of  $f$  vanish and  $G(0) = G(1) = 0$ ; hence (4.21) immediately recovers  $\rho(1/2) = 2f(1/2) = 4$ .

#### 4.4 Conditional transport form

**Proposition 4.6** (Canonical velocity field and conditional versions). *Let  $\tilde{b}(x) = \mathbb{E}[V \mid X = x]$  be any regular conditional version. The half-base density is strictly positive on  $(0, 1)$ , and every such version agrees Lebesgue-a.e. there with the canonical representative*

$$b_*(x) := -\frac{G(x)}{f(x)}. \quad (4.22)$$

The function  $b_*$  is smooth and odd on  $(0, 1)$ , and it satisfies the pointwise identities

$$b_*(x)f(x) = -G(x), \quad \rho(x) = -\frac{d}{dx}(b_*(x)f(x)), \quad b_*(1-x) = -b_*(x). \quad (4.23)$$

For an arbitrary regular conditional version, the product identity holds only a.e. and the continuity equation only distributionally:

$$\tilde{b}f = -G \quad \text{a.e. on } (0, 1), \quad \rho = -(\tilde{b}f)' \quad \text{in } \mathcal{D}'(\mathbb{R}).$$

Thus the  $q$ -response is a one-dimensional continuity equation with canonical velocity  $b_*$ ; null-set changes to a conditional version carry no pointwise content.

*Proof.* For a smooth compactly supported test function,

$$\int \varphi(x)\rho(x) dx = \mathbb{E}[\varphi'(X)V] = \int \varphi'(x)\tilde{b}(x)f(x) dx.$$

Hence  $\rho = -(\tilde{b}f)'$  in distributions. Extend  $\tilde{b}f$  by zero off  $[0, 1]$ . Since  $V$  is bounded, this product is integrable and compactly supported;  $G$  is also compactly supported because  $\int \rho = 0$ . Therefore  $(\tilde{b}f + G)' = 0$  on  $\mathbb{R}$ , and compact support forces  $\tilde{b}f = -G$  a.e.

At half base,  $f(x) = 2(F(2x) - F(2x - 1))$  by (2.5), and this is strictly positive for  $0 < x < 1$ : if  $x \leq \frac{1}{2}$  then  $F(2x - 1) = 0 < F(2x)$ , and if  $x \geq \frac{1}{2}$  then  $F(2x) = 1 > F(2x - 1)$ , because  $F$  is strictly increasing on  $[0, 1]$  and constant outside it (a repository input listed in [appendix 1](#); equivalently,  $\mu > 0$  on  $(-1, 1)$ ). Thus  $\mu$  and Lebesgue measure have the same null sets in the interior, and every conditional version equals  $b_* = -G/f$  there a.e. Smoothness follows from smoothness of  $G$ ,  $f$  and interior positivity. Finally,  $G(1 - x) = -G(x)$  and  $f(1 - x) = f(x)$ , so the canonical representative is odd. Its product with  $f$  equals  $-G$  pointwise, and differentiation yields the pointwise continuity equation.  $\square$

## 5 Fourier, Bernoulli–Bell, and Legendre faces of the response

### 5.1 A marked infinite-sinc product

At  $q = 1/2$ , write  $c_j = 2^{-j-1}$ . The centered Fourier transform of the tangent density is the parameter derivative of  $\Phi_q$ .

**Theorem 5.1** (Tangent sinc product). *For every real or complex  $t$ ,*

$$\dot{\Phi}(t) := \partial_q \Phi_q(t)|_{q=1/2} = \sum_{j \geq 0} d_j \frac{t}{2} \operatorname{sinc}'_{\text{rad}}\left(\frac{t}{2^{j+2}}\right) \prod_{k \neq j} \operatorname{sinc}_{\text{rad}}\left(\frac{t}{2^{k+2}}\right), \quad (5.1)$$

where  $d_j$  is given by (4.1). *Off the zero set of  $\Phi$ , this is equivalently*

$$\dot{\Phi}(t) = \Phi(t) \sum_{j \geq 0} d_j \frac{t}{2} \left[ \cot\left(\frac{c_j t}{2}\right) - \frac{2}{c_j t} \right]. \quad (5.2)$$

*The apparent poles in (5.2) are removable after multiplication by  $\Phi$ , as certified by the entire expression (5.1). The noncentered Fourier transform is  $\hat{\rho}_{\text{ang}}(-t) = e^{it/2} \dot{\Phi}(t)$ .*

*Proof.* Allow  $q$  and  $t$  to be complex, with  $|q| \leq \varrho < 1$  and  $|t| \leq R$ . From the Taylor series,  $|\operatorname{sinc}_{\text{rad}} z - 1| \leq \sum_{k \geq 1} |z|^{2k} / (2k+1)! \leq \frac{1}{6} |z|^2 e^{|z|}$  and  $|\operatorname{sinc}'_{\text{rad}}(z)| \leq \sum_{k \geq 1} 2k |z|^{2k-1} / (2k+1)! \leq \frac{1}{3} |z| e^{|z|}$  for all complex  $z$ . With  $z_j = c_j(q)t/2$  and  $|c_j(q)| \leq 2\varrho^j$  one gets  $\sum_j |\operatorname{sinc}_{\text{rad}} z_j - 1| \leq \frac{R^2}{6} e^R \sum_j \varrho^{2j} < \infty$ , uniformly on the set considered. By the Weierstrass criterion the product (2.3) converges uniformly there, so  $\Phi_q(t)$  is holomorphic in  $(q, t)$  on  $\{|q| < 1\} \times \mathbb{C}$  and, by Weierstrass' theorem on uniformly convergent sequences of holomorphic functions,  $\partial_q$  of the limit is the locally uniform limit of  $\partial_q$  of the partial products. The latter is, by the product rule,  $\sum_{j < M} c'_j(q) \frac{t}{2} \operatorname{sinc}'_{\text{rad}}(c_j t/2) \prod_{k < M, k \neq j} \operatorname{sinc}_{\text{rad}}(c_k t/2)$ , which at  $q = \frac{1}{2}$  is the  $M$ -th partial sum of (5.1) because  $c'_j(\frac{1}{2}) = d_j$  and  $c_j(\frac{1}{2})t/2 = t/2^{j+2}$ . The series converges locally uniformly in  $t \in \mathbb{C}$  on its own: since  $\prod_{k \neq j} |\operatorname{sinc}_{\text{rad}}(t/2^{k+2})| \leq \prod_k (1 + \frac{R^2}{6 \cdot 4^{k+2}} e^R) \leq e^{R^2 e^R / 72}$ , its  $j$ -th term is bounded by  $|d_j| \frac{R}{2} \cdot \frac{R}{3 \cdot 2^{j+2}} e^R e^{R^2 e^R / 72} = O(j 4^{-j})$ . Hence (5.1) holds for every complex  $t$ , and both sides are entire. The logarithmic form (5.2) uses  $\operatorname{sinc}'_{\text{rad}}(z) / \operatorname{sinc}_{\text{rad}}(z) = \cot z - z^{-1}$  off the zeros of  $\Phi$ , and the statement about  $\hat{\rho}_{\text{ang}}$  is the  $q$ -derivative of the relation  $\hat{f}_{q \text{ ang}}(-t) = e^{it/2} \Phi_q(t)$  from the proof of theorem 3.1.  $\square$

## 5.2 Closed cumulants and first moment jets

Let  $B_r$  denote the Bernoulli numbers with  $B_1 = -1/2$ , and let

$$[r]_q = \frac{1 - q^r}{1 - q}$$

be the  $q$ -integer.

**Theorem 5.2** (Bernoulli- $q$ -integer cumulants). *For  $r \geq 2$ , the centered cumulants of  $Z_q$  are*

$$\kappa_r(q) = \frac{B_r}{r} \frac{(1 - q)^r}{1 - q^r} = \frac{B_r}{r} \frac{(1 - q)^{r-1}}{[r]_q}. \quad (5.3)$$

*At the Fabius point,*

$$\kappa_r(1/2) = \frac{B_r}{r(2^r - 1)}, \quad (5.4)$$

$$\dot{\kappa}_r := \partial_q \kappa_r(q)|_{q=1/2} = -2B_r \frac{2^r - 2}{(2^r - 1)^2}. \quad (5.5)$$

*All odd values with  $r > 1$  vanish.*

*Proof.* The centered uniform digit has cumulant generating function

$$\log \frac{\sinh(z/2)}{z/2} = \sum_{r \geq 2} \frac{B_r}{r} \frac{z^r}{r!}.$$

Since  $Z_q = X_q^\circ$ , (5.3) is [theorem 0.4.2](#) in the letters of this part; the interchange of the sum over  $j$  with the cumulant expansion is justified there. Concretely, with  $\sinh w = \sinh w/w$  one has  $|\log \sinh w| \leq |w|^2$  for complex  $|w| \leq 1$ , so  $K_q(z) = \log \mathbb{E} e^{zZ_q} = \sum_j \log \sinh(c_j(q)z/2)$  converges uniformly on  $|z| \leq 2/(1-q)$  and its Taylor coefficients at 0 are the sums of the termwise coefficients; summing  $c_j(q)^r$  over  $j$  gives (5.3). At  $q = \frac{1}{2}$ ,  $(1-q)^r/(1-q^r) = 1/(2^r - 1)$ , which is (5.4); and

$$\frac{d}{dq} \frac{(1-q)^r}{1-q^r} = \frac{-r(1-q)^{r-1}(1-q^r) + rq^{r-1}(1-q)^r}{(1-q^r)^2} \Big|_{q=1/2} = \frac{r 2^{1-2r} (2-2^r)}{(2^r-1)^2/2^{2r}} = -\frac{2r(2^r-2)}{(2^r-1)^2},$$

which multiplied by  $B_r/r$  is (5.5).  $\square$

Let  $\mu_n(q) = \mathbb{E} Z_q^n$ ,  $\mu_n = \mu_n(1/2)$ , and  $\dot{\mu}_n = \partial_q \mu_n(q)|_{1/2}$ .

**Corollary 5.3** (Bell moment response). *For every  $n \geq 1$ ,*

$$\dot{\mu}_n = \sum_{r=1}^n \binom{n}{r} \dot{\kappa}_r \mu_{n-r}. \quad (5.6)$$

*In particular, for even orders,*

$$\dot{\mu}_{2n} = \sum_{m=1}^n \binom{2n}{2m} \dot{\kappa}_{2m} \mu_{2n-2m}. \quad (5.7)$$

*The first values are*

| $n$ | $\mu_n$                 | $\dot{\mu}_n$                       |
|-----|-------------------------|-------------------------------------|
| 2   | 1/36                    | -2/27                               |
| 4   | 19/10800                | -83/10125                           |
| 6   | 583/3810240             | -599/617400                         |
| 8   | 132809/8328096000       | -57910967/464551605000              |
| 10  | 46840699/24990950476800 | -32576528018233/1901452086371376000 |

*Proof.* Let  $M_q(z) = \mathbb{E} e^{zZ_q} = \exp K_q(z)$ . Since  $|Z_q| \leq \frac{1}{2}$ , one has  $|\mu_n(q)| \leq 2^{-n}$ , so  $M_q(z) = \sum_n \mu_n(q) z^n/n!$  is entire; and by the pathwise derivative of [theorem 4.1](#),  $\partial_q \mu_n(q) = n \mathbb{E}[Z_q^{n-1} \partial_q Z_q]$  with  $|\partial_q \mu_n(q)| \leq n 2^{1-n} \sum_j \sup_K |c'_j|$ , so  $\partial_q M_q(z)$  is obtained by differentiating the series termwise in  $q$ . On the disc  $|z| < 2/(1-q)$  of the previous proof,  $M_q$  has no zeros and  $K_q = \log M_q$  is analytic in  $z$  and  $C^1$  in  $q$ , with  $\partial_q K_q(z) = \partial_q M_q(z)/M_q(z) = \sum_r \partial_q \kappa_r(q) z^r/r!$ , the last series converging because  $|\partial_q \kappa_r(q)| \leq C_K r |B_r| (1-\alpha)^{r-1}$  on  $K = [\alpha, \beta]$  and  $|B_r| \leq 4r!/(2\pi)^r$ . Hence  $\partial_q M_q = M_q \partial_q K_q$  is an identity between convergent power series in  $z$ , and comparing the coefficients of  $z^n/n!$  gives (5.6).  $\square$

**Proposition 5.4** (All parameter jets and complete Bell polynomials). *For  $s \geq 1$ , put  $K_j(z) = \partial_q^j \log M_q(z)$ . Then*

$$\partial_q^s M_q(z) = M_q(z) \mathcal{B}_s(K_1(z), \dots, K_s(z)), \quad (5.8)$$

*where  $\mathcal{B}_s$  is the complete exponential Bell polynomial. At  $q = 1/2$ , every coefficient is rational. More precisely, the odd part of the denominator of  $\partial_q^s \kappa_r(1/2)$  divides the odd part of*

$$\text{den} \left( \frac{B_r}{r} \right) (2^r - 1)^{s+1}. \quad (5.9)$$

*Proof.* Equation (5.8) is the Faà di Bruno formula for the exponential. By induction on  $s$ ,  $\partial_q^s \kappa_r(q) = \frac{B_r}{r} P_{r,s}(q)/(1-q^r)^{s+1}$  with  $P_{r,s} \in \mathbb{Z}[q]$ : indeed  $P_{r,0} = (1-q)^r$ , and  $\partial_q [P/(1-q^r)^{s+1}] = [P'(1-q^r) + (s+1)r q^{r-1} P]/(1-q^r)^{s+2}$  keeps integer coefficients. At  $q = \frac{1}{2}$ ,  $P_{r,s}(\frac{1}{2})$  is an integer divided by a power of two, and  $(1-2^{-r})^{s+1} = (2^r-1)^{s+1} 2^{-r(s+1)}$ ; hence the odd part of the denominator of  $\partial_q^s \kappa_r(\frac{1}{2})$  divides the odd part of (5.9).  $\square$

### 5.3 An exact rational shifted-Legendre spectrum

Let  $P_k$  be the ordinary Legendre polynomial and  $P_k^*(x) = P_k(2x - 1)$ . Orthogonality on  $[0, 1]$  gives

$$\int_0^1 P_k^*(x) P_\ell^*(x) dx = \frac{\delta_{k\ell}}{2k+1}.$$

Since  $\rho$  is symmetric about  $1/2$ , only even degrees occur.

**Theorem 5.5** (Legendre coefficients of the tangent density). *Write*

$$\rho(x) = \sum_{n \geq 1} a_{2n} P_{2n}(2x - 1). \quad (5.10)$$

The expansion converges in  $C^m[0, 1]$  for every  $m$ , and the coefficients are rational. If  $P_{2n}(y) = \sum_{j=0}^n p_{n,j} y^{2j}$ , then

$$a_{2n} = (4n+1) \sum_{j=1}^n p_{n,j} 2^{2j} \dot{\mu}_{2j}. \quad (5.11)$$

The first coefficients are

$$\begin{aligned} a_2 &= -\frac{20}{9}, & a_4 &= \frac{1088}{225}, & a_6 &= -\frac{37076}{11025}, \\ a_8 &= \frac{49280096}{75907125}, & a_{10} &= -\frac{65166997444}{756561977325}, & a_{12} &= \frac{10886281949792992}{31325448671141625}. \end{aligned} \quad (5.12)$$

Moreover  $a_{2n} = \mathcal{O}_A(n^{-A})$  for every  $A > 0$ .

*Proof.* Orthogonality gives

$$a_k = (2k+1) \int_0^1 \rho(x) P_k(2x-1) dx = (2k+1) \partial_q \mathbb{E} P_k(2Z_q) |_{q=1/2}.$$

Expanding the polynomial proves (5.11). Rationality follows from [corollary 5.3](#). For the decay put  $u(y) = \rho((1+y)/2)$ , a  $C^\infty$  function on  $[-1, 1]$  (indeed on  $\mathbb{R}$ ), so that  $a_k = \frac{2k+1}{2} \int_{-1}^1 u(y) P_k(y) dy$ , and let  $\mathcal{L}v = ((1-y^2)v')'$  be the Legendre operator, with  $\mathcal{L}P_k = -k(k+1)P_k$ . For any  $v, w \in C^2[-1, 1]$ , two integrations by parts give  $\int_{-1}^1 v \mathcal{L}w = \int_{-1}^1 (\mathcal{L}v) w$ , the boundary terms  $[v(1-y^2)w']_{-1}^1$  and  $[(1-y^2)v'w]_{-1}^1$  vanishing because the weight  $1-y^2$  does — no endpoint flatness of  $u$  is needed. Iterating,

$$|a_k| = \frac{2k+1}{2(k(k+1))^m} \left| \int_{-1}^1 (\mathcal{L}^m u) P_k \right| \leq \frac{(2k+1) \|\mathcal{L}^m u\|_{L^1[-1,1]}}{2(k(k+1))^m} \quad (m \geq 0),$$

which is  $a_k = \mathcal{O}_A(k^{-A})$  for every  $A$ . For the  $C^m$  convergence, the Markov inequality  $\|p'\|_{L^\infty[-1,1]} \leq (\deg p)^2 \|p\|_{L^\infty[-1,1]}$ , iterated  $m$  times with  $\|P_k\|_\infty = 1$ , gives  $\|P_k^{(m)}\|_{L^\infty[-1,1]} \leq k^{2m}$ , so  $\|\frac{d^m}{dx^m} P_{2n}(2x-1)\|_{L^\infty[0,1]} \leq 2^m (2n)^{2m}$  and  $\sum_n |a_{2n}| 2^m (2n)^{2m} < \infty$ ; the series (5.10) therefore converges in  $C^m[0, 1]$  for every  $m$ , and its limit is  $\rho$  because it is the  $L^2$  limit.  $\square$

**Remark 5.6** (A useful failed pattern). The signs alternate through degrees 2, 4,  $\dots$ , 14, which might suggest  $(-1)^n a_{2n} > 0$ . Exact arithmetic disproves this at degree 16:

$$a_{16} = -\frac{20997028161526723805395541511872}{64367687808725429829005095415625} \approx -0.3262044805.$$

Thus any eventual sign theorem must involve a more subtle phase, and the low-degree pattern should not be promoted to a conjecture without additional asymptotic analysis.

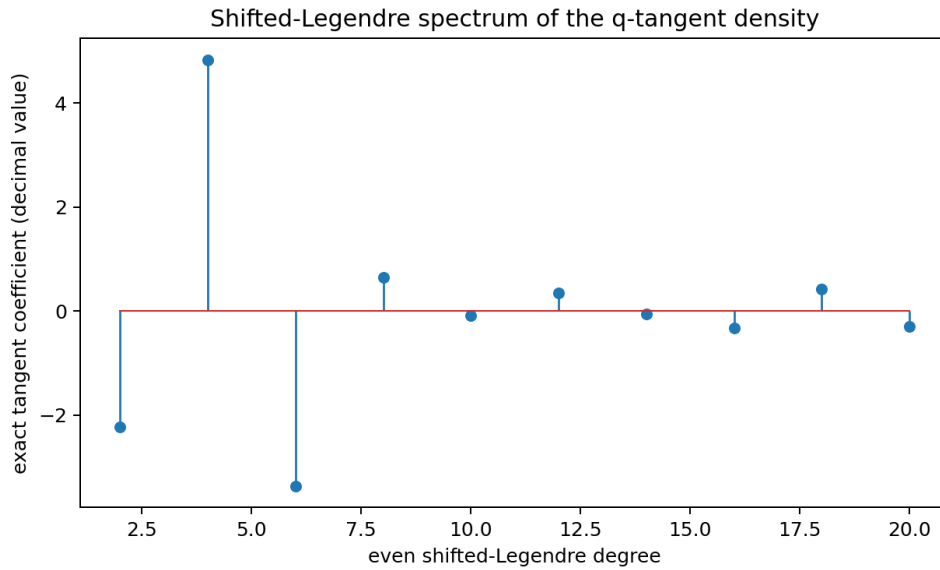


Figure 52: Exact shifted-Legendre coefficients of the tangent density through degree 20, shown in decimal form. The same-sign pair at degrees 14 and 16 is visible.

## 6 A global smooth Schröder coordinate for Fabius iteration

Let

$$p = \frac{1}{2}, \quad Q = F^{-1} : (0, 1) \rightarrow (0, 1).$$

The primary exposition already supplies the smooth inverse, strict curvature, and the exact midpoint defect. After the dynamical preliminaries, the local linearization statement (K) is isolated and proved, and the global composition claims follow from it.

### 6.1 The global basin of the midpoint

**Lemma 6.1** (Diagonal inequalities and inverse attraction). *For  $0 < x < p$ ,  $F(x) < x$ ; for  $p < x < 1$ ,  $F(x) > x$ . Consequently*

$$Q^n(x) \longrightarrow p \quad (x \in (0, 1)). \quad (6.1)$$

*Proof.* On  $[0, p]$ , strict convexity places the graph strictly below the chord joining  $(0, 0)$  and  $(p, p)$ , which is the diagonal. Reflection gives the right inequality. If  $x < p$ , then  $Q(x) > x$  and  $Q(x) < p$ ; the increasing bounded orbit converges to a fixed point. The only fixed points in  $(0, p]$  are  $p$ , because the diagonal inequality excludes all others. The right half is symmetric.  $\square$

### 6.2 A flat local contraction

Center the inverse at  $p$ :

$$g(y) = Q(p + y) - p, \quad |y| < p. \quad (6.2)$$

The midpoint transfer (1.1) implies, for  $\delta \geq 0$ ,

$$g(\delta) = \frac{\delta}{2} + \frac{1}{2}F(g(\delta)). \quad (6.3)$$

Reflection makes  $g$  odd.



**Lemma 6.2** (Flat local normal form). *There is a smooth flat function  $r$  at zero such that*

$$g(y) = \frac{y}{2} + r(y), \quad r^{(k)}(0) = 0 \quad (k \geq 0). \quad (6.4)$$

*The function  $r$  is odd and nonzero; for  $y > 0$ ,  $r(y) > 0$ .*

*Proof.* Equation (6.3) gives  $r(\delta) = F(g(\delta))/2$  for  $\delta \geq 0$ . Endpoint flatness says  $F(t) = o(t^N)$  as  $t \downarrow 0$  for every  $N$ , while  $g(\delta) = \mathcal{O}(\delta)$ . Hence  $r$  is flat from the right, and oddness gives the full statement. Positivity follows from  $F(t) > 0$  for  $t > 0$ .  $\square$

The local  $C^\infty$  linearization is the one analytic step that the absorbed report did not complete: it isolated the statement as a hypothesis (K), and the later results were stated conditionally on it. The statement is kept under its name, because the later results refer to it, and is proved in [theorem 6.4](#).

**Definition 6.3** (Flat Koenigs statement (K)). For the map  $g$  in (6.2), statement (K) is the assertion that on a neighborhood of zero the limit

$$\theta(y) = \lim_{n \rightarrow \infty} 2^n g^n(y) \quad (6.5)$$

exists in  $C^\infty$ , is a local increasing diffeomorphism, and satisfies

$$\theta(g(y)) = \frac{1}{2}\theta(y), \quad \theta'(0) = 1, \quad \theta(y) - y \text{ is flat at } 0. \quad (6.6)$$

It also asserts uniqueness among local  $C^1$  solutions with derivative one at zero.

**Theorem 6.4** (Flat Koenigs linearization). *Statement (K) of [definition 6.3](#) holds.*

*Proof.* *A contracting interval.* By [lemma 6.2](#),  $g = \frac{1}{2}\text{id} + r$  with  $r$  smooth and flat at 0; choose  $\delta_0 \in (0, p)$  with  $|r'| \leq \frac{1}{4}$  on  $J = [-\delta_0, \delta_0]$ . Then  $\frac{1}{4} \leq g' \leq \frac{3}{4}$  on  $J$ , so  $g$  is increasing, maps  $J$  into itself, and

$$|g^k(y)| \leq (3/4)^k |y| \quad (y \in J, k \geq 0). \quad (6.7)$$

Flatness means that for every  $j \geq 0$  and  $N \geq 0$  there is  $C_{j,N}$  with  $|r^{(j)}(u)| \leq C_{j,N}|u|^N$  on  $J$ .

*Derivatives of the iterates.* We claim that for every  $m \geq 1$  there is  $A_m$  with

$$\sup_{y \in J} |(g^k)^{(m)}(y)| \leq A_m (3/4)^k \quad (k \geq 0). \quad (6.8)$$

For  $m = 1$ ,  $(g^k)'(y) = \prod_{i < k} g'(g^i(y)) \leq (3/4)^k$ . Let  $m \geq 2$  and assume (6.8) for all orders below  $m$ . By the Faà di Bruno formula applied to  $g^k = g \circ g^{k-1}$ ,

$$(g^k)^{(m)} = g'(g^{k-1}) (g^{k-1})^{(m)} + \sum_{j=2}^m g^{(j)}(g^{k-1}) B_{m,j}((g^{k-1})', \dots, (g^{k-1})^{(m-j+1)}),$$

where  $B_{m,j}$  is the partial Bell polynomial. For  $j \geq 2$ ,  $g^{(j)} = r^{(j)}$ , so by flatness of order 2 and (6.7),  $|g^{(j)}(g^{k-1}(y))| \leq C_{j,2}\delta_0^2(3/4)^{2(k-1)}$ ; the arguments of  $B_{m,j}$  have orders at most  $m-1$ , hence are bounded by the induction hypothesis. Writing  $a_k = \sup_J |(g^k)^{(m)}|$  this gives  $a_k \leq \frac{3}{4}a_{k-1} + C(3/4)^{2k}$  with  $a_0 = 0$ , and unrolling,  $a_k \leq C \sum_{i=1}^k (3/4)^{k-i} (3/4)^{2i} \leq C(3/4)^k \sum_{i \geq 1} (3/4)^i$ , which is (6.8).

*The telescoped series.* Since  $g(y) - \frac{1}{2}y = r(y)$ , induction on  $n$  gives the telescoping identity

$$2^n g^n(y) = y + \sum_{k=0}^{n-1} 2^{k+1} r(g^k(y)), \quad y \in J. \quad (6.9)$$

Put  $T_k(y) = 2^{k+1}r(g^k(y))$ . By (6.7) and flatness of order 3,  $|T_k(y)| \leq 2^{k+1}C_{0,3}\delta_0^3(3/4)^{3k} = C(27/32)^k$ . For  $m \geq 1$ , the Faà di Bruno formula gives

$$T_k^{(m)} = 2^{k+1} \sum_{j=1}^m r^{(j)}(g^k) B_{m,j}((g^k)', \dots, (g^k)^{(m-j+1)}),$$

in which  $|r^{(j)}(g^k(y))| \leq C_{j,3}\delta_0^3(3/4)^{3k}$  and, by (6.8),  $|B_{m,j}| \leq C_m(3/4)^{jk} \leq C_m(3/4)^k$ ; hence

$$\sup_J |T_k^{(m)}| \leq C'_m 2^{k+1} (3/4)^{4k} = 2C'_m (81/128)^k.$$

Every derivative series  $\sum_k T_k^{(m)}$ ,  $m \geq 0$ , therefore converges uniformly on  $J$ , so  $\theta(y) = \lim_n 2^n g^n(y) = y + \sum_{k \geq 0} T_k(y)$  exists, is  $C^\infty$  on  $J$ , and the convergence in (6.5) holds in every  $C^m(J)$ .

*The properties in (6.6).* Passing to the limit in  $2^n g^n(g(y)) = \frac{1}{2} \cdot 2^{n+1} g^{n+1}(y)$  gives  $\theta(g(y)) = \frac{1}{2}\theta(y)$ . Since  $(g^n)'(0) = g'(0)^n = 2^{-n}$ , differentiating the limit termwise gives  $\theta'(0) = 1$ , so  $\theta$  is an increasing diffeomorphism near 0. Each  $T_k$  is flat at 0, because  $g^k(0) = 0$  and every derivative of  $r \circ g^k$  at 0 is a sum of products containing some  $r^{(j)}(0) = 0$ ; as  $\sum_k T_k^{(m)}$  converges uniformly, the  $m$ -th derivative of  $\theta - \text{id} = \sum_k T_k$  at 0 is  $\sum_k T_k^{(m)}(0) = 0$ , so  $\theta - \text{id}$  is flat. Finally, if  $\psi$  is  $C^1$  near 0 with  $\psi(g(y)) = \frac{1}{2}\psi(y)$  and  $\psi'(0) = 1$ , then for  $y \neq 0$  in  $J$ ,  $g^n(y) \neq 0$  and

$$\psi(y) = 2^n \psi(g^n(y)) = 2^n g^n(y) \cdot \frac{\psi(g^n(y))}{g^n(y)} \longrightarrow \theta(y) \cdot \psi'(0) = \theta(y),$$

which is the uniqueness assertion.  $\square$

---

**Status of (K).** The absorbed report assumed statement (K) as a hypothesis and stated everything involving  $\Theta$ , its inverse-iterate expansion, its escape clock, and the periodic quotient with  $B$  conditionally. [theorem 6.4](#) proves it, so those results are unconditional in this volume; the phrase “statement (K)” below is a pointer to that theorem. No current Lean theorem covers it.

---

### 6.3 Globalization and exact conjugacy

**Theorem 6.5** (Global Fabius Schröder coordinate). *There is a unique  $C^\infty$  increasing diffeomorphism*

$$\Theta : (0, 1) \longrightarrow \mathbb{R} \tag{6.10}$$

with

$$\Theta(p) = 0, \quad \Theta'(p) = 1, \tag{6.11}$$

that satisfies either, hence both, of the conjugacy equations

$$\boxed{\Theta(Q(x)) = \frac{1}{2}\Theta(x), \quad \Theta(F(x)) = 2\Theta(x).} \tag{6.12}$$

It is odd under reflection,

$$\Theta(1-x) = -\Theta(x), \tag{6.13}$$

and has the global limit representation

$$\boxed{\Theta(x) = \lim_{n \rightarrow \infty} 2^n \left( Q^n(x) - \frac{1}{2} \right).} \tag{6.14}$$

At the midpoint,

$$\Theta(p+y) = y + \text{Flat}(y), \quad \Theta^{-1}(t) = p + t + \text{Flat}(t), \tag{6.15}$$

where each displayed remainder is nonzero and flat.

*Proof.* [theorem 6.4](#) supplies the local coordinate  $\theta$  of statement (K). By [lemma 6.1](#), every interior point enters the local neighborhood under some inverse iterate. If  $m$  is large enough, define

$$\Theta(x) = 2^m \theta(Q^m(x) - p). \quad (6.16)$$

Since  $Q^{m+1}(x) - p = g(Q^m(x) - p)$  by (6.2), the local equation  $\theta(g(y)) = \theta(y)/2$  gives  $2^{m+1}\theta(Q^{m+1}(x) - p) = 2^m\theta(Q^m(x) - p)$ , so the value is independent of the admissible  $m$ . If  $Q^m(x_0) - p$  lies in the open interval on which  $\theta$  is defined, then by continuity of  $Q^m$  so does  $Q^m(x) - p$  for all  $x$  in a neighborhood of  $x_0$ ; on that neighborhood  $\Theta$  is given by (6.16) with the fixed  $m$ , hence is  $C^\infty$ . Its derivative is positive because  $Q^m$  and  $\theta$  are increasing. The conjugacy and normalization are immediate, and the limit formula follows from the local limit after a finite initial segment.

The image is an open interval containing zero. It is invariant under multiplication by two because  $\Theta(F(x)) = 2\Theta(x)$ . Since it contains both positive and negative values, it must be all of  $\mathbb{R}$ . Reflection and local uniqueness give oddness. For global uniqueness, let  $\tilde{\Theta}$  be a  $C^1$  solution of  $\tilde{\Theta} \circ Q = \frac{1}{2}\tilde{\Theta}$  with  $\tilde{\Theta}(p) = 0$  and  $\tilde{\Theta}'(p) = 1$ . Iterating,  $\tilde{\Theta}(x) = 2^n \tilde{\Theta}(Q^n(x))$ , and differentiability at  $p$  gives  $\tilde{\Theta}(Q^n(x)) = (Q^n(x) - p)(1 + o(1))$  as  $n \rightarrow \infty$ , because  $Q^n(x) \rightarrow p$  by [lemma 6.1](#). Hence

$$\tilde{\Theta}(x) = 2^n(Q^n(x) - p)(1 + o(1)) \rightarrow \Theta(x)$$

by (6.14), the sequence  $2^n(Q^n(x) - p)$  being convergent and therefore bounded; so  $\tilde{\Theta} = \Theta$ . Flatness is part of statement (K); flatness of the inverse follows by composition. The remainder is not identically zero because that would make  $F$  exactly affine near  $p$ , contradicting (1.1) and  $F(h) > 0$  for  $h > 0$ .  $\square$

**Corollary 6.6** (Exact iterate formula and inverse asymptotic). *For every  $x \in (0, 1)$  and  $n \geq 0$ ,*

$$F^n(x) = \Theta^{-1}(2^n \Theta(x)), \quad Q^n(x) = \Theta^{-1}(2^{-n} \Theta(x)). \quad (6.17)$$

*For every integer  $N \geq 2$ ,*

$$\boxed{Q^n(x) = \frac{1}{2} + 2^{-n} \Theta(x) + \mathcal{O}_{x,N}(2^{-Nn}) \quad (n \rightarrow \infty).} \quad (6.18)$$

*Thus every algebraic correction between  $2^{-n}$  and  $2^{-Nn}$  vanishes: the first correction is beyond all orders in the contraction scale.*

*Proof.* Equation (6.17) is iteration of the conjugacy. Since  $\Theta^{-1}(t) - p - t$  is flat, it is  $\mathcal{O}(t^N)$  for every  $N$ . Substitute  $t = 2^{-n} \Theta(x)$ .  $\square$

**Corollary 6.7** (Nonanalyticity of the linearizing coordinate). *Neither  $\Theta$  nor  $\Theta^{-1}$  is real analytic at the midpoint. Their Taylor series there are exactly  $x - p$  and  $p + t$ , respectively.*

*Proof.* If  $\Theta$  were analytic, its flat difference from  $x - p$  would vanish locally. The conjugacy would then force  $F(p + h) = p + 2h$  locally, contradicting  $F(p + h) = p + 2h - F(h)$ . The inverse case is equivalent.  $\square$

## 6.4 Exact escape clocks

**Corollary 6.8** (Dyadic escape time). *Fix  $a \in (0, 1/2)$ . For  $x \in (p, p + a)$ , let*

$$\tau_a(x) = \min\{n \geq 0 : F^n(x) \geq p + a\}.$$

*Then*

$$\tau_a(x) = \left\lceil \log_2 \frac{\Theta(p + a)}{\Theta(x)} \right\rceil. \quad (6.19)$$

*The left-half formula is obtained by absolute values. As  $x \rightarrow p$ ,*

$$\log_2 \frac{\Theta(p + a)}{|\Theta(x)|} = -\log_2 |x - p| + \log_2 \Theta(p + a) + \text{Flat}(x - p). \quad (6.20)$$

*Proof.* In the  $\Theta$ -coordinate, one forward iterate is multiplication by two. Therefore the first crossing of  $p + a$  is the first  $n$  with  $2^n \Theta(x) \geq \Theta(p + a)$ . Flatness gives the final expansion.  $\square$

## 7 A logarithmic Böttcher height at the endpoints

The coordinate  $\Theta$  solves the hyperbolic dynamics globally, but its endpoint inverse  $\Theta^{-1}(t)$  as  $|t| \rightarrow \infty$  is highly non-elementary. Independently of the linearization, the leading endpoint law implies a natural doubling height adapted to the endpoint logarithm.

Put

$$L = \log 2, \quad a = \frac{1}{2L}. \quad (7.1)$$

Equation (1.2) is equivalent to

$$-\log F(e^{-s}) = as^2(1 + o(1)) \quad (s \rightarrow \infty). \quad (7.2)$$

For  $0 < x < p$ , define

$$z(x) = \log(a[-\log x]), \quad \varepsilon(x) = z(F(x)) - 2z(x). \quad (7.3)$$

Then  $\varepsilon(x) \rightarrow 0$  as  $x \downarrow 0$ : with  $s = -\log x \rightarrow \infty$ , (7.2) gives  $z(F(x)) = \log(a \cdot as^2(1 + o(1))) = 2\log(as) + o(1) = 2z(x) + o(1)$ . The function  $\varepsilon$  is continuous on  $(0, p)$ , being composed of continuous functions.

**Theorem 7.1** (Endpoint dynamical height). *For every  $x \in (0, p)$ , the limit*

$$B(x) = \lim_{n \rightarrow \infty} 2^{-n} \log(a[-\log F^n(x)]) \quad (7.4)$$

*exists, is positive, and is continuous in  $x$ . It has the convergent cocycle series*

$$B(x) = z(x) + \sum_{k=0}^{\infty} 2^{-k-1} \varepsilon(F^k(x)) \quad (7.5)$$

*and satisfies the exact Böttcher-type equation*

$$B(F(x)) = 2B(x). \quad (7.6)$$

*Moreover, for every fixed  $x \in (0, p)$ ,*

$$-\log F^n(x) = 2L \exp(2^n B(x) + o(1)), \quad (7.7)$$

*or equivalently*

$$F^n(x) = \exp\{-2L \exp(2^n B(x) + o(1))\}. \quad (7.8)$$

*For  $p < x < 1$ , define  $B_R(x) = B(1 - x)$ . Reflection gives the explicit right-endpoint formula*

$$1 - F^n(x) = \exp\{-2L \exp(2^n B_R(x) + o(1))\}. \quad (7.9)$$

*Proof.* Let  $x_n = F^n(x)$ ,  $s_n = -\log x_n$ , and  $z_n = \log(as_n)$ . Since  $F(x) < x$  on the left half,  $x_n \downarrow 0$ . Equation (7.2) gives

$$z_{n+1} = 2z_n + \varepsilon(x_n), \quad \varepsilon(x_n) \rightarrow 0. \quad (7.10)$$

Thus

$$2^{-n} z_n = z_0 + \sum_{k=0}^{n-1} 2^{-k-1} \varepsilon(x_k),$$

and the series converges absolutely because  $\varepsilon(x_k) \rightarrow 0$ , so that its terms are eventually bounded by  $2^{-k-1}$ . This proves (7.4) and (7.5). On a compact  $K \subseteq (0, p)$ , the continuous functions  $F^k|_K$  decrease pointwise to the continuous limit zero. Dini's theorem makes this convergence uniform, i.e.  $\delta_k = \sup_K F^k \rightarrow 0$ . Consequently  $\sup_{x \in K} |\varepsilon(F^k(x))| \leq \sup_{0 < y \leq \delta_k} |\varepsilon(y)| \rightarrow 0$  by the limit  $\varepsilon(y) \rightarrow 0$  established above, and the geometrically weighted series in (7.5) converges uniformly on  $K$ . This proves continuity of  $B$ . Shifting the orbit proves (7.6).

The tail identity

$$2^n B(x) - z_n = \sum_{k=n}^{\infty} 2^{n-k-1} \varepsilon(x_k)$$

tends to zero. Therefore  $z_n = 2^n B(x) + o(1)$ , which gives (7.7) and (7.8). Also  $z_n \rightarrow +\infty$ . If  $B(x) < 0$ , the tail identity would force  $z_n \rightarrow -\infty$ ; if  $B(x) = 0$ , it would force  $z_n \rightarrow 0$ . Both alternatives are impossible, so  $B(x) > 0$ .

Finally, reflection iterates exactly:  $1 - F^n(x) = F^n(1 - x)$ . Applying (7.8) to  $1 - x \in (0, p)$  proves (7.9).  $\square$

## 7.1 A periodic quotient of the two dynamical coordinates

Let  $\Theta$  be the coordinate of theorem 6.5. Both  $B$  and  $-\Theta$  double under  $F$  on the left half. Their quotient is invariant along each orbit, and in the linear coordinate invariance becomes periodicity.

**Theorem 7.2** (Dynamical one-periodic factor). *Define, for  $u \in \mathbb{R}$ ,*

$$\mathcal{P}_{\text{dyn}}(u) = 2^{-u} B(\Theta^{-1}(-2^u)). \quad (7.11)$$

*Then  $\mathcal{P}_{\text{dyn}}$  is positive, continuous, and one-periodic. For every  $x \in (0, p)$ ,*

$$\boxed{B(x) = (-\Theta(x)) \mathcal{P}_{\text{dyn}}(\log_2[-\Theta(x)])}. \quad (7.12)$$

*Proof.* If  $x_u = \Theta^{-1}(-2^u)$ , then  $x_{u+1} = F(x_u)$  by theorem 6.5. Hence

$$\mathcal{P}_{\text{dyn}}(u+1) = 2^{-u-1} B(F(x_u)) = 2^{-u} B(x_u).$$

The representation (7.12) is the definition with  $u = \log_2[-\Theta(x)]$ .  $\square$

**Remark 7.3.** The repository's sharp endpoint expansion contains a nonconstant periodic function  $\Psi$  of a lower-Lambert phase. The periodic factor  $\mathcal{P}_{\text{dyn}}$  is different in construction: it is extracted from composition orbits and the global Schröder coordinate. Determining the transform between the two periodic objects is one of the main open problems proposed below.

## 8 Numerical experiments and exact symbolic checks

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**Role of computation.** All theorems above are proved independently of numerics. The included Python program uses exact rational arithmetic for cumulants, moments, and Legendre coefficients, and common-random-number Monte Carlo only as a consistency check of the pathwise derivative formula.

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## 8.1 Reproducibility package

This part's verification material is kept in the volume directory under `assets/q-fabiús-parameter-deformations/Fabiús_Flat_Parameter_Response_Dynamics/`, which holds

- `assets/q-fabiús-parameter-deformations/Fabiús_Flat_Parameter_Response_Dynamics/code/fabiús_q_response_experiments.py`, with detailed comments;
- exact CSV tables under `assets/q-fabiús-parameter-deformations/Fabiús_Flat_Parameter_Response_Dynamics/data/`;
- Monte Carlo validation tables in the same `data/` directory;
- the figures under `assets/q-fabiús-parameter-deformations/Fabiús_Flat_Parameter_Response_Dynamics/figures/`; and
- `assets/q-fabiús-parameter-deformations/Fabiús_Flat_Parameter_Response_Dynamics/NUMERICAL_README.txt` with the run parameters.

The reported run used  $10^6$  pathwise samples,  $1.5 \times 10^6$  common-random-number CDF samples, 32 geometric digits, and a central-difference step  $h = 0.005$ . The omitted tail weight near  $q = 1/2$  is approximately  $2^{-32}$ , far below the sampling resolution.

## 8.2 Pathwise moment checks

For even  $m$ , [theorem 4.1](#) gives

$$\dot{\mu}_m = m\mathbb{E}[Z^{m-1}V]. \quad (8.1)$$

The table compares exact values with the Monte Carlo estimator. The quoted uncertainty is one standard error.

| order | exact $\dot{\mu}_m$         | estimate                   | standard error        |
|-------|-----------------------------|----------------------------|-----------------------|
| 2     | $-7.4074074 \times 10^{-2}$ | $-7.400607 \times 10^{-2}$ | $8.98 \times 10^{-5}$ |
| 4     | $-8.1975309 \times 10^{-3}$ | $-8.188966 \times 10^{-3}$ | $1.54 \times 10^{-5}$ |
| 6     | $-9.7019760 \times 10^{-4}$ | $-9.693966 \times 10^{-4}$ | $2.61 \times 10^{-6}$ |
| 8     | $-1.2465992 \times 10^{-4}$ | $-1.246502 \times 10^{-4}$ | $4.52 \times 10^{-7}$ |
| 10    | $-1.7132447 \times 10^{-5}$ | $-1.714973 \times 10^{-5}$ | $8.07 \times 10^{-8}$ |
| 12    | $-2.4860408 \times 10^{-6}$ | $-2.491708 \times 10^{-6}$ | $1.48 \times 10^{-8}$ |
| 14    | $-3.7717453 \times 10^{-7}$ | $-3.785464 \times 10^{-7}$ | $2.78 \times 10^{-9}$ |

All discrepancies are below one standard error in this run.

## 8.3 Tangent CDF and antisymmetry

A common set of digits was evaluated at  $q = 1/2 + h$  and  $q = 1/2 - h$ , and the two empirical CDFs were differenced. The exact theory predicts  $G(1 - x) = -G(x)$ . Across the displayed grid the root-mean-square antisymmetry residual was approximately  $5.8 \times 10^{-3}$ , compared with a peak response magnitude of about 0.43 (the grid maximum of the estimate is 0.4337).

# 9 Further deductions and frontier conjectures

## 9.1 Uniform transseries at the flat fronts

The exact edge defect (3.8) transports the small-argument asymptotic of  $F_q$  into a two-variable boundary layer. At a left-front point  $(q_0, x_0)$  with  $q_0 = x_0 < 1/2$ , put  $q = x_0 + \epsilon$ . Then

$$\Delta(q, x_0) = \frac{1}{1 - q} F_q\left(\frac{\epsilon}{q}\right).$$

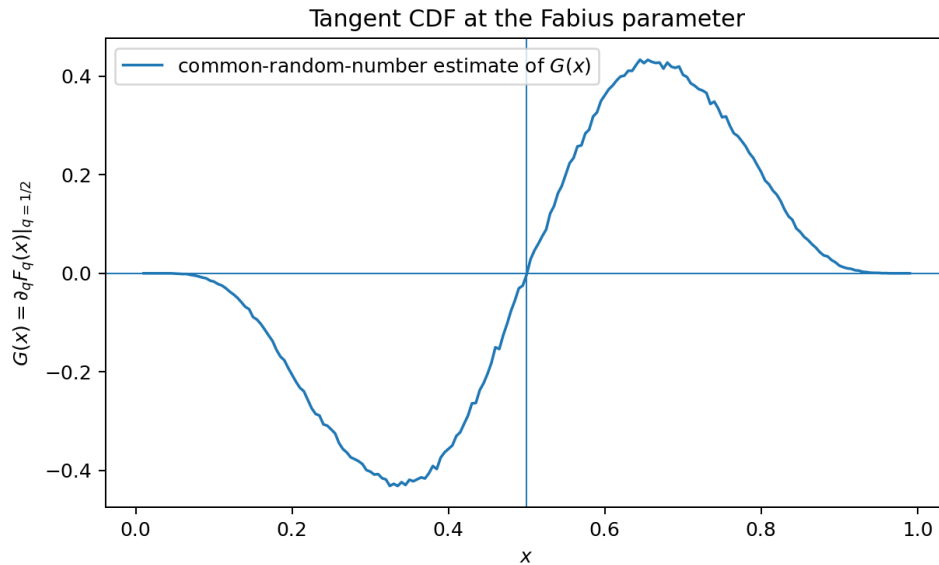


Figure 53: Common-random-number estimate of the tangent CDF. The exact function is antisymmetric about  $x = 1/2$ ; the visible small mismatch is sampling noise and central-difference error.

The flatness theorem says this is smaller than every power of  $\epsilon$ . General-base atomic-function results in the repository strongly suggest the following sharper form.

**Conjecture 9.1** (Uniform Lambert–periodic front expansion). *For every compact  $K \Subset (0, 1/2]$ , the endpoint expansion of  $F_q(y)$  admits a version uniform for  $q \in K$ , including a lower-Lambert phase and a nonconstant smooth periodic correction. Consequently, as  $\epsilon \downarrow 0$ ,*

$$\log \Delta(q_0 + \epsilon, q_0) \sim -\frac{\log^2(1/\epsilon)}{2 \log(1/q_0)}, \quad (9.1)$$

*with a complete expansion in logarithms, inverse logarithms, and a periodic phase whose coefficients vary smoothly with  $q_0$ .*

A proof would turn the geometric plateau boundary into a laboratory for beyond-all-orders parameter asymptotics. The dyadic point is especially interesting because two fronts meet there and the defect acquires the factor two in (3.9).

## 9.2 Analyticity above the critical ratio

**Conjecture 9.2** (Analytic supercritical parameter regime). *The map*

$$q \mapsto f_q$$

*is real analytic as a map into  $C^\infty(\mathbb{R})$  on  $(1/2, 1)$ , in the sense fixed in the proof of [theorem 3.4](#). On  $(0, 1/2]$ , the flat fronts of [theorem 3.4](#) are the complete obstruction to such function-space analyticity.*

The Fourier product is termwise analytic in  $q$ , but complexifying  $q$  destroys the simple real-frequency bounds because sinc factors can grow exponentially in imaginary directions. A successful proof will likely use a transfer-operator scale adapted to the overlapping regime, rather than a naive complex neighborhood of the Fourier product.



### 9.3 Asymptotics of the tangent Legendre spectrum

**Open problem 9.3** (Legendre phase and sign changes). *Determine the sharp asymptotic of  $a_{2n}$  in (5.10). In particular:*

1. *Does the sequence have infinitely many sign changes and infinitely many same-sign adjacent pairs?*
2. *Is its leading phase inherited from the endpoint Lambert-periodic correction, from the real-frequency shell profile of the sinc product, or from a new stationary phase?*
3. *Can one express its asymptotic through a saddle point involving the logarithmic derivative in (5.2)?*

The exact degree-16 sign failure is useful here: it rules out the simplest total-positivity ansatz and forces any asymptotic argument to retain a phase.

### 9.4 The dynamical periodic factor

**Conjecture 9.4** (Nonconstant dynamical cocycle). *The function  $\mathcal{P}_{\text{dyn}}$  in (7.11) is nonconstant and real analytic. Its Fourier coefficients can be obtained by an explicit cohomological transform of the Gamma–zeta Fourier coefficients of the repository’s endpoint periodic function  $\Psi$ .*

One route is to insert the sharp endpoint expansion into the cocycle  $z(F(x)) - 2z(x)$ , sum it along the orbit as in (7.5), and then convert the orbit phase to  $u = \log_2[-\Theta(x)]$ . No Fourier multiplier is asserted here: it must be derived only after the lower-Lambert phase has been converted to the dynamical phase. An unvalidated candidate in the submitted draft simplified to a non-small constant and is withdrawn.

### 9.5 A $q$ -family of global dynamical coordinates

For  $q < 1/2$ , the density is exactly constant near the midpoint, so  $F_q$  itself is exactly affine there with multiplier  $(1 - q)^{-1}$ . At  $q = 1/2$ , exact local affinity collapses to an affine Taylor jet plus a flat defect. For  $q > 1/2$ , the midpoint is no longer inside a plateau.

**Open problem 9.5** (Parameter-dependent Schröder coordinates). *Construct and analyze normalized coordinates  $\Theta_q$  satisfying*

$$\Theta_q(F_q(x)) = \lambda_q \Theta_q(x), \quad \lambda_q = f_q(1/2), \quad \Theta'_q(1/2) = 1. \quad (9.2)$$

*Determine the regularity of  $(q, x) \mapsto \Theta_q(x)$  across  $q = 1/2$ . The front theorem predicts a  $C^\infty$ , nonanalytic transition in the multiplier and potentially in the coordinate itself.*

For  $q < 1/2$ , the local coordinate can be chosen literally linear throughout the plateau. At the critical dyadic ratio, it is linear only to infinite order. This makes  $q = 1/2$  a natural bifurcation point between exact and asymptotic linearization.

### 9.6 Arithmetic of higher response jets

**Conjecture 9.6** (Cyclotomic denominator sharpness). *After removing the von Staudt–Clausen denominator of  $B_{2m}$ , the odd denominator of  $\partial_q^s \kappa_{2m}(1/2)$  contains the full cyclotomic contribution predicted by  $(2^{2m} - 1)^{s+1}$ , except for cancellations governed by explicit divisibility of a universal derivative polynomial. The corresponding moment and Legendre jets admit a finite prime-by-prime valuation recursion through complete Bell polynomials.*

This would extend the repository’s  $q$ -integer and Gaussian-binomial arithmetic from static moments to parameter response.

## 9.7 Thue–Morse finite models for susceptibility

The existing finite spline approximants have highest derivatives equal to finite Thue–Morse words. Differentiating their geometric weights with respect to  $q$  marks a digit position. This suggests a finite combinatorial model for  $\rho$ .

**Open problem 9.7** (Marked Thue–Morse spline approximation). *Construct finite signed splines  $\rho_N$  by differentiating the  $N$ -digit geometric-uniform box spline at  $q = 1/2$ . Prove:*

1. *an exact truncated-power formula whose coefficients are Thue–Morse signs weighted by a binary-position statistic;*
2. *sharp  $C^r$  convergence rates to  $\rho$ ;*
3. *a finite-product Richardson acceleration in the quarter-base  $q = 1/4$  scale; and*
4. *exact formulas for the top derivative plateaux and their signed value distribution.*

This program would unite the response theory with the repository’s strongest exact finite models rather than treating  $\rho$  only through infinite products.

## 10 Lean formalization roadmap

The new results divide naturally into formalization tranches.

### 10.1 Tranche A: parameter calculus for the probability law

1. Prove uniform summability of  $\partial_q^r[(1 - q)q^j]$  on compact parameter intervals.
2. Define the coupled random variable  $X(q, \omega)$  and its iterated parameter derivatives as uniformly convergent series.
3. Establish the pathwise test-function formula  $\partial_q \mathbb{E}\varphi(X_q) = \mathbb{E}[\varphi'(X_q)\partial_q X_q]$ .
4. Differentiate the affine independent-copy fixed point and package [theorem 4.2](#) in the zero-mass Kantorovich–Rubinstein space.

The repository already has the affine independent-copy operator and uniqueness theorem, so the new work is principally differentiability and the signed-measure norm.

### 10.2 Tranche B: joint smoothness and flat fronts

1. Extend the geometric sinc factorization with uniform compact-parameter decay estimates.
2. Obtain joint  $C^\infty$  density regularity by Fourier inversion.
3. Formalize the exact plateau and edge-defect identities.
4. Prove that a jointly smooth function vanishing on one side of a smooth boundary has all mixed derivatives zero on the boundary.
5. Combine strict positivity of the edge CDF with the analytic identity theorem to prove nonanalyticity.

The front-jet formulas (3.12)–(3.14) should then be short corollaries.

### 10.3 Tranche C: exact response algebra

1. Differentiate the CDF integral equation and verify (4.19)–(4.21).
2. Add the marked sinc-product identity.
3. Reuse the existing Bernoulli cumulant and complete Bell infrastructure for (5.5) and (5.6).
4. Define shifted-Legendre response coefficients and prove (5.11) by polynomial coefficient extraction.

### 10.4 Tranche D: smooth dynamics

1. Derive the diagonal inequalities from strict convexity and prove global attraction of the inverse midpoint.
2. Formalize the flat Koenigs linearization theorem 6.4, including the all-orders derivative estimates (6.8), first in a local one-dimensional real setting.
3. Globalize by finite inverse iteration and prove image  $\mathbb{R}$ , uniqueness, reflection, and the limit formula.
4. Deduce exact iterate conjugacy and the beyond-all-orders estimate.
5. Convert the endpoint quadratic law into the cocycle series (7.5) and the double-exponential orbit asymptotic.

The local linearization is the only substantial general-purpose analytic component; its proof in theorem 6.4 is elementary but is the largest single formalization item here. It is independent of Fabius-specific arithmetic and may be useful elsewhere in the library.

## 11 Conclusions

The geometric-uniform family contains two distinct kinds of smooth nonanalyticity. In the space variable, the classical Fabius function is smooth and nonanalytic because dyadic self-similarity transports endpoint flatness throughout its structure. In the parameter variable, the same endpoint flatness is pulled across the moving plateau boundary, producing a whole curve of smooth nonanalytic fronts. The critical dyadic point  $(q, x) = (1/2, 1/2)$  is the meeting point of these mechanisms.

The tangent measure  $\nu$  makes that critical point computable. It has a contractive susceptibility expansion, an exact refinement system, a marked infinite-sinc product, closed Bernoulli cumulants, Bell moment recurrences, and a rational Legendre spectrum. It is therefore a natural object for extending the repository’s exact arithmetic and finite Thue–Morse approximants into parameter space.

The global coordinate  $\Theta$ , now unconditional through the flat Koenigs linearization theorem 6.4, reveals a simple composition law beneath the nonanalytic graph: the whole open unit interval is smoothly conjugate to the real line under multiplication by two. The logarithmic height  $B$  captures the singular endpoint geometry, and its exact periodic quotient with  $\Theta$  gives a concrete target for connecting composition dynamics to the known Lambert–periodic endpoint expansion.

These results suggest a coherent next phase: formalize parameter response, construct marked Thue–Morse finite models, solve the tangent Legendre asymptotics, and identify the Fourier content of the dynamical periodic factor.

## A Derivation of selected exact coefficients

For illustration,  $P_2(y) = (3y^2 - 1)/2$ . Since the tangent measure has zero mass,

$$a_2 = 5 \cdot \frac{3}{2} \cdot 4\dot{\mu}_2 = 30 \left( -\frac{2}{27} \right) = -\frac{20}{9}.$$

For  $P_4(y) = (35y^4 - 30y^2 + 3)/8$ ,

$$\begin{aligned} a_4 &= 9(70\dot{\mu}_4 - 15\dot{\mu}_2) \\ &= 9 \left[ -70 \left( \frac{83}{10125} \right) + 15 \left( \frac{2}{27} \right) \right] = \frac{1088}{225}. \end{aligned}$$

Higher coefficients are obtained by the same triangular formula. The included script keeps all operations in exact rational arithmetic.

## B Numerical file inventory

The files below are kept in the volume directory under `assets/q-fabius-parameter-deformations/Fabius_Flat_Parameter_Response_Dynamics/`; the absorbed report's own source and PDF were deleted when it was merged, and git history is their archive.

- `assets/q-fabius-parameter-deformations/Fabius_Flat_Parameter_Response_Dynamics/code/fabius_q_response_experiments.py` — exact symbolic tables, pathwise Monte Carlo checks, common-random-number CDF experiment, and figure generation.
- `assets/q-fabius-parameter-deformations/Fabius_Flat_Parameter_Response_Dynamics/data/exact_cumulant_and_moment_jets.csv` — exact  $\kappa_n, \dot{\kappa}_n, \mu_n, \dot{\mu}_n$  through the configured order.
- `assets/q-fabius-parameter-deformations/Fabius_Flat_Parameter_Response_Dynamics/data/exact_tangent_legendre_coefficients.csv` — exact shifted-Legendre coefficients through degree 20.
- `assets/q-fabius-parameter-deformations/Fabius_Flat_Parameter_Response_Dynamics/data/monte_carlo_moment_derivative_validation.csv` — estimates, standard errors, and  $z$ -scores for  $\dot{\mu}_{2m}$ .
- `assets/q-fabius-parameter-deformations/Fabius_Flat_Parameter_Response_Dynamics/data/monte_carlo_legendre_validation.csv` — pathwise estimates and standard errors for the exact Legendre coefficients.
- `assets/q-fabius-parameter-deformations/Fabius_Flat_Parameter_Response_Dynamics/data/tangent_cdf_common_random_numbers.csv` — grid values of the central-difference estimate of  $G$  and its antisymmetry residual.
- `assets/q-fabius-parameter-deformations/Fabius_Flat_Parameter_Response_Dynamics/figures/tangent_cdf.png` — plot used in [figure 53](#).
- `assets/q-fabius-parameter-deformations/Fabius_Flat_Parameter_Response_Dynamics/figures/tangent_legendre_coefficients.png` — plot used in [figure 52](#).
- `assets/q-fabius-parameter-deformations/Fabius_Flat_Parameter_Response_Dynamics/NUMERICAL_README.txt` — reproduction parameters and a concise explanation of the numerical layer.

# References

The twelve absorbed reports each carried their own reference list. They are consolidated here: entries that name the same work are stated once, in the fullest wording any of the reports used, and every citation in the volume points at this single list.

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